

# GUV\_Analysis\_summary

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## Details

### DESCRIPTION

Loads the reorganised data from  
1Data/colour\_dance\_reorg.csv, saved from  
GUV\_organisedata.R, and generates plots to show the how  
dances changed as a function of light stimulus.

### INPUTS

colour\_dance\_reorg.csv,  
Edrich1979\_brightness-accuracy.csv

### OUTPUTS

2Results/phototaxis\_modlist.RData,  
2Results/MLE\_paired\_diffs.csv,  
2Results/MLEdetails\_paired\_diffs.RData,  
2Results/LRT\_paired\_diffs.csv,  
2Results/BF\_paired\_diffs.csv,

### REFERENCES

Edrich, W., Neumeyer, C. and von Helversen, O. (1979). "Anti-sun orientation" of bees with regard to a field of ultraviolet light. *J. Comp. Physiol.* 134, 151–157.

Rossel, S. and Wehner, R. (1984). Celestial orientation in bees: the use of spectral cues. *J. Comp. Physiol. A* 155, 605–613.

## Setting up

First load the functions and packages required for this project. These can all be found in `GUV_functions.R`, within the same Github repository.

```
source('GUV_functions.R')
```

Then load the data that will be used. This includes our own dance data, reorganised by `GUV_organisedata.R` as `colour_dance_reorg.csv`, as well as data from Edrich *et al.* (1979) on the relationship between accuracy (mean vector length), wavelength and intensity. This was manually extracted from Fig. 2 and saved in `"Edrich1979_brightness-accuracy.csv"`.

```
#Dance angle data
cd = read.table(file = '1Data/colour_dance_reorg.csv',
                header = T,
                sep = ',')
#Accuracy data
ed = read.table(file = '1Data/Edrich1979_brightness-accuracy.csv',
                header = T,
                sep = ',')
#Inspect each dataset
head(cd)
```

```
## measured_angle run      date bearing      angle      ID br
## 1           270   1 2016.05.31      90  1.57079633 2016.05.31.14.24
## 2            5   2 2016.05.31     355 -0.08726646 2016.05.31.14.24
## 3           20   3 2016.05.31     340 -0.34906585 2016.05.31.14.24
## 4          320   4 2016.05.31      40  0.69813170 2016.05.31.14.24
## 5           20   5 2016.05.31     340 -0.34906585 2016.05.31.14.24
## 6           35   6 2016.05.31     325 -0.61086524 2016.05.31.14.24
## colour      cet_time  sun_az  sun_el  sun_el_rad  sun_az_rad
## 1      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556 -2.647205
## 2      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556 -2.647205
## 3      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556 -2.647205
## 4      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556 -2.647205
## 5      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556 -2.647205
## 6      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556 -2.647205
```

```
head(ed)
```

```
##      xx intensity      nn accuracy wavelength
## 1 -31.208 2.15e+09 13.010      0.14      354
## 2 -20.263 3.69e+09 14.249      0.15      535
## 3  -8.217 6.67e+09 19.135      0.20      354
## 4  -1.342 9.36e+09 15.350      0.16      535
## 5  15.118 2.10e+10 30.769      0.33      354
## 6  24.548 3.35e+10 14.731      0.16      535
```

## Format data

Ensure that each variable has the correct data format for subsequent analysis.

```
cd = within(cd,
  {
    ID = as.factor(ID) # beedance identifier as a factor
    date = as.factor(date) # date as a factor
    signed_angle = Mod360.180(bearing) # bearing between -180 and 180
    angle = circular(rad(signed_angle), # bearing between -pi and pi
      rotation = 'clock') # circular format suppression
  }
)
u_id = with(cd, unique(ID)) # unique beedances
length(u_id)#169 beedances
```

```
## [1] 169
```

Identify dances in which all four stimulus conditions were used.

```
luc = sapply(u_id,
  FUN = IndCond,
  dt = cd)

#Most individuals that made it to bright green
full_ids = u_id[luc == 4]
length(full_ids)#19 individuals
```

```
## [1] 19
```

```
#extract just those individuals

#find data for full condition individuals
full_cd = subset(cd, ID %in% full_ids)
```

## Check dance length

Ensure that each variable has the correct data format for subsequent analysis.

```
mruns = aggregate(run ~ ID,
  data = cd,
```

```

FUN = max)
summary(mruns$run)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    14.00   26.00   33.00   36.83   43.00   100.00

```

## Plot the raw dance data

### Plot an example individual to show effect

For example we can look at the dance recorded on 2016.06.20 at 13:17.

```

par(pty = 's')
par(mfrow = c(1,1), mar = c(0,0,0,0))
PltDance(ids = '2016.06.20.13.17',
         dtf = cd,
         cx = 0.7)
with(subset(cd, ID %in% '2016.06.20.13.17' & run %in% c(10, 20, 40)),
     {
text(x = run*sin(as.numeric(angle)),
     y = run*cos(as.numeric(angle)),
     labels = paste(' run', run),
     col = c('green', 'darkgreen', 'magenta', 'purple')
        [ifelse(colour %in% 'g',
                 yes = ifelse(brightn %in% 'h', yes = 1, no = 2),
                 no = ifelse(brightn %in% 'h', yes = 3, no = 4))],
     cex = 0.7,
     adj = c(0,0))
})
legend(x = 'bottomleft',
       legend = c('Green Bright',
                  'Green Dim',
                  'UV Bright',
                  'UV Dim',
                  'Sun azimuth (North = Up)'),
       col = c('green',
               'darkgreen',
               'magenta',
               'purple',
               'darkred'),
       pch = c(21,21,21,21,NA),
       lty = c(NA, NA, NA, NA, 1),
       lwd = c(2,2,2,2,3),

```



```

rayleigh_tests = aggregate(angle~ID*brightn*colour+sun_az+sun_el,
                           data = cd,
                           FUN = Rayleigh_p
                           )
#add to the summary table and
#calculate inverse softplus kappa
mean_vectors = within(mean_vectors,
                      {
                        mu = deg(mle_estimates$angle[, 'mu'])
                        kappa = mle_estimates$angle[, 'kappa']
                        iskappa = inv_softplus(kappa)
                      }
)
#find the full condition individuals
mean_vectors_full = subset(mean_vectors,
                           subset = ID %in% full_ids)
rayleigh_tests_full = subset(rayleigh_tests,
                             subset = ID %in% full_ids)
#check how many dances would count as "oriented"
#all dances
with(rayleigh_tests,
     round(mean(angle <0.05) *100,digits = 2)
)

```

```
## [1] 88.43
```

```

#all dances with balanced data
with(rayleigh_tests_full,
     round(mean(angle <0.05) *100,digits = 2)
)

```

```
## [1] 88.24
```

```

#bright green (sun spot) dances
with(subset(rayleigh_tests,
            subset = brightn %in% 'h' & colour %in% 'g'),
     round(mean(angle <0.05) *100,digits = 2)
)

```

```
## [1] 94.44
```

```
#bright green (sun spot) dances with balanced data
with(subset(rayleigh_tests_full,
           subset = brightn %in% 'h' & colour %in% 'g'),
     round(mean(angle <0.05) *100,digits = 2)
)
```

```
## [1] 88.24
```

## Effect of brightness on mean vector length

### Entire dataset

The first thing we notice is that in the low intensity conditions the accuracy is generally lower than for the high intensity conditions.

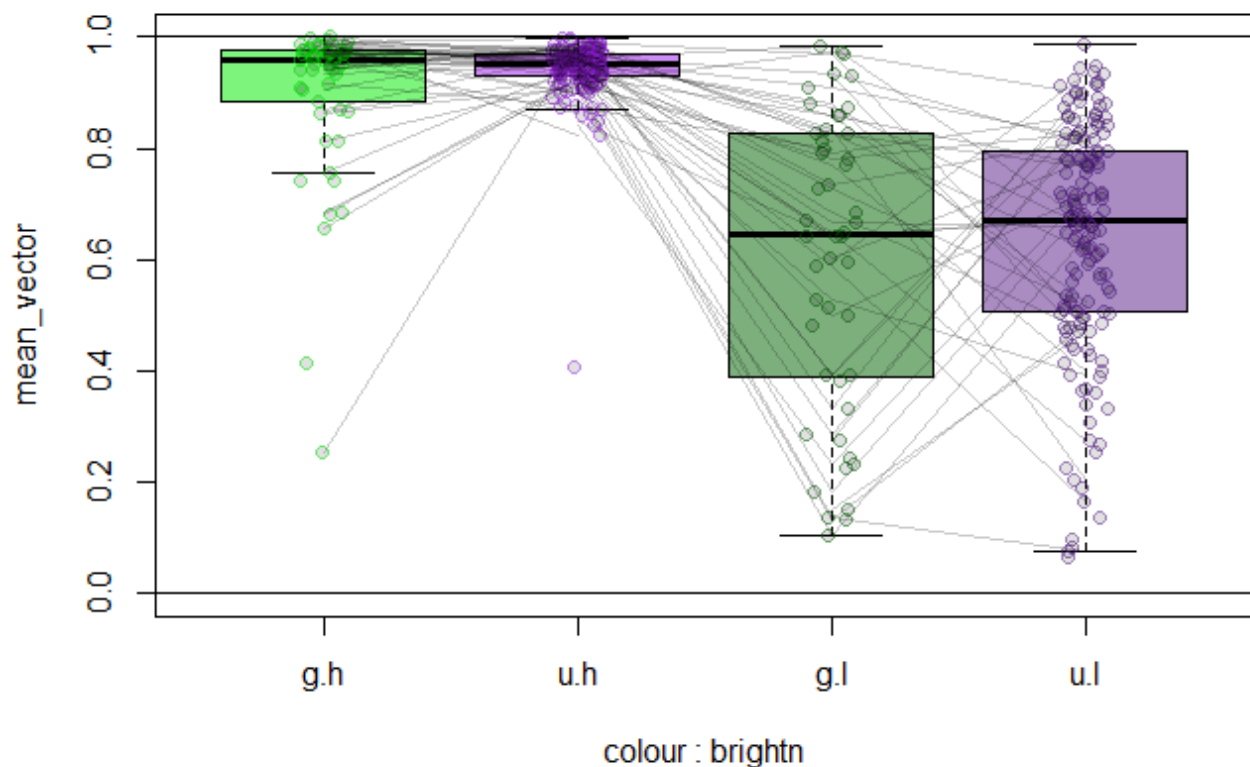
```
boxplot(mean_vector~colour*brightn,
        data = mean_vectors,
        ylim = c(0,1),
        col = adjustcolor(c('green2',
                             'purple',
                             'darkgreen',
                             'purple4'), alpha.f = 0.5),
        outline = FALSE)
stripchart(mean_vector~colour*brightn,
           data = mean_vectors,
           add = TRUE,
           vertical = TRUE,
           method = 'jitter',
           col = adjustcolor(c('green2',
                              'purple',
                              'darkgreen',
                              'purple4'), alpha.f = 0.5),
           bg = gray(level = 0,
                     alpha = 0.1),
           pch = 21)
#Identify stimuli
stim = apply(X = expand.grid(c = c('g','u'),b = c('h','l')),
            FUN = paste,
            collapse = '',
            MARGIN = 1)
#Plot lines linking dances by the same individual
invisible(
  {
    lapply(X = u_id,
          FUN = Plt_mvec
    )
  }
)
```



```

}))
abline(h = c(0,1))

```



## Print summary statistics

Use the quantile function to summarise the distribution of mean vectors within each combination of stimulus colour and brightness.

```

mv_agg = aggregate(mean_vector~colour*brightn,
                    data = mean_vectors,
                    FUN = quantile,
                    probs = c(0.5, 0.25, 0.75))
rownames(mv_agg) = stim
mv_agg = within(mv_agg, rm(colour, brightn))
mv_agg = t(apply(mv_agg, MARGIN = 1, unlist))
print(mv_agg,
      digits = 3)

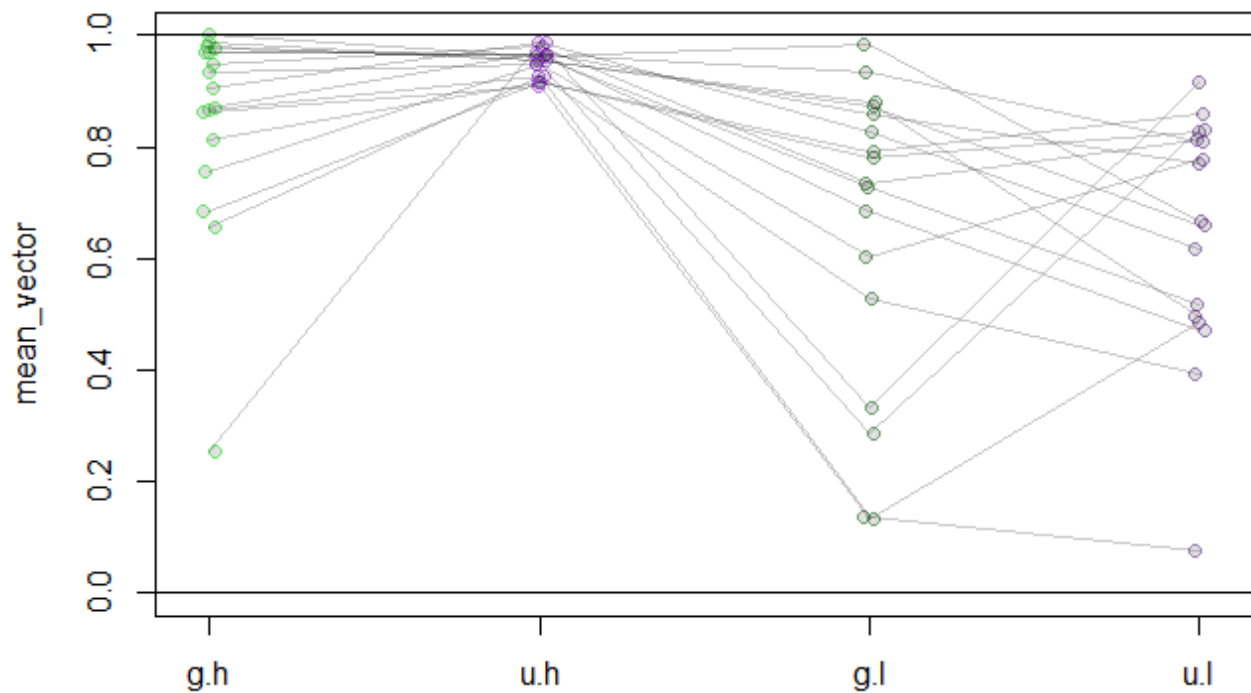
```

| ##    | mean_vector.50% | mean_vector.25% | mean_vector.75% |
|-------|-----------------|-----------------|-----------------|
| ## gh | 0.959           | 0.890           | 0.977           |
| ## uh | 0.952           | 0.930           | 0.971           |
| ## gl | 0.646           | 0.391           | 0.827           |
| ## ul | 0.671           | 0.505           | 0.795           |

## Within individuals (balanced cases only)

This effect is even clearer if we look at just the full condition dances.

```
stripchart(mean_vector~colour*brightn,
            data = mean_vectors_full,
            # add = TRUE,
            vertical = TRUE,
            method = 'jitter',
            jitter = 0.02,
            ylim = c(0,1),
            col = adjustcolor(c('green2',
                                'purple',
                                'darkgreen',
                                'purple4'), alpha.f = 0.5),
            bg = gray(level = 0,
                      alpha = 0.1),
            pch = 21)
#Plot lines linking dances by the same individual
invisible(
  {
    lapply(X = full_ids,
           FUN = Plt_mvec
          )
  })
abline(h = c(0,1))
```

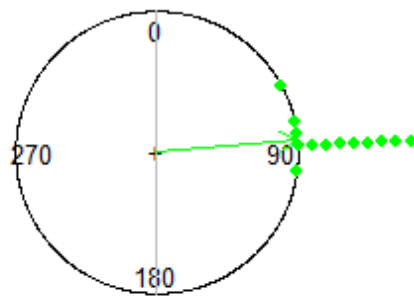


## Plot an example individual to show effect

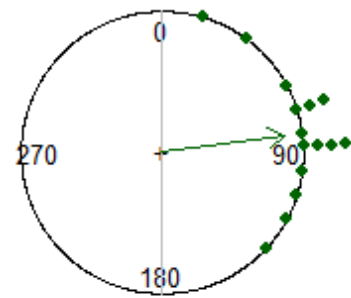
For example we can look at the dance recorded on 2016.06.20 at 13:17.

```
example_subs = subset(x = cd,
                      ID %in% '2016.06.20.13.17')
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'g' & brightn %in% 'h')$bearing,
      col = 'green',
      shrink = 1.5,
      title = 'Green Bright')
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'g' & brightn %in% 'l')$bearing,
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim')
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'u' & brightn %in% 'h')$bearing,
      col = 'magenta',
      shrink = 1.5,
      title = 'UV Bright')
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'u' & brightn %in% 'l')$bearing,
      col = 'purple',
```

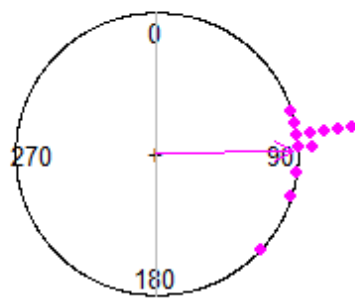
```
shrink = 1.5,  
title = 'UV Dim')
```



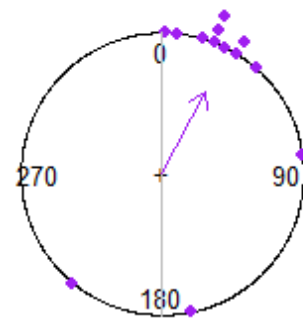
Green Bright



Green Dim



UV Bright



UV Dim

For this individual orientation is at a fixed angle to the light source ( $0^\circ$ ), but changes in consistency (green, dim), mean direction and bimodality (UV, dim) in response to changes in light source.

## Non-parametric tests on accuracy

Here non-parametric or parametric statistics could be revealing.

```
#Identify all effects as factors for aligned ranks transform  
mean_vectors_full = within(mean_vectors_full,  
    {  
        colour = as.factor(colour)  
        brightn = as.factor(brightn)  
        ID = as.factor(ID)  
    }  
)  
mean_vectors = within(mean_vectors,  
    {  
        colour = as.factor(colour)  
        brightn = as.factor(brightn)  
        ID = as.factor(ID)  
    })
```

```

    )
mv_full_art = art(formula = mean_vector~colour*brightn +
                  (1+colour+brightn|ID),#insufficient DF to es
                  data = mean_vectors_full)
sm_mv_full_art = summary(mv_full_art)

```

```

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

```

```

#check if rank alignment was successful
message('\nSuccessful rank alignment:\n',
      all(sapply(sm_mv_full_art[11], round) == 0),
      '\n')

```

```

##
## Successful rank alignment:
## TRUE

```

```

print(anova(mv_full_art), digits = 3)

```

```

## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

```

```

## Analysis of Variance of Aligned Rank Transformed Data
##
## Table Type: Analysis of Deviance Table (Type III Wald F tests with Kenw
## Model: Mixed Effects (lmer)
## Response: art(mean_vector)
##
##              F Df Df.res      Pr(>F)
## 1 colour      0.00452  1      16    0.94724
## 2 brightn     56.97631  1      16 1.1689e-06 ***
## 3 colour:brightn 2.99074  1      16    0.10298
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

We can successfully fit an aligned ranks model to compare across individuals. Note that the “boundary” level random effects indicate that individual effects were too small to estimate. In the individual linear models (e.g. `summary(artlm(mv_full_art, "brightn"))$varcor`) there may still be

substantial individual differences. The primary effect appears to be that of brightness, but there may also be effects of interest in the post-hoc contrasts.

```
cn_mv_full_art = sapply(X = c('brightn', 'colour', 'colour:brightn'),  
                        FUN = ART_extract,  
                        mod = mv_full_art,  
                        adjust = 'sidak')
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
print(cn_mv_full_art)
```

```
## $brightn  
## contrast estimate SE df t.ratio p.value  
## h - l 25.5 3.37 16 7.548 <.0001  
##  
## Results are averaged over the levels of: colour  
## Degrees-of-freedom method: kenward-roger  
##  
## $colour  
## contrast estimate SE df t.ratio p.value  
## g - u -0.353 5.25 16 -0.067 0.9472  
##  
## Results are averaged over the levels of: brightn  
## Degrees-of-freedom method: kenward-roger  
##  
## $`colour:brightn`  
## contrast estimate SE df t.ratio p.value  
## g,h - g,l 17.71 3.51 28.3 5.038 0.0001  
## g,h - u,h -11.35 4.84 22.2 -2.347 0.1579  
## g,h - u,l 22.00 6.29 16.0 3.497 0.0178  
## g,l - u,h -29.06 4.02 16.0 -7.231 <.0001  
## g,l - u,l 4.29 4.84 22.2 0.888 0.9455  
## u,h - u,l 33.35 3.51 28.3 9.490 <.0001  
##  
## Degrees-of-freedom method: kenward-roger  
## P value adjustment: sidak method for 6 tests
```

The post-hoc tests indicate a large difference between bright and dim conditions, as expected, but very little evidence for differences between colours. The interactions (for which there was low support in the overall model) reflect the pattern we expect, that all bright conditions have higher accuracy than all dim conditions, and that while the two dim conditions have similar accuracy, the bright UV bright condition may be slightly more accurate than the bright green condition ( $p = 0.158$ ).

## Parametric beta-distributed model of accuracy

To investigate this with greater statistical power we would need an appropriate parametric model for bounded variables. One approach is to use the beta distribution, which is defined between (but not at) 0 and 1.

### Beta model for accuracy

```
### Maximal formula with all random effects -----
#set up model formula with indiv effects on light type
frm_bmod_maxRE = bf(mean_vector~brightn*colour + # light types affect accu
                    (1 + brightn*colour|ID), # each individual is
                    phi ~ brightn*colour + # light types affect variance in accu
                    (1 + brightn*colour|ID), # some individuals vary mor
                    family = brms::Beta())

#suggest prior values
pr_mu_Intercept = list(dist = 'normal',
                       mu = 0,
                       sd = 1)#deconstrain to beat divergences?
pr_phi_Intercept = list(dist = 'normal',
                       mu = 1,
                       sd = 1)
pr_mu_b = list(dist = 'normal',
              mu = 0,
              sd = 1)#deconstrain to beat divergences?
pr_phi_b = list(dist = 'normal',
              mu = 0,
              sd = 1)
pr_mu_sd_Intercept = list(dist = 'student_t',
                         df = 3,
                         mu = 0,
                         sd = 0.1)#constrain to beat divergences?
pr_phi_sd_Intercept = list(dist = 'student_t',
                          df = 3,
                          mu = 0,
                          sd = 0.1)#constrain to beat divergences?
pr_mu_sd = list(dist = 'student_t',
```

```

        df = 3,
        mu = 0,
        sd = 0.1)#constrain to beat divergences?
pr_phi_sd = list(dist = 'student_t',
                df = 3,
                mu = 0,
                sd = 0.1)#constrain to beat divergences?

### Assign priors -----

#set priors on:
# - all coefficients to a normal(0,1) = 95% probability of values 0.12-0.8
# - all grouping paramters to a student_t(df = 3, mu = 0, sigma = 2.5) = v
# - all other parameters (?) to a normal(0,10)= 95% probability of values

pr_expression = expression({
  prior = sapply(class,
    switch,
    Intercept = with(pr_mu_Intercept,
      {paste0(dist,'(',mu,',',sd,')')}),
    #the general prior for coefficients (should centre on 0)
    b = with(pr_mu_b,
      {paste0(dist,'(',mu,',',sd,')')}),
    #the general prior for standard deviations (should be Lon
    sd = with(pr_mu_sd,
      {paste0(dist,'(',df,',',mu,',',sd,')')}),
    cor = 'lkj(1)', # The only supported prior for correlatio
    'normal(0,1)' # with this many params need more informati
  )
#special priors
#mu sd intercept
prior[dpar %in% '' & class %in% 'sd' & coef %in% ''] =
  with(pr_mu_sd_Intercept,
    {paste0(dist,'(',df,',',mu,',',sd,')')})
#phi b
prior[dpar %in% 'phi' & class %in% 'b'] =
  with(pr_phi_b,
    {paste0(dist,'(',mu,',',sd,')')})
#phi intercept
prior[dpar %in% 'phi' & class %in% 'b' & coef %in% ''] =
  with(pr_phi_Intercept,
    {paste0(dist,'(',mu,',',sd,')')})
#phi sd
prior[dpar %in% 'phi' & class %in% 'sd'] =
  with(pr_phi_sd,
    {paste0(dist,'(',df,',',mu,',',sd,')')})
#phi sd intercept
prior[dpar %in% 'phi' & class %in% 'sd' & coef %in% ''] =

```





```
## Run the models -----
system.time( #takes about 2 minutes
{
md_beta = brm(formula = frm_bmod_maxRE,
               prior = priors_bmod_mxRE,
               data = within(mean_vectors_full,
                             {
                               mean_vector = (mean_vector * (length(mean_v
                             }
                             ),
               family = Beta(link = 'logit',
                             link_phi = 'log'),
               iter = 2e3, # only small number necessary
               control = list( adapt_delta = 0.99 ), # divergences are h
               cores = 4,
               backend = 'cmdstanr',
               silent = 2, #don't print lots of iteration information
               refresh = 0
)
}
)
```

```
## Loading required namespace: rstan
```

```
##      user  system elapsed
##      3.91    0.50    87.87
```

```
#some divergent transitions, but rhat and ESS look good
rh_beta = rhat(md_beta)
sm_beta = summary(md_beta)
print(summary(rh_beta), digits= 4)#all  $\leq 1.01$ 
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.9994  1.0003  1.0009  1.0012  1.0017  1.0048
```

```
print(sm_beta$fixed[, c(1, 3:4)], digits = 3)
```

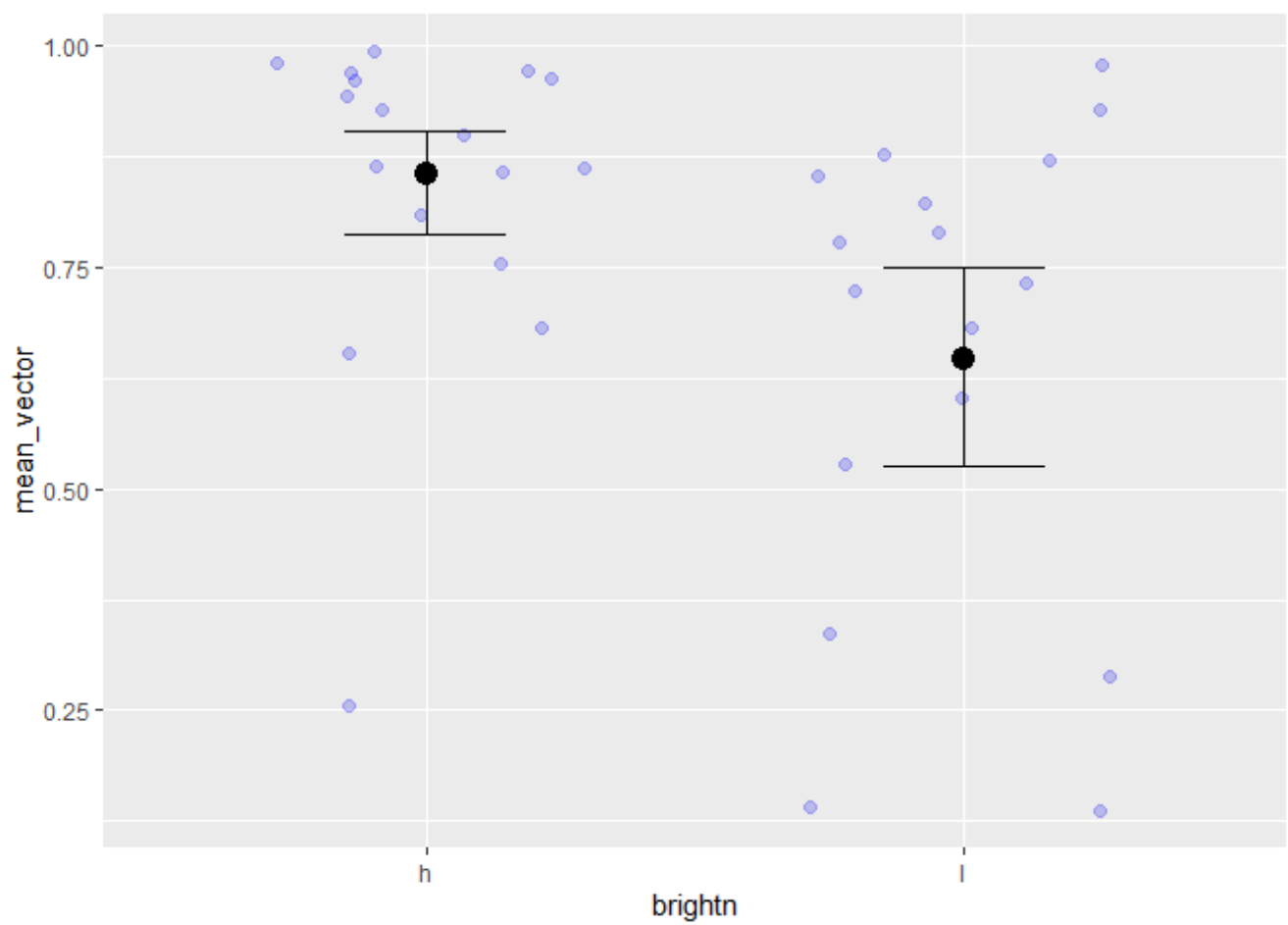
```
##              Estimate 1-95% CI u-95% CI
## Intercept          1.774    1.303    2.2218
## phi_Intercept       1.812    1.203    2.3751
## brightn1            -1.169   -1.787   -0.5245
```

|                         |        |        |         |
|-------------------------|--------|--------|---------|
| ## colouru              | 0.961  | 0.425  | 1.4764  |
| ## brightnl:colouru     | -0.944 | -1.659 | -0.2101 |
| ## phi_brightnl         | -0.755 | -1.557 | 0.0683  |
| ## phi_colouru          | 2.022  | 0.972  | 3.0021  |
| ## phi_brightnl:colouru | -1.132 | -2.314 | 0.2044  |

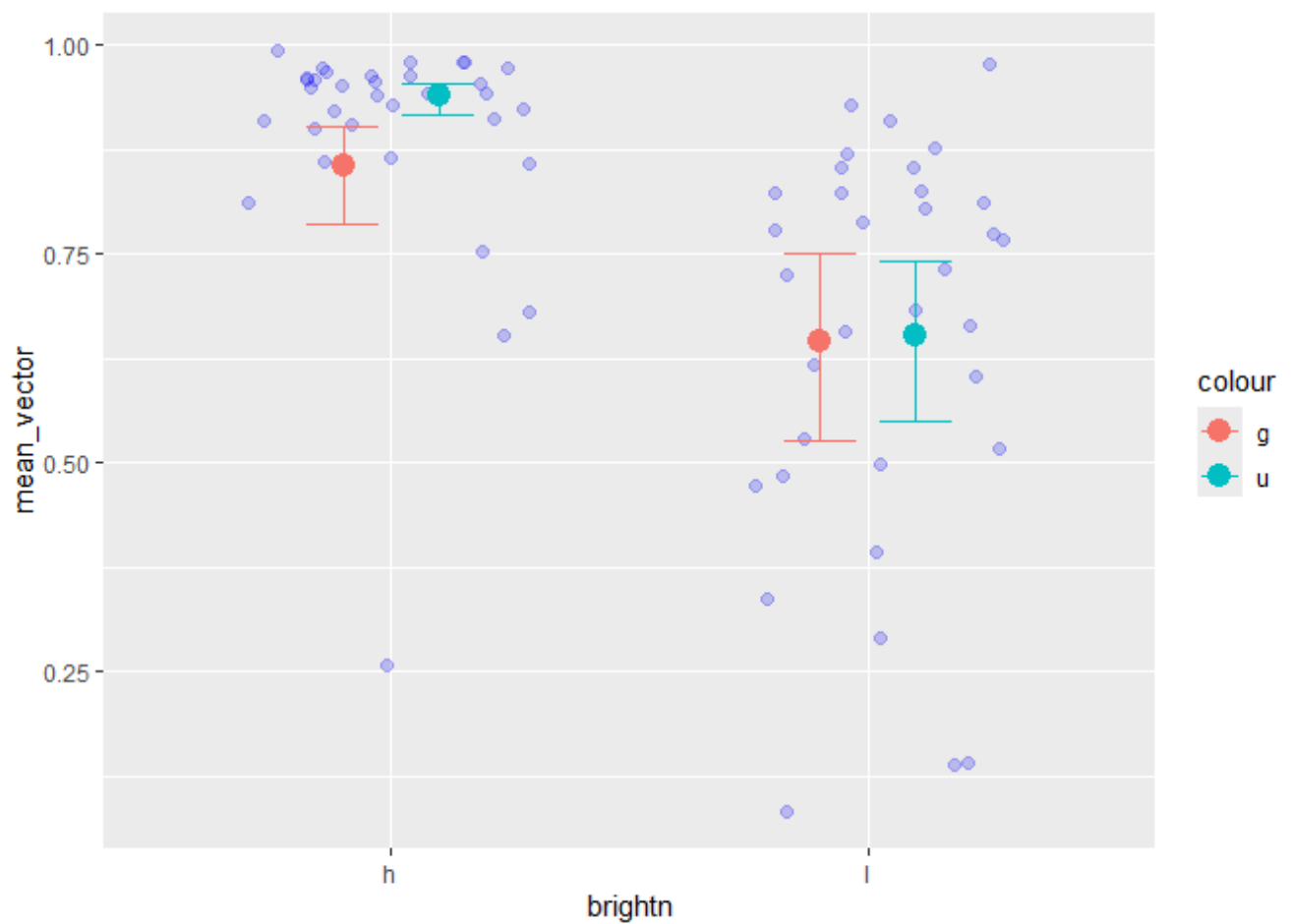
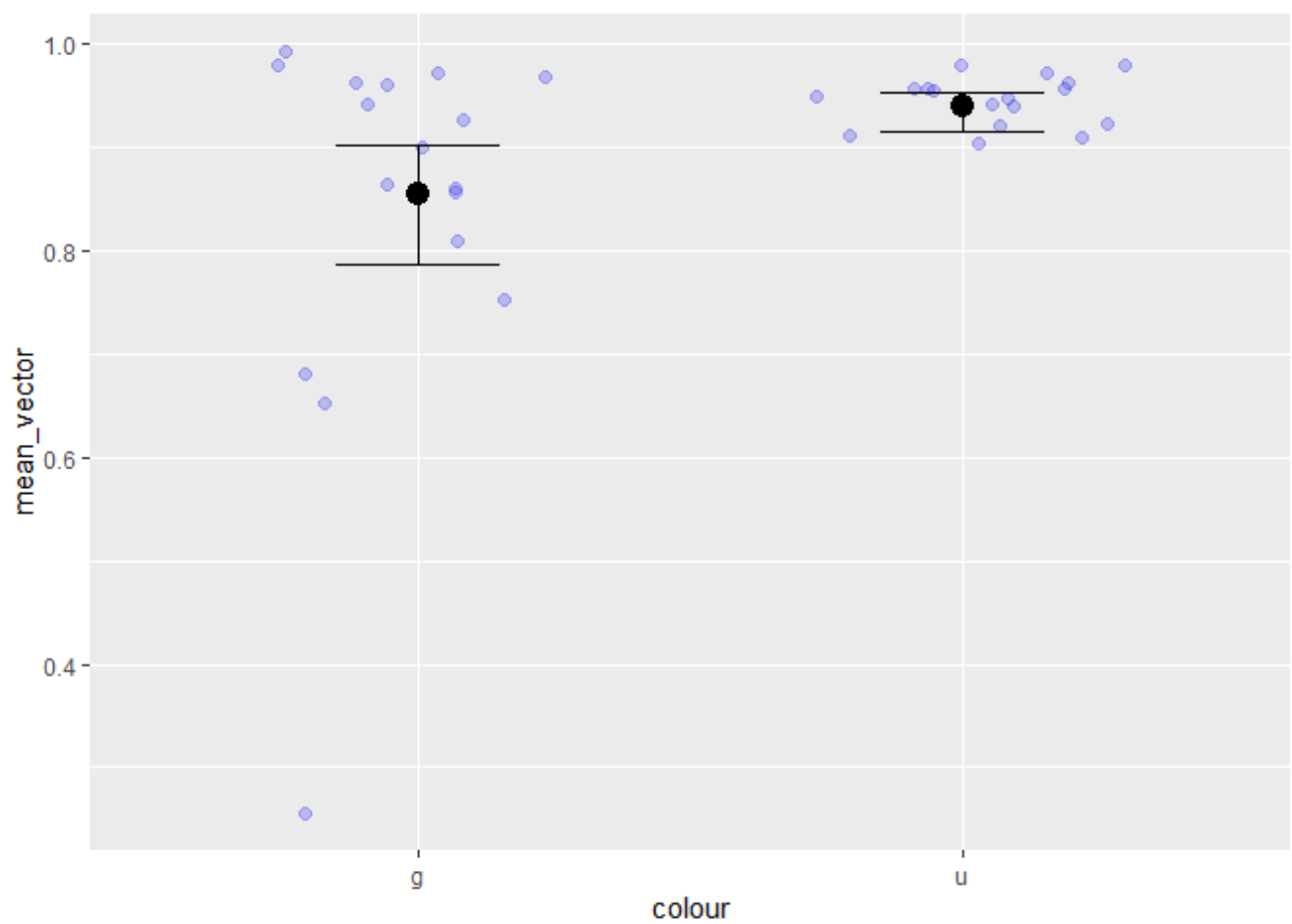
Extract the model's predictions and compare with the original data.

```
# Extract model predictions -----
beta_cond_eff = conditional_effects(md_beta)
#Plot including individual data points
plot(x = beta_cond_eff,
     points = TRUE,
     point_args = list(width = 0.3,
                       col = adjustcolor('blue',
                                          alpha.f = 0.2),
                       width = 0.3
                     ),
     ylim = c(0,1),
     ask = FALSE
  )
```

```
## Ignoring unknown labels:
## • fill : "NA"
## • colour : "NA"
## Ignoring unknown labels:
## • fill : "NA"
## • colour : "NA"
```



```
## Ignoring unknown labels:
## • fill : "NA"
## • colour : "NA"
## Ignoring unknown labels:
## • fill : "NA"
## • colour : "NA"
```



This seems to suggest that dances were more accurate in the bright UV condition, which the aligned-ranks test was sceptical of. We can quantify this difference by extracting and comparing the parameter estimates from each (post-warmup) iteration of the model.

```

draws_beta = prepare_predictions(md_beta,
                                re_formula = NA)#just population level ef
#for the average vector length
draws_mu = draws_beta$dparams$mu$fe$b
#for variance in vector length
draws_phi = draws_beta$dparams$phi$fe$b
#N.B. Could also just calculate predictions, then subtract
contr_mu = with(data.frame(draws_mu),
{
  data.frame(
    #bright green -> dim green = low brightness effect
    gh_gl = b_brightnl,
    #bright UV -> dim UV = dim + UVdim - UV effect
    uvh_uvl = b_brightnl + `b_brightnl.colouru` - b_colouru
    #bright green -> bright UV = UV effect
    gh_uvh = b_colouru,
    #bright green -> dim UV = UV + dim + UVdim
    gh_uvl = b_brightnl + b_colouru + `b_brightnl.colouru`
    #dim green -> dim UV = UV effect + UVdim - dim effect
    gl_uvl = b_colouru + `b_brightnl.colouru` - b_brightnl
  )
})
#On the original scale
contr_rho = with(data.frame(draws_mu),
{
  data.frame(
    #bright green -> dim green = low brightness effect
    gh_gl = plogis(b_Intercept + b_brightnl) -
      plogis(b_Intercept),
    #bright UV -> dim UV = dim + UVdim - UV effect
    uvh_uvl = plogis(b_Intercept + b_brightnl + `b_brightnl.colouru`) -
      plogis(b_Intercept + b_colouru),
    #bright green -> bright UV = UV effect
    gh_uvh = plogis(b_Intercept + b_colouru) -
      plogis(b_Intercept),
    #bright green -> dim UV = UV + dim + UVdim
    gh_uvl = plogis(b_Intercept + b_brightnl +
      b_colouru + `b_brightnl.colouru`) -
      plogis(b_Intercept),
    #dim green -> dim UV = UV effect + UVdim - dim effect
    gl_uvl = plogis(b_Intercept + b_colouru +
      `b_brightnl.colouru`) -
      plogis(b_Intercept + b_brightnl)
  )
})

```

```

contr_phi = with(data.frame(draws_phi),
  {
    data.frame(
      #bright green -> dim green = low brightness effect
      gh_gl = b_phi_brightnl,
      #bright UV -> dim UV = dim + UVdim - UV effect
      uvh_uvl = b_phi_brightnl + `b_phi_brightnl.colouru` -
      #bright green -> bright UV = UV effect
      gh_uvh = b_phi_colouru,
      #bright green -> dim UV = UV + dim + UVdim
      gh_uvl = b_phi_brightnl + b_phi_colouru + b_phi_bright
      #dim green -> dim UV = UV effect + UVdim - dim effect
      gl_uvl = b_phi_colouru + `b_phi_brightnl.colouru` - b_
    )
  }
)

contr_all = cbind(contr_mu, contr_phi)
quant_all = apply(X = contr_all,
  MARGIN = 2,
  FUN = quantile,
  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
)

quant_mu = apply(X = contr_mu,
  MARGIN = 2,
  FUN = quantile,
  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
)

quant_rho = apply(X = contr_rho,
  MARGIN = 2,
  FUN = quantile,
  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
)

quant_phi = apply(X = contr_phi,
  MARGIN = 2,
  FUN = quantile,
  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
)

par(mfrow = c(1, length(contr_mu)))
for(ct in 1:length(contr_mu) )
{
  VertHist(data = contr_mu[,ct],
    main = gsub(pattern = '_',
      replacement = ' → ',
      x = names(contr_mu)[ct]),
    axes = FALSE,
    col = gray(0.75),
    ylim = c(-1, 1)*4,

```

```

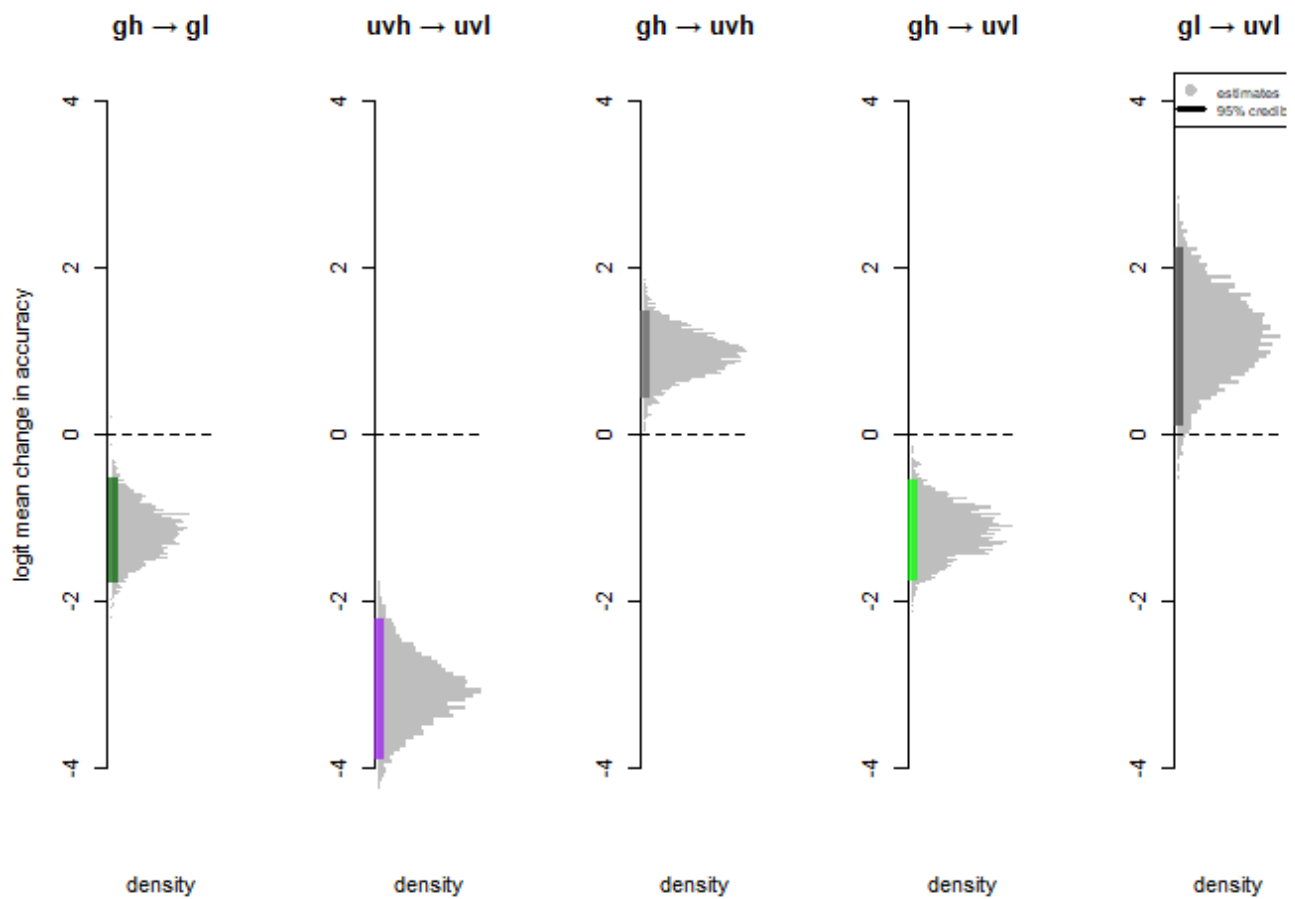
        ylab = if(ct ==1){'logit mean change in accuracy'}else{''}
    )
axis(2)

abline(h = 0,
       lty = 2)
with(data.frame(t(quant_mu[,ct])),
     {
        arrows(x0 = 0,
              y0 = X2.5.,
              y1 = X97.5.,
              length = 0,
              angle = 90,
              lwd = 5,
              col = adjustcolor(c('darkgreen',
                                'purple',
                                'gray40',
                                'green',
                                'gray25'
                                ), alpha = 0.7)[ct],
              lend = 'butt'
        )
    }
)
}

legend(x = 'topleft',
       legend = c('estimates',
                  '95% credible interval'),
       pch = c(20, NA),
       lty = c(NA, 1),
       lwd = c(5, 3),
       col = c('gray', 'black'),
       cex = 0.7
       )

```





```

par(mfrow = c(1, length(contr_phi)))
for(ct in 1:length(contr_phi) )
{
  VertHist(data = contr_phi[,ct],
            main = gsub(pattern = '_',
                        replacement = ' → ',
                        x = names(contr_phi)[ct]),

            axes = FALSE,
            col = gray(0.75),
            ylim = c(-1, 1)*6,
            ylab = if(ct == 1){'log change in accuracy precision'}else{'')
            )

  axis(2)

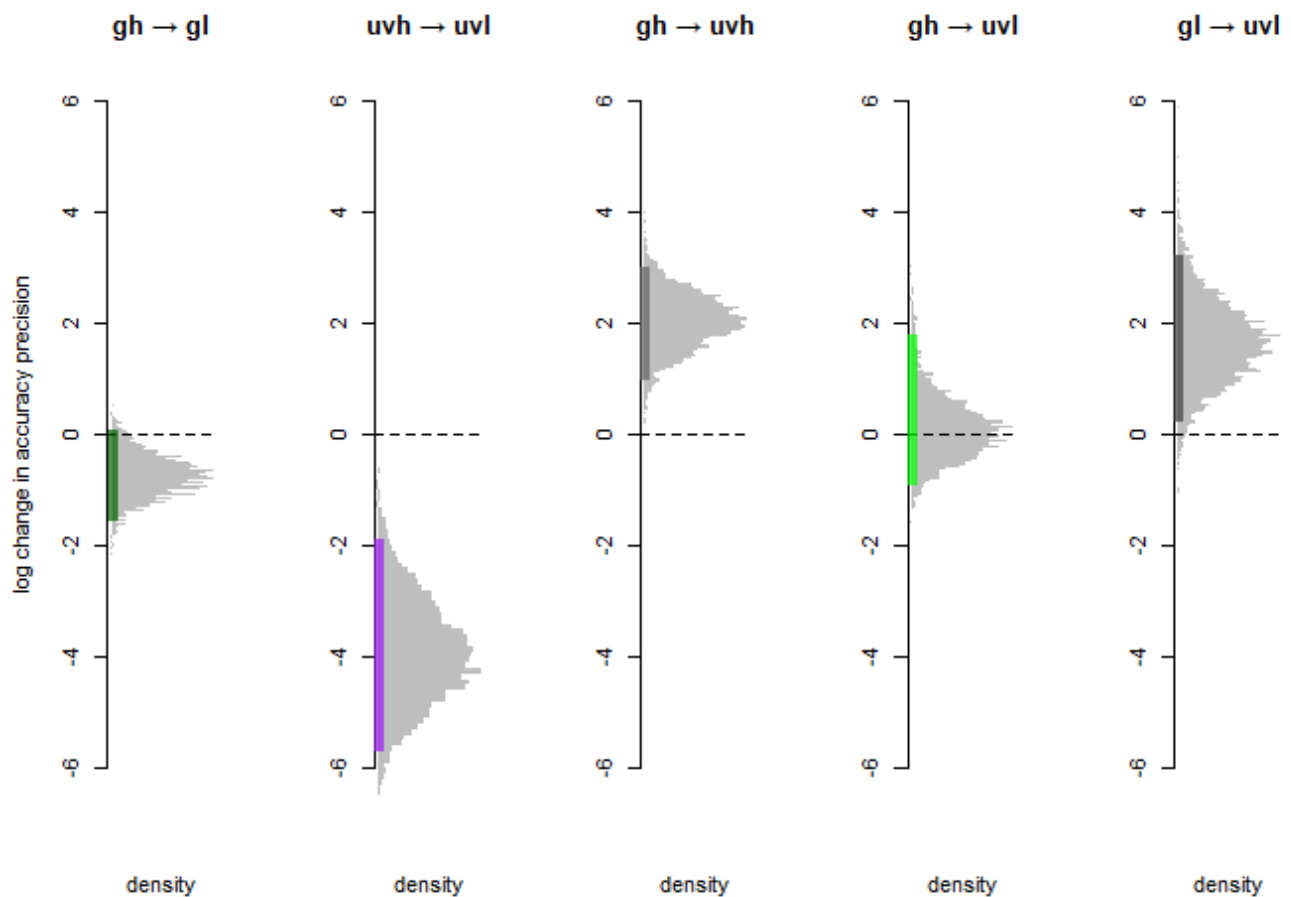
  abline(h = 0,
          lty = 2)
  with(data.frame(t(quant_phi[,ct])),
        {
          arrows(x0 = 0,
                 y0 = X2.5.,
                 y1 = X97.5.,
                 length = 0,
                 angle = 90,
                 lwd = 5,
                 col = adjustcolor(c('darkgreen',
                                     'purple',

```

```

        'gray40',
        'green',
        'gray25'
    ), alpha = 0.7)[ct],
    lend = 'butt'
)
}

```



```

#can also calculate these with emmeans
em_beta = emmeans(object = md_beta,
                  specs = ~brightn*colour)
ct_beta_mu = contrast(object = em_beta,
                    specs = ~brightn*colour,
                    method = 'pairwise',
                    side = c(1,-1,1,-1,-1,1)
                    )
print(ct_beta_mu, digits = 3)

```

```

## contrast estimate lower.HPD upper.HPD
## h g - l g 1.1648 0.547 1.81
## h g - h u -0.9635 -1.509 -0.46
## h g - l u 1.1502 0.555 1.77

```

```
##      l g - h u   -2.1301    -2.720    -1.54
##      l g - l u   -0.0117    -0.615     0.57
##      h u - l u    2.1141     1.575     2.59
##
## Point estimate displayed: median
## Results are given on the log odds ratio (not the response) scale.
## HPD interval probability: 0.95
```

```
#collect and estimate p values from samples
empirical.p = function(x, tails = 2){ifelse(test = median(x)>0,
                                             yes = mean(x<0)*tails,
                                             no = mean(x>0)*tails)
}

p_beta_mu = apply(X = contr_mu, MARGIN = 2, FUN = empirical.p, tails = 2)
p_beta_phi = apply(X = contr_phi, MARGIN = 2, FUN = empirical.p, tails = 2)
beta_all_res = data.frame(rho_lower = quant_rho[1,],
                           rho_median = quant_rho[2,],
                           rho_upper = quant_rho[3,],
                           mu_lower = quant_mu[1,],
                           mu_median = quant_mu[2,],
                           mu_upper = quant_mu[3,],
                           p_mu = p_beta_mu,
                           p_mu_adj = p.adjust(p = p_beta_mu,
                                                  method = 'BH'),
                           phi_lower = quant_phi[1,],
                           phi_median = quant_phi[2,],
                           phi_upper = quant_phi[3,],
                           p_phi = p_beta_phi,
                           p_phi_adj = p.adjust(p = p_beta_phi,
                                                  method = 'BH')
                           )
print(beta_all_res[,c(2,5,7,10)], digits = 3)
```

```
##      rho_median mu_median  p_mu phi_median
## gh_g1      -0.2066    -1.165 0.0020    -0.7592
## uvh_uvl     -0.5228    -3.078 0.0000    -3.9487
## gh_uvh       0.0838     0.964 0.0005     2.0306
## gh_uvl      -0.2025    -1.150 0.0005     0.0758
## gl_uvl       0.2087     1.181 0.0290     1.6318
```

Investigate effect of sun elevation

```

#set up model formula with indiv effects on light type
frm_elemod_maxRE = bf(mean_vector~ cos(sun_el_rad) + # accuracy is affected
                        (1 |ID), # each individual is more or less ori
                        phi ~ cos(sun_el_rad) + # accuracy is affected by projection
                        (1 |ID), # some individuals vary more in accuracy th
                        family = brms::Beta())

```

*#suggest prior values*

```

pr_mu_Intercept = list(dist = 'normal',
                        mu = 0,
                        sd = 1)#deconstrain to beat divergences?
pr_phi_Intercept = list(dist = 'normal',
                        mu = 1,
                        sd = 1)
pr_mu_b = list(dist = 'normal',
               mu = 0,
               sd = 1)#deconstrain to beat divergences?
pr_phi_b = list(dist = 'normal',
                mu = 0,
                sd = 1)
pr_mu_sd_Intercept = list(dist = 'student_t',
                          df = 3,
                          mu = 0,
                          sd = 0.1)#constrain to beat divergences?
pr_phi_sd_Intercept = list(dist = 'student_t',
                           df = 3,
                           mu = 0,
                           sd = 0.1)#constrain to beat divergences?
pr_mu_sd = list(dist = 'student_t',
                df = 3,
                mu = 0,
                sd = 0.1)#constrain to beat divergences?
pr_phi_sd = list(dist = 'student_t',
                 df = 3,
                 mu = 0,
                 sd = 0.1)#constrain to beat divergences?

```

### Assign priors -----

*#set priors on:*

```

# - all coefficients to a normal(0,1) = 95% probability of values 0.12-0.8
# - all grouping paramters to a student_t(df = 3, mu = 0, sigma = 2.5) = v
# - all other parameters (?) to a normal(0,10)= 95% probability of values

```

```

pr_expression = expression({
  prior = sapply(class,
                 switch,

```

```

Intercept = with(pr_mu_Intercept,
  {paste0(dist, '(', mu, ', ', sd, ')')}),
#the general prior for coefficients (should centre on 0)
b = with(pr_mu_b,
  {paste0(dist, '(', mu, ', ', sd, ')')}),
#the general prior for standard deviations (should be long)
sd = with(pr_mu_sd,
  {paste0(dist, '(', df, ', ', mu, ', ', sd, ')')}),
cor = 'lkj(1)', # The only supported prior for correlation
'normal(0,1)' # with this many params need more information
)
#special priors
#mu sd intercept
prior[dpar %in% '' & class %in% 'sd' & coef %in% ''] =
  with(pr_mu_sd_Intercept,
    {paste0(dist, '(', df, ', ', mu, ', ', sd, ')')}),
#phi b
prior[dpar %in% 'phi' & class %in% 'b'] =
  with(pr_phi_b,
    {paste0(dist, '(', mu, ', ', sd, ')')}),
#phi intercept
prior[dpar %in% 'phi' & class %in% 'b' & coef %in% ''] =
  with(pr_phi_Intercept,
    {paste0(dist, '(', mu, ', ', sd, ')')}),
#phi sd
prior[dpar %in% 'phi' & class %in% 'sd'] =
  with(pr_phi_sd,
    {paste0(dist, '(', df, ', ', mu, ', ', sd, ')')}),
#phi sd intercept
prior[dpar %in% 'phi' & class %in% 'sd' & coef %in% ''] =
  with(pr_phi_sd_Intercept,
    {paste0(dist, '(', df, ', ', mu, ', ', sd, ')')}),
}
)

#model with all random effects
priors_elemod_mxRE = within(get_prior(formula = frm_elemod_maxRE,
  data = mean_vectors_full),
  expr = eval(pr_expression))
#remove additional priors added by get_prior but not used by brms
priors_elemod_mxRE = subset(priors_elemod_mxRE, !(coef %in% ''))
print(priors_elemod_mxRE)#all parameters are accounted for

```

```

##           prior class      coef group resp dpar nlpar lb ub tag
##      normal(0,1)      b cossun_el_rad
## student_t(3,0,0.1)    sd      Intercept    ID

```

```
##          normal(0,1)          b cossun_el_rad          phi
## student t(3.0.0.1)          sd          Intercept          ID          phi
```

```
system.time( #takes about 2 minutes
{
md_ele = brm(formula = frm_elemod_maxRE,
              prior = priors_elemod_mxRE,
              data = within(mean_vectors_full,
                             {
                               mean_vector = (mean_vector * (length(mean_v
                               }
                             ),
              family = Beta(link = 'logit',
                            link_phi = 'log'),
              iter = 2e3, # only small number necessary
              control = list( adapt_delta = 0.99 ), # divergences are h
              cores = 4,
              backend = 'cmdstanr',
              silent = 2, #don't print lots of iteration information
              refresh = 0
)
}
)
```

```
##      user  system elapsed
##      0.64    0.37    32.56
```

```
#too many divergent transitions, check chains
rh_ele = rhat(md_ele)
sm_ele = summary(md_ele)
print(summary(rh_ele), digits= 4)#all ≤1.01
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.9997  1.0002  1.0007  1.0009  1.0013  1.0039
```

```
print(sm_ele$fixed[, c(1, 3:4)], digits = 3)
```

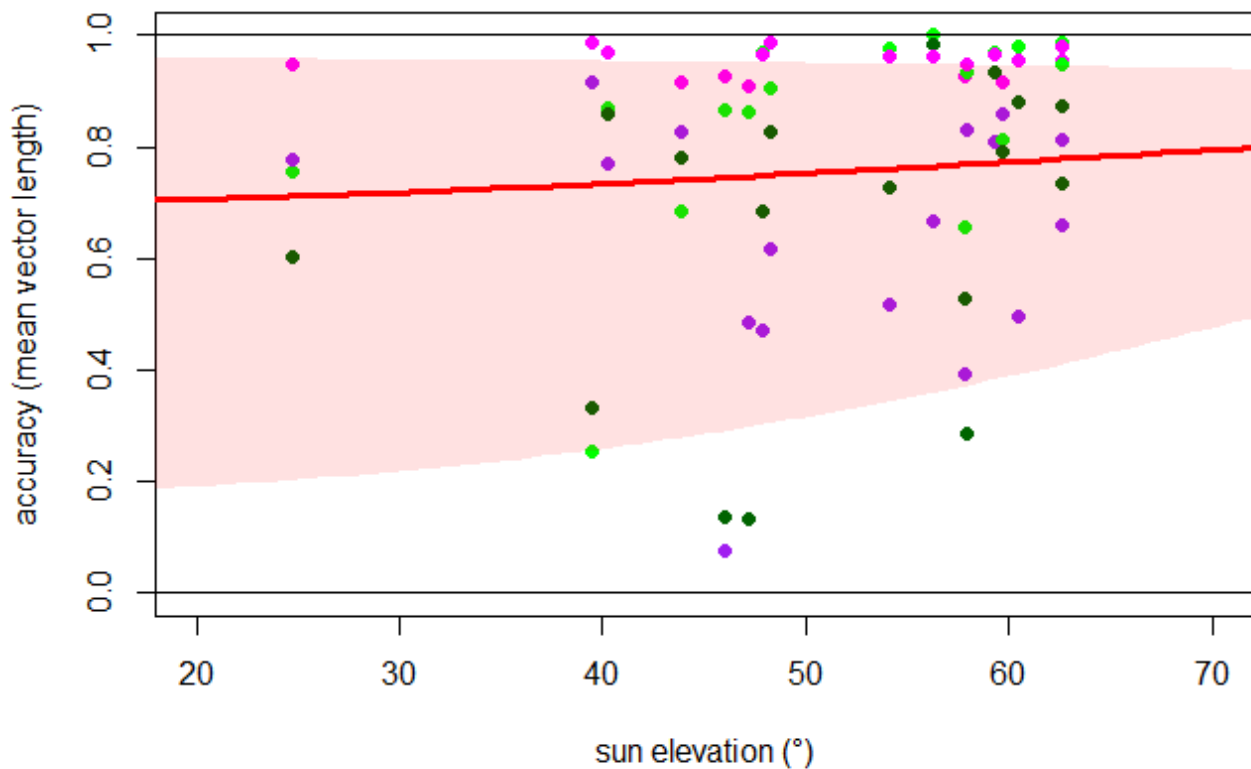
```
##
##      Estimate 1-95% CI u-95% CI
## Intercept      1.618    0.670    2.537
## phi_Intercept   1.364    0.246    2.471
```

|                       |        |        |       |
|-----------------------|--------|--------|-------|
| ## cos_sun_el_rad     | -0.786 | -2.246 | 0.638 |
| ## phi_cos_sun_el_rad | -0.351 | -2.001 | 1.311 |

Extract the model's predictions and compare with the original data.

```
fx_ele = sm_ele$fixed[, c(1, 3:4)]
with(mean_vectors_full,
  {
    plot(x = as.numeric(deg(sun_el_rad)),
         y = mean_vector,
         xlab = 'sun elevation (°)',
         ylab = 'accuracy (mean vector length)',
         main = 'general effect of cos(sun elevation) on accuracy',
         xlim = c(20,70),
         ylim = c(0,1),
         pch = 19,
         col = sapply(X = paste0(brightn, colour),
                      FUN = switch,
                      hg = 'green',
                      lg = 'darkgreen',
                      hu = 'magenta',
                      lu = 'purple'
                      )
        )
  }
)
abline(h = c(0,1))
polygon(x = c(0:85, 85:0),
       y = plogis(
         c(fx_ele$`u-95% CI`[1] + fx_ele$`u-95% CI`[3]*
           cos(rad(0:85)),
           rev( fx_ele$`l-95% CI`[1] + fx_ele$`l-95% CI`[3]*
               cos(rad(0:85)))
         )
       ),
       col = adjustcolor('red', alpha.f = 0.1),
       border = NA)
lines(x = 0:85,
      y = plogis(
        fx_ele$Estimate[1] + fx_ele$Estimate[3]*
          cos(rad(0:85))
      ),
      col = 'red',
      lwd = 3)
```

## general effect of cos(sun elevation) on accuracy



```
loo_beta = loo(md_beta)
```

```
## Warning: Found 4 observations with a pareto_k > 0.7 in model 'md_beta'.  
## recommend to set 'moment_match = TRUE' in order to perform moment match  
## problematic observations.
```

```
loo_ele = loo(md_ele)  
loo_compare(loo_beta, loo_ele)
```

```
##           elpd_diff se_diff  
## md_beta    0.0      0.0  
## md_ele   -23.1      7.1
```

## Comparison with Edrich *et al.* (1979)

How does this compare with previous experiments in which a light spot (2.3°) was presented through bandpass filters and ND filters to produce different intensities? In that data, there was a clear sigmoidal relationship between intensity and accuracy (mean vector length). They mention that the particular relationship with light intensity was variable, but that the relative relationships remained constant, so we may also observe the same relationship but shifted.



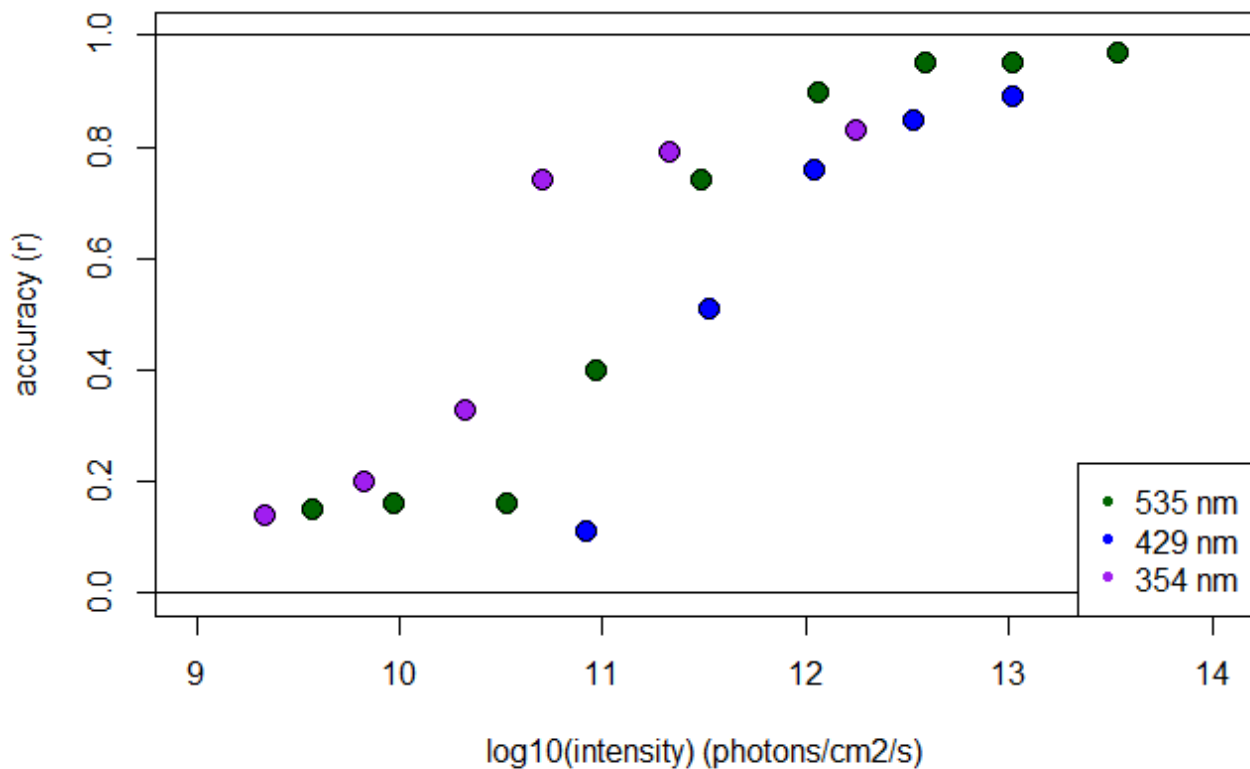
Unfortunately, the scatter of the dance directions as a function of light intensity proved to be unstable over long periods of time; the absolute values changed from day to day. However, these changes were such that curves for adjacent wavelengths were shifted in parallel - the relative sensitivities remained constant. — Edrich et al. (1979), p154

Plot raw data from Edrich *et al.* (1979) Fig. 2

```
with(ed,
  {
    plot(x = log10(intensity),
         y = accuracy,
         xlim = c(9,14),
         ylim = c(0,1),
         main = 'Data from Edrich et al. 1979, Fig. 2',
         xlab = 'log10(intensity) (photons/cm2/s)',
         ylab = 'accuracy (r)',
         pch = 21,
         cex = 1.5,
         col = 'black',
         bg = sapply(paste(wavelength),
                     FUN = switch,
                     `354` = 'purple',
                     `429` = 'blue',
                     `535` = 'darkgreen',
                     'red')
        )
    abline(h = c(0,1))
  }
)

legend(x = 'bottomright',
       legend = paste(sort(unique(ed$wavelength),
                             decreasing = TRUE), 'nm'),
       col = c('darkgreen', 'blue', 'purple'),
       pch = c(20, 20, 20))
```

Data from Edrich et al. 1979, Fig. 2



This is best modelled with a nonlinear (psychometric model) following the beta distribution (for variables bounded between zero and one).

### Reorganise data

```
ed = within(ed,
  {
    logit_accuracy = qlogis(accuracy)
    log10_intensity = log10(intensity)
    colour = sapply(paste(wavelength),
      FUN = switch,
      `354` = 'UV',
      `429` = 'blue',
      `535` = '0green',
      NA)
  }
)
```

### Fit psychometric curves

For each light stimulus, fit a sigmoid curve bounded between baseline accuracy (shared across stimuli) and maximum accuracy ( $1 - \text{lapses}$ ). Inflection point and 80%-rise-region width vary between stimuli, maximising posterior probability for a beta distribution in  $(0,1)$ . To do this, we first define a nonlinear formula in `brms`.

```

#set up model fit
formula_beta = bf(
  formula = accuracy ~
    inv_logit(LogitBase) + (1 - inv_logit(LogitLapse) - inv_logit(LogitBas
    inv_logit( 4.39*(log10_intensity - Inflex) / exp(LogWidth) ) , #inflec
# for each of these parameters, we can set up a separate formula
# that describes how to predict them from the data
#Base rate of correct choices: "
LogitBase ~ 1, #Base rate of correct choices: "~ 1" gives the instructio
#Lapse rate on a log(odds) scale:
LogitLapse ~ colour, #this is similar to the formula in our LMM example
#inflection point of the initial curve:
Inflex ~ colour, #N.B. this is similar to the intercept, so it does not
#Log 80% width of the curve:
LogWidth ~ colour, #N.B. this is similar to the slope, so all of its eff
family = Beta(link = "identity", link_phi = 'log'),
nl = TRUE)#the joint distribution for these parameters is undefined, and

```

Since this is a nonlinear model (there are terms both inside and outside the logistic transform) it is important to establish prior distributions for all parameters. These ensure that parameter estimates explore credible ranges during fitting. The baseline is expected to be around  $r = 0.27$ , while the maximum accuracy should be around  $r = 0.95$ , and their estimates should not converge (that would flatten the curve, or reverse its direction). The inflection points are expected around  $10^{11}$  photons  $\text{cm}^{-2} \text{s}^{-1}$ , with a rise-region width of  $10^3$  photons  $\text{cm}^{-2} \text{s}^{-1}$ . All colour effects may vary by a few orders of magnitude.

```

#set up priors
prior_beta = get_prior(formula = formula_beta,
  data = ed)
#set the prior distribution
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitBase' &
    coef %in% 'Intercept'
  ] = 'normal(-1,1)' #a normal distribution centred on plo
  })
#this prior is automatically bounded between 0 and 1,
#Lapse rate priors -----
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitLapse' &
    coef %in% 'Intercept'
  ] = 'normal(-1,1)' #a normal distribution centred on plo
  })

```

```

    ] = 'normal(-3,2)' #a normal distribution centred on -3
  })

#for all other fixed effects on lapse rate, we'll suggest values around 0
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitLapse' &
    coef %in% 'colourblue'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitLapse' &
    coef %in% 'colourUV'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

#Inflection point priors -----
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'Inflex' &
    coef %in% 'Intercept'
  ] = 'normal(11,2)' #a normal distribution:mean 11, sd 2
  })

#for all coefficients, we'll suggest values around 0 (no effect)
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'Inflex' &
    coef %in% 'colourblue'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 2
  })

prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'Inflex' &
    coef %in% 'colourUV'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 2
  })

#Rise region width priors -----
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects

```

```

        nlpar %in% 'LogWidth' &
        coef %in% 'Intercept'
      ] = 'normal(1,2)' #a normal distribution:mean 1, sd 3
    })
#for all coefficients, we'll suggest values around 0 (no effect)
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogWidth' &
    coef %in% 'colourblue'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogWidth' &
    coef %in% 'colourUV'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

```

Fit a psychometric model to the data using the nonlinear formula and specified prior distributions. 4000 iterations on 4 chains takes less than one minute.

```

#takes less than 1 minute
#parameters for the brms runs
wup = 2000
sam = 2000
refr = 0
system.time(
  {
    beta_fit = brm( formula = formula_beta, # using our nonlinear formula
      data = ed, # our data
      prior = prior_beta, # our priors
      iter = wup+sam, # long run for 4000 iterations
      chains = 4, # 4 chains in parallel
      cores = 4, # on 4 CPUs
      refresh = refr,
      silent = 0, # don't echo chain progress
      control = list(adapt_delta = 0.95), #finer sampling
      backend = 'cmdstanr') # use cmdstanr (other compilers
  }
)

```

## Start sampling

```
## Running MCMC with 4 parallel chains...
```

```
## Chain 3 Informational Message: The current Metropolis proposal is about
```

```
## Chain 3 Exception: beta_lpdf: Second shape parameter[3] is 0, but must
```

```
## Chain 3 If this warning occurs sporadically, such as for highly constr
```

```
## Chain 3 but if this warning occurs often then your model may be either
```

```
## Chain 3
```

```
## Chain 1 finished in 4.4 seconds.  
## Chain 3 finished in 4.5 seconds.  
## Chain 2 finished in 5.0 seconds.  
## Chain 4 finished in 5.0 seconds.  
##  
## All 4 chains finished successfully.  
## Mean chain execution time: 4.7 seconds.  
## Total execution time: 5.1 seconds.
```

```
## Warning: 1 of 8000 (0.0%) transitions ended with a divergence.  
## See https://mc-stan.org/misc/warnings for details.
```

```
##      user  system elapsed  
##    0.77    0.44    25.76
```

## Plot and compare data from Edrich and Kühn.

Summarise the data from full cases in the Kühn data using quantiles (minimum, 25%, median, 75% and maximum).

```
#Examine quantiles  
meanQ = aggregate(mean_vector ~ colour*brightn,  
                  data = mean_vectors_full,  
                  FUN = quantile)
```

```
#Extract different quantiles as separate columns in a data frame
meanQ = data.frame(cbind(meanQ[1:2], meanQ$mean_vector) )
#Rename
names(meanQ) = c('colour', 'brightn', paste0('q',0:4))
#Subset by condition
gh_meanQ = subset(meanQ, colour %in% 'g' & brightn %in% 'h')
gl_meanQ = subset(meanQ, colour %in% 'g' & brightn %in% 'l')
uh_meanQ = subset(meanQ, colour %in% 'u' & brightn %in% 'h')
ul_meanQ = subset(meanQ, colour %in% 'u' & brightn %in% 'l')
```

Extract the model predictions by summarising parameter estimates from the sampling phases of all chains. To identify specific combinations of parameter values, use the `conditional_effects` function, which organises all simultaneous estimates into a data frame.

[illegible]

```
## Warning: There were 1 divergent transitions after warmup. Increasing
## adapt_delta above 0.95 may help. See
## http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmu
```

[illegible]

```

    effects = c('log10_intensity:colo
    plot = FALSE
  )
}
)

```

```

##      user  system elapsed
##      2.48    0.19    2.65

```

```

#extract predictions
pred_data_beta = full_cond_beta$`log10_intensity:colour`

#Green light curve
g_curve =
with(data.frame(t(full_estimates_beta)),
  {
    list(Base = plogis(LogitBase_Intercept),
         Maxima = 1- plogis(LogitLapse_Intercept),
         Inflection = Inflex_Intercept,
         Width = exp(LogWidth_Intercept)
    )
  }
)

#UV light curve
u_curve =
with(data.frame(t(full_estimates_beta)),
  {
    list(Base = plogis(LogitBase_Intercept),
         Maxima = 1- plogis(LogitLapse_Intercept + LogitLapse_colourUV),
         Inflection = Inflex_Intercept + Inflex_colourUV,
         Width = exp(LogWidth_Intercept + LogWidth_colourUV)
    )
  }
)

```

Plot the raw data alongside the predictions for the data from Edrich *et al.*, (1979) and a boxplot summary of the data data from Kühn (2017).

```

#Prepare a sequence for plotting
xx1 = seq(from = 9, to = 16,
          length.out = 1e3)

#plot only the stimulus types also used by Kühn.

```



```

ed_ug = subset(ed, wavelength != 429)

#plot each stimulus type
with(ed_ug,
{
  plot(x = log10_intensity,
       y = accuracy,
       main =
'Comparison of data from Edrich et al. 1979
with Kühn, 2017',
       xlab = 'log10(intensity) (photons/cm2/s)',
       ylab = 'accuracy (r)',
       ylim = c(0,1),
       xlim = c(9,16),
       pch = 21,
       cex = 1.5,
       col = 'black',
       bg = sapply(paste(wavelength),
                   FUN = switch,
                   `354` = 'purple',
                   `429` = 'blue',
                   `535` = 'darkgreen',
                   'red')
       )
  abline(h = c(0,1))
  abline(v = log10(4*c(1e12,1e14)) )
}
)
#Add the estimated range of the curve for the green stimulus
with(data.frame(t(full_estimates_beta)),
{
  abline(h = c(0,1,
              plogis(LogitBase_Intercept),
              1- plogis(LogitLapse_Intercept) ),
        lty = c(1,1,3,3)
        )
}
)

#plot total 95% prediction intervals
with(subset(pred_data_beta, colour == '0green'),
{
  polygon(x = c(sort(log10_intensity), rev(sort(log10_intensity))),
         y = c(lower__[order(log10_intensity)],
               rev(upper__[order(log10_intensity)]))
         ),
         col = adjustcolor('green', alpha.f = 25/256),
         border = NA,

```

```

        lwd = 0.1
    )
}
)
with(subset(pred_data_beta, colour == 'UV'),
{
    polygon(x = c(sort(log10_intensity), rev(sort(log10_intensity))),
            y = c(lower__[order(log10_intensity)],
                  rev(upper__[order(log10_intensity)]))
            ),
            col = adjustcolor('magenta', alpha.f = 25/256),
            border = NA,
            lwd = 0.1
    )
}
)

```

*#plot the median prediction lines*

```

with(g_curve,
    lines(x = xx1,
           y = Base + (Maxima - Base) *
               plogis(4.39*(xx1 - Inflection)/Width),
           col = 'darkgreen',
           lwd = 3,
           lty = 3)
)
with(u_curve,
    lines(x = xx1,
           y = Base + (Maxima - Base) *
               plogis(4.39*(xx1 - Inflection)/Width),
           col = 'purple4',
           lwd = 3,
           lty = 3)
)

```

*#plot observed data from Kühn as boxplots showing the quartiles, median and  
#Green High intensity*

```

QBox(quant = subset(meanQ, colour %in% 'g' & brightn %in% 'h'),
      x = log10(4e14),
      offset = -0.1,
      cols = c('green', 'darkgreen')
)

```

*#Green Low intensity*

```

QBox(quant = subset(meanQ, colour %in% 'g' & brightn %in% 'l'),
      x = log10(4e12),
      offset = -0.1,
      cols = c('green', 'darkgreen')
)

```

```
)
```

```
#UV High intensity
```

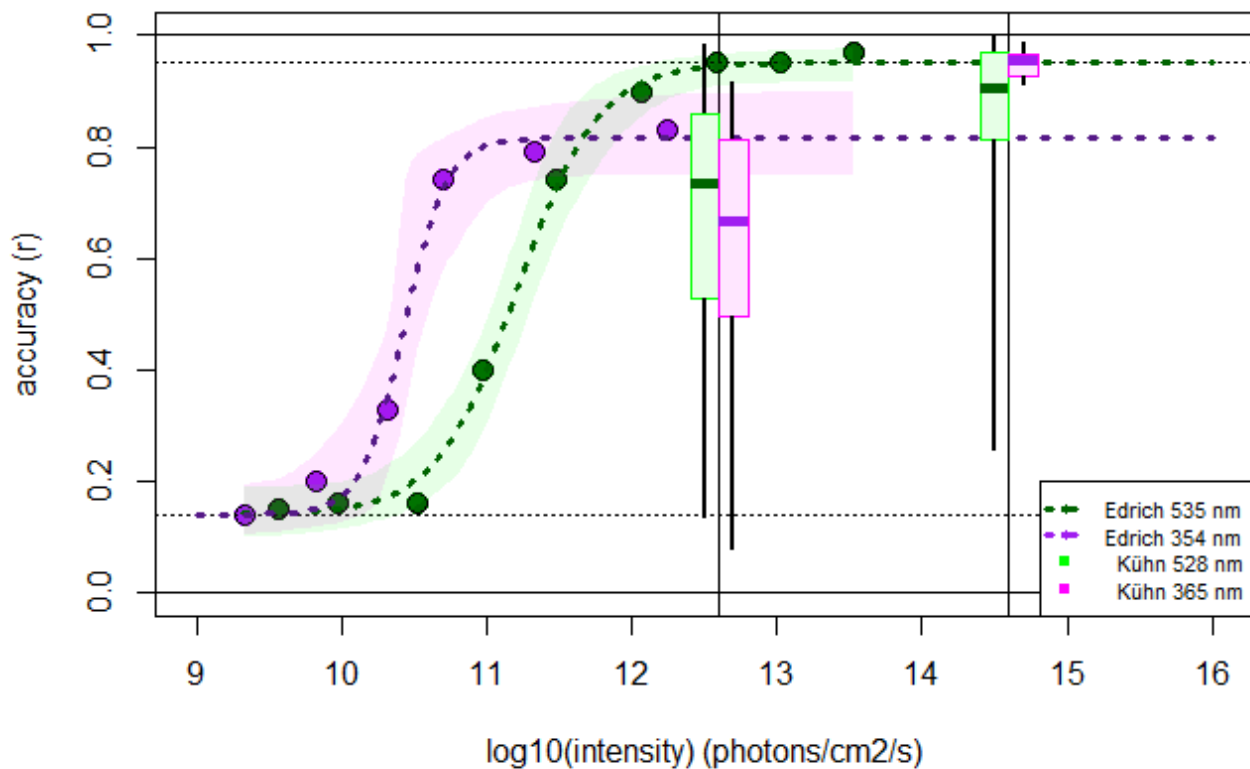
```
QBox(quant = subset(meanQ, colour %in% 'u' & brightn %in% 'h'),  
      x = log10(4e14),  
      offset = 0.1,  
      cols = c('magenta', 'purple'))  
)
```

```
#UV Low intensity
```

```
QBox(quant = subset(meanQ, colour %in% 'u' & brightn %in% 'l'),  
      x = log10(4e12),  
      offset = 0.1,  
      cols = c('magenta', 'purple'))  
)
```

```
legend(x = 'bottomright',  
       legend = c(  
         paste('Edrich',  
               sort(unique(ed_ug$wavelength),  
                       decreasing = TRUE), 'nm'),  
         paste('Kühn', c(528, 365), 'nm' )  
       ),  
       col = c('darkgreen', 'purple', 'green', 'magenta'),  
       pch = c(20, 20, 15, 15),  
       lty = c(3, 3, NA, NA),  
       lwd = 3,  
       cex = 0.7)
```

## Comparison of data from Edrich et al. 1979 with Kühn, 2017



It would be difficult to explain the orientation accuracy for the high intensity ( $4 \times 10^{14}$ ) UV stimulus from the accuracy curve for Edrich's UV data, but it would fit well with the maximum accuracy observed for green light. In Edrich's study, the green light was interpreted as the sun (dances at correct angle between sun and feeder), whereas the UV light was interpreted as a patch of sky near the anti-sun. In Kühn's data the stimuli were larger (assuming  $10^\circ$  across) than Edrich's ( $2.3^\circ$ ), which may have resulted in them being perceived as dimmer (left shift in this graph).

## Sun compass orientation

Since most dances were at least somewhat oriented ( $r > 0.5$ ) we can assume that the mean angles of these dances typically align with an intended dance direction. If most dances are for the same geographic location, we should observe dances as the same fixed angle to the stimulus, interpreted either as the sun or the anti-sun.

```
sun_diff_gh = sapply(X = u_id,
                     FUN = SunDiff,
                     dt = mean_vectors,
                     cl = 'g',
                     br = 'h')
sun_diff_gl = sapply(X = u_id,
```

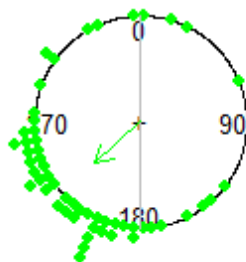
```

        FUN = SunDiff,
        dt = mean_vectors,
        cl = 'g',
        br = 'l')
sun_diff_uh = sapply(X = u_id,
        FUN = SunDiff,
        dt = mean_vectors,
        cl = 'u',
        br = 'h')
sun_diff_ul = sapply(X = u_id,
        FUN = SunDiff,
        dt = mean_vectors,
        cl = 'u',
        br = 'l')

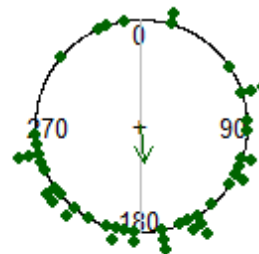
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(sun_diff_gh),
      col = 'green',
      shrink = 2.0,
      side = 3,
      title = 'Green Bright')
PCfun(angles = unlist(sun_diff_gl),
      col = 'darkgreen',
      shrink = 2.0,
      side = 3,
      title = 'Green Dim')
PCfun(angles = unlist(sun_diff_uh),
      col = 'magenta',
      shrink = 2.0,
      side = 3,
      title = 'UV Bright')
PCfun(angles = unlist(sun_diff_ul),
      col = 'purple',
      shrink = 2.0,
      side = 3,
      title = '          UV Dim')

```

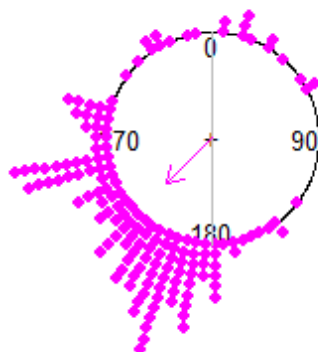
Green Bright



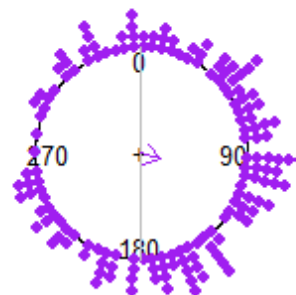
Green Dim



UV Bright



UV Dim



For the bright stimuli the dances generally indicate South, but with a very wide range. (or possibly clusters to the South-West and South-East). For the dim stimuli there is less of a clear pattern, particularly for UV dim.

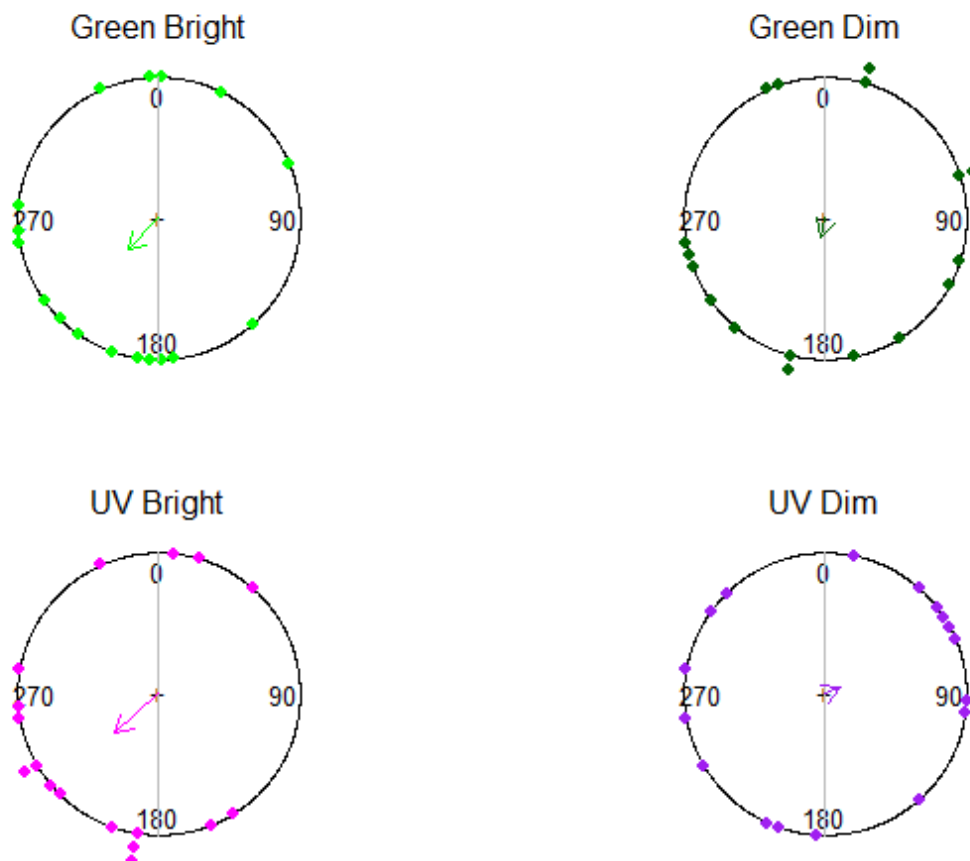
We can also restrict this analysis to just the dances with all conditions.

```
full_sun_diff_gh = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'g',
                          br = 'h')
full_sun_diff_gl = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'g',
                          br = 'l')
full_sun_diff_uh = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'u',
                          br = 'h')
full_sun_diff_ul = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'u',
                          br = 'l')
```

```

par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(full_sun_diff_gh),
      col = 'green',
      shrink = 1.5,
      side = 3,
      title = 'Green Bright')
PCfun(angles = unlist(full_sun_diff_gl),
      col = 'darkgreen',
      shrink = 1.5,
      side = 3,
      title = 'Green Dim')
PCfun(angles = unlist(full_sun_diff_uh),
      col = 'magenta',
      shrink = 1.5,
      side = 3,
      title = 'UV Bright')
PCfun(angles = unlist(full_sun_diff_ul),
      col = 'purple',
      shrink = 1.5,
      side = 3,
      title = 'UV Dim')

```



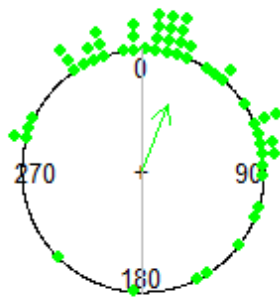
When restricting to only the balanced cases, the pattern becomes clearer. For the most oriented dances to the bright UV stimulus there are two clusters to the

South-West and South-East. These are likely also present for the bright green stimulus. The dim green stimulus appears to align somewhat with the bright stimuli. The dim UV stimulus shows no apparent pattern, though a slight southwards bias.

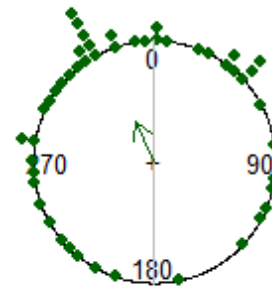
## Inspect dance angles relative to the stimulus

```
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'g' & brightn %in% 'h')$mu,
      col = 'green',
      shrink = 1.75,
      title = 'Green Bright')
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'g' & brightn %in% 'l')$mu,
      col = 'darkgreen',
      shrink = 1.75,
      title = 'Green Dim')
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'u' & brightn %in% 'h')$mu,
      col = 'magenta',
      shrink = 1.75,
      title = 'UV Bright')
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'u' & brightn %in% 'l')$mu,
      col = 'purple',
      shrink = 1.75,
      title = 'UV Dim')
```

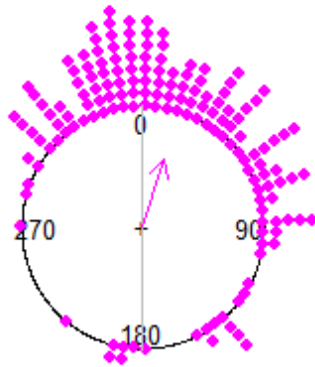




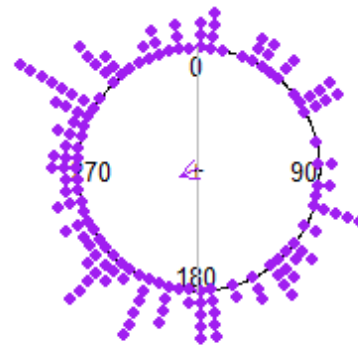
Green Bright



Green Dim



UV Bright



UV Dim

```
cd_sd = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                  FUN = sd.circular,
                  data = mean_vectors)
cd_rho = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                   FUN = rho.circular,
                   data = mean_vectors)
names(cd_sd)[3] = 'circular_sd'
names(cd_rho)[3] = 'rho'
cd_desc = merge(cd_sd, cd_rho)
cd_desc = within(cd_desc,
                 {sd_degrees = deg(circular_sd)})
print(cd_desc,digits = 2)
```

| ##   | colour | brightn | circular_sd | rho | sd_degrees |
|------|--------|---------|-------------|-----|------------|
| ## 1 | g      | h       | 1.0 0.58    |     | 60         |
| ## 2 | g      | l       | 1.4 0.36    |     | 82         |
| ## 3 | u      | h       | 1.0 0.59    |     | 59         |
| ## 4 | u      | l       | 1.9 0.16    |     | 110        |

## Compare Sun-Spot & Phototaxis Hypotheses

It is likely that a plurality of dancers had visited locations in the Marburg Botanic Gardens, perhaps in the centre of the main growing area (50°48'03.3"N

8°48'24.1"E). The hive is at 50°48'22.7"N 8°48'40.8"E. That would give the following bearing:

```
# Define the coordinates

hive_coords = c(longitude = 8.811333,
                 latitude = 50.806306)

#the 150m circle intersecting with the nearest edge
garden_coords = list(left =
                      c(longitude = 8.808956,
                        latitude = 50.805509),
                      centre =
                        c(longitude = 8.810290,
                          latitude = 50.805565),
                      right =
                        c(longitude = 8.810110,
                          latitude = 50.804854)
                      )

#find the bearing from the hive
garden_bearing = sapply(garden_coords,
                        FUN = bearing,
                        p1 = hive_coords
                        )

#standard bearing
print(garden_bearing %% 360)
```

```
##      left  centre  right
## 242.1153 221.7301 208.0902
```

```
garden_bearing_rad = rad(as.numeric(garden_bearing[2]))
```

```
#save and load models instead of running them each time
load_model = FALSE
# Custom family for "unwrap" von Mises distribution
unwrap_von_mises = custom_family(
  "unwrap_von_mises", dpars = c("mu", "kappa"),
  links = c('identity', #brms cannot accept custom link functions, do via n
            "softplus"),
  lb = c(-pi, 0),
  ub = c(pi, NA),
  type = "real"
)
```

## ## Stan variables

### ### Functions

```
stan_unwrap_fun = stanvar(scode = "
  real unwrap_von_mises_lpdf(real y, real mu, real kappa) {
    return von_mises_lpdf(y | mod_circular(mu), kappa);
  }
  real unwrap_von_mises_rng(real mu, real kappa) {
    return von_mises_rng(mod_circular(mu), kappa);
  }
  real unwrap_von_mises_vect_lpdf(vector y, real mu, real kappa) {
    real tmp = 0;
    for(i in 1:size(y))
    {
      tmp = tmp + unwrap_von_mises_lpdf(y[i] | mu, kappa);
    }
    return tmp;
  }
",
                        block = 'functions') +
stanvar(scode = "
  real mod_circular(real y) {
    return fmod(y + pi(), 2*pi()) - pi();
  }

  real inv_mod_circular(real y) {
    return mod_circular(y);
  }
",
                        block = 'functions')
```

### ### Parameters

*#concentration of random effects on fixed effect on mean angle*

```
kappamu_param = stanvar(scode = "
  real kappamu;
",
                        block = "parameters") +
stanvar(scode = "
  real kappa_mu = log1p_exp(kappamu);
",
                        block = 'genquant')
```

```
stanvars_unwrap = stan_unwrap_fun + kappamu_param
```

```

#subset to just gh data
cd_gh = subset(cd,
               subset = brightn %in% 'h' &
                      colour %in% 'g' &
                      !is.na(sun_az_rad) # for fair comparison only inc
               )

dt_photo = within(cd_gh,
                  {
                    BR = brightn
                    CL = colour
                    DT = date
                    RN = run
                    sunangle = angle + sun_az_rad
                  }
)

dt_sun = within(dt_photo,
                {bgangle = sunangle - garden_bearing_rad}
                )

# Model formulae
#model assuming mean direction at 0°, with individual biases
formula_photo = bf(
  formula = angle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 0, # fixed effects intercept == 0
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
  family = unwrap_von_mises,#mod mu, kappa via the softplus
  nl = TRUE)#to accept user-defined extra parameters (mui) we need to treat

#model allowing any angle
formula_nophoto = bf(
  formula = angle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 1, # fixed effects intercept estimated
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
  family = unwrap_von_mises,#mod mu, kappa via the softplus
  nl = TRUE)#to accept user-defined extra parameters (mui) we need to treat

formula_sunspot = bf(
  formula = bgangle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 0, # fixed effects intercept estimated

```

```

      mui ~ 0 + ID,
      kappa ~ 1 + (1|ID),
family = unwrap_von_mises,#mod mu, kappa via the softplus
nl = TRUE)#to accept user-defined extra parameters (mui) we need to treat

formula_nosunspot = bf(
  formula = bgangle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 1, # fixed effects intercept estimated
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
family = unwrap_von_mises,#mod mu, kappa via the softplus
nl = TRUE)#to accept user-defined extra parameters (mui) we need to treat

# Priors
#priors for mu
pr_photo =
  set_prior(paste("target +=",
    'unwrap_von_mises_vect_lpdf(b_mui | 0, log1p_exp(kappamu
    '+ normal_lpdf(b_mui | 0, 2*pi()))'# additional prior to
  ),
  check = FALSE) +
  set_prior("target += lognormal_lpdf(log1p_exp(kappamu) | log(15), 0.6)",
    check = FALSE) +
    prior(normal(3.0,3.0), class = Intercept, dpar = 'kappa') + # bias to
    prior(student_t(3, 0, 3.0), class = sd, dpar = 'kappa') # weak bias no

#for models with a mean we also need an extra parameter
pr_nophoto = pr_photo +
  prior(unwrap_von_mises(0,1e-16), class = b, nlpar = 'mu0', coef = 'Intercept')

#parameters for the brms runs
wup = 1000 #number of warmup iterations used to tune the sampler
sam = 1000 #number of iterations to sample
refr = 0 #print progress every n iterations
siln = 2#print only very important messages

#these are pretty slow! About 7 minutes in total
if(load_model)
{ load(file = '2Results/phototaxis_modlist.RData')
}else
{
  cglm_photo = brm(formula = formula_photo,
    data = dt_photo,
    prior = pr_photo,
    stanvars = stanvars_unwrap,
    warmup = wup,#may be necessary
    iter = wup+sam, #doesn't take a lot of runs
  )
}

```

```

chains = 4, # 4 chains in parallel
cores = 4, # on 4 CPUs
# control = list(adapt_delta = 0.90),
refresh = refr, # echo chain progress every n iterations
silent = siln, # echo some Stan messages
save_pars = save_pars(all = TRUE), # <--- Crucial for mo
backend = 'cmdstanr')

```

```

cglm_nophoto = brm(formula = formula_nophoto,
  data = dt_photo,
  prior = pr_nophoto,
  stanvars = stanvars_unwrap,
  warmup = wup, #may be necessary
  iter = wup+sam, #doesn't take a lot of runs
  chains = 4, # 4 chains in parallel
  cores = 4, # on 4 CPUs
  # control = list(adapt_delta = 0.90),
  refresh = refr, # echo chain progress every n iterations
  silent = siln, # echo some Stan messages
  save_pars = save_pars(all = TRUE), # <--- Crucial for mo
  backend = 'cmdstanr')

```

```

cglm_sunspot = brm(formula = formula_sunspot,
  data = dt_sun,
  prior = pr_photo, #different response, same predictors
  stanvars = stanvars_unwrap,
  warmup = wup, #may be necessary
  iter = wup+sam, #doesn't take a lot of runs
  chains = 4, # 4 chains in parallel
  cores = 4, # on 4 CPUs
  control = list(adapt_delta = 0.90),
  refresh = refr, # echo chain progress every n iterations
  silent = siln, # echo some Stan messages
  save_pars = save_pars(all = TRUE), # <--- Crucial for mo
  backend = 'cmdstanr')

```

```

cglm_nosunspot = brm(formula = formula_nosunspot,
  data = dt_sun,
  prior = pr_nophoto, #different response, same predictors
  stanvars = stanvars_unwrap,
  warmup = wup, #may be necessary
  iter = wup+sam, #doesn't take a lot of runs
  chains = 4, # 4 chains in parallel
  cores = 4, # on 4 CPUs
  control = list(adapt_delta = 0.90),
  refresh = refr, # echo chain progress every n iterations
  silent = siln, # echo some Stan messages
  save_pars = save_pars(all = TRUE), # <--- Crucial for mo

```

```

      backend = 'cmdstanr')
#combine into a single list
modlist = list(phototaxis = cglm_photo,
               no_phototaxis = cglm_nophoto,
               sunspot = cglm_sunspot,
               no_sunspot = cglm_nosunspot)
}
#summarise
suppressWarnings({ #for circular models rhats may be smaller than they app
sm_modlist = lapply(X = modlist,
                    FUN = summary)
})
Sm_only = function(x,
                   type = 'fixed',
                   var = 'kappa_Intercept')
{
  x[[type]][var,]
}

#average kappa
print(lapply(X = sm_modlist, Sm_only), digits = 3 )

```

```

## $phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail
## kappa_Intercept      8.95      1.01    7.05      11      1    1450
##
## $no_phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail
## kappa_Intercept      8.91      1.03      7      11      1    1087
##
## $sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail
## kappa_Intercept      8.89      1.01    6.98    10.9      1    1433
##
## $no_sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail
## kappa_Intercept      8.92      1.04    6.95    11.1      1    1215

```

```

#across individuals kappa
print(lapply(X = sm_modlist, Sm_only, type = 'spec_pars', var = 'kappa_mu'

```

```

## $phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS

```

```
## kappa_mu      1.74      0.289      1.22      2.34      1      4889      2826
##
## $no_phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.85      0.307      1.29      2.49      1      3754      2297
##
## $sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.91      0.313      1.33      2.55      1      3527      3163
##
## $no_sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.88      0.317      1.3      2.55      1      3109      2580
```

```
#check population means
lapply(X = modlist[c(2,4)],
      FUN = UnwrapRhats,
      variable = '^b_mu0')
```

```
## $no_phototaxis
## b_mu0_Intercept
##           1.00855
##
## $no_sunspot
## b_mu0_Intercept
##           1.00234
```

```
#Summarise Rhats
lapply(X = modlist,
      FUN = SumUwRhats)
```

```
## $phototaxis
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 0.9999  1.0001  1.0003  1.0004  1.0033
##
## $no_phototaxis
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 1.0000  1.0005  1.0017  1.0030  1.0097
##
## $sunspot
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 0.9999  1.0001  1.0003  1.0006  1.0034
##
## $no_sunspot
```



```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998  1.0000  1.0003  1.0007  1.0011  1.0068
```

```
#calculate loo
if(!load_model)
{
modlist = lapply(modlist,
  FUN = add_criterion,
  'loo',
  moment_match = TRUE)

save(modlist,
  file = '2Results/phototaxis_modlist.RData')
}
```

```
## Recompiling the model with 'rstan'
```

```
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto
```



```
## Warning: Found 8 observations with a pareto_k > 0.7 in model 'X[[i]]'.
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without
## assumption that these observations are negligible. This will refit the
## times to compute the ELPDs for the problematic observations directly.
```



```
## Recompiling the model with 'rstan'
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto
```



```
## Warning: Found 5 observations with a pareto_k > 0.7 in model 'X[[i]]'.
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without
## assumption that these observations are negligible. This will refit the
## times to compute the ELPDs for the problematic observations directly.
```



```
## Recompiling the model with 'rstan'
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto
```

```
## Warning: Found 8 observations with a pareto_k > 0.7 in model 'X[[i]]'.
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without
## assumption that these observations are negligible. This will refit the
## times to compute the ELPDs for the problematic observations directly.
```

```
## Recompiling the model with 'rstan'
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto
```

```
## Warning: Found 6 observations with a pareto_k > 0.7 in model 'X[[i]]'.
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without
## assumption that these observations are negligible. This will refit the
## times to compute the ELPDs for the problematic observations directly.
```

```
#compare the phototaxis models
with(modlist, loo_compare(phototaxis, no_phototaxis))
```

```
##               elpd_diff se_diff
## phototaxis      0.0         0.0
## no_phototaxis -0.4         0.8
```

```
#compare the sun-spot orientation models
with(modlist, loo_compare(sunspot, no_sunspot))
```

```
##               elpd_diff se_diff
## sunspot        0.0         0.0
## no_sunspot    -1.0         0.7
```

```
#compare LOO IC
LooIC = function(lmod)
{
  loo(lmod)$estimates['looic', 'Estimate']
}

modLIC = lapply(X = modlist,
  FUN = LooIC)
print(sort(unlist(modLIC)), digits = 1)
```

| ## | sunspot | phototaxis | no_phototaxis | no_sunspot |
|----|---------|------------|---------------|------------|
| ## | 642     | 643        | 644           | 644        |

There is very little separation between our different hypotheses here, suggesting each would do well to explain the data. We can also look at the specific predictions to see how well each hypothesis matches.

## Hypothesis test

```
#since all models are similarly credible, what about the hypotheses?
#check if the sector surrounding the expected direction is equally full

#for each individual, calculate the likelihood of foraging in the expected

PostROPE = function(id, rope, drs, mu = 'b_mu0_Intercept')
{
  mu = if(!is.na(mu)){with(drs, get(mu))}else{0}
  dr_id = with(drs,
    {
      mod_circular(
        mu +
          get(paste0('b_mui_ID', id))
      )
    }
  )

  #centre both
  circ_rope = circular(rope, template = 'none')
  circ_dr = circular(dr_id, template = 'none')
  mean_rope = mean.circular(circ_rope)
  rope_ctr = as.numeric(mod_circular(circ_rope - mean_rope))
  dr_ctr = as.numeric(mod_circular(dr_id - mean_rope))
  postp = mean( rope_ctr[1] < dr_id &
    dr_id < rope_ctr[2])

  return(postp)
```

```

}

photo_id = unique(dt_photo$ID)

#width of sector
rope_width = abs(mod_circular(rad(diff(garden_bearing[-2])))) # 35°
rope_photo = c(-1,1)*rope_width/2
rope_sunspot = sort( mod_circular(rad(garden_bearing[-2] - garden_bearing[

#extract draws
dr_modlist = lapply(modlist, as_draws_df)

postl_nophoto = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_photo,
  drs = dr_modlist$no_phototaxis
)
postl_photo = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_photo,
  mu = NA,
  drs = dr_modlist$phototaxis
)

postl_nosunspot = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_sunspot, # relative to centre as in model
  drs = dr_modlist$no_sunspot
)
postl_sunspot = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_sunspot, # relative to centre as in model
  drs = dr_modlist$sunspot,
  mu = NA
)

postl_dt = data.frame(phototaxis = postl_photo,
  no_phototaxis = postl_nophoto,
  sunspot = postl_sunspot,
  no_sunspot = postl_nosunspot,
  contrast = postl_sunspot -
    postl_photo)

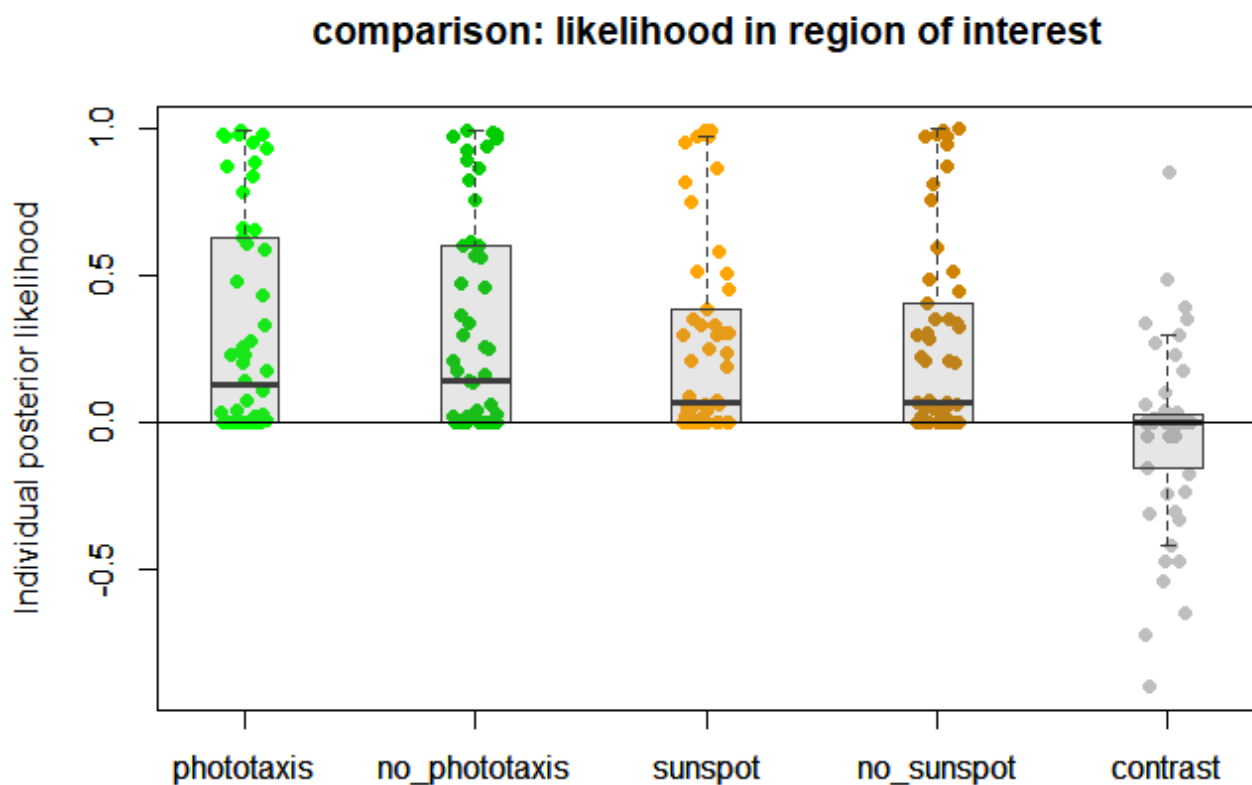
stripchart(x = postl_dt,
  vertical = TRUE,
  method = 'jitter',
  pch = 19,
  ylab = 'Individual posterior likelihood',

```

```

    main = 'comparison: likelihood in region of interest',
    col = c('green','green3', 'orange','orange3', 'gray')
  )
boxplot(x = postl_dt,
  add = TRUE,
  col = gray(0.5, alpha = 0.2),
  border = 'gray25',
  pars = list(boxwex = 0.3, staplewex = 0.2, outwex = 0.1),
  outline = FALSE
)
abline(h = 0)

```



It is hard to say for the individuals whether the majority orient towards the sector containing the Botanic Garden, or a similar sized sector containing the stimulus. What we can say is that the population average falls directly towards the Botanic Garden centre in real-world bearings, while the population average relative to the light source has a leftwards bias.

## Plot of sun compass versus phototaxis

```
with(dr_modlist$no_phototaxis, paste0(mean(b_mu0_Intercept > 0)*100, '%')
```

```
## [1] "99.425%"
```

```
with(dr_modlist$no_sunspot, paste0(mean(b_mu0_Intercept > 0)*100, '%') ) #
```

```
## [1] "77.95%"
```

```
par(pty = 's')
par(mfrow = c(2,2),
    mar = c(0,0,0,0))
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'g' &
                                brightn %in% 'h' &
                                !is.na(sun_az))$mu,

      col = 'green',
      shrink = 1.5,
      plot_rho = FALSE,
      title = '\n\n\n\nRelative to stimulus')
with(dr_modlist$phototaxis,
     arrows(x0 = 0, x1 = 0,
            y0 = 0,
            y1 = A1(softplus(median(kappamu))),
            col = 'orange',
            lwd = 5,
            length = 0.1
           )
)

with(dr_modlist$no_phototaxis,
     arrows(x0 = 0,
            x1 = A1(softplus(median(kappamu)))*
                  sin(median(mod_circular(b_mu0_Intercept))),
            y0 = 0,
            y1 = A1(softplus(median(kappamu)))*
                  cos(median(mod_circular(b_mu0_Intercept))),
            col = 'green3',
            lwd = 5,
            length = 0.1
           )
)
arrows(x0 = c(0,0),
      x1 = sin(c(-1,1)*rope_width/2),
      y0 = c(0,0),
      y1 = cos(c(-1,1)*rope_width/2),
      col = 'orange',
      lty = 2,
      length = 0,
      lwd = 5
```

```

    )
  for(id in unique(dt_photo$ID))
  {
    with(dr_modlist$no_phototaxis,
      arrows(x0 = 0,
             x1 = A1(softplus(median(
               b_kappa_Intercept +
               get(paste0('r_ID__kappa[' , id, ',Intercept]'))
             ))) *
             sin(median(mod_circular(
               b_mu0_Intercept +
               get(paste0('b_mui_ID', id))
             ))),
             y0 = 0,
             y1 = A1(softplus(median(
               b_kappa_Intercept +
               get(paste0('r_ID__kappa[' , id, ',Intercept]'))
             )))) *
             cos(median(mod_circular(
               b_mu0_Intercept +
               get(paste0('b_mui_ID', id))
             ))),
             col = adjustcolor('green4', alpha = 0.5),
             lwd = 0.5,
             length = 0
            )
    )
  }
  legend(x = 'topleft',
        legend = c('expected mean',
                    'expected region',
                    'alternative mean',
                    'individual biases'
                  ),
        col = c('orange',
                 'orange',
                 'green3',
                 'green4'
                ),
        lty = c(1,3,1,1),
        lwd = c(3,3,3,1),
        cex = 0.5,
        bg = NA,
        bty = 'n',
        inset = c(0, 0)
        )

```

```

PCfun(angles = with(subset(mean_vectors,
                           subset = colour %in% 'g' &
                           brightn %in% 'h' &
                           !is.na(sun_az)),
                           mu + sun_az),
      col = 'green',
      shrink = 1.5,
      plot_rho = FALSE,
      title = '\n\n\n\nRelative to North')
with(as_draws_df(modlist$sunspot),
      arrows(x0 = 0,
             x1 = A1(softplus(median(kappamu))) *
             sin(garden_bearing_rad),
             y0 = 0,
             y1 = A1(softplus(median(kappamu))) *
             cos(garden_bearing_rad),
             col = 'orange',
             lwd = 5,
             length = 0.1
            )
)
with(as_draws_df(modlist$no_sunspot),
      arrows(x0 = 0,
             x1 = A1(softplus(median(kappamu))) *
             sin(median(mod_circular(b_mu0_Intercept + garden_bearing_rad
             y0 = 0,
             y1 = A1(softplus(median(kappamu))) *
             cos(median(mod_circular(b_mu0_Intercept + garden_bearing_rad
                                   ))),
             col = 'green3',
             lwd = 5,
             length = 0.1
            )
)
)
arrows(x0 = c(0,0),
      x1 = sin(rad(garden_bearing[-2])),
      y0 = c(0,0),
      y1 = cos(rad(garden_bearing[-2])),
      col = 'orange',
      lty = 2,
      length = 0,
      lwd = 5
)
)
for(id in unique(dt_photo$ID))
with(as_draws_df(modlist$no_sunspot),
      arrows(x0 = 0,

```



```

x1 = A1(softplus(median(
  b_kappa_Intercept +
  get(paste0('r_ID__kappa[' , id , ',Intercept']'))))) *
sin(median(mod_circular(
  b_mu0_Intercept +
  garden_bearing_rad +
  get(paste0('b_mui_ID', id))
))),

y0 = 0,
y1 = A1(softplus(median(
  b_kappa_Intercept +
  get(paste0('r_ID__kappa[' , id , ',Intercept']'))))) *
cos(median(mod_circular(
  b_mu0_Intercept +
  garden_bearing_rad +
  get(paste0('b_mui_ID', id))
))),

col = adjustcolor('green4', alpha = 0.5),
lwd = 0.5,
length = 0
)

)

par(mar = c(0,3,0,0))

with(dr_modlist$no_phototaxis,
{
  VertHist(data = deg(mod_circular(b_mu0_Intercept)),
    main = '\nPopulation mean angle',
    ylab = 'model estimates',
    ylim = c(-30, 90),
    col = adjustcolor('green3', alpha.f = 100/255),
    cex.axis = 0.7)
  abline(h = 0,
    col = 'orange',
    lwd = 7)
  arrows(x0 = 0, x1 = 0,
    y0 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 0.025)
    y1 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 1-0.02)
    col = 'green4',
    lwd = 5,
    code = 3,
    length = 0.05,
    angle = 90)
}
)

legend(x = 'bottomright',

```

```

legend = c('stimulus angle',
           '95% CI'
           ),
col = c('orange',
        'green4'
        ),
lty = c(1,1),
lwd = c(3,3),
cex = 0.5,
bg = NA,
bty = 'n',
inset = c(0, 0)
)

```

```
gb_pos = garden_bearing[2] %% 360
```

```

with(dr_modlist$no_sunspot,
{
  VertHist(data = deg(mod_circular(b_mu0_Intercept)) + gb_pos,
            main = '\nPopulation mean bearing',
            ylab = 'model estimates',
            ylim = c(-30, 90) + gb_pos,
            col = adjustcolor('green3', alpha.f = 100/255),
            cex.axis = 0.7)
  abline(h = garden_bearing %% 360,
         col = 'orange',
         lwd = c(1,7,1),
         lty = c(2,1,2) )
  arrows(x0 = 0, x1 = 0,
         y0 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 0.025)
         gb_pos,
         y1 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 1-0.02)
         gb_pos,
         col = 'green4',
         lwd = 5,
         code = 3,
         length = 0.05,
         angle = 90)
}
)

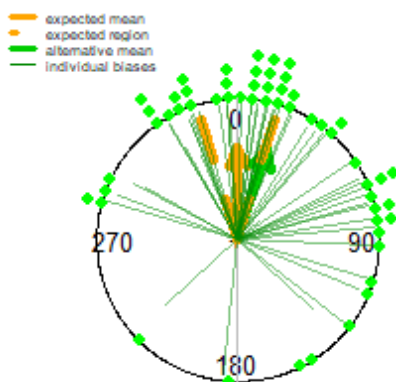
```

```

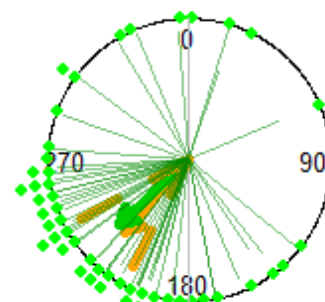
legend(x = 'bottomright',
       legend = c('Botanic Gardens centre',
                  'sector',
                  '95% CI'
                  ),

```

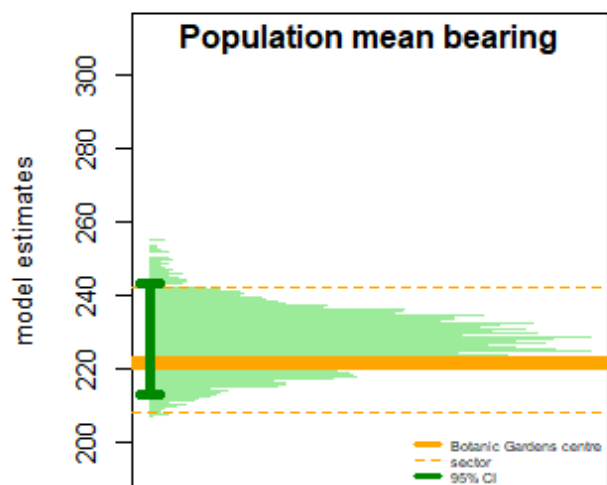
```
col = c('orange',
        'orange',
        'green4'
        ),
lty = c(1,2,1),
lwd = c(3,0.5,3),
cex = 0.5,
bg = NA,
bty = 'n',
inset = c(0, 0)
)
```



Relative to stimulus



Relative to North



```
#How much do individual dance angles vary according to kappa_mu?
with(sm_modlist$no_sunspot,
     paste0(round(deg(MardiaSD(spec_pars['kappa_mu' , 'Estimate']))), '°'))
```

```
## [1] "51°"
```

```
#Where is the population centred?
with(dr_modlist$no_sunspot,
     {
       paste0(
```

```

round(
quantile(x = deg(b_mu0_Intercept),
        p = c(0,0.5,1)+c(1,0,-1)*(0.05/2) ) +
        gb_pos
        ) %% 360, # as a bearing
'°' )
}
)

```

```
## [1] "213°" "228°" "243°"
```

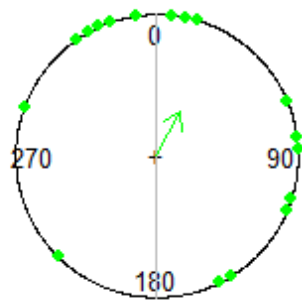
## Mean dance angles relative to the stimulus (full conditions)

Since dances do not point to a specific geographic location, it is possible that they advertise a variety of directions relative to the sun, or are not menotactic but phototactic or uniformly oriented. We can identify phototactic orientation from the angle relative to the stimulus.

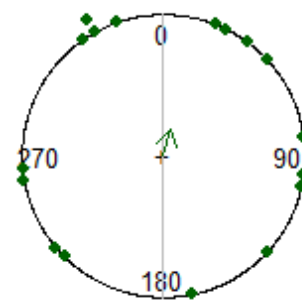
```

par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'g' & brightn %in% 'h')$mu,
      col = 'green',
      shrink = 1.5,
      title = 'Green Bright')
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'g' & brightn %in% 'l')$mu,
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim')
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'u' & brightn %in% 'h')$mu,
      col = 'magenta',
      shrink = 1.5,
      title = 'UV Bright')
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'u' & brightn %in% 'l')$mu,
      col = 'purple',
      shrink = 1.5,
      title = 'UV Dim')

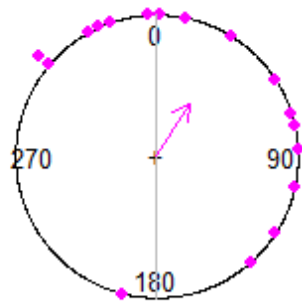
```



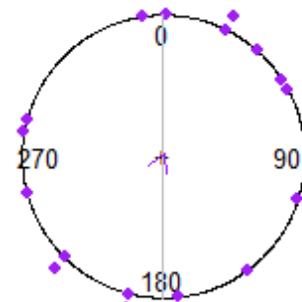
Green Bright



Green Dim



UV Bright



UV Dim

While there are some dances towards or near to the stimulus for most conditions, there is a wide spread, suggesting a factor other than phototaxis sets dance directions. N.B. for the dim stimuli where the widest spread is observed, the dances are also less accurate, which may increase variation in mean dance angle.

```
cd_sd_full = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                        FUN = sd.circular,
                        data = mean_vectors_full)
cd_rho_full = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                        FUN = rho.circular,
                        data = mean_vectors_full)
names(cd_sd_full)[3] = 'circular_sd'
names(cd_rho_full)[3] = 'rho'
cd_desc_full = merge(cd_sd_full, cd_rho_full)
cd_desc_full = within(cd_desc_full,
                      {sd_degrees = deg(circular_sd)})
print(cd_desc_full,digits = 2)
```

| ##   | colour | brightn | circular_sd | rho | sd_degrees |
|------|--------|---------|-------------|-----|------------|
| ## 1 | g      | h       | 1.4 0.351   |     | 83         |
| ## 2 | g      | l       | 1.8 0.200   |     | 103        |
| ## 3 | u      | h       | 1.3 0.438   |     | 74         |
| ## 4 | u      | l       | 2.9 0.016   |     | 164        |

# Change in mean dance angle between stimuli

## Change arcs

To determine whether these dances interpret the stimuli as the sun, anti-sun or something else, we would need to check the differences in dance angle between the conditions. It is also worth considering any differences in accuracy between dances to each condition, since this would influence the variation in angular change. These can be visualised as arcs.

```
par(mfrow = c(2,2), mar = c(0,0,0,0))
OpenCplot(n_sample = with(full_cd, median(run)/4) )
mtext(text = 'Green Bright to Green Dim',
      side = 1,
      line = -1,
      cex = 0.5)

#open the loop
for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
      {
        DiffArc(a1 = mu[colour %in% 'g' &
                        brightn %in% 'h'],
                a2 = mu[colour %in% 'g' &
                        brightn %in% 'l'],
                r1 = mean_vector[colour %in% 'g' &
                                brightn %in% 'h'],
                r2 = mean_vector[colour %in% 'g' &
                                brightn %in% 'l'],
                col1 = 'green',
                col2 = 'darkgreen'
                )
      }
  )
}

OpenCplot(n_sample = with(full_cd, median(run)/4) )
mtext(text = 'UV Bright to UV Dim',
      side = 1,
      line = -1,
      cex = 0.5)

for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
```

```

{
  DiffArc(a1 = mu[colour %in% 'u' &
            brightn %in% 'h'],
          a2 = mu[colour %in% 'u' &
            brightn %in% 'l'],
          r1 = mean_vector[colour %in% 'u' &
            brightn %in% 'h'],
          r2 = mean_vector[colour %in% 'u' &
            brightn %in% 'l'],
          col1 = 'magenta',
          col2 = 'purple'
        )
    }
  )
}

OpenCplot(n_sample = with(full_cd, median(run)/4) )
mtext(text = 'Green Bright to UV Bright',
      side = 1,
      line = -1,
      cex = 0.5)
for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
        {
          DiffArc(a1 = mu[colour %in% 'g' &
                        brightn %in% 'h'],
                  a2 = mu[colour %in% 'u' &
                        brightn %in% 'h'],
                  r1 = mean_vector[colour %in% 'g' &
                        brightn %in% 'h'],
                  r2 = mean_vector[colour %in% 'u' &
                        brightn %in% 'h'],
                  col1 = 'green',
                  col2 = 'magenta'
                )
        }
      )
}

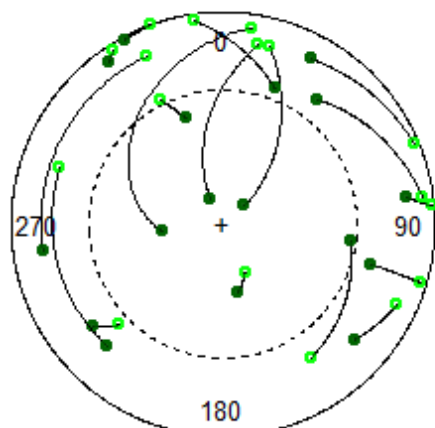
OpenCplot(n_sample = with(full_cd, median(run)/4) )
for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
        {
          DiffArc(a1 = mu[colour %in% 'g' &
                        brightn %in% 'l'],
                  a2 = mu[colour %in% 'u' &

```

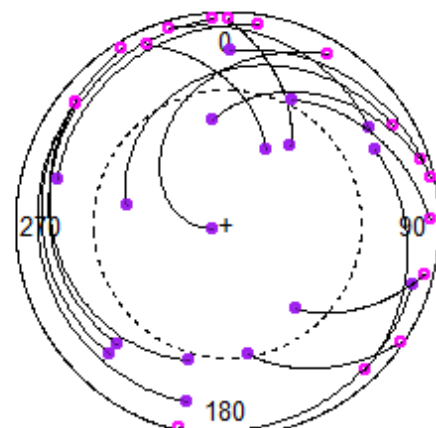
```

        brightn %in% 'l'],
    r1 = mean_vector[colour %in% 'g' &
        brightn %in% 'l'],
    r2 = mean_vector[colour %in% 'u' &
        brightn %in% 'l'],
    col1 = 'darkgreen',
    col2 = 'purple'
  )
}
)
}
mtext(text = 'Green Dim to UV Dim',
      side = 1,
      line = -1,
      cex = 0.5)

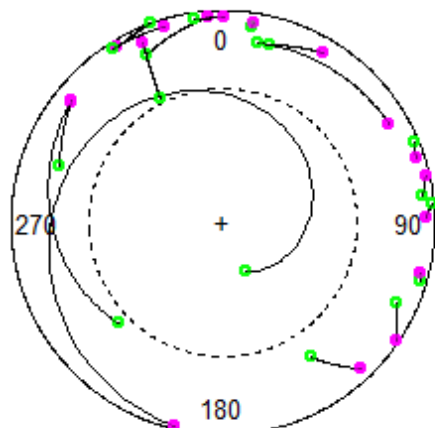
```



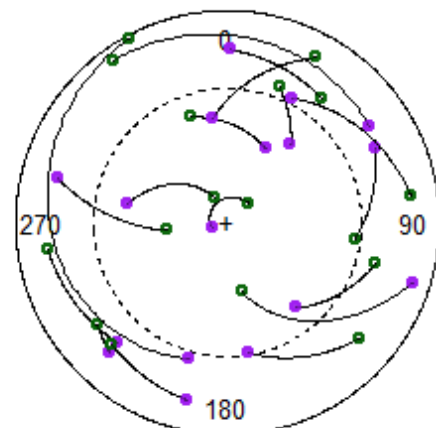
Green Bright to Green Dim



UV Bright to UV Dim



Green Bright to UV Bright



Green Dim to UV Dim

## Differences in mean angle

```

mu_diff_g1 = sapply(X = full_ids,
                    FUN = MuDiff,
                    dt = mean_vectors_full,
                    cl = 'g',
                    br = 'l')
mu_diff_uh = sapply(X = full_ids,
                    FUN = MuDiff,

```



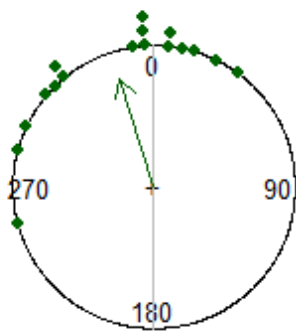
```

        dt = mean_vectors_full,
        cl = 'u',
        br = 'h')
mu_diff_ul = sapply(X = full_ids,
                    FUN = MuDiff,
                    dt = mean_vectors_full,
                    cl = 'u',
                    br = 'l')

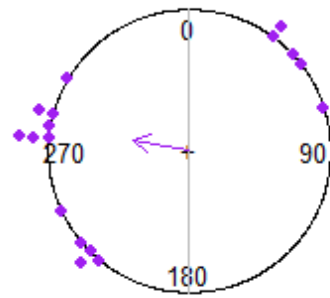
#Other contrasts
mu_diff_uhl = sapply(X = full_ids,
                    FUN = MuDiff,
                    dt = mean_vectors_full,
                    cl = 'u',
                    br = 'l',
                    ref_cl = 'u',
                    ref_br = 'h')
mu_diff_gul = sapply(X = full_ids,
                    FUN = MuDiff,
                    dt = mean_vectors_full,
                    cl = 'u',
                    br = 'l',
                    ref_cl = 'g',
                    ref_br = 'h')

par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(mu_diff_gl),
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim - Green Bright')
PCfun(angles = unlist(mu_diff_uhl),
      col = 'purple',
      shrink = 1.5,
      title = 'UV Dim - UV Bright')
PCfun(angles = unlist(mu_diff_uh),
      col = 'gray40',
      shrink = 1.5,
      title = 'UV Bright - Green Bright')
PCfun(angles = unlist(mu_diff_gul),
      col = 'gray25',
      shrink = 1.5,
      title = 'UV Dim - Green Dim')

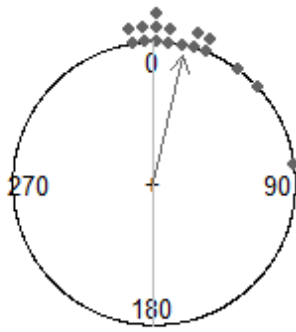
```



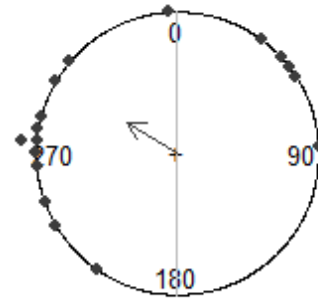
Green Dim - Green Bright



UV Dim - UV Bright



UV Bright - Green Bright



UV Dim - Green Dim

## Estimate turn effects

### Find fit maximum likelihood turn distributions

For each change distribution, find the maximum likelihood von Mises distribution. For the UV high to UV low change, this will be a bimodal distribution.

```
#collect all heading differences
pair_diffs_lst = lapply(X = list(#bright vs dim green
                                green_hl = mu_diff_gl,
                                #bright vs dim UV
                                uv_hl = mu_diff_uhl,
                                #bright green vs bright uv
                                uvg_h = mu_diff_uh,
                                #dim green vs bright UV
                                uvg_l = mu_diff_gul ),
                        FUN = unlist)

#Provide appropriate data format
angle_unit = 'degrees'
angle_rot = 'clock'
#make sure they are in circular format
pair_diffs_lst = lapply(X = pair_diffs_lst,
                        FUN = circular,
```

```
units = angle_unit,  
rotation = angle_rot)
```

Fit the maximum likelihood distribution for no difference between pairs, differences between pairs, and differences between pairs with two means. If angles shift in one direction between trials, the “difference” distribution should have significantly higher likelihood. If angles shift in two directions between pairs, then the “two means” distribution should be the best fit.

```
#TODO neaten up!  
#Distribution for same means  
ml_same_lst = lapply(X = pair_diffs_lst,  
                      FUN = mle.vonmises,  
                      bias = TRUE,  
                      mu = circular(x = 0,  
                                   units = angle_unit,  
                                   rotation = angle_rot)  
                      )  
#Distribution for different means  
ml_diff_lst = lapply(X = pair_diffs_lst,  
                     FUN = mle.vonmises,  
                     bias = TRUE)  
#Distribution for multiple different means  
mod_diff_lst = lapply(X = pair_diffs_lst,  
                      FUN = circ_mle, #converts circular format data to rad  
                      niter = 1e4) #some stochastic variability for green s
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
#Forced multiple different means
```

```
bim_diff_lst = lapply(X = pair_diffs_lst,  
                      FUN = circ_mle,#converts circular format data to rad  
                      exclude = c('M1', 'M2A', 'M2B', 'M2C'),#no uniform or  
                      niter = 1e4)#some stochastic variability for green s
```



```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
## Warning in check_data(data): Converting from degrees to radians...
```

```
## Warning in check_data(data): Converting to modulo 2pi...
```

```
#Extract the parameters of the model with the Lowest AIC
```

```
mod_diff_par = t( sapply(X = mod_diff_lst,  
                        FUN = MD_extract) )
```

```
bim_diff_par = t( sapply(X = bim_diff_lst,  
                        FUN = MD_extract) )
```

```
# Calculate Likelihood
```

```
#calculate all in list
```

```
all_ll = mapply(FUN = LLcalc,  
                m1 = c(m1_same_lst, m1_diff_lst), #apply to both sets of M  
                angles = pair_diffs_lst, #repeat datasets  
                au = angle_unit, #important, use the same frame of referen  
                ar = angle_rot)
```

```

#inspect resulting parameters
mod_diff_results = do.call(rbind, c(ml_same_lst, ml_diff_lst))
#add easy to parse names
rownames(mod_diff_results) = paste( sort( #distinguish same and different
                                     rep(x = c( 'same', 'diff'),
                                             times = length(pair_diffs_lst) ),
                                     decreasing = TRUE
                                   ),
                                   rep(x = names(pair_diffs_lst),
                                       times = 2)
                                   )

#display version
rspar = mod_diff_results[,c('mu', 'kappa', 'se.mu', 'se.kappa')]
rspar = cbind(rspar, loglikelihood = all_ll)
rspar = apply(X = rspar, MARGIN = 2, FUN = unlist)
#reorder for easy comparison
rspar = rspar[c(rbind(1:4, 1:4+4)),]#put the 'diff' directly after 'same'
rspar[,3:4] = NA #overwrite columns for secondary parameters
rspar = cbind(rspar, weight1 = 1.0)
colnames(rspar) = colnames(bim_diff_par)[c(1:4, 6:5)]
all_res = rbind(data.frame(rspar),
                  data.frame(bim_diff_par) )
all_res = apply(all_res, MARGIN = 2, FUN = unlist)
rn_res = rownames(all_res)#NB in R4.5 need to make a copy of rownames
rn_res[9:12] = paste('multi', rownames(bim_diff_par)) #identify bimodal ro
rownames(all_res) = rn_res #
#reorder to place unimodal and bimodal distributions of the same type toge
all_res = all_res[c(1:2, 9, 3:4, 10, 5:6, 11, 7:8, 12),]

all_res = within(data = data.frame(all_res),
{
  model = c('same',
            'diff',
            'multi')
  dataset = c(sapply(X = names(pair_diffs_lst),
                     FUN = rep,
                     times = 3) )
})
)
nprint(all res. digits = 3)

```

```

##           mu1 kappa1    mu2 kappa2 loglikelihood weight1  datase
## same green_h1    0.00  2.046     NA     NA        -19.3   1.000 green_h
## diff green_h1 -17.46  2.361     NA     NA        -17.7   1.000 green_h
## multi green_h1 359.99  5.450 298.3    5.45        -16.3   0.571 green_h
## same uv_h1      0.00  0.000     NA     NA        -31.2   1.000   uv_h

```

|                |        |       |       |       |       |       |      |
|----------------|--------|-------|-------|-------|-------|-------|------|
| ## diff uv_h1  | -78.76 | 0.763 | NA    | NA    | -28.3 | 1.000 | uv_h |
| ## multi uv_h1 | 260.12 | 4.800 | 49.3  | 20.94 | -18.2 | 0.706 | uv_h |
| ## same uvg_h  | 0.00   | 4.464 | NA    | NA    | -10.9 | 1.000 | uvg_ |
| ## diff uvg_h  | 13.11  | 5.827 | NA    | NA    | -8.4  | 1.000 | uvg_ |
| ## multi uvg_h | 8.48   | 8.854 | 188.5 | 0.00  | -13.2 | 0.500 | uvg_ |
| ## same uvg_l  | 0.00   | 0.184 | NA    | NA    | -30.7 | 1.000 | uvg_ |
| ## diff uvg_l  | -58.39 | 0.786 | NA    | NA    | -28.2 | 1.000 | uvg_ |
| ## multi uvg l | 47.27  | 4.979 | 270.7 | 4.98  | -22.1 | 0.347 | uvg  |

```
#save results
write.table(x = all_res,
            file = '2Results/MLE_paired_diffs.csv',
            sep = ',',
            row.names = TRUE)
```

As expected, the bimodal models fitted by `circ_mle` have the highest likelihood, followed by the models for differences and then the models for no difference. The difference models require at  $\geq 1$  more parameter, so should increase likelihood by  $>1$  order of magnitude to be appropriate. The bimodal model require  $\geq 2$  more parameters, so should increase likelihood by  $>2$  log units.

## Estimate confidence intervals

For each mean, simulate a distribution of mean estimates

```
#estimate CI for unimodal von Mises models
mle_CI = with(data.frame(mod_diff_results),
              {
                mapply(FUN = CI_vM,
                       angles = pair_diffs_lst,
                       m1 = mu,
                       k1 = kappa,
                       w1 = 1,
                       alternative = 'two.sided', #not a hypothesis test (sym
                       n = 1e4) #takes ~30s for 10 000 simulations
              })
#reorder for comparison like the original models
mle_CI = t(mle_CI)[c(rbind(1:4, 1:4+4)),]
rownames(mle_CI) = rownames(rspar)
colnames(mle_CI) = c('lower', 'median', 'upper')

#estimate CI for unimodal and bimodal von Mises models
mod_CI = with(data.frame(mod_diff_par),
              {
```

```

    mapply(FUN = CI_vM,
           angles = pair_diffs_lst,
           m1 = mu1,
           k1 = kappa1,
           m2 = mu2,
           k2 = kappa2,
           w1 = weight1,
           alternative = 'two.sided', #not a hypothesis test (sym
           n = 1e4) #takes ≈15 for 10 000 simulations
    }
  )
)
#estimate CI for unimodal and bimodal von Mises models
bim_CI = with(data.frame(bim_diff_par),
{
  mapply(FUN = CI_vM,
         angles = pair_diffs_lst,
         m1 = mu1,
         k1 = kappa1,
         m2 = mu2,
         k2 = kappa2,
         w1 = weight1,
         SIMPLIFY = FALSE,
         alternative = 'two.sided', #not a hypothesis test (sym
         n = 1e4) #takes ≈15 for 10 000 simulations
  }
)
#collapse to printable table
mod_CI_tb = Table_vCI(mod_CI)
bim_CI_tb = Table_vCI(bim_CI)
print(mle CI. digits = 3)

```

```

##           lower    median    upper
## same green_h1 22.541   -0.0877 -23.06
## diff green_h1  3.617  -17.4676 -38.14
## same uv_h1    20.907 -150.3497  39.09
## diff uv_h1   -15.131  -78.4741 -140.41
## same uvg_h    13.889   0.0445  -13.56
## diff uvg_h    24.941  13.2806   1.28
## same uvg_l    160.076  -0.5201 -160.34
## diff uvg_l    -0.875  -59.0572 -120.37

```

```
print(mod_CI_tb, digits = 3)
```

```

##           m1.lower m1.median m1.upper m2.lower m2.median m2.upper
## green_h1      17.5      9.07   0.607   -38.2     -56.4    -75.0

```

|           |       |        |          |       |       |        |
|-----------|-------|--------|----------|-------|-------|--------|
| ## uv_h1  | -84.3 | -99.91 | -116.135 | 60.3  | 49.2  | 38.3   |
| ## uv_g_h | 23.7  | 13.03  | 2.376    | NA    | NA    | NA     |
| ## uv_g_l | 68.8  | 46.97  | 25.603   | -73.3 | -89.1 | -105.3 |

We see here that while the confidence interval for the difference distribution between the two green conditions and the two bright conditions nearly overlap with  $0^\circ$  (and the  $0^\circ$  distribution CI overlap with the mean difference), for differences between green lights and dim UV there is a bimodal distribution with CI that do not overlap with  $0^\circ$ . N.B. For *one-sided* calculations, the CI do overlap with  $0^\circ$  for green conditions, bright conditions and the leftmost mean for dim conditions. ### Plot difference CI

```
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(mu_diff_g1),
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim - Green Bright')
PlotCI_vM(mle_CI['diff green_h1'],,
          col = 'darkgreen',
          radius = 1+0.3)

plot.circular(x = circular(unlist(mu_diff_uh1),
                          units = angle_unit,
                          rot = angle_rot
                          ),
             col = 'purple',
             stack = TRUE,
             sep = 0.1,
             bins = 355/5,
             units = angle_unit,
             rotation = angle_rot,
             zero = pi/2,
             shrink = 1.5)
mtext(text = 'UV Dim - Green Bright',
      side = 1,
      line = -2)
lines(x = c(0,0),
      y = c(-1,1),
      col = 'gray')
PlotMV_circMLE(mod_par = mod_diff_par['uv_h1'],,
               col1 = 'purple')
PlotCI_vM(mod_CI_tb['uv_h1'],,
          col = 'purple',
          radius = 1+0.3)

PCfun(angles = unlist(mu_diff_uh),
```

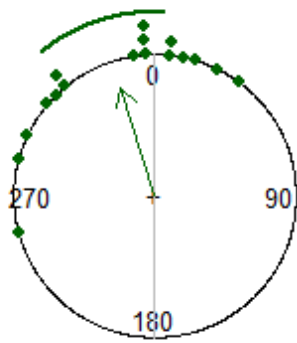


```

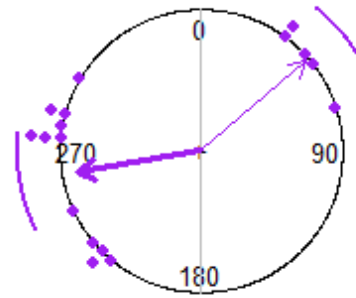
    col = 'gray40',
    shrink = 1.5,
    title = 'UV Bright - Green Bright')
PlotCI_vM(mod_CI_tb['uv_g_h'],,
    col = 'gray40',
    radius = 1+0.3)
#according to model comparison, bim_diff_par['uv_g_h',] is sometimes better
#but is not truly bimodal, kappa2 = 0

plot.circular(x = circular(unlist(mu_diff_gul),
    units = angle_unit,
    rot = angle_rot
    ),
    col = 'gray25',
    stack = TRUE,
    sep = 0.1,
    bins = 355/5,
    units = angle_unit,
    rotation = angle_rot,
    zero = pi/2,
    shrink = 1.5)
mtext(text = 'UV Dim - Green Dim',
    side = 1,
    line = -2)
lines(x = c(0,0),
    y = c(-1,1),
    col = 'gray')
PlotMV_circMLE(mod_par = mod_diff_par['uv_g_l'],,
    col1 = 'gray25')
PlotCI_vM(mod_CI_tb['uv_g_l'],,
    col = 'gray25',
    radius = 1+0.3)

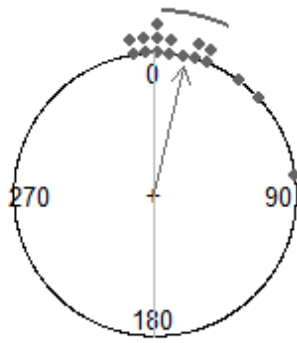
```



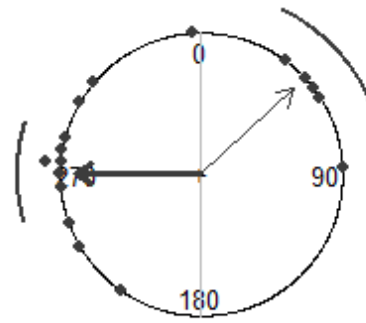
Green Dim - Green Bright



UV Dim - Green Bright



UV Bright - Green Bright



UV Dim - Green Dim

Here we see that the confidence interval for the difference between the green conditions overlaps with  $0^\circ$ , and the one for the two bright conditions almost does (the one sided CI does, but looks odd due it its asymmetry). To confirm whether these models better explain the data than the null hypothesis of the same mean for each condition (no difference), we can perform a likelihood-ratio test.

## Likelihood ratio tests

For each condition, compare the models for same mean, different mean and multiple means. First perform all tests separately, then collect them together and finally adjust for multiple tests before reporting.

```
mod_details= lapply(X = names(pair_diffs_lst),
                    FUN = CollectDetails,
                    ll = all_res,
                    bimod = TRUE)
names(mod_details) = names(pair_diffs_lst)

#save details
save(x = mod_details,
     file = '2Results/MLEdetails_paired_diffs.RData')

#set up tests for hypotheses
```

```

lr_tests = c('pairs_diff_zero', #is the turn different from zero?
             'pairs_multi_diff', #is the turn in one or two directions?
             'pairs_multi_zero' #are the two directions different from zero?
             )#multiple tests

#Perform collect likelihood ratio tests for each hypothesis
all_lr_results_lst = lapply(X = mod_details,
                           FUN = LR_calc_lst,
                           tst = lr_tests)
all_lr_results = do.call(what = rbind,
                        args = all_lr_results_lst
                        )
all_lr_results = within(data.frame(all_lr_results),
                        {
                            comparison = c( sapply(X = names(mod_details),
                                                    FUN = rep,
                                                    times = length(lr_tests)
                                                    ) )
                            test = rep(lr_tests,
                                       times = length(names(mod_details))
                                       )
                        }
                        )

#add labels to interpret each test according to p value and sign of comparison
all_lr_results = within(all_lr_results,
                        {
                            result = mapply(tst = lr_tests,
                                             dev0 = unlist(dev0),
                                             dev1 = unlist(dev1),
                                             pa = p_adjusted,
                                             FUN = H1label
                                             )
                        }
                        )

print_lr_results = all_lr_results
print_lr_results = within(print_lr_results,
                        {
                            rm(dev0, dev1, p)
                            test = sub(sub(sub(test,
                                                  pattern = '_',
                                                  replacement = ' '),
                                          pattern = '_',
                                          replacement = ' '),
                                      pattern = 'pairs ',
                                      replacement = '')
                            mu1 = round(all_res$mu1, 1)
                        }
                        )

```

```

    )
print_lr_results = within(print_lr_results,
                           {mu2 = round(all_res$mu2, 1)}
                           )
rownames(print_lr_results) = with(print_lr_results, paste(comparison, test
#Round numeric columns
num_col = sapply(X = print_lr_results,
                 FUN = is.numeric)
print_lr_results[,num_col] = apply(X = print_lr_results[,num_col],
                                   MARGIN = 2,
                                   FUN = round,
                                   digits = 3)

#Print the results
print(print_lr_results[,-6],
      digits = 3)

```

```

##              d.f. chi_squared p_adjusted      test comparison
## green_hl diff zero      1         3.31      0.206 diff zero green_hl
## green_hl multi diff     3         2.78      0.427 multi diff green_hl -
## green_hl multi zero     4         6.09      0.289 multi zero green_hl 3
## uv_hl diff zero        1         5.82      0.016 diff zero uv_hl
## uv_hl multi diff       3        20.21      0.000 multi diff uv_hl -
## uv_hl multi zero       4        26.03      0.000 multi zero uv_hl 2
## uvg_h diff zero        1         5.04      0.037 diff zero uvg_h
## uvg_h multi diff       3         9.56      0.037 multi diff uvg_h
## uvg_h multi zero       4         4.52      0.340 multi zero uvg_h
## uvg_l diff zero        1         4.98      0.026 diff zero uvg_l
## uvg_l multi diff       3        12.27      0.010 multi diff uvg_l -
## uvg_l multi zero       4        17.26      0.005 multi zero uvg_l
##              mu2
## green_hl diff zero      NA
## green_hl multi diff      NA
## green_hl multi zero 298.3
## uv_hl diff zero         NA
## uv_hl multi diff         NA
## uv_hl multi zero      49.3
## uvg_h diff zero         NA
## uvg_h multi diff         NA
## uvg_h multi zero     188.5
## uvg_l diff zero         NA
## uvg_l multi diff         NA
## uvg_l multi zero     270.7

```

```
#save results
write.table(x = print_lr_results,
            file = '2Results/LRT_paired_diffs.csv',
            sep = ',',
            row.names = TRUE)
```

These tests confirm, that there is no significant difference between the two green conditions, a marginally-significant 13° left turn between the bright conditions, and significant bimodal differences between the two UV conditions and the two dim conditions, driven by left and right turns for the dim UV condition.

## Bayes-factor equivalent

```
# Initialize an empty list to store row segments
table_rows = list()

# Loop through each dataset in your list
for (dts in names(pair_diffs_lst)) {

  mdt_metrics = BFDetails(all_res,
                          dts = dts,
                          bimod = TRUE,
                          n_obs = length(pair_diffs_lst[[dts]]))

  # Define the 3 tests to run per dataset to match your layout
  test_types = c("pairs_diff_zero",
                 "pairs_multi_diff",
                 "pairs_multi_zero")
  display_names = c("diff zero",
                    "multi diff",
                    "multi zero")
  dfs = c(1, 3, 4) # structural degrees of freedom for the tests

  for (i in 1:length(test_types)) {

    # Calculate Bayesian metrics for this specific test
    bf_out = BF_calc(tst = test_types[i],
                     mdt = mdt_metrics)
    label_out = BFlabel(test_types[i],
                        bf_out$Bayes_factor_1.0 )

    # Construct unique row name (e.g., "green_hl diff zero")
    row_id = paste(dts, display_names[i])

    # Assemble the individual row without mu1 and mu2 columns
    table_rows[[row_id]] = with(bf_out,
```

```

    {
      data.frame(
        df          = dfs[i],
        delta_BIC   = round(delta_bic, 3),
        Bayes_factor_1.0 = round(Bayes_factor_1.0 , 4),
        test        = display_names[i],
        comparison   = dts,
        result       = label_out,
        stringsAsFactors = FALSE
      )
    }
  )
}

```

```

# Bind all collected rows into a unified final master table
bf_summary_table = do.call(rbind, table_rows)
#Print the results
print(bf_summary_table,
      digits = 3)

```

```

##           df delta_BIC Bayes_factor_1.0      test comparison
## green_hl diff zero    1      0.479      1.27e+00 diff zero  green_hl
## green_hl multi diff    3     -5.723      5.72e-02 multi diff  green_hl
## green_hl multi zero    4     -5.244      7.26e-02 multi zero  green_hl
## uv_hl diff zero        1      2.982      4.44e+00 diff zero   uv_hl
## uv_hl multi diff        3     11.711      3.49e+02 multi diff   uv_hl
## uv_hl multi zero        4     14.693      1.55e+03 multi zero   uv_hl
## uvg_h diff zero         1      2.210      3.02e+00 diff zero   uvg_h
## uvg_h multi diff        3    -18.065      1.00e-04 multi diff   uvg_h
## uvg_h multi zero        4    -15.855      4.00e-04 multi zero   uvg_h
## uvg_l diff zero         1      2.150      2.93e+00 diff zero   uvg_l
## uvg_l multi diff        3      3.774      6.60e+00 multi diff   uvg_l
## uvg_l multi zero        4      5.925      1.93e+01 multi zero   uvg_l
##                                     result
## green_hl diff zero Anecdotal evidence for H1
## green_hl multi diff Strong evidence for H0
## green_hl multi zero Strong evidence for H0
## uv_hl diff zero    *Moderate evidence for H1
## uv_hl multi diff   Decisive evidence for H1
## uv_hl multi zero   Decisive evidence for H1
## uvg_h diff zero    *Moderate evidence for H1
## uvg_h multi diff   Decisive evidence for H0
## uvg_h multi zero   Decisive evidence for H0
## uvg_l diff zero    Anecdotal evidence for H1

```

```
## uvgl multi diff      *Moderate evidence for H1
## uvgl multi zero      *Strong evidence for H1
```

```
#save results
write.table(x = bf_summary_table,
            file = '2Results/BF_paired_diffs.csv',
            sep = ',',
            row.names = TRUE)
```

## Are the two means for UV dim equally likely

```
wt_uv_h1 = all_res['multi uv_h1', 'weight1']
est_no_m1 = round(length(full_ids)*wt_uv_h1)
binom.test(x = est_no_m1,
            n = length(full_ids),
            p = 0.5,
            alternative = 'two.sided')
```

```
##
## Exact binomial test
##
## data: est_no_m1 and length(full_ids)
## number of successes = 13, number of trials = 19, p-value = 0.1671
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.4344984 0.8742394
## sample estimates:
## probability of success
##          0.6842105
```



```
#CI for relative weight
#circMLE default from Hessians
with(mod_diff_lst,
     ci_circmle(circmle = uv_h1,
                model = uv_h1$bestmodel)
)
```

```
## Parameter      MLE      SE      CI
## 1      q1      4.54    0.14    4.266 - 4.814
```

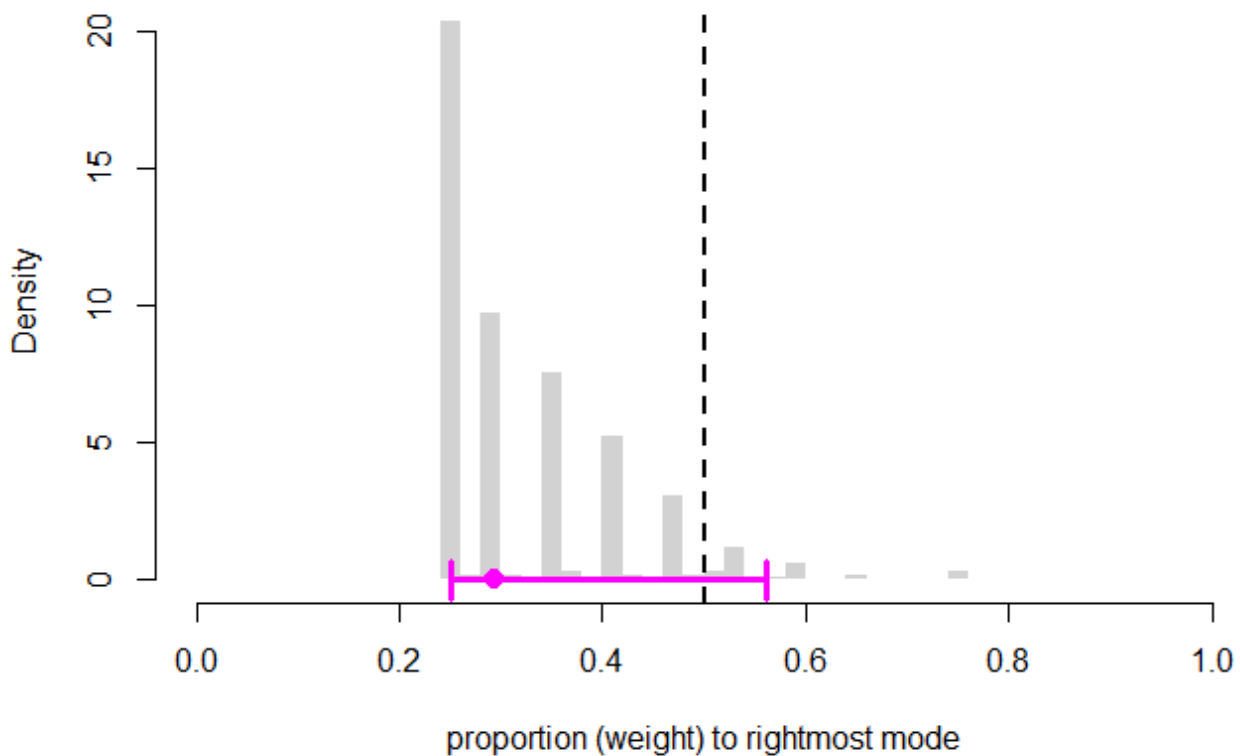
```
## 2      k1  4.798 1.824   1.223 - 8.373
## 3      q2   0.86 0.099   0.666 - 1.054
## 4      k2 20.941 13.08 -4.697 - 46.579
## 5     lambda  0.706 0.111   0.489 - 0.923
```

*#we can also bootstrap it*

```
lambda_uhl = BootBimod(angles = unlist(mu_diff_uhl),
                        model = with(mod_diff_1st, uv_hl$bestmodel),
                        speedup_parallel = TRUE,
                        calc_q = FALSE,
                        n = 1e3) #save all for plotting
q_uhl = quantile(lambda_uhl,
                  probs = c(0,0.5,1)+c(1,0,-1)*((1-0.95)/2)
                  )
hist(lambda_uhl,
     breaks = 20,
     xlim = c(0,1),
     border = NA,
     probability = TRUE,
     main = 'boostrapped M5B model',
     xlab = 'proportion (weight) to rightmost mode'
     )
abline(v = 0.5,
       lty = 2,
       col = 'black',
       lwd = 2)
arrows(x0 = q_uhl[1],
       x1 = q_uhl[3],
       y0 = 0,
       y1 = 0,
       length = 0.1,
       angle = 90,
       code = 3,
       lwd = 3,
       col = 'magenta'
       )
points(x = q_uhl[2],
       y = 0,
       pch = 20,
       col = 'magenta',
       cex = 2)
```



## boostrapped M5B model



```
print(q_uhl,  
      digits = 3)
```

```
## 2.5% 50% 97.5%  
## 0.250 0.294 0.560
```

## Angle between the means and $0^\circ$

```
abs_ang_uhl = abs(unlist(mu_diff_uhl))  
dt_ang_uhl = data.frame(betaresp = abs_ang_uhl/180)  
beta_ang_uhl = brm(bf(betaresp~1,  
                      phi~1),  
                  data = dt_ang_uhl,  
                  family = Beta(link = 'logit',  
                                link_phi = 'log'),  
                  backend = 'cmdstanr')
```

```
## Start sampling
```

```
## Running MCMC with 4 sequential chains...  
##  
## Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
```

```
## Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
## Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)
## Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
## Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
## Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
## Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
## Chain 1 Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 1 Iteration: 1300 / 2000 [ 65%] (Sampling)
## Chain 1 Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 1 Iteration: 1500 / 2000 [ 75%] (Sampling)
## Chain 1 Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 1 Iteration: 1700 / 2000 [ 85%] (Sampling)
## Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 1 finished in 0.1 seconds.
## Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
## Chain 2 Iteration: 400 / 2000 [ 20%] (Warmup)
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## Chain 2 Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 2 Iteration: 1001 / 2000 [ 50%] (Sampling)
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## Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
## Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
## Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2 finished in 0.1 seconds.
## Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
```

```
## Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
## Chain 3 Iteration: 400 / 2000 [ 20%] (Warmup)
## Chain 3 Iteration: 500 / 2000 [ 25%] (Warmup)
## Chain 3 Iteration: 600 / 2000 [ 30%] (Warmup)
## Chain 3 Iteration: 700 / 2000 [ 35%] (Warmup)
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## Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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## Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3 finished in 0.1 seconds.
## Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 4 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
## Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)
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## Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
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## Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
## Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
## Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
## Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4 finished in 0.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.1 seconds.
## Total execution time: 0.8 seconds.
```

```
beta_ang_uhl_null = brm(bf(betaresp~0,
                           phi~1),
                        data = dt_ang_uhl,
                        family = Beta(link = 'logit',
                                     link_phi = 'log'),
                        backend = 'cmdstanr')
```

```
## Start sampling
```

```
## Running MCMC with 4 sequential chains...
##
## Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)
## Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
## Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
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## Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
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## Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
## Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
## Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 1 finished in 0.1 seconds.
## Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
## Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
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## Chain 2 Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 2 Iteration:   900 / 2000 [ 45%] (Warmup)
## Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
```

```
## Chain 2 Iteration: 1100 / 2000 [ 55%] (Sampling)
## Chain 2 Iteration: 1200 / 2000 [ 60%] (Sampling)
## Chain 2 Iteration: 1300 / 2000 [ 65%] (Sampling)
## Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
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## Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2 finished in 0.1 seconds.
## Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 3 Iteration:   100 / 2000 [  5%] (Warmup)
## Chain 3 Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 3 Iteration:   300 / 2000 [ 15%] (Warmup)
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## Chain 3 Iteration: 1700 / 2000 [ 85%] (Sampling)
## Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3 finished in 0.1 seconds.
## Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
## Chain 4 Iteration:   100 / 2000 [  5%] (Warmup)
## Chain 4 Iteration:   200 / 2000 [ 10%] (Warmup)
## Chain 4 Iteration:   300 / 2000 [ 15%] (Warmup)
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## Chain 4 Iteration:   700 / 2000 [ 35%] (Warmup)
## Chain 4 Iteration:   800 / 2000 [ 40%] (Warmup)
## Chain 4 Iteration:   900 / 2000 [ 45%] (Warmup)
## Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)
## Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
## Chain 4 Iteration: 1200 / 2000 [ 60%] (Sampling)
```

```
## Chain 4 Iteration: 1300 / 2000 [ 65%] (Sampling)
## Chain 4 Iteration: 1400 / 2000 [ 70%] (Sampling)
## Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
## Chain 4 Iteration: 1600 / 2000 [ 80%] (Sampling)
## Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
## Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4 finished in 0.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.1 seconds.
## Total execution time: 0.9 seconds.
```

```
summary(beta_ang_uhl)
```

```
## Family: beta
## Links: mu = logit; phi = log
## Formula: betaresp ~ 1
##          phi ~ 1
## Data: dt_ang_uhl (Number of observations: 17)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##        total post-warmup draws = 4000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_E
## Intercept      -0.09      0.19   -0.46    0.28 1.00    2748    19
## phi_Intercept   1.79      0.33    1.11    2.40 1.00    2788    26
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potent
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
loo_beta_uhl = loo(beta_ang_uhl)
loo_beta_uhl_null = loo(beta_ang_uhl_null)
loo_compare(loo_beta_uhl, loo_beta_uhl_null)
```

```
##           elpd_diff se_diff
## beta_ang_uhl_null  0.0      0.0
## beta_ang_uhl      -0.9      0.6
```

```
#the distribution is concave ( $\phi > 2$ ) in 99.9% of draws
mean(
  exp(
    as_draws_df(beta_ang_uhl_null,
                 dpar = 'Intercept_phi')$Intercept_phi
  ) < 2) * 100
```

```
## [1] 0.15
```

```
#estimate mean angular difference
emm_ang_uhl = emmeans(beta_ang_uhl,
                      specs = '1')
print(
  summary(emm_ang_uhl,
          type = 'response')[2:4]*180,
  digits = 1)
```

```
## response lower.HPD upper.HPD
## 1      86      69      102
```

## Plot all raw data

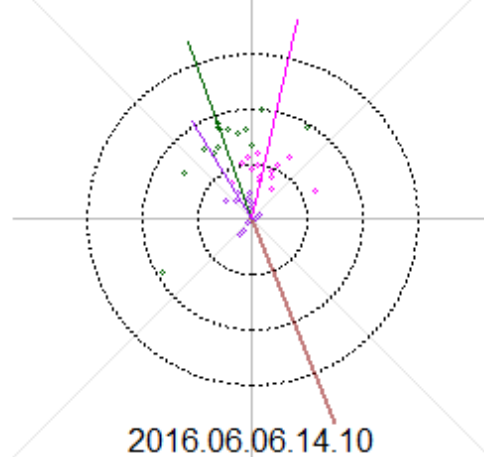
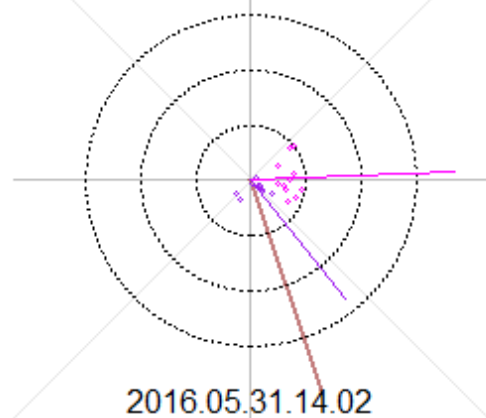
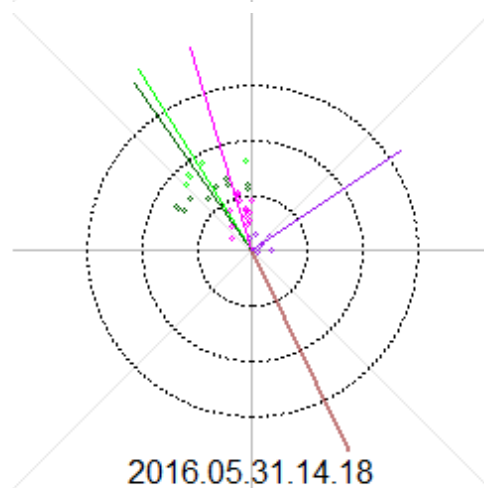
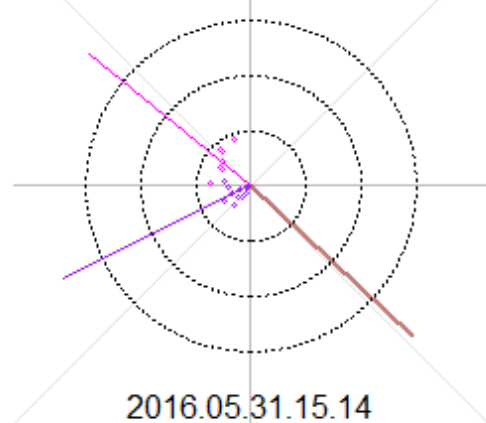
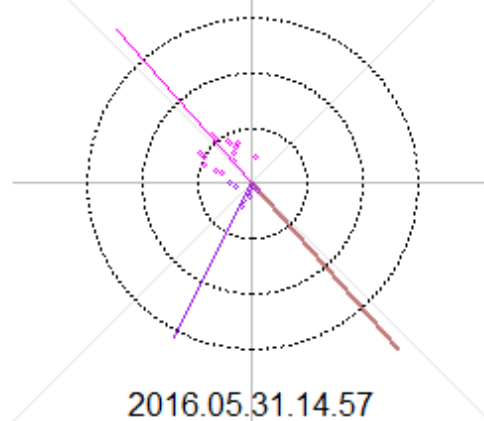
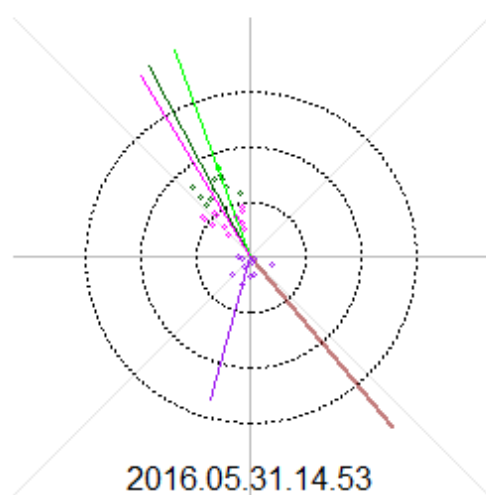
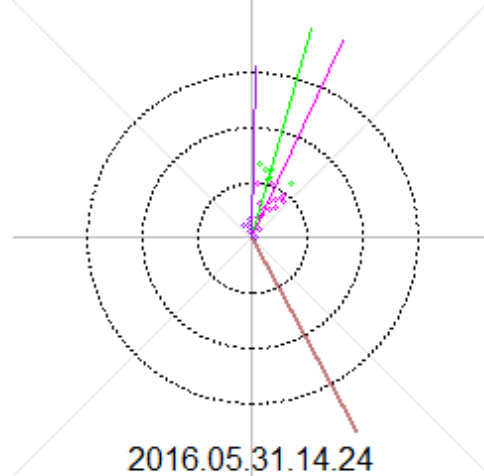
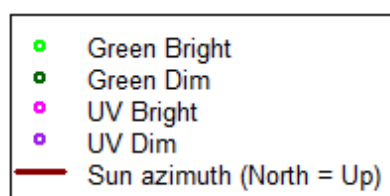
Plot all 169 dances

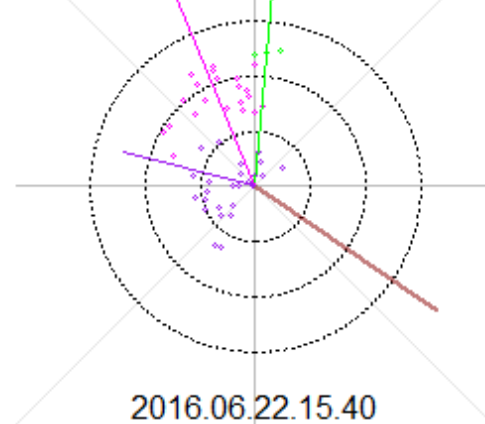
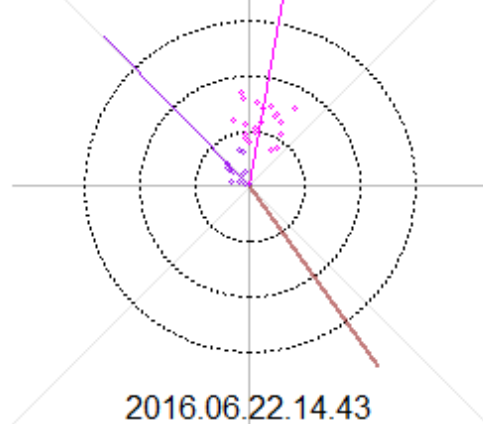
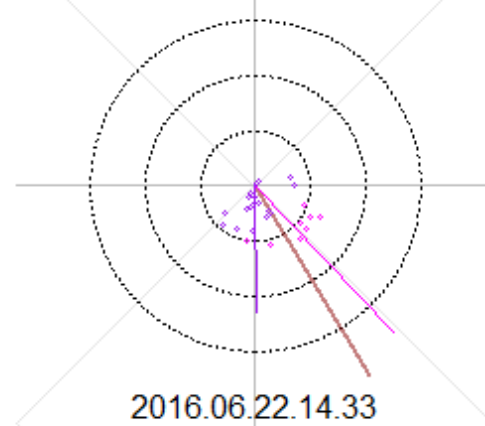
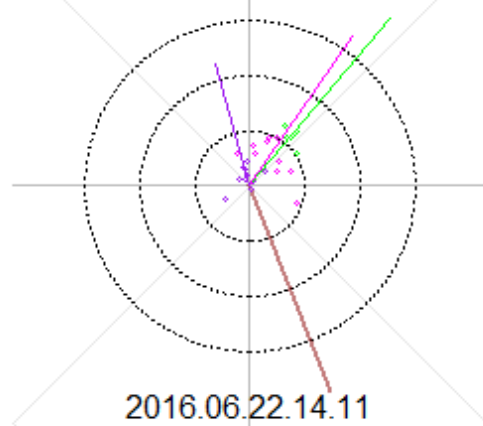
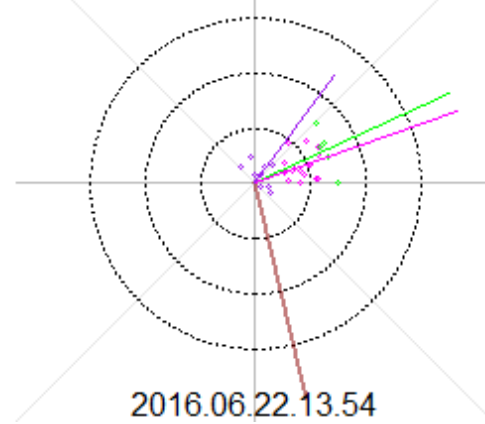
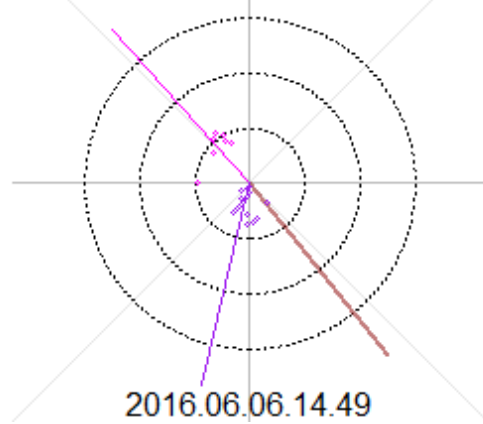
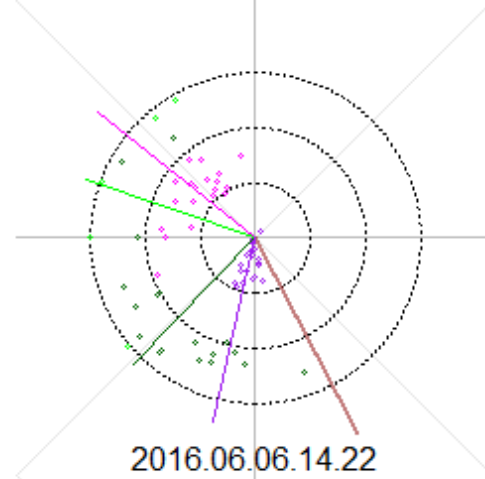
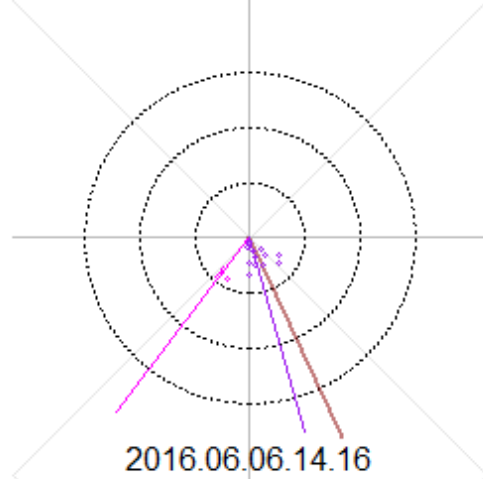
```
#set up a sequence for the axes
xc = seq(from = -pi, to = pi-1e-16, length.out = 1e3)
par(mfrow = c(2, 2),
    mar = c(0,0,0,0))
par(pty = 's')

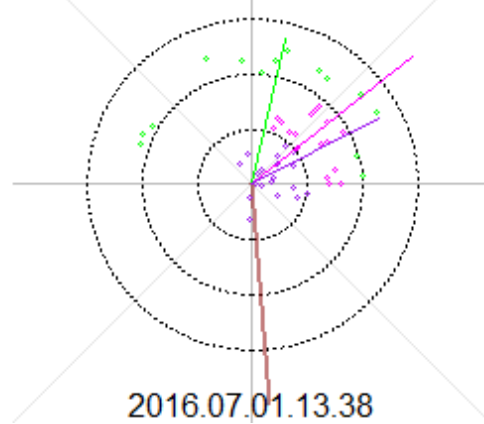
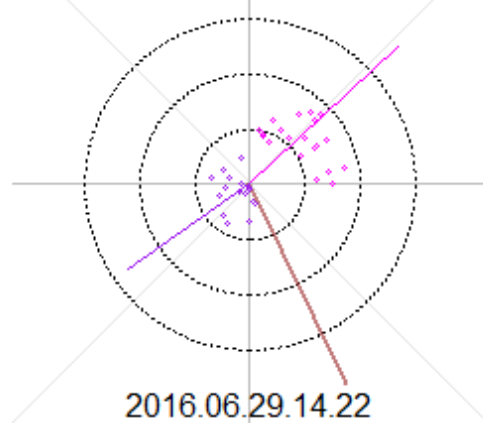
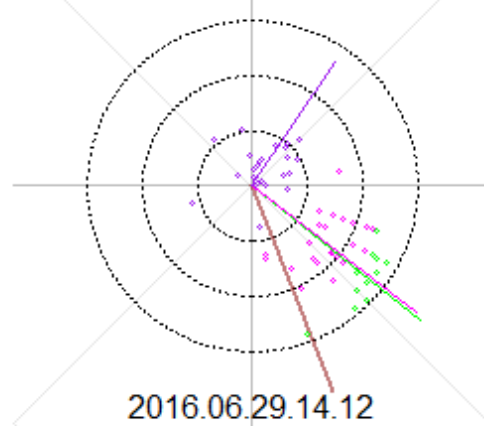
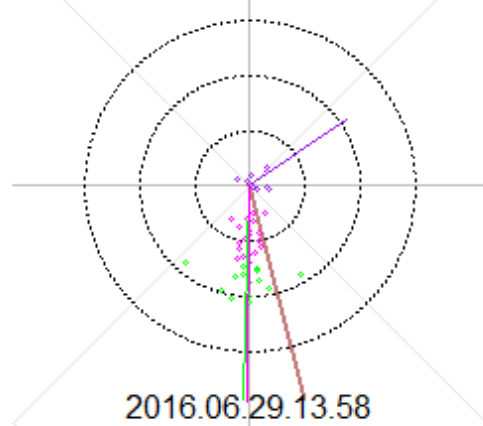
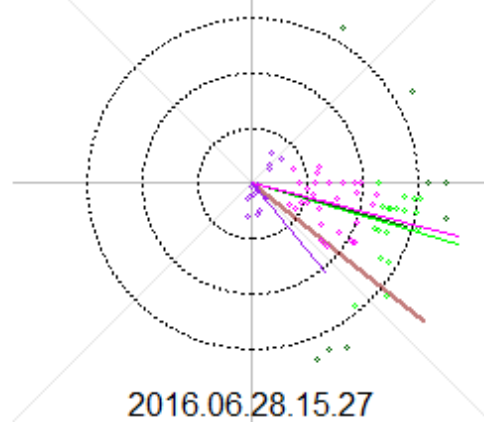
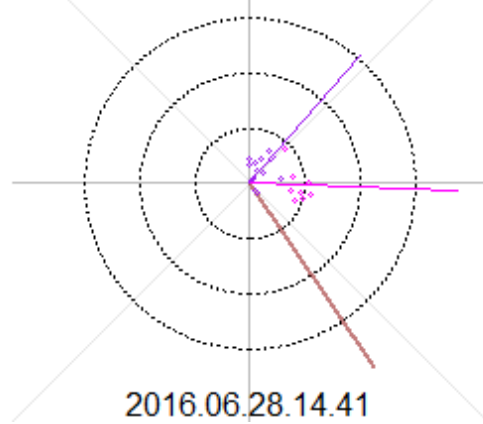
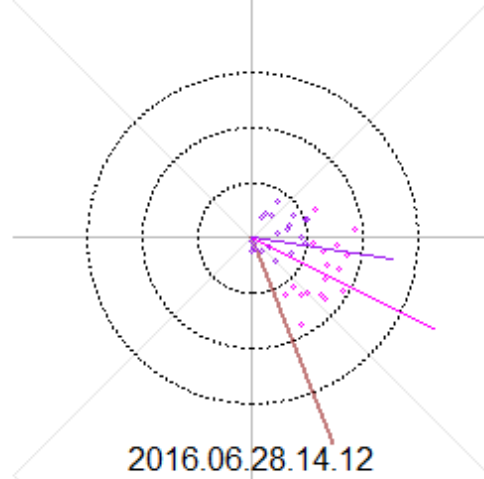
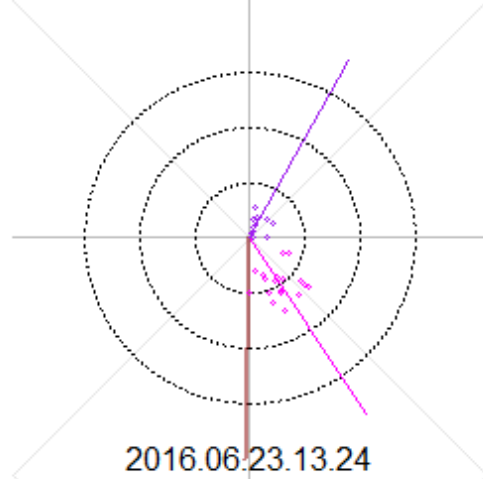
#add legend to 1st page
plot(x = NULL,
     xlim = c(-1,1),
     ylim = c(-1,1),
     pch = 19,
     axes = FALSE,
     xlab = '',
     ylab = '',
     main = ''
)
legend(x = 'center',
      legend = c('Green Bright',
                  'Green Dim',
```

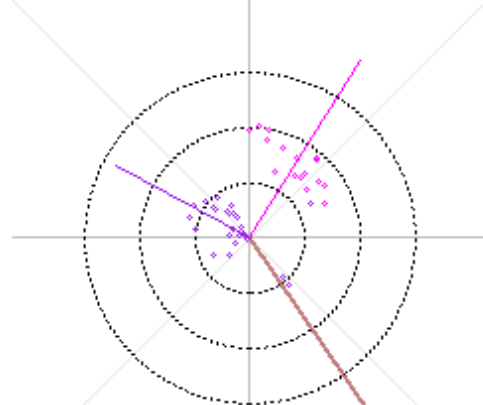
```
        'UV Bright',
        'UV Dim',
        'Sun azimuth (North = Up)'),
col = c('green',
        'darkgreen',
        'magenta',
        'purple',
        'darkred'),
pch = c(21,21,21,21,NA),
lty = c(NA, NA, NA, NA, 1),
lwd = c(2,2,2,2,3)
)
#loop through individuals
PltDance(ids = u_id,
        dtf = cd,
        cx = 1.0)
```



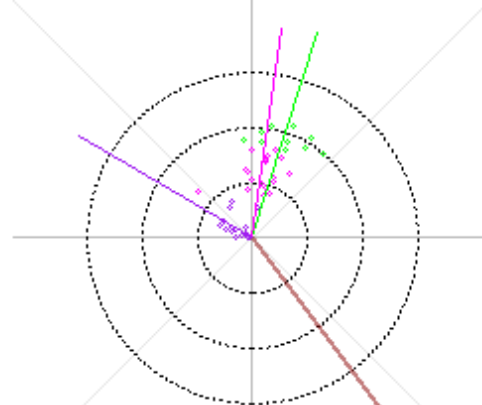




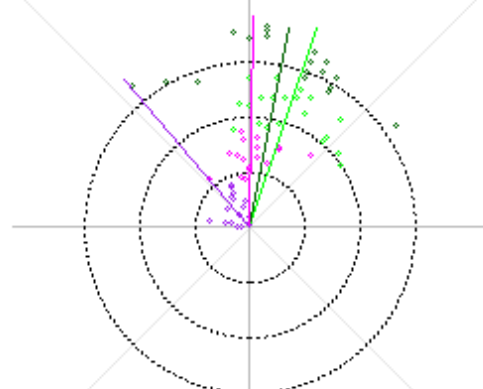




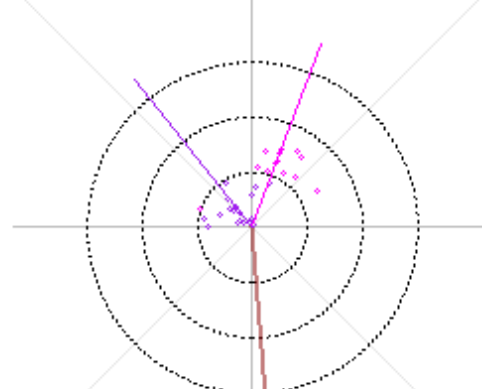
2016.07.01.14.43



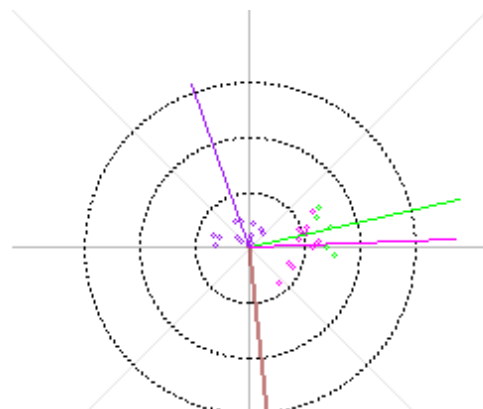
2016.07.01.14.49



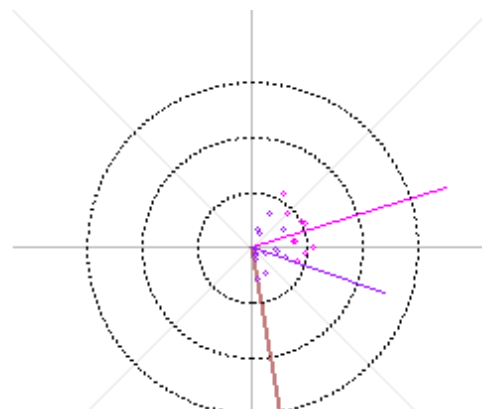
2016.07.04.\_\_\_\_



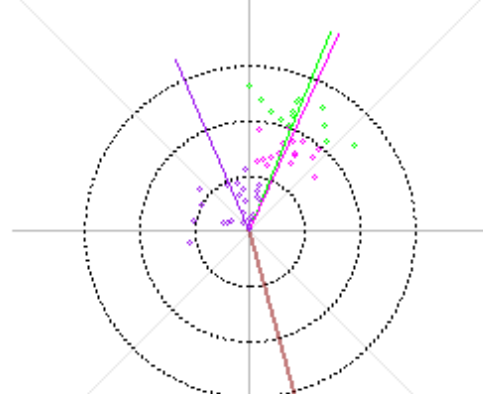
2016.07.05.13.38



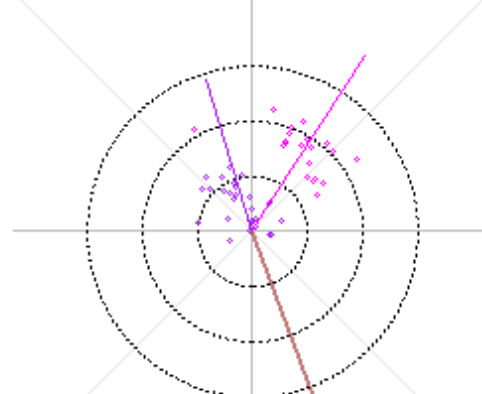
2016.07.05.13.42



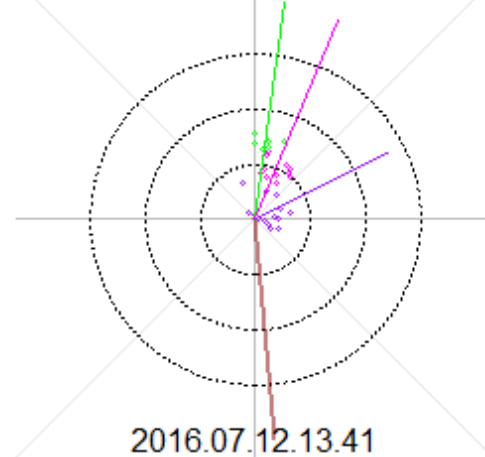
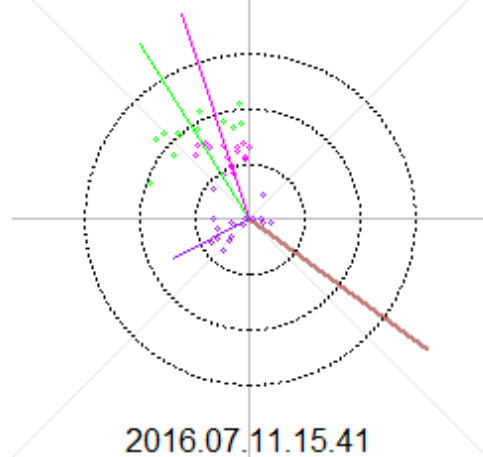
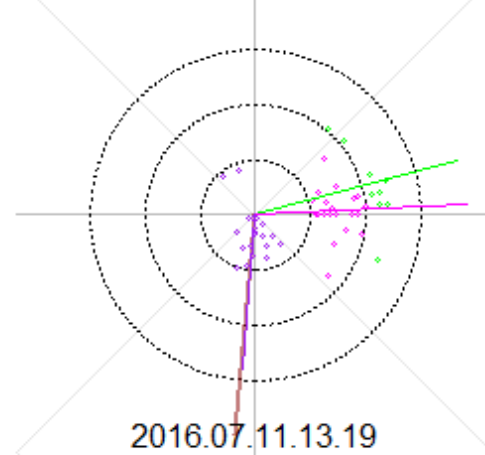
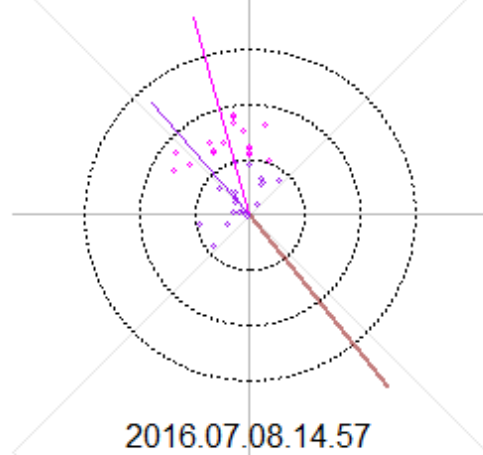
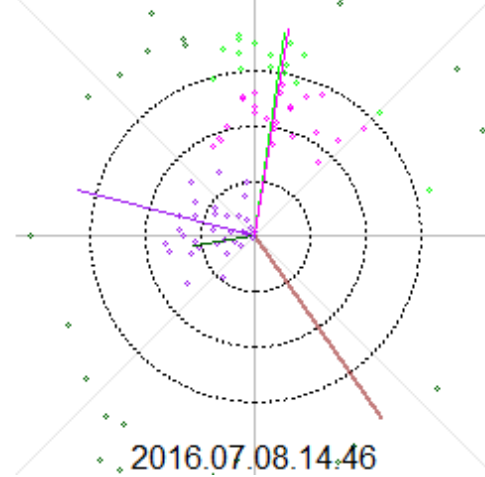
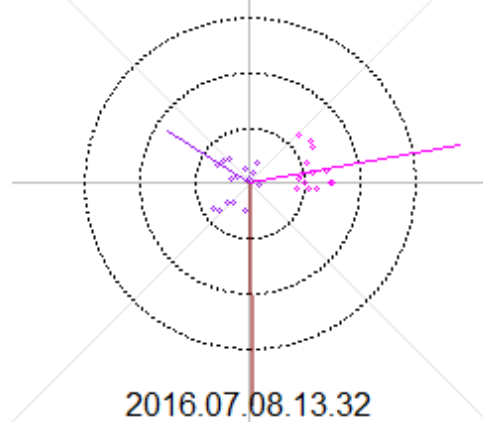
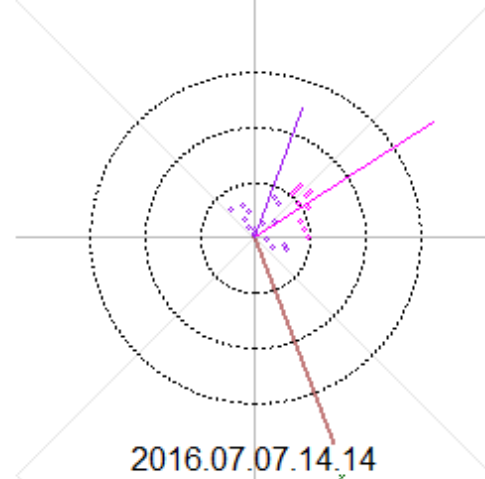
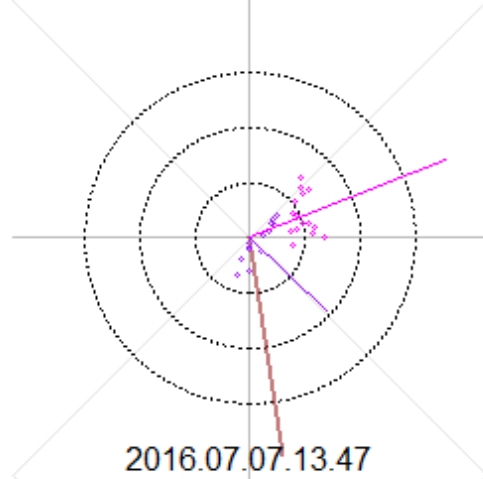
2016.07.05.13.49

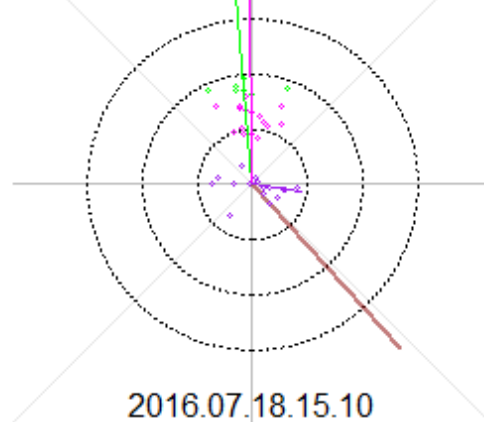
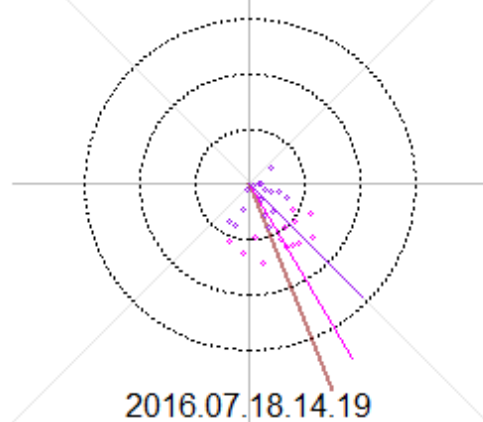
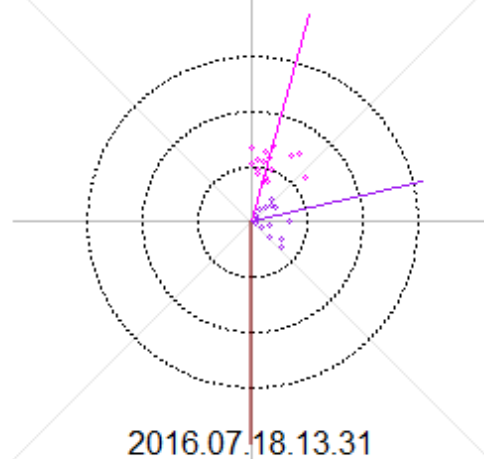
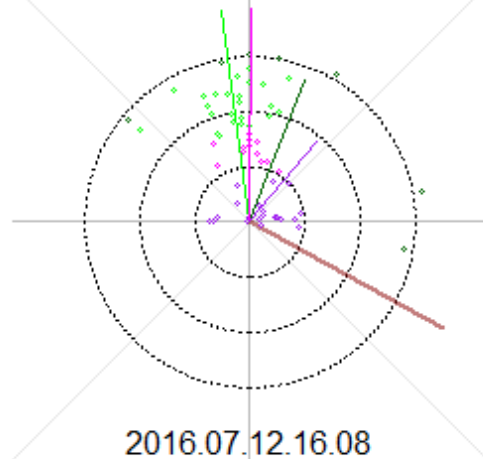
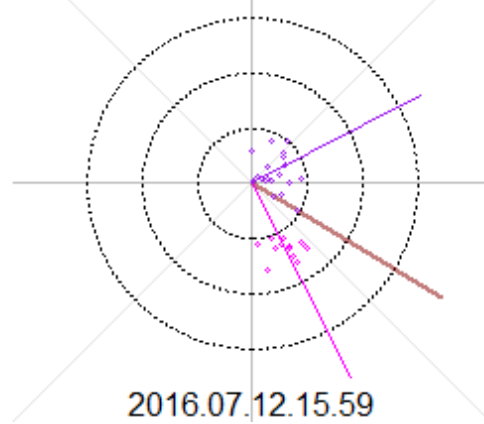
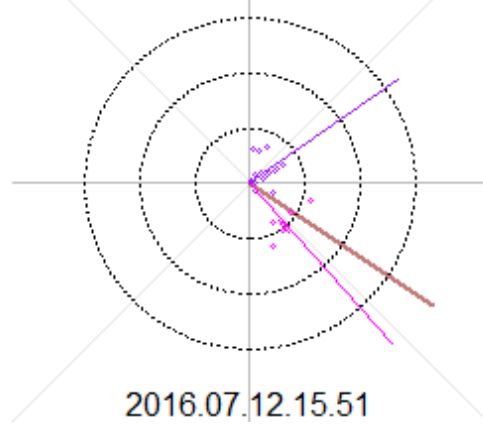
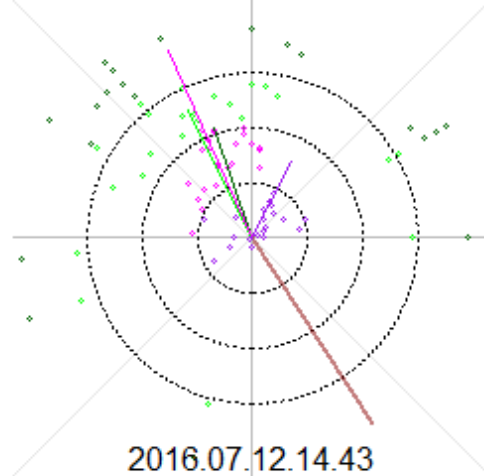
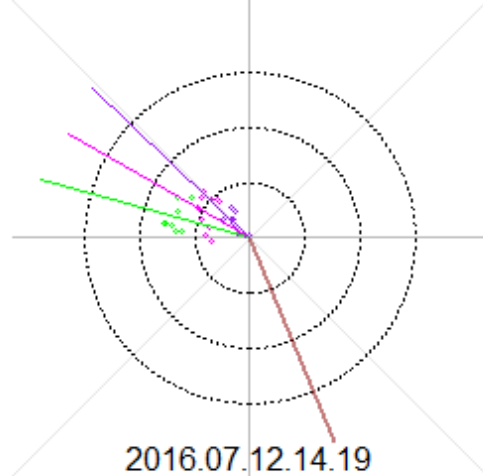


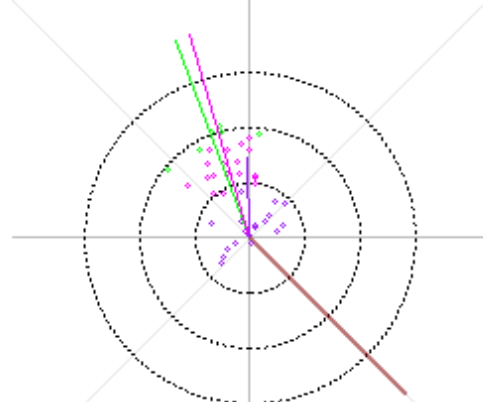
2016.07.05.14.01



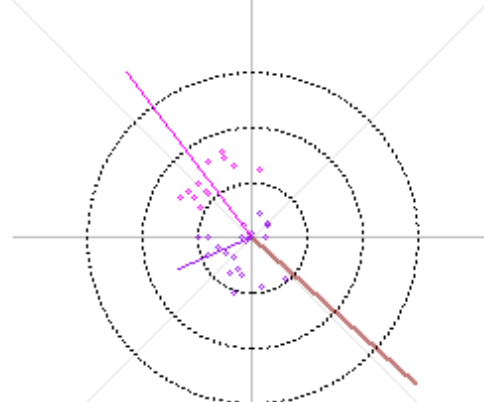
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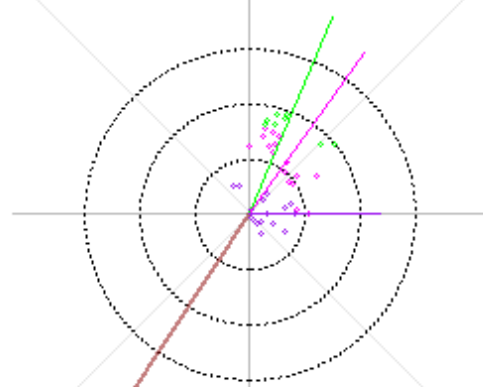




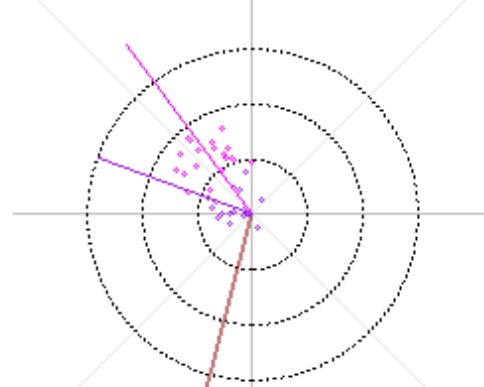
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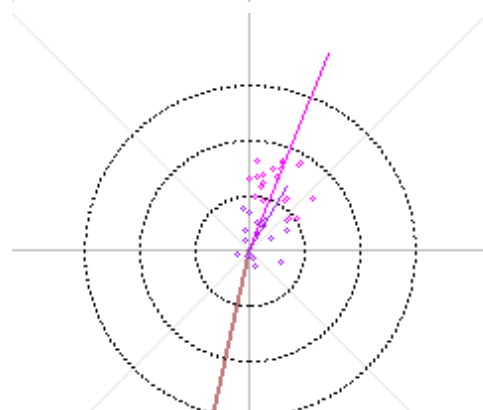
2016.07.18.15.28



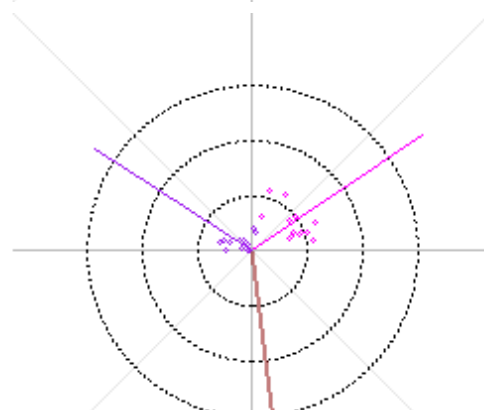
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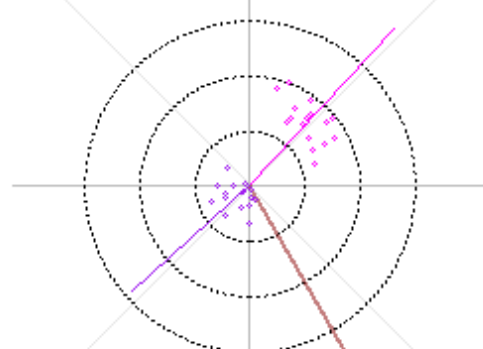
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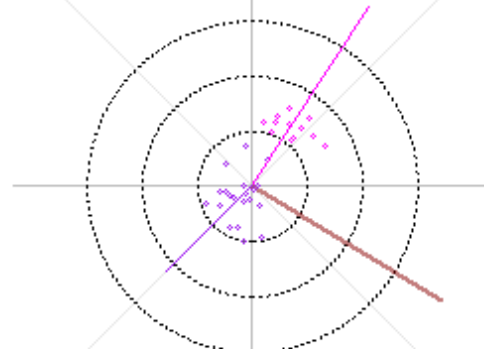
2016.07.19.13.04



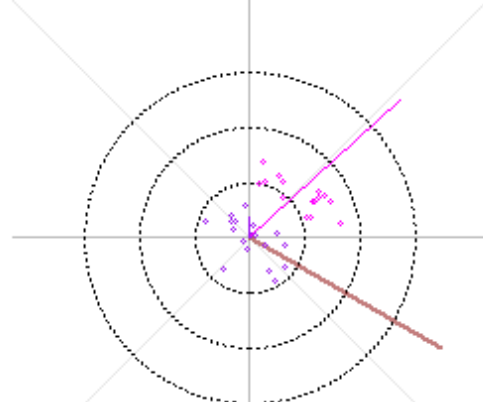
2016.08.17.13.46



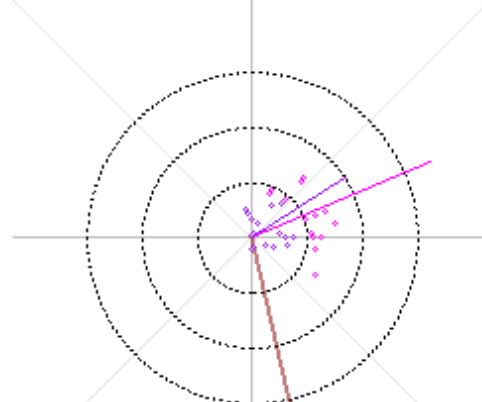
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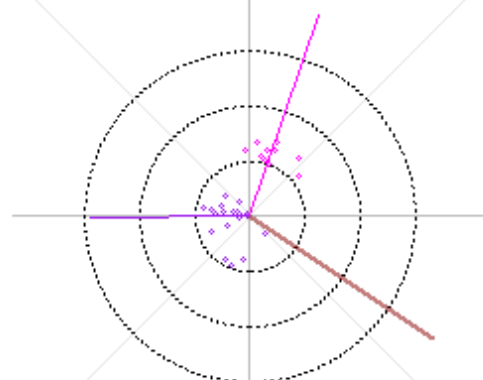
2016.08.17.16.22



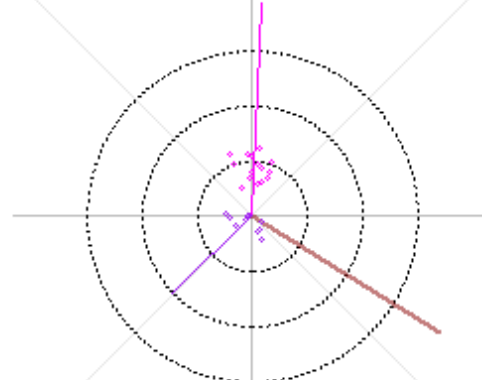
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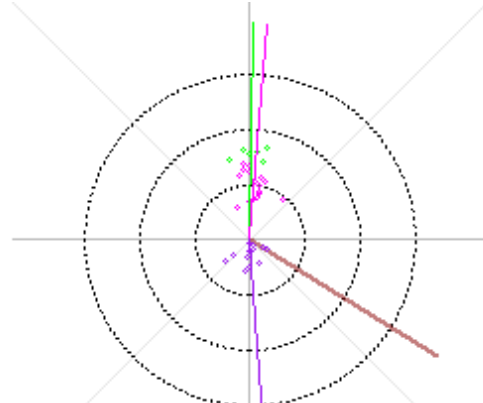
2016.08.18.14.01



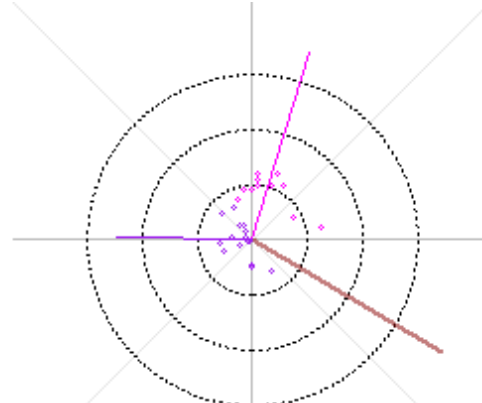
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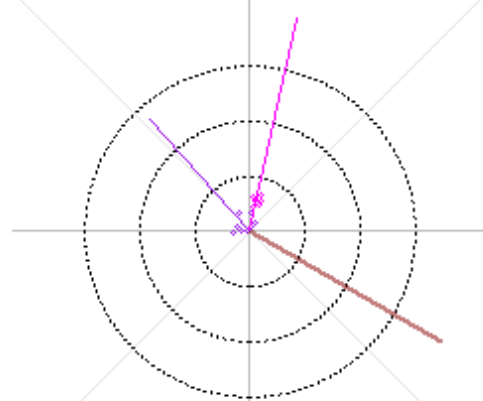
2016.08.18.16.20\_2



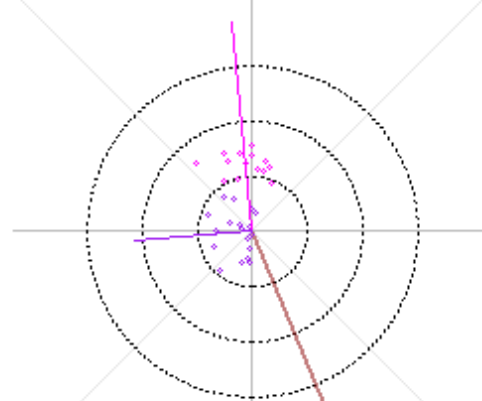
2016.08.18.16.20\_1



2016.08.18.16.26\_1

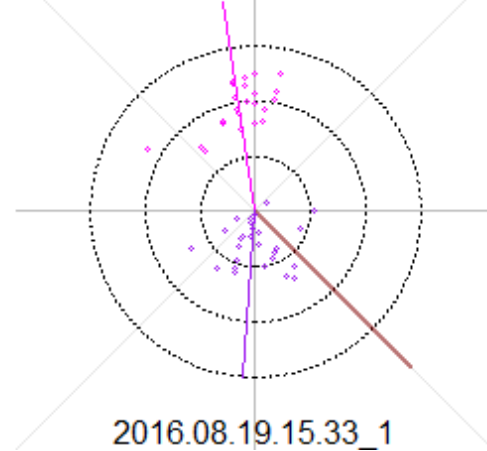
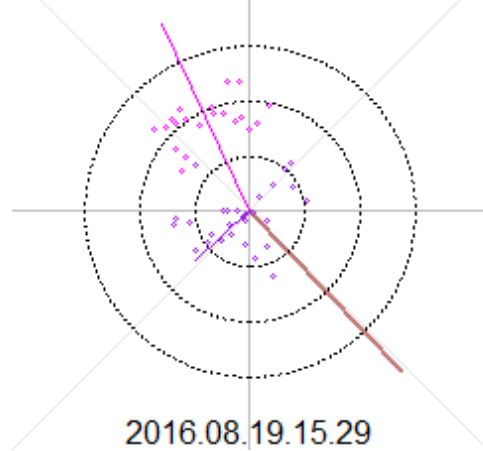
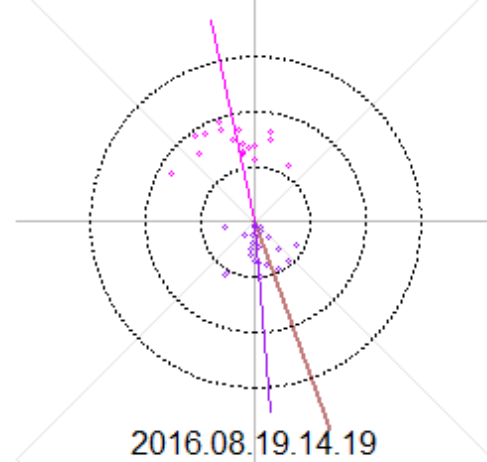
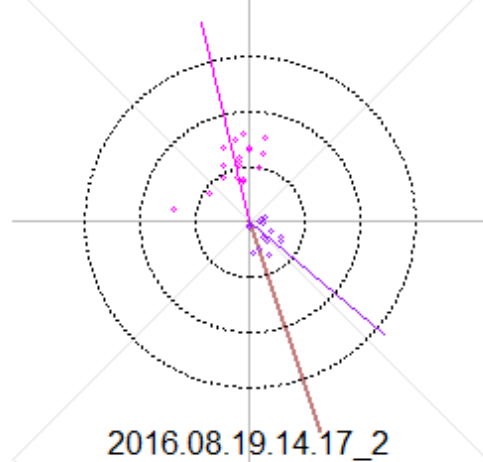
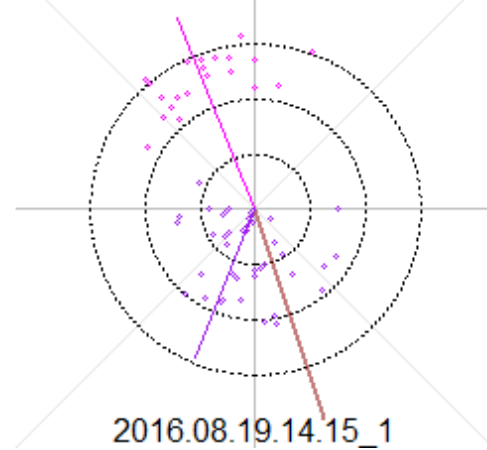
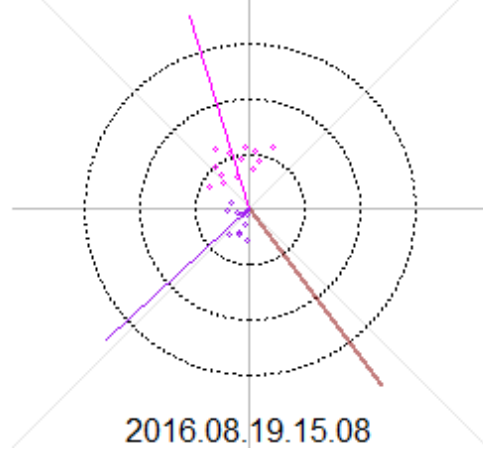
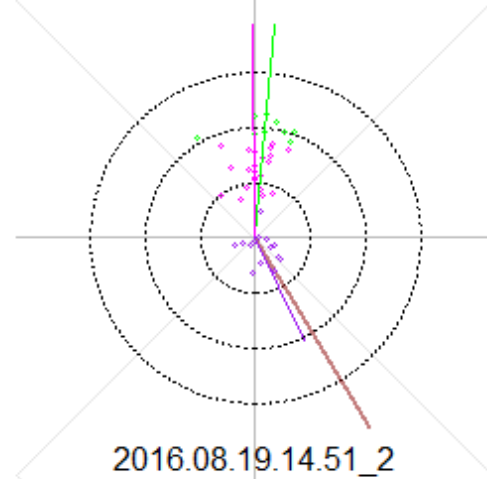
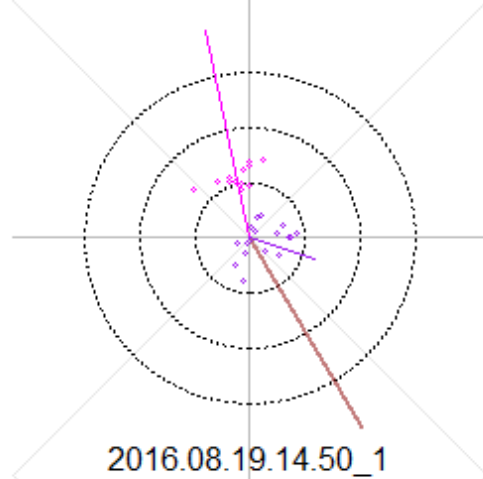


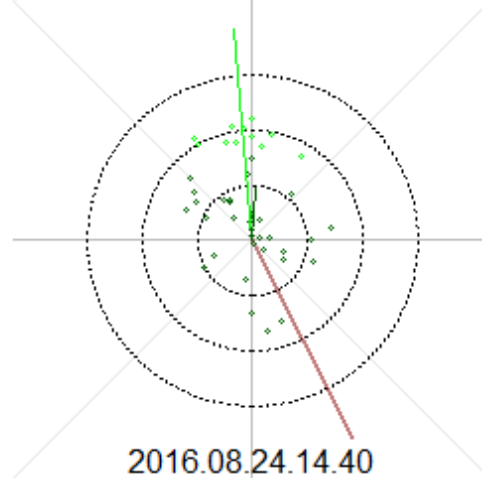
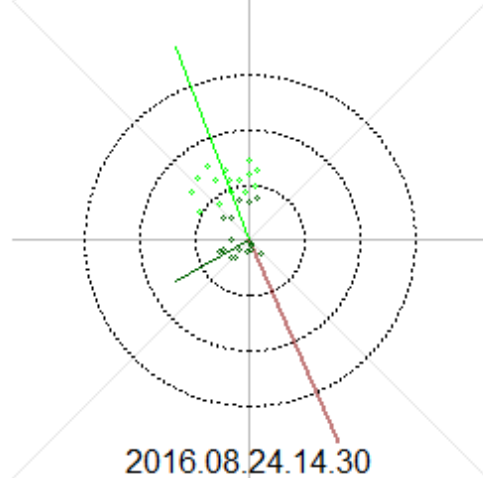
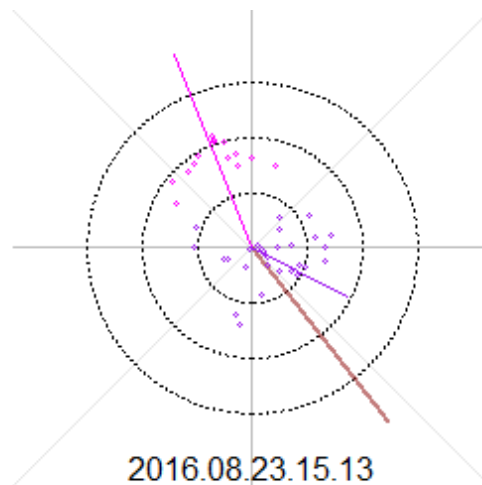
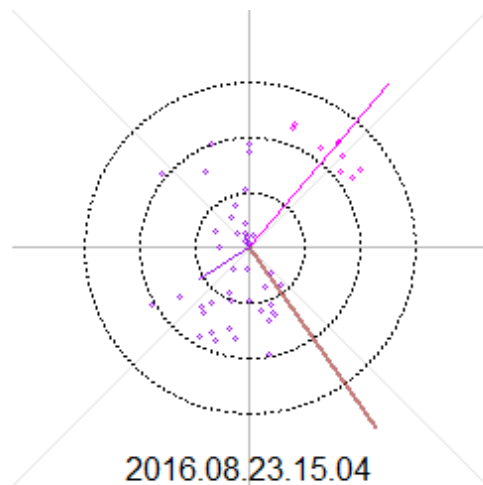
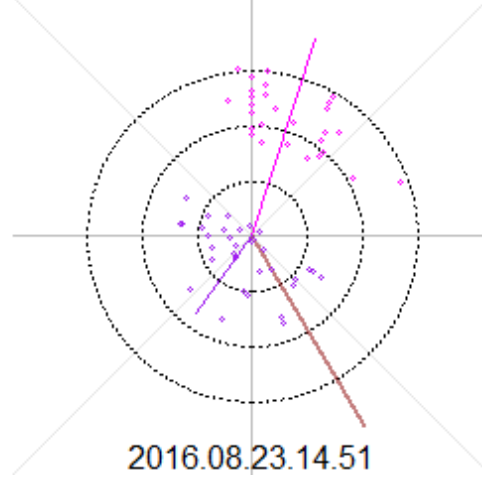
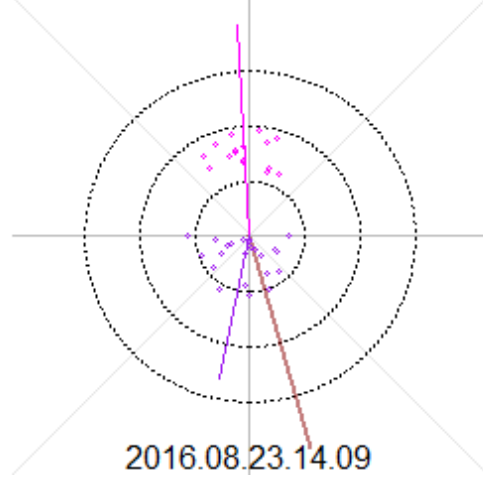
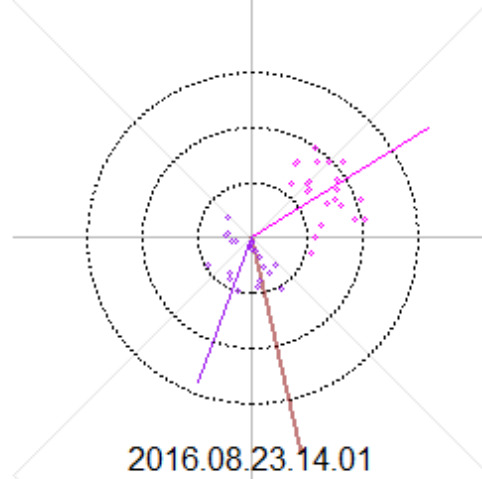
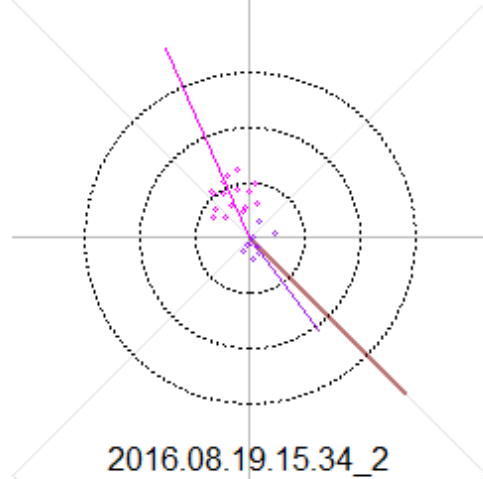
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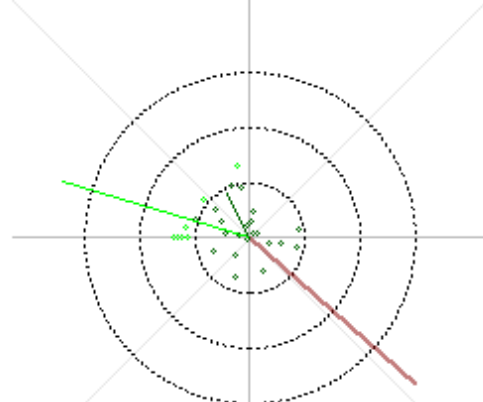


2016.08.19.14.26

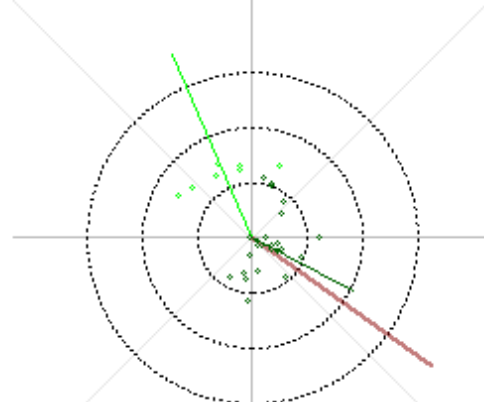




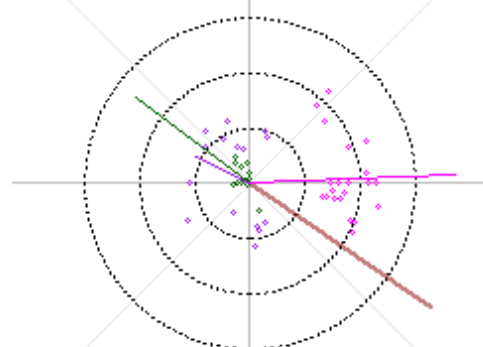




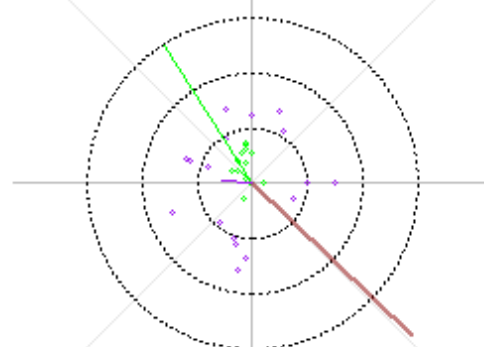
2016.08.24.15.48



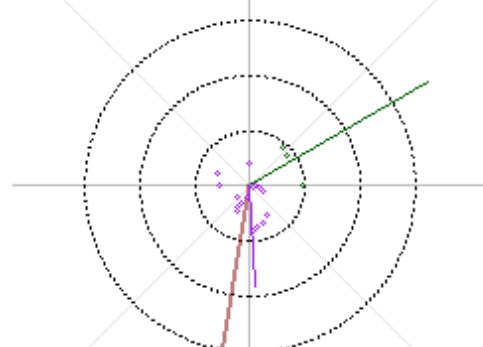
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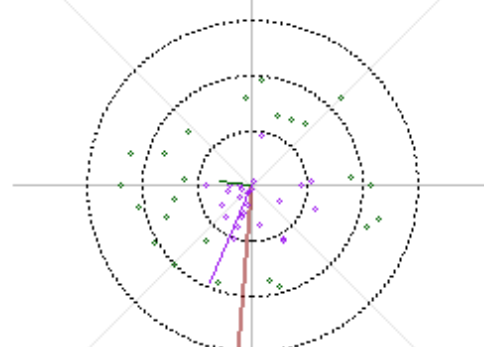
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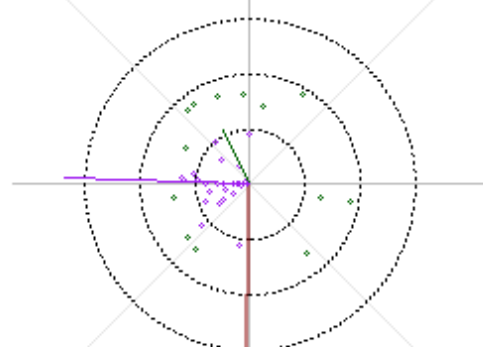
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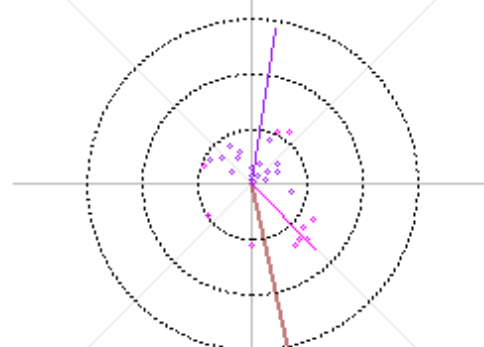
2016.08.25.13.01



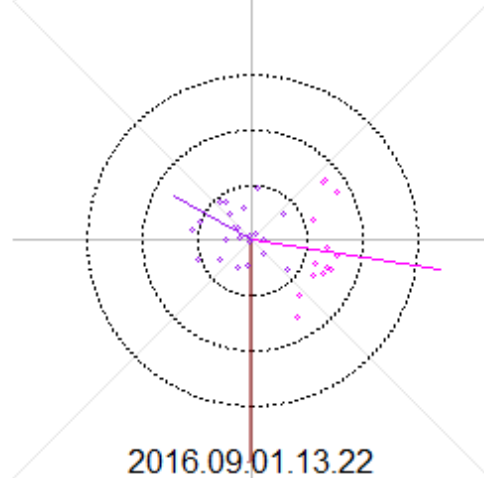
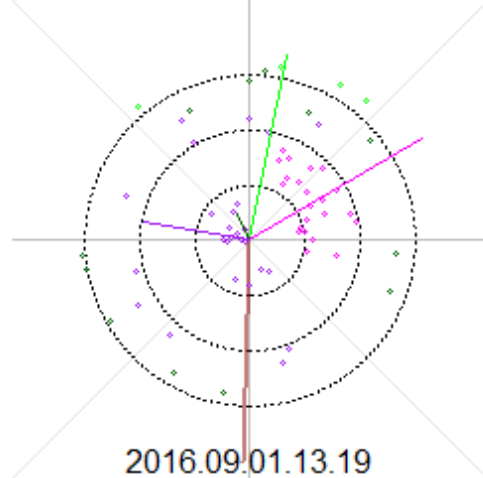
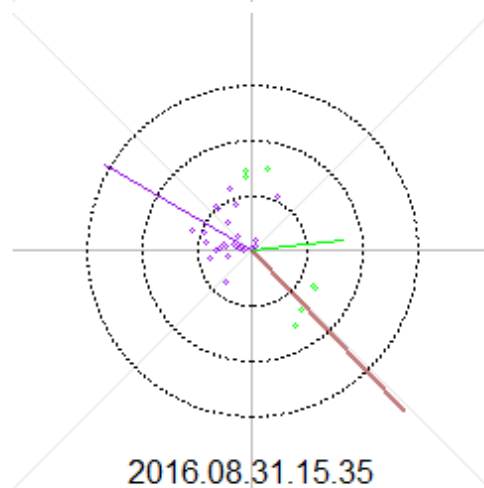
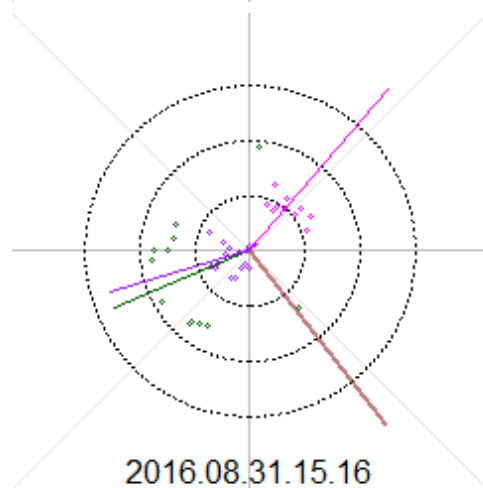
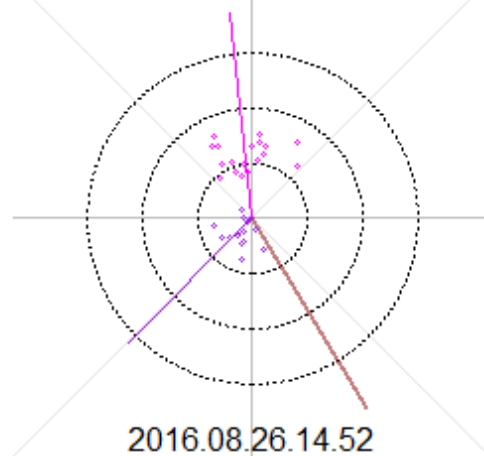
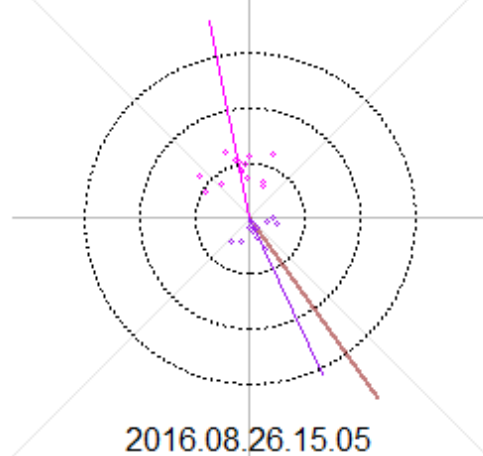
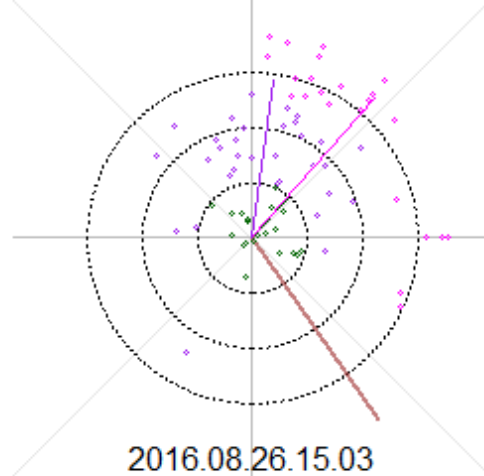
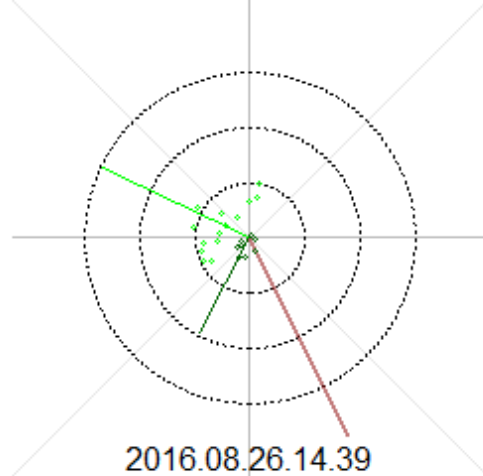
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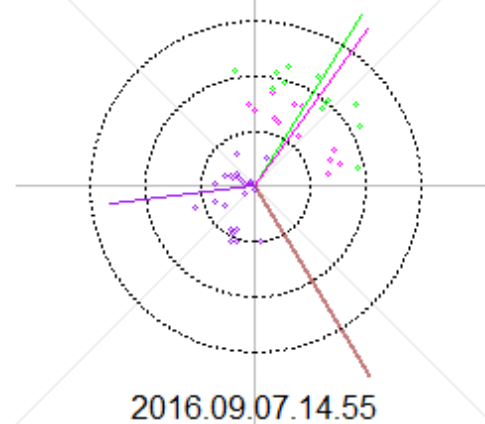
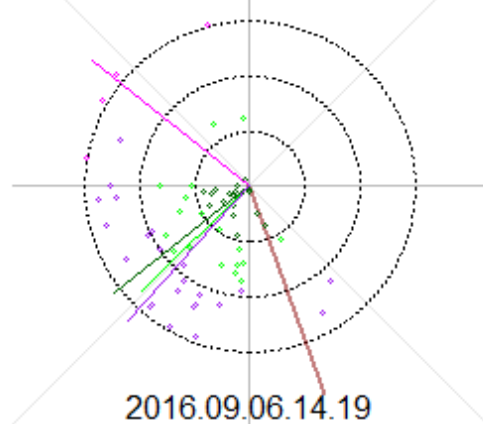
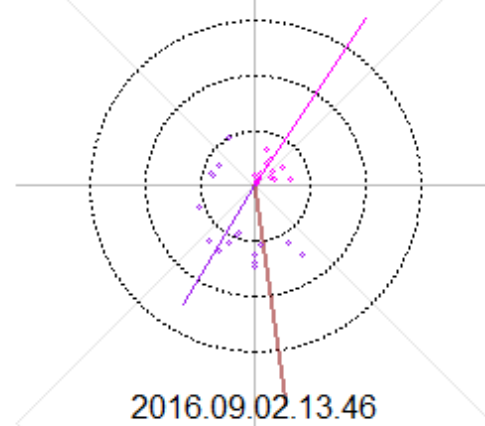
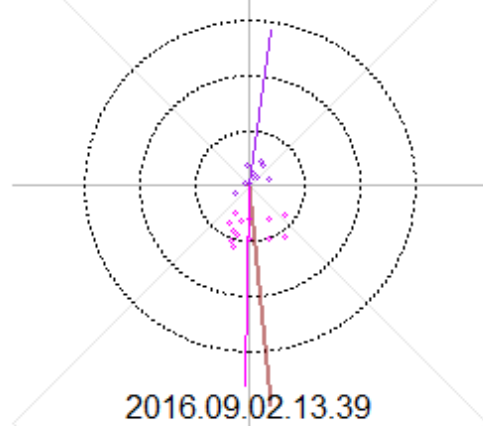
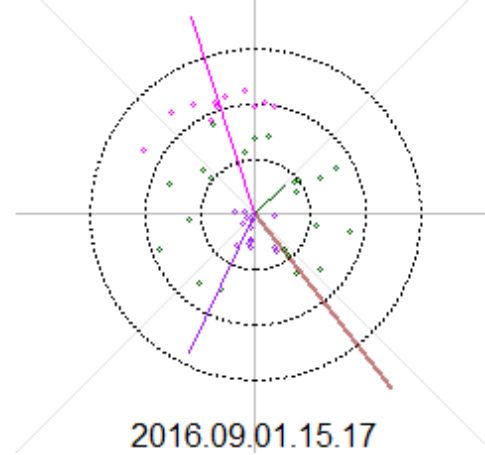
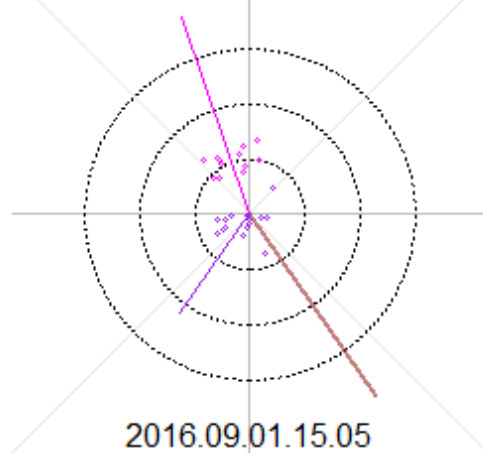
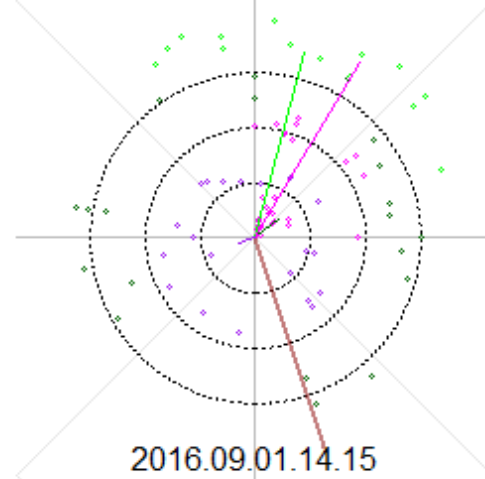
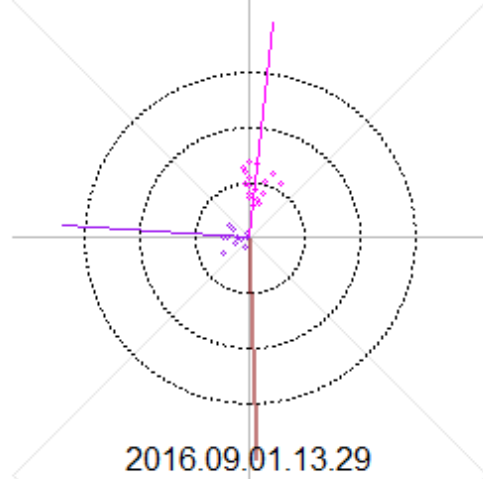


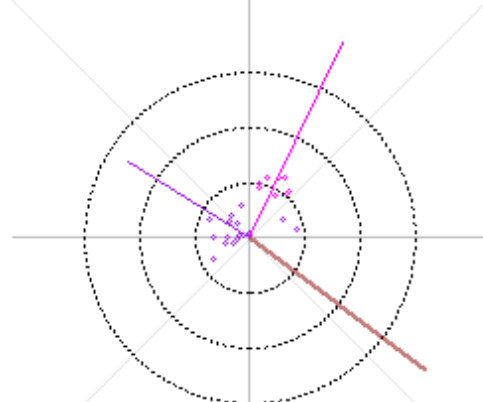
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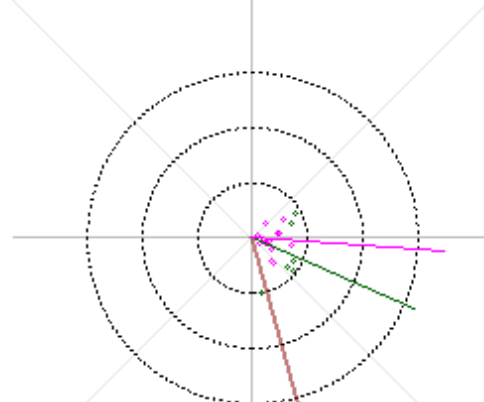
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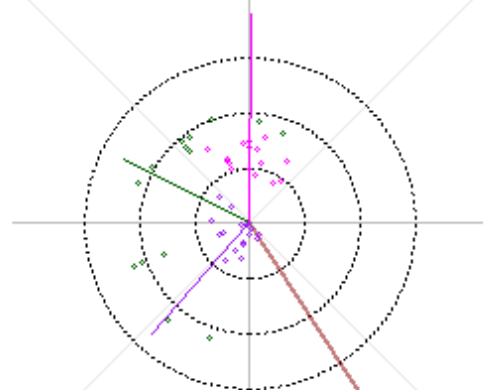




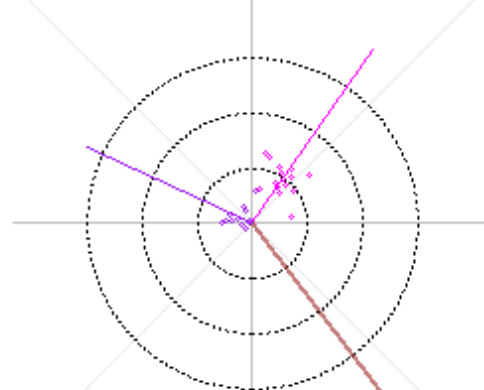
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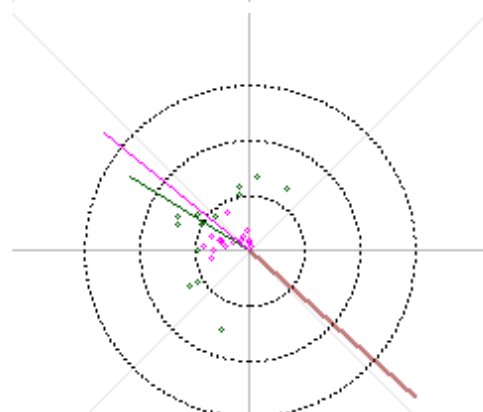
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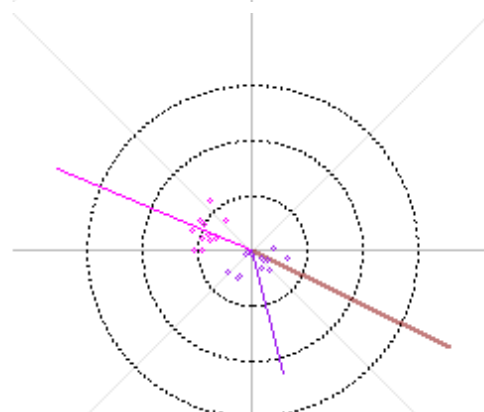
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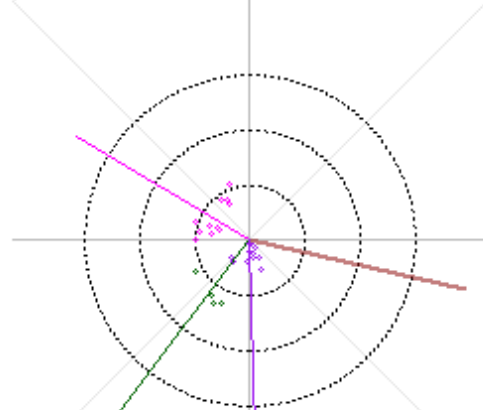
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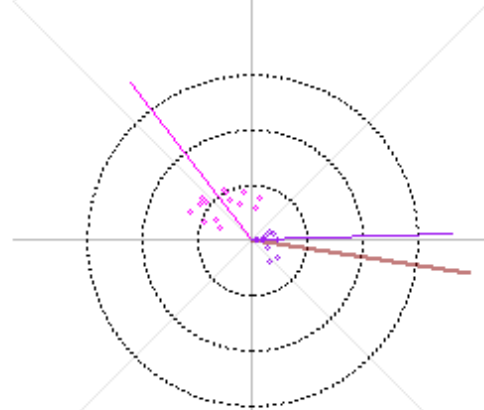
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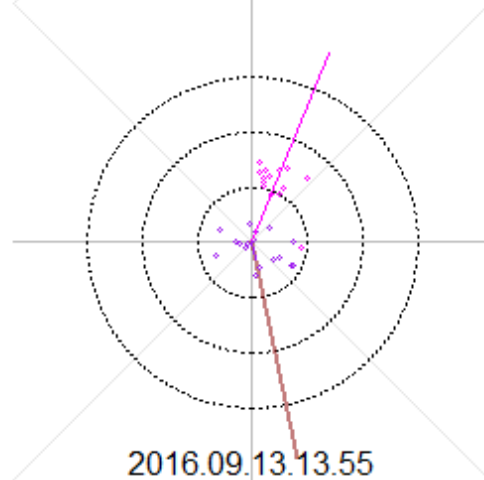
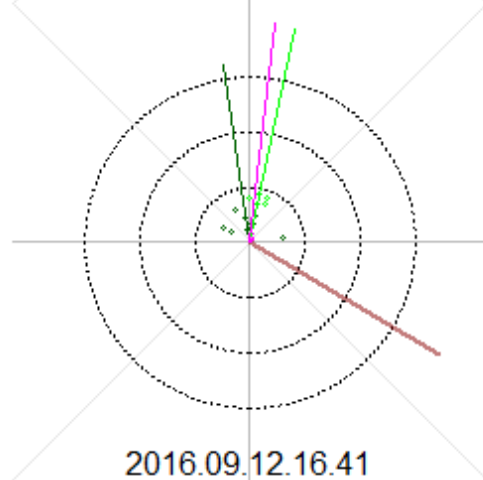
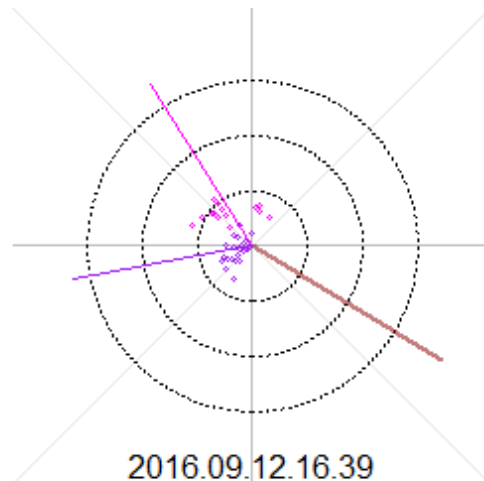
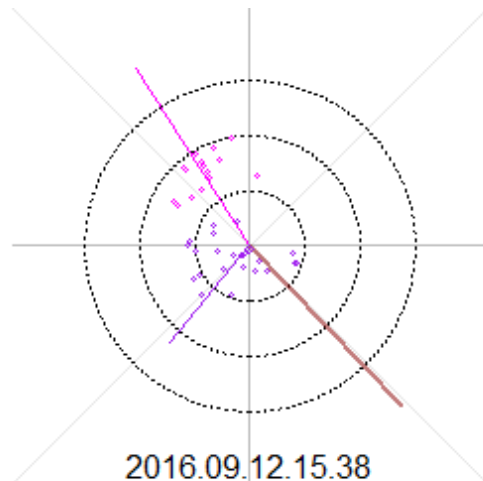
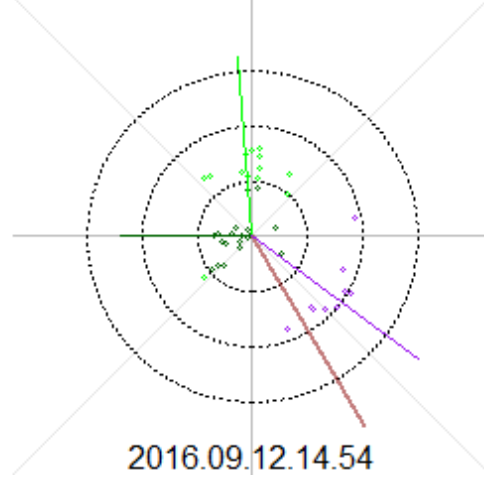
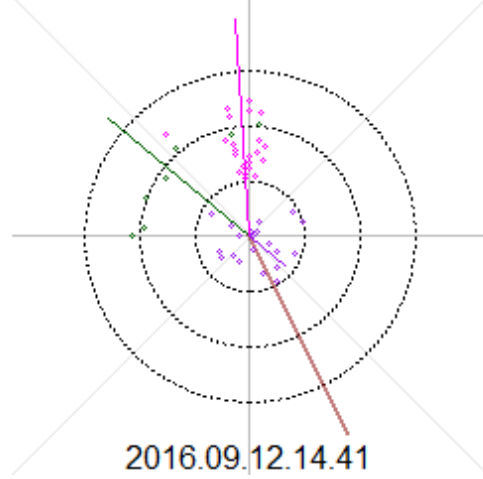
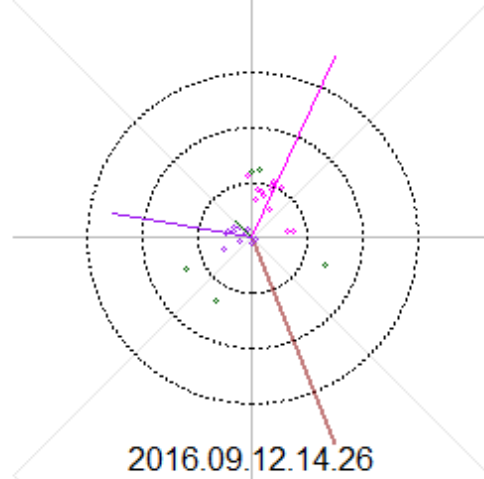
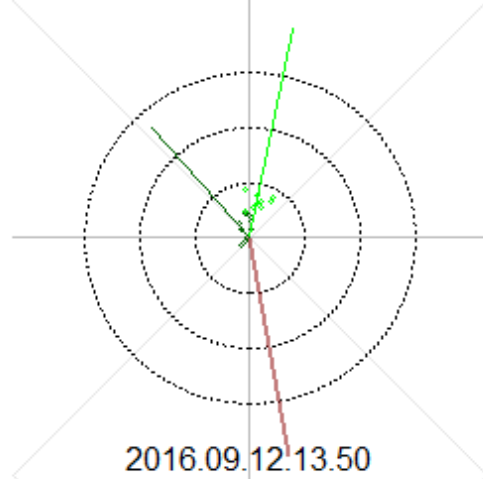
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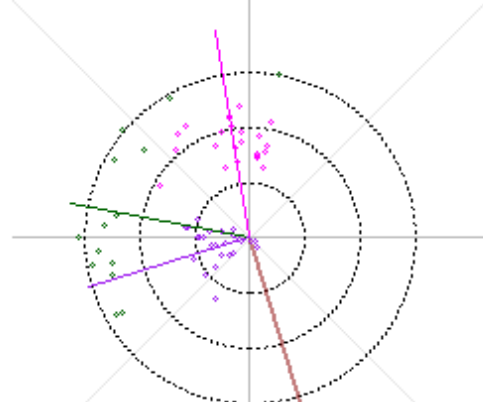


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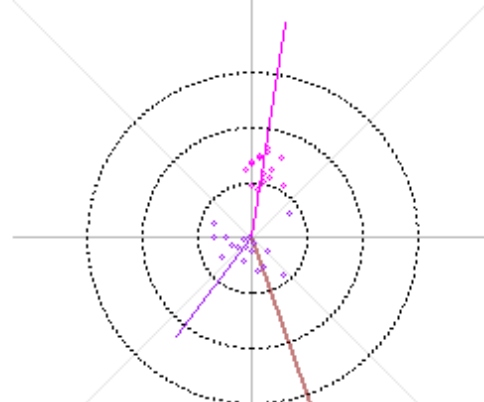


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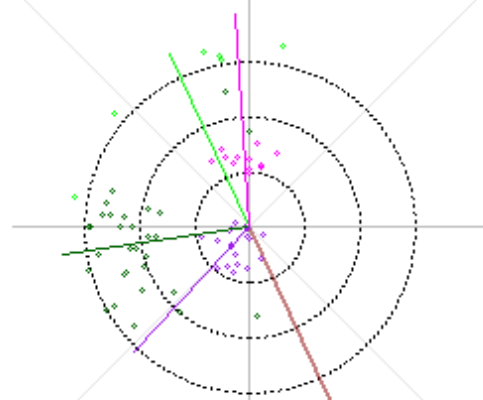




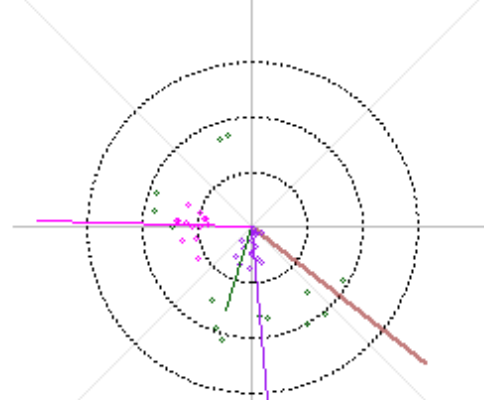
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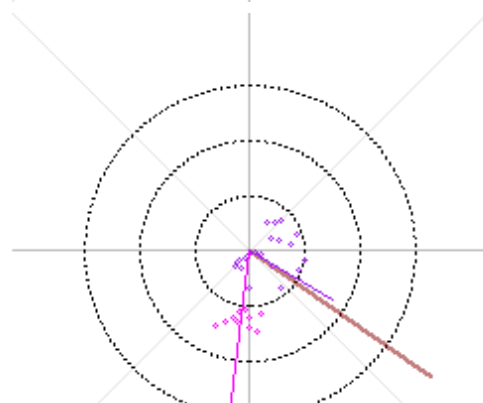
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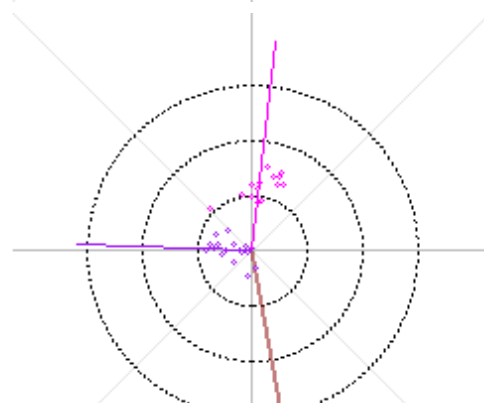
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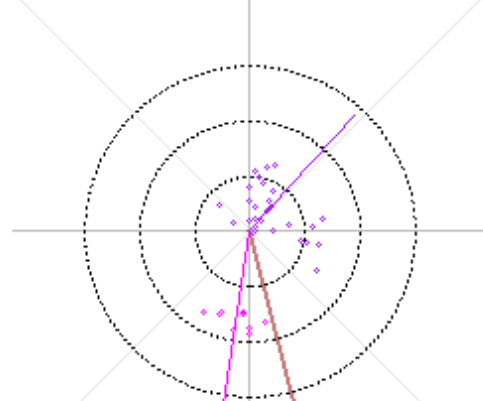
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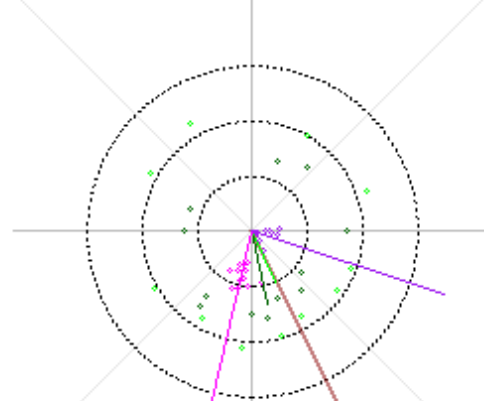
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2016.09.14.13.51



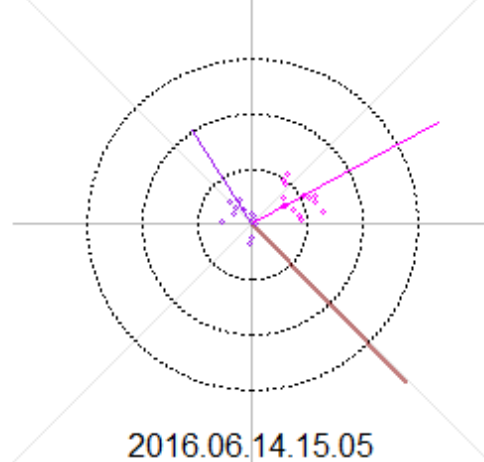
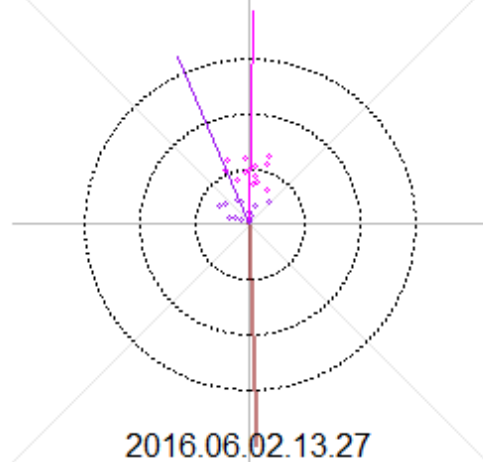
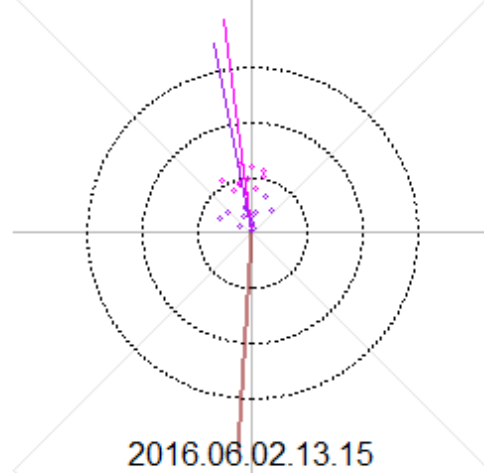
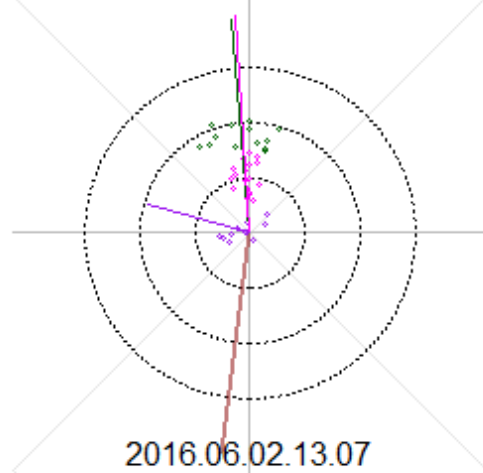
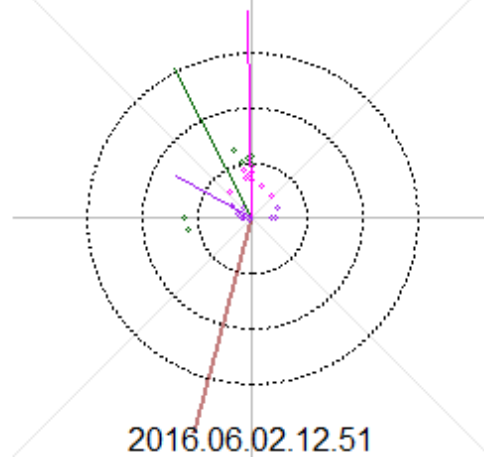
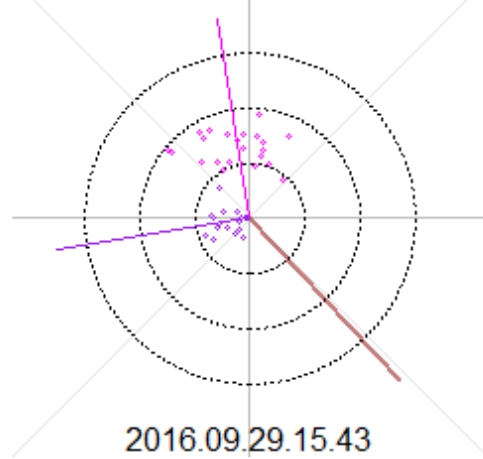
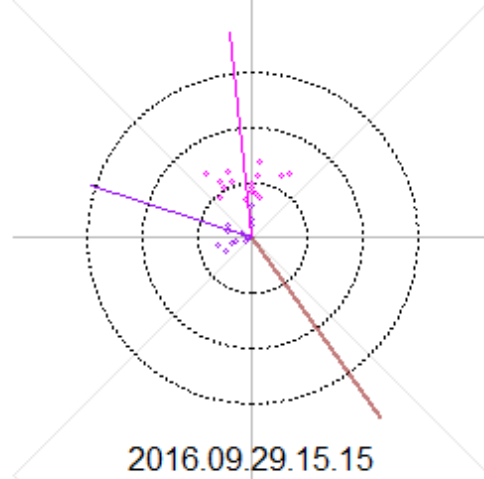
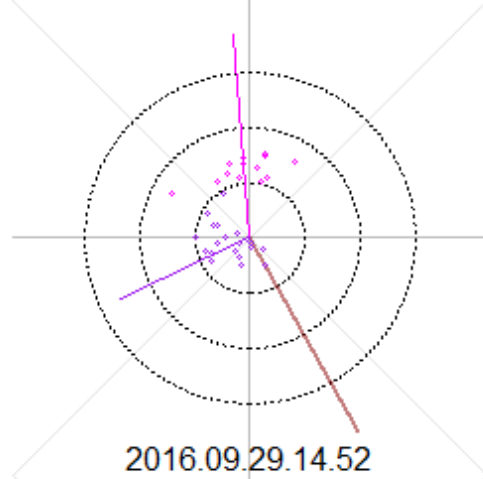
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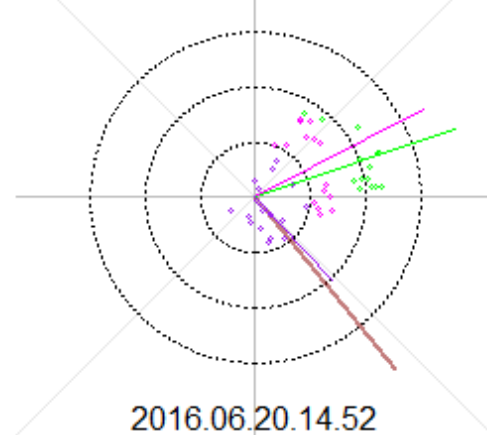
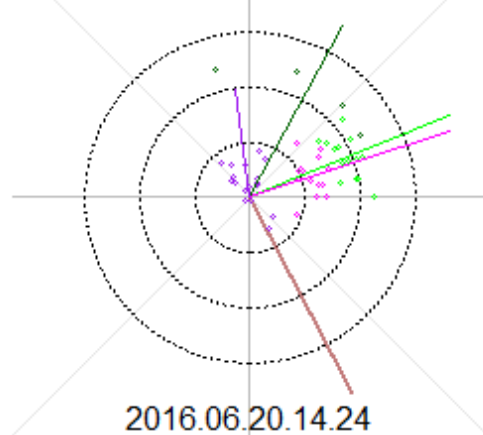
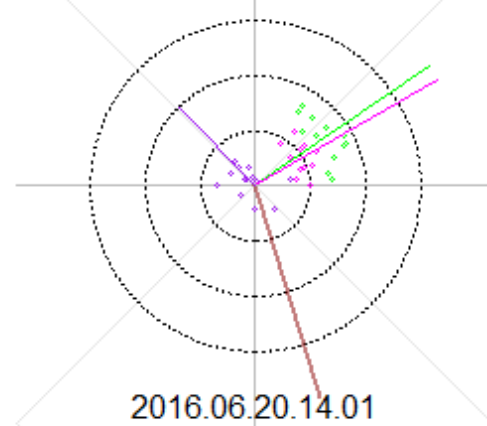
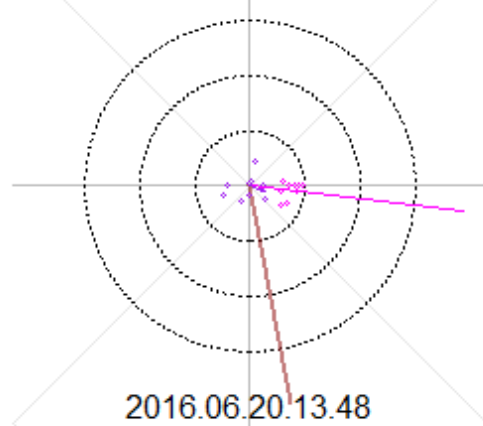
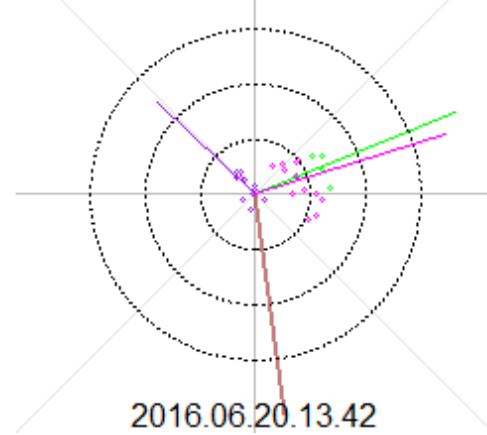
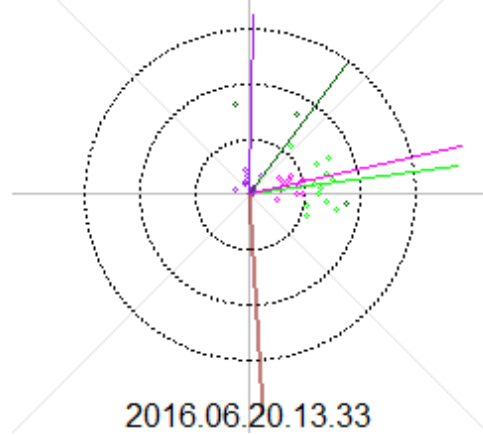
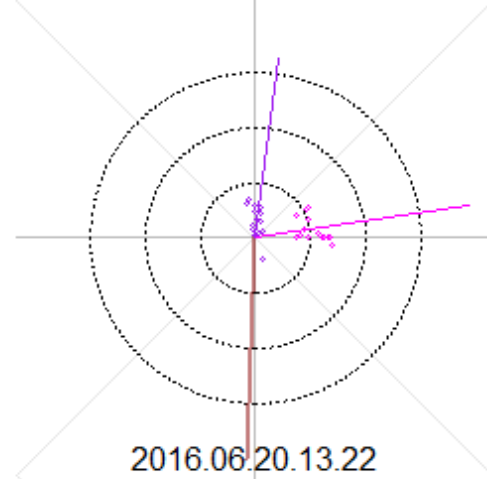
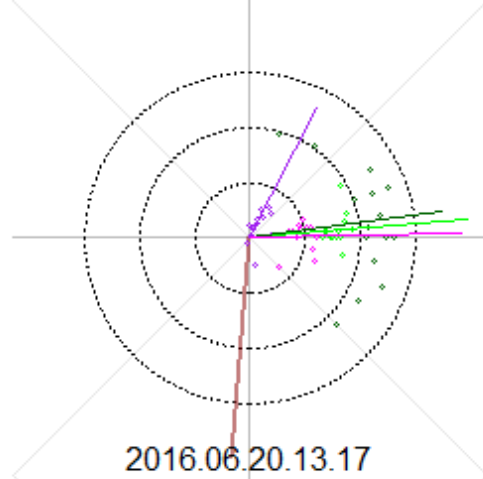


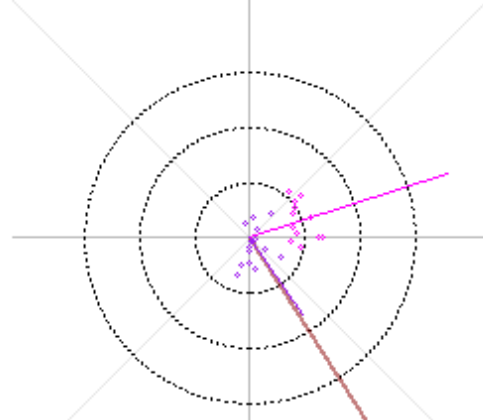
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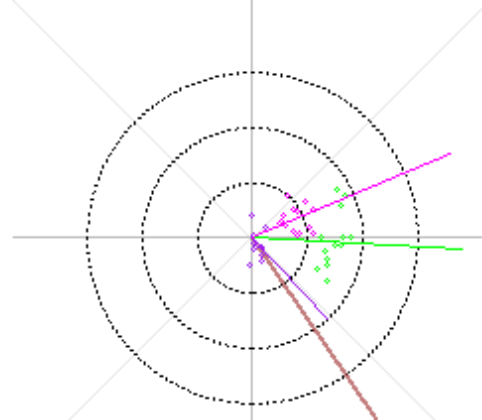




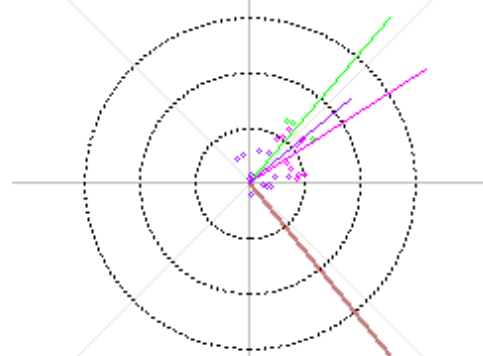




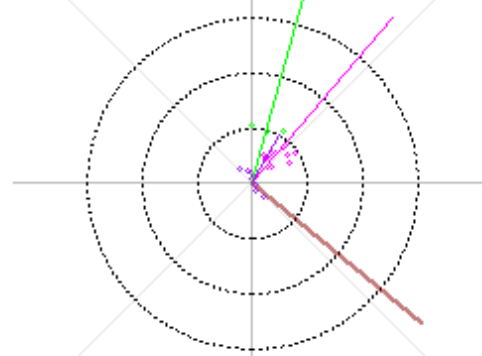
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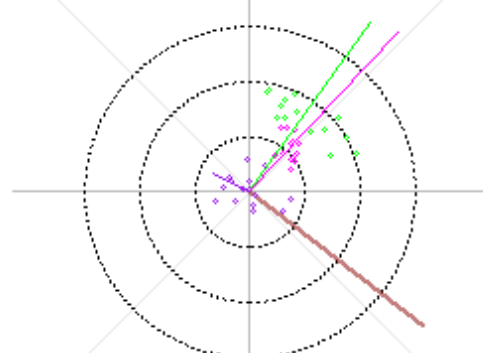
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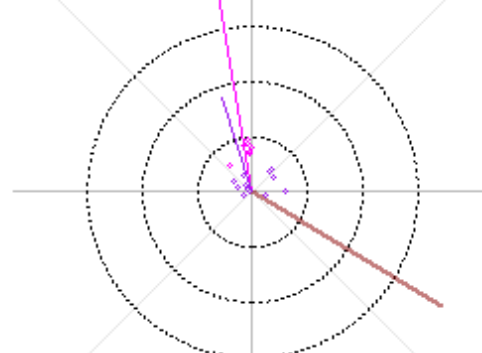
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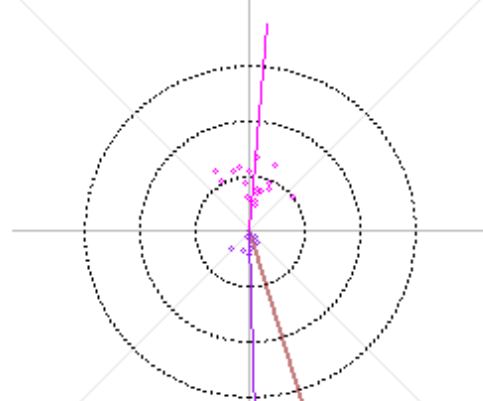
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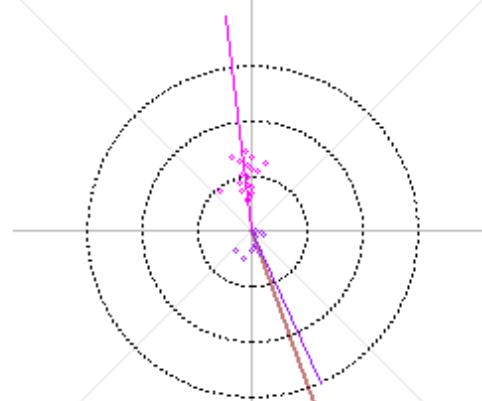
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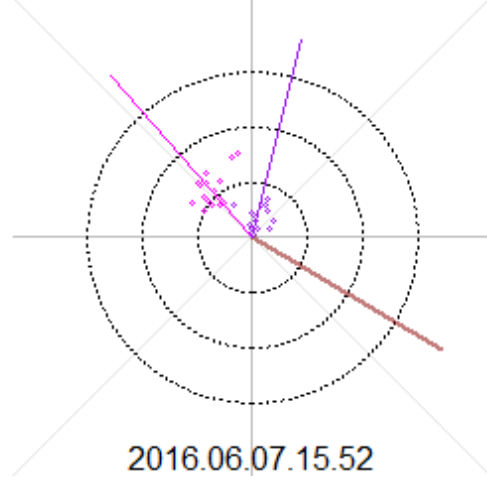
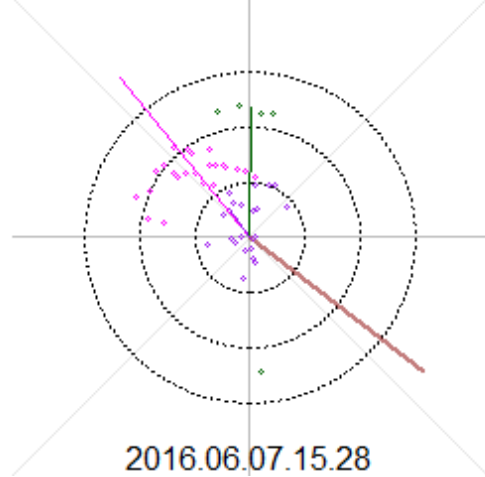
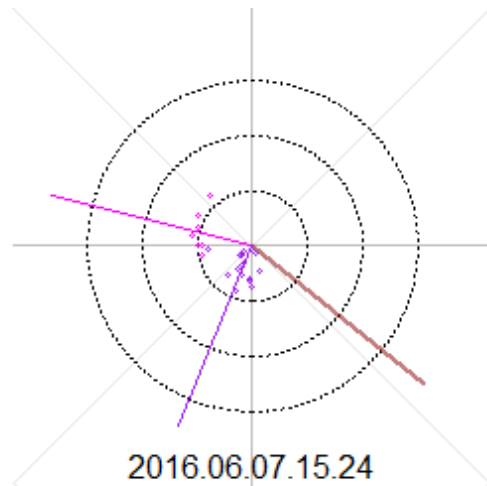
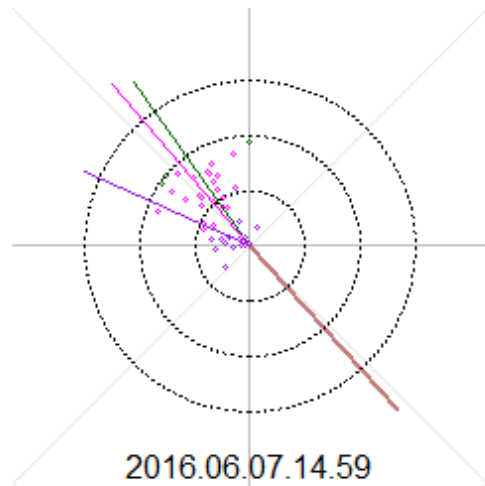
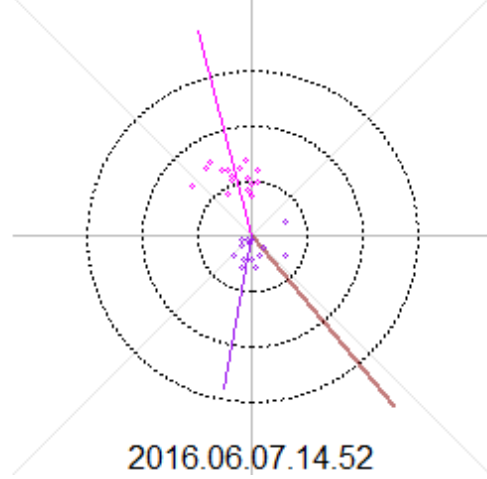
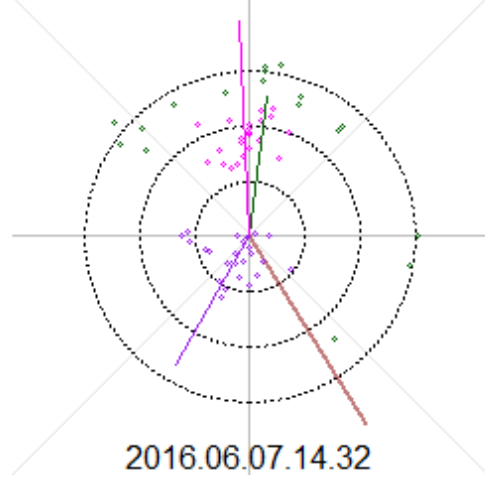
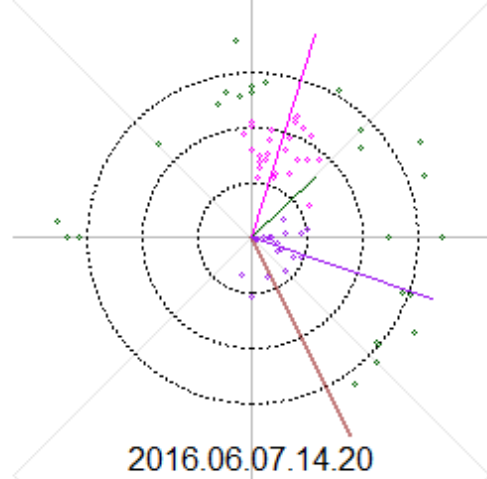
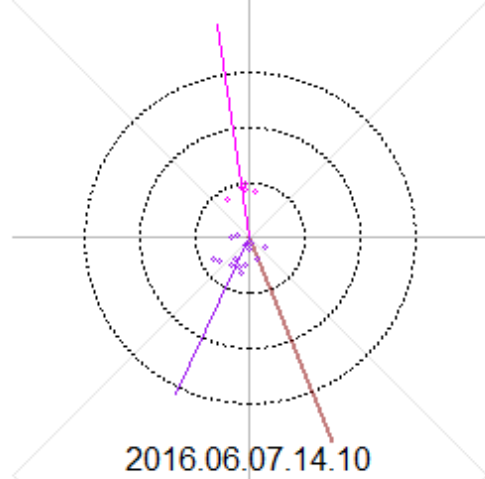
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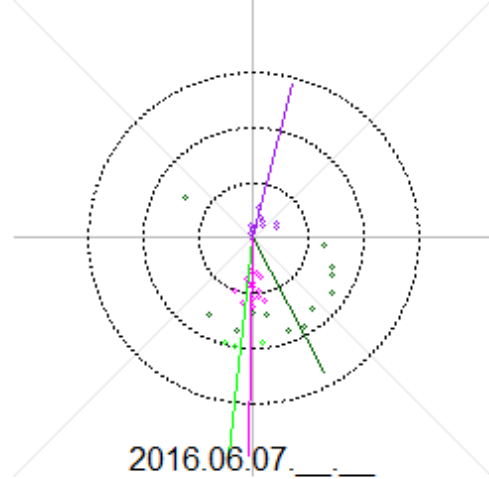
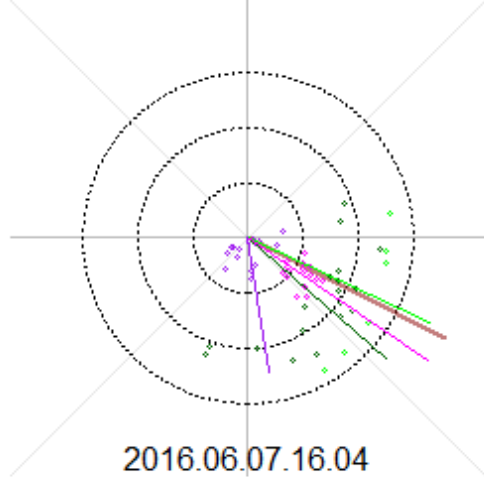


2016.06.07.13.59



2016.06.07.14.05





## Appendix

### Print raw data

Print all dance data loaded for this analysis.

```
#our dance data
print(cd)
```

| ##    | measured_angle | run | date       | bearing | angle       |                |
|-------|----------------|-----|------------|---------|-------------|----------------|
| ## 1  | 270            | 1   | 2016.05.31 | 90      | 1.57079633  | 2016.05.31.14. |
| ## 2  | 5              | 2   | 2016.05.31 | 355     | -0.08726646 | 2016.05.31.14. |
| ## 3  | 20             | 3   | 2016.05.31 | 340     | -0.34906585 | 2016.05.31.14. |
| ## 4  | 320            | 4   | 2016.05.31 | 40      | 0.69813170  | 2016.05.31.14. |
| ## 5  | 20             | 5   | 2016.05.31 | 340     | -0.34906585 | 2016.05.31.14. |
| ## 6  | 35             | 6   | 2016.05.31 | 325     | -0.61086524 | 2016.05.31.14. |
| ## 7  | 10             | 7   | 2016.05.31 | 350     | -0.17453293 | 2016.05.31.14. |
| ## 8  | 345            | 8   | 2016.05.31 | 15      | 0.26179939  | 2016.05.31.14. |
| ## 9  | 340            | 9   | 2016.05.31 | 20      | 0.34906585  | 2016.05.31.14. |
| ## 10 | 340            | 10  | 2016.05.31 | 20      | 0.34906585  | 2016.05.31.14. |
| ## 11 | 340            | 11  | 2016.05.31 | 20      | 0.34906585  | 2016.05.31.14. |
| ## 12 | 330            | 12  | 2016.05.31 | 30      | 0.52359878  | 2016.05.31.14. |
| ## 13 | 350            | 13  | 2016.05.31 | 10      | 0.17453293  | 2016.05.31.14. |
| ## 14 | 325            | 14  | 2016.05.31 | 35      | 0.61086524  | 2016.05.31.14. |

|       |     |    |            |     |             |                |
|-------|-----|----|------------|-----|-------------|----------------|
| ## 15 | 335 | 15 | 2016.05.31 | 25  | 0.43633231  | 2016.05.31.14. |
| ## 16 | 330 | 16 | 2016.05.31 | 30  | 0.52359878  | 2016.05.31.14. |
| ## 17 | 320 | 17 | 2016.05.31 | 40  | 0.69813170  | 2016.05.31.14. |
| ## 18 | 325 | 18 | 2016.05.31 | 35  | 0.61086524  | 2016.05.31.14. |
| ## 19 | 325 | 19 | 2016.05.31 | 35  | 0.61086524  | 2016.05.31.14. |
| ## 20 | 355 | 20 | 2016.05.31 | 5   | 0.08726646  | 2016.05.31.14. |
| ## 21 | 340 | 21 | 2016.05.31 | 20  | 0.34906585  | 2016.05.31.14. |
| ## 22 | 345 | 22 | 2016.05.31 | 15  | 0.26179939  | 2016.05.31.14. |
| ## 23 | 345 | 23 | 2016.05.31 | 15  | 0.26179939  | 2016.05.31.14. |
| ## 24 | 325 | 24 | 2016.05.31 | 35  | 0.61086524  | 2016.05.31.14. |
| ## 25 | 350 | 25 | 2016.05.31 | 10  | 0.17453293  | 2016.05.31.14. |
| ## 26 | 345 | 26 | 2016.05.31 | 15  | 0.26179939  | 2016.05.31.14. |
| ## 27 | 355 | 27 | 2016.05.31 | 5   | 0.08726646  | 2016.05.31.14. |
| ## 28 | 235 | 1  | 2016.05.31 | 125 | 2.18166156  | 2016.05.31.14. |
| ## 29 | 185 | 2  | 2016.05.31 | 175 | 3.05432619  | 2016.05.31.14. |
| ## 30 | 100 | 3  | 2016.05.31 | 260 | -1.74532925 | 2016.05.31.14. |
| ## 31 | 140 | 4  | 2016.05.31 | 220 | -2.44346095 | 2016.05.31.14. |
| ## 32 | 90  | 5  | 2016.05.31 | 270 | -1.57079633 | 2016.05.31.14. |
| ## 33 | 190 | 6  | 2016.05.31 | 170 | 2.96705973  | 2016.05.31.14. |
| ## 34 | 175 | 7  | 2016.05.31 | 185 | -3.05432619 | 2016.05.31.14. |
| ## 35 | 250 | 8  | 2016.05.31 | 110 | 1.91986218  | 2016.05.31.14. |
| ## 36 | 130 | 9  | 2016.05.31 | 230 | -2.26892803 | 2016.05.31.14. |
| ## 37 | 160 | 10 | 2016.05.31 | 200 | -2.79252680 | 2016.05.31.14. |
| ## 38 | 15  | 11 | 2016.05.31 | 345 | -0.26179939 | 2016.05.31.14. |
| ## 39 | 45  | 12 | 2016.05.31 | 315 | -0.78539816 | 2016.05.31.14. |
| ## 40 | 15  | 13 | 2016.05.31 | 345 | -0.26179939 | 2016.05.31.14. |
| ## 41 | 20  | 14 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 42 | 40  | 15 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 43 | 20  | 16 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 44 | 10  | 17 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 45 | 50  | 18 | 2016.05.31 | 310 | -0.87266463 | 2016.05.31.14. |
| ## 46 | 10  | 19 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 47 | 40  | 20 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 48 | 40  | 21 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 49 | 50  | 22 | 2016.05.31 | 310 | -0.87266463 | 2016.05.31.14. |
| ## 50 | 50  | 23 | 2016.05.31 | 310 | -0.87266463 | 2016.05.31.14. |
| ## 51 | 10  | 24 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 52 | 40  | 25 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 53 | 35  | 26 | 2016.05.31 | 325 | -0.61086524 | 2016.05.31.14. |
| ## 54 | 20  | 27 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 55 | 30  | 28 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 56 | 40  | 29 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 57 | 20  | 30 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 58 | 25  | 31 | 2016.05.31 | 335 | -0.43633231 | 2016.05.31.14. |
| ## 59 | 20  | 32 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 60 | 40  | 33 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 61 | 20  | 34 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 62 | 20  | 35 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 63  | 200 | 1  | 2016.05.31 | 160 | 2.79252680  | 2016.05.31.14. |
| ## 64  | 140 | 2  | 2016.05.31 | 220 | -2.44346095 | 2016.05.31.14. |
| ## 65  | 220 | 3  | 2016.05.31 | 140 | 2.44346095  | 2016.05.31.14. |
| ## 66  | 150 | 4  | 2016.05.31 | 210 | -2.61799388 | 2016.05.31.14. |
| ## 67  | 170 | 5  | 2016.05.31 | 190 | -2.96705973 | 2016.05.31.14. |
| ## 68  | 100 | 6  | 2016.05.31 | 260 | -1.74532925 | 2016.05.31.14. |
| ## 69  | 155 | 7  | 2016.05.31 | 205 | -2.70526034 | 2016.05.31.14. |
| ## 70  | 90  | 8  | 2016.05.31 | 270 | -1.57079633 | 2016.05.31.14. |
| ## 71  | 155 | 9  | 2016.05.31 | 205 | -2.70526034 | 2016.05.31.14. |
| ## 72  | 355 | 10 | 2016.05.31 | 5   | 0.08726646  | 2016.05.31.14. |
| ## 73  | 40  | 11 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 74  | 70  | 12 | 2016.05.31 | 290 | -1.22173048 | 2016.05.31.14. |
| ## 75  | 30  | 13 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 76  | 70  | 14 | 2016.05.31 | 290 | -1.22173048 | 2016.05.31.14. |
| ## 77  | 25  | 15 | 2016.05.31 | 335 | -0.43633231 | 2016.05.31.14. |
| ## 78  | 20  | 16 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 79  | 30  | 17 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 80  | 30  | 18 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 81  | 70  | 19 | 2016.05.31 | 290 | -1.22173048 | 2016.05.31.14. |
| ## 82  | 60  | 20 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.14. |
| ## 83  | 40  | 21 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 84  | 60  | 22 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.14. |
| ## 85  | 40  | 23 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 86  | 100 | 1  | 2016.05.31 | 260 | -1.74532925 | 2016.05.31.15. |
| ## 87  | 110 | 2  | 2016.05.31 | 250 | -1.91986218 | 2016.05.31.15. |
| ## 88  | 140 | 3  | 2016.05.31 | 220 | -2.44346095 | 2016.05.31.15. |
| ## 89  | 110 | 4  | 2016.05.31 | 250 | -1.91986218 | 2016.05.31.15. |
| ## 90  | 140 | 5  | 2016.05.31 | 220 | -2.44346095 | 2016.05.31.15. |
| ## 91  | 130 | 6  | 2016.05.31 | 230 | -2.26892803 | 2016.05.31.15. |
| ## 92  | 110 | 7  | 2016.05.31 | 250 | -1.91986218 | 2016.05.31.15. |
| ## 93  | 95  | 8  | 2016.05.31 | 265 | -1.65806279 | 2016.05.31.15. |
| ## 94  | 140 | 9  | 2016.05.31 | 220 | -2.44346095 | 2016.05.31.15. |
| ## 95  | 80  | 10 | 2016.05.31 | 280 | -1.39626340 | 2016.05.31.15. |
| ## 96  | 120 | 11 | 2016.05.31 | 240 | -2.09439510 | 2016.05.31.15. |
| ## 97  | 60  | 12 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.15. |
| ## 98  | 60  | 13 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.15. |
| ## 99  | 50  | 14 | 2016.05.31 | 310 | -0.87266463 | 2016.05.31.15. |
| ## 100 | 85  | 15 | 2016.05.31 | 275 | -1.48352986 | 2016.05.31.15. |
| ## 101 | 40  | 16 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.15. |
| ## 102 | 40  | 17 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.15. |
| ## 103 | 20  | 18 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.15. |
| ## 104 | 270 | 1  | 2016.05.31 | 90  | 1.57079633  | 2016.05.31.14. |
| ## 105 | 260 | 2  | 2016.05.31 | 100 | 1.74532925  | 2016.05.31.14. |
| ## 106 | 310 | 3  | 2016.05.31 | 50  | 0.87266463  | 2016.05.31.14. |
| ## 107 | 305 | 4  | 2016.05.31 | 55  | 0.95993109  | 2016.05.31.14. |
| ## 108 | 10  | 5  | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 109 | 350 | 6  | 2016.05.31 | 10  | 0.17453293  | 2016.05.31.14. |
| ## 110 | 270 | 7  | 2016.05.31 | 90  | 1.57079633  | 2016.05.31.14. |



|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 111 | 310 | 8  | 2016.05.31 | 50  | 0.87266463  | 2016.05.31.14. |
| ## 112 | 60  | 9  | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.14. |
| ## 113 | 10  | 10 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 114 | 5   | 11 | 2016.05.31 | 355 | -0.08726646 | 2016.05.31.14. |
| ## 115 | 40  | 12 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 116 | 10  | 13 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 117 | 5   | 14 | 2016.05.31 | 355 | -0.08726646 | 2016.05.31.14. |
| ## 118 | 10  | 15 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 119 | 10  | 16 | 2016.05.31 | 350 | -0.17453293 | 2016.05.31.14. |
| ## 120 | 30  | 17 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 121 | 0   | 18 | 2016.05.31 | 0   | 0.00000000  | 2016.05.31.14. |
| ## 122 | 15  | 19 | 2016.05.31 | 345 | -0.26179939 | 2016.05.31.14. |
| ## 123 | 15  | 20 | 2016.05.31 | 345 | -0.26179939 | 2016.05.31.14. |
| ## 124 | 15  | 21 | 2016.05.31 | 345 | -0.26179939 | 2016.05.31.14. |
| ## 125 | 15  | 22 | 2016.05.31 | 345 | -0.26179939 | 2016.05.31.14. |
| ## 126 | 5   | 23 | 2016.05.31 | 355 | -0.08726646 | 2016.05.31.14. |
| ## 127 | 5   | 24 | 2016.05.31 | 355 | -0.08726646 | 2016.05.31.14. |
| ## 128 | 40  | 25 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 129 | 20  | 26 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 130 | 30  | 27 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 131 | 20  | 28 | 2016.05.31 | 340 | -0.34906585 | 2016.05.31.14. |
| ## 132 | 60  | 29 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.14. |
| ## 133 | 50  | 30 | 2016.05.31 | 310 | -0.87266463 | 2016.05.31.14. |
| ## 134 | 60  | 31 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.14. |
| ## 135 | 60  | 32 | 2016.05.31 | 300 | -1.04719755 | 2016.05.31.14. |
| ## 136 | 5   | 33 | 2016.05.31 | 355 | -0.08726646 | 2016.05.31.14. |
| ## 137 | 45  | 34 | 2016.05.31 | 315 | -0.78539816 | 2016.05.31.14. |
| ## 138 | 40  | 35 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 139 | 40  | 36 | 2016.05.31 | 320 | -0.69813170 | 2016.05.31.14. |
| ## 140 | 30  | 37 | 2016.05.31 | 330 | -0.52359878 | 2016.05.31.14. |
| ## 141 | 200 | 1  | 2016.05.31 | 160 | 2.79252680  | 2016.05.31.14. |
| ## 142 | 310 | 2  | 2016.05.31 | 50  | 0.87266463  | 2016.05.31.14. |
| ## 143 | 230 | 3  | 2016.05.31 | 130 | 2.26892803  | 2016.05.31.14. |
| ## 144 | 240 | 4  | 2016.05.31 | 120 | 2.09439510  | 2016.05.31.14. |
| ## 145 | 230 | 5  | 2016.05.31 | 130 | 2.26892803  | 2016.05.31.14. |
| ## 146 | 225 | 6  | 2016.05.31 | 135 | 2.35619449  | 2016.05.31.14. |
| ## 147 | 130 | 7  | 2016.05.31 | 230 | -2.26892803 | 2016.05.31.14. |
| ## 148 | 150 | 8  | 2016.05.31 | 210 | -2.61799388 | 2016.05.31.14. |
| ## 149 | 240 | 9  | 2016.05.31 | 120 | 2.09439510  | 2016.05.31.14. |
| ## 150 | 265 | 10 | 2016.05.31 | 95  | 1.65806279  | 2016.05.31.14. |
| ## 151 | 300 | 11 | 2016.05.31 | 60  | 1.04719755  | 2016.05.31.14. |
| ## 152 | 260 | 12 | 2016.05.31 | 100 | 1.74532925  | 2016.05.31.14. |
| ## 153 | 255 | 13 | 2016.05.31 | 105 | 1.83259571  | 2016.05.31.14. |
| ## 154 | 270 | 14 | 2016.05.31 | 90  | 1.57079633  | 2016.05.31.14. |
| ## 155 | 240 | 15 | 2016.05.31 | 120 | 2.09439510  | 2016.05.31.14. |
| ## 156 | 280 | 16 | 2016.05.31 | 80  | 1.39626340  | 2016.05.31.14. |
| ## 157 | 250 | 17 | 2016.05.31 | 110 | 1.91986218  | 2016.05.31.14. |
| ## 158 | 310 | 18 | 2016.05.31 | 50  | 0.87266463  | 2016.05.31.14. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 159 | 260 | 19 | 2016.05.31 | 100 | 1.74532925  | 2016.05.31.14. |
| ## 160 | 310 | 20 | 2016.05.31 | 50  | 0.87266463  | 2016.05.31.14. |
| ## 161 | 305 | 1  | 2016.06.06 | 55  | 0.95993109  | 2016.06.06.14. |
| ## 162 | 115 | 2  | 2016.06.06 | 245 | -2.00712864 | 2016.06.06.14. |
| ## 163 | 310 | 3  | 2016.06.06 | 50  | 0.87266463  | 2016.06.06.14. |
| ## 164 | 20  | 4  | 2016.06.06 | 340 | -0.34906585 | 2016.06.06.14. |
| ## 165 | 140 | 5  | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## 166 | 0   | 6  | 2016.06.06 | 0   | 0.00000000  | 2016.06.06.14. |
| ## 167 | 140 | 7  | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## 168 | 15  | 8  | 2016.06.06 | 345 | -0.26179939 | 2016.06.06.14. |
| ## 169 | 40  | 9  | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## 170 | 5   | 10 | 2016.06.06 | 355 | -0.08726646 | 2016.06.06.14. |
| ## 171 | 30  | 11 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## 172 | 55  | 12 | 2016.06.06 | 305 | -0.95993109 | 2016.06.06.14. |
| ## 173 | 330 | 13 | 2016.06.06 | 30  | 0.52359878  | 2016.06.06.14. |
| ## 174 | 350 | 14 | 2016.06.06 | 10  | 0.17453293  | 2016.06.06.14. |
| ## 175 | 30  | 15 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## 176 | 350 | 16 | 2016.06.06 | 10  | 0.17453293  | 2016.06.06.14. |
| ## 177 | 335 | 17 | 2016.06.06 | 25  | 0.43633231  | 2016.06.06.14. |
| ## 178 | 0   | 18 | 2016.06.06 | 0   | 0.00000000  | 2016.06.06.14. |
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| ## 180 | 355 | 20 | 2016.06.06 | 5   | 0.08726646  | 2016.06.06.14. |
| ## 181 | 10  | 21 | 2016.06.06 | 350 | -0.17453293 | 2016.06.06.14. |
| ## 182 | 335 | 22 | 2016.06.06 | 25  | 0.43633231  | 2016.06.06.14. |
| ## 183 | 5   | 23 | 2016.06.06 | 355 | -0.08726646 | 2016.06.06.14. |
| ## 184 | 355 | 24 | 2016.06.06 | 5   | 0.08726646  | 2016.06.06.14. |
| ## 185 | 295 | 25 | 2016.06.06 | 65  | 1.13446401  | 2016.06.06.14. |
| ## 186 | 330 | 26 | 2016.06.06 | 30  | 0.52359878  | 2016.06.06.14. |
| ## 187 | 0   | 27 | 2016.06.06 | 0   | 0.00000000  | 2016.06.06.14. |
| ## 188 | 30  | 28 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## 189 | 25  | 29 | 2016.06.06 | 335 | -0.43633231 | 2016.06.06.14. |
| ## 190 | 55  | 30 | 2016.06.06 | 305 | -0.95993109 | 2016.06.06.14. |
| ## 191 | 35  | 31 | 2016.06.06 | 325 | -0.61086524 | 2016.06.06.14. |
| ## 192 | 10  | 32 | 2016.06.06 | 350 | -0.17453293 | 2016.06.06.14. |
| ## 193 | 5   | 33 | 2016.06.06 | 355 | -0.08726646 | 2016.06.06.14. |
| ## 194 | 15  | 34 | 2016.06.06 | 345 | -0.26179939 | 2016.06.06.14. |
| ## 195 | 20  | 35 | 2016.06.06 | 340 | -0.34906585 | 2016.06.06.14. |
| ## 196 | 20  | 36 | 2016.06.06 | 340 | -0.34906585 | 2016.06.06.14. |
| ## 197 | 20  | 37 | 2016.06.06 | 340 | -0.34906585 | 2016.06.06.14. |
| ## 198 | 120 | 38 | 2016.06.06 | 240 | -2.09439510 | 2016.06.06.14. |
| ## 199 | 330 | 39 | 2016.06.06 | 30  | 0.52359878  | 2016.06.06.14. |
| ## 200 | 355 | 40 | 2016.06.06 | 5   | 0.08726646  | 2016.06.06.14. |
| ## 201 | 155 | 1  | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## 202 | 180 | 2  | 2016.06.06 | 180 | 3.14159265  | 2016.06.06.14. |
| ## 203 | 165 | 3  | 2016.06.06 | 195 | -2.87979327 | 2016.06.06.14. |
| ## 204 | 185 | 4  | 2016.06.06 | 175 | 3.05432619  | 2016.06.06.14. |
| ## 205 | 190 | 5  | 2016.06.06 | 170 | 2.96705973  | 2016.06.06.14. |
| ## 206 | 225 | 6  | 2016.06.06 | 135 | 2.35619449  | 2016.06.06.14. |

|    |     |     |    |            |     |             |                |
|----|-----|-----|----|------------|-----|-------------|----------------|
| ## | 207 | 195 | 7  | 2016.06.06 | 165 | 2.87979327  | 2016.06.06.14. |
| ## | 208 | 220 | 8  | 2016.06.06 | 140 | 2.44346095  | 2016.06.06.14. |
| ## | 209 | 180 | 9  | 2016.06.06 | 180 | 3.14159265  | 2016.06.06.14. |
| ## | 210 | 190 | 10 | 2016.06.06 | 170 | 2.96705973  | 2016.06.06.14. |
| ## | 211 | 205 | 11 | 2016.06.06 | 155 | 2.70526034  | 2016.06.06.14. |
| ## | 212 | 240 | 12 | 2016.06.06 | 120 | 2.09439510  | 2016.06.06.14. |
| ## | 213 | 180 | 13 | 2016.06.06 | 180 | 3.14159265  | 2016.06.06.14. |
| ## | 214 | 230 | 14 | 2016.06.06 | 130 | 2.26892803  | 2016.06.06.14. |
| ## | 215 | 140 | 15 | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## | 216 | 140 | 16 | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## | 217 | 150 | 17 | 2016.06.06 | 210 | -2.61799388 | 2016.06.06.14. |
| ## | 218 | 140 | 18 | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## | 219 | 120 | 1  | 2016.06.06 | 240 | -2.09439510 | 2016.06.06.14. |
| ## | 220 | 145 | 2  | 2016.06.06 | 215 | -2.53072742 | 2016.06.06.14. |
| ## | 221 | 320 | 3  | 2016.06.06 | 40  | 0.69813170  | 2016.06.06.14. |
| ## | 222 | 170 | 4  | 2016.06.06 | 190 | -2.96705973 | 2016.06.06.14. |
| ## | 223 | 160 | 5  | 2016.06.06 | 200 | -2.79252680 | 2016.06.06.14. |
| ## | 224 | 170 | 6  | 2016.06.06 | 190 | -2.96705973 | 2016.06.06.14. |
| ## | 225 | 155 | 7  | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## | 226 | 190 | 8  | 2016.06.06 | 170 | 2.96705973  | 2016.06.06.14. |
| ## | 227 | 185 | 9  | 2016.06.06 | 175 | 3.05432619  | 2016.06.06.14. |
| ## | 228 | 185 | 10 | 2016.06.06 | 175 | 3.05432619  | 2016.06.06.14. |
| ## | 229 | 150 | 11 | 2016.06.06 | 210 | -2.61799388 | 2016.06.06.14. |
| ## | 230 | 165 | 12 | 2016.06.06 | 195 | -2.87979327 | 2016.06.06.14. |
| ## | 231 | 155 | 13 | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## | 232 | 180 | 14 | 2016.06.06 | 180 | 3.14159265  | 2016.06.06.14. |
| ## | 233 | 175 | 15 | 2016.06.06 | 185 | -3.05432619 | 2016.06.06.14. |
| ## | 234 | 190 | 16 | 2016.06.06 | 170 | 2.96705973  | 2016.06.06.14. |
| ## | 235 | 165 | 17 | 2016.06.06 | 195 | -2.87979327 | 2016.06.06.14. |
| ## | 236 | 155 | 18 | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## | 237 | 160 | 19 | 2016.06.06 | 200 | -2.79252680 | 2016.06.06.14. |
| ## | 238 | 35  | 20 | 2016.06.06 | 325 | -0.61086524 | 2016.06.06.14. |
| ## | 239 | 30  | 21 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## | 240 | 45  | 22 | 2016.06.06 | 315 | -0.78539816 | 2016.06.06.14. |
| ## | 241 | 40  | 23 | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## | 242 | 80  | 24 | 2016.06.06 | 280 | -1.39626340 | 2016.06.06.14. |
| ## | 243 | 35  | 25 | 2016.06.06 | 325 | -0.61086524 | 2016.06.06.14. |
| ## | 244 | 60  | 26 | 2016.06.06 | 300 | -1.04719755 | 2016.06.06.14. |
| ## | 245 | 30  | 27 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## | 246 | 40  | 28 | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## | 247 | 50  | 29 | 2016.06.06 | 310 | -0.87266463 | 2016.06.06.14. |
| ## | 248 | 10  | 30 | 2016.06.06 | 350 | -0.17453293 | 2016.06.06.14. |
| ## | 249 | 50  | 31 | 2016.06.06 | 310 | -0.87266463 | 2016.06.06.14. |
| ## | 250 | 65  | 32 | 2016.06.06 | 295 | -1.13446401 | 2016.06.06.14. |
| ## | 251 | 90  | 33 | 2016.06.06 | 270 | -1.57079633 | 2016.06.06.14. |
| ## | 252 | 85  | 34 | 2016.06.06 | 275 | -1.48352986 | 2016.06.06.14. |
| ## | 253 | 35  | 35 | 2016.06.06 | 325 | -0.61086524 | 2016.06.06.14. |
| ## | 254 | 55  | 36 | 2016.06.06 | 305 | -0.95993109 | 2016.06.06.14. |

|    |     |     |    |            |     |             |                |
|----|-----|-----|----|------------|-----|-------------|----------------|
| ## | 255 | 40  | 37 | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## | 256 | 110 | 38 | 2016.06.06 | 250 | -1.91986218 | 2016.06.06.14. |
| ## | 257 | 165 | 39 | 2016.06.06 | 195 | -2.87979327 | 2016.06.06.14. |
| ## | 258 | 120 | 40 | 2016.06.06 | 240 | -2.09439510 | 2016.06.06.14. |
| ## | 259 | 120 | 41 | 2016.06.06 | 240 | -2.09439510 | 2016.06.06.14. |
| ## | 260 | 170 | 42 | 2016.06.06 | 190 | -2.96705973 | 2016.06.06.14. |
| ## | 261 | 90  | 43 | 2016.06.06 | 270 | -1.57079633 | 2016.06.06.14. |
| ## | 262 | 150 | 44 | 2016.06.06 | 210 | -2.61799388 | 2016.06.06.14. |
| ## | 263 | 160 | 45 | 2016.06.06 | 200 | -2.79252680 | 2016.06.06.14. |
| ## | 264 | 175 | 46 | 2016.06.06 | 185 | -3.05432619 | 2016.06.06.14. |
| ## | 265 | 40  | 47 | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## | 266 | 160 | 48 | 2016.06.06 | 200 | -2.79252680 | 2016.06.06.14. |
| ## | 267 | 155 | 49 | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## | 268 | 120 | 50 | 2016.06.06 | 240 | -2.09439510 | 2016.06.06.14. |
| ## | 269 | 110 | 51 | 2016.06.06 | 250 | -1.91986218 | 2016.06.06.14. |
| ## | 270 | 200 | 52 | 2016.06.06 | 160 | 2.79252680  | 2016.06.06.14. |
| ## | 271 | 140 | 53 | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## | 272 | 140 | 54 | 2016.06.06 | 220 | -2.44346095 | 2016.06.06.14. |
| ## | 273 | 130 | 55 | 2016.06.06 | 230 | -2.26892803 | 2016.06.06.14. |
| ## | 274 | 60  | 56 | 2016.06.06 | 300 | -1.04719755 | 2016.06.06.14. |
| ## | 275 | 40  | 57 | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## | 276 | 30  | 58 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## | 277 | 70  | 59 | 2016.06.06 | 290 | -1.22173048 | 2016.06.06.14. |
| ## | 278 | 90  | 60 | 2016.06.06 | 270 | -1.57079633 | 2016.06.06.14. |
| ## | 279 | 130 | 61 | 2016.06.06 | 230 | -2.26892803 | 2016.06.06.14. |
| ## | 280 | 170 | 1  | 2016.06.06 | 190 | -2.96705973 | 2016.06.06.14. |
| ## | 281 | 155 | 2  | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## | 282 | 170 | 3  | 2016.06.06 | 190 | -2.96705973 | 2016.06.06.14. |
| ## | 283 | 130 | 4  | 2016.06.06 | 230 | -2.26892803 | 2016.06.06.14. |
| ## | 284 | 165 | 5  | 2016.06.06 | 195 | -2.87979327 | 2016.06.06.14. |
| ## | 285 | 150 | 6  | 2016.06.06 | 210 | -2.61799388 | 2016.06.06.14. |
| ## | 286 | 160 | 7  | 2016.06.06 | 200 | -2.79252680 | 2016.06.06.14. |
| ## | 287 | 155 | 8  | 2016.06.06 | 205 | -2.70526034 | 2016.06.06.14. |
| ## | 288 | 220 | 9  | 2016.06.06 | 140 | 2.44346095  | 2016.06.06.14. |
| ## | 289 | 150 | 10 | 2016.06.06 | 210 | -2.61799388 | 2016.06.06.14. |
| ## | 290 | 175 | 11 | 2016.06.06 | 185 | -3.05432619 | 2016.06.06.14. |
| ## | 291 | 150 | 12 | 2016.06.06 | 210 | -2.61799388 | 2016.06.06.14. |
| ## | 292 | 190 | 13 | 2016.06.06 | 170 | 2.96705973  | 2016.06.06.14. |
| ## | 293 | 185 | 14 | 2016.06.06 | 175 | 3.05432619  | 2016.06.06.14. |
| ## | 294 | 175 | 15 | 2016.06.06 | 185 | -3.05432619 | 2016.06.06.14. |
| ## | 295 | 25  | 16 | 2016.06.06 | 335 | -0.43633231 | 2016.06.06.14. |
| ## | 296 | 50  | 17 | 2016.06.06 | 310 | -0.87266463 | 2016.06.06.14. |
| ## | 297 | 30  | 18 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## | 298 | 90  | 19 | 2016.06.06 | 270 | -1.57079633 | 2016.06.06.14. |
| ## | 299 | 30  | 20 | 2016.06.06 | 330 | -0.52359878 | 2016.06.06.14. |
| ## | 300 | 40  | 21 | 2016.06.06 | 320 | -0.69813170 | 2016.06.06.14. |
| ## | 301 | 35  | 22 | 2016.06.06 | 325 | -0.61086524 | 2016.06.06.14. |
| ## | 302 | 355 | 1  | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.13. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 303 | 230 | 2  | 2016.06.22 | 130 | 2.26892803  | 2016.06.22.13. |
| ## 304 | 0   | 3  | 2016.06.22 | 0   | 0.00000000  | 2016.06.22.13. |
| ## 305 | 330 | 4  | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.13. |
| ## 306 | 260 | 5  | 2016.06.22 | 100 | 1.74532925  | 2016.06.22.13. |
| ## 307 | 240 | 6  | 2016.06.22 | 120 | 2.09439510  | 2016.06.22.13. |
| ## 308 | 330 | 7  | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.13. |
| ## 309 | 40  | 8  | 2016.06.22 | 320 | -0.69813170 | 2016.06.22.13. |
| ## 310 | 320 | 9  | 2016.06.22 | 40  | 0.69813170  | 2016.06.22.13. |
| ## 311 | 10  | 10 | 2016.06.22 | 350 | -0.17453293 | 2016.06.22.13. |
| ## 312 | 290 | 11 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.13. |
| ## 313 | 275 | 12 | 2016.06.22 | 85  | 1.48352986  | 2016.06.22.13. |
| ## 314 | 305 | 13 | 2016.06.22 | 55  | 0.95993109  | 2016.06.22.13. |
| ## 315 | 290 | 14 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.13. |
| ## 316 | 290 | 15 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.13. |
| ## 317 | 270 | 16 | 2016.06.22 | 90  | 1.57079633  | 2016.06.22.13. |
| ## 318 | 285 | 17 | 2016.06.22 | 75  | 1.30899694  | 2016.06.22.13. |
| ## 319 | 280 | 18 | 2016.06.22 | 80  | 1.39626340  | 2016.06.22.13. |
| ## 320 | 320 | 19 | 2016.06.22 | 40  | 0.69813170  | 2016.06.22.13. |
| ## 321 | 290 | 20 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.13. |
| ## 322 | 290 | 21 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.13. |
| ## 323 | 275 | 22 | 2016.06.22 | 85  | 1.48352986  | 2016.06.22.13. |
| ## 324 | 275 | 23 | 2016.06.22 | 85  | 1.48352986  | 2016.06.22.13. |
| ## 325 | 310 | 24 | 2016.06.22 | 50  | 0.87266463  | 2016.06.22.13. |
| ## 326 | 295 | 25 | 2016.06.22 | 65  | 1.13446401  | 2016.06.22.13. |
| ## 327 | 300 | 26 | 2016.06.22 | 60  | 1.04719755  | 2016.06.22.13. |
| ## 328 | 300 | 27 | 2016.06.22 | 60  | 1.04719755  | 2016.06.22.13. |
| ## 329 | 290 | 28 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.13. |
| ## 330 | 300 | 29 | 2016.06.22 | 60  | 1.04719755  | 2016.06.22.13. |
| ## 331 | 270 | 30 | 2016.06.22 | 90  | 1.57079633  | 2016.06.22.13. |
| ## 332 | 315 | 31 | 2016.06.22 | 45  | 0.78539816  | 2016.06.22.13. |
| ## 333 | 210 | 1  | 2016.06.22 | 150 | 2.61799388  | 2016.06.22.14. |
| ## 334 | 335 | 2  | 2016.06.22 | 25  | 0.43633231  | 2016.06.22.14. |
| ## 335 | 35  | 3  | 2016.06.22 | 325 | -0.61086524 | 2016.06.22.14. |
| ## 336 | 15  | 4  | 2016.06.22 | 345 | -0.26179939 | 2016.06.22.14. |
| ## 337 | 55  | 5  | 2016.06.22 | 305 | -0.95993109 | 2016.06.22.14. |
| ## 338 | 15  | 6  | 2016.06.22 | 345 | -0.26179939 | 2016.06.22.14. |
| ## 339 | 20  | 7  | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.14. |
| ## 340 | 315 | 8  | 2016.06.22 | 45  | 0.78539816  | 2016.06.22.14. |
| ## 341 | 5   | 9  | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.14. |
| ## 342 | 120 | 10 | 2016.06.22 | 240 | -2.09439510 | 2016.06.22.14. |
| ## 343 | 300 | 11 | 2016.06.22 | 60  | 1.04719755  | 2016.06.22.14. |
| ## 344 | 350 | 12 | 2016.06.22 | 10  | 0.17453293  | 2016.06.22.14. |
| ## 345 | 20  | 13 | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.14. |
| ## 346 | 310 | 14 | 2016.06.22 | 50  | 0.87266463  | 2016.06.22.14. |
| ## 347 | 355 | 15 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.14. |
| ## 348 | 290 | 16 | 2016.06.22 | 70  | 1.22173048  | 2016.06.22.14. |
| ## 349 | 340 | 17 | 2016.06.22 | 20  | 0.34906585  | 2016.06.22.14. |
| ## 350 | 250 | 18 | 2016.06.22 | 110 | 1.91986218  | 2016.06.22.14. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 351 | 340 | 19 | 2016.06.22 | 20  | 0.34906585  | 2016.06.22.14. |
| ## 352 | 330 | 20 | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.14. |
| ## 353 | 305 | 21 | 2016.06.22 | 55  | 0.95993109  | 2016.06.22.14. |
| ## 354 | 325 | 22 | 2016.06.22 | 35  | 0.61086524  | 2016.06.22.14. |
| ## 355 | 320 | 23 | 2016.06.22 | 40  | 0.69813170  | 2016.06.22.14. |
| ## 356 | 320 | 24 | 2016.06.22 | 40  | 0.69813170  | 2016.06.22.14. |
| ## 357 | 330 | 25 | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.14. |
| ## 358 | 320 | 26 | 2016.06.22 | 40  | 0.69813170  | 2016.06.22.14. |
| ## 359 | 350 | 1  | 2016.06.22 | 10  | 0.17453293  | 2016.06.22.14. |
| ## 360 | 330 | 2  | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.14. |
| ## 361 | 140 | 3  | 2016.06.22 | 220 | -2.44346095 | 2016.06.22.14. |
| ## 362 | 170 | 4  | 2016.06.22 | 190 | -2.96705973 | 2016.06.22.14. |
| ## 363 | 145 | 5  | 2016.06.22 | 215 | -2.53072742 | 2016.06.22.14. |
| ## 364 | 190 | 6  | 2016.06.22 | 170 | 2.96705973  | 2016.06.22.14. |
| ## 365 | 170 | 7  | 2016.06.22 | 190 | -2.96705973 | 2016.06.22.14. |
| ## 366 | 165 | 8  | 2016.06.22 | 195 | -2.87979327 | 2016.06.22.14. |
| ## 367 | 160 | 9  | 2016.06.22 | 200 | -2.79252680 | 2016.06.22.14. |
| ## 368 | 210 | 10 | 2016.06.22 | 150 | 2.61799388  | 2016.06.22.14. |
| ## 369 | 210 | 11 | 2016.06.22 | 150 | 2.61799388  | 2016.06.22.14. |
| ## 370 | 200 | 12 | 2016.06.22 | 160 | 2.79252680  | 2016.06.22.14. |
| ## 371 | 285 | 13 | 2016.06.22 | 75  | 1.30899694  | 2016.06.22.14. |
| ## 372 | 270 | 14 | 2016.06.22 | 90  | 1.57079633  | 2016.06.22.14. |
| ## 373 | 130 | 15 | 2016.06.22 | 230 | -2.26892803 | 2016.06.22.14. |
| ## 374 | 175 | 16 | 2016.06.22 | 185 | -3.05432619 | 2016.06.22.14. |
| ## 375 | 155 | 17 | 2016.06.22 | 205 | -2.70526034 | 2016.06.22.14. |
| ## 376 | 140 | 18 | 2016.06.22 | 220 | -2.44346095 | 2016.06.22.14. |
| ## 377 | 250 | 19 | 2016.06.22 | 110 | 1.91986218  | 2016.06.22.14. |
| ## 378 | 170 | 20 | 2016.06.22 | 190 | -2.96705973 | 2016.06.22.14. |
| ## 379 | 230 | 21 | 2016.06.22 | 130 | 2.26892803  | 2016.06.22.14. |
| ## 380 | 195 | 22 | 2016.06.22 | 165 | 2.87979327  | 2016.06.22.14. |
| ## 381 | 240 | 23 | 2016.06.22 | 120 | 2.09439510  | 2016.06.22.14. |
| ## 382 | 230 | 24 | 2016.06.22 | 130 | 2.26892803  | 2016.06.22.14. |
| ## 383 | 220 | 25 | 2016.06.22 | 140 | 2.44346095  | 2016.06.22.14. |
| ## 384 | 245 | 26 | 2016.06.22 | 115 | 2.00712864  | 2016.06.22.14. |
| ## 385 | 35  | 1  | 2016.06.22 | 325 | -0.61086524 | 2016.06.22.14. |
| ## 386 | 60  | 2  | 2016.06.22 | 300 | -1.04719755 | 2016.06.22.14. |
| ## 387 | 60  | 3  | 2016.06.22 | 300 | -1.04719755 | 2016.06.22.14. |
| ## 388 | 65  | 4  | 2016.06.22 | 295 | -1.13446401 | 2016.06.22.14. |
| ## 389 | 30  | 5  | 2016.06.22 | 330 | -0.52359878 | 2016.06.22.14. |
| ## 390 | 20  | 6  | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.14. |
| ## 391 | 75  | 7  | 2016.06.22 | 285 | -1.30899694 | 2016.06.22.14. |
| ## 392 | 50  | 8  | 2016.06.22 | 310 | -0.87266463 | 2016.06.22.14. |
| ## 393 | 50  | 9  | 2016.06.22 | 310 | -0.87266463 | 2016.06.22.14. |
| ## 394 | 50  | 10 | 2016.06.22 | 310 | -0.87266463 | 2016.06.22.14. |
| ## 395 | 50  | 11 | 2016.06.22 | 310 | -0.87266463 | 2016.06.22.14. |
| ## 396 | 45  | 12 | 2016.06.22 | 315 | -0.78539816 | 2016.06.22.14. |
| ## 397 | 10  | 13 | 2016.06.22 | 350 | -0.17453293 | 2016.06.22.14. |
| ## 398 | 15  | 14 | 2016.06.22 | 345 | -0.26179939 | 2016.06.22.14. |



|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 399 | 330 | 15 | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.14. |
| ## 400 | 0   | 16 | 2016.06.22 | 0   | 0.00000000  | 2016.06.22.14. |
| ## 401 | 325 | 17 | 2016.06.22 | 35  | 0.61086524  | 2016.06.22.14. |
| ## 402 | 5   | 18 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.14. |
| ## 403 | 5   | 19 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.14. |
| ## 404 | 355 | 20 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.14. |
| ## 405 | 355 | 21 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.14. |
| ## 406 | 330 | 22 | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.14. |
| ## 407 | 5   | 23 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.14. |
| ## 408 | 350 | 24 | 2016.06.22 | 10  | 0.17453293  | 2016.06.22.14. |
| ## 409 | 15  | 25 | 2016.06.22 | 345 | -0.26179939 | 2016.06.22.14. |
| ## 410 | 335 | 26 | 2016.06.22 | 25  | 0.43633231  | 2016.06.22.14. |
| ## 411 | 340 | 27 | 2016.06.22 | 20  | 0.34906585  | 2016.06.22.14. |
| ## 412 | 340 | 28 | 2016.06.22 | 20  | 0.34906585  | 2016.06.22.14. |
| ## 413 | 350 | 29 | 2016.06.22 | 10  | 0.17453293  | 2016.06.22.14. |
| ## 414 | 345 | 30 | 2016.06.22 | 15  | 0.26179939  | 2016.06.22.14. |
| ## 415 | 355 | 31 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.14. |
| ## 416 | 5   | 32 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.14. |
| ## 417 | 330 | 33 | 2016.06.22 | 30  | 0.52359878  | 2016.06.22.14. |
| ## 418 | 5   | 34 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.14. |
| ## 419 | 55  | 1  | 2016.06.22 | 305 | -0.95993109 | 2016.06.22.15. |
| ## 420 | 50  | 2  | 2016.06.22 | 310 | -0.87266463 | 2016.06.22.15. |
| ## 421 | 55  | 3  | 2016.06.22 | 305 | -0.95993109 | 2016.06.22.15. |
| ## 422 | 5   | 4  | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.15. |
| ## 423 | 325 | 5  | 2016.06.22 | 35  | 0.61086524  | 2016.06.22.15. |
| ## 424 | 85  | 6  | 2016.06.22 | 275 | -1.48352986 | 2016.06.22.15. |
| ## 425 | 50  | 7  | 2016.06.22 | 310 | -0.87266463 | 2016.06.22.15. |
| ## 426 | 90  | 8  | 2016.06.22 | 270 | -1.57079633 | 2016.06.22.15. |
| ## 427 | 350 | 9  | 2016.06.22 | 10  | 0.17453293  | 2016.06.22.15. |
| ## 428 | 35  | 10 | 2016.06.22 | 325 | -0.61086524 | 2016.06.22.15. |
| ## 429 | 130 | 11 | 2016.06.22 | 230 | -2.26892803 | 2016.06.22.15. |
| ## 430 | 305 | 12 | 2016.06.22 | 55  | 0.95993109  | 2016.06.22.15. |
| ## 431 | 355 | 13 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.15. |
| ## 432 | 140 | 14 | 2016.06.22 | 220 | -2.44346095 | 2016.06.22.15. |
| ## 433 | 120 | 15 | 2016.06.22 | 240 | -2.09439510 | 2016.06.22.15. |
| ## 434 | 130 | 16 | 2016.06.22 | 230 | -2.26892803 | 2016.06.22.15. |
| ## 435 | 85  | 17 | 2016.06.22 | 275 | -1.48352986 | 2016.06.22.15. |
| ## 436 | 95  | 18 | 2016.06.22 | 265 | -1.65806279 | 2016.06.22.15. |
| ## 437 | 105 | 19 | 2016.06.22 | 255 | -1.83259571 | 2016.06.22.15. |
| ## 438 | 115 | 20 | 2016.06.22 | 245 | -2.00712864 | 2016.06.22.15. |
| ## 439 | 40  | 21 | 2016.06.22 | 320 | -0.69813170 | 2016.06.22.15. |
| ## 440 | 100 | 22 | 2016.06.22 | 260 | -1.74532925 | 2016.06.22.15. |
| ## 441 | 80  | 23 | 2016.06.22 | 280 | -1.39626340 | 2016.06.22.15. |
| ## 442 | 55  | 24 | 2016.06.22 | 305 | -0.95993109 | 2016.06.22.15. |
| ## 443 | 150 | 25 | 2016.06.22 | 210 | -2.61799388 | 2016.06.22.15. |
| ## 444 | 145 | 26 | 2016.06.22 | 215 | -2.53072742 | 2016.06.22.15. |
| ## 445 | 0   | 27 | 2016.06.22 | 0   | 0.00000000  | 2016.06.22.15. |
| ## 446 | 10  | 28 | 2016.06.22 | 350 | -0.17453293 | 2016.06.22.15. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 447 | 355 | 29 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.15. |
| ## 448 | 20  | 30 | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.15. |
| ## 449 | 10  | 31 | 2016.06.22 | 350 | -0.17453293 | 2016.06.22.15. |
| ## 450 | 70  | 32 | 2016.06.22 | 290 | -1.22173048 | 2016.06.22.15. |
| ## 451 | 5   | 33 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.15. |
| ## 452 | 40  | 34 | 2016.06.22 | 320 | -0.69813170 | 2016.06.22.15. |
| ## 453 | 5   | 35 | 2016.06.22 | 355 | -0.08726646 | 2016.06.22.15. |
| ## 454 | 30  | 36 | 2016.06.22 | 330 | -0.52359878 | 2016.06.22.15. |
| ## 455 | 10  | 37 | 2016.06.22 | 350 | -0.17453293 | 2016.06.22.15. |
| ## 456 | 55  | 38 | 2016.06.22 | 305 | -0.95993109 | 2016.06.22.15. |
| ## 457 | 60  | 39 | 2016.06.22 | 300 | -1.04719755 | 2016.06.22.15. |
| ## 458 | 10  | 40 | 2016.06.22 | 350 | -0.17453293 | 2016.06.22.15. |
| ## 459 | 40  | 41 | 2016.06.22 | 320 | -0.69813170 | 2016.06.22.15. |
| ## 460 | 20  | 42 | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.15. |
| ## 461 | 30  | 43 | 2016.06.22 | 330 | -0.52359878 | 2016.06.22.15. |
| ## 462 | 0   | 44 | 2016.06.22 | 0   | 0.00000000  | 2016.06.22.15. |
| ## 463 | 20  | 45 | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.15. |
| ## 464 | 20  | 46 | 2016.06.22 | 340 | -0.34906585 | 2016.06.22.15. |
| ## 465 | 30  | 47 | 2016.06.22 | 330 | -0.52359878 | 2016.06.22.15. |
| ## 466 | 0   | 48 | 2016.06.22 | 0   | 0.00000000  | 2016.06.22.15. |
| ## 467 | 355 | 49 | 2016.06.22 | 5   | 0.08726646  | 2016.06.22.15. |
| ## 468 | 350 | 50 | 2016.06.22 | 10  | 0.17453293  | 2016.06.22.15. |
| ## 469 | 310 | 1  | 2016.06.23 | 50  | 0.87266463  | 2016.06.23.13. |
| ## 470 | 355 | 2  | 2016.06.23 | 5   | 0.08726646  | 2016.06.23.13. |
| ## 471 | 340 | 3  | 2016.06.23 | 20  | 0.34906585  | 2016.06.23.13. |
| ## 472 | 345 | 4  | 2016.06.23 | 15  | 0.26179939  | 2016.06.23.13. |
| ## 473 | 340 | 5  | 2016.06.23 | 20  | 0.34906585  | 2016.06.23.13. |
| ## 474 | 275 | 6  | 2016.06.23 | 85  | 1.48352986  | 2016.06.23.13. |
| ## 475 | 350 | 7  | 2016.06.23 | 10  | 0.17453293  | 2016.06.23.13. |
| ## 476 | 340 | 8  | 2016.06.23 | 20  | 0.34906585  | 2016.06.23.13. |
| ## 477 | 320 | 9  | 2016.06.23 | 40  | 0.69813170  | 2016.06.23.13. |
| ## 478 | 305 | 10 | 2016.06.23 | 55  | 0.95993109  | 2016.06.23.13. |
| ## 479 | 350 | 11 | 2016.06.23 | 10  | 0.17453293  | 2016.06.23.13. |
| ## 480 | 190 | 12 | 2016.06.23 | 170 | 2.96705973  | 2016.06.23.13. |
| ## 481 | 245 | 13 | 2016.06.23 | 115 | 2.00712864  | 2016.06.23.13. |
| ## 482 | 200 | 14 | 2016.06.23 | 160 | 2.79252680  | 2016.06.23.13. |
| ## 483 | 250 | 15 | 2016.06.23 | 110 | 1.91986218  | 2016.06.23.13. |
| ## 484 | 200 | 16 | 2016.06.23 | 160 | 2.79252680  | 2016.06.23.13. |
| ## 485 | 215 | 17 | 2016.06.23 | 145 | 2.53072742  | 2016.06.23.13. |
| ## 486 | 210 | 18 | 2016.06.23 | 150 | 2.61799388  | 2016.06.23.13. |
| ## 487 | 220 | 19 | 2016.06.23 | 140 | 2.44346095  | 2016.06.23.13. |
| ## 488 | 180 | 20 | 2016.06.23 | 180 | 3.14159265  | 2016.06.23.13. |
| ## 489 | 200 | 21 | 2016.06.23 | 160 | 2.79252680  | 2016.06.23.13. |
| ## 490 | 210 | 22 | 2016.06.23 | 150 | 2.61799388  | 2016.06.23.13. |
| ## 491 | 210 | 23 | 2016.06.23 | 150 | 2.61799388  | 2016.06.23.13. |
| ## 492 | 230 | 24 | 2016.06.23 | 130 | 2.26892803  | 2016.06.23.13. |
| ## 493 | 200 | 25 | 2016.06.23 | 160 | 2.79252680  | 2016.06.23.13. |
| ## 494 | 230 | 26 | 2016.06.23 | 130 | 2.26892803  | 2016.06.23.13. |



|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 495 | 220 | 27 | 2016.06.23 | 140 | 2.44346095  | 2016.06.23.13. |
| ## 496 | 230 | 28 | 2016.06.23 | 130 | 2.26892803  | 2016.06.23.13. |
| ## 497 | 205 | 29 | 2016.06.23 | 155 | 2.70526034  | 2016.06.23.13. |
| ## 498 | 165 | 1  | 2016.06.28 | 195 | -2.87979327 | 2016.06.28.14. |
| ## 499 | 200 | 2  | 2016.06.28 | 160 | 2.79252680  | 2016.06.28.14. |
| ## 500 | 185 | 3  | 2016.06.28 | 175 | 3.05432619  | 2016.06.28.14. |
| ## 501 | 200 | 4  | 2016.06.28 | 160 | 2.79252680  | 2016.06.28.14. |
| ## 502 | 180 | 5  | 2016.06.28 | 180 | 3.14159265  | 2016.06.28.14. |
| ## 503 | 215 | 6  | 2016.06.28 | 145 | 2.53072742  | 2016.06.28.14. |
| ## 504 | 240 | 7  | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.14. |
| ## 505 | 335 | 8  | 2016.06.28 | 25  | 0.43633231  | 2016.06.28.14. |
| ## 506 | 280 | 9  | 2016.06.28 | 80  | 1.39626340  | 2016.06.28.14. |
| ## 507 | 330 | 10 | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.14. |
| ## 508 | 320 | 11 | 2016.06.28 | 40  | 0.69813170  | 2016.06.28.14. |
| ## 509 | 225 | 12 | 2016.06.28 | 135 | 2.35619449  | 2016.06.28.14. |
| ## 510 | 285 | 13 | 2016.06.28 | 75  | 1.30899694  | 2016.06.28.14. |
| ## 511 | 290 | 14 | 2016.06.28 | 70  | 1.22173048  | 2016.06.28.14. |
| ## 512 | 250 | 15 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.14. |
| ## 513 | 325 | 16 | 2016.06.28 | 35  | 0.61086524  | 2016.06.28.14. |
| ## 514 | 300 | 17 | 2016.06.28 | 60  | 1.04719755  | 2016.06.28.14. |
| ## 515 | 270 | 18 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.14. |
| ## 516 | 265 | 19 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.14. |
| ## 517 | 290 | 20 | 2016.06.28 | 70  | 1.22173048  | 2016.06.28.14. |
| ## 518 | 290 | 21 | 2016.06.28 | 70  | 1.22173048  | 2016.06.28.14. |
| ## 519 | 265 | 22 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.14. |
| ## 520 | 220 | 23 | 2016.06.28 | 140 | 2.44346095  | 2016.06.28.14. |
| ## 521 | 210 | 24 | 2016.06.28 | 150 | 2.61799388  | 2016.06.28.14. |
| ## 522 | 295 | 25 | 2016.06.28 | 65  | 1.13446401  | 2016.06.28.14. |
| ## 523 | 260 | 26 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.14. |
| ## 524 | 220 | 27 | 2016.06.28 | 140 | 2.44346095  | 2016.06.28.14. |
| ## 525 | 225 | 28 | 2016.06.28 | 135 | 2.35619449  | 2016.06.28.14. |
| ## 526 | 250 | 29 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.14. |
| ## 527 | 240 | 30 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.14. |
| ## 528 | 265 | 31 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.14. |
| ## 529 | 230 | 32 | 2016.06.28 | 130 | 2.26892803  | 2016.06.28.14. |
| ## 530 | 250 | 33 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.14. |
| ## 531 | 230 | 34 | 2016.06.28 | 130 | 2.26892803  | 2016.06.28.14. |
| ## 532 | 260 | 35 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.14. |
| ## 533 | 210 | 36 | 2016.06.28 | 150 | 2.61799388  | 2016.06.28.14. |
| ## 534 | 275 | 37 | 2016.06.28 | 85  | 1.48352986  | 2016.06.28.14. |
| ## 535 | 240 | 38 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.14. |
| ## 536 | 330 | 1  | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.14. |
| ## 537 | 330 | 2  | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.14. |
| ## 538 | 220 | 3  | 2016.06.28 | 140 | 2.44346095  | 2016.06.28.14. |
| ## 539 | 220 | 4  | 2016.06.28 | 140 | 2.44346095  | 2016.06.28.14. |
| ## 540 | 330 | 5  | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.14. |
| ## 541 | 310 | 6  | 2016.06.28 | 50  | 0.87266463  | 2016.06.28.14. |
| ## 542 | 0   | 7  | 2016.06.28 | 0   | 0.00000000  | 2016.06.28.14. |

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|--------|-----|----|------------|-----|-------------|----------------|
| ## 543 | 345 | 8  | 2016.06.28 | 15  | 0.26179939  | 2016.06.28.14. |
| ## 544 | 0   | 9  | 2016.06.28 | 0   | 0.00000000  | 2016.06.28.14. |
| ## 545 | 335 | 10 | 2016.06.28 | 25  | 0.43633231  | 2016.06.28.14. |
| ## 546 | 280 | 11 | 2016.06.28 | 80  | 1.39626340  | 2016.06.28.14. |
| ## 547 | 320 | 12 | 2016.06.28 | 40  | 0.69813170  | 2016.06.28.14. |
| ## 548 | 320 | 13 | 2016.06.28 | 40  | 0.69813170  | 2016.06.28.14. |
| ## 549 | 330 | 14 | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.14. |
| ## 550 | 260 | 15 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.14. |
| ## 551 | 280 | 16 | 2016.06.28 | 80  | 1.39626340  | 2016.06.28.14. |
| ## 552 | 250 | 17 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.14. |
| ## 553 | 315 | 18 | 2016.06.28 | 45  | 0.78539816  | 2016.06.28.14. |
| ## 554 | 260 | 19 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.14. |
| ## 555 | 255 | 20 | 2016.06.28 | 105 | 1.83259571  | 2016.06.28.14. |
| ## 556 | 270 | 21 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.14. |
| ## 557 | 260 | 22 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.14. |
| ## 558 | 170 | 1  | 2016.06.28 | 190 | -2.96705973 | 2016.06.28.15. |
| ## 559 | 265 | 2  | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 560 | 210 | 3  | 2016.06.28 | 150 | 2.61799388  | 2016.06.28.15. |
| ## 561 | 190 | 4  | 2016.06.28 | 170 | 2.96705973  | 2016.06.28.15. |
| ## 562 | 170 | 5  | 2016.06.28 | 190 | -2.96705973 | 2016.06.28.15. |
| ## 563 | 160 | 6  | 2016.06.28 | 200 | -2.79252680 | 2016.06.28.15. |
| ## 564 | 220 | 7  | 2016.06.28 | 140 | 2.44346095  | 2016.06.28.15. |
| ## 565 | 310 | 8  | 2016.06.28 | 50  | 0.87266463  | 2016.06.28.15. |
| ## 566 | 320 | 9  | 2016.06.28 | 40  | 0.69813170  | 2016.06.28.15. |
| ## 567 | 195 | 10 | 2016.06.28 | 165 | 2.87979327  | 2016.06.28.15. |
| ## 568 | 190 | 11 | 2016.06.28 | 170 | 2.96705973  | 2016.06.28.15. |
| ## 569 | 170 | 12 | 2016.06.28 | 190 | -2.96705973 | 2016.06.28.15. |
| ## 570 | 330 | 13 | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.15. |
| ## 571 | 310 | 14 | 2016.06.28 | 50  | 0.87266463  | 2016.06.28.15. |
| ## 572 | 240 | 15 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 573 | 290 | 16 | 2016.06.28 | 70  | 1.22173048  | 2016.06.28.15. |
| ## 574 | 270 | 17 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 575 | 240 | 18 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 576 | 250 | 19 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
| ## 577 | 265 | 20 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 578 | 250 | 21 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
| ## 579 | 260 | 22 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 580 | 270 | 23 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 581 | 280 | 24 | 2016.06.28 | 80  | 1.39626340  | 2016.06.28.15. |
| ## 582 | 285 | 25 | 2016.06.28 | 75  | 1.30899694  | 2016.06.28.15. |
| ## 583 | 250 | 26 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
| ## 584 | 255 | 27 | 2016.06.28 | 105 | 1.83259571  | 2016.06.28.15. |
| ## 585 | 270 | 28 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 586 | 240 | 29 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 587 | 255 | 30 | 2016.06.28 | 105 | 1.83259571  | 2016.06.28.15. |
| ## 588 | 260 | 31 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 589 | 230 | 32 | 2016.06.28 | 130 | 2.26892803  | 2016.06.28.15. |
| ## 590 | 270 | 33 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |

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| ## 591 | 230 | 34 | 2016.06.28 | 130 | 2.26892803  | 2016.06.28.15. |
| ## 592 | 230 | 35 | 2016.06.28 | 130 | 2.26892803  | 2016.06.28.15. |
| ## 593 | 240 | 36 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 594 | 270 | 37 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 595 | 265 | 38 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 596 | 270 | 39 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 597 | 250 | 40 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
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| ## 599 | 240 | 42 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 600 | 240 | 43 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 601 | 260 | 44 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 602 | 265 | 45 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 603 | 270 | 46 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 604 | 250 | 47 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
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| ## 606 | 250 | 49 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
| ## 607 | 260 | 50 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 608 | 260 | 51 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 609 | 250 | 52 | 2016.06.28 | 110 | 1.91986218  | 2016.06.28.15. |
| ## 610 | 260 | 53 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 611 | 240 | 54 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 612 | 265 | 55 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 613 | 260 | 56 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 614 | 240 | 57 | 2016.06.28 | 120 | 2.09439510  | 2016.06.28.15. |
| ## 615 | 220 | 58 | 2016.06.28 | 140 | 2.44346095  | 2016.06.28.15. |
| ## 616 | 265 | 59 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 617 | 260 | 60 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 618 | 265 | 61 | 2016.06.28 | 95  | 1.65806279  | 2016.06.28.15. |
| ## 619 | 255 | 62 | 2016.06.28 | 105 | 1.83259571  | 2016.06.28.15. |
| ## 620 | 230 | 63 | 2016.06.28 | 130 | 2.26892803  | 2016.06.28.15. |
| ## 621 | 270 | 64 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 622 | 330 | 65 | 2016.06.28 | 30  | 0.52359878  | 2016.06.28.15. |
| ## 623 | 205 | 66 | 2016.06.28 | 155 | 2.70526034  | 2016.06.28.15. |
| ## 624 | 300 | 67 | 2016.06.28 | 60  | 1.04719755  | 2016.06.28.15. |
| ## 625 | 200 | 68 | 2016.06.28 | 160 | 2.79252680  | 2016.06.28.15. |
| ## 626 | 210 | 69 | 2016.06.28 | 150 | 2.61799388  | 2016.06.28.15. |
| ## 627 | 270 | 70 | 2016.06.28 | 90  | 1.57079633  | 2016.06.28.15. |
| ## 628 | 260 | 71 | 2016.06.28 | 100 | 1.74532925  | 2016.06.28.15. |
| ## 629 | 250 | 1  | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.13. |
| ## 630 | 30  | 2  | 2016.06.29 | 330 | -0.52359878 | 2016.06.29.13. |
| ## 631 | 245 | 3  | 2016.06.29 | 115 | 2.00712864  | 2016.06.29.13. |
| ## 632 | 355 | 4  | 2016.06.29 | 5   | 0.08726646  | 2016.06.29.13. |
| ## 633 | 60  | 5  | 2016.06.29 | 300 | -1.04719755 | 2016.06.29.13. |
| ## 634 | 265 | 6  | 2016.06.29 | 95  | 1.65806279  | 2016.06.29.13. |
| ## 635 | 260 | 7  | 2016.06.29 | 100 | 1.74532925  | 2016.06.29.13. |
| ## 636 | 310 | 8  | 2016.06.29 | 50  | 0.87266463  | 2016.06.29.13. |
| ## 637 | 320 | 9  | 2016.06.29 | 40  | 0.69813170  | 2016.06.29.13. |
| ## 638 | 185 | 10 | 2016.06.29 | 175 | 3.05432619  | 2016.06.29.13. |

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|--------|-----|----|------------|-----|-------------|----------------|
| ## 639 | 210 | 11 | 2016.06.29 | 150 | 2.61799388  | 2016.06.29.13. |
| ## 640 | 175 | 12 | 2016.06.29 | 185 | -3.05432619 | 2016.06.29.13. |
| ## 641 | 185 | 13 | 2016.06.29 | 175 | 3.05432619  | 2016.06.29.13. |
| ## 642 | 150 | 14 | 2016.06.29 | 210 | -2.61799388 | 2016.06.29.13. |
| ## 643 | 165 | 15 | 2016.06.29 | 195 | -2.87979327 | 2016.06.29.13. |
| ## 644 | 180 | 16 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 645 | 190 | 17 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.13. |
| ## 646 | 180 | 18 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 647 | 175 | 19 | 2016.06.29 | 185 | -3.05432619 | 2016.06.29.13. |
| ## 648 | 190 | 20 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.13. |
| ## 649 | 170 | 21 | 2016.06.29 | 190 | -2.96705973 | 2016.06.29.13. |
| ## 650 | 190 | 22 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.13. |
| ## 651 | 170 | 23 | 2016.06.29 | 190 | -2.96705973 | 2016.06.29.13. |
| ## 652 | 185 | 24 | 2016.06.29 | 175 | 3.05432619  | 2016.06.29.13. |
| ## 653 | 180 | 25 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 654 | 175 | 26 | 2016.06.29 | 185 | -3.05432619 | 2016.06.29.13. |
| ## 655 | 170 | 27 | 2016.06.29 | 190 | -2.96705973 | 2016.06.29.13. |
| ## 656 | 180 | 28 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 657 | 175 | 29 | 2016.06.29 | 185 | -3.05432619 | 2016.06.29.13. |
| ## 658 | 185 | 30 | 2016.06.29 | 175 | 3.05432619  | 2016.06.29.13. |
| ## 659 | 185 | 31 | 2016.06.29 | 175 | 3.05432619  | 2016.06.29.13. |
| ## 660 | 175 | 32 | 2016.06.29 | 185 | -3.05432619 | 2016.06.29.13. |
| ## 661 | 170 | 33 | 2016.06.29 | 190 | -2.96705973 | 2016.06.29.13. |
| ## 662 | 185 | 34 | 2016.06.29 | 175 | 3.05432619  | 2016.06.29.13. |
| ## 663 | 180 | 35 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 664 | 140 | 36 | 2016.06.29 | 220 | -2.44346095 | 2016.06.29.13. |
| ## 665 | 210 | 37 | 2016.06.29 | 150 | 2.61799388  | 2016.06.29.13. |
| ## 666 | 190 | 38 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.13. |
| ## 667 | 165 | 39 | 2016.06.29 | 195 | -2.87979327 | 2016.06.29.13. |
| ## 668 | 180 | 40 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 669 | 170 | 41 | 2016.06.29 | 190 | -2.96705973 | 2016.06.29.13. |
| ## 670 | 180 | 42 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.13. |
| ## 671 | 310 | 1  | 2016.06.29 | 50  | 0.87266463  | 2016.06.29.14. |
| ## 672 | 310 | 2  | 2016.06.29 | 50  | 0.87266463  | 2016.06.29.14. |
| ## 673 | 350 | 3  | 2016.06.29 | 10  | 0.17453293  | 2016.06.29.14. |
| ## 674 | 300 | 4  | 2016.06.29 | 60  | 1.04719755  | 2016.06.29.14. |
| ## 675 | 270 | 5  | 2016.06.29 | 90  | 1.57079633  | 2016.06.29.14. |
| ## 676 | 355 | 6  | 2016.06.29 | 5   | 0.08726646  | 2016.06.29.14. |
| ## 677 | 55  | 7  | 2016.06.29 | 305 | -0.95993109 | 2016.06.29.14. |
| ## 678 | 345 | 8  | 2016.06.29 | 15  | 0.26179939  | 2016.06.29.14. |
| ## 679 | 345 | 9  | 2016.06.29 | 15  | 0.26179939  | 2016.06.29.14. |
| ## 680 | 340 | 10 | 2016.06.29 | 20  | 0.34906585  | 2016.06.29.14. |
| ## 681 | 5   | 11 | 2016.06.29 | 355 | -0.08726646 | 2016.06.29.14. |
| ## 682 | 290 | 12 | 2016.06.29 | 70  | 1.22173048  | 2016.06.29.14. |
| ## 683 | 265 | 13 | 2016.06.29 | 95  | 1.65806279  | 2016.06.29.14. |
| ## 684 | 290 | 14 | 2016.06.29 | 70  | 1.22173048  | 2016.06.29.14. |
| ## 685 | 190 | 15 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.14. |
| ## 686 | 310 | 16 | 2016.06.29 | 50  | 0.87266463  | 2016.06.29.14. |

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| ## 687 | 330 | 17 | 2016.06.29 | 30  | 0.52359878  | 2016.06.29.14. |
| ## 688 | 320 | 18 | 2016.06.29 | 40  | 0.69813170  | 2016.06.29.14. |
| ## 689 | 300 | 19 | 2016.06.29 | 60  | 1.04719755  | 2016.06.29.14. |
| ## 690 | 320 | 20 | 2016.06.29 | 40  | 0.69813170  | 2016.06.29.14. |
| ## 691 | 10  | 21 | 2016.06.29 | 350 | -0.17453293 | 2016.06.29.14. |
| ## 692 | 40  | 22 | 2016.06.29 | 320 | -0.69813170 | 2016.06.29.14. |
| ## 693 | 105 | 23 | 2016.06.29 | 255 | -1.83259571 | 2016.06.29.14. |
| ## 694 | 315 | 24 | 2016.06.29 | 45  | 0.78539816  | 2016.06.29.14. |
| ## 695 | 190 | 25 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.14. |
| ## 696 | 250 | 26 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 697 | 190 | 27 | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.14. |
| ## 698 | 240 | 28 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 699 | 225 | 29 | 2016.06.29 | 135 | 2.35619449  | 2016.06.29.14. |
| ## 700 | 240 | 30 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 701 | 250 | 31 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 702 | 280 | 32 | 2016.06.29 | 80  | 1.39626340  | 2016.06.29.14. |
| ## 703 | 205 | 33 | 2016.06.29 | 155 | 2.70526034  | 2016.06.29.14. |
| ## 704 | 220 | 34 | 2016.06.29 | 140 | 2.44346095  | 2016.06.29.14. |
| ## 705 | 250 | 35 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 706 | 220 | 36 | 2016.06.29 | 140 | 2.44346095  | 2016.06.29.14. |
| ## 707 | 230 | 37 | 2016.06.29 | 130 | 2.26892803  | 2016.06.29.14. |
| ## 708 | 230 | 38 | 2016.06.29 | 130 | 2.26892803  | 2016.06.29.14. |
| ## 709 | 250 | 39 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 710 | 230 | 40 | 2016.06.29 | 130 | 2.26892803  | 2016.06.29.14. |
| ## 711 | 205 | 41 | 2016.06.29 | 155 | 2.70526034  | 2016.06.29.14. |
| ## 712 | 240 | 42 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 713 | 230 | 43 | 2016.06.29 | 130 | 2.26892803  | 2016.06.29.14. |
| ## 714 | 250 | 44 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 715 | 220 | 45 | 2016.06.29 | 140 | 2.44346095  | 2016.06.29.14. |
| ## 716 | 250 | 46 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 717 | 240 | 47 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 718 | 250 | 48 | 2016.06.29 | 110 | 1.91986218  | 2016.06.29.14. |
| ## 719 | 230 | 49 | 2016.06.29 | 130 | 2.26892803  | 2016.06.29.14. |
| ## 720 | 240 | 50 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 721 | 240 | 51 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 722 | 240 | 52 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 723 | 225 | 53 | 2016.06.29 | 135 | 2.35619449  | 2016.06.29.14. |
| ## 724 | 230 | 54 | 2016.06.29 | 130 | 2.26892803  | 2016.06.29.14. |
| ## 725 | 235 | 55 | 2016.06.29 | 125 | 2.18166156  | 2016.06.29.14. |
| ## 726 | 240 | 56 | 2016.06.29 | 120 | 2.09439510  | 2016.06.29.14. |
| ## 727 | 200 | 57 | 2016.06.29 | 160 | 2.79252680  | 2016.06.29.14. |
| ## 728 | 220 | 58 | 2016.06.29 | 140 | 2.44346095  | 2016.06.29.14. |
| ## 729 | 225 | 59 | 2016.06.29 | 135 | 2.35619449  | 2016.06.29.14. |
| ## 730 | 145 | 1  | 2016.06.29 | 215 | -2.53072742 | 2016.06.29.14. |
| ## 731 | 165 | 2  | 2016.06.29 | 195 | -2.87979327 | 2016.06.29.14. |
| ## 732 | 80  | 3  | 2016.06.29 | 280 | -1.39626340 | 2016.06.29.14. |
| ## 733 | 150 | 4  | 2016.06.29 | 210 | -2.61799388 | 2016.06.29.14. |
| ## 734 | 125 | 5  | 2016.06.29 | 235 | -2.18166156 | 2016.06.29.14. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 735 | 190 | 6  | 2016.06.29 | 170 | 2.96705973  | 2016.06.29.14. |
| ## 736 | 195 | 7  | 2016.06.29 | 165 | 2.87979327  | 2016.06.29.14. |
| ## 737 | 70  | 8  | 2016.06.29 | 290 | -1.22173048 | 2016.06.29.14. |
| ## 738 | 95  | 9  | 2016.06.29 | 265 | -1.65806279 | 2016.06.29.14. |
| ## 739 | 20  | 10 | 2016.06.29 | 340 | -0.34906585 | 2016.06.29.14. |
| ## 740 | 60  | 11 | 2016.06.29 | 300 | -1.04719755 | 2016.06.29.14. |
| ## 741 | 110 | 12 | 2016.06.29 | 250 | -1.91986218 | 2016.06.29.14. |
| ## 742 | 180 | 13 | 2016.06.29 | 180 | 3.14159265  | 2016.06.29.14. |
| ## 743 | 80  | 14 | 2016.06.29 | 280 | -1.39626340 | 2016.06.29.14. |
| ## 744 | 140 | 15 | 2016.06.29 | 220 | -2.44346095 | 2016.06.29.14. |
| ## 745 | 150 | 16 | 2016.06.29 | 210 | -2.61799388 | 2016.06.29.14. |
| ## 746 | 335 | 17 | 2016.06.29 | 25  | 0.43633231  | 2016.06.29.14. |
| ## 747 | 345 | 18 | 2016.06.29 | 15  | 0.26179939  | 2016.06.29.14. |
| ## 748 | 345 | 19 | 2016.06.29 | 15  | 0.26179939  | 2016.06.29.14. |
| ## 749 | 350 | 20 | 2016.06.29 | 10  | 0.17453293  | 2016.06.29.14. |
| ## 750 | 300 | 21 | 2016.06.29 | 60  | 1.04719755  | 2016.06.29.14. |
| ## 751 | 320 | 22 | 2016.06.29 | 40  | 0.69813170  | 2016.06.29.14. |
| ## 752 | 330 | 23 | 2016.06.29 | 30  | 0.52359878  | 2016.06.29.14. |
| ## 753 | 275 | 24 | 2016.06.29 | 85  | 1.48352986  | 2016.06.29.14. |
| ## 754 | 340 | 25 | 2016.06.29 | 20  | 0.34906585  | 2016.06.29.14. |
| ## 755 | 310 | 26 | 2016.06.29 | 50  | 0.87266463  | 2016.06.29.14. |
| ## 756 | 300 | 27 | 2016.06.29 | 60  | 1.04719755  | 2016.06.29.14. |
| ## 757 | 300 | 28 | 2016.06.29 | 60  | 1.04719755  | 2016.06.29.14. |
| ## 758 | 280 | 29 | 2016.06.29 | 80  | 1.39626340  | 2016.06.29.14. |
| ## 759 | 270 | 30 | 2016.06.29 | 90  | 1.57079633  | 2016.06.29.14. |
| ## 760 | 325 | 31 | 2016.06.29 | 35  | 0.61086524  | 2016.06.29.14. |
| ## 761 | 300 | 32 | 2016.06.29 | 60  | 1.04719755  | 2016.06.29.14. |
| ## 762 | 315 | 33 | 2016.06.29 | 45  | 0.78539816  | 2016.06.29.14. |
| ## 763 | 320 | 34 | 2016.06.29 | 40  | 0.69813170  | 2016.06.29.14. |
| ## 764 | 280 | 35 | 2016.06.29 | 80  | 1.39626340  | 2016.06.29.14. |
| ## 765 | 315 | 36 | 2016.06.29 | 45  | 0.78539816  | 2016.06.29.14. |
| ## 766 | 310 | 1  | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.13. |
| ## 767 | 345 | 2  | 2016.07.01 | 15  | 0.26179939  | 2016.07.01.13. |
| ## 768 | 265 | 3  | 2016.07.01 | 95  | 1.65806279  | 2016.07.01.13. |
| ## 769 | 335 | 4  | 2016.07.01 | 25  | 0.43633231  | 2016.07.01.13. |
| ## 770 | 170 | 5  | 2016.07.01 | 190 | -2.96705973 | 2016.07.01.13. |
| ## 771 | 330 | 6  | 2016.07.01 | 30  | 0.52359878  | 2016.07.01.13. |
| ## 772 | 280 | 7  | 2016.07.01 | 80  | 1.39626340  | 2016.07.01.13. |
| ## 773 | 290 | 8  | 2016.07.01 | 70  | 1.22173048  | 2016.07.01.13. |
| ## 774 | 30  | 9  | 2016.07.01 | 330 | -0.52359878 | 2016.07.01.13. |
| ## 775 | 245 | 10 | 2016.07.01 | 115 | 2.00712864  | 2016.07.01.13. |
| ## 776 | 10  | 11 | 2016.07.01 | 350 | -0.17453293 | 2016.07.01.13. |
| ## 777 | 310 | 12 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.13. |
| ## 778 | 175 | 13 | 2016.07.01 | 185 | -3.05432619 | 2016.07.01.13. |
| ## 779 | 320 | 14 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 780 | 265 | 15 | 2016.07.01 | 95  | 1.65806279  | 2016.07.01.13. |
| ## 781 | 295 | 16 | 2016.07.01 | 65  | 1.13446401  | 2016.07.01.13. |
| ## 782 | 255 | 17 | 2016.07.01 | 105 | 1.83259571  | 2016.07.01.13. |



|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 783 | 320 | 18 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 784 | 310 | 19 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.13. |
| ## 785 | 260 | 20 | 2016.07.01 | 100 | 1.74532925  | 2016.07.01.13. |
| ## 786 | 310 | 21 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.13. |
| ## 787 | 340 | 22 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.13. |
| ## 788 | 325 | 23 | 2016.07.01 | 35  | 0.61086524  | 2016.07.01.13. |
| ## 789 | 320 | 24 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 790 | 335 | 25 | 2016.07.01 | 25  | 0.43633231  | 2016.07.01.13. |
| ## 791 | 340 | 26 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.13. |
| ## 792 | 275 | 27 | 2016.07.01 | 85  | 1.48352986  | 2016.07.01.13. |
| ## 793 | 270 | 28 | 2016.07.01 | 90  | 1.57079633  | 2016.07.01.13. |
| ## 794 | 300 | 29 | 2016.07.01 | 60  | 1.04719755  | 2016.07.01.13. |
| ## 795 | 280 | 30 | 2016.07.01 | 80  | 1.39626340  | 2016.07.01.13. |
| ## 796 | 300 | 31 | 2016.07.01 | 60  | 1.04719755  | 2016.07.01.13. |
| ## 797 | 270 | 32 | 2016.07.01 | 90  | 1.57079633  | 2016.07.01.13. |
| ## 798 | 320 | 33 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 799 | 320 | 34 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 800 | 310 | 35 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.13. |
| ## 801 | 320 | 36 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 802 | 320 | 37 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.13. |
| ## 803 | 300 | 38 | 2016.07.01 | 60  | 1.04719755  | 2016.07.01.13. |
| ## 804 | 285 | 39 | 2016.07.01 | 75  | 1.30899694  | 2016.07.01.13. |
| ## 805 | 275 | 40 | 2016.07.01 | 85  | 1.48352986  | 2016.07.01.13. |
| ## 806 | 355 | 41 | 2016.07.01 | 5   | 0.08726646  | 2016.07.01.13. |
| ## 807 | 60  | 42 | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.13. |
| ## 808 | 70  | 43 | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.13. |
| ## 809 | 65  | 44 | 2016.07.01 | 295 | -1.13446401 | 2016.07.01.13. |
| ## 810 | 5   | 45 | 2016.07.01 | 355 | -0.08726646 | 2016.07.01.13. |
| ## 811 | 350 | 46 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.13. |
| ## 812 | 325 | 47 | 2016.07.01 | 35  | 0.61086524  | 2016.07.01.13. |
| ## 813 | 330 | 48 | 2016.07.01 | 30  | 0.52359878  | 2016.07.01.13. |
| ## 814 | 20  | 49 | 2016.07.01 | 340 | -0.34906585 | 2016.07.01.13. |
| ## 815 | 345 | 50 | 2016.07.01 | 15  | 0.26179939  | 2016.07.01.13. |
| ## 816 | 310 | 51 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.13. |
| ## 817 | 300 | 52 | 2016.07.01 | 60  | 1.04719755  | 2016.07.01.13. |
| ## 818 | 95  | 1  | 2016.07.01 | 265 | -1.65806279 | 2016.07.01.14. |
| ## 819 | 50  | 2  | 2016.07.01 | 310 | -0.87266463 | 2016.07.01.14. |
| ## 820 | 60  | 3  | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.14. |
| ## 821 | 75  | 4  | 2016.07.01 | 285 | -1.30899694 | 2016.07.01.14. |
| ## 822 | 40  | 5  | 2016.07.01 | 320 | -0.69813170 | 2016.07.01.14. |
| ## 823 | 110 | 6  | 2016.07.01 | 250 | -1.91986218 | 2016.07.01.14. |
| ## 824 | 60  | 7  | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.14. |
| ## 825 | 80  | 8  | 2016.07.01 | 280 | -1.39626340 | 2016.07.01.14. |
| ## 826 | 30  | 9  | 2016.07.01 | 330 | -0.52359878 | 2016.07.01.14. |
| ## 827 | 130 | 10 | 2016.07.01 | 230 | -2.26892803 | 2016.07.01.14. |
| ## 828 | 35  | 11 | 2016.07.01 | 325 | -0.61086524 | 2016.07.01.14. |
| ## 829 | 40  | 12 | 2016.07.01 | 320 | -0.69813170 | 2016.07.01.14. |
| ## 830 | 40  | 13 | 2016.07.01 | 320 | -0.69813170 | 2016.07.01.14. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 831 | 30  | 14 | 2016.07.01 | 330 | -0.52359878 | 2016.07.01.14. |
| ## 832 | 115 | 15 | 2016.07.01 | 245 | -2.00712864 | 2016.07.01.14. |
| ## 833 | 50  | 16 | 2016.07.01 | 310 | -0.87266463 | 2016.07.01.14. |
| ## 834 | 50  | 17 | 2016.07.01 | 310 | -0.87266463 | 2016.07.01.14. |
| ## 835 | 220 | 18 | 2016.07.01 | 140 | 2.44346095  | 2016.07.01.14. |
| ## 836 | 40  | 19 | 2016.07.01 | 320 | -0.69813170 | 2016.07.01.14. |
| ## 837 | 80  | 20 | 2016.07.01 | 280 | -1.39626340 | 2016.07.01.14. |
| ## 838 | 50  | 21 | 2016.07.01 | 310 | -0.87266463 | 2016.07.01.14. |
| ## 839 | 220 | 22 | 2016.07.01 | 140 | 2.44346095  | 2016.07.01.14. |
| ## 840 | 70  | 23 | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.14. |
| ## 841 | 60  | 24 | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.14. |
| ## 842 | 300 | 25 | 2016.07.01 | 60  | 1.04719755  | 2016.07.01.14. |
| ## 843 | 340 | 26 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 844 | 310 | 27 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.14. |
| ## 845 | 325 | 28 | 2016.07.01 | 35  | 0.61086524  | 2016.07.01.14. |
| ## 846 | 320 | 29 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.14. |
| ## 847 | 295 | 30 | 2016.07.01 | 65  | 1.13446401  | 2016.07.01.14. |
| ## 848 | 320 | 31 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.14. |
| ## 849 | 310 | 32 | 2016.07.01 | 50  | 0.87266463  | 2016.07.01.14. |
| ## 850 | 305 | 33 | 2016.07.01 | 55  | 0.95993109  | 2016.07.01.14. |
| ## 851 | 330 | 34 | 2016.07.01 | 30  | 0.52359878  | 2016.07.01.14. |
| ## 852 | 340 | 35 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 853 | 350 | 36 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 854 | 320 | 37 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.14. |
| ## 855 | 320 | 38 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.14. |
| ## 856 | 0   | 39 | 2016.07.01 | 0   | 0.00000000  | 2016.07.01.14. |
| ## 857 | 350 | 40 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 858 | 355 | 41 | 2016.07.01 | 5   | 0.08726646  | 2016.07.01.14. |
| ## 859 | 85  | 1  | 2016.07.01 | 275 | -1.48352986 | 2016.07.01.14. |
| ## 860 | 85  | 2  | 2016.07.01 | 275 | -1.48352986 | 2016.07.01.14. |
| ## 861 | 60  | 3  | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.14. |
| ## 862 | 70  | 4  | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.14. |
| ## 863 | 35  | 5  | 2016.07.01 | 325 | -0.61086524 | 2016.07.01.14. |
| ## 864 | 85  | 6  | 2016.07.01 | 275 | -1.48352986 | 2016.07.01.14. |
| ## 865 | 60  | 7  | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.14. |
| ## 866 | 70  | 8  | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.14. |
| ## 867 | 70  | 9  | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.14. |
| ## 868 | 70  | 10 | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.14. |
| ## 869 | 350 | 11 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 870 | 60  | 12 | 2016.07.01 | 300 | -1.04719755 | 2016.07.01.14. |
| ## 871 | 70  | 13 | 2016.07.01 | 290 | -1.22173048 | 2016.07.01.14. |
| ## 872 | 35  | 14 | 2016.07.01 | 325 | -0.61086524 | 2016.07.01.14. |
| ## 873 | 30  | 15 | 2016.07.01 | 330 | -0.52359878 | 2016.07.01.14. |
| ## 874 | 345 | 16 | 2016.07.01 | 15  | 0.26179939  | 2016.07.01.14. |
| ## 875 | 340 | 17 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 876 | 5   | 18 | 2016.07.01 | 355 | -0.08726646 | 2016.07.01.14. |
| ## 877 | 345 | 19 | 2016.07.01 | 15  | 0.26179939  | 2016.07.01.14. |
| ## 878 | 350 | 20 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |



|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 879 | 0   | 21 | 2016.07.01 | 0   | 0.00000000  | 2016.07.01.14. |
| ## 880 | 340 | 22 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 881 | 340 | 23 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 882 | 5   | 24 | 2016.07.01 | 355 | -0.08726646 | 2016.07.01.14. |
| ## 883 | 5   | 25 | 2016.07.01 | 355 | -0.08726646 | 2016.07.01.14. |
| ## 884 | 50  | 26 | 2016.07.01 | 310 | -0.87266463 | 2016.07.01.14. |
| ## 885 | 330 | 27 | 2016.07.01 | 30  | 0.52359878  | 2016.07.01.14. |
| ## 886 | 350 | 28 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 887 | 350 | 29 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 888 | 350 | 30 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 889 | 340 | 31 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 890 | 0   | 32 | 2016.07.01 | 0   | 0.00000000  | 2016.07.01.14. |
| ## 891 | 345 | 33 | 2016.07.01 | 15  | 0.26179939  | 2016.07.01.14. |
| ## 892 | 340 | 34 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 893 | 355 | 35 | 2016.07.01 | 5   | 0.08726646  | 2016.07.01.14. |
| ## 894 | 5   | 36 | 2016.07.01 | 355 | -0.08726646 | 2016.07.01.14. |
| ## 895 | 340 | 37 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 896 | 330 | 38 | 2016.07.01 | 30  | 0.52359878  | 2016.07.01.14. |
| ## 897 | 355 | 39 | 2016.07.01 | 5   | 0.08726646  | 2016.07.01.14. |
| ## 898 | 320 | 40 | 2016.07.01 | 40  | 0.69813170  | 2016.07.01.14. |
| ## 899 | 350 | 41 | 2016.07.01 | 10  | 0.17453293  | 2016.07.01.14. |
| ## 900 | 330 | 42 | 2016.07.01 | 30  | 0.52359878  | 2016.07.01.14. |
| ## 901 | 340 | 43 | 2016.07.01 | 20  | 0.34906585  | 2016.07.01.14. |
| ## 902 | 20  | 1  | 2016.07.04 | 340 | -0.34906585 | 2016.07.04.__. |
| ## 903 | 10  | 2  | 2016.07.04 | 350 | -0.17453293 | 2016.07.04.__. |
| ## 904 | 90  | 3  | 2016.07.04 | 270 | -1.57079633 | 2016.07.04.__. |
| ## 905 | 5   | 4  | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |
| ## 906 | 85  | 5  | 2016.07.04 | 275 | -1.48352986 | 2016.07.04.__. |
| ## 907 | 40  | 6  | 2016.07.04 | 320 | -0.69813170 | 2016.07.04.__. |
| ## 908 | 75  | 7  | 2016.07.04 | 285 | -1.30899694 | 2016.07.04.__. |
| ## 909 | 20  | 8  | 2016.07.04 | 340 | -0.34906585 | 2016.07.04.__. |
| ## 910 | 80  | 9  | 2016.07.04 | 280 | -1.39626340 | 2016.07.04.__. |
| ## 911 | 10  | 10 | 2016.07.04 | 350 | -0.17453293 | 2016.07.04.__. |
| ## 912 | 50  | 11 | 2016.07.04 | 310 | -0.87266463 | 2016.07.04.__. |
| ## 913 | 30  | 12 | 2016.07.04 | 330 | -0.52359878 | 2016.07.04.__. |
| ## 914 | 30  | 13 | 2016.07.04 | 330 | -0.52359878 | 2016.07.04.__. |
| ## 915 | 25  | 14 | 2016.07.04 | 335 | -0.43633231 | 2016.07.04.__. |
| ## 916 | 80  | 15 | 2016.07.04 | 280 | -1.39626340 | 2016.07.04.__. |
| ## 917 | 25  | 16 | 2016.07.04 | 335 | -0.43633231 | 2016.07.04.__. |
| ## 918 | 25  | 17 | 2016.07.04 | 335 | -0.43633231 | 2016.07.04.__. |
| ## 919 | 5   | 18 | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |
| ## 920 | 10  | 19 | 2016.07.04 | 350 | -0.17453293 | 2016.07.04.__. |
| ## 921 | 0   | 20 | 2016.07.04 | 0   | 0.00000000  | 2016.07.04.__. |
| ## 922 | 0   | 21 | 2016.07.04 | 0   | 0.00000000  | 2016.07.04.__. |
| ## 923 | 0   | 22 | 2016.07.04 | 0   | 0.00000000  | 2016.07.04.__. |
| ## 924 | 40  | 23 | 2016.07.04 | 320 | -0.69813170 | 2016.07.04.__. |
| ## 925 | 345 | 24 | 2016.07.04 | 15  | 0.26179939  | 2016.07.04.__. |
| ## 926 | 5   | 25 | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |

|        |     |    |            |     |             |                |
|--------|-----|----|------------|-----|-------------|----------------|
| ## 927 | 355 | 26 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 928 | 10  | 27 | 2016.07.04 | 350 | -0.17453293 | 2016.07.04.__. |
| ## 929 | 15  | 28 | 2016.07.04 | 345 | -0.26179939 | 2016.07.04.__. |
| ## 930 | 355 | 29 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 931 | 340 | 30 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 932 | 340 | 31 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 933 | 5   | 32 | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |
| ## 934 | 355 | 33 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 935 | 320 | 34 | 2016.07.04 | 40  | 0.69813170  | 2016.07.04.__. |
| ## 936 | 5   | 35 | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |
| ## 937 | 10  | 36 | 2016.07.04 | 350 | -0.17453293 | 2016.07.04.__. |
| ## 938 | 350 | 37 | 2016.07.04 | 10  | 0.17453293  | 2016.07.04.__. |
| ## 939 | 355 | 38 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 940 | 345 | 39 | 2016.07.04 | 15  | 0.26179939  | 2016.07.04.__. |
| ## 941 | 305 | 40 | 2016.07.04 | 55  | 0.95993109  | 2016.07.04.__. |
| ## 942 | 320 | 41 | 2016.07.04 | 40  | 0.69813170  | 2016.07.04.__. |
| ## 943 | 310 | 42 | 2016.07.04 | 50  | 0.87266463  | 2016.07.04.__. |
| ## 944 | 320 | 43 | 2016.07.04 | 40  | 0.69813170  | 2016.07.04.__. |
| ## 945 | 0   | 44 | 2016.07.04 | 0   | 0.00000000  | 2016.07.04.__. |
| ## 946 | 5   | 45 | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |
| ## 947 | 335 | 46 | 2016.07.04 | 25  | 0.43633231  | 2016.07.04.__. |
| ## 948 | 355 | 47 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 949 | 350 | 48 | 2016.07.04 | 10  | 0.17453293  | 2016.07.04.__. |
| ## 950 | 345 | 49 | 2016.07.04 | 15  | 0.26179939  | 2016.07.04.__. |
| ## 951 | 340 | 50 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 952 | 320 | 51 | 2016.07.04 | 40  | 0.69813170  | 2016.07.04.__. |
| ## 953 | 335 | 52 | 2016.07.04 | 25  | 0.43633231  | 2016.07.04.__. |
| ## 954 | 340 | 53 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 955 | 0   | 54 | 2016.07.04 | 0   | 0.00000000  | 2016.07.04.__. |
| ## 956 | 350 | 55 | 2016.07.04 | 10  | 0.17453293  | 2016.07.04.__. |
| ## 957 | 20  | 56 | 2016.07.04 | 340 | -0.34906585 | 2016.07.04.__. |
| ## 958 | 330 | 57 | 2016.07.04 | 30  | 0.52359878  | 2016.07.04.__. |
| ## 959 | 340 | 58 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 960 | 330 | 59 | 2016.07.04 | 30  | 0.52359878  | 2016.07.04.__. |
| ## 961 | 340 | 60 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 962 | 30  | 61 | 2016.07.04 | 330 | -0.52359878 | 2016.07.04.__. |
| ## 963 | 335 | 62 | 2016.07.04 | 25  | 0.43633231  | 2016.07.04.__. |
| ## 964 | 330 | 63 | 2016.07.04 | 30  | 0.52359878  | 2016.07.04.__. |
| ## 965 | 340 | 64 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 966 | 305 | 65 | 2016.07.04 | 55  | 0.95993109  | 2016.07.04.__. |
| ## 967 | 335 | 66 | 2016.07.04 | 25  | 0.43633231  | 2016.07.04.__. |
| ## 968 | 40  | 67 | 2016.07.04 | 320 | -0.69813170 | 2016.07.04.__. |
| ## 969 | 340 | 68 | 2016.07.04 | 20  | 0.34906585  | 2016.07.04.__. |
| ## 970 | 0   | 69 | 2016.07.04 | 0   | 0.00000000  | 2016.07.04.__. |
| ## 971 | 355 | 70 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 972 | 5   | 71 | 2016.07.04 | 355 | -0.08726646 | 2016.07.04.__. |
| ## 973 | 355 | 72 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |
| ## 974 | 355 | 73 | 2016.07.04 | 5   | 0.08726646  | 2016.07.04.__. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 975  | 350 | 1  | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.13. |
| ## 976  | 35  | 2  | 2016.07.05 | 325 | -0.61086524 | 2016.07.05.13. |
| ## 977  | 5   | 3  | 2016.07.05 | 355 | -0.08726646 | 2016.07.05.13. |
| ## 978  | 60  | 4  | 2016.07.05 | 300 | -1.04719755 | 2016.07.05.13. |
| ## 979  | 60  | 5  | 2016.07.05 | 300 | -1.04719755 | 2016.07.05.13. |
| ## 980  | 70  | 6  | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.13. |
| ## 981  | 40  | 7  | 2016.07.05 | 320 | -0.69813170 | 2016.07.05.13. |
| ## 982  | 40  | 8  | 2016.07.05 | 320 | -0.69813170 | 2016.07.05.13. |
| ## 983  | 50  | 9  | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.13. |
| ## 984  | 40  | 10 | 2016.07.05 | 320 | -0.69813170 | 2016.07.05.13. |
| ## 985  | 50  | 11 | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.13. |
| ## 986  | 0   | 12 | 2016.07.05 | 0   | 0.00000000  | 2016.07.05.13. |
| ## 987  | 70  | 13 | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.13. |
| ## 988  | 40  | 14 | 2016.07.05 | 320 | -0.69813170 | 2016.07.05.13. |
| ## 989  | 355 | 15 | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.13. |
| ## 990  | 90  | 16 | 2016.07.05 | 270 | -1.57079633 | 2016.07.05.13. |
| ## 991  | 340 | 17 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.13. |
| ## 992  | 80  | 18 | 2016.07.05 | 280 | -1.39626340 | 2016.07.05.13. |
| ## 993  | 30  | 19 | 2016.07.05 | 330 | -0.52359878 | 2016.07.05.13. |
| ## 994  | 70  | 20 | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.13. |
| ## 995  | 345 | 21 | 2016.07.05 | 15  | 0.26179939  | 2016.07.05.13. |
| ## 996  | 355 | 22 | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.13. |
| ## 997  | 330 | 23 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.13. |
| ## 998  | 320 | 24 | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.13. |
| ## 999  | 340 | 25 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.13. |
| ## 1000 | 340 | 26 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.13. |
| ## 1001 | 300 | 27 | 2016.07.05 | 60  | 1.04719755  | 2016.07.05.13. |
| ## 1002 | 350 | 28 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.13. |
| ## 1003 | 340 | 29 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.13. |
| ## 1004 | 340 | 30 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.13. |
| ## 1005 | 325 | 31 | 2016.07.05 | 35  | 0.61086524  | 2016.07.05.13. |
| ## 1006 | 330 | 32 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.13. |
| ## 1007 | 5   | 1  | 2016.07.05 | 355 | -0.08726646 | 2016.07.05.13. |
| ## 1008 | 340 | 2  | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.13. |
| ## 1009 | 5   | 3  | 2016.07.05 | 355 | -0.08726646 | 2016.07.05.13. |
| ## 1010 | 50  | 4  | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.13. |
| ## 1011 | 355 | 5  | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.13. |
| ## 1012 | 50  | 6  | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.13. |
| ## 1013 | 320 | 7  | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.13. |
| ## 1014 | 330 | 8  | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.13. |
| ## 1015 | 355 | 9  | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.13. |
| ## 1016 | 20  | 10 | 2016.07.05 | 340 | -0.34906585 | 2016.07.05.13. |
| ## 1017 | 30  | 11 | 2016.07.05 | 330 | -0.52359878 | 2016.07.05.13. |
| ## 1018 | 70  | 12 | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.13. |
| ## 1019 | 85  | 13 | 2016.07.05 | 275 | -1.48352986 | 2016.07.05.13. |
| ## 1020 | 70  | 14 | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.13. |
| ## 1021 | 250 | 15 | 2016.07.05 | 110 | 1.91986218  | 2016.07.05.13. |
| ## 1022 | 220 | 16 | 2016.07.05 | 140 | 2.44346095  | 2016.07.05.13. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1023 | 245 | 17 | 2016.07.05 | 115 | 2.00712864  | 2016.07.05.13. |
| ## 1024 | 280 | 18 | 2016.07.05 | 80  | 1.39626340  | 2016.07.05.13. |
| ## 1025 | 290 | 19 | 2016.07.05 | 70  | 1.22173048  | 2016.07.05.13. |
| ## 1026 | 285 | 20 | 2016.07.05 | 75  | 1.30899694  | 2016.07.05.13. |
| ## 1027 | 285 | 21 | 2016.07.05 | 75  | 1.30899694  | 2016.07.05.13. |
| ## 1028 | 290 | 22 | 2016.07.05 | 70  | 1.22173048  | 2016.07.05.13. |
| ## 1029 | 270 | 23 | 2016.07.05 | 90  | 1.57079633  | 2016.07.05.13. |
| ## 1030 | 275 | 24 | 2016.07.05 | 85  | 1.48352986  | 2016.07.05.13. |
| ## 1031 | 275 | 25 | 2016.07.05 | 85  | 1.48352986  | 2016.07.05.13. |
| ## 1032 | 300 | 26 | 2016.07.05 | 60  | 1.04719755  | 2016.07.05.13. |
| ## 1033 | 295 | 27 | 2016.07.05 | 65  | 1.13446401  | 2016.07.05.13. |
| ## 1034 | 270 | 28 | 2016.07.05 | 90  | 1.57079633  | 2016.07.05.13. |
| ## 1035 | 300 | 29 | 2016.07.05 | 60  | 1.04719755  | 2016.07.05.13. |
| ## 1036 | 285 | 30 | 2016.07.05 | 75  | 1.30899694  | 2016.07.05.13. |
| ## 1037 | 265 | 31 | 2016.07.05 | 95  | 1.65806279  | 2016.07.05.13. |
| ## 1038 | 235 | 1  | 2016.07.05 | 125 | 2.18166156  | 2016.07.05.13. |
| ## 1039 | 210 | 2  | 2016.07.05 | 150 | 2.61799388  | 2016.07.05.13. |
| ## 1040 | 200 | 3  | 2016.07.05 | 160 | 2.79252680  | 2016.07.05.13. |
| ## 1041 | 195 | 4  | 2016.07.05 | 165 | 2.87979327  | 2016.07.05.13. |
| ## 1042 | 245 | 5  | 2016.07.05 | 115 | 2.00712864  | 2016.07.05.13. |
| ## 1043 | 335 | 6  | 2016.07.05 | 25  | 0.43633231  | 2016.07.05.13. |
| ## 1044 | 345 | 7  | 2016.07.05 | 15  | 0.26179939  | 2016.07.05.13. |
| ## 1045 | 265 | 8  | 2016.07.05 | 95  | 1.65806279  | 2016.07.05.13. |
| ## 1046 | 265 | 9  | 2016.07.05 | 95  | 1.65806279  | 2016.07.05.13. |
| ## 1047 | 210 | 10 | 2016.07.05 | 150 | 2.61799388  | 2016.07.05.13. |
| ## 1048 | 190 | 11 | 2016.07.05 | 170 | 2.96705973  | 2016.07.05.13. |
| ## 1049 | 255 | 12 | 2016.07.05 | 105 | 1.83259571  | 2016.07.05.13. |
| ## 1050 | 300 | 13 | 2016.07.05 | 60  | 1.04719755  | 2016.07.05.13. |
| ## 1051 | 335 | 14 | 2016.07.05 | 25  | 0.43633231  | 2016.07.05.13. |
| ## 1052 | 280 | 15 | 2016.07.05 | 80  | 1.39626340  | 2016.07.05.13. |
| ## 1053 | 280 | 16 | 2016.07.05 | 80  | 1.39626340  | 2016.07.05.13. |
| ## 1054 | 255 | 17 | 2016.07.05 | 105 | 1.83259571  | 2016.07.05.13. |
| ## 1055 | 315 | 18 | 2016.07.05 | 45  | 0.78539816  | 2016.07.05.13. |
| ## 1056 | 265 | 19 | 2016.07.05 | 95  | 1.65806279  | 2016.07.05.13. |
| ## 1057 | 300 | 20 | 2016.07.05 | 60  | 1.04719755  | 2016.07.05.13. |
| ## 1058 | 295 | 21 | 2016.07.05 | 65  | 1.13446401  | 2016.07.05.13. |
| ## 1059 | 270 | 22 | 2016.07.05 | 90  | 1.57079633  | 2016.07.05.13. |
| ## 1060 | 330 | 23 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.13. |
| ## 1061 | 20  | 1  | 2016.07.05 | 340 | -0.34906585 | 2016.07.05.14. |
| ## 1062 | 15  | 2  | 2016.07.05 | 345 | -0.26179939 | 2016.07.05.14. |
| ## 1063 | 355 | 3  | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.14. |
| ## 1064 | 35  | 4  | 2016.07.05 | 325 | -0.61086524 | 2016.07.05.14. |
| ## 1065 | 0   | 5  | 2016.07.05 | 0   | 0.00000000  | 2016.07.05.14. |
| ## 1066 | 350 | 6  | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## 1067 | 20  | 7  | 2016.07.05 | 340 | -0.34906585 | 2016.07.05.14. |
| ## 1068 | 60  | 8  | 2016.07.05 | 300 | -1.04719755 | 2016.07.05.14. |
| ## 1069 | 70  | 9  | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.14. |
| ## 1070 | 70  | 10 | 2016.07.05 | 290 | -1.22173048 | 2016.07.05.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1071 | 10  | 11 | 2016.07.05 | 350 | -0.17453293 | 2016.07.05.14. |
| ## | 1072 | 340 | 12 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1073 | 350 | 13 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## | 1074 | 15  | 14 | 2016.07.05 | 345 | -0.26179939 | 2016.07.05.14. |
| ## | 1075 | 350 | 15 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## | 1076 | 10  | 16 | 2016.07.05 | 350 | -0.17453293 | 2016.07.05.14. |
| ## | 1077 | 350 | 17 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## | 1078 | 15  | 18 | 2016.07.05 | 345 | -0.26179939 | 2016.07.05.14. |
| ## | 1079 | 25  | 19 | 2016.07.05 | 335 | -0.43633231 | 2016.07.05.14. |
| ## | 1080 | 60  | 20 | 2016.07.05 | 300 | -1.04719755 | 2016.07.05.14. |
| ## | 1081 | 80  | 21 | 2016.07.05 | 280 | -1.39626340 | 2016.07.05.14. |
| ## | 1082 | 100 | 22 | 2016.07.05 | 260 | -1.74532925 | 2016.07.05.14. |
| ## | 1083 | 5   | 23 | 2016.07.05 | 355 | -0.08726646 | 2016.07.05.14. |
| ## | 1084 | 50  | 24 | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.14. |
| ## | 1085 | 345 | 25 | 2016.07.05 | 15  | 0.26179939  | 2016.07.05.14. |
| ## | 1086 | 355 | 26 | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.14. |
| ## | 1087 | 350 | 27 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## | 1088 | 345 | 28 | 2016.07.05 | 15  | 0.26179939  | 2016.07.05.14. |
| ## | 1089 | 330 | 29 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1090 | 340 | 30 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1091 | 310 | 31 | 2016.07.05 | 50  | 0.87266463  | 2016.07.05.14. |
| ## | 1092 | 330 | 32 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1093 | 330 | 33 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1094 | 340 | 34 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1095 | 320 | 35 | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.14. |
| ## | 1096 | 335 | 36 | 2016.07.05 | 25  | 0.43633231  | 2016.07.05.14. |
| ## | 1097 | 355 | 37 | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.14. |
| ## | 1098 | 330 | 38 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1099 | 320 | 39 | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.14. |
| ## | 1100 | 340 | 40 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1101 | 340 | 41 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1102 | 345 | 42 | 2016.07.05 | 15  | 0.26179939  | 2016.07.05.14. |
| ## | 1103 | 320 | 43 | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.14. |
| ## | 1104 | 350 | 44 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## | 1105 | 340 | 45 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1106 | 340 | 46 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1107 | 325 | 47 | 2016.07.05 | 35  | 0.61086524  | 2016.07.05.14. |
| ## | 1108 | 355 | 48 | 2016.07.05 | 5   | 0.08726646  | 2016.07.05.14. |
| ## | 1109 | 310 | 49 | 2016.07.05 | 50  | 0.87266463  | 2016.07.05.14. |
| ## | 1110 | 340 | 50 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1111 | 340 | 51 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1112 | 330 | 52 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1113 | 0   | 53 | 2016.07.05 | 0   | 0.00000000  | 2016.07.05.14. |
| ## | 1114 | 50  | 1  | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.14. |
| ## | 1115 | 330 | 2  | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1116 | 0   | 3  | 2016.07.05 | 0   | 0.00000000  | 2016.07.05.14. |
| ## | 1117 | 0   | 4  | 2016.07.05 | 0   | 0.00000000  | 2016.07.05.14. |
| ## | 1118 | 350 | 5  | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1119 | 260 | 6  | 2016.07.05 | 100 | 1.74532925  | 2016.07.05.14. |
| ## | 1120 | 260 | 7  | 2016.07.05 | 100 | 1.74532925  | 2016.07.05.14. |
| ## | 1121 | 5   | 8  | 2016.07.05 | 355 | -0.08726646 | 2016.07.05.14. |
| ## | 1122 | 110 | 9  | 2016.07.05 | 250 | -1.91986218 | 2016.07.05.14. |
| ## | 1123 | 60  | 10 | 2016.07.05 | 300 | -1.04719755 | 2016.07.05.14. |
| ## | 1124 | 290 | 11 | 2016.07.05 | 70  | 1.22173048  | 2016.07.05.14. |
| ## | 1125 | 330 | 12 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1126 | 5   | 13 | 2016.07.05 | 355 | -0.08726646 | 2016.07.05.14. |
| ## | 1127 | 30  | 14 | 2016.07.05 | 330 | -0.52359878 | 2016.07.05.14. |
| ## | 1128 | 25  | 15 | 2016.07.05 | 335 | -0.43633231 | 2016.07.05.14. |
| ## | 1129 | 30  | 16 | 2016.07.05 | 330 | -0.52359878 | 2016.07.05.14. |
| ## | 1130 | 20  | 17 | 2016.07.05 | 340 | -0.34906585 | 2016.07.05.14. |
| ## | 1131 | 35  | 18 | 2016.07.05 | 325 | -0.61086524 | 2016.07.05.14. |
| ## | 1132 | 20  | 19 | 2016.07.05 | 340 | -0.34906585 | 2016.07.05.14. |
| ## | 1133 | 80  | 20 | 2016.07.05 | 280 | -1.39626340 | 2016.07.05.14. |
| ## | 1134 | 10  | 21 | 2016.07.05 | 350 | -0.17453293 | 2016.07.05.14. |
| ## | 1135 | 45  | 22 | 2016.07.05 | 315 | -0.78539816 | 2016.07.05.14. |
| ## | 1136 | 30  | 23 | 2016.07.05 | 330 | -0.52359878 | 2016.07.05.14. |
| ## | 1137 | 50  | 24 | 2016.07.05 | 310 | -0.87266463 | 2016.07.05.14. |
| ## | 1138 | 20  | 25 | 2016.07.05 | 340 | -0.34906585 | 2016.07.05.14. |
| ## | 1139 | 40  | 26 | 2016.07.05 | 320 | -0.69813170 | 2016.07.05.14. |
| ## | 1140 | 300 | 27 | 2016.07.05 | 60  | 1.04719755  | 2016.07.05.14. |
| ## | 1141 | 315 | 28 | 2016.07.05 | 45  | 0.78539816  | 2016.07.05.14. |
| ## | 1142 | 310 | 29 | 2016.07.05 | 50  | 0.87266463  | 2016.07.05.14. |
| ## | 1143 | 310 | 30 | 2016.07.05 | 50  | 0.87266463  | 2016.07.05.14. |
| ## | 1144 | 305 | 31 | 2016.07.05 | 55  | 0.95993109  | 2016.07.05.14. |
| ## | 1145 | 320 | 32 | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.14. |
| ## | 1146 | 340 | 33 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1147 | 340 | 34 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1148 | 340 | 35 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1149 | 330 | 36 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1150 | 325 | 37 | 2016.07.05 | 35  | 0.61086524  | 2016.07.05.14. |
| ## | 1151 | 340 | 38 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1152 | 330 | 39 | 2016.07.05 | 30  | 0.52359878  | 2016.07.05.14. |
| ## | 1153 | 340 | 40 | 2016.07.05 | 20  | 0.34906585  | 2016.07.05.14. |
| ## | 1154 | 315 | 41 | 2016.07.05 | 45  | 0.78539816  | 2016.07.05.14. |
| ## | 1155 | 320 | 42 | 2016.07.05 | 40  | 0.69813170  | 2016.07.05.14. |
| ## | 1156 | 30  | 43 | 2016.07.05 | 330 | -0.52359878 | 2016.07.05.14. |
| ## | 1157 | 335 | 44 | 2016.07.05 | 25  | 0.43633231  | 2016.07.05.14. |
| ## | 1158 | 350 | 45 | 2016.07.05 | 10  | 0.17453293  | 2016.07.05.14. |
| ## | 1159 | 305 | 46 | 2016.07.05 | 55  | 0.95993109  | 2016.07.05.14. |
| ## | 1160 | 190 | 1  | 2016.07.07 | 170 | 2.96705973  | 2016.07.07.13. |
| ## | 1161 | 170 | 2  | 2016.07.07 | 190 | -2.96705973 | 2016.07.07.13. |
| ## | 1162 | 180 | 3  | 2016.07.07 | 180 | 3.14159265  | 2016.07.07.13. |
| ## | 1163 | 170 | 4  | 2016.07.07 | 190 | -2.96705973 | 2016.07.07.13. |
| ## | 1164 | 285 | 5  | 2016.07.07 | 75  | 1.30899694  | 2016.07.07.13. |
| ## | 1165 | 220 | 6  | 2016.07.07 | 140 | 2.44346095  | 2016.07.07.13. |
| ## | 1166 | 290 | 7  | 2016.07.07 | 70  | 1.22173048  | 2016.07.07.13. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1167 | 155 | 8  | 2016.07.07 | 205 | -2.70526034 | 2016.07.07.13. |
| ## 1168 | 305 | 9  | 2016.07.07 | 55  | 0.95993109  | 2016.07.07.13. |
| ## 1169 | 300 | 10 | 2016.07.07 | 60  | 1.04719755  | 2016.07.07.13. |
| ## 1170 | 310 | 11 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.13. |
| ## 1171 | 180 | 12 | 2016.07.07 | 180 | 3.14159265  | 2016.07.07.13. |
| ## 1172 | 310 | 13 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.13. |
| ## 1173 | 160 | 14 | 2016.07.07 | 200 | -2.79252680 | 2016.07.07.13. |
| ## 1174 | 280 | 15 | 2016.07.07 | 80  | 1.39626340  | 2016.07.07.13. |
| ## 1175 | 260 | 16 | 2016.07.07 | 100 | 1.74532925  | 2016.07.07.13. |
| ## 1176 | 280 | 17 | 2016.07.07 | 80  | 1.39626340  | 2016.07.07.13. |
| ## 1177 | 300 | 18 | 2016.07.07 | 60  | 1.04719755  | 2016.07.07.13. |
| ## 1178 | 295 | 19 | 2016.07.07 | 65  | 1.13446401  | 2016.07.07.13. |
| ## 1179 | 285 | 20 | 2016.07.07 | 75  | 1.30899694  | 2016.07.07.13. |
| ## 1180 | 310 | 21 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.13. |
| ## 1181 | 285 | 22 | 2016.07.07 | 75  | 1.30899694  | 2016.07.07.13. |
| ## 1182 | 275 | 23 | 2016.07.07 | 85  | 1.48352986  | 2016.07.07.13. |
| ## 1183 | 280 | 24 | 2016.07.07 | 80  | 1.39626340  | 2016.07.07.13. |
| ## 1184 | 310 | 25 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.13. |
| ## 1185 | 315 | 26 | 2016.07.07 | 45  | 0.78539816  | 2016.07.07.13. |
| ## 1186 | 270 | 27 | 2016.07.07 | 90  | 1.57079633  | 2016.07.07.13. |
| ## 1187 | 310 | 28 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.13. |
| ## 1188 | 320 | 29 | 2016.07.07 | 40  | 0.69813170  | 2016.07.07.13. |
| ## 1189 | 5   | 1  | 2016.07.07 | 355 | -0.08726646 | 2016.07.07.14. |
| ## 1190 | 0   | 2  | 2016.07.07 | 0   | 0.00000000  | 2016.07.07.14. |
| ## 1191 | 10  | 3  | 2016.07.07 | 350 | -0.17453293 | 2016.07.07.14. |
| ## 1192 | 260 | 4  | 2016.07.07 | 100 | 1.74532925  | 2016.07.07.14. |
| ## 1193 | 35  | 5  | 2016.07.07 | 325 | -0.61086524 | 2016.07.07.14. |
| ## 1194 | 335 | 6  | 2016.07.07 | 25  | 0.43633231  | 2016.07.07.14. |
| ## 1195 | 245 | 7  | 2016.07.07 | 115 | 2.00712864  | 2016.07.07.14. |
| ## 1196 | 30  | 8  | 2016.07.07 | 330 | -0.52359878 | 2016.07.07.14. |
| ## 1197 | 310 | 9  | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.14. |
| ## 1198 | 15  | 10 | 2016.07.07 | 345 | -0.26179939 | 2016.07.07.14. |
| ## 1199 | 255 | 11 | 2016.07.07 | 105 | 1.83259571  | 2016.07.07.14. |
| ## 1200 | 250 | 12 | 2016.07.07 | 110 | 1.91986218  | 2016.07.07.14. |
| ## 1201 | 20  | 13 | 2016.07.07 | 340 | -0.34906585 | 2016.07.07.14. |
| ## 1202 | 40  | 14 | 2016.07.07 | 320 | -0.69813170 | 2016.07.07.14. |
| ## 1203 | 325 | 15 | 2016.07.07 | 35  | 0.61086524  | 2016.07.07.14. |
| ## 1204 | 335 | 16 | 2016.07.07 | 25  | 0.43633231  | 2016.07.07.14. |
| ## 1205 | 290 | 17 | 2016.07.07 | 70  | 1.22173048  | 2016.07.07.14. |
| ## 1206 | 280 | 18 | 2016.07.07 | 80  | 1.39626340  | 2016.07.07.14. |
| ## 1207 | 270 | 19 | 2016.07.07 | 90  | 1.57079633  | 2016.07.07.14. |
| ## 1208 | 310 | 20 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.14. |
| ## 1209 | 320 | 21 | 2016.07.07 | 40  | 0.69813170  | 2016.07.07.14. |
| ## 1210 | 300 | 22 | 2016.07.07 | 60  | 1.04719755  | 2016.07.07.14. |
| ## 1211 | 320 | 23 | 2016.07.07 | 40  | 0.69813170  | 2016.07.07.14. |
| ## 1212 | 310 | 24 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.14. |
| ## 1213 | 320 | 25 | 2016.07.07 | 40  | 0.69813170  | 2016.07.07.14. |
| ## 1214 | 310 | 26 | 2016.07.07 | 50  | 0.87266463  | 2016.07.07.14. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1215 | 350 | 1  | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.13. |
| ## 1216 | 5   | 2  | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.13. |
| ## 1217 | 265 | 3  | 2016.07.08 | 95  | 1.65806279  | 2016.07.08.13. |
| ## 1218 | 345 | 4  | 2016.07.08 | 15  | 0.26179939  | 2016.07.08.13. |
| ## 1219 | 60  | 5  | 2016.07.08 | 300 | -1.04719755 | 2016.07.08.13. |
| ## 1220 | 20  | 6  | 2016.07.08 | 340 | -0.34906585 | 2016.07.08.13. |
| ## 1221 | 75  | 7  | 2016.07.08 | 285 | -1.30899694 | 2016.07.08.13. |
| ## 1222 | 340 | 8  | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.13. |
| ## 1223 | 140 | 9  | 2016.07.08 | 220 | -2.44346095 | 2016.07.08.13. |
| ## 1224 | 170 | 10 | 2016.07.08 | 190 | -2.96705973 | 2016.07.08.13. |
| ## 1225 | 130 | 11 | 2016.07.08 | 230 | -2.26892803 | 2016.07.08.13. |
| ## 1226 | 40  | 12 | 2016.07.08 | 320 | -0.69813170 | 2016.07.08.13. |
| ## 1227 | 50  | 13 | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.13. |
| ## 1228 | 60  | 14 | 2016.07.08 | 300 | -1.04719755 | 2016.07.08.13. |
| ## 1229 | 130 | 15 | 2016.07.08 | 230 | -2.26892803 | 2016.07.08.13. |
| ## 1230 | 125 | 16 | 2016.07.08 | 235 | -2.18166156 | 2016.07.08.13. |
| ## 1231 | 265 | 17 | 2016.07.08 | 95  | 1.65806279  | 2016.07.08.13. |
| ## 1232 | 275 | 18 | 2016.07.08 | 85  | 1.48352986  | 2016.07.08.13. |
| ## 1233 | 280 | 19 | 2016.07.08 | 80  | 1.39626340  | 2016.07.08.13. |
| ## 1234 | 270 | 20 | 2016.07.08 | 90  | 1.57079633  | 2016.07.08.13. |
| ## 1235 | 265 | 21 | 2016.07.08 | 95  | 1.65806279  | 2016.07.08.13. |
| ## 1236 | 290 | 22 | 2016.07.08 | 70  | 1.22173048  | 2016.07.08.13. |
| ## 1237 | 280 | 23 | 2016.07.08 | 80  | 1.39626340  | 2016.07.08.13. |
| ## 1238 | 265 | 24 | 2016.07.08 | 95  | 1.65806279  | 2016.07.08.13. |
| ## 1239 | 315 | 25 | 2016.07.08 | 45  | 0.78539816  | 2016.07.08.13. |
| ## 1240 | 300 | 26 | 2016.07.08 | 60  | 1.04719755  | 2016.07.08.13. |
| ## 1241 | 305 | 27 | 2016.07.08 | 55  | 0.95993109  | 2016.07.08.13. |
| ## 1242 | 280 | 28 | 2016.07.08 | 80  | 1.39626340  | 2016.07.08.13. |
| ## 1243 | 270 | 29 | 2016.07.08 | 90  | 1.57079633  | 2016.07.08.13. |
| ## 1244 | 270 | 30 | 2016.07.08 | 90  | 1.57079633  | 2016.07.08.13. |
| ## 1245 | 105 | 1  | 2016.07.08 | 255 | -1.83259571 | 2016.07.08.14. |
| ## 1246 | 0   | 2  | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1247 | 75  | 3  | 2016.07.08 | 285 | -1.30899694 | 2016.07.08.14. |
| ## 1248 | 30  | 4  | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## 1249 | 70  | 5  | 2016.07.08 | 290 | -1.22173048 | 2016.07.08.14. |
| ## 1250 | 120 | 6  | 2016.07.08 | 240 | -2.09439510 | 2016.07.08.14. |
| ## 1251 | 30  | 7  | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## 1252 | 105 | 8  | 2016.07.08 | 255 | -1.83259571 | 2016.07.08.14. |
| ## 1253 | 80  | 9  | 2016.07.08 | 280 | -1.39626340 | 2016.07.08.14. |
| ## 1254 | 40  | 10 | 2016.07.08 | 320 | -0.69813170 | 2016.07.08.14. |
| ## 1255 | 95  | 11 | 2016.07.08 | 265 | -1.65806279 | 2016.07.08.14. |
| ## 1256 | 120 | 12 | 2016.07.08 | 240 | -2.09439510 | 2016.07.08.14. |
| ## 1257 | 55  | 13 | 2016.07.08 | 305 | -0.95993109 | 2016.07.08.14. |
| ## 1258 | 100 | 14 | 2016.07.08 | 260 | -1.74532925 | 2016.07.08.14. |
| ## 1259 | 15  | 15 | 2016.07.08 | 345 | -0.26179939 | 2016.07.08.14. |
| ## 1260 | 80  | 16 | 2016.07.08 | 280 | -1.39626340 | 2016.07.08.14. |
| ## 1261 | 60  | 17 | 2016.07.08 | 300 | -1.04719755 | 2016.07.08.14. |
| ## 1262 | 55  | 18 | 2016.07.08 | 305 | -0.95993109 | 2016.07.08.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1263 | 140 | 19 | 2016.07.08 | 220 | -2.44346095 | 2016.07.08.14. |
| ## | 1264 | 10  | 20 | 2016.07.08 | 350 | -0.17453293 | 2016.07.08.14. |
| ## | 1265 | 100 | 21 | 2016.07.08 | 260 | -1.74532925 | 2016.07.08.14. |
| ## | 1266 | 95  | 22 | 2016.07.08 | 265 | -1.65806279 | 2016.07.08.14. |
| ## | 1267 | 70  | 23 | 2016.07.08 | 290 | -1.22173048 | 2016.07.08.14. |
| ## | 1268 | 105 | 24 | 2016.07.08 | 255 | -1.83259571 | 2016.07.08.14. |
| ## | 1269 | 60  | 25 | 2016.07.08 | 300 | -1.04719755 | 2016.07.08.14. |
| ## | 1270 | 90  | 26 | 2016.07.08 | 270 | -1.57079633 | 2016.07.08.14. |
| ## | 1271 | 30  | 27 | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## | 1272 | 85  | 28 | 2016.07.08 | 275 | -1.48352986 | 2016.07.08.14. |
| ## | 1273 | 75  | 29 | 2016.07.08 | 285 | -1.30899694 | 2016.07.08.14. |
| ## | 1274 | 125 | 30 | 2016.07.08 | 235 | -2.18166156 | 2016.07.08.14. |
| ## | 1275 | 50  | 31 | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.14. |
| ## | 1276 | 100 | 32 | 2016.07.08 | 260 | -1.74532925 | 2016.07.08.14. |
| ## | 1277 | 95  | 33 | 2016.07.08 | 265 | -1.65806279 | 2016.07.08.14. |
| ## | 1278 | 350 | 34 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1279 | 320 | 35 | 2016.07.08 | 40  | 0.69813170  | 2016.07.08.14. |
| ## | 1280 | 25  | 36 | 2016.07.08 | 335 | -0.43633231 | 2016.07.08.14. |
| ## | 1281 | 20  | 37 | 2016.07.08 | 340 | -0.34906585 | 2016.07.08.14. |
| ## | 1282 | 20  | 38 | 2016.07.08 | 340 | -0.34906585 | 2016.07.08.14. |
| ## | 1283 | 340 | 39 | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## | 1284 | 350 | 40 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1285 | 15  | 41 | 2016.07.08 | 345 | -0.26179939 | 2016.07.08.14. |
| ## | 1286 | 350 | 42 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1287 | 355 | 43 | 2016.07.08 | 5   | 0.08726646  | 2016.07.08.14. |
| ## | 1288 | 330 | 44 | 2016.07.08 | 30  | 0.52359878  | 2016.07.08.14. |
| ## | 1289 | 0   | 45 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## | 1290 | 320 | 46 | 2016.07.08 | 40  | 0.69813170  | 2016.07.08.14. |
| ## | 1291 | 0   | 47 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## | 1292 | 345 | 48 | 2016.07.08 | 15  | 0.26179939  | 2016.07.08.14. |
| ## | 1293 | 345 | 49 | 2016.07.08 | 15  | 0.26179939  | 2016.07.08.14. |
| ## | 1294 | 5   | 50 | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## | 1295 | 5   | 51 | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## | 1296 | 0   | 52 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## | 1297 | 350 | 53 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1298 | 340 | 54 | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## | 1299 | 315 | 55 | 2016.07.08 | 45  | 0.78539816  | 2016.07.08.14. |
| ## | 1300 | 350 | 56 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1301 | 335 | 57 | 2016.07.08 | 25  | 0.43633231  | 2016.07.08.14. |
| ## | 1302 | 345 | 58 | 2016.07.08 | 15  | 0.26179939  | 2016.07.08.14. |
| ## | 1303 | 15  | 59 | 2016.07.08 | 345 | -0.26179939 | 2016.07.08.14. |
| ## | 1304 | 350 | 60 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1305 | 5   | 61 | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## | 1306 | 355 | 62 | 2016.07.08 | 5   | 0.08726646  | 2016.07.08.14. |
| ## | 1307 | 350 | 63 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## | 1308 | 315 | 64 | 2016.07.08 | 45  | 0.78539816  | 2016.07.08.14. |
| ## | 1309 | 285 | 65 | 2016.07.08 | 75  | 1.30899694  | 2016.07.08.14. |
| ## | 1310 | 355 | 66 | 2016.07.08 | 5   | 0.08726646  | 2016.07.08.14. |

|         |     |     |            |     |             |                |
|---------|-----|-----|------------|-----|-------------|----------------|
| ## 1311 | 5   | 67  | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## 1312 | 345 | 68  | 2016.07.08 | 15  | 0.26179939  | 2016.07.08.14. |
| ## 1313 | 10  | 69  | 2016.07.08 | 350 | -0.17453293 | 2016.07.08.14. |
| ## 1314 | 0   | 70  | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1315 | 350 | 71  | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## 1316 | 5   | 72  | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## 1317 | 5   | 73  | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## 1318 | 20  | 74  | 2016.07.08 | 340 | -0.34906585 | 2016.07.08.14. |
| ## 1319 | 115 | 75  | 2016.07.08 | 245 | -2.00712864 | 2016.07.08.14. |
| ## 1320 | 20  | 76  | 2016.07.08 | 340 | -0.34906585 | 2016.07.08.14. |
| ## 1321 | 40  | 77  | 2016.07.08 | 320 | -0.69813170 | 2016.07.08.14. |
| ## 1322 | 30  | 78  | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## 1323 | 50  | 79  | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.14. |
| ## 1324 | 130 | 80  | 2016.07.08 | 230 | -2.26892803 | 2016.07.08.14. |
| ## 1325 | 175 | 81  | 2016.07.08 | 185 | -3.05432619 | 2016.07.08.14. |
| ## 1326 | 90  | 82  | 2016.07.08 | 270 | -1.57079633 | 2016.07.08.14. |
| ## 1327 | 145 | 83  | 2016.07.08 | 215 | -2.53072742 | 2016.07.08.14. |
| ## 1328 | 205 | 84  | 2016.07.08 | 155 | 2.70526034  | 2016.07.08.14. |
| ## 1329 | 140 | 85  | 2016.07.08 | 220 | -2.44346095 | 2016.07.08.14. |
| ## 1330 | 230 | 86  | 2016.07.08 | 130 | 2.26892803  | 2016.07.08.14. |
| ## 1331 | 200 | 87  | 2016.07.08 | 160 | 2.79252680  | 2016.07.08.14. |
| ## 1332 | 170 | 88  | 2016.07.08 | 190 | -2.96705973 | 2016.07.08.14. |
| ## 1333 | 25  | 89  | 2016.07.08 | 335 | -0.43633231 | 2016.07.08.14. |
| ## 1334 | 340 | 90  | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## 1335 | 295 | 91  | 2016.07.08 | 65  | 1.13446401  | 2016.07.08.14. |
| ## 1336 | 340 | 92  | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## 1337 | 90  | 93  | 2016.07.08 | 270 | -1.57079633 | 2016.07.08.14. |
| ## 1338 | 160 | 94  | 2016.07.08 | 200 | -2.79252680 | 2016.07.08.14. |
| ## 1339 | 310 | 95  | 2016.07.08 | 50  | 0.87266463  | 2016.07.08.14. |
| ## 1340 | 105 | 96  | 2016.07.08 | 255 | -1.83259571 | 2016.07.08.14. |
| ## 1341 | 90  | 97  | 2016.07.08 | 270 | -1.57079633 | 2016.07.08.14. |
| ## 1342 | 150 | 98  | 2016.07.08 | 210 | -2.61799388 | 2016.07.08.14. |
| ## 1343 | 145 | 99  | 2016.07.08 | 215 | -2.53072742 | 2016.07.08.14. |
| ## 1344 | 0   | 100 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1345 | 95  | 1   | 2016.07.08 | 265 | -1.65806279 | 2016.07.08.14. |
| ## 1346 | 70  | 2   | 2016.07.08 | 290 | -1.22173048 | 2016.07.08.14. |
| ## 1347 | 60  | 3   | 2016.07.08 | 300 | -1.04719755 | 2016.07.08.14. |
| ## 1348 | 70  | 4   | 2016.07.08 | 290 | -1.22173048 | 2016.07.08.14. |
| ## 1349 | 325 | 5   | 2016.07.08 | 35  | 0.61086524  | 2016.07.08.14. |
| ## 1350 | 80  | 6   | 2016.07.08 | 280 | -1.39626340 | 2016.07.08.14. |
| ## 1351 | 50  | 7   | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.14. |
| ## 1352 | 40  | 8   | 2016.07.08 | 320 | -0.69813170 | 2016.07.08.14. |
| ## 1353 | 110 | 9   | 2016.07.08 | 250 | -1.91986218 | 2016.07.08.14. |
| ## 1354 | 35  | 10  | 2016.07.08 | 325 | -0.61086524 | 2016.07.08.14. |
| ## 1355 | 40  | 11  | 2016.07.08 | 320 | -0.69813170 | 2016.07.08.14. |
| ## 1356 | 340 | 12  | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## 1357 | 340 | 13  | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## 1358 | 345 | 14  | 2016.07.08 | 15  | 0.26179939  | 2016.07.08.14. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1359 | 50  | 15 | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.14. |
| ## 1360 | 320 | 16 | 2016.07.08 | 40  | 0.69813170  | 2016.07.08.14. |
| ## 1361 | 130 | 17 | 2016.07.08 | 230 | -2.26892803 | 2016.07.08.14. |
| ## 1362 | 0   | 18 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1363 | 100 | 19 | 2016.07.08 | 260 | -1.74532925 | 2016.07.08.14. |
| ## 1364 | 15  | 20 | 2016.07.08 | 345 | -0.26179939 | 2016.07.08.14. |
| ## 1365 | 340 | 21 | 2016.07.08 | 20  | 0.34906585  | 2016.07.08.14. |
| ## 1366 | 0   | 22 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1367 | 0   | 23 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1368 | 0   | 24 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1369 | 0   | 25 | 2016.07.08 | 0   | 0.00000000  | 2016.07.08.14. |
| ## 1370 | 30  | 26 | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## 1371 | 30  | 27 | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## 1372 | 20  | 28 | 2016.07.08 | 340 | -0.34906585 | 2016.07.08.14. |
| ## 1373 | 50  | 29 | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.14. |
| ## 1374 | 30  | 30 | 2016.07.08 | 330 | -0.52359878 | 2016.07.08.14. |
| ## 1375 | 5   | 31 | 2016.07.08 | 355 | -0.08726646 | 2016.07.08.14. |
| ## 1376 | 60  | 32 | 2016.07.08 | 300 | -1.04719755 | 2016.07.08.14. |
| ## 1377 | 350 | 33 | 2016.07.08 | 10  | 0.17453293  | 2016.07.08.14. |
| ## 1378 | 10  | 34 | 2016.07.08 | 350 | -0.17453293 | 2016.07.08.14. |
| ## 1379 | 50  | 35 | 2016.07.08 | 310 | -0.87266463 | 2016.07.08.14. |
| ## 1380 | 10  | 36 | 2016.07.08 | 350 | -0.17453293 | 2016.07.08.14. |
| ## 1381 | 10  | 37 | 2016.07.08 | 350 | -0.17453293 | 2016.07.08.14. |
| ## 1382 | 220 | 1  | 2016.07.11 | 140 | 2.44346095  | 2016.07.11.13. |
| ## 1383 | 135 | 2  | 2016.07.11 | 225 | -2.35619449 | 2016.07.11.13. |
| ## 1384 | 115 | 3  | 2016.07.11 | 245 | -2.00712864 | 2016.07.11.13. |
| ## 1385 | 215 | 4  | 2016.07.11 | 145 | 2.53072742  | 2016.07.11.13. |
| ## 1386 | 175 | 5  | 2016.07.11 | 185 | -3.05432619 | 2016.07.11.13. |
| ## 1387 | 180 | 6  | 2016.07.11 | 180 | 3.14159265  | 2016.07.11.13. |
| ## 1388 | 175 | 7  | 2016.07.11 | 185 | -3.05432619 | 2016.07.11.13. |
| ## 1389 | 200 | 8  | 2016.07.11 | 160 | 2.79252680  | 2016.07.11.13. |
| ## 1390 | 130 | 9  | 2016.07.11 | 230 | -2.26892803 | 2016.07.11.13. |
| ## 1391 | 220 | 10 | 2016.07.11 | 140 | 2.44346095  | 2016.07.11.13. |
| ## 1392 | 170 | 11 | 2016.07.11 | 190 | -2.96705973 | 2016.07.11.13. |
| ## 1393 | 200 | 12 | 2016.07.11 | 160 | 2.79252680  | 2016.07.11.13. |
| ## 1394 | 160 | 13 | 2016.07.11 | 200 | -2.79252680 | 2016.07.11.13. |
| ## 1395 | 220 | 14 | 2016.07.11 | 140 | 2.44346095  | 2016.07.11.13. |
| ## 1396 | 175 | 15 | 2016.07.11 | 185 | -3.05432619 | 2016.07.11.13. |
| ## 1397 | 195 | 16 | 2016.07.11 | 165 | 2.87979327  | 2016.07.11.13. |
| ## 1398 | 20  | 17 | 2016.07.11 | 340 | -0.34906585 | 2016.07.11.13. |
| ## 1399 | 40  | 18 | 2016.07.11 | 320 | -0.69813170 | 2016.07.11.13. |
| ## 1400 | 170 | 19 | 2016.07.11 | 190 | -2.96705973 | 2016.07.11.13. |
| ## 1401 | 160 | 20 | 2016.07.11 | 200 | -2.79252680 | 2016.07.11.13. |
| ## 1402 | 285 | 21 | 2016.07.11 | 75  | 1.30899694  | 2016.07.11.13. |
| ## 1403 | 270 | 22 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |
| ## 1404 | 270 | 23 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |
| ## 1405 | 290 | 24 | 2016.07.11 | 70  | 1.22173048  | 2016.07.11.13. |
| ## 1406 | 270 | 25 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1407 | 280 | 26 | 2016.07.11 | 80  | 1.39626340  | 2016.07.11.13. |
| ## 1408 | 270 | 27 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |
| ## 1409 | 275 | 28 | 2016.07.11 | 85  | 1.48352986  | 2016.07.11.13. |
| ## 1410 | 270 | 29 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |
| ## 1411 | 250 | 30 | 2016.07.11 | 110 | 1.91986218  | 2016.07.11.13. |
| ## 1412 | 290 | 31 | 2016.07.11 | 70  | 1.22173048  | 2016.07.11.13. |
| ## 1413 | 310 | 32 | 2016.07.11 | 50  | 0.87266463  | 2016.07.11.13. |
| ## 1414 | 260 | 33 | 2016.07.11 | 100 | 1.74532925  | 2016.07.11.13. |
| ## 1415 | 230 | 34 | 2016.07.11 | 130 | 2.26892803  | 2016.07.11.13. |
| ## 1416 | 270 | 35 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |
| ## 1417 | 280 | 36 | 2016.07.11 | 80  | 1.39626340  | 2016.07.11.13. |
| ## 1418 | 270 | 37 | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.13. |
| ## 1419 | 280 | 38 | 2016.07.11 | 80  | 1.39626340  | 2016.07.11.13. |
| ## 1420 | 260 | 39 | 2016.07.11 | 100 | 1.74532925  | 2016.07.11.13. |
| ## 1421 | 275 | 40 | 2016.07.11 | 85  | 1.48352986  | 2016.07.11.13. |
| ## 1422 | 320 | 41 | 2016.07.11 | 40  | 0.69813170  | 2016.07.11.13. |
| ## 1423 | 310 | 42 | 2016.07.11 | 50  | 0.87266463  | 2016.07.11.13. |
| ## 1424 | 280 | 43 | 2016.07.11 | 80  | 1.39626340  | 2016.07.11.13. |
| ## 1425 | 290 | 44 | 2016.07.11 | 70  | 1.22173048  | 2016.07.11.13. |
| ## 1426 | 275 | 45 | 2016.07.11 | 85  | 1.48352986  | 2016.07.11.13. |
| ## 1427 | 280 | 46 | 2016.07.11 | 80  | 1.39626340  | 2016.07.11.13. |
| ## 1428 | 250 | 47 | 2016.07.11 | 110 | 1.91986218  | 2016.07.11.13. |
| ## 1429 | 275 | 48 | 2016.07.11 | 85  | 1.48352986  | 2016.07.11.13. |
| ## 1430 | 285 | 49 | 2016.07.11 | 75  | 1.30899694  | 2016.07.11.13. |
| ## 1431 | 280 | 1  | 2016.07.11 | 80  | 1.39626340  | 2016.07.11.15. |
| ## 1432 | 90  | 2  | 2016.07.11 | 270 | -1.57079633 | 2016.07.11.15. |
| ## 1433 | 120 | 3  | 2016.07.11 | 240 | -2.09439510 | 2016.07.11.15. |
| ## 1434 | 270 | 4  | 2016.07.11 | 90  | 1.57079633  | 2016.07.11.15. |
| ## 1435 | 260 | 5  | 2016.07.11 | 100 | 1.74532925  | 2016.07.11.15. |
| ## 1436 | 105 | 6  | 2016.07.11 | 255 | -1.83259571 | 2016.07.11.15. |
| ## 1437 | 100 | 7  | 2016.07.11 | 260 | -1.74532925 | 2016.07.11.15. |
| ## 1438 | 260 | 8  | 2016.07.11 | 100 | 1.74532925  | 2016.07.11.15. |
| ## 1439 | 130 | 9  | 2016.07.11 | 230 | -2.26892803 | 2016.07.11.15. |
| ## 1440 | 330 | 10 | 2016.07.11 | 30  | 0.52359878  | 2016.07.11.15. |
| ## 1441 | 135 | 11 | 2016.07.11 | 225 | -2.35619449 | 2016.07.11.15. |
| ## 1442 | 105 | 12 | 2016.07.11 | 255 | -1.83259571 | 2016.07.11.15. |
| ## 1443 | 90  | 13 | 2016.07.11 | 270 | -1.57079633 | 2016.07.11.15. |
| ## 1444 | 120 | 14 | 2016.07.11 | 240 | -2.09439510 | 2016.07.11.15. |
| ## 1445 | 140 | 15 | 2016.07.11 | 220 | -2.44346095 | 2016.07.11.15. |
| ## 1446 | 120 | 16 | 2016.07.11 | 240 | -2.09439510 | 2016.07.11.15. |
| ## 1447 | 50  | 17 | 2016.07.11 | 310 | -0.87266463 | 2016.07.11.15. |
| ## 1448 | 20  | 18 | 2016.07.11 | 340 | -0.34906585 | 2016.07.11.15. |
| ## 1449 | 30  | 19 | 2016.07.11 | 330 | -0.52359878 | 2016.07.11.15. |
| ## 1450 | 20  | 20 | 2016.07.11 | 340 | -0.34906585 | 2016.07.11.15. |
| ## 1451 | 20  | 21 | 2016.07.11 | 340 | -0.34906585 | 2016.07.11.15. |
| ## 1452 | 5   | 22 | 2016.07.11 | 355 | -0.08726646 | 2016.07.11.15. |
| ## 1453 | 5   | 23 | 2016.07.11 | 355 | -0.08726646 | 2016.07.11.15. |
| ## 1454 | 20  | 24 | 2016.07.11 | 340 | -0.34906585 | 2016.07.11.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1455 | 10  | 25 | 2016.07.11 | 350 | -0.17453293 | 2016.07.11.15. |
| ## 1456 | 0   | 26 | 2016.07.11 | 0   | 0.00000000  | 2016.07.11.15. |
| ## 1457 | 10  | 27 | 2016.07.11 | 350 | -0.17453293 | 2016.07.11.15. |
| ## 1458 | 5   | 28 | 2016.07.11 | 355 | -0.08726646 | 2016.07.11.15. |
| ## 1459 | 20  | 29 | 2016.07.11 | 340 | -0.34906585 | 2016.07.11.15. |
| ## 1460 | 30  | 30 | 2016.07.11 | 330 | -0.52359878 | 2016.07.11.15. |
| ## 1461 | 40  | 31 | 2016.07.11 | 320 | -0.69813170 | 2016.07.11.15. |
| ## 1462 | 30  | 32 | 2016.07.11 | 330 | -0.52359878 | 2016.07.11.15. |
| ## 1463 | 35  | 33 | 2016.07.11 | 325 | -0.61086524 | 2016.07.11.15. |
| ## 1464 | 10  | 34 | 2016.07.11 | 350 | -0.17453293 | 2016.07.11.15. |
| ## 1465 | 5   | 35 | 2016.07.11 | 355 | -0.08726646 | 2016.07.11.15. |
| ## 1466 | 50  | 36 | 2016.07.11 | 310 | -0.87266463 | 2016.07.11.15. |
| ## 1467 | 15  | 37 | 2016.07.11 | 345 | -0.26179939 | 2016.07.11.15. |
| ## 1468 | 30  | 38 | 2016.07.11 | 330 | -0.52359878 | 2016.07.11.15. |
| ## 1469 | 70  | 39 | 2016.07.11 | 290 | -1.22173048 | 2016.07.11.15. |
| ## 1470 | 40  | 40 | 2016.07.11 | 320 | -0.69813170 | 2016.07.11.15. |
| ## 1471 | 40  | 41 | 2016.07.11 | 320 | -0.69813170 | 2016.07.11.15. |
| ## 1472 | 5   | 42 | 2016.07.11 | 355 | -0.08726646 | 2016.07.11.15. |
| ## 1473 | 25  | 43 | 2016.07.11 | 335 | -0.43633231 | 2016.07.11.15. |
| ## 1474 | 45  | 44 | 2016.07.11 | 315 | -0.78539816 | 2016.07.11.15. |
| ## 1475 | 50  | 45 | 2016.07.11 | 310 | -0.87266463 | 2016.07.11.15. |
| ## 1476 | 300 | 1  | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.13. |
| ## 1477 | 0   | 2  | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.13. |
| ## 1478 | 260 | 3  | 2016.07.12 | 100 | 1.74532925  | 2016.07.12.13. |
| ## 1479 | 45  | 4  | 2016.07.12 | 315 | -0.78539816 | 2016.07.12.13. |
| ## 1480 | 245 | 5  | 2016.07.12 | 115 | 2.00712864  | 2016.07.12.13. |
| ## 1481 | 240 | 6  | 2016.07.12 | 120 | 2.09439510  | 2016.07.12.13. |
| ## 1482 | 280 | 7  | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.13. |
| ## 1483 | 270 | 8  | 2016.07.12 | 90  | 1.57079633  | 2016.07.12.13. |
| ## 1484 | 250 | 9  | 2016.07.12 | 110 | 1.91986218  | 2016.07.12.13. |
| ## 1485 | 295 | 10 | 2016.07.12 | 65  | 1.13446401  | 2016.07.12.13. |
| ## 1486 | 340 | 11 | 2016.07.12 | 20  | 0.34906585  | 2016.07.12.13. |
| ## 1487 | 320 | 12 | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.13. |
| ## 1488 | 280 | 13 | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.13. |
| ## 1489 | 20  | 14 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.13. |
| ## 1490 | 330 | 15 | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.13. |
| ## 1491 | 345 | 16 | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.13. |
| ## 1492 | 355 | 17 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.13. |
| ## 1493 | 335 | 18 | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.13. |
| ## 1494 | 350 | 19 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.13. |
| ## 1495 | 320 | 20 | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.13. |
| ## 1496 | 325 | 21 | 2016.07.12 | 35  | 0.61086524  | 2016.07.12.13. |
| ## 1497 | 325 | 22 | 2016.07.12 | 35  | 0.61086524  | 2016.07.12.13. |
| ## 1498 | 330 | 23 | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.13. |
| ## 1499 | 350 | 24 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.13. |
| ## 1500 | 350 | 25 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.13. |
| ## 1501 | 355 | 26 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.13. |
| ## 1502 | 350 | 27 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.13. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1503 | 0   | 28 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.13. |
| ## 1504 | 350 | 29 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.13. |
| ## 1505 | 340 | 30 | 2016.07.12 | 20  | 0.34906585  | 2016.07.12.13. |
| ## 1506 | 0   | 31 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.13. |
| ## 1507 | 40  | 1  | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## 1508 | 50  | 2  | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1509 | 50  | 3  | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1510 | 50  | 4  | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1511 | 65  | 5  | 2016.07.12 | 295 | -1.13446401 | 2016.07.12.14. |
| ## 1512 | 50  | 6  | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1513 | 50  | 7  | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1514 | 50  | 8  | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1515 | 40  | 9  | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## 1516 | 45  | 10 | 2016.07.12 | 315 | -0.78539816 | 2016.07.12.14. |
| ## 1517 | 30  | 11 | 2016.07.12 | 330 | -0.52359878 | 2016.07.12.14. |
| ## 1518 | 55  | 12 | 2016.07.12 | 305 | -0.95993109 | 2016.07.12.14. |
| ## 1519 | 30  | 13 | 2016.07.12 | 330 | -0.52359878 | 2016.07.12.14. |
| ## 1520 | 95  | 14 | 2016.07.12 | 265 | -1.65806279 | 2016.07.12.14. |
| ## 1521 | 75  | 15 | 2016.07.12 | 285 | -1.30899694 | 2016.07.12.14. |
| ## 1522 | 85  | 16 | 2016.07.12 | 275 | -1.48352986 | 2016.07.12.14. |
| ## 1523 | 40  | 17 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## 1524 | 40  | 18 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## 1525 | 70  | 19 | 2016.07.12 | 290 | -1.22173048 | 2016.07.12.14. |
| ## 1526 | 45  | 20 | 2016.07.12 | 315 | -0.78539816 | 2016.07.12.14. |
| ## 1527 | 60  | 21 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## 1528 | 60  | 22 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## 1529 | 50  | 23 | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## 1530 | 45  | 24 | 2016.07.12 | 315 | -0.78539816 | 2016.07.12.14. |
| ## 1531 | 85  | 25 | 2016.07.12 | 275 | -1.48352986 | 2016.07.12.14. |
| ## 1532 | 55  | 26 | 2016.07.12 | 305 | -0.95993109 | 2016.07.12.14. |
| ## 1533 | 85  | 27 | 2016.07.12 | 275 | -1.48352986 | 2016.07.12.14. |
| ## 1534 | 70  | 28 | 2016.07.12 | 290 | -1.22173048 | 2016.07.12.14. |
| ## 1535 | 80  | 29 | 2016.07.12 | 280 | -1.39626340 | 2016.07.12.14. |
| ## 1536 | 60  | 30 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## 1537 | 80  | 31 | 2016.07.12 | 280 | -1.39626340 | 2016.07.12.14. |
| ## 1538 | 80  | 32 | 2016.07.12 | 280 | -1.39626340 | 2016.07.12.14. |
| ## 1539 | 115 | 1  | 2016.07.12 | 245 | -2.00712864 | 2016.07.12.14. |
| ## 1540 | 310 | 2  | 2016.07.12 | 50  | 0.87266463  | 2016.07.12.14. |
| ## 1541 | 175 | 3  | 2016.07.12 | 185 | -3.05432619 | 2016.07.12.14. |
| ## 1542 | 270 | 4  | 2016.07.12 | 90  | 1.57079633  | 2016.07.12.14. |
| ## 1543 | 305 | 5  | 2016.07.12 | 55  | 0.95993109  | 2016.07.12.14. |
| ## 1544 | 310 | 6  | 2016.07.12 | 50  | 0.87266463  | 2016.07.12.14. |
| ## 1545 | 90  | 7  | 2016.07.12 | 270 | -1.57079633 | 2016.07.12.14. |
| ## 1546 | 330 | 8  | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.14. |
| ## 1547 | 110 | 9  | 2016.07.12 | 250 | -1.91986218 | 2016.07.12.14. |
| ## 1548 | 40  | 10 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## 1549 | 340 | 11 | 2016.07.12 | 20  | 0.34906585  | 2016.07.12.14. |
| ## 1550 | 320 | 12 | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1551 | 300 | 13 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.14. |
| ## | 1552 | 330 | 14 | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.14. |
| ## | 1553 | 335 | 15 | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.14. |
| ## | 1554 | 120 | 16 | 2016.07.12 | 240 | -2.09439510 | 2016.07.12.14. |
| ## | 1555 | 280 | 17 | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.14. |
| ## | 1556 | 335 | 18 | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.14. |
| ## | 1557 | 70  | 19 | 2016.07.12 | 290 | -1.22173048 | 2016.07.12.14. |
| ## | 1558 | 290 | 20 | 2016.07.12 | 70  | 1.22173048  | 2016.07.12.14. |
| ## | 1559 | 60  | 21 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## | 1560 | 85  | 22 | 2016.07.12 | 275 | -1.48352986 | 2016.07.12.14. |
| ## | 1561 | 35  | 23 | 2016.07.12 | 325 | -0.61086524 | 2016.07.12.14. |
| ## | 1562 | 55  | 24 | 2016.07.12 | 305 | -0.95993109 | 2016.07.12.14. |
| ## | 1563 | 45  | 25 | 2016.07.12 | 315 | -0.78539816 | 2016.07.12.14. |
| ## | 1564 | 355 | 26 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.14. |
| ## | 1565 | 20  | 27 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.14. |
| ## | 1566 | 15  | 28 | 2016.07.12 | 345 | -0.26179939 | 2016.07.12.14. |
| ## | 1567 | 25  | 29 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1568 | 25  | 30 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1569 | 50  | 31 | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## | 1570 | 355 | 32 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.14. |
| ## | 1571 | 355 | 33 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.14. |
| ## | 1572 | 0   | 34 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.14. |
| ## | 1573 | 10  | 35 | 2016.07.12 | 350 | -0.17453293 | 2016.07.12.14. |
| ## | 1574 | 25  | 36 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1575 | 30  | 37 | 2016.07.12 | 330 | -0.52359878 | 2016.07.12.14. |
| ## | 1576 | 5   | 38 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.14. |
| ## | 1577 | 25  | 39 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1578 | 5   | 40 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.14. |
| ## | 1579 | 20  | 41 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.14. |
| ## | 1580 | 20  | 42 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.14. |
| ## | 1581 | 25  | 43 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1582 | 5   | 44 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.14. |
| ## | 1583 | 35  | 45 | 2016.07.12 | 325 | -0.61086524 | 2016.07.12.14. |
| ## | 1584 | 60  | 46 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## | 1585 | 25  | 47 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1586 | 50  | 48 | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## | 1587 | 10  | 49 | 2016.07.12 | 350 | -0.17453293 | 2016.07.12.14. |
| ## | 1588 | 25  | 50 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1589 | 30  | 51 | 2016.07.12 | 330 | -0.52359878 | 2016.07.12.14. |
| ## | 1590 | 350 | 52 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.14. |
| ## | 1591 | 15  | 53 | 2016.07.12 | 345 | -0.26179939 | 2016.07.12.14. |
| ## | 1592 | 70  | 54 | 2016.07.12 | 290 | -1.22173048 | 2016.07.12.14. |
| ## | 1593 | 355 | 55 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.14. |
| ## | 1594 | 0   | 56 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.14. |
| ## | 1595 | 300 | 57 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.14. |
| ## | 1596 | 270 | 58 | 2016.07.12 | 90  | 1.57079633  | 2016.07.12.14. |
| ## | 1597 | 40  | 59 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## | 1598 | 25  | 60 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1599 | 300 | 61 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.14. |
| ## | 1600 | 165 | 62 | 2016.07.12 | 195 | -2.87979327 | 2016.07.12.14. |
| ## | 1601 | 40  | 63 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## | 1602 | 95  | 64 | 2016.07.12 | 265 | -1.65806279 | 2016.07.12.14. |
| ## | 1603 | 60  | 65 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## | 1604 | 110 | 66 | 2016.07.12 | 250 | -1.91986218 | 2016.07.12.14. |
| ## | 1605 | 40  | 67 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## | 1606 | 60  | 68 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## | 1607 | 345 | 69 | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.14. |
| ## | 1608 | 305 | 70 | 2016.07.12 | 55  | 0.95993109  | 2016.07.12.14. |
| ## | 1609 | 350 | 71 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.14. |
| ## | 1610 | 300 | 72 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.14. |
| ## | 1611 | 40  | 73 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## | 1612 | 50  | 74 | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.14. |
| ## | 1613 | 45  | 75 | 2016.07.12 | 315 | -0.78539816 | 2016.07.12.14. |
| ## | 1614 | 0   | 76 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.14. |
| ## | 1615 | 300 | 77 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.14. |
| ## | 1616 | 270 | 78 | 2016.07.12 | 90  | 1.57079633  | 2016.07.12.14. |
| ## | 1617 | 40  | 79 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## | 1618 | 25  | 80 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.14. |
| ## | 1619 | 300 | 81 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.14. |
| ## | 1620 | 165 | 82 | 2016.07.12 | 195 | -2.87979327 | 2016.07.12.14. |
| ## | 1621 | 40  | 83 | 2016.07.12 | 320 | -0.69813170 | 2016.07.12.14. |
| ## | 1622 | 95  | 84 | 2016.07.12 | 265 | -1.65806279 | 2016.07.12.14. |
| ## | 1623 | 60  | 85 | 2016.07.12 | 300 | -1.04719755 | 2016.07.12.14. |
| ## | 1624 | 110 | 86 | 2016.07.12 | 250 | -1.91986218 | 2016.07.12.14. |
| ## | 1625 | 345 | 1  | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.15. |
| ## | 1626 | 270 | 2  | 2016.07.12 | 90  | 1.57079633  | 2016.07.12.15. |
| ## | 1627 | 215 | 3  | 2016.07.12 | 145 | 2.53072742  | 2016.07.12.15. |
| ## | 1628 | 330 | 4  | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.15. |
| ## | 1629 | 295 | 5  | 2016.07.12 | 65  | 1.13446401  | 2016.07.12.15. |
| ## | 1630 | 315 | 6  | 2016.07.12 | 45  | 0.78539816  | 2016.07.12.15. |
| ## | 1631 | 295 | 7  | 2016.07.12 | 65  | 1.13446401  | 2016.07.12.15. |
| ## | 1632 | 300 | 8  | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.15. |
| ## | 1633 | 250 | 9  | 2016.07.12 | 110 | 1.91986218  | 2016.07.12.15. |
| ## | 1634 | 300 | 10 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.15. |
| ## | 1635 | 300 | 11 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.15. |
| ## | 1636 | 345 | 12 | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.15. |
| ## | 1637 | 355 | 13 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.15. |
| ## | 1638 | 300 | 14 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.15. |
| ## | 1639 | 335 | 15 | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.15. |
| ## | 1640 | 210 | 16 | 2016.07.12 | 150 | 2.61799388  | 2016.07.12.15. |
| ## | 1641 | 220 | 17 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |
| ## | 1642 | 235 | 18 | 2016.07.12 | 125 | 2.18166156  | 2016.07.12.15. |
| ## | 1643 | 220 | 19 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |
| ## | 1644 | 220 | 20 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |
| ## | 1645 | 215 | 21 | 2016.07.12 | 145 | 2.53072742  | 2016.07.12.15. |
| ## | 1646 | 220 | 22 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1647 | 255 | 23 | 2016.07.12 | 105 | 1.83259571  | 2016.07.12.15. |
| ## 1648 | 200 | 24 | 2016.07.12 | 160 | 2.79252680  | 2016.07.12.15. |
| ## 1649 | 335 | 1  | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.15. |
| ## 1650 | 320 | 2  | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.15. |
| ## 1651 | 320 | 3  | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.15. |
| ## 1652 | 300 | 4  | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.15. |
| ## 1653 | 280 | 5  | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.15. |
| ## 1654 | 290 | 6  | 2016.07.12 | 70  | 1.22173048  | 2016.07.12.15. |
| ## 1655 | 280 | 7  | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.15. |
| ## 1656 | 320 | 8  | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.15. |
| ## 1657 | 240 | 9  | 2016.07.12 | 120 | 2.09439510  | 2016.07.12.15. |
| ## 1658 | 290 | 10 | 2016.07.12 | 70  | 1.22173048  | 2016.07.12.15. |
| ## 1659 | 250 | 11 | 2016.07.12 | 110 | 1.91986218  | 2016.07.12.15. |
| ## 1660 | 0   | 12 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.15. |
| ## 1661 | 270 | 13 | 2016.07.12 | 90  | 1.57079633  | 2016.07.12.15. |
| ## 1662 | 300 | 14 | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.15. |
| ## 1663 | 310 | 15 | 2016.07.12 | 50  | 0.87266463  | 2016.07.12.15. |
| ## 1664 | 315 | 16 | 2016.07.12 | 45  | 0.78539816  | 2016.07.12.15. |
| ## 1665 | 335 | 17 | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.15. |
| ## 1666 | 275 | 18 | 2016.07.12 | 85  | 1.48352986  | 2016.07.12.15. |
| ## 1667 | 240 | 19 | 2016.07.12 | 120 | 2.09439510  | 2016.07.12.15. |
| ## 1668 | 320 | 20 | 2016.07.12 | 40  | 0.69813170  | 2016.07.12.15. |
| ## 1669 | 200 | 21 | 2016.07.12 | 160 | 2.79252680  | 2016.07.12.15. |
| ## 1670 | 185 | 22 | 2016.07.12 | 175 | 3.05432619  | 2016.07.12.15. |
| ## 1671 | 210 | 23 | 2016.07.12 | 150 | 2.61799388  | 2016.07.12.15. |
| ## 1672 | 205 | 24 | 2016.07.12 | 155 | 2.70526034  | 2016.07.12.15. |
| ## 1673 | 200 | 25 | 2016.07.12 | 160 | 2.79252680  | 2016.07.12.15. |
| ## 1674 | 210 | 26 | 2016.07.12 | 150 | 2.61799388  | 2016.07.12.15. |
| ## 1675 | 210 | 27 | 2016.07.12 | 150 | 2.61799388  | 2016.07.12.15. |
| ## 1676 | 220 | 28 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |
| ## 1677 | 220 | 29 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |
| ## 1678 | 210 | 30 | 2016.07.12 | 150 | 2.61799388  | 2016.07.12.15. |
| ## 1679 | 220 | 31 | 2016.07.12 | 140 | 2.44346095  | 2016.07.12.15. |
| ## 1680 | 190 | 32 | 2016.07.12 | 170 | 2.96705973  | 2016.07.12.15. |
| ## 1681 | 210 | 33 | 2016.07.12 | 150 | 2.61799388  | 2016.07.12.15. |
| ## 1682 | 75  | 1  | 2016.07.12 | 285 | -1.30899694 | 2016.07.12.16. |
| ## 1683 | 15  | 2  | 2016.07.12 | 345 | -0.26179939 | 2016.07.12.16. |
| ## 1684 | 285 | 3  | 2016.07.12 | 75  | 1.30899694  | 2016.07.12.16. |
| ## 1685 | 250 | 4  | 2016.07.12 | 110 | 1.91986218  | 2016.07.12.16. |
| ## 1686 | 300 | 5  | 2016.07.12 | 60  | 1.04719755  | 2016.07.12.16. |
| ## 1687 | 310 | 6  | 2016.07.12 | 50  | 0.87266463  | 2016.07.12.16. |
| ## 1688 | 335 | 7  | 2016.07.12 | 25  | 0.43633231  | 2016.07.12.16. |
| ## 1689 | 35  | 8  | 2016.07.12 | 325 | -0.61086524 | 2016.07.12.16. |
| ## 1690 | 280 | 9  | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.16. |
| ## 1691 | 280 | 10 | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.16. |
| ## 1692 | 275 | 11 | 2016.07.12 | 85  | 1.48352986  | 2016.07.12.16. |
| ## 1693 | 80  | 12 | 2016.07.12 | 280 | -1.39626340 | 2016.07.12.16. |
| ## 1694 | 90  | 13 | 2016.07.12 | 270 | -1.57079633 | 2016.07.12.16. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1695 | 20  | 14 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.16. |
| ## 1696 | 90  | 15 | 2016.07.12 | 270 | -1.57079633 | 2016.07.12.16. |
| ## 1697 | 275 | 16 | 2016.07.12 | 85  | 1.48352986  | 2016.07.12.16. |
| ## 1698 | 330 | 17 | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.16. |
| ## 1699 | 265 | 18 | 2016.07.12 | 95  | 1.65806279  | 2016.07.12.16. |
| ## 1700 | 280 | 19 | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.16. |
| ## 1701 | 315 | 20 | 2016.07.12 | 45  | 0.78539816  | 2016.07.12.16. |
| ## 1702 | 330 | 21 | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.16. |
| ## 1703 | 350 | 22 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.16. |
| ## 1704 | 345 | 23 | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.16. |
| ## 1705 | 30  | 24 | 2016.07.12 | 330 | -0.52359878 | 2016.07.12.16. |
| ## 1706 | 355 | 25 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.16. |
| ## 1707 | 5   | 26 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1708 | 5   | 27 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1709 | 0   | 28 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |
| ## 1710 | 355 | 29 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.16. |
| ## 1711 | 0   | 30 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |
| ## 1712 | 25  | 31 | 2016.07.12 | 335 | -0.43633231 | 2016.07.12.16. |
| ## 1713 | 0   | 32 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |
| ## 1714 | 20  | 33 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.16. |
| ## 1715 | 20  | 34 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.16. |
| ## 1716 | 0   | 35 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |
| ## 1717 | 5   | 36 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1718 | 10  | 37 | 2016.07.12 | 350 | -0.17453293 | 2016.07.12.16. |
| ## 1719 | 5   | 38 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1720 | 350 | 39 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.16. |
| ## 1721 | 10  | 40 | 2016.07.12 | 350 | -0.17453293 | 2016.07.12.16. |
| ## 1722 | 345 | 41 | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.16. |
| ## 1723 | 20  | 42 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.16. |
| ## 1724 | 350 | 43 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.16. |
| ## 1725 | 5   | 44 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1726 | 20  | 45 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.16. |
| ## 1727 | 5   | 46 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1728 | 10  | 47 | 2016.07.12 | 350 | -0.17453293 | 2016.07.12.16. |
| ## 1729 | 15  | 48 | 2016.07.12 | 345 | -0.26179939 | 2016.07.12.16. |
| ## 1730 | 20  | 49 | 2016.07.12 | 340 | -0.34906585 | 2016.07.12.16. |
| ## 1731 | 355 | 50 | 2016.07.12 | 5   | 0.08726646  | 2016.07.12.16. |
| ## 1732 | 0   | 51 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |
| ## 1733 | 50  | 52 | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.16. |
| ## 1734 | 5   | 53 | 2016.07.12 | 355 | -0.08726646 | 2016.07.12.16. |
| ## 1735 | 345 | 54 | 2016.07.12 | 15  | 0.26179939  | 2016.07.12.16. |
| ## 1736 | 30  | 55 | 2016.07.12 | 330 | -0.52359878 | 2016.07.12.16. |
| ## 1737 | 0   | 56 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |
| ## 1738 | 260 | 57 | 2016.07.12 | 100 | 1.74532925  | 2016.07.12.16. |
| ## 1739 | 50  | 58 | 2016.07.12 | 310 | -0.87266463 | 2016.07.12.16. |
| ## 1740 | 10  | 59 | 2016.07.12 | 350 | -0.17453293 | 2016.07.12.16. |
| ## 1741 | 350 | 60 | 2016.07.12 | 10  | 0.17453293  | 2016.07.12.16. |
| ## 1742 | 0   | 61 | 2016.07.12 | 0   | 0.00000000  | 2016.07.12.16. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1743 | 330 | 62 | 2016.07.12 | 30  | 0.52359878  | 2016.07.12.16. |
| ## 1744 | 280 | 63 | 2016.07.12 | 80  | 1.39626340  | 2016.07.12.16. |
| ## 1745 | 260 | 1  | 2016.07.18 | 100 | 1.74532925  | 2016.07.18.13. |
| ## 1746 | 310 | 2  | 2016.07.18 | 50  | 0.87266463  | 2016.07.18.13. |
| ## 1747 | 340 | 3  | 2016.07.18 | 20  | 0.34906585  | 2016.07.18.13. |
| ## 1748 | 240 | 4  | 2016.07.18 | 120 | 2.09439510  | 2016.07.18.13. |
| ## 1749 | 330 | 5  | 2016.07.18 | 30  | 0.52359878  | 2016.07.18.13. |
| ## 1750 | 260 | 6  | 2016.07.18 | 100 | 1.74532925  | 2016.07.18.13. |
| ## 1751 | 320 | 7  | 2016.07.18 | 40  | 0.69813170  | 2016.07.18.13. |
| ## 1752 | 230 | 8  | 2016.07.18 | 130 | 2.26892803  | 2016.07.18.13. |
| ## 1753 | 310 | 9  | 2016.07.18 | 50  | 0.87266463  | 2016.07.18.13. |
| ## 1754 | 305 | 10 | 2016.07.18 | 55  | 0.95993109  | 2016.07.18.13. |
| ## 1755 | 320 | 11 | 2016.07.18 | 40  | 0.69813170  | 2016.07.18.13. |
| ## 1756 | 240 | 12 | 2016.07.18 | 120 | 2.09439510  | 2016.07.18.13. |
| ## 1757 | 270 | 13 | 2016.07.18 | 90  | 1.57079633  | 2016.07.18.13. |
| ## 1758 | 230 | 14 | 2016.07.18 | 130 | 2.26892803  | 2016.07.18.13. |
| ## 1759 | 345 | 15 | 2016.07.18 | 15  | 0.26179939  | 2016.07.18.13. |
| ## 1760 | 340 | 16 | 2016.07.18 | 20  | 0.34906585  | 2016.07.18.13. |
| ## 1761 | 345 | 17 | 2016.07.18 | 15  | 0.26179939  | 2016.07.18.13. |
| ## 1762 | 355 | 18 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.13. |
| ## 1763 | 355 | 19 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.13. |
| ## 1764 | 340 | 20 | 2016.07.18 | 20  | 0.34906585  | 2016.07.18.13. |
| ## 1765 | 0   | 21 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.13. |
| ## 1766 | 350 | 22 | 2016.07.18 | 10  | 0.17453293  | 2016.07.18.13. |
| ## 1767 | 355 | 23 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.13. |
| ## 1768 | 345 | 24 | 2016.07.18 | 15  | 0.26179939  | 2016.07.18.13. |
| ## 1769 | 310 | 25 | 2016.07.18 | 50  | 0.87266463  | 2016.07.18.13. |
| ## 1770 | 350 | 26 | 2016.07.18 | 10  | 0.17453293  | 2016.07.18.13. |
| ## 1771 | 0   | 27 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.13. |
| ## 1772 | 330 | 28 | 2016.07.18 | 30  | 0.52359878  | 2016.07.18.13. |
| ## 1773 | 345 | 29 | 2016.07.18 | 15  | 0.26179939  | 2016.07.18.13. |
| ## 1774 | 325 | 30 | 2016.07.18 | 35  | 0.61086524  | 2016.07.18.13. |
| ## 1775 | 260 | 1  | 2016.07.18 | 100 | 1.74532925  | 2016.07.18.14. |
| ## 1776 | 150 | 2  | 2016.07.18 | 210 | -2.61799388 | 2016.07.18.14. |
| ## 1777 | 270 | 3  | 2016.07.18 | 90  | 1.57079633  | 2016.07.18.14. |
| ## 1778 | 270 | 4  | 2016.07.18 | 90  | 1.57079633  | 2016.07.18.14. |
| ## 1779 | 220 | 5  | 2016.07.18 | 140 | 2.44346095  | 2016.07.18.14. |
| ## 1780 | 250 | 6  | 2016.07.18 | 110 | 1.91986218  | 2016.07.18.14. |
| ## 1781 | 215 | 7  | 2016.07.18 | 145 | 2.53072742  | 2016.07.18.14. |
| ## 1782 | 250 | 8  | 2016.07.18 | 110 | 1.91986218  | 2016.07.18.14. |
| ## 1783 | 165 | 9  | 2016.07.18 | 195 | -2.87979327 | 2016.07.18.14. |
| ## 1784 | 310 | 10 | 2016.07.18 | 50  | 0.87266463  | 2016.07.18.14. |
| ## 1785 | 255 | 11 | 2016.07.18 | 105 | 1.83259571  | 2016.07.18.14. |
| ## 1786 | 205 | 12 | 2016.07.18 | 155 | 2.70526034  | 2016.07.18.14. |
| ## 1787 | 220 | 13 | 2016.07.18 | 140 | 2.44346095  | 2016.07.18.14. |
| ## 1788 | 250 | 14 | 2016.07.18 | 110 | 1.91986218  | 2016.07.18.14. |
| ## 1789 | 150 | 15 | 2016.07.18 | 210 | -2.61799388 | 2016.07.18.14. |
| ## 1790 | 160 | 16 | 2016.07.18 | 200 | -2.79252680 | 2016.07.18.14. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1791 | 205 | 17 | 2016.07.18 | 155 | 2.70526034  | 2016.07.18.14. |
| ## 1792 | 240 | 18 | 2016.07.18 | 120 | 2.09439510  | 2016.07.18.14. |
| ## 1793 | 185 | 19 | 2016.07.18 | 175 | 3.05432619  | 2016.07.18.14. |
| ## 1794 | 220 | 20 | 2016.07.18 | 140 | 2.44346095  | 2016.07.18.14. |
| ## 1795 | 230 | 21 | 2016.07.18 | 130 | 2.26892803  | 2016.07.18.14. |
| ## 1796 | 160 | 22 | 2016.07.18 | 200 | -2.79252680 | 2016.07.18.14. |
| ## 1797 | 210 | 23 | 2016.07.18 | 150 | 2.61799388  | 2016.07.18.14. |
| ## 1798 | 245 | 24 | 2016.07.18 | 115 | 2.00712864  | 2016.07.18.14. |
| ## 1799 | 175 | 25 | 2016.07.18 | 185 | -3.05432619 | 2016.07.18.14. |
| ## 1800 | 210 | 26 | 2016.07.18 | 150 | 2.61799388  | 2016.07.18.14. |
| ## 1801 | 215 | 27 | 2016.07.18 | 145 | 2.53072742  | 2016.07.18.14. |
| ## 1802 | 220 | 28 | 2016.07.18 | 140 | 2.44346095  | 2016.07.18.14. |
| ## 1803 | 190 | 29 | 2016.07.18 | 170 | 2.96705973  | 2016.07.18.14. |
| ## 1804 | 230 | 30 | 2016.07.18 | 130 | 2.26892803  | 2016.07.18.14. |
| ## 1805 | 50  | 1  | 2016.07.18 | 310 | -0.87266463 | 2016.07.18.15. |
| ## 1806 | 300 | 2  | 2016.07.18 | 60  | 1.04719755  | 2016.07.18.15. |
| ## 1807 | 340 | 3  | 2016.07.18 | 20  | 0.34906585  | 2016.07.18.15. |
| ## 1808 | 250 | 4  | 2016.07.18 | 110 | 1.91986218  | 2016.07.18.15. |
| ## 1809 | 230 | 5  | 2016.07.18 | 130 | 2.26892803  | 2016.07.18.15. |
| ## 1810 | 260 | 6  | 2016.07.18 | 100 | 1.74532925  | 2016.07.18.15. |
| ## 1811 | 90  | 7  | 2016.07.18 | 270 | -1.57079633 | 2016.07.18.15. |
| ## 1812 | 30  | 8  | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## 1813 | 220 | 9  | 2016.07.18 | 140 | 2.44346095  | 2016.07.18.15. |
| ## 1814 | 240 | 10 | 2016.07.18 | 120 | 2.09439510  | 2016.07.18.15. |
| ## 1815 | 260 | 11 | 2016.07.18 | 100 | 1.74532925  | 2016.07.18.15. |
| ## 1816 | 260 | 12 | 2016.07.18 | 100 | 1.74532925  | 2016.07.18.15. |
| ## 1817 | 80  | 13 | 2016.07.18 | 280 | -1.39626340 | 2016.07.18.15. |
| ## 1818 | 145 | 14 | 2016.07.18 | 215 | -2.53072742 | 2016.07.18.15. |
| ## 1819 | 90  | 15 | 2016.07.18 | 270 | -1.57079633 | 2016.07.18.15. |
| ## 1820 | 265 | 16 | 2016.07.18 | 95  | 1.65806279  | 2016.07.18.15. |
| ## 1821 | 355 | 17 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## 1822 | 0   | 18 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.15. |
| ## 1823 | 10  | 19 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## 1824 | 20  | 20 | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## 1825 | 10  | 21 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## 1826 | 345 | 22 | 2016.07.18 | 15  | 0.26179939  | 2016.07.18.15. |
| ## 1827 | 350 | 23 | 2016.07.18 | 10  | 0.17453293  | 2016.07.18.15. |
| ## 1828 | 335 | 24 | 2016.07.18 | 25  | 0.43633231  | 2016.07.18.15. |
| ## 1829 | 355 | 25 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## 1830 | 0   | 26 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.15. |
| ## 1831 | 5   | 27 | 2016.07.18 | 355 | -0.08726646 | 2016.07.18.15. |
| ## 1832 | 10  | 28 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## 1833 | 10  | 29 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## 1834 | 340 | 30 | 2016.07.18 | 20  | 0.34906585  | 2016.07.18.15. |
| ## 1835 | 25  | 31 | 2016.07.18 | 335 | -0.43633231 | 2016.07.18.15. |
| ## 1836 | 5   | 32 | 2016.07.18 | 355 | -0.08726646 | 2016.07.18.15. |
| ## 1837 | 0   | 33 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.15. |
| ## 1838 | 5   | 34 | 2016.07.18 | 355 | -0.08726646 | 2016.07.18.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1839 | 10  | 35 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## | 1840 | 10  | 36 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## | 1841 | 340 | 37 | 2016.07.18 | 20  | 0.34906585  | 2016.07.18.15. |
| ## | 1842 | 25  | 38 | 2016.07.18 | 335 | -0.43633231 | 2016.07.18.15. |
| ## | 1843 | 5   | 39 | 2016.07.18 | 355 | -0.08726646 | 2016.07.18.15. |
| ## | 1844 | 20  | 1  | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## | 1845 | 190 | 2  | 2016.07.18 | 170 | 2.96705973  | 2016.07.18.15. |
| ## | 1846 | 30  | 3  | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## | 1847 | 335 | 4  | 2016.07.18 | 25  | 0.43633231  | 2016.07.18.15. |
| ## | 1848 | 335 | 5  | 2016.07.18 | 25  | 0.43633231  | 2016.07.18.15. |
| ## | 1849 | 110 | 6  | 2016.07.18 | 250 | -1.91986218 | 2016.07.18.15. |
| ## | 1850 | 25  | 7  | 2016.07.18 | 335 | -0.43633231 | 2016.07.18.15. |
| ## | 1851 | 320 | 8  | 2016.07.18 | 40  | 0.69813170  | 2016.07.18.15. |
| ## | 1852 | 115 | 9  | 2016.07.18 | 245 | -2.00712864 | 2016.07.18.15. |
| ## | 1853 | 285 | 10 | 2016.07.18 | 75  | 1.30899694  | 2016.07.18.15. |
| ## | 1854 | 320 | 11 | 2016.07.18 | 40  | 0.69813170  | 2016.07.18.15. |
| ## | 1855 | 125 | 12 | 2016.07.18 | 235 | -2.18166156 | 2016.07.18.15. |
| ## | 1856 | 290 | 13 | 2016.07.18 | 70  | 1.22173048  | 2016.07.18.15. |
| ## | 1857 | 130 | 14 | 2016.07.18 | 230 | -2.26892803 | 2016.07.18.15. |
| ## | 1858 | 70  | 15 | 2016.07.18 | 290 | -1.22173048 | 2016.07.18.15. |
| ## | 1859 | 325 | 16 | 2016.07.18 | 35  | 0.61086524  | 2016.07.18.15. |
| ## | 1860 | 10  | 17 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## | 1861 | 315 | 18 | 2016.07.18 | 45  | 0.78539816  | 2016.07.18.15. |
| ## | 1862 | 30  | 19 | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## | 1863 | 355 | 20 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## | 1864 | 40  | 21 | 2016.07.18 | 320 | -0.69813170 | 2016.07.18.15. |
| ## | 1865 | 355 | 22 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## | 1866 | 355 | 23 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## | 1867 | 10  | 24 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## | 1868 | 20  | 25 | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## | 1869 | 30  | 26 | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## | 1870 | 35  | 27 | 2016.07.18 | 325 | -0.61086524 | 2016.07.18.15. |
| ## | 1871 | 10  | 28 | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## | 1872 | 20  | 29 | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## | 1873 | 50  | 30 | 2016.07.18 | 310 | -0.87266463 | 2016.07.18.15. |
| ## | 1874 | 30  | 31 | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## | 1875 | 0   | 32 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.15. |
| ## | 1876 | 15  | 33 | 2016.07.18 | 345 | -0.26179939 | 2016.07.18.15. |
| ## | 1877 | 5   | 34 | 2016.07.18 | 355 | -0.08726646 | 2016.07.18.15. |
| ## | 1878 | 25  | 35 | 2016.07.18 | 335 | -0.43633231 | 2016.07.18.15. |
| ## | 1879 | 0   | 36 | 2016.07.18 | 0   | 0.00000000  | 2016.07.18.15. |
| ## | 1880 | 30  | 37 | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## | 1881 | 355 | 38 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## | 1882 | 50  | 39 | 2016.07.18 | 310 | -0.87266463 | 2016.07.18.15. |
| ## | 1883 | 15  | 40 | 2016.07.18 | 345 | -0.26179939 | 2016.07.18.15. |
| ## | 1884 | 20  | 41 | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## | 1885 | 15  | 42 | 2016.07.18 | 345 | -0.26179939 | 2016.07.18.15. |
| ## | 1886 | 70  | 1  | 2016.07.18 | 290 | -1.22173048 | 2016.07.18.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 1887 | 10  | 2  | 2016.07.18 | 350 | -0.17453293 | 2016.07.18.15. |
| ## 1888 | 110 | 3  | 2016.07.18 | 250 | -1.91986218 | 2016.07.18.15. |
| ## 1889 | 80  | 4  | 2016.07.18 | 280 | -1.39626340 | 2016.07.18.15. |
| ## 1890 | 270 | 5  | 2016.07.18 | 90  | 1.57079633  | 2016.07.18.15. |
| ## 1891 | 35  | 6  | 2016.07.18 | 325 | -0.61086524 | 2016.07.18.15. |
| ## 1892 | 310 | 7  | 2016.07.18 | 50  | 0.87266463  | 2016.07.18.15. |
| ## 1893 | 315 | 8  | 2016.07.18 | 45  | 0.78539816  | 2016.07.18.15. |
| ## 1894 | 345 | 9  | 2016.07.18 | 15  | 0.26179939  | 2016.07.18.15. |
| ## 1895 | 135 | 10 | 2016.07.18 | 225 | -2.35619449 | 2016.07.18.15. |
| ## 1896 | 120 | 11 | 2016.07.18 | 240 | -2.09439510 | 2016.07.18.15. |
| ## 1897 | 155 | 12 | 2016.07.18 | 205 | -2.70526034 | 2016.07.18.15. |
| ## 1898 | 105 | 13 | 2016.07.18 | 255 | -1.83259571 | 2016.07.18.15. |
| ## 1899 | 165 | 14 | 2016.07.18 | 195 | -2.87979327 | 2016.07.18.15. |
| ## 1900 | 145 | 15 | 2016.07.18 | 215 | -2.53072742 | 2016.07.18.15. |
| ## 1901 | 90  | 16 | 2016.07.18 | 270 | -1.57079633 | 2016.07.18.15. |
| ## 1902 | 110 | 17 | 2016.07.18 | 250 | -1.91986218 | 2016.07.18.15. |
| ## 1903 | 190 | 18 | 2016.07.18 | 170 | 2.96705973  | 2016.07.18.15. |
| ## 1904 | 220 | 19 | 2016.07.18 | 140 | 2.44346095  | 2016.07.18.15. |
| ## 1905 | 90  | 20 | 2016.07.18 | 270 | -1.57079633 | 2016.07.18.15. |
| ## 1906 | 160 | 21 | 2016.07.18 | 200 | -2.79252680 | 2016.07.18.15. |
| ## 1907 | 60  | 22 | 2016.07.18 | 300 | -1.04719755 | 2016.07.18.15. |
| ## 1908 | 45  | 23 | 2016.07.18 | 315 | -0.78539816 | 2016.07.18.15. |
| ## 1909 | 45  | 24 | 2016.07.18 | 315 | -0.78539816 | 2016.07.18.15. |
| ## 1910 | 355 | 25 | 2016.07.18 | 5   | 0.08726646  | 2016.07.18.15. |
| ## 1911 | 55  | 26 | 2016.07.18 | 305 | -0.95993109 | 2016.07.18.15. |
| ## 1912 | 15  | 27 | 2016.07.18 | 345 | -0.26179939 | 2016.07.18.15. |
| ## 1913 | 45  | 28 | 2016.07.18 | 315 | -0.78539816 | 2016.07.18.15. |
| ## 1914 | 55  | 29 | 2016.07.18 | 305 | -0.95993109 | 2016.07.18.15. |
| ## 1915 | 60  | 30 | 2016.07.18 | 300 | -1.04719755 | 2016.07.18.15. |
| ## 1916 | 20  | 31 | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## 1917 | 30  | 32 | 2016.07.18 | 330 | -0.52359878 | 2016.07.18.15. |
| ## 1918 | 20  | 33 | 2016.07.18 | 340 | -0.34906585 | 2016.07.18.15. |
| ## 1919 | 215 | 1  | 2016.07.19 | 145 | 2.53072742  | 2016.07.19.12. |
| ## 1920 | 355 | 2  | 2016.07.19 | 5   | 0.08726646  | 2016.07.19.12. |
| ## 1921 | 210 | 3  | 2016.07.19 | 150 | 2.61799388  | 2016.07.19.12. |
| ## 1922 | 220 | 4  | 2016.07.19 | 140 | 2.44346095  | 2016.07.19.12. |
| ## 1923 | 235 | 5  | 2016.07.19 | 125 | 2.18166156  | 2016.07.19.12. |
| ## 1924 | 270 | 6  | 2016.07.19 | 90  | 1.57079633  | 2016.07.19.12. |
| ## 1925 | 320 | 7  | 2016.07.19 | 40  | 0.69813170  | 2016.07.19.12. |
| ## 1926 | 210 | 8  | 2016.07.19 | 150 | 2.61799388  | 2016.07.19.12. |
| ## 1927 | 250 | 9  | 2016.07.19 | 110 | 1.91986218  | 2016.07.19.12. |
| ## 1928 | 320 | 10 | 2016.07.19 | 40  | 0.69813170  | 2016.07.19.12. |
| ## 1929 | 20  | 11 | 2016.07.19 | 340 | -0.34906585 | 2016.07.19.12. |
| ## 1930 | 30  | 12 | 2016.07.19 | 330 | -0.52359878 | 2016.07.19.12. |
| ## 1931 | 280 | 13 | 2016.07.19 | 80  | 1.39626340  | 2016.07.19.12. |
| ## 1932 | 245 | 14 | 2016.07.19 | 115 | 2.00712864  | 2016.07.19.12. |
| ## 1933 | 285 | 15 | 2016.07.19 | 75  | 1.30899694  | 2016.07.19.12. |
| ## 1934 | 270 | 16 | 2016.07.19 | 90  | 1.57079633  | 2016.07.19.12. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1935 | 275 | 17 | 2016.07.19 | 85  | 1.48352986  | 2016.07.19.12. |
| ## | 1936 | 310 | 18 | 2016.07.19 | 50  | 0.87266463  | 2016.07.19.12. |
| ## | 1937 | 305 | 19 | 2016.07.19 | 55  | 0.95993109  | 2016.07.19.12. |
| ## | 1938 | 320 | 20 | 2016.07.19 | 40  | 0.69813170  | 2016.07.19.12. |
| ## | 1939 | 270 | 21 | 2016.07.19 | 90  | 1.57079633  | 2016.07.19.12. |
| ## | 1940 | 310 | 22 | 2016.07.19 | 50  | 0.87266463  | 2016.07.19.12. |
| ## | 1941 | 325 | 23 | 2016.07.19 | 35  | 0.61086524  | 2016.07.19.12. |
| ## | 1942 | 340 | 24 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.12. |
| ## | 1943 | 0   | 25 | 2016.07.19 | 0   | 0.00000000  | 2016.07.19.12. |
| ## | 1944 | 345 | 26 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.12. |
| ## | 1945 | 340 | 27 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.12. |
| ## | 1946 | 300 | 28 | 2016.07.19 | 60  | 1.04719755  | 2016.07.19.12. |
| ## | 1947 | 350 | 29 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.12. |
| ## | 1948 | 340 | 30 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.12. |
| ## | 1949 | 345 | 31 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.12. |
| ## | 1950 | 350 | 32 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.12. |
| ## | 1951 | 350 | 33 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.12. |
| ## | 1952 | 345 | 34 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.12. |
| ## | 1953 | 350 | 35 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.12. |
| ## | 1954 | 315 | 36 | 2016.07.19 | 45  | 0.78539816  | 2016.07.19.12. |
| ## | 1955 | 340 | 37 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.12. |
| ## | 1956 | 345 | 38 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.12. |
| ## | 1957 | 340 | 39 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.12. |
| ## | 1958 | 310 | 40 | 2016.07.19 | 50  | 0.87266463  | 2016.07.19.12. |
| ## | 1959 | 50  | 1  | 2016.07.19 | 310 | -0.87266463 | 2016.07.19.12. |
| ## | 1960 | 80  | 2  | 2016.07.19 | 280 | -1.39626340 | 2016.07.19.12. |
| ## | 1961 | 100 | 3  | 2016.07.19 | 260 | -1.74532925 | 2016.07.19.12. |
| ## | 1962 | 60  | 4  | 2016.07.19 | 300 | -1.04719755 | 2016.07.19.12. |
| ## | 1963 | 200 | 5  | 2016.07.19 | 160 | 2.79252680  | 2016.07.19.12. |
| ## | 1964 | 330 | 6  | 2016.07.19 | 30  | 0.52359878  | 2016.07.19.12. |
| ## | 1965 | 80  | 7  | 2016.07.19 | 280 | -1.39626340 | 2016.07.19.12. |
| ## | 1966 | 90  | 8  | 2016.07.19 | 270 | -1.57079633 | 2016.07.19.12. |
| ## | 1967 | 110 | 9  | 2016.07.19 | 250 | -1.91986218 | 2016.07.19.12. |
| ## | 1968 | 30  | 10 | 2016.07.19 | 330 | -0.52359878 | 2016.07.19.12. |
| ## | 1969 | 90  | 11 | 2016.07.19 | 270 | -1.57079633 | 2016.07.19.12. |
| ## | 1970 | 35  | 12 | 2016.07.19 | 325 | -0.61086524 | 2016.07.19.12. |
| ## | 1971 | 95  | 13 | 2016.07.19 | 265 | -1.65806279 | 2016.07.19.12. |
| ## | 1972 | 70  | 14 | 2016.07.19 | 290 | -1.22173048 | 2016.07.19.12. |
| ## | 1973 | 80  | 15 | 2016.07.19 | 280 | -1.39626340 | 2016.07.19.12. |
| ## | 1974 | 10  | 16 | 2016.07.19 | 350 | -0.17453293 | 2016.07.19.12. |
| ## | 1975 | 70  | 17 | 2016.07.19 | 290 | -1.22173048 | 2016.07.19.12. |
| ## | 1976 | 60  | 18 | 2016.07.19 | 300 | -1.04719755 | 2016.07.19.12. |
| ## | 1977 | 0   | 19 | 2016.07.19 | 0   | 0.00000000  | 2016.07.19.12. |
| ## | 1978 | 35  | 20 | 2016.07.19 | 325 | -0.61086524 | 2016.07.19.12. |
| ## | 1979 | 20  | 21 | 2016.07.19 | 340 | -0.34906585 | 2016.07.19.12. |
| ## | 1980 | 20  | 22 | 2016.07.19 | 340 | -0.34906585 | 2016.07.19.12. |
| ## | 1981 | 25  | 23 | 2016.07.19 | 335 | -0.43633231 | 2016.07.19.12. |
| ## | 1982 | 25  | 24 | 2016.07.19 | 335 | -0.43633231 | 2016.07.19.12. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 1983 | 70  | 25 | 2016.07.19 | 290 | -1.22173048 | 2016.07.19.12. |
| ## | 1984 | 20  | 26 | 2016.07.19 | 340 | -0.34906585 | 2016.07.19.12. |
| ## | 1985 | 50  | 27 | 2016.07.19 | 310 | -0.87266463 | 2016.07.19.12. |
| ## | 1986 | 30  | 28 | 2016.07.19 | 330 | -0.52359878 | 2016.07.19.12. |
| ## | 1987 | 60  | 29 | 2016.07.19 | 300 | -1.04719755 | 2016.07.19.12. |
| ## | 1988 | 30  | 30 | 2016.07.19 | 330 | -0.52359878 | 2016.07.19.12. |
| ## | 1989 | 40  | 31 | 2016.07.19 | 320 | -0.69813170 | 2016.07.19.12. |
| ## | 1990 | 60  | 32 | 2016.07.19 | 300 | -1.04719755 | 2016.07.19.12. |
| ## | 1991 | 20  | 33 | 2016.07.19 | 340 | -0.34906585 | 2016.07.19.12. |
| ## | 1992 | 50  | 34 | 2016.07.19 | 310 | -0.87266463 | 2016.07.19.12. |
| ## | 1993 | 40  | 35 | 2016.07.19 | 320 | -0.69813170 | 2016.07.19.12. |
| ## | 1994 | 40  | 36 | 2016.07.19 | 320 | -0.69813170 | 2016.07.19.12. |
| ## | 1995 | 160 | 1  | 2016.07.19 | 200 | -2.79252680 | 2016.07.19.13. |
| ## | 1996 | 150 | 2  | 2016.07.19 | 210 | -2.61799388 | 2016.07.19.13. |
| ## | 1997 | 200 | 3  | 2016.07.19 | 160 | 2.79252680  | 2016.07.19.13. |
| ## | 1998 | 25  | 4  | 2016.07.19 | 335 | -0.43633231 | 2016.07.19.13. |
| ## | 1999 | 100 | 5  | 2016.07.19 | 260 | -1.74532925 | 2016.07.19.13. |
| ## | 2000 | 200 | 6  | 2016.07.19 | 160 | 2.79252680  | 2016.07.19.13. |
| ## | 2001 | 340 | 7  | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.13. |
| ## | 2002 | 15  | 8  | 2016.07.19 | 345 | -0.26179939 | 2016.07.19.13. |
| ## | 2003 | 300 | 9  | 2016.07.19 | 60  | 1.04719755  | 2016.07.19.13. |
| ## | 2004 | 330 | 10 | 2016.07.19 | 30  | 0.52359878  | 2016.07.19.13. |
| ## | 2005 | 345 | 11 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.13. |
| ## | 2006 | 250 | 12 | 2016.07.19 | 110 | 1.91986218  | 2016.07.19.13. |
| ## | 2007 | 340 | 13 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.13. |
| ## | 2008 | 0   | 14 | 2016.07.19 | 0   | 0.00000000  | 2016.07.19.13. |
| ## | 2009 | 300 | 15 | 2016.07.19 | 60  | 1.04719755  | 2016.07.19.13. |
| ## | 2010 | 10  | 16 | 2016.07.19 | 350 | -0.17453293 | 2016.07.19.13. |
| ## | 2011 | 310 | 17 | 2016.07.19 | 50  | 0.87266463  | 2016.07.19.13. |
| ## | 2012 | 310 | 18 | 2016.07.19 | 50  | 0.87266463  | 2016.07.19.13. |
| ## | 2013 | 345 | 19 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.13. |
| ## | 2014 | 355 | 20 | 2016.07.19 | 5   | 0.08726646  | 2016.07.19.13. |
| ## | 2015 | 305 | 21 | 2016.07.19 | 55  | 0.95993109  | 2016.07.19.13. |
| ## | 2016 | 325 | 22 | 2016.07.19 | 35  | 0.61086524  | 2016.07.19.13. |
| ## | 2017 | 325 | 23 | 2016.07.19 | 35  | 0.61086524  | 2016.07.19.13. |
| ## | 2018 | 350 | 24 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.13. |
| ## | 2019 | 350 | 25 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.13. |
| ## | 2020 | 0   | 26 | 2016.07.19 | 0   | 0.00000000  | 2016.07.19.13. |
| ## | 2021 | 355 | 27 | 2016.07.19 | 5   | 0.08726646  | 2016.07.19.13. |
| ## | 2022 | 350 | 28 | 2016.07.19 | 10  | 0.17453293  | 2016.07.19.13. |
| ## | 2023 | 340 | 29 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.13. |
| ## | 2024 | 310 | 30 | 2016.07.19 | 50  | 0.87266463  | 2016.07.19.13. |
| ## | 2025 | 345 | 31 | 2016.07.19 | 15  | 0.26179939  | 2016.07.19.13. |
| ## | 2026 | 340 | 32 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.13. |
| ## | 2027 | 355 | 33 | 2016.07.19 | 5   | 0.08726646  | 2016.07.19.13. |
| ## | 2028 | 340 | 34 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.13. |
| ## | 2029 | 340 | 35 | 2016.07.19 | 20  | 0.34906585  | 2016.07.19.13. |
| ## | 2030 | 330 | 36 | 2016.07.19 | 30  | 0.52359878  | 2016.07.19.13. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2031 | 330 | 37 | 2016.07.19 | 30  | 0.52359878  | 2016.07.19.13. |
| ## | 2032 | 90  | 1  | 2016.08.17 | 270 | -1.57079633 | 2016.08.17.13. |
| ## | 2033 | 80  | 2  | 2016.08.17 | 280 | -1.39626340 | 2016.08.17.13. |
| ## | 2034 | 40  | 3  | 2016.08.17 | 320 | -0.69813170 | 2016.08.17.13. |
| ## | 2035 | 70  | 4  | 2016.08.17 | 290 | -1.22173048 | 2016.08.17.13. |
| ## | 2036 | 40  | 5  | 2016.08.17 | 320 | -0.69813170 | 2016.08.17.13. |
| ## | 2037 | 50  | 6  | 2016.08.17 | 310 | -0.87266463 | 2016.08.17.13. |
| ## | 2038 | 350 | 7  | 2016.08.17 | 10  | 0.17453293  | 2016.08.17.13. |
| ## | 2039 | 355 | 8  | 2016.08.17 | 5   | 0.08726646  | 2016.08.17.13. |
| ## | 2040 | 70  | 9  | 2016.08.17 | 290 | -1.22173048 | 2016.08.17.13. |
| ## | 2041 | 90  | 10 | 2016.08.17 | 270 | -1.57079633 | 2016.08.17.13. |
| ## | 2042 | 70  | 11 | 2016.08.17 | 290 | -1.22173048 | 2016.08.17.13. |
| ## | 2043 | 75  | 12 | 2016.08.17 | 285 | -1.30899694 | 2016.08.17.13. |
| ## | 2044 | 345 | 13 | 2016.08.17 | 15  | 0.26179939  | 2016.08.17.13. |
| ## | 2045 | 290 | 14 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.13. |
| ## | 2046 | 290 | 15 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.13. |
| ## | 2047 | 295 | 16 | 2016.08.17 | 65  | 1.13446401  | 2016.08.17.13. |
| ## | 2048 | 310 | 17 | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.13. |
| ## | 2049 | 290 | 18 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.13. |
| ## | 2050 | 290 | 19 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.13. |
| ## | 2051 | 310 | 20 | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.13. |
| ## | 2052 | 290 | 21 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.13. |
| ## | 2053 | 280 | 22 | 2016.08.17 | 80  | 1.39626340  | 2016.08.17.13. |
| ## | 2054 | 345 | 23 | 2016.08.17 | 15  | 0.26179939  | 2016.08.17.13. |
| ## | 2055 | 330 | 24 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.13. |
| ## | 2056 | 295 | 25 | 2016.08.17 | 65  | 1.13446401  | 2016.08.17.13. |
| ## | 2057 | 185 | 1  | 2016.08.17 | 175 | 3.05432619  | 2016.08.17.14. |
| ## | 2058 | 65  | 2  | 2016.08.17 | 295 | -1.13446401 | 2016.08.17.14. |
| ## | 2059 | 120 | 3  | 2016.08.17 | 240 | -2.09439510 | 2016.08.17.14. |
| ## | 2060 | 185 | 4  | 2016.08.17 | 175 | 3.05432619  | 2016.08.17.14. |
| ## | 2061 | 200 | 5  | 2016.08.17 | 160 | 2.79252680  | 2016.08.17.14. |
| ## | 2062 | 90  | 6  | 2016.08.17 | 270 | -1.57079633 | 2016.08.17.14. |
| ## | 2063 | 175 | 7  | 2016.08.17 | 185 | -3.05432619 | 2016.08.17.14. |
| ## | 2064 | 155 | 8  | 2016.08.17 | 205 | -2.70526034 | 2016.08.17.14. |
| ## | 2065 | 105 | 9  | 2016.08.17 | 255 | -1.83259571 | 2016.08.17.14. |
| ## | 2066 | 115 | 10 | 2016.08.17 | 245 | -2.00712864 | 2016.08.17.14. |
| ## | 2067 | 50  | 11 | 2016.08.17 | 310 | -0.87266463 | 2016.08.17.14. |
| ## | 2068 | 90  | 12 | 2016.08.17 | 270 | -1.57079633 | 2016.08.17.14. |
| ## | 2069 | 180 | 13 | 2016.08.17 | 180 | 3.14159265  | 2016.08.17.14. |
| ## | 2070 | 140 | 14 | 2016.08.17 | 220 | -2.44346095 | 2016.08.17.14. |
| ## | 2071 | 110 | 15 | 2016.08.17 | 250 | -1.91986218 | 2016.08.17.14. |
| ## | 2072 | 290 | 25 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.14. |
| ## | 2073 | 300 | 26 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.14. |
| ## | 2074 | 330 | 27 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.14. |
| ## | 2075 | 310 | 28 | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.14. |
| ## | 2076 | 330 | 29 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.14. |
| ## | 2077 | 320 | 30 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.14. |
| ## | 2078 | 300 | 31 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2079 | 320 | 32 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.14. |
| ## | 2080 | 330 | 33 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.14. |
| ## | 2081 | 320 | 34 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.14. |
| ## | 2082 | 300 | 35 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.14. |
| ## | 2083 | 310 | 36 | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.14. |
| ## | 2084 | 345 | 37 | 2016.08.17 | 15  | 0.26179939  | 2016.08.17.14. |
| ## | 2085 | 325 | 38 | 2016.08.17 | 35  | 0.61086524  | 2016.08.17.14. |
| ## | 2086 | 310 | 39 | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.14. |
| ## | 2087 | 340 | 40 | 2016.08.17 | 20  | 0.34906585  | 2016.08.17.14. |
| ## | 2088 | 190 | 1  | 2016.08.17 | 170 | 2.96705973  | 2016.08.17.16. |
| ## | 2089 | 280 | 2  | 2016.08.17 | 80  | 1.39626340  | 2016.08.17.16. |
| ## | 2090 | 90  | 3  | 2016.08.17 | 270 | -1.57079633 | 2016.08.17.16. |
| ## | 2091 | 135 | 4  | 2016.08.17 | 225 | -2.35619449 | 2016.08.17.16. |
| ## | 2092 | 170 | 5  | 2016.08.17 | 190 | -2.96705973 | 2016.08.17.16. |
| ## | 2093 | 150 | 6  | 2016.08.17 | 210 | -2.61799388 | 2016.08.17.16. |
| ## | 2094 | 200 | 7  | 2016.08.17 | 160 | 2.79252680  | 2016.08.17.16. |
| ## | 2095 | 120 | 8  | 2016.08.17 | 240 | -2.09439510 | 2016.08.17.16. |
| ## | 2096 | 110 | 9  | 2016.08.17 | 250 | -1.91986218 | 2016.08.17.16. |
| ## | 2097 | 100 | 10 | 2016.08.17 | 260 | -1.74532925 | 2016.08.17.16. |
| ## | 2098 | 330 | 11 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.16. |
| ## | 2099 | 100 | 12 | 2016.08.17 | 260 | -1.74532925 | 2016.08.17.16. |
| ## | 2100 | 50  | 13 | 2016.08.17 | 310 | -0.87266463 | 2016.08.17.16. |
| ## | 2101 | 120 | 14 | 2016.08.17 | 240 | -2.09439510 | 2016.08.17.16. |
| ## | 2102 | 10  | 15 | 2016.08.17 | 350 | -0.17453293 | 2016.08.17.16. |
| ## | 2103 | 160 | 16 | 2016.08.17 | 200 | -2.79252680 | 2016.08.17.16. |
| ## | 2104 | 150 | 17 | 2016.08.17 | 210 | -2.61799388 | 2016.08.17.16. |
| ## | 2105 | 110 | 18 | 2016.08.17 | 250 | -1.91986218 | 2016.08.17.16. |
| ## | 2106 | 190 | 19 | 2016.08.17 | 170 | 2.96705973  | 2016.08.17.16. |
| ## | 2107 | 170 | 20 | 2016.08.17 | 190 | -2.96705973 | 2016.08.17.16. |
| ## | 2108 | 340 | 21 | 2016.08.17 | 20  | 0.34906585  | 2016.08.17.16. |
| ## | 2109 | 320 | 22 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.16. |
| ## | 2110 | 320 | 23 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.16. |
| ## | 2111 | 350 | 24 | 2016.08.17 | 10  | 0.17453293  | 2016.08.17.16. |
| ## | 2112 | 340 | 25 | 2016.08.17 | 20  | 0.34906585  | 2016.08.17.16. |
| ## | 2113 | 330 | 26 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.16. |
| ## | 2114 | 340 | 27 | 2016.08.17 | 20  | 0.34906585  | 2016.08.17.16. |
| ## | 2115 | 320 | 28 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.16. |
| ## | 2116 | 310 | 29 | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.16. |
| ## | 2117 | 300 | 30 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.16. |
| ## | 2118 | 335 | 31 | 2016.08.17 | 25  | 0.43633231  | 2016.08.17.16. |
| ## | 2119 | 320 | 32 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.16. |
| ## | 2120 | 340 | 1  | 2016.08.17 | 20  | 0.34906585  | 2016.08.17.16. |
| ## | 2121 | 310 | 2  | 2016.08.17 | 50  | 0.87266463  | 2016.08.17.16. |
| ## | 2122 | 110 | 3  | 2016.08.17 | 250 | -1.91986218 | 2016.08.17.16. |
| ## | 2123 | 160 | 4  | 2016.08.17 | 200 | -2.79252680 | 2016.08.17.16. |
| ## | 2124 | 350 | 5  | 2016.08.17 | 10  | 0.17453293  | 2016.08.17.16. |
| ## | 2125 | 245 | 6  | 2016.08.17 | 115 | 2.00712864  | 2016.08.17.16. |
| ## | 2126 | 60  | 7  | 2016.08.17 | 300 | -1.04719755 | 2016.08.17.16. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2127 | 40  | 8  | 2016.08.17 | 320 | -0.69813170 | 2016.08.17.16. |
| ## | 2128 | 50  | 9  | 2016.08.17 | 310 | -0.87266463 | 2016.08.17.16. |
| ## | 2129 | 280 | 10 | 2016.08.17 | 80  | 1.39626340  | 2016.08.17.16. |
| ## | 2130 | 40  | 11 | 2016.08.17 | 320 | -0.69813170 | 2016.08.17.16. |
| ## | 2131 | 10  | 12 | 2016.08.17 | 350 | -0.17453293 | 2016.08.17.16. |
| ## | 2132 | 260 | 13 | 2016.08.17 | 100 | 1.74532925  | 2016.08.17.16. |
| ## | 2133 | 210 | 14 | 2016.08.17 | 150 | 2.61799388  | 2016.08.17.16. |
| ## | 2134 | 140 | 15 | 2016.08.17 | 220 | -2.44346095 | 2016.08.17.16. |
| ## | 2135 | 230 | 16 | 2016.08.17 | 130 | 2.26892803  | 2016.08.17.16. |
| ## | 2136 | 70  | 17 | 2016.08.17 | 290 | -1.22173048 | 2016.08.17.16. |
| ## | 2137 | 210 | 18 | 2016.08.17 | 150 | 2.61799388  | 2016.08.17.16. |
| ## | 2138 | 320 | 19 | 2016.08.17 | 40  | 0.69813170  | 2016.08.17.16. |
| ## | 2139 | 350 | 20 | 2016.08.17 | 10  | 0.17453293  | 2016.08.17.16. |
| ## | 2140 | 345 | 21 | 2016.08.17 | 15  | 0.26179939  | 2016.08.17.16. |
| ## | 2141 | 290 | 22 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.16. |
| ## | 2142 | 290 | 23 | 2016.08.17 | 70  | 1.22173048  | 2016.08.17.16. |
| ## | 2143 | 330 | 24 | 2016.08.17 | 30  | 0.52359878  | 2016.08.17.16. |
| ## | 2144 | 335 | 25 | 2016.08.17 | 25  | 0.43633231  | 2016.08.17.16. |
| ## | 2145 | 300 | 26 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.16. |
| ## | 2146 | 300 | 27 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.16. |
| ## | 2147 | 350 | 28 | 2016.08.17 | 10  | 0.17453293  | 2016.08.17.16. |
| ## | 2148 | 300 | 29 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.16. |
| ## | 2149 | 305 | 30 | 2016.08.17 | 55  | 0.95993109  | 2016.08.17.16. |
| ## | 2150 | 300 | 31 | 2016.08.17 | 60  | 1.04719755  | 2016.08.17.16. |
| ## | 2151 | 295 | 32 | 2016.08.17 | 65  | 1.13446401  | 2016.08.17.16. |
| ## | 2152 | 280 | 33 | 2016.08.17 | 80  | 1.39626340  | 2016.08.17.16. |
| ## | 2153 | 30  | 1  | 2016.08.18 | 330 | -0.52359878 | 2016.08.18.14. |
| ## | 2154 | 345 | 2  | 2016.08.18 | 15  | 0.26179939  | 2016.08.18.14. |
| ## | 2155 | 190 | 3  | 2016.08.18 | 170 | 2.96705973  | 2016.08.18.14. |
| ## | 2156 | 180 | 4  | 2016.08.18 | 180 | 3.14159265  | 2016.08.18.14. |
| ## | 2157 | 240 | 5  | 2016.08.18 | 120 | 2.09439510  | 2016.08.18.14. |
| ## | 2158 | 340 | 6  | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.14. |
| ## | 2159 | 0   | 7  | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.14. |
| ## | 2160 | 245 | 8  | 2016.08.18 | 115 | 2.00712864  | 2016.08.18.14. |
| ## | 2161 | 10  | 9  | 2016.08.18 | 350 | -0.17453293 | 2016.08.18.14. |
| ## | 2162 | 280 | 10 | 2016.08.18 | 80  | 1.39626340  | 2016.08.18.14. |
| ## | 2163 | 15  | 11 | 2016.08.18 | 345 | -0.26179939 | 2016.08.18.14. |
| ## | 2164 | 270 | 12 | 2016.08.18 | 90  | 1.57079633  | 2016.08.18.14. |
| ## | 2165 | 260 | 13 | 2016.08.18 | 100 | 1.74532925  | 2016.08.18.14. |
| ## | 2166 | 330 | 14 | 2016.08.18 | 30  | 0.52359878  | 2016.08.18.14. |
| ## | 2167 | 270 | 15 | 2016.08.18 | 90  | 1.57079633  | 2016.08.18.14. |
| ## | 2168 | 320 | 16 | 2016.08.18 | 40  | 0.69813170  | 2016.08.18.14. |
| ## | 2169 | 340 | 17 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.14. |
| ## | 2170 | 320 | 18 | 2016.08.18 | 40  | 0.69813170  | 2016.08.18.14. |
| ## | 2171 | 340 | 19 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.14. |
| ## | 2172 | 290 | 20 | 2016.08.18 | 70  | 1.22173048  | 2016.08.18.14. |
| ## | 2173 | 275 | 21 | 2016.08.18 | 85  | 1.48352986  | 2016.08.18.14. |
| ## | 2174 | 270 | 22 | 2016.08.18 | 90  | 1.57079633  | 2016.08.18.14. |

|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2175 | 260 | 23 | 2016.08.18 | 100 | 1.74532925  | 2016.08.18.14.   |
| ## | 2176 | 290 | 24 | 2016.08.18 | 70  | 1.22173048  | 2016.08.18.14.   |
| ## | 2177 | 270 | 25 | 2016.08.18 | 90  | 1.57079633  | 2016.08.18.14.   |
| ## | 2178 | 240 | 26 | 2016.08.18 | 120 | 2.09439510  | 2016.08.18.14.   |
| ## | 2179 | 320 | 27 | 2016.08.18 | 40  | 0.69813170  | 2016.08.18.14.   |
| ## | 2180 | 290 | 28 | 2016.08.18 | 70  | 1.22173048  | 2016.08.18.14.   |
| ## | 2181 | 320 | 29 | 2016.08.18 | 40  | 0.69813170  | 2016.08.18.14.   |
| ## | 2182 | 280 | 30 | 2016.08.18 | 80  | 1.39626340  | 2016.08.18.14.   |
| ## | 2183 | 50  | 1  | 2016.08.18 | 310 | -0.87266463 | 2016.08.18.16.   |
| ## | 2184 | 45  | 2  | 2016.08.18 | 315 | -0.78539816 | 2016.08.18.16.   |
| ## | 2185 | 90  | 3  | 2016.08.18 | 270 | -1.57079633 | 2016.08.18.16.   |
| ## | 2186 | 100 | 4  | 2016.08.18 | 260 | -1.74532925 | 2016.08.18.16.   |
| ## | 2187 | 70  | 5  | 2016.08.18 | 290 | -1.22173048 | 2016.08.18.16.   |
| ## | 2188 | 75  | 6  | 2016.08.18 | 285 | -1.30899694 | 2016.08.18.16.   |
| ## | 2189 | 35  | 7  | 2016.08.18 | 325 | -0.61086524 | 2016.08.18.16.   |
| ## | 2190 | 220 | 8  | 2016.08.18 | 140 | 2.44346095  | 2016.08.18.16.   |
| ## | 2191 | 110 | 9  | 2016.08.18 | 250 | -1.91986218 | 2016.08.18.16.   |
| ## | 2192 | 80  | 10 | 2016.08.18 | 280 | -1.39626340 | 2016.08.18.16.   |
| ## | 2193 | 70  | 11 | 2016.08.18 | 290 | -1.22173048 | 2016.08.18.16.   |
| ## | 2194 | 50  | 12 | 2016.08.18 | 310 | -0.87266463 | 2016.08.18.16.   |
| ## | 2195 | 85  | 13 | 2016.08.18 | 275 | -1.48352986 | 2016.08.18.16.   |
| ## | 2196 | 80  | 14 | 2016.08.18 | 280 | -1.39626340 | 2016.08.18.16.   |
| ## | 2197 | 110 | 15 | 2016.08.18 | 250 | -1.91986218 | 2016.08.18.16.   |
| ## | 2198 | 170 | 16 | 2016.08.18 | 190 | -2.96705973 | 2016.08.18.16.   |
| ## | 2199 | 80  | 17 | 2016.08.18 | 280 | -1.39626340 | 2016.08.18.16.   |
| ## | 2200 | 150 | 18 | 2016.08.18 | 210 | -2.61799388 | 2016.08.18.16.   |
| ## | 2201 | 160 | 19 | 2016.08.18 | 200 | -2.79252680 | 2016.08.18.16.   |
| ## | 2202 | 340 | 20 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.   |
| ## | 2203 | 345 | 21 | 2016.08.18 | 15  | 0.26179939  | 2016.08.18.16.   |
| ## | 2204 | 350 | 22 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.   |
| ## | 2205 | 310 | 23 | 2016.08.18 | 50  | 0.87266463  | 2016.08.18.16.   |
| ## | 2206 | 5   | 24 | 2016.08.18 | 355 | -0.08726646 | 2016.08.18.16.   |
| ## | 2207 | 345 | 25 | 2016.08.18 | 15  | 0.26179939  | 2016.08.18.16.   |
| ## | 2208 | 340 | 26 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.   |
| ## | 2209 | 355 | 27 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.   |
| ## | 2210 | 320 | 28 | 2016.08.18 | 40  | 0.69813170  | 2016.08.18.16.   |
| ## | 2211 | 340 | 29 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.   |
| ## | 2212 | 70  | 1  | 2016.08.18 | 290 | -1.22173048 | 2016.08.18.16.20 |
| ## | 2213 | 75  | 2  | 2016.08.18 | 285 | -1.30899694 | 2016.08.18.16.20 |
| ## | 2214 | 110 | 3  | 2016.08.18 | 250 | -1.91986218 | 2016.08.18.16.20 |
| ## | 2215 | 240 | 4  | 2016.08.18 | 120 | 2.09439510  | 2016.08.18.16.20 |
| ## | 2216 | 210 | 5  | 2016.08.18 | 150 | 2.61799388  | 2016.08.18.16.20 |
| ## | 2217 | 200 | 6  | 2016.08.18 | 160 | 2.79252680  | 2016.08.18.16.20 |
| ## | 2218 | 120 | 7  | 2016.08.18 | 240 | -2.09439510 | 2016.08.18.16.20 |
| ## | 2219 | 95  | 8  | 2016.08.18 | 265 | -1.65806279 | 2016.08.18.16.20 |
| ## | 2220 | 200 | 9  | 2016.08.18 | 160 | 2.79252680  | 2016.08.18.16.20 |
| ## | 2221 | 85  | 10 | 2016.08.18 | 275 | -1.48352986 | 2016.08.18.16.20 |
| ## | 2222 | 20  | 11 | 2016.08.18 | 340 | -0.34906585 | 2016.08.18.16.20 |

|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2223 | 350 | 12 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2224 | 345 | 13 | 2016.08.18 | 15  | 0.26179939  | 2016.08.18.16.20 |
| ## | 2225 | 5   | 14 | 2016.08.18 | 355 | -0.08726646 | 2016.08.18.16.20 |
| ## | 2226 | 340 | 15 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.20 |
| ## | 2227 | 0   | 16 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.20 |
| ## | 2228 | 340 | 17 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.20 |
| ## | 2229 | 350 | 18 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2230 | 355 | 19 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.20 |
| ## | 2231 | 20  | 20 | 2016.08.18 | 340 | -0.34906585 | 2016.08.18.16.20 |
| ## | 2232 | 340 | 21 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.20 |
| ## | 2233 | 0   | 22 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.20 |
| ## | 2234 | 5   | 23 | 2016.08.18 | 355 | -0.08726646 | 2016.08.18.16.20 |
| ## | 2235 | 20  | 24 | 2016.08.18 | 340 | -0.34906585 | 2016.08.18.16.20 |
| ## | 2236 | 355 | 25 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.20 |
| ## | 2237 | 180 | 1  | 2016.08.18 | 180 | 3.14159265  | 2016.08.18.16.20 |
| ## | 2238 | 205 | 2  | 2016.08.18 | 155 | 2.70526034  | 2016.08.18.16.20 |
| ## | 2239 | 195 | 3  | 2016.08.18 | 165 | 2.87979327  | 2016.08.18.16.20 |
| ## | 2240 | 170 | 4  | 2016.08.18 | 190 | -2.96705973 | 2016.08.18.16.20 |
| ## | 2241 | 240 | 5  | 2016.08.18 | 120 | 2.09439510  | 2016.08.18.16.20 |
| ## | 2242 | 170 | 6  | 2016.08.18 | 190 | -2.96705973 | 2016.08.18.16.20 |
| ## | 2243 | 240 | 7  | 2016.08.18 | 120 | 2.09439510  | 2016.08.18.16.20 |
| ## | 2244 | 130 | 8  | 2016.08.18 | 230 | -2.26892803 | 2016.08.18.16.20 |
| ## | 2245 | 200 | 9  | 2016.08.18 | 160 | 2.79252680  | 2016.08.18.16.20 |
| ## | 2246 | 180 | 10 | 2016.08.18 | 180 | 3.14159265  | 2016.08.18.16.20 |
| ## | 2247 | 170 | 11 | 2016.08.18 | 190 | -2.96705973 | 2016.08.18.16.20 |
| ## | 2248 | 130 | 12 | 2016.08.18 | 230 | -2.26892803 | 2016.08.18.16.20 |
| ## | 2249 | 20  | 13 | 2016.08.18 | 340 | -0.34906585 | 2016.08.18.16.20 |
| ## | 2250 | 0   | 14 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.20 |
| ## | 2251 | 355 | 15 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.20 |
| ## | 2252 | 350 | 16 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2253 | 350 | 17 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2254 | 350 | 18 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2255 | 320 | 19 | 2016.08.18 | 40  | 0.69813170  | 2016.08.18.16.20 |
| ## | 2256 | 350 | 20 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2257 | 355 | 21 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.20 |
| ## | 2258 | 345 | 22 | 2016.08.18 | 15  | 0.26179939  | 2016.08.18.16.20 |
| ## | 2259 | 350 | 23 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2260 | 10  | 24 | 2016.08.18 | 350 | -0.17453293 | 2016.08.18.16.20 |
| ## | 2261 | 0   | 25 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.20 |
| ## | 2262 | 5   | 26 | 2016.08.18 | 355 | -0.08726646 | 2016.08.18.16.20 |
| ## | 2263 | 0   | 27 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.20 |
| ## | 2264 | 5   | 28 | 2016.08.18 | 355 | -0.08726646 | 2016.08.18.16.20 |
| ## | 2265 | 350 | 29 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |
| ## | 2266 | 15  | 30 | 2016.08.18 | 345 | -0.26179939 | 2016.08.18.16.20 |
| ## | 2267 | 0   | 31 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.20 |
| ## | 2268 | 355 | 32 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.20 |
| ## | 2269 | 5   | 33 | 2016.08.18 | 355 | -0.08726646 | 2016.08.18.16.20 |
| ## | 2270 | 350 | 34 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.20 |

|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2271 | 100 | 1  | 2016.08.18 | 260 | -1.74532925 | 2016.08.18.16.26 |
| ## | 2272 | 110 | 2  | 2016.08.18 | 250 | -1.91986218 | 2016.08.18.16.26 |
| ## | 2273 | 70  | 3  | 2016.08.18 | 290 | -1.22173048 | 2016.08.18.16.26 |
| ## | 2274 | 40  | 4  | 2016.08.18 | 320 | -0.69813170 | 2016.08.18.16.26 |
| ## | 2275 | 110 | 5  | 2016.08.18 | 250 | -1.91986218 | 2016.08.18.16.26 |
| ## | 2276 | 30  | 6  | 2016.08.18 | 330 | -0.52359878 | 2016.08.18.16.26 |
| ## | 2277 | 40  | 7  | 2016.08.18 | 320 | -0.69813170 | 2016.08.18.16.26 |
| ## | 2278 | 80  | 8  | 2016.08.18 | 280 | -1.39626340 | 2016.08.18.16.26 |
| ## | 2279 | 180 | 9  | 2016.08.18 | 180 | 3.14159265  | 2016.08.18.16.26 |
| ## | 2280 | 180 | 10 | 2016.08.18 | 180 | 3.14159265  | 2016.08.18.16.26 |
| ## | 2281 | 110 | 11 | 2016.08.18 | 250 | -1.91986218 | 2016.08.18.16.26 |
| ## | 2282 | 95  | 12 | 2016.08.18 | 265 | -1.65806279 | 2016.08.18.16.26 |
| ## | 2283 | 210 | 13 | 2016.08.18 | 150 | 2.61799388  | 2016.08.18.16.26 |
| ## | 2284 | 30  | 14 | 2016.08.18 | 330 | -0.52359878 | 2016.08.18.16.26 |
| ## | 2285 | 50  | 15 | 2016.08.18 | 310 | -0.87266463 | 2016.08.18.16.26 |
| ## | 2286 | 20  | 16 | 2016.08.18 | 340 | -0.34906585 | 2016.08.18.16.26 |
| ## | 2287 | 300 | 17 | 2016.08.18 | 60  | 1.04719755  | 2016.08.18.16.26 |
| ## | 2288 | 0   | 18 | 2016.08.18 | 0   | 0.00000000  | 2016.08.18.16.26 |
| ## | 2289 | 10  | 19 | 2016.08.18 | 350 | -0.17453293 | 2016.08.18.16.26 |
| ## | 2290 | 355 | 20 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.26 |
| ## | 2291 | 340 | 21 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.26 |
| ## | 2292 | 355 | 22 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.26 |
| ## | 2293 | 330 | 23 | 2016.08.18 | 30  | 0.52359878  | 2016.08.18.16.26 |
| ## | 2294 | 355 | 24 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.26 |
| ## | 2295 | 280 | 25 | 2016.08.18 | 80  | 1.39626340  | 2016.08.18.16.26 |
| ## | 2296 | 340 | 26 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.26 |
| ## | 2297 | 90  | 1  | 2016.08.18 | 270 | -1.57079633 | 2016.08.18.16.27 |
| ## | 2298 | 350 | 2  | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.27 |
| ## | 2299 | 90  | 3  | 2016.08.18 | 270 | -1.57079633 | 2016.08.18.16.27 |
| ## | 2300 | 330 | 4  | 2016.08.18 | 30  | 0.52359878  | 2016.08.18.16.27 |
| ## | 2301 | 65  | 5  | 2016.08.18 | 295 | -1.13446401 | 2016.08.18.16.27 |
| ## | 2302 | 95  | 6  | 2016.08.18 | 265 | -1.65806279 | 2016.08.18.16.27 |
| ## | 2303 | 355 | 7  | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.27 |
| ## | 2304 | 30  | 8  | 2016.08.18 | 330 | -0.52359878 | 2016.08.18.16.27 |
| ## | 2305 | 355 | 9  | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.27 |
| ## | 2306 | 340 | 10 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.27 |
| ## | 2307 | 350 | 11 | 2016.08.18 | 10  | 0.17453293  | 2016.08.18.16.27 |
| ## | 2308 | 340 | 12 | 2016.08.18 | 20  | 0.34906585  | 2016.08.18.16.27 |
| ## | 2309 | 355 | 13 | 2016.08.18 | 5   | 0.08726646  | 2016.08.18.16.27 |
| ## | 2310 | 345 | 14 | 2016.08.18 | 15  | 0.26179939  | 2016.08.18.16.27 |
| ## | 2311 | 145 | 1  | 2016.08.19 | 215 | -2.53072742 | 2016.08.19.14.   |
| ## | 2312 | 30  | 2  | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.   |
| ## | 2313 | 150 | 3  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.   |
| ## | 2314 | 70  | 4  | 2016.08.19 | 290 | -1.22173048 | 2016.08.19.14.   |
| ## | 2315 | 60  | 5  | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.14.   |
| ## | 2316 | 170 | 6  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.   |
| ## | 2317 | 350 | 7  | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.   |
| ## | 2318 | 0   | 8  | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.   |



|    |      |     |    |            |     |             |                   |
|----|------|-----|----|------------|-----|-------------|-------------------|
| ## | 2319 | 70  | 9  | 2016.08.19 | 290 | -1.22173048 | 2016.08.19.14.    |
| ## | 2320 | 170 | 10 | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.    |
| ## | 2321 | 175 | 11 | 2016.08.19 | 185 | -3.05432619 | 2016.08.19.14.    |
| ## | 2322 | 160 | 12 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.14.    |
| ## | 2323 | 90  | 13 | 2016.08.19 | 270 | -1.57079633 | 2016.08.19.14.    |
| ## | 2324 | 30  | 14 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.    |
| ## | 2325 | 110 | 15 | 2016.08.19 | 250 | -1.91986218 | 2016.08.19.14.    |
| ## | 2326 | 40  | 16 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.    |
| ## | 2327 | 70  | 17 | 2016.08.19 | 290 | -1.22173048 | 2016.08.19.14.    |
| ## | 2328 | 140 | 18 | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.    |
| ## | 2329 | 340 | 19 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.    |
| ## | 2330 | 15  | 20 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.14.    |
| ## | 2331 | 30  | 21 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.    |
| ## | 2332 | 350 | 22 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.    |
| ## | 2333 | 355 | 23 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.14.    |
| ## | 2334 | 345 | 24 | 2016.08.19 | 15  | 0.26179939  | 2016.08.19.14.    |
| ## | 2335 | 5   | 25 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14.    |
| ## | 2336 | 350 | 26 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.    |
| ## | 2337 | 20  | 27 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.    |
| ## | 2338 | 0   | 28 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.    |
| ## | 2339 | 10  | 29 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.    |
| ## | 2340 | 20  | 30 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.    |
| ## | 2341 | 0   | 31 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.    |
| ## | 2342 | 40  | 32 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.    |
| ## | 2343 | 145 | 1  | 2016.08.19 | 215 | -2.53072742 | 2016.08.19.14.50. |
| ## | 2344 | 140 | 2  | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.50. |
| ## | 2345 | 320 | 3  | 2016.08.19 | 40  | 0.69813170  | 2016.08.19.14.50. |
| ## | 2346 | 350 | 4  | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.50. |
| ## | 2347 | 110 | 5  | 2016.08.19 | 250 | -1.91986218 | 2016.08.19.14.50. |
| ## | 2348 | 160 | 6  | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.14.50. |
| ## | 2349 | 230 | 7  | 2016.08.19 | 130 | 2.26892803  | 2016.08.19.14.50. |
| ## | 2350 | 340 | 8  | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.50. |
| ## | 2351 | 335 | 9  | 2016.08.19 | 25  | 0.43633231  | 2016.08.19.14.50. |
| ## | 2352 | 280 | 10 | 2016.08.19 | 80  | 1.39626340  | 2016.08.19.14.50. |
| ## | 2353 | 150 | 11 | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.50. |
| ## | 2354 | 240 | 12 | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.14.50. |
| ## | 2355 | 290 | 13 | 2016.08.19 | 70  | 1.22173048  | 2016.08.19.14.50. |
| ## | 2356 | 270 | 14 | 2016.08.19 | 90  | 1.57079633  | 2016.08.19.14.50. |
| ## | 2357 | 270 | 15 | 2016.08.19 | 90  | 1.57079633  | 2016.08.19.14.50. |
| ## | 2358 | 170 | 16 | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.50. |
| ## | 2359 | 275 | 17 | 2016.08.19 | 85  | 1.48352986  | 2016.08.19.14.50. |
| ## | 2360 | 10  | 18 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.50. |
| ## | 2361 | 0   | 19 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.50. |
| ## | 2362 | 10  | 20 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.50. |
| ## | 2363 | 15  | 21 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.14.50. |
| ## | 2364 | 20  | 22 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.50. |
| ## | 2365 | 20  | 23 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.50. |
| ## | 2366 | 30  | 24 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.50. |

|         |     |    |            |     |             |                  |
|---------|-----|----|------------|-----|-------------|------------------|
| ## 2367 | 5   | 25 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14.50 |
| ## 2368 | 0   | 26 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.50 |
| ## 2369 | 50  | 27 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.14.50 |
| ## 2370 | 0   | 28 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.50 |
| ## 2371 | 350 | 29 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.50 |
| ## 2372 | 270 | 1  | 2016.08.19 | 90  | 1.57079633  | 2016.08.19.14.51 |
| ## 2373 | 140 | 2  | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.51 |
| ## 2374 | 150 | 3  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.51 |
| ## 2375 | 260 | 4  | 2016.08.19 | 100 | 1.74532925  | 2016.08.19.14.51 |
| ## 2376 | 110 | 5  | 2016.08.19 | 250 | -1.91986218 | 2016.08.19.14.51 |
| ## 2377 | 240 | 6  | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.14.51 |
| ## 2378 | 250 | 7  | 2016.08.19 | 110 | 1.91986218  | 2016.08.19.14.51 |
| ## 2379 | 110 | 8  | 2016.08.19 | 250 | -1.91986218 | 2016.08.19.14.51 |
| ## 2380 | 190 | 9  | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.14.51 |
| ## 2381 | 350 | 10 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |
| ## 2382 | 230 | 11 | 2016.08.19 | 130 | 2.26892803  | 2016.08.19.14.51 |
| ## 2383 | 230 | 12 | 2016.08.19 | 130 | 2.26892803  | 2016.08.19.14.51 |
| ## 2384 | 175 | 13 | 2016.08.19 | 185 | -3.05432619 | 2016.08.19.14.51 |
| ## 2385 | 210 | 14 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.14.51 |
| ## 2386 | 20  | 15 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.51 |
| ## 2387 | 350 | 16 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |
| ## 2388 | 340 | 17 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.51 |
| ## 2389 | 355 | 18 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.14.51 |
| ## 2390 | 10  | 19 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.51 |
| ## 2391 | 40  | 20 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.51 |
| ## 2392 | 0   | 21 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## 2393 | 0   | 22 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## 2394 | 355 | 23 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.14.51 |
| ## 2395 | 0   | 24 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## 2396 | 5   | 25 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14.51 |
| ## 2397 | 0   | 26 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## 2398 | 20  | 27 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.51 |
| ## 2399 | 350 | 28 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |
| ## 2400 | 355 | 29 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.14.51 |
| ## 2401 | 350 | 30 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |
| ## 2402 | 0   | 31 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## 2403 | 5   | 32 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14.51 |
| ## 2404 | 350 | 33 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |
| ## 2405 | 340 | 34 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.51 |
| ## 2406 | 350 | 35 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |
| ## 2407 | 20  | 36 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.51 |
| ## 2408 | 340 | 37 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.51 |
| ## 2409 | 0   | 38 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## 2410 | 355 | 39 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.14.51 |
| ## 2411 | 345 | 40 | 2016.08.19 | 15  | 0.26179939  | 2016.08.19.14.51 |
| ## 2412 | 340 | 41 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.51 |
| ## 2413 | 30  | 42 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.51 |
| ## 2414 | 350 | 43 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.51 |



|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2415 | 0   | 44 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.51 |
| ## | 2416 | 355 | 45 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.14.51 |
| ## | 2417 | 120 | 1  | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15.   |
| ## | 2418 | 150 | 2  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.15.   |
| ## | 2419 | 130 | 3  | 2016.08.19 | 230 | -2.26892803 | 2016.08.19.15.   |
| ## | 2420 | 120 | 4  | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15.   |
| ## | 2421 | 105 | 5  | 2016.08.19 | 255 | -1.83259571 | 2016.08.19.15.   |
| ## | 2422 | 160 | 6  | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.15.   |
| ## | 2423 | 70  | 7  | 2016.08.19 | 290 | -1.22173048 | 2016.08.19.15.   |
| ## | 2424 | 95  | 8  | 2016.08.19 | 265 | -1.65806279 | 2016.08.19.15.   |
| ## | 2425 | 155 | 9  | 2016.08.19 | 205 | -2.70526034 | 2016.08.19.15.   |
| ## | 2426 | 155 | 10 | 2016.08.19 | 205 | -2.70526034 | 2016.08.19.15.   |
| ## | 2427 | 175 | 11 | 2016.08.19 | 185 | -3.05432619 | 2016.08.19.15.   |
| ## | 2428 | 140 | 12 | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.15.   |
| ## | 2429 | 20  | 13 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.   |
| ## | 2430 | 45  | 14 | 2016.08.19 | 315 | -0.78539816 | 2016.08.19.15.   |
| ## | 2431 | 355 | 15 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.15.   |
| ## | 2432 | 40  | 16 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.   |
| ## | 2433 | 60  | 17 | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.15.   |
| ## | 2434 | 350 | 18 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.15.   |
| ## | 2435 | 10  | 19 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.   |
| ## | 2436 | 40  | 20 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.   |
| ## | 2437 | 355 | 21 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.15.   |
| ## | 2438 | 20  | 22 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.   |
| ## | 2439 | 5   | 23 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.15.   |
| ## | 2440 | 340 | 24 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.15.   |
| ## | 2441 | 30  | 25 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.15.   |
| ## | 2442 | 125 | 1  | 2016.08.19 | 235 | -2.18166156 | 2016.08.19.14.15 |
| ## | 2443 | 115 | 2  | 2016.08.19 | 245 | -2.00712864 | 2016.08.19.14.15 |
| ## | 2444 | 145 | 3  | 2016.08.19 | 215 | -2.53072742 | 2016.08.19.14.15 |
| ## | 2445 | 140 | 4  | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.15 |
| ## | 2446 | 170 | 5  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.15 |
| ## | 2447 | 240 | 6  | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.14.15 |
| ## | 2448 | 150 | 7  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.15 |
| ## | 2449 | 155 | 8  | 2016.08.19 | 205 | -2.70526034 | 2016.08.19.14.15 |
| ## | 2450 | 150 | 9  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.15 |
| ## | 2451 | 90  | 10 | 2016.08.19 | 270 | -1.57079633 | 2016.08.19.14.15 |
| ## | 2452 | 95  | 11 | 2016.08.19 | 265 | -1.65806279 | 2016.08.19.14.15 |
| ## | 2453 | 100 | 12 | 2016.08.19 | 260 | -1.74532925 | 2016.08.19.14.15 |
| ## | 2454 | 130 | 13 | 2016.08.19 | 230 | -2.26892803 | 2016.08.19.14.15 |
| ## | 2455 | 210 | 14 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.14.15 |
| ## | 2456 | 130 | 15 | 2016.08.19 | 230 | -2.26892803 | 2016.08.19.14.15 |
| ## | 2457 | 140 | 16 | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.15 |
| ## | 2458 | 90  | 17 | 2016.08.19 | 270 | -1.57079633 | 2016.08.19.14.15 |
| ## | 2459 | 120 | 18 | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.14.15 |
| ## | 2460 | 210 | 19 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.14.15 |
| ## | 2461 | 190 | 20 | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.14.15 |
| ## | 2462 | 185 | 21 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.14.15 |

|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2463 | 180 | 22 | 2016.08.19 | 180 | 3.14159265  | 2016.08.19.14.15 |
| ## | 2464 | 65  | 23 | 2016.08.19 | 295 | -1.13446401 | 2016.08.19.14.15 |
| ## | 2465 | 180 | 24 | 2016.08.19 | 180 | 3.14159265  | 2016.08.19.14.15 |
| ## | 2466 | 160 | 25 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.14.15 |
| ## | 2467 | 165 | 26 | 2016.08.19 | 195 | -2.87979327 | 2016.08.19.14.15 |
| ## | 2468 | 210 | 27 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.14.15 |
| ## | 2469 | 95  | 28 | 2016.08.19 | 265 | -1.65806279 | 2016.08.19.14.15 |
| ## | 2470 | 100 | 29 | 2016.08.19 | 260 | -1.74532925 | 2016.08.19.14.15 |
| ## | 2471 | 270 | 30 | 2016.08.19 | 90  | 1.57079633  | 2016.08.19.14.15 |
| ## | 2472 | 140 | 31 | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.15 |
| ## | 2473 | 230 | 32 | 2016.08.19 | 130 | 2.26892803  | 2016.08.19.14.15 |
| ## | 2474 | 170 | 33 | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.15 |
| ## | 2475 | 240 | 34 | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.14.15 |
| ## | 2476 | 160 | 35 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.14.15 |
| ## | 2477 | 160 | 36 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.14.15 |
| ## | 2478 | 150 | 37 | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.15 |
| ## | 2479 | 220 | 38 | 2016.08.19 | 140 | 2.44346095  | 2016.08.19.14.15 |
| ## | 2480 | 190 | 39 | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.14.15 |
| ## | 2481 | 140 | 40 | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.15 |
| ## | 2482 | 185 | 41 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.14.15 |
| ## | 2483 | 190 | 42 | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.14.15 |
| ## | 2484 | 40  | 43 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.15 |
| ## | 2485 | 0   | 44 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.15 |
| ## | 2486 | 60  | 45 | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.14.15 |
| ## | 2487 | 350 | 46 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.15 |
| ## | 2488 | 45  | 47 | 2016.08.19 | 315 | -0.78539816 | 2016.08.19.14.15 |
| ## | 2489 | 40  | 48 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.15 |
| ## | 2490 | 30  | 49 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.15 |
| ## | 2491 | 35  | 50 | 2016.08.19 | 325 | -0.61086524 | 2016.08.19.14.15 |
| ## | 2492 | 10  | 51 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.15 |
| ## | 2493 | 20  | 52 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.15 |
| ## | 2494 | 40  | 53 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.15 |
| ## | 2495 | 0   | 54 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.15 |
| ## | 2496 | 20  | 55 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.15 |
| ## | 2497 | 10  | 56 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.15 |
| ## | 2498 | 15  | 57 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.14.15 |
| ## | 2499 | 20  | 58 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.15 |
| ## | 2500 | 25  | 59 | 2016.08.19 | 335 | -0.43633231 | 2016.08.19.14.15 |
| ## | 2501 | 40  | 60 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.15 |
| ## | 2502 | 340 | 61 | 2016.08.19 | 20  | 0.34906585  | 2016.08.19.14.15 |
| ## | 2503 | 40  | 62 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14.15 |
| ## | 2504 | 5   | 63 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14.15 |
| ## | 2505 | 175 | 1  | 2016.08.19 | 185 | -3.05432619 | 2016.08.19.14.17 |
| ## | 2506 | 170 | 2  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.17 |
| ## | 2507 | 270 | 3  | 2016.08.19 | 90  | 1.57079633  | 2016.08.19.14.17 |
| ## | 2508 | 290 | 4  | 2016.08.19 | 70  | 1.22173048  | 2016.08.19.14.17 |
| ## | 2509 | 265 | 5  | 2016.08.19 | 95  | 1.65806279  | 2016.08.19.14.17 |
| ## | 2510 | 290 | 6  | 2016.08.19 | 70  | 1.22173048  | 2016.08.19.14.17 |

|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2511 | 220 | 7  | 2016.08.19 | 140 | 2.44346095  | 2016.08.19.14.17 |
| ## | 2512 | 245 | 8  | 2016.08.19 | 115 | 2.00712864  | 2016.08.19.14.17 |
| ## | 2513 | 220 | 9  | 2016.08.19 | 140 | 2.44346095  | 2016.08.19.14.17 |
| ## | 2514 | 200 | 10 | 2016.08.19 | 160 | 2.79252680  | 2016.08.19.14.17 |
| ## | 2515 | 185 | 11 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.14.17 |
| ## | 2516 | 245 | 12 | 2016.08.19 | 115 | 2.00712864  | 2016.08.19.14.17 |
| ## | 2517 | 240 | 13 | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.14.17 |
| ## | 2518 | 210 | 14 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.14.17 |
| ## | 2519 | 10  | 15 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.17 |
| ## | 2520 | 10  | 16 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.17 |
| ## | 2521 | 15  | 17 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.14.17 |
| ## | 2522 | 55  | 18 | 2016.08.19 | 305 | -0.95993109 | 2016.08.19.14.17 |
| ## | 2523 | 30  | 19 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14.17 |
| ## | 2524 | 350 | 20 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.17 |
| ## | 2525 | 10  | 21 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.17 |
| ## | 2526 | 10  | 22 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.17 |
| ## | 2527 | 25  | 23 | 2016.08.19 | 335 | -0.43633231 | 2016.08.19.14.17 |
| ## | 2528 | 10  | 24 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.17 |
| ## | 2529 | 350 | 25 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.17 |
| ## | 2530 | 0   | 26 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.17 |
| ## | 2531 | 0   | 27 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14.17 |
| ## | 2532 | 80  | 28 | 2016.08.19 | 280 | -1.39626340 | 2016.08.19.14.17 |
| ## | 2533 | 20  | 29 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14.17 |
| ## | 2534 | 10  | 30 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14.17 |
| ## | 2535 | 350 | 31 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14.17 |
| ## | 2536 | 5   | 32 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14.17 |
| ## | 2537 | 180 | 1  | 2016.08.19 | 180 | 3.14159265  | 2016.08.19.14.   |
| ## | 2538 | 170 | 2  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.   |
| ## | 2539 | 225 | 3  | 2016.08.19 | 135 | 2.35619449  | 2016.08.19.14.   |
| ## | 2540 | 210 | 4  | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.14.   |
| ## | 2541 | 170 | 5  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.   |
| ## | 2542 | 140 | 6  | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.14.   |
| ## | 2543 | 220 | 7  | 2016.08.19 | 140 | 2.44346095  | 2016.08.19.14.   |
| ## | 2544 | 170 | 8  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.   |
| ## | 2545 | 185 | 9  | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.14.   |
| ## | 2546 | 170 | 10 | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.   |
| ## | 2547 | 100 | 11 | 2016.08.19 | 260 | -1.74532925 | 2016.08.19.14.   |
| ## | 2548 | 170 | 12 | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.14.   |
| ## | 2549 | 220 | 13 | 2016.08.19 | 140 | 2.44346095  | 2016.08.19.14.   |
| ## | 2550 | 180 | 14 | 2016.08.19 | 180 | 3.14159265  | 2016.08.19.14.   |
| ## | 2551 | 185 | 15 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.14.   |
| ## | 2552 | 195 | 16 | 2016.08.19 | 165 | 2.87979327  | 2016.08.19.14.   |
| ## | 2553 | 240 | 17 | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.14.   |
| ## | 2554 | 220 | 18 | 2016.08.19 | 140 | 2.44346095  | 2016.08.19.14.   |
| ## | 2555 | 205 | 19 | 2016.08.19 | 155 | 2.70526034  | 2016.08.19.14.   |
| ## | 2556 | 185 | 20 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.14.   |
| ## | 2557 | 150 | 21 | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.   |
| ## | 2558 | 150 | 22 | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.14.   |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2559 | 0   | 23 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14. |
| ## | 2560 | 330 | 24 | 2016.08.19 | 30  | 0.52359878  | 2016.08.19.14. |
| ## | 2561 | 10  | 25 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14. |
| ## | 2562 | 10  | 26 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14. |
| ## | 2563 | 5   | 27 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.14. |
| ## | 2564 | 0   | 28 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.14. |
| ## | 2565 | 10  | 29 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14. |
| ## | 2566 | 350 | 30 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14. |
| ## | 2567 | 15  | 31 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.14. |
| ## | 2568 | 40  | 32 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.14. |
| ## | 2569 | 350 | 33 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.14. |
| ## | 2570 | 10  | 34 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.14. |
| ## | 2571 | 60  | 35 | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.14. |
| ## | 2572 | 20  | 36 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14. |
| ## | 2573 | 30  | 37 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.14. |
| ## | 2574 | 35  | 38 | 2016.08.19 | 325 | -0.61086524 | 2016.08.19.14. |
| ## | 2575 | 20  | 39 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.14. |
| ## | 2576 | 245 | 1  | 2016.08.19 | 115 | 2.00712864  | 2016.08.19.15. |
| ## | 2577 | 115 | 2  | 2016.08.19 | 245 | -2.00712864 | 2016.08.19.15. |
| ## | 2578 | 120 | 3  | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15. |
| ## | 2579 | 150 | 4  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.15. |
| ## | 2580 | 90  | 5  | 2016.08.19 | 270 | -1.57079633 | 2016.08.19.15. |
| ## | 2581 | 330 | 6  | 2016.08.19 | 30  | 0.52359878  | 2016.08.19.15. |
| ## | 2582 | 240 | 7  | 2016.08.19 | 120 | 2.09439510  | 2016.08.19.15. |
| ## | 2583 | 90  | 8  | 2016.08.19 | 270 | -1.57079633 | 2016.08.19.15. |
| ## | 2584 | 120 | 9  | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15. |
| ## | 2585 | 90  | 10 | 2016.08.19 | 270 | -1.57079633 | 2016.08.19.15. |
| ## | 2586 | 140 | 11 | 2016.08.19 | 220 | -2.44346095 | 2016.08.19.15. |
| ## | 2587 | 170 | 12 | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.15. |
| ## | 2588 | 320 | 13 | 2016.08.19 | 40  | 0.69813170  | 2016.08.19.15. |
| ## | 2589 | 205 | 14 | 2016.08.19 | 155 | 2.70526034  | 2016.08.19.15. |
| ## | 2590 | 135 | 15 | 2016.08.19 | 225 | -2.35619449 | 2016.08.19.15. |
| ## | 2591 | 120 | 16 | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15. |
| ## | 2592 | 185 | 17 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.15. |
| ## | 2593 | 300 | 18 | 2016.08.19 | 60  | 1.04719755  | 2016.08.19.15. |
| ## | 2594 | 125 | 19 | 2016.08.19 | 235 | -2.18166156 | 2016.08.19.15. |
| ## | 2595 | 320 | 20 | 2016.08.19 | 40  | 0.69813170  | 2016.08.19.15. |
| ## | 2596 | 280 | 21 | 2016.08.19 | 80  | 1.39626340  | 2016.08.19.15. |
| ## | 2597 | 100 | 22 | 2016.08.19 | 260 | -1.74532925 | 2016.08.19.15. |
| ## | 2598 | 320 | 23 | 2016.08.19 | 40  | 0.69813170  | 2016.08.19.15. |
| ## | 2599 | 125 | 24 | 2016.08.19 | 235 | -2.18166156 | 2016.08.19.15. |
| ## | 2600 | 200 | 25 | 2016.08.19 | 160 | 2.79252680  | 2016.08.19.15. |
| ## | 2601 | 50  | 26 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.15. |
| ## | 2602 | 95  | 27 | 2016.08.19 | 265 | -1.65806279 | 2016.08.19.15. |
| ## | 2603 | 100 | 28 | 2016.08.19 | 260 | -1.74532925 | 2016.08.19.15. |
| ## | 2604 | 60  | 29 | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.15. |
| ## | 2605 | 0   | 30 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.15. |
| ## | 2606 | 50  | 31 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.15. |

|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2607 | 355 | 32 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.15.   |
| ## | 2608 | 10  | 33 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.   |
| ## | 2609 | 5   | 34 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.15.   |
| ## | 2610 | 50  | 35 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.15.   |
| ## | 2611 | 30  | 36 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.15.   |
| ## | 2612 | 15  | 37 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.15.   |
| ## | 2613 | 20  | 38 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.   |
| ## | 2614 | 350 | 39 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.15.   |
| ## | 2615 | 20  | 40 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.   |
| ## | 2616 | 35  | 41 | 2016.08.19 | 325 | -0.61086524 | 2016.08.19.15.   |
| ## | 2617 | 40  | 42 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.   |
| ## | 2618 | 45  | 43 | 2016.08.19 | 315 | -0.78539816 | 2016.08.19.15.   |
| ## | 2619 | 40  | 44 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.   |
| ## | 2620 | 35  | 45 | 2016.08.19 | 325 | -0.61086524 | 2016.08.19.15.   |
| ## | 2621 | 50  | 46 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.15.   |
| ## | 2622 | 5   | 47 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.15.   |
| ## | 2623 | 10  | 48 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.   |
| ## | 2624 | 180 | 1  | 2016.08.19 | 180 | 3.14159265  | 2016.08.19.15.33 |
| ## | 2625 | 150 | 2  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.15.33 |
| ## | 2626 | 145 | 3  | 2016.08.19 | 215 | -2.53072742 | 2016.08.19.15.33 |
| ## | 2627 | 155 | 4  | 2016.08.19 | 205 | -2.70526034 | 2016.08.19.15.33 |
| ## | 2628 | 310 | 5  | 2016.08.19 | 50  | 0.87266463  | 2016.08.19.15.33 |
| ## | 2629 | 170 | 6  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.15.33 |
| ## | 2630 | 110 | 7  | 2016.08.19 | 250 | -1.91986218 | 2016.08.19.15.33 |
| ## | 2631 | 190 | 8  | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.15.33 |
| ## | 2632 | 170 | 9  | 2016.08.19 | 190 | -2.96705973 | 2016.08.19.15.33 |
| ## | 2633 | 150 | 10 | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.15.33 |
| ## | 2634 | 150 | 11 | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.15.33 |
| ## | 2635 | 185 | 12 | 2016.08.19 | 175 | 3.05432619  | 2016.08.19.15.33 |
| ## | 2636 | 120 | 13 | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15.33 |
| ## | 2637 | 155 | 14 | 2016.08.19 | 205 | -2.70526034 | 2016.08.19.15.33 |
| ## | 2638 | 210 | 15 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.15.33 |
| ## | 2639 | 205 | 16 | 2016.08.19 | 155 | 2.70526034  | 2016.08.19.15.33 |
| ## | 2640 | 250 | 17 | 2016.08.19 | 110 | 1.91986218  | 2016.08.19.15.33 |
| ## | 2641 | 200 | 18 | 2016.08.19 | 160 | 2.79252680  | 2016.08.19.15.33 |
| ## | 2642 | 160 | 19 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.15.33 |
| ## | 2643 | 190 | 20 | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.15.33 |
| ## | 2644 | 270 | 21 | 2016.08.19 | 90  | 1.57079633  | 2016.08.19.15.33 |
| ## | 2645 | 160 | 22 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.15.33 |
| ## | 2646 | 160 | 23 | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.15.33 |
| ## | 2647 | 215 | 24 | 2016.08.19 | 145 | 2.53072742  | 2016.08.19.15.33 |
| ## | 2648 | 145 | 25 | 2016.08.19 | 215 | -2.53072742 | 2016.08.19.15.33 |
| ## | 2649 | 205 | 26 | 2016.08.19 | 155 | 2.70526034  | 2016.08.19.15.33 |
| ## | 2650 | 120 | 27 | 2016.08.19 | 240 | -2.09439510 | 2016.08.19.15.33 |
| ## | 2651 | 210 | 28 | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.15.33 |
| ## | 2652 | 40  | 29 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.33 |
| ## | 2653 | 10  | 30 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.33 |
| ## | 2654 | 40  | 31 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.33 |



|    |      |     |    |            |     |             |                  |
|----|------|-----|----|------------|-----|-------------|------------------|
| ## | 2655 | 0   | 32 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.15.33 |
| ## | 2656 | 355 | 33 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.15.33 |
| ## | 2657 | 20  | 34 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.33 |
| ## | 2658 | 20  | 35 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.33 |
| ## | 2659 | 10  | 36 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.33 |
| ## | 2660 | 355 | 37 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.15.33 |
| ## | 2661 | 10  | 38 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.33 |
| ## | 2662 | 0   | 39 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.15.33 |
| ## | 2663 | 5   | 40 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.15.33 |
| ## | 2664 | 350 | 41 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.15.33 |
| ## | 2665 | 10  | 42 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.33 |
| ## | 2666 | 0   | 43 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.15.33 |
| ## | 2667 | 350 | 44 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.15.33 |
| ## | 2668 | 60  | 45 | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.15.33 |
| ## | 2669 | 5   | 46 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.15.33 |
| ## | 2670 | 10  | 47 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.33 |
| ## | 2671 | 10  | 48 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.33 |
| ## | 2672 | 5   | 49 | 2016.08.19 | 355 | -0.08726646 | 2016.08.19.15.33 |
| ## | 2673 | 0   | 50 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.15.33 |
| ## | 2674 | 350 | 51 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.15.33 |
| ## | 2675 | 310 | 1  | 2016.08.19 | 50  | 0.87266463  | 2016.08.19.15.34 |
| ## | 2676 | 190 | 2  | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.15.34 |
| ## | 2677 | 160 | 3  | 2016.08.19 | 200 | -2.79252680 | 2016.08.19.15.34 |
| ## | 2678 | 215 | 4  | 2016.08.19 | 145 | 2.53072742  | 2016.08.19.15.34 |
| ## | 2679 | 150 | 5  | 2016.08.19 | 210 | -2.61799388 | 2016.08.19.15.34 |
| ## | 2680 | 210 | 6  | 2016.08.19 | 150 | 2.61799388  | 2016.08.19.15.34 |
| ## | 2681 | 330 | 7  | 2016.08.19 | 30  | 0.52359878  | 2016.08.19.15.34 |
| ## | 2682 | 190 | 8  | 2016.08.19 | 170 | 2.96705973  | 2016.08.19.15.34 |
| ## | 2683 | 280 | 9  | 2016.08.19 | 80  | 1.39626340  | 2016.08.19.15.34 |
| ## | 2684 | 15  | 10 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.15.34 |
| ## | 2685 | 10  | 11 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.34 |
| ## | 2686 | 50  | 12 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.15.34 |
| ## | 2687 | 350 | 13 | 2016.08.19 | 10  | 0.17453293  | 2016.08.19.15.34 |
| ## | 2688 | 30  | 14 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.15.34 |
| ## | 2689 | 60  | 15 | 2016.08.19 | 300 | -1.04719755 | 2016.08.19.15.34 |
| ## | 2690 | 50  | 16 | 2016.08.19 | 310 | -0.87266463 | 2016.08.19.15.34 |
| ## | 2691 | 0   | 17 | 2016.08.19 | 0   | 0.00000000  | 2016.08.19.15.34 |
| ## | 2692 | 15  | 18 | 2016.08.19 | 345 | -0.26179939 | 2016.08.19.15.34 |
| ## | 2693 | 30  | 19 | 2016.08.19 | 330 | -0.52359878 | 2016.08.19.15.34 |
| ## | 2694 | 355 | 20 | 2016.08.19 | 5   | 0.08726646  | 2016.08.19.15.34 |
| ## | 2695 | 40  | 21 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.34 |
| ## | 2696 | 40  | 22 | 2016.08.19 | 320 | -0.69813170 | 2016.08.19.15.34 |
| ## | 2697 | 25  | 23 | 2016.08.19 | 335 | -0.43633231 | 2016.08.19.15.34 |
| ## | 2698 | 20  | 24 | 2016.08.19 | 340 | -0.34906585 | 2016.08.19.15.34 |
| ## | 2699 | 10  | 25 | 2016.08.19 | 350 | -0.17453293 | 2016.08.19.15.34 |
| ## | 2700 | 145 | 1  | 2016.08.23 | 215 | -2.53072742 | 2016.08.23.14.   |
| ## | 2701 | 180 | 2  | 2016.08.23 | 180 | 3.14159265  | 2016.08.23.14.   |
| ## | 2702 | 160 | 3  | 2016.08.23 | 200 | -2.79252680 | 2016.08.23.14.   |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 2703 | 175 | 4  | 2016.08.23 | 185 | -3.05432619 | 2016.08.23.14. |
| ## 2704 | 190 | 5  | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.14. |
| ## 2705 | 100 | 6  | 2016.08.23 | 260 | -1.74532925 | 2016.08.23.14. |
| ## 2706 | 200 | 7  | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.14. |
| ## 2707 | 100 | 8  | 2016.08.23 | 260 | -1.74532925 | 2016.08.23.14. |
| ## 2708 | 80  | 9  | 2016.08.23 | 280 | -1.39626340 | 2016.08.23.14. |
| ## 2709 | 85  | 10 | 2016.08.23 | 275 | -1.48352986 | 2016.08.23.14. |
| ## 2710 | 200 | 11 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.14. |
| ## 2711 | 50  | 12 | 2016.08.23 | 310 | -0.87266463 | 2016.08.23.14. |
| ## 2712 | 220 | 13 | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.14. |
| ## 2713 | 205 | 14 | 2016.08.23 | 155 | 2.70526034  | 2016.08.23.14. |
| ## 2714 | 145 | 15 | 2016.08.23 | 215 | -2.53072742 | 2016.08.23.14. |
| ## 2715 | 190 | 16 | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.14. |
| ## 2716 | 150 | 17 | 2016.08.23 | 210 | -2.61799388 | 2016.08.23.14. |
| ## 2717 | 185 | 18 | 2016.08.23 | 175 | 3.05432619  | 2016.08.23.14. |
| ## 2718 | 120 | 19 | 2016.08.23 | 240 | -2.09439510 | 2016.08.23.14. |
| ## 2719 | 165 | 20 | 2016.08.23 | 195 | -2.87979327 | 2016.08.23.14. |
| ## 2720 | 210 | 21 | 2016.08.23 | 150 | 2.61799388  | 2016.08.23.14. |
| ## 2721 | 255 | 22 | 2016.08.23 | 105 | 1.83259571  | 2016.08.23.14. |
| ## 2722 | 270 | 23 | 2016.08.23 | 90  | 1.57079633  | 2016.08.23.14. |
| ## 2723 | 325 | 24 | 2016.08.23 | 35  | 0.61086524  | 2016.08.23.14. |
| ## 2724 | 280 | 25 | 2016.08.23 | 80  | 1.39626340  | 2016.08.23.14. |
| ## 2725 | 310 | 26 | 2016.08.23 | 50  | 0.87266463  | 2016.08.23.14. |
| ## 2726 | 310 | 27 | 2016.08.23 | 50  | 0.87266463  | 2016.08.23.14. |
| ## 2727 | 315 | 28 | 2016.08.23 | 45  | 0.78539816  | 2016.08.23.14. |
| ## 2728 | 315 | 29 | 2016.08.23 | 45  | 0.78539816  | 2016.08.23.14. |
| ## 2729 | 295 | 30 | 2016.08.23 | 65  | 1.13446401  | 2016.08.23.14. |
| ## 2730 | 330 | 31 | 2016.08.23 | 30  | 0.52359878  | 2016.08.23.14. |
| ## 2731 | 330 | 32 | 2016.08.23 | 30  | 0.52359878  | 2016.08.23.14. |
| ## 2732 | 295 | 33 | 2016.08.23 | 65  | 1.13446401  | 2016.08.23.14. |
| ## 2733 | 290 | 34 | 2016.08.23 | 70  | 1.22173048  | 2016.08.23.14. |
| ## 2734 | 300 | 35 | 2016.08.23 | 60  | 1.04719755  | 2016.08.23.14. |
| ## 2735 | 320 | 36 | 2016.08.23 | 40  | 0.69813170  | 2016.08.23.14. |
| ## 2736 | 305 | 37 | 2016.08.23 | 55  | 0.95993109  | 2016.08.23.14. |
| ## 2737 | 280 | 38 | 2016.08.23 | 80  | 1.39626340  | 2016.08.23.14. |
| ## 2738 | 315 | 39 | 2016.08.23 | 45  | 0.78539816  | 2016.08.23.14. |
| ## 2739 | 325 | 40 | 2016.08.23 | 35  | 0.61086524  | 2016.08.23.14. |
| ## 2740 | 280 | 41 | 2016.08.23 | 80  | 1.39626340  | 2016.08.23.14. |
| ## 2741 | 290 | 42 | 2016.08.23 | 70  | 1.22173048  | 2016.08.23.14. |
| ## 2742 | 310 | 43 | 2016.08.23 | 50  | 0.87266463  | 2016.08.23.14. |
| ## 2743 | 175 | 1  | 2016.08.23 | 185 | -3.05432619 | 2016.08.23.14. |
| ## 2744 | 180 | 2  | 2016.08.23 | 180 | 3.14159265  | 2016.08.23.14. |
| ## 2745 | 110 | 3  | 2016.08.23 | 250 | -1.91986218 | 2016.08.23.14. |
| ## 2746 | 190 | 4  | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.14. |
| ## 2747 | 200 | 5  | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.14. |
| ## 2748 | 165 | 6  | 2016.08.23 | 195 | -2.87979327 | 2016.08.23.14. |
| ## 2749 | 110 | 7  | 2016.08.23 | 250 | -1.91986218 | 2016.08.23.14. |
| ## 2750 | 210 | 8  | 2016.08.23 | 150 | 2.61799388  | 2016.08.23.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2751 | 110 | 9  | 2016.08.23 | 250 | -1.91986218 | 2016.08.23.14. |
| ## | 2752 | 240 | 10 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.14. |
| ## | 2753 | 240 | 11 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.14. |
| ## | 2754 | 120 | 12 | 2016.08.23 | 240 | -2.09439510 | 2016.08.23.14. |
| ## | 2755 | 95  | 13 | 2016.08.23 | 265 | -1.65806279 | 2016.08.23.14. |
| ## | 2756 | 270 | 14 | 2016.08.23 | 90  | 1.57079633  | 2016.08.23.14. |
| ## | 2757 | 205 | 15 | 2016.08.23 | 155 | 2.70526034  | 2016.08.23.14. |
| ## | 2758 | 220 | 16 | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.14. |
| ## | 2759 | 130 | 17 | 2016.08.23 | 230 | -2.26892803 | 2016.08.23.14. |
| ## | 2760 | 175 | 18 | 2016.08.23 | 185 | -3.05432619 | 2016.08.23.14. |
| ## | 2761 | 110 | 19 | 2016.08.23 | 250 | -1.91986218 | 2016.08.23.14. |
| ## | 2762 | 200 | 20 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.14. |
| ## | 2763 | 180 | 21 | 2016.08.23 | 180 | 3.14159265  | 2016.08.23.14. |
| ## | 2764 | 150 | 22 | 2016.08.23 | 210 | -2.61799388 | 2016.08.23.14. |
| ## | 2765 | 90  | 23 | 2016.08.23 | 270 | -1.57079633 | 2016.08.23.14. |
| ## | 2766 | 345 | 24 | 2016.08.23 | 15  | 0.26179939  | 2016.08.23.14. |
| ## | 2767 | 335 | 25 | 2016.08.23 | 25  | 0.43633231  | 2016.08.23.14. |
| ## | 2768 | 345 | 26 | 2016.08.23 | 15  | 0.26179939  | 2016.08.23.14. |
| ## | 2769 | 5   | 27 | 2016.08.23 | 355 | -0.08726646 | 2016.08.23.14. |
| ## | 2770 | 5   | 28 | 2016.08.23 | 355 | -0.08726646 | 2016.08.23.14. |
| ## | 2771 | 30  | 29 | 2016.08.23 | 330 | -0.52359878 | 2016.08.23.14. |
| ## | 2772 | 15  | 30 | 2016.08.23 | 345 | -0.26179939 | 2016.08.23.14. |
| ## | 2773 | 10  | 31 | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.14. |
| ## | 2774 | 10  | 32 | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.14. |
| ## | 2775 | 5   | 33 | 2016.08.23 | 355 | -0.08726646 | 2016.08.23.14. |
| ## | 2776 | 30  | 34 | 2016.08.23 | 330 | -0.52359878 | 2016.08.23.14. |
| ## | 2777 | 350 | 35 | 2016.08.23 | 10  | 0.17453293  | 2016.08.23.14. |
| ## | 2778 | 20  | 36 | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.14. |
| ## | 2779 | 345 | 37 | 2016.08.23 | 15  | 0.26179939  | 2016.08.23.14. |
| ## | 2780 | 10  | 38 | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.14. |
| ## | 2781 | 355 | 39 | 2016.08.23 | 5   | 0.08726646  | 2016.08.23.14. |
| ## | 2782 | 190 | 1  | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.14. |
| ## | 2783 | 165 | 2  | 2016.08.23 | 195 | -2.87979327 | 2016.08.23.14. |
| ## | 2784 | 310 | 3  | 2016.08.23 | 50  | 0.87266463  | 2016.08.23.14. |
| ## | 2785 | 20  | 4  | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.14. |
| ## | 2786 | 60  | 5  | 2016.08.23 | 300 | -1.04719755 | 2016.08.23.14. |
| ## | 2787 | 220 | 6  | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.14. |
| ## | 2788 | 120 | 7  | 2016.08.23 | 240 | -2.09439510 | 2016.08.23.14. |
| ## | 2789 | 95  | 8  | 2016.08.23 | 265 | -1.65806279 | 2016.08.23.14. |
| ## | 2790 | 130 | 9  | 2016.08.23 | 230 | -2.26892803 | 2016.08.23.14. |
| ## | 2791 | 140 | 10 | 2016.08.23 | 220 | -2.44346095 | 2016.08.23.14. |
| ## | 2792 | 75  | 11 | 2016.08.23 | 285 | -1.30899694 | 2016.08.23.14. |
| ## | 2793 | 50  | 12 | 2016.08.23 | 310 | -0.87266463 | 2016.08.23.14. |
| ## | 2794 | 190 | 13 | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.14. |
| ## | 2795 | 210 | 14 | 2016.08.23 | 150 | 2.61799388  | 2016.08.23.14. |
| ## | 2796 | 105 | 15 | 2016.08.23 | 255 | -1.83259571 | 2016.08.23.14. |
| ## | 2797 | 90  | 16 | 2016.08.23 | 270 | -1.57079633 | 2016.08.23.14. |
| ## | 2798 | 120 | 17 | 2016.08.23 | 240 | -2.09439510 | 2016.08.23.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2799 | 65  | 18 | 2016.08.23 | 295 | -1.13446401 | 2016.08.23.14. |
| ## | 2800 | 80  | 19 | 2016.08.23 | 280 | -1.39626340 | 2016.08.23.14. |
| ## | 2801 | 170 | 20 | 2016.08.23 | 190 | -2.96705973 | 2016.08.23.14. |
| ## | 2802 | 175 | 21 | 2016.08.23 | 185 | -3.05432619 | 2016.08.23.14. |
| ## | 2803 | 225 | 22 | 2016.08.23 | 135 | 2.35619449  | 2016.08.23.14. |
| ## | 2804 | 220 | 23 | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.14. |
| ## | 2805 | 240 | 24 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.14. |
| ## | 2806 | 240 | 25 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.14. |
| ## | 2807 | 80  | 26 | 2016.08.23 | 280 | -1.39626340 | 2016.08.23.14. |
| ## | 2808 | 80  | 27 | 2016.08.23 | 280 | -1.39626340 | 2016.08.23.14. |
| ## | 2809 | 60  | 28 | 2016.08.23 | 300 | -1.04719755 | 2016.08.23.14. |
| ## | 2810 | 240 | 29 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.14. |
| ## | 2811 | 130 | 30 | 2016.08.23 | 230 | -2.26892803 | 2016.08.23.14. |
| ## | 2812 | 200 | 31 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.14. |
| ## | 2813 | 160 | 32 | 2016.08.23 | 200 | -2.79252680 | 2016.08.23.14. |
| ## | 2814 | 200 | 33 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.14. |
| ## | 2815 | 355 | 34 | 2016.08.23 | 5   | 0.08726646  | 2016.08.23.14. |
| ## | 2816 | 325 | 35 | 2016.08.23 | 35  | 0.61086524  | 2016.08.23.14. |
| ## | 2817 | 340 | 36 | 2016.08.23 | 20  | 0.34906585  | 2016.08.23.14. |
| ## | 2818 | 0   | 37 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2819 | 320 | 38 | 2016.08.23 | 40  | 0.69813170  | 2016.08.23.14. |
| ## | 2820 | 0   | 39 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2821 | 320 | 40 | 2016.08.23 | 40  | 0.69813170  | 2016.08.23.14. |
| ## | 2822 | 355 | 41 | 2016.08.23 | 5   | 0.08726646  | 2016.08.23.14. |
| ## | 2823 | 300 | 42 | 2016.08.23 | 60  | 1.04719755  | 2016.08.23.14. |
| ## | 2824 | 325 | 43 | 2016.08.23 | 35  | 0.61086524  | 2016.08.23.14. |
| ## | 2825 | 340 | 44 | 2016.08.23 | 20  | 0.34906585  | 2016.08.23.14. |
| ## | 2826 | 0   | 45 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2827 | 325 | 46 | 2016.08.23 | 35  | 0.61086524  | 2016.08.23.14. |
| ## | 2828 | 350 | 47 | 2016.08.23 | 10  | 0.17453293  | 2016.08.23.14. |
| ## | 2829 | 0   | 48 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2830 | 320 | 49 | 2016.08.23 | 40  | 0.69813170  | 2016.08.23.14. |
| ## | 2831 | 10  | 50 | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.14. |
| ## | 2832 | 0   | 51 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2833 | 355 | 52 | 2016.08.23 | 5   | 0.08726646  | 2016.08.23.14. |
| ## | 2834 | 0   | 53 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2835 | 330 | 54 | 2016.08.23 | 30  | 0.52359878  | 2016.08.23.14. |
| ## | 2836 | 355 | 55 | 2016.08.23 | 5   | 0.08726646  | 2016.08.23.14. |
| ## | 2837 | 330 | 56 | 2016.08.23 | 30  | 0.52359878  | 2016.08.23.14. |
| ## | 2838 | 290 | 57 | 2016.08.23 | 70  | 1.22173048  | 2016.08.23.14. |
| ## | 2839 | 0   | 58 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.14. |
| ## | 2840 | 330 | 59 | 2016.08.23 | 30  | 0.52359878  | 2016.08.23.14. |
| ## | 2841 | 355 | 60 | 2016.08.23 | 5   | 0.08726646  | 2016.08.23.14. |
| ## | 2842 | 5   | 61 | 2016.08.23 | 355 | -0.08726646 | 2016.08.23.14. |
| ## | 2843 | 0   | 1  | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.15. |
| ## | 2844 | 5   | 2  | 2016.08.23 | 355 | -0.08726646 | 2016.08.23.15. |
| ## | 2845 | 30  | 3  | 2016.08.23 | 330 | -0.52359878 | 2016.08.23.15. |
| ## | 2846 | 15  | 4  | 2016.08.23 | 345 | -0.26179939 | 2016.08.23.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2847 | 345 | 5  | 2016.08.23 | 15  | 0.26179939  | 2016.08.23.15. |
| ## | 2848 | 20  | 6  | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.15. |
| ## | 2849 | 40  | 7  | 2016.08.23 | 320 | -0.69813170 | 2016.08.23.15. |
| ## | 2850 | 170 | 8  | 2016.08.23 | 190 | -2.96705973 | 2016.08.23.15. |
| ## | 2851 | 10  | 9  | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.15. |
| ## | 2852 | 140 | 10 | 2016.08.23 | 220 | -2.44346095 | 2016.08.23.15. |
| ## | 2853 | 90  | 11 | 2016.08.23 | 270 | -1.57079633 | 2016.08.23.15. |
| ## | 2854 | 220 | 12 | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.15. |
| ## | 2855 | 30  | 13 | 2016.08.23 | 330 | -0.52359878 | 2016.08.23.15. |
| ## | 2856 | 65  | 14 | 2016.08.23 | 295 | -1.13446401 | 2016.08.23.15. |
| ## | 2857 | 200 | 15 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.15. |
| ## | 2858 | 19  | 16 | 2016.08.23 | 341 | -0.33161256 | 2016.08.23.15. |
| ## | 2859 | 220 | 17 | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.15. |
| ## | 2860 | 155 | 18 | 2016.08.23 | 205 | -2.70526034 | 2016.08.23.15. |
| ## | 2861 | 180 | 19 | 2016.08.23 | 180 | 3.14159265  | 2016.08.23.15. |
| ## | 2862 | 160 | 20 | 2016.08.23 | 200 | -2.79252680 | 2016.08.23.15. |
| ## | 2863 | 5   | 21 | 2016.08.23 | 355 | -0.08726646 | 2016.08.23.15. |
| ## | 2864 | 200 | 22 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.15. |
| ## | 2865 | 190 | 23 | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.15. |
| ## | 2866 | 145 | 24 | 2016.08.23 | 215 | -2.53072742 | 2016.08.23.15. |
| ## | 2867 | 200 | 25 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.15. |
| ## | 2868 | 200 | 26 | 2016.08.23 | 160 | 2.79252680  | 2016.08.23.15. |
| ## | 2869 | 195 | 27 | 2016.08.23 | 165 | 2.87979327  | 2016.08.23.15. |
| ## | 2870 | 140 | 28 | 2016.08.23 | 220 | -2.44346095 | 2016.08.23.15. |
| ## | 2871 | 145 | 29 | 2016.08.23 | 215 | -2.53072742 | 2016.08.23.15. |
| ## | 2872 | 165 | 30 | 2016.08.23 | 195 | -2.87979327 | 2016.08.23.15. |
| ## | 2873 | 125 | 31 | 2016.08.23 | 235 | -2.18166156 | 2016.08.23.15. |
| ## | 2874 | 30  | 32 | 2016.08.23 | 330 | -0.52359878 | 2016.08.23.15. |
| ## | 2875 | 170 | 33 | 2016.08.23 | 190 | -2.96705973 | 2016.08.23.15. |
| ## | 2876 | 155 | 34 | 2016.08.23 | 205 | -2.70526034 | 2016.08.23.15. |
| ## | 2877 | 0   | 35 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.15. |
| ## | 2878 | 160 | 36 | 2016.08.23 | 200 | -2.79252680 | 2016.08.23.15. |
| ## | 2879 | 150 | 37 | 2016.08.23 | 210 | -2.61799388 | 2016.08.23.15. |
| ## | 2880 | 0   | 38 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.15. |
| ## | 2881 | 190 | 39 | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.15. |
| ## | 2882 | 20  | 40 | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.15. |
| ## | 2883 | 120 | 41 | 2016.08.23 | 240 | -2.09439510 | 2016.08.23.15. |
| ## | 2884 | 50  | 42 | 2016.08.23 | 310 | -0.87266463 | 2016.08.23.15. |
| ## | 2885 | 310 | 43 | 2016.08.23 | 50  | 0.87266463  | 2016.08.23.15. |
| ## | 2886 | 325 | 44 | 2016.08.23 | 35  | 0.61086524  | 2016.08.23.15. |
| ## | 2887 | 305 | 45 | 2016.08.23 | 55  | 0.95993109  | 2016.08.23.15. |
| ## | 2888 | 340 | 46 | 2016.08.23 | 20  | 0.34906585  | 2016.08.23.15. |
| ## | 2889 | 315 | 47 | 2016.08.23 | 45  | 0.78539816  | 2016.08.23.15. |
| ## | 2890 | 340 | 48 | 2016.08.23 | 20  | 0.34906585  | 2016.08.23.15. |
| ## | 2891 | 305 | 49 | 2016.08.23 | 55  | 0.95993109  | 2016.08.23.15. |
| ## | 2892 | 320 | 50 | 2016.08.23 | 40  | 0.69813170  | 2016.08.23.15. |
| ## | 2893 | 100 | 1  | 2016.08.23 | 260 | -1.74532925 | 2016.08.23.15. |
| ## | 2894 | 310 | 2  | 2016.08.23 | 50  | 0.87266463  | 2016.08.23.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2895 | 260 | 3  | 2016.08.23 | 100 | 1.74532925  | 2016.08.23.15. |
| ## | 2896 | 250 | 4  | 2016.08.23 | 110 | 1.91986218  | 2016.08.23.15. |
| ## | 2897 | 240 | 5  | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.15. |
| ## | 2898 | 240 | 6  | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.15. |
| ## | 2899 | 160 | 7  | 2016.08.23 | 200 | -2.79252680 | 2016.08.23.15. |
| ## | 2900 | 220 | 8  | 2016.08.23 | 140 | 2.44346095  | 2016.08.23.15. |
| ## | 2901 | 270 | 9  | 2016.08.23 | 90  | 1.57079633  | 2016.08.23.15. |
| ## | 2902 | 115 | 10 | 2016.08.23 | 245 | -2.00712864 | 2016.08.23.15. |
| ## | 2903 | 110 | 11 | 2016.08.23 | 250 | -1.91986218 | 2016.08.23.15. |
| ## | 2904 | 305 | 12 | 2016.08.23 | 55  | 0.95993109  | 2016.08.23.15. |
| ## | 2905 | 230 | 13 | 2016.08.23 | 130 | 2.26892803  | 2016.08.23.15. |
| ## | 2906 | 275 | 14 | 2016.08.23 | 85  | 1.48352986  | 2016.08.23.15. |
| ## | 2907 | 320 | 15 | 2016.08.23 | 40  | 0.69813170  | 2016.08.23.15. |
| ## | 2908 | 240 | 16 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.15. |
| ## | 2909 | 190 | 17 | 2016.08.23 | 170 | 2.96705973  | 2016.08.23.15. |
| ## | 2910 | 250 | 18 | 2016.08.23 | 110 | 1.91986218  | 2016.08.23.15. |
| ## | 2911 | 240 | 19 | 2016.08.23 | 120 | 2.09439510  | 2016.08.23.15. |
| ## | 2912 | 250 | 20 | 2016.08.23 | 110 | 1.91986218  | 2016.08.23.15. |
| ## | 2913 | 90  | 21 | 2016.08.23 | 270 | -1.57079633 | 2016.08.23.15. |
| ## | 2914 | 70  | 22 | 2016.08.23 | 290 | -1.22173048 | 2016.08.23.15. |
| ## | 2915 | 280 | 23 | 2016.08.23 | 80  | 1.39626340  | 2016.08.23.15. |
| ## | 2916 | 300 | 24 | 2016.08.23 | 60  | 1.04719755  | 2016.08.23.15. |
| ## | 2917 | 165 | 25 | 2016.08.23 | 195 | -2.87979327 | 2016.08.23.15. |
| ## | 2918 | 270 | 26 | 2016.08.23 | 90  | 1.57079633  | 2016.08.23.15. |
| ## | 2919 | 260 | 27 | 2016.08.23 | 100 | 1.74532925  | 2016.08.23.15. |
| ## | 2920 | 170 | 28 | 2016.08.23 | 190 | -2.96705973 | 2016.08.23.15. |
| ## | 2921 | 280 | 29 | 2016.08.23 | 80  | 1.39626340  | 2016.08.23.15. |
| ## | 2922 | 10  | 30 | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.15. |
| ## | 2923 | 345 | 31 | 2016.08.23 | 15  | 0.26179939  | 2016.08.23.15. |
| ## | 2924 | 60  | 32 | 2016.08.23 | 300 | -1.04719755 | 2016.08.23.15. |
| ## | 2925 | 0   | 33 | 2016.08.23 | 0   | 0.00000000  | 2016.08.23.15. |
| ## | 2926 | 15  | 34 | 2016.08.23 | 345 | -0.26179939 | 2016.08.23.15. |
| ## | 2927 | 10  | 35 | 2016.08.23 | 350 | -0.17453293 | 2016.08.23.15. |
| ## | 2928 | 40  | 36 | 2016.08.23 | 320 | -0.69813170 | 2016.08.23.15. |
| ## | 2929 | 35  | 37 | 2016.08.23 | 325 | -0.61086524 | 2016.08.23.15. |
| ## | 2930 | 50  | 38 | 2016.08.23 | 310 | -0.87266463 | 2016.08.23.15. |
| ## | 2931 | 30  | 39 | 2016.08.23 | 330 | -0.52359878 | 2016.08.23.15. |
| ## | 2932 | 15  | 40 | 2016.08.23 | 345 | -0.26179939 | 2016.08.23.15. |
| ## | 2933 | 20  | 41 | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.15. |
| ## | 2934 | 20  | 42 | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.15. |
| ## | 2935 | 20  | 43 | 2016.08.23 | 340 | -0.34906585 | 2016.08.23.15. |
| ## | 2936 | 210 | 1  | 2016.08.24 | 150 | 2.61799388  | 2016.08.24.14. |
| ## | 2937 | 190 | 2  | 2016.08.24 | 170 | 2.96705973  | 2016.08.24.14. |
| ## | 2938 | 190 | 3  | 2016.08.24 | 170 | 2.96705973  | 2016.08.24.14. |
| ## | 2939 | 160 | 4  | 2016.08.24 | 200 | -2.79252680 | 2016.08.24.14. |
| ## | 2940 | 130 | 5  | 2016.08.24 | 230 | -2.26892803 | 2016.08.24.14. |
| ## | 2941 | 220 | 6  | 2016.08.24 | 140 | 2.44346095  | 2016.08.24.14. |
| ## | 2942 | 90  | 7  | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2943 | 140 | 8  | 2016.08.24 | 220 | -2.44346095 | 2016.08.24.14. |
| ## | 2944 | 130 | 9  | 2016.08.24 | 230 | -2.26892803 | 2016.08.24.14. |
| ## | 2945 | 110 | 10 | 2016.08.24 | 250 | -1.91986218 | 2016.08.24.14. |
| ## | 2946 | 40  | 11 | 2016.08.24 | 320 | -0.69813170 | 2016.08.24.14. |
| ## | 2947 | 110 | 12 | 2016.08.24 | 250 | -1.91986218 | 2016.08.24.14. |
| ## | 2948 | 50  | 13 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.14. |
| ## | 2949 | 0   | 14 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 2950 | 15  | 15 | 2016.08.24 | 345 | -0.26179939 | 2016.08.24.14. |
| ## | 2951 | 350 | 16 | 2016.08.24 | 10  | 0.17453293  | 2016.08.24.14. |
| ## | 2952 | 40  | 17 | 2016.08.24 | 320 | -0.69813170 | 2016.08.24.14. |
| ## | 2953 | 5   | 18 | 2016.08.24 | 355 | -0.08726646 | 2016.08.24.14. |
| ## | 2954 | 20  | 19 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.14. |
| ## | 2955 | 355 | 20 | 2016.08.24 | 5   | 0.08726646  | 2016.08.24.14. |
| ## | 2956 | 60  | 21 | 2016.08.24 | 300 | -1.04719755 | 2016.08.24.14. |
| ## | 2957 | 10  | 22 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.14. |
| ## | 2958 | 20  | 23 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.14. |
| ## | 2959 | 0   | 24 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 2960 | 30  | 25 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.14. |
| ## | 2961 | 355 | 26 | 2016.08.24 | 5   | 0.08726646  | 2016.08.24.14. |
| ## | 2962 | 20  | 27 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.14. |
| ## | 2963 | 50  | 28 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.14. |
| ## | 2964 | 0   | 29 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 2965 | 40  | 30 | 2016.08.24 | 320 | -0.69813170 | 2016.08.24.14. |
| ## | 2966 | 30  | 31 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.14. |
| ## | 2967 | 215 | 1  | 2016.08.24 | 145 | 2.53072742  | 2016.08.24.14. |
| ## | 2968 | 20  | 2  | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.14. |
| ## | 2969 | 290 | 3  | 2016.08.24 | 70  | 1.22173048  | 2016.08.24.14. |
| ## | 2970 | 10  | 4  | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.14. |
| ## | 2971 | 230 | 5  | 2016.08.24 | 130 | 2.26892803  | 2016.08.24.14. |
| ## | 2972 | 280 | 6  | 2016.08.24 | 80  | 1.39626340  | 2016.08.24.14. |
| ## | 2973 | 10  | 7  | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.14. |
| ## | 2974 | 340 | 8  | 2016.08.24 | 20  | 0.34906585  | 2016.08.24.14. |
| ## | 2975 | 0   | 9  | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 2976 | 0   | 10 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 2977 | 40  | 11 | 2016.08.24 | 320 | -0.69813170 | 2016.08.24.14. |
| ## | 2978 | 250 | 12 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.14. |
| ## | 2979 | 240 | 13 | 2016.08.24 | 120 | 2.09439510  | 2016.08.24.14. |
| ## | 2980 | 170 | 14 | 2016.08.24 | 190 | -2.96705973 | 2016.08.24.14. |
| ## | 2981 | 110 | 15 | 2016.08.24 | 250 | -1.91986218 | 2016.08.24.14. |
| ## | 2982 | 30  | 16 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.14. |
| ## | 2983 | 30  | 17 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.14. |
| ## | 2984 | 35  | 18 | 2016.08.24 | 325 | -0.61086524 | 2016.08.24.14. |
| ## | 2985 | 65  | 19 | 2016.08.24 | 295 | -1.13446401 | 2016.08.24.14. |
| ## | 2986 | 120 | 20 | 2016.08.24 | 240 | -2.09439510 | 2016.08.24.14. |
| ## | 2987 | 270 | 21 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.14. |
| ## | 2988 | 320 | 22 | 2016.08.24 | 40  | 0.69813170  | 2016.08.24.14. |
| ## | 2989 | 250 | 23 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.14. |
| ## | 2990 | 5   | 24 | 2016.08.24 | 355 | -0.08726646 | 2016.08.24.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 2991 | 55  | 25 | 2016.08.24 | 305 | -0.95993109 | 2016.08.24.14. |
| ## | 2992 | 180 | 26 | 2016.08.24 | 180 | 3.14159265  | 2016.08.24.14. |
| ## | 2993 | 65  | 27 | 2016.08.24 | 295 | -1.13446401 | 2016.08.24.14. |
| ## | 2994 | 50  | 28 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.14. |
| ## | 2995 | 280 | 29 | 2016.08.24 | 80  | 1.39626340  | 2016.08.24.14. |
| ## | 2996 | 0   | 30 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 2997 | 200 | 31 | 2016.08.24 | 160 | 2.79252680  | 2016.08.24.14. |
| ## | 2998 | 45  | 32 | 2016.08.24 | 315 | -0.78539816 | 2016.08.24.14. |
| ## | 2999 | 190 | 33 | 2016.08.24 | 170 | 2.96705973  | 2016.08.24.14. |
| ## | 3000 | 355 | 34 | 2016.08.24 | 5   | 0.08726646  | 2016.08.24.14. |
| ## | 3001 | 330 | 35 | 2016.08.24 | 30  | 0.52359878  | 2016.08.24.14. |
| ## | 3002 | 10  | 36 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.14. |
| ## | 3003 | 15  | 37 | 2016.08.24 | 345 | -0.26179939 | 2016.08.24.14. |
| ## | 3004 | 0   | 38 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 3005 | 350 | 39 | 2016.08.24 | 10  | 0.17453293  | 2016.08.24.14. |
| ## | 3006 | 30  | 40 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.14. |
| ## | 3007 | 5   | 41 | 2016.08.24 | 355 | -0.08726646 | 2016.08.24.14. |
| ## | 3008 | 10  | 42 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.14. |
| ## | 3009 | 30  | 43 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.14. |
| ## | 3010 | 0   | 44 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.14. |
| ## | 3011 | 130 | 1  | 2016.08.24 | 230 | -2.26892803 | 2016.08.24.15. |
| ## | 3012 | 330 | 2  | 2016.08.24 | 30  | 0.52359878  | 2016.08.24.15. |
| ## | 3013 | 300 | 3  | 2016.08.24 | 60  | 1.04719755  | 2016.08.24.15. |
| ## | 3014 | 70  | 4  | 2016.08.24 | 290 | -1.22173048 | 2016.08.24.15. |
| ## | 3015 | 15  | 5  | 2016.08.24 | 345 | -0.26179939 | 2016.08.24.15. |
| ## | 3016 | 355 | 6  | 2016.08.24 | 5   | 0.08726646  | 2016.08.24.15. |
| ## | 3017 | 255 | 7  | 2016.08.24 | 105 | 1.83259571  | 2016.08.24.15. |
| ## | 3018 | 140 | 8  | 2016.08.24 | 220 | -2.44346095 | 2016.08.24.15. |
| ## | 3019 | 80  | 9  | 2016.08.24 | 280 | -1.39626340 | 2016.08.24.15. |
| ## | 3020 | 355 | 10 | 2016.08.24 | 5   | 0.08726646  | 2016.08.24.15. |
| ## | 3021 | 260 | 11 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.15. |
| ## | 3022 | 60  | 12 | 2016.08.24 | 300 | -1.04719755 | 2016.08.24.15. |
| ## | 3023 | 200 | 13 | 2016.08.24 | 160 | 2.79252680  | 2016.08.24.15. |
| ## | 3024 | 110 | 14 | 2016.08.24 | 250 | -1.91986218 | 2016.08.24.15. |
| ## | 3025 | 160 | 15 | 2016.08.24 | 200 | -2.79252680 | 2016.08.24.15. |
| ## | 3026 | 50  | 16 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.15. |
| ## | 3027 | 260 | 17 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.15. |
| ## | 3028 | 280 | 18 | 2016.08.24 | 80  | 1.39626340  | 2016.08.24.15. |
| ## | 3029 | 10  | 19 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.15. |
| ## | 3030 | 20  | 20 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.15. |
| ## | 3031 | 70  | 21 | 2016.08.24 | 290 | -1.22173048 | 2016.08.24.15. |
| ## | 3032 | 50  | 22 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.15. |
| ## | 3033 | 90  | 23 | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.15. |
| ## | 3034 | 80  | 24 | 2016.08.24 | 280 | -1.39626340 | 2016.08.24.15. |
| ## | 3035 | 90  | 25 | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.15. |
| ## | 3036 | 90  | 26 | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.15. |
| ## | 3037 | 10  | 27 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.15. |
| ## | 3038 | 90  | 28 | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 3039 | 55  | 1  | 2016.08.24 | 305 | -0.95993109 | 2016.08.24.16. |
| ## 3040 | 240 | 2  | 2016.08.24 | 120 | 2.09439510  | 2016.08.24.16. |
| ## 3041 | 220 | 3  | 2016.08.24 | 140 | 2.44346095  | 2016.08.24.16. |
| ## 3042 | 230 | 4  | 2016.08.24 | 130 | 2.26892803  | 2016.08.24.16. |
| ## 3043 | 270 | 5  | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3044 | 170 | 6  | 2016.08.24 | 190 | -2.96705973 | 2016.08.24.16. |
| ## 3045 | 250 | 7  | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.16. |
| ## 3046 | 240 | 8  | 2016.08.24 | 120 | 2.09439510  | 2016.08.24.16. |
| ## 3047 | 260 | 9  | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.16. |
| ## 3048 | 240 | 10 | 2016.08.24 | 120 | 2.09439510  | 2016.08.24.16. |
| ## 3049 | 250 | 11 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.16. |
| ## 3050 | 190 | 12 | 2016.08.24 | 170 | 2.96705973  | 2016.08.24.16. |
| ## 3051 | 165 | 13 | 2016.08.24 | 195 | -2.87979327 | 2016.08.24.16. |
| ## 3052 | 310 | 14 | 2016.08.24 | 50  | 0.87266463  | 2016.08.24.16. |
| ## 3053 | 170 | 15 | 2016.08.24 | 190 | -2.96705973 | 2016.08.24.16. |
| ## 3054 | 150 | 16 | 2016.08.24 | 210 | -2.61799388 | 2016.08.24.16. |
| ## 3055 | 320 | 17 | 2016.08.24 | 40  | 0.69813170  | 2016.08.24.16. |
| ## 3056 | 220 | 18 | 2016.08.24 | 140 | 2.44346095  | 2016.08.24.16. |
| ## 3057 | 250 | 19 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.16. |
| ## 3058 | 340 | 20 | 2016.08.24 | 20  | 0.34906585  | 2016.08.24.16. |
| ## 3059 | 340 | 21 | 2016.08.24 | 20  | 0.34906585  | 2016.08.24.16. |
| ## 3060 | 350 | 22 | 2016.08.24 | 10  | 0.17453293  | 2016.08.24.16. |
| ## 3061 | 175 | 23 | 2016.08.24 | 185 | -3.05432619 | 2016.08.24.16. |
| ## 3062 | 270 | 24 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3063 | 10  | 25 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.16. |
| ## 3064 | 30  | 26 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.16. |
| ## 3065 | 10  | 27 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.16. |
| ## 3066 | 340 | 28 | 2016.08.24 | 20  | 0.34906585  | 2016.08.24.16. |
| ## 3067 | 50  | 29 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.16. |
| ## 3068 | 25  | 30 | 2016.08.24 | 335 | -0.43633231 | 2016.08.24.16. |
| ## 3069 | 60  | 31 | 2016.08.24 | 300 | -1.04719755 | 2016.08.24.16. |
| ## 3070 | 100 | 1  | 2016.08.24 | 260 | -1.74532925 | 2016.08.24.16. |
| ## 3071 | 10  | 2  | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.16. |
| ## 3072 | 90  | 3  | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.16. |
| ## 3073 | 10  | 4  | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.16. |
| ## 3074 | 90  | 5  | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.16. |
| ## 3075 | 95  | 6  | 2016.08.24 | 265 | -1.65806279 | 2016.08.24.16. |
| ## 3076 | 25  | 7  | 2016.08.24 | 335 | -0.43633231 | 2016.08.24.16. |
| ## 3077 | 10  | 8  | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.16. |
| ## 3078 | 35  | 9  | 2016.08.24 | 325 | -0.61086524 | 2016.08.24.16. |
| ## 3079 | 200 | 10 | 2016.08.24 | 160 | 2.79252680  | 2016.08.24.16. |
| ## 3080 | 30  | 11 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.16. |
| ## 3081 | 150 | 12 | 2016.08.24 | 210 | -2.61799388 | 2016.08.24.16. |
| ## 3082 | 10  | 13 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.16. |
| ## 3083 | 20  | 14 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.16. |
| ## 3084 | 200 | 15 | 2016.08.24 | 160 | 2.79252680  | 2016.08.24.16. |
| ## 3085 | 190 | 16 | 2016.08.24 | 170 | 2.96705973  | 2016.08.24.16. |
| ## 3086 | 190 | 17 | 2016.08.24 | 170 | 2.96705973  | 2016.08.24.16. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 3087 | 340 | 18 | 2016.08.24 | 20  | 0.34906585  | 2016.08.24.16. |
| ## 3088 | 30  | 19 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.16. |
| ## 3089 | 345 | 20 | 2016.08.24 | 15  | 0.26179939  | 2016.08.24.16. |
| ## 3090 | 50  | 21 | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.16. |
| ## 3091 | 90  | 22 | 2016.08.24 | 270 | -1.57079633 | 2016.08.24.16. |
| ## 3092 | 185 | 23 | 2016.08.24 | 175 | 3.05432619  | 2016.08.24.16. |
| ## 3093 | 20  | 24 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.16. |
| ## 3094 | 40  | 25 | 2016.08.24 | 320 | -0.69813170 | 2016.08.24.16. |
| ## 3095 | 120 | 26 | 2016.08.24 | 240 | -2.09439510 | 2016.08.24.16. |
| ## 3096 | 260 | 27 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.16. |
| ## 3097 | 260 | 28 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.16. |
| ## 3098 | 270 | 29 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3099 | 265 | 30 | 2016.08.24 | 95  | 1.65806279  | 2016.08.24.16. |
| ## 3100 | 260 | 31 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.16. |
| ## 3101 | 270 | 32 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3102 | 260 | 33 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.16. |
| ## 3103 | 265 | 34 | 2016.08.24 | 95  | 1.65806279  | 2016.08.24.16. |
| ## 3104 | 310 | 35 | 2016.08.24 | 50  | 0.87266463  | 2016.08.24.16. |
| ## 3105 | 270 | 36 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3106 | 320 | 37 | 2016.08.24 | 40  | 0.69813170  | 2016.08.24.16. |
| ## 3107 | 290 | 38 | 2016.08.24 | 70  | 1.22173048  | 2016.08.24.16. |
| ## 3108 | 250 | 39 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.16. |
| ## 3109 | 250 | 40 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.16. |
| ## 3110 | 245 | 41 | 2016.08.24 | 115 | 2.00712864  | 2016.08.24.16. |
| ## 3111 | 275 | 42 | 2016.08.24 | 85  | 1.48352986  | 2016.08.24.16. |
| ## 3112 | 270 | 43 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3113 | 320 | 44 | 2016.08.24 | 40  | 0.69813170  | 2016.08.24.16. |
| ## 3114 | 290 | 45 | 2016.08.24 | 70  | 1.22173048  | 2016.08.24.16. |
| ## 3115 | 270 | 46 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.16. |
| ## 3116 | 260 | 47 | 2016.08.24 | 100 | 1.74532925  | 2016.08.24.16. |
| ## 3117 | 50  | 1  | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.15. |
| ## 3118 | 60  | 2  | 2016.08.24 | 300 | -1.04719755 | 2016.08.24.15. |
| ## 3119 | 60  | 3  | 2016.08.24 | 300 | -1.04719755 | 2016.08.24.15. |
| ## 3120 | 270 | 4  | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.15. |
| ## 3121 | 40  | 5  | 2016.08.24 | 320 | -0.69813170 | 2016.08.24.15. |
| ## 3122 | 150 | 6  | 2016.08.24 | 210 | -2.61799388 | 2016.08.24.15. |
| ## 3123 | 50  | 7  | 2016.08.24 | 310 | -0.87266463 | 2016.08.24.15. |
| ## 3124 | 20  | 8  | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.15. |
| ## 3125 | 60  | 9  | 2016.08.24 | 300 | -1.04719755 | 2016.08.24.15. |
| ## 3126 | 35  | 10 | 2016.08.24 | 325 | -0.61086524 | 2016.08.24.15. |
| ## 3127 | 0   | 11 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.15. |
| ## 3128 | 20  | 12 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.15. |
| ## 3129 | 10  | 13 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.15. |
| ## 3130 | 10  | 14 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.15. |
| ## 3131 | 10  | 15 | 2016.08.24 | 350 | -0.17453293 | 2016.08.24.15. |
| ## 3132 | 250 | 16 | 2016.08.24 | 110 | 1.91986218  | 2016.08.24.15. |
| ## 3133 | 70  | 17 | 2016.08.24 | 290 | -1.22173048 | 2016.08.24.15. |
| ## 3134 | 140 | 18 | 2016.08.24 | 220 | -2.44346095 | 2016.08.24.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3135 | 30  | 19 | 2016.08.24 | 330 | -0.52359878 | 2016.08.24.15. |
| ## | 3136 | 270 | 20 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.15. |
| ## | 3137 | 160 | 21 | 2016.08.24 | 200 | -2.79252680 | 2016.08.24.15. |
| ## | 3138 | 330 | 22 | 2016.08.24 | 30  | 0.52359878  | 2016.08.24.15. |
| ## | 3139 | 165 | 23 | 2016.08.24 | 195 | -2.87979327 | 2016.08.24.15. |
| ## | 3140 | 70  | 24 | 2016.08.24 | 290 | -1.22173048 | 2016.08.24.15. |
| ## | 3141 | 0   | 25 | 2016.08.24 | 0   | 0.00000000  | 2016.08.24.15. |
| ## | 3142 | 70  | 26 | 2016.08.24 | 290 | -1.22173048 | 2016.08.24.15. |
| ## | 3143 | 175 | 27 | 2016.08.24 | 185 | -3.05432619 | 2016.08.24.15. |
| ## | 3144 | 340 | 28 | 2016.08.24 | 20  | 0.34906585  | 2016.08.24.15. |
| ## | 3145 | 20  | 29 | 2016.08.24 | 340 | -0.34906585 | 2016.08.24.15. |
| ## | 3146 | 270 | 30 | 2016.08.24 | 90  | 1.57079633  | 2016.08.24.15. |
| ## | 3147 | 110 | 31 | 2016.08.24 | 250 | -1.91986218 | 2016.08.24.15. |
| ## | 3148 | 170 | 32 | 2016.08.24 | 190 | -2.96705973 | 2016.08.24.15. |
| ## | 3149 | 260 | 1  | 2016.08.25 | 100 | 1.74532925  | 2016.08.25.13. |
| ## | 3150 | 270 | 2  | 2016.08.25 | 90  | 1.57079633  | 2016.08.25.13. |
| ## | 3151 | 260 | 3  | 2016.08.25 | 100 | 1.74532925  | 2016.08.25.13. |
| ## | 3152 | 160 | 4  | 2016.08.25 | 200 | -2.79252680 | 2016.08.25.13. |
| ## | 3153 | 250 | 5  | 2016.08.25 | 110 | 1.91986218  | 2016.08.25.13. |
| ## | 3154 | 130 | 6  | 2016.08.25 | 230 | -2.26892803 | 2016.08.25.13. |
| ## | 3155 | 150 | 7  | 2016.08.25 | 210 | -2.61799388 | 2016.08.25.13. |
| ## | 3156 | 5   | 8  | 2016.08.25 | 355 | -0.08726646 | 2016.08.25.13. |
| ## | 3157 | 150 | 9  | 2016.08.25 | 210 | -2.61799388 | 2016.08.25.13. |
| ## | 3158 | 150 | 10 | 2016.08.25 | 210 | -2.61799388 | 2016.08.25.13. |
| ## | 3159 | 90  | 11 | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3160 | 210 | 12 | 2016.08.25 | 150 | 2.61799388  | 2016.08.25.13. |
| ## | 3161 | 70  | 13 | 2016.08.25 | 290 | -1.22173048 | 2016.08.25.13. |
| ## | 3162 | 200 | 14 | 2016.08.25 | 160 | 2.79252680  | 2016.08.25.13. |
| ## | 3163 | 190 | 15 | 2016.08.25 | 170 | 2.96705973  | 2016.08.25.13. |
| ## | 3164 | 185 | 16 | 2016.08.25 | 175 | 3.05432619  | 2016.08.25.13. |
| ## | 3165 | 310 | 17 | 2016.08.25 | 50  | 0.87266463  | 2016.08.25.13. |
| ## | 3166 | 320 | 18 | 2016.08.25 | 40  | 0.69813170  | 2016.08.25.13. |
| ## | 3167 | 270 | 19 | 2016.08.25 | 90  | 1.57079633  | 2016.08.25.13. |
| ## | 3168 | 150 | 1  | 2016.08.25 | 210 | -2.61799388 | 2016.08.25.13. |
| ## | 3169 | 355 | 2  | 2016.08.25 | 5   | 0.08726646  | 2016.08.25.13. |
| ## | 3170 | 150 | 3  | 2016.08.25 | 210 | -2.61799388 | 2016.08.25.13. |
| ## | 3171 | 110 | 4  | 2016.08.25 | 250 | -1.91986218 | 2016.08.25.13. |
| ## | 3172 | 90  | 5  | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3173 | 130 | 6  | 2016.08.25 | 230 | -2.26892803 | 2016.08.25.13. |
| ## | 3174 | 165 | 7  | 2016.08.25 | 195 | -2.87979327 | 2016.08.25.13. |
| ## | 3175 | 90  | 8  | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3176 | 100 | 9  | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |
| ## | 3177 | 160 | 10 | 2016.08.25 | 200 | -2.79252680 | 2016.08.25.13. |
| ## | 3178 | 240 | 11 | 2016.08.25 | 120 | 2.09439510  | 2016.08.25.13. |
| ## | 3179 | 160 | 12 | 2016.08.25 | 200 | -2.79252680 | 2016.08.25.13. |
| ## | 3180 | 120 | 13 | 2016.08.25 | 240 | -2.09439510 | 2016.08.25.13. |
| ## | 3181 | 190 | 14 | 2016.08.25 | 170 | 2.96705973  | 2016.08.25.13. |
| ## | 3182 | 140 | 15 | 2016.08.25 | 220 | -2.44346095 | 2016.08.25.13. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3183 | 160 | 16 | 2016.08.25 | 200 | -2.79252680 | 2016.08.25.13. |
| ## | 3184 | 90  | 17 | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3185 | 270 | 18 | 2016.08.25 | 90  | 1.57079633  | 2016.08.25.13. |
| ## | 3186 | 350 | 19 | 2016.08.25 | 10  | 0.17453293  | 2016.08.25.13. |
| ## | 3187 | 160 | 20 | 2016.08.25 | 200 | -2.79252680 | 2016.08.25.13. |
| ## | 3188 | 275 | 21 | 2016.08.25 | 85  | 1.48352986  | 2016.08.25.13. |
| ## | 3189 | 210 | 22 | 2016.08.25 | 150 | 2.61799388  | 2016.08.25.13. |
| ## | 3190 | 210 | 23 | 2016.08.25 | 150 | 2.61799388  | 2016.08.25.13. |
| ## | 3191 | 250 | 24 | 2016.08.25 | 110 | 1.91986218  | 2016.08.25.13. |
| ## | 3192 | 85  | 25 | 2016.08.25 | 275 | -1.48352986 | 2016.08.25.13. |
| ## | 3193 | 140 | 26 | 2016.08.25 | 220 | -2.44346095 | 2016.08.25.13. |
| ## | 3194 | 340 | 27 | 2016.08.25 | 20  | 0.34906585  | 2016.08.25.13. |
| ## | 3195 | 330 | 28 | 2016.08.25 | 30  | 0.52359878  | 2016.08.25.13. |
| ## | 3196 | 100 | 29 | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |
| ## | 3197 | 320 | 30 | 2016.08.25 | 40  | 0.69813170  | 2016.08.25.13. |
| ## | 3198 | 50  | 31 | 2016.08.25 | 310 | -0.87266463 | 2016.08.25.13. |
| ## | 3199 | 5   | 32 | 2016.08.25 | 355 | -0.08726646 | 2016.08.25.13. |
| ## | 3200 | 110 | 33 | 2016.08.25 | 250 | -1.91986218 | 2016.08.25.13. |
| ## | 3201 | 70  | 34 | 2016.08.25 | 290 | -1.22173048 | 2016.08.25.13. |
| ## | 3202 | 190 | 35 | 2016.08.25 | 170 | 2.96705973  | 2016.08.25.13. |
| ## | 3203 | 275 | 36 | 2016.08.25 | 85  | 1.48352986  | 2016.08.25.13. |
| ## | 3204 | 160 | 37 | 2016.08.25 | 200 | -2.79252680 | 2016.08.25.13. |
| ## | 3205 | 195 | 38 | 2016.08.25 | 165 | 2.87979327  | 2016.08.25.13. |
| ## | 3206 | 355 | 39 | 2016.08.25 | 5   | 0.08726646  | 2016.08.25.13. |
| ## | 3207 | 135 | 40 | 2016.08.25 | 225 | -2.35619449 | 2016.08.25.13. |
| ## | 3208 | 120 | 41 | 2016.08.25 | 240 | -2.09439510 | 2016.08.25.13. |
| ## | 3209 | 100 | 42 | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |
| ## | 3210 | 270 | 43 | 2016.08.25 | 90  | 1.57079633  | 2016.08.25.13. |
| ## | 3211 | 250 | 44 | 2016.08.25 | 110 | 1.91986218  | 2016.08.25.13. |
| ## | 3212 | 315 | 45 | 2016.08.25 | 45  | 0.78539816  | 2016.08.25.13. |
| ## | 3213 | 75  | 46 | 2016.08.25 | 285 | -1.30899694 | 2016.08.25.13. |
| ## | 3214 | 255 | 47 | 2016.08.25 | 105 | 1.83259571  | 2016.08.25.13. |
| ## | 3215 | 90  | 48 | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3216 | 60  | 1  | 2016.08.25 | 300 | -1.04719755 | 2016.08.25.13. |
| ## | 3217 | 60  | 2  | 2016.08.25 | 300 | -1.04719755 | 2016.08.25.13. |
| ## | 3218 | 100 | 3  | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |
| ## | 3219 | 90  | 4  | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3220 | 90  | 5  | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3221 | 90  | 6  | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3222 | 120 | 7  | 2016.08.25 | 240 | -2.09439510 | 2016.08.25.13. |
| ## | 3223 | 30  | 8  | 2016.08.25 | 330 | -0.52359878 | 2016.08.25.13. |
| ## | 3224 | 100 | 9  | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |
| ## | 3225 | 90  | 10 | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3226 | 120 | 11 | 2016.08.25 | 240 | -2.09439510 | 2016.08.25.13. |
| ## | 3227 | 90  | 12 | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3228 | 120 | 13 | 2016.08.25 | 240 | -2.09439510 | 2016.08.25.13. |
| ## | 3229 | 50  | 14 | 2016.08.25 | 310 | -0.87266463 | 2016.08.25.13. |
| ## | 3230 | 100 | 15 | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3231 | 90  | 16 | 2016.08.25 | 270 | -1.57079633 | 2016.08.25.13. |
| ## | 3232 | 110 | 17 | 2016.08.25 | 250 | -1.91986218 | 2016.08.25.13. |
| ## | 3233 | 0   | 18 | 2016.08.25 | 0   | 0.00000000  | 2016.08.25.13. |
| ## | 3234 | 85  | 19 | 2016.08.25 | 275 | -1.48352986 | 2016.08.25.13. |
| ## | 3235 | 40  | 20 | 2016.08.25 | 320 | -0.69813170 | 2016.08.25.13. |
| ## | 3236 | 80  | 21 | 2016.08.25 | 280 | -1.39626340 | 2016.08.25.13. |
| ## | 3237 | 170 | 22 | 2016.08.25 | 190 | -2.96705973 | 2016.08.25.13. |
| ## | 3238 | 130 | 23 | 2016.08.25 | 230 | -2.26892803 | 2016.08.25.13. |
| ## | 3239 | 85  | 24 | 2016.08.25 | 275 | -1.48352986 | 2016.08.25.13. |
| ## | 3240 | 85  | 25 | 2016.08.25 | 275 | -1.48352986 | 2016.08.25.13. |
| ## | 3241 | 260 | 26 | 2016.08.25 | 100 | 1.74532925  | 2016.08.25.13. |
| ## | 3242 | 60  | 27 | 2016.08.25 | 300 | -1.04719755 | 2016.08.25.13. |
| ## | 3243 | 100 | 28 | 2016.08.25 | 260 | -1.74532925 | 2016.08.25.13. |
| ## | 3244 | 350 | 29 | 2016.08.25 | 10  | 0.17453293  | 2016.08.25.13. |
| ## | 3245 | 130 | 30 | 2016.08.25 | 230 | -2.26892803 | 2016.08.25.13. |
| ## | 3246 | 140 | 31 | 2016.08.25 | 220 | -2.44346095 | 2016.08.25.13. |
| ## | 3247 | 220 | 32 | 2016.08.25 | 140 | 2.44346095  | 2016.08.25.13. |
| ## | 3248 | 5   | 33 | 2016.08.25 | 355 | -0.08726646 | 2016.08.25.13. |
| ## | 3249 | 20  | 34 | 2016.08.25 | 340 | -0.34906585 | 2016.08.25.13. |
| ## | 3250 | 40  | 35 | 2016.08.25 | 320 | -0.69813170 | 2016.08.25.13. |
| ## | 3251 | 35  | 36 | 2016.08.25 | 325 | -0.61086524 | 2016.08.25.13. |
| ## | 3252 | 260 | 37 | 2016.08.25 | 100 | 1.74532925  | 2016.08.25.13. |
| ## | 3253 | 330 | 38 | 2016.08.25 | 30  | 0.52359878  | 2016.08.25.13. |
| ## | 3254 | 320 | 1  | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.13. |
| ## | 3255 | 20  | 2  | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.13. |
| ## | 3256 | 5   | 3  | 2016.08.26 | 355 | -0.08726646 | 2016.08.26.13. |
| ## | 3257 | 330 | 4  | 2016.08.26 | 30  | 0.52359878  | 2016.08.26.13. |
| ## | 3258 | 290 | 5  | 2016.08.26 | 70  | 1.22173048  | 2016.08.26.13. |
| ## | 3259 | 5   | 6  | 2016.08.26 | 355 | -0.08726646 | 2016.08.26.13. |
| ## | 3260 | 310 | 7  | 2016.08.26 | 50  | 0.87266463  | 2016.08.26.13. |
| ## | 3261 | 340 | 8  | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.13. |
| ## | 3262 | 60  | 9  | 2016.08.26 | 300 | -1.04719755 | 2016.08.26.13. |
| ## | 3263 | 300 | 10 | 2016.08.26 | 60  | 1.04719755  | 2016.08.26.13. |
| ## | 3264 | 30  | 11 | 2016.08.26 | 330 | -0.52359878 | 2016.08.26.13. |
| ## | 3265 | 310 | 12 | 2016.08.26 | 50  | 0.87266463  | 2016.08.26.13. |
| ## | 3266 | 20  | 13 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.13. |
| ## | 3267 | 260 | 14 | 2016.08.26 | 100 | 1.74532925  | 2016.08.26.13. |
| ## | 3268 | 50  | 15 | 2016.08.26 | 310 | -0.87266463 | 2016.08.26.13. |
| ## | 3269 | 30  | 16 | 2016.08.26 | 330 | -0.52359878 | 2016.08.26.13. |
| ## | 3270 | 340 | 17 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.13. |
| ## | 3271 | 60  | 18 | 2016.08.26 | 300 | -1.04719755 | 2016.08.26.13. |
| ## | 3272 | 70  | 19 | 2016.08.26 | 290 | -1.22173048 | 2016.08.26.13. |
| ## | 3273 | 125 | 20 | 2016.08.26 | 235 | -2.18166156 | 2016.08.26.13. |
| ## | 3274 | 335 | 21 | 2016.08.26 | 25  | 0.43633231  | 2016.08.26.13. |
| ## | 3275 | 180 | 22 | 2016.08.26 | 180 | 3.14159265  | 2016.08.26.13. |
| ## | 3276 | 325 | 23 | 2016.08.26 | 35  | 0.61086524  | 2016.08.26.13. |
| ## | 3277 | 230 | 24 | 2016.08.26 | 130 | 2.26892803  | 2016.08.26.13. |
| ## | 3278 | 240 | 25 | 2016.08.26 | 120 | 2.09439510  | 2016.08.26.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3279 | 220 | 26 | 2016.08.26 | 140 | 2.44346095  | 2016.08.26.13. |
| ## | 3280 | 215 | 27 | 2016.08.26 | 145 | 2.53072742  | 2016.08.26.13. |
| ## | 3281 | 225 | 28 | 2016.08.26 | 135 | 2.35619449  | 2016.08.26.13. |
| ## | 3282 | 355 | 1  | 2016.08.26 | 5   | 0.08726646  | 2016.08.26.14. |
| ## | 3283 | 260 | 2  | 2016.08.26 | 100 | 1.74532925  | 2016.08.26.14. |
| ## | 3284 | 105 | 3  | 2016.08.26 | 255 | -1.83259571 | 2016.08.26.14. |
| ## | 3285 | 130 | 4  | 2016.08.26 | 230 | -2.26892803 | 2016.08.26.14. |
| ## | 3286 | 205 | 5  | 2016.08.26 | 155 | 2.70526034  | 2016.08.26.14. |
| ## | 3287 | 125 | 6  | 2016.08.26 | 235 | -2.18166156 | 2016.08.26.14. |
| ## | 3288 | 165 | 7  | 2016.08.26 | 195 | -2.87979327 | 2016.08.26.14. |
| ## | 3289 | 150 | 8  | 2016.08.26 | 210 | -2.61799388 | 2016.08.26.14. |
| ## | 3290 | 30  | 9  | 2016.08.26 | 330 | -0.52359878 | 2016.08.26.14. |
| ## | 3291 | 60  | 10 | 2016.08.26 | 300 | -1.04719755 | 2016.08.26.14. |
| ## | 3292 | 80  | 11 | 2016.08.26 | 280 | -1.39626340 | 2016.08.26.14. |
| ## | 3293 | 95  | 12 | 2016.08.26 | 265 | -1.65806279 | 2016.08.26.14. |
| ## | 3294 | 0   | 13 | 2016.08.26 | 0   | 0.00000000  | 2016.08.26.14. |
| ## | 3295 | 50  | 14 | 2016.08.26 | 310 | -0.87266463 | 2016.08.26.14. |
| ## | 3296 | 350 | 15 | 2016.08.26 | 10  | 0.17453293  | 2016.08.26.14. |
| ## | 3297 | 120 | 16 | 2016.08.26 | 240 | -2.09439510 | 2016.08.26.14. |
| ## | 3298 | 95  | 17 | 2016.08.26 | 265 | -1.65806279 | 2016.08.26.14. |
| ## | 3299 | 105 | 18 | 2016.08.26 | 255 | -1.83259571 | 2016.08.26.14. |
| ## | 3300 | 115 | 19 | 2016.08.26 | 245 | -2.00712864 | 2016.08.26.14. |
| ## | 3301 | 350 | 20 | 2016.08.26 | 10  | 0.17453293  | 2016.08.26.14. |
| ## | 3302 | 80  | 21 | 2016.08.26 | 280 | -1.39626340 | 2016.08.26.14. |
| ## | 3303 | 60  | 22 | 2016.08.26 | 300 | -1.04719755 | 2016.08.26.14. |
| ## | 3304 | 185 | 1  | 2016.08.26 | 175 | 3.05432619  | 2016.08.26.15. |
| ## | 3305 | 305 | 2  | 2016.08.26 | 55  | 0.95993109  | 2016.08.26.15. |
| ## | 3306 | 125 | 3  | 2016.08.26 | 235 | -2.18166156 | 2016.08.26.15. |
| ## | 3307 | 130 | 4  | 2016.08.26 | 230 | -2.26892803 | 2016.08.26.15. |
| ## | 3308 | 290 | 5  | 2016.08.26 | 70  | 1.22173048  | 2016.08.26.15. |
| ## | 3309 | 15  | 6  | 2016.08.26 | 345 | -0.26179939 | 2016.08.26.15. |
| ## | 3310 | 15  | 7  | 2016.08.26 | 345 | -0.26179939 | 2016.08.26.15. |
| ## | 3311 | 80  | 8  | 2016.08.26 | 280 | -1.39626340 | 2016.08.26.15. |
| ## | 3312 | 290 | 9  | 2016.08.26 | 70  | 1.22173048  | 2016.08.26.15. |
| ## | 3313 | 25  | 10 | 2016.08.26 | 335 | -0.43633231 | 2016.08.26.15. |
| ## | 3314 | 240 | 11 | 2016.08.26 | 120 | 2.09439510  | 2016.08.26.15. |
| ## | 3315 | 40  | 12 | 2016.08.26 | 320 | -0.69813170 | 2016.08.26.15. |
| ## | 3316 | 330 | 13 | 2016.08.26 | 30  | 0.52359878  | 2016.08.26.15. |
| ## | 3317 | 170 | 14 | 2016.08.26 | 190 | -2.96705973 | 2016.08.26.15. |
| ## | 3318 | 310 | 15 | 2016.08.26 | 50  | 0.87266463  | 2016.08.26.15. |
| ## | 3319 | 250 | 16 | 2016.08.26 | 110 | 1.91986218  | 2016.08.26.15. |
| ## | 3320 | 250 | 17 | 2016.08.26 | 110 | 1.91986218  | 2016.08.26.15. |
| ## | 3321 | 255 | 18 | 2016.08.26 | 105 | 1.83259571  | 2016.08.26.15. |
| ## | 3322 | 50  | 19 | 2016.08.26 | 310 | -0.87266463 | 2016.08.26.15. |
| ## | 3323 | 335 | 20 | 2016.08.26 | 25  | 0.43633231  | 2016.08.26.15. |
| ## | 3324 | 80  | 21 | 2016.08.26 | 280 | -1.39626340 | 2016.08.26.15. |
| ## | 3325 | 335 | 22 | 2016.08.26 | 25  | 0.43633231  | 2016.08.26.15. |
| ## | 3326 | 335 | 23 | 2016.08.26 | 25  | 0.43633231  | 2016.08.26.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3327 | 20  | 24 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.15. |
| ## | 3328 | 290 | 25 | 2016.08.26 | 70  | 1.22173048  | 2016.08.26.15. |
| ## | 3329 | 15  | 26 | 2016.08.26 | 345 | -0.26179939 | 2016.08.26.15. |
| ## | 3330 | 260 | 27 | 2016.08.26 | 100 | 1.74532925  | 2016.08.26.15. |
| ## | 3331 | 85  | 28 | 2016.08.26 | 275 | -1.48352986 | 2016.08.26.15. |
| ## | 3332 | 0   | 29 | 2016.08.26 | 0   | 0.00000000  | 2016.08.26.15. |
| ## | 3333 | 10  | 30 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.15. |
| ## | 3334 | 345 | 31 | 2016.08.26 | 15  | 0.26179939  | 2016.08.26.15. |
| ## | 3335 | 300 | 32 | 2016.08.26 | 60  | 1.04719755  | 2016.08.26.15. |
| ## | 3336 | 30  | 33 | 2016.08.26 | 330 | -0.52359878 | 2016.08.26.15. |
| ## | 3337 | 320 | 34 | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.15. |
| ## | 3338 | 20  | 35 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.15. |
| ## | 3339 | 10  | 36 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.15. |
| ## | 3340 | 315 | 37 | 2016.08.26 | 45  | 0.78539816  | 2016.08.26.15. |
| ## | 3341 | 20  | 38 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.15. |
| ## | 3342 | 290 | 39 | 2016.08.26 | 70  | 1.22173048  | 2016.08.26.15. |
| ## | 3343 | 5   | 40 | 2016.08.26 | 355 | -0.08726646 | 2016.08.26.15. |
| ## | 3344 | 335 | 41 | 2016.08.26 | 25  | 0.43633231  | 2016.08.26.15. |
| ## | 3345 | 345 | 42 | 2016.08.26 | 15  | 0.26179939  | 2016.08.26.15. |
| ## | 3346 | 325 | 43 | 2016.08.26 | 35  | 0.61086524  | 2016.08.26.15. |
| ## | 3347 | 10  | 44 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.15. |
| ## | 3348 | 340 | 45 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3349 | 50  | 46 | 2016.08.26 | 310 | -0.87266463 | 2016.08.26.15. |
| ## | 3350 | 340 | 47 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3351 | 150 | 48 | 2016.08.26 | 210 | -2.61799388 | 2016.08.26.15. |
| ## | 3352 | 345 | 49 | 2016.08.26 | 15  | 0.26179939  | 2016.08.26.15. |
| ## | 3353 | 35  | 50 | 2016.08.26 | 325 | -0.61086524 | 2016.08.26.15. |
| ## | 3354 | 310 | 51 | 2016.08.26 | 50  | 0.87266463  | 2016.08.26.15. |
| ## | 3355 | 0   | 52 | 2016.08.26 | 0   | 0.00000000  | 2016.08.26.15. |
| ## | 3356 | 345 | 53 | 2016.08.26 | 15  | 0.26179939  | 2016.08.26.15. |
| ## | 3357 | 285 | 54 | 2016.08.26 | 75  | 1.30899694  | 2016.08.26.15. |
| ## | 3358 | 340 | 55 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3359 | 330 | 56 | 2016.08.26 | 30  | 0.52359878  | 2016.08.26.15. |
| ## | 3360 | 250 | 57 | 2016.08.26 | 110 | 1.91986218  | 2016.08.26.15. |
| ## | 3361 | 335 | 58 | 2016.08.26 | 25  | 0.43633231  | 2016.08.26.15. |
| ## | 3362 | 245 | 59 | 2016.08.26 | 115 | 2.00712864  | 2016.08.26.15. |
| ## | 3363 | 345 | 60 | 2016.08.26 | 15  | 0.26179939  | 2016.08.26.15. |
| ## | 3364 | 320 | 61 | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.15. |
| ## | 3365 | 340 | 62 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3366 | 270 | 63 | 2016.08.26 | 90  | 1.57079633  | 2016.08.26.15. |
| ## | 3367 | 330 | 64 | 2016.08.26 | 30  | 0.52359878  | 2016.08.26.15. |
| ## | 3368 | 320 | 65 | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.15. |
| ## | 3369 | 355 | 66 | 2016.08.26 | 5   | 0.08726646  | 2016.08.26.15. |
| ## | 3370 | 310 | 67 | 2016.08.26 | 50  | 0.87266463  | 2016.08.26.15. |
| ## | 3371 | 320 | 68 | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.15. |
| ## | 3372 | 270 | 69 | 2016.08.26 | 90  | 1.57079633  | 2016.08.26.15. |
| ## | 3373 | 340 | 70 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3374 | 270 | 71 | 2016.08.26 | 90  | 1.57079633  | 2016.08.26.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3375 | 350 | 72 | 2016.08.26 | 10  | 0.17453293  | 2016.08.26.15. |
| ## | 3376 | 355 | 73 | 2016.08.26 | 5   | 0.08726646  | 2016.08.26.15. |
| ## | 3377 | 340 | 74 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3378 | 320 | 75 | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.15. |
| ## | 3379 | 185 | 1  | 2016.08.26 | 175 | 3.05432619  | 2016.08.26.15. |
| ## | 3380 | 210 | 2  | 2016.08.26 | 150 | 2.61799388  | 2016.08.26.15. |
| ## | 3381 | 170 | 3  | 2016.08.26 | 190 | -2.96705973 | 2016.08.26.15. |
| ## | 3382 | 220 | 4  | 2016.08.26 | 140 | 2.44346095  | 2016.08.26.15. |
| ## | 3383 | 190 | 5  | 2016.08.26 | 170 | 2.96705973  | 2016.08.26.15. |
| ## | 3384 | 260 | 6  | 2016.08.26 | 100 | 1.74532925  | 2016.08.26.15. |
| ## | 3385 | 200 | 7  | 2016.08.26 | 160 | 2.79252680  | 2016.08.26.15. |
| ## | 3386 | 270 | 8  | 2016.08.26 | 90  | 1.57079633  | 2016.08.26.15. |
| ## | 3387 | 160 | 9  | 2016.08.26 | 200 | -2.79252680 | 2016.08.26.15. |
| ## | 3388 | 260 | 10 | 2016.08.26 | 100 | 1.74532925  | 2016.08.26.15. |
| ## | 3389 | 140 | 11 | 2016.08.26 | 220 | -2.44346095 | 2016.08.26.15. |
| ## | 3390 | 205 | 12 | 2016.08.26 | 155 | 2.70526034  | 2016.08.26.15. |
| ## | 3391 | 340 | 13 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3392 | 340 | 14 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3393 | 5   | 15 | 2016.08.26 | 355 | -0.08726646 | 2016.08.26.15. |
| ## | 3394 | 40  | 16 | 2016.08.26 | 320 | -0.69813170 | 2016.08.26.15. |
| ## | 3395 | 10  | 17 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.15. |
| ## | 3396 | 10  | 18 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.15. |
| ## | 3397 | 60  | 19 | 2016.08.26 | 300 | -1.04719755 | 2016.08.26.15. |
| ## | 3398 | 5   | 20 | 2016.08.26 | 355 | -0.08726646 | 2016.08.26.15. |
| ## | 3399 | 10  | 21 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.15. |
| ## | 3400 | 15  | 22 | 2016.08.26 | 345 | -0.26179939 | 2016.08.26.15. |
| ## | 3401 | 0   | 23 | 2016.08.26 | 0   | 0.00000000  | 2016.08.26.15. |
| ## | 3402 | 50  | 24 | 2016.08.26 | 310 | -0.87266463 | 2016.08.26.15. |
| ## | 3403 | 340 | 25 | 2016.08.26 | 20  | 0.34906585  | 2016.08.26.15. |
| ## | 3404 | 20  | 26 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.15. |
| ## | 3405 | 105 | 1  | 2016.08.26 | 255 | -1.83259571 | 2016.08.26.14. |
| ## | 3406 | 125 | 2  | 2016.08.26 | 235 | -2.18166156 | 2016.08.26.14. |
| ## | 3407 | 85  | 3  | 2016.08.26 | 275 | -1.48352986 | 2016.08.26.14. |
| ## | 3408 | 200 | 4  | 2016.08.26 | 160 | 2.79252680  | 2016.08.26.14. |
| ## | 3409 | 50  | 5  | 2016.08.26 | 310 | -0.87266463 | 2016.08.26.14. |
| ## | 3410 | 145 | 6  | 2016.08.26 | 215 | -2.53072742 | 2016.08.26.14. |
| ## | 3411 | 135 | 7  | 2016.08.26 | 225 | -2.35619449 | 2016.08.26.14. |
| ## | 3412 | 140 | 8  | 2016.08.26 | 220 | -2.44346095 | 2016.08.26.14. |
| ## | 3413 | 160 | 9  | 2016.08.26 | 200 | -2.79252680 | 2016.08.26.14. |
| ## | 3414 | 155 | 10 | 2016.08.26 | 205 | -2.70526034 | 2016.08.26.14. |
| ## | 3415 | 130 | 11 | 2016.08.26 | 230 | -2.26892803 | 2016.08.26.14. |
| ## | 3416 | 200 | 12 | 2016.08.26 | 160 | 2.79252680  | 2016.08.26.14. |
| ## | 3417 | 120 | 13 | 2016.08.26 | 240 | -2.09439510 | 2016.08.26.14. |
| ## | 3418 | 100 | 14 | 2016.08.26 | 260 | -1.74532925 | 2016.08.26.14. |
| ## | 3419 | 165 | 15 | 2016.08.26 | 195 | -2.87979327 | 2016.08.26.14. |
| ## | 3420 | 15  | 16 | 2016.08.26 | 345 | -0.26179939 | 2016.08.26.14. |
| ## | 3421 | 5   | 17 | 2016.08.26 | 355 | -0.08726646 | 2016.08.26.14. |
| ## | 3422 | 20  | 18 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3423 | 40  | 19 | 2016.08.26 | 320 | -0.69813170 | 2016.08.26.14. |
| ## | 3424 | 10  | 20 | 2016.08.26 | 350 | -0.17453293 | 2016.08.26.14. |
| ## | 3425 | 355 | 21 | 2016.08.26 | 5   | 0.08726646  | 2016.08.26.14. |
| ## | 3426 | 20  | 22 | 2016.08.26 | 340 | -0.34906585 | 2016.08.26.14. |
| ## | 3427 | 30  | 23 | 2016.08.26 | 330 | -0.52359878 | 2016.08.26.14. |
| ## | 3428 | 350 | 24 | 2016.08.26 | 10  | 0.17453293  | 2016.08.26.14. |
| ## | 3429 | 320 | 25 | 2016.08.26 | 40  | 0.69813170  | 2016.08.26.14. |
| ## | 3430 | 0   | 26 | 2016.08.26 | 0   | 0.00000000  | 2016.08.26.14. |
| ## | 3431 | 350 | 27 | 2016.08.26 | 10  | 0.17453293  | 2016.08.26.14. |
| ## | 3432 | 355 | 28 | 2016.08.26 | 5   | 0.08726646  | 2016.08.26.14. |
| ## | 3433 | 25  | 29 | 2016.08.26 | 335 | -0.43633231 | 2016.08.26.14. |
| ## | 3434 | 30  | 30 | 2016.08.26 | 330 | -0.52359878 | 2016.08.26.14. |
| ## | 3435 | 355 | 31 | 2016.08.26 | 5   | 0.08726646  | 2016.08.26.14. |
| ## | 3436 | 330 | 32 | 2016.08.26 | 30  | 0.52359878  | 2016.08.26.14. |
| ## | 3437 | 25  | 33 | 2016.08.26 | 335 | -0.43633231 | 2016.08.26.14. |
| ## | 3438 | 85  | 1  | 2016.08.31 | 275 | -1.48352986 | 2016.08.31.15. |
| ## | 3439 | 10  | 2  | 2016.08.31 | 350 | -0.17453293 | 2016.08.31.15. |
| ## | 3440 | 320 | 3  | 2016.08.31 | 40  | 0.69813170  | 2016.08.31.15. |
| ## | 3441 | 95  | 4  | 2016.08.31 | 265 | -1.65806279 | 2016.08.31.15. |
| ## | 3442 | 155 | 5  | 2016.08.31 | 205 | -2.70526034 | 2016.08.31.15. |
| ## | 3443 | 180 | 6  | 2016.08.31 | 180 | 3.14159265  | 2016.08.31.15. |
| ## | 3444 | 150 | 7  | 2016.08.31 | 210 | -2.61799388 | 2016.08.31.15. |
| ## | 3445 | 80  | 8  | 2016.08.31 | 280 | -1.39626340 | 2016.08.31.15. |
| ## | 3446 | 95  | 9  | 2016.08.31 | 265 | -1.65806279 | 2016.08.31.15. |
| ## | 3447 | 70  | 10 | 2016.08.31 | 290 | -1.22173048 | 2016.08.31.15. |
| ## | 3448 | 150 | 11 | 2016.08.31 | 210 | -2.61799388 | 2016.08.31.15. |
| ## | 3449 | 145 | 12 | 2016.08.31 | 215 | -2.53072742 | 2016.08.31.15. |
| ## | 3450 | 110 | 13 | 2016.08.31 | 250 | -1.91986218 | 2016.08.31.15. |
| ## | 3451 | 115 | 14 | 2016.08.31 | 245 | -2.00712864 | 2016.08.31.15. |
| ## | 3452 | 110 | 15 | 2016.08.31 | 250 | -1.91986218 | 2016.08.31.15. |
| ## | 3453 | 65  | 16 | 2016.08.31 | 295 | -1.13446401 | 2016.08.31.15. |
| ## | 3454 | 330 | 17 | 2016.08.31 | 30  | 0.52359878  | 2016.08.31.15. |
| ## | 3455 | 340 | 18 | 2016.08.31 | 20  | 0.34906585  | 2016.08.31.15. |
| ## | 3456 | 330 | 19 | 2016.08.31 | 30  | 0.52359878  | 2016.08.31.15. |
| ## | 3457 | 320 | 20 | 2016.08.31 | 40  | 0.69813170  | 2016.08.31.15. |
| ## | 3458 | 310 | 21 | 2016.08.31 | 50  | 0.87266463  | 2016.08.31.15. |
| ## | 3459 | 290 | 22 | 2016.08.31 | 70  | 1.22173048  | 2016.08.31.15. |
| ## | 3460 | 325 | 23 | 2016.08.31 | 35  | 0.61086524  | 2016.08.31.15. |
| ## | 3461 | 310 | 24 | 2016.08.31 | 50  | 0.87266463  | 2016.08.31.15. |
| ## | 3462 | 300 | 25 | 2016.08.31 | 60  | 1.04719755  | 2016.08.31.15. |
| ## | 3463 | 340 | 26 | 2016.08.31 | 20  | 0.34906585  | 2016.08.31.15. |
| ## | 3464 | 220 | 27 | 2016.08.31 | 140 | 2.44346095  | 2016.08.31.15. |
| ## | 3465 | 80  | 28 | 2016.08.31 | 280 | -1.39626340 | 2016.08.31.15. |
| ## | 3466 | 70  | 29 | 2016.08.31 | 290 | -1.22173048 | 2016.08.31.15. |
| ## | 3467 | 90  | 30 | 2016.08.31 | 270 | -1.57079633 | 2016.08.31.15. |
| ## | 3468 | 150 | 31 | 2016.08.31 | 210 | -2.61799388 | 2016.08.31.15. |
| ## | 3469 | 145 | 32 | 2016.08.31 | 215 | -2.53072742 | 2016.08.31.15. |
| ## | 3470 | 140 | 33 | 2016.08.31 | 220 | -2.44346095 | 2016.08.31.15. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3471 | 140 | 34 | 2016.08.31 | 220 | -2.44346095 | 2016.08.31.15. |
| ## | 3472 | 90  | 35 | 2016.08.31 | 270 | -1.57079633 | 2016.08.31.15. |
| ## | 3473 | 95  | 36 | 2016.08.31 | 265 | -1.65806279 | 2016.08.31.15. |
| ## | 3474 | 120 | 37 | 2016.08.31 | 240 | -2.09439510 | 2016.08.31.15. |
| ## | 3475 | 355 | 38 | 2016.08.31 | 5   | 0.08726646  | 2016.08.31.15. |
| ## | 3476 | 40  | 1  | 2016.08.31 | 320 | -0.69813170 | 2016.08.31.15. |
| ## | 3477 | 330 | 2  | 2016.08.31 | 30  | 0.52359878  | 2016.08.31.15. |
| ## | 3478 | 90  | 3  | 2016.08.31 | 270 | -1.57079633 | 2016.08.31.15. |
| ## | 3479 | 340 | 4  | 2016.08.31 | 20  | 0.34906585  | 2016.08.31.15. |
| ## | 3480 | 80  | 5  | 2016.08.31 | 280 | -1.39626340 | 2016.08.31.15. |
| ## | 3481 | 75  | 6  | 2016.08.31 | 285 | -1.30899694 | 2016.08.31.15. |
| ## | 3482 | 70  | 7  | 2016.08.31 | 290 | -1.22173048 | 2016.08.31.15. |
| ## | 3483 | 45  | 8  | 2016.08.31 | 315 | -0.78539816 | 2016.08.31.15. |
| ## | 3484 | 100 | 9  | 2016.08.31 | 260 | -1.74532925 | 2016.08.31.15. |
| ## | 3485 | 80  | 10 | 2016.08.31 | 280 | -1.39626340 | 2016.08.31.15. |
| ## | 3486 | 75  | 11 | 2016.08.31 | 285 | -1.30899694 | 2016.08.31.15. |
| ## | 3487 | 85  | 12 | 2016.08.31 | 275 | -1.48352986 | 2016.08.31.15. |
| ## | 3488 | 90  | 13 | 2016.08.31 | 270 | -1.57079633 | 2016.08.31.15. |
| ## | 3489 | 40  | 14 | 2016.08.31 | 320 | -0.69813170 | 2016.08.31.15. |
| ## | 3490 | 140 | 15 | 2016.08.31 | 220 | -2.44346095 | 2016.08.31.15. |
| ## | 3491 | 100 | 16 | 2016.08.31 | 260 | -1.74532925 | 2016.08.31.15. |
| ## | 3492 | 80  | 17 | 2016.08.31 | 280 | -1.39626340 | 2016.08.31.15. |
| ## | 3493 | 20  | 18 | 2016.08.31 | 340 | -0.34906585 | 2016.08.31.15. |
| ## | 3494 | 70  | 19 | 2016.08.31 | 290 | -1.22173048 | 2016.08.31.15. |
| ## | 3495 | 40  | 20 | 2016.08.31 | 320 | -0.69813170 | 2016.08.31.15. |
| ## | 3496 | 40  | 21 | 2016.08.31 | 320 | -0.69813170 | 2016.08.31.15. |
| ## | 3497 | 335 | 22 | 2016.08.31 | 25  | 0.43633231  | 2016.08.31.15. |
| ## | 3498 | 70  | 23 | 2016.08.31 | 290 | -1.22173048 | 2016.08.31.15. |
| ## | 3499 | 20  | 24 | 2016.08.31 | 340 | -0.34906585 | 2016.08.31.15. |
| ## | 3500 | 240 | 25 | 2016.08.31 | 120 | 2.09439510  | 2016.08.31.15. |
| ## | 3501 | 240 | 26 | 2016.08.31 | 120 | 2.09439510  | 2016.08.31.15. |
| ## | 3502 | 5   | 27 | 2016.08.31 | 355 | -0.08726646 | 2016.08.31.15. |
| ## | 3503 | 220 | 28 | 2016.08.31 | 140 | 2.44346095  | 2016.08.31.15. |
| ## | 3504 | 5   | 29 | 2016.08.31 | 355 | -0.08726646 | 2016.08.31.15. |
| ## | 3505 | 350 | 30 | 2016.08.31 | 10  | 0.17453293  | 2016.08.31.15. |
| ## | 3506 | 210 | 31 | 2016.08.31 | 150 | 2.61799388  | 2016.08.31.15. |
| ## | 3507 | 85  | 1  | 2016.09.01 | 275 | -1.48352986 | 2016.09.01.13. |
| ## | 3508 | 100 | 2  | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.13. |
| ## | 3509 | 90  | 3  | 2016.09.01 | 270 | -1.57079633 | 2016.09.01.13. |
| ## | 3510 | 25  | 4  | 2016.09.01 | 335 | -0.43633231 | 2016.09.01.13. |
| ## | 3511 | 60  | 5  | 2016.09.01 | 300 | -1.04719755 | 2016.09.01.13. |
| ## | 3512 | 70  | 6  | 2016.09.01 | 290 | -1.22173048 | 2016.09.01.13. |
| ## | 3513 | 80  | 7  | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.13. |
| ## | 3514 | 95  | 8  | 2016.09.01 | 265 | -1.65806279 | 2016.09.01.13. |
| ## | 3515 | 60  | 9  | 2016.09.01 | 300 | -1.04719755 | 2016.09.01.13. |
| ## | 3516 | 90  | 10 | 2016.09.01 | 270 | -1.57079633 | 2016.09.01.13. |
| ## | 3517 | 200 | 11 | 2016.09.01 | 160 | 2.79252680  | 2016.09.01.13. |
| ## | 3518 | 30  | 12 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3519 | 210 | 13 | 2016.09.01 | 150 | 2.61799388  | 2016.09.01.13. |
| ## | 3520 | 20  | 14 | 2016.09.01 | 340 | -0.34906585 | 2016.09.01.13. |
| ## | 3521 | 160 | 15 | 2016.09.01 | 200 | -2.79252680 | 2016.09.01.13. |
| ## | 3522 | 180 | 16 | 2016.09.01 | 180 | 3.14159265  | 2016.09.01.13. |
| ## | 3523 | 55  | 17 | 2016.09.01 | 305 | -0.95993109 | 2016.09.01.13. |
| ## | 3524 | 280 | 18 | 2016.09.01 | 80  | 1.39626340  | 2016.09.01.13. |
| ## | 3525 | 285 | 19 | 2016.09.01 | 75  | 1.30899694  | 2016.09.01.13. |
| ## | 3526 | 280 | 20 | 2016.09.01 | 80  | 1.39626340  | 2016.09.01.13. |
| ## | 3527 | 260 | 21 | 2016.09.01 | 100 | 1.74532925  | 2016.09.01.13. |
| ## | 3528 | 290 | 22 | 2016.09.01 | 70  | 1.22173048  | 2016.09.01.13. |
| ## | 3529 | 270 | 23 | 2016.09.01 | 90  | 1.57079633  | 2016.09.01.13. |
| ## | 3530 | 330 | 24 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.13. |
| ## | 3531 | 300 | 25 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.13. |
| ## | 3532 | 330 | 26 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.13. |
| ## | 3533 | 310 | 27 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.13. |
| ## | 3534 | 320 | 28 | 2016.09.01 | 40  | 0.69813170  | 2016.09.01.13. |
| ## | 3535 | 290 | 29 | 2016.09.01 | 70  | 1.22173048  | 2016.09.01.13. |
| ## | 3536 | 300 | 30 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.13. |
| ## | 3537 | 340 | 31 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.13. |
| ## | 3538 | 260 | 32 | 2016.09.01 | 100 | 1.74532925  | 2016.09.01.13. |
| ## | 3539 | 335 | 33 | 2016.09.01 | 25  | 0.43633231  | 2016.09.01.13. |
| ## | 3540 | 320 | 34 | 2016.09.01 | 40  | 0.69813170  | 2016.09.01.13. |
| ## | 3541 | 340 | 35 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.13. |
| ## | 3542 | 300 | 36 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.13. |
| ## | 3543 | 315 | 37 | 2016.09.01 | 45  | 0.78539816  | 2016.09.01.13. |
| ## | 3544 | 285 | 38 | 2016.09.01 | 75  | 1.30899694  | 2016.09.01.13. |
| ## | 3545 | 280 | 39 | 2016.09.01 | 80  | 1.39626340  | 2016.09.01.13. |
| ## | 3546 | 350 | 40 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.13. |
| ## | 3547 | 30  | 41 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.13. |
| ## | 3548 | 200 | 42 | 2016.09.01 | 160 | 2.79252680  | 2016.09.01.13. |
| ## | 3549 | 105 | 43 | 2016.09.01 | 255 | -1.83259571 | 2016.09.01.13. |
| ## | 3550 | 0   | 44 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3551 | 140 | 45 | 2016.09.01 | 220 | -2.44346095 | 2016.09.01.13. |
| ## | 3552 | 195 | 46 | 2016.09.01 | 165 | 2.87979327  | 2016.09.01.13. |
| ## | 3553 | 120 | 47 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.13. |
| ## | 3554 | 70  | 48 | 2016.09.01 | 290 | -1.22173048 | 2016.09.01.13. |
| ## | 3555 | 330 | 49 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.13. |
| ## | 3556 | 30  | 50 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.13. |
| ## | 3557 | 335 | 51 | 2016.09.01 | 25  | 0.43633231  | 2016.09.01.13. |
| ## | 3558 | 25  | 52 | 2016.09.01 | 335 | -0.43633231 | 2016.09.01.13. |
| ## | 3559 | 265 | 53 | 2016.09.01 | 95  | 1.65806279  | 2016.09.01.13. |
| ## | 3560 | 250 | 54 | 2016.09.01 | 110 | 1.91986218  | 2016.09.01.13. |
| ## | 3561 | 150 | 55 | 2016.09.01 | 210 | -2.61799388 | 2016.09.01.13. |
| ## | 3562 | 170 | 56 | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.13. |
| ## | 3563 | 310 | 57 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.13. |
| ## | 3564 | 0   | 58 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3565 | 120 | 59 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.13. |
| ## | 3566 | 100 | 60 | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.13. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3567 | 95  | 61 | 2016.09.01 | 265 | -1.65806279 | 2016.09.01.13. |
| ## | 3568 | 355 | 62 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.13. |
| ## | 3569 | 40  | 63 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.13. |
| ## | 3570 | 350 | 64 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.13. |
| ## | 3571 | 330 | 65 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.13. |
| ## | 3572 | 320 | 66 | 2016.09.01 | 40  | 0.69813170  | 2016.09.01.13. |
| ## | 3573 | 105 | 1  | 2016.09.01 | 255 | -1.83259571 | 2016.09.01.13. |
| ## | 3574 | 30  | 2  | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.13. |
| ## | 3575 | 335 | 3  | 2016.09.01 | 25  | 0.43633231  | 2016.09.01.13. |
| ## | 3576 | 280 | 4  | 2016.09.01 | 80  | 1.39626340  | 2016.09.01.13. |
| ## | 3577 | 75  | 5  | 2016.09.01 | 285 | -1.30899694 | 2016.09.01.13. |
| ## | 3578 | 220 | 6  | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.13. |
| ## | 3579 | 50  | 7  | 2016.09.01 | 310 | -0.87266463 | 2016.09.01.13. |
| ## | 3580 | 55  | 8  | 2016.09.01 | 305 | -0.95993109 | 2016.09.01.13. |
| ## | 3581 | 170 | 9  | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.13. |
| ## | 3582 | 90  | 10 | 2016.09.01 | 270 | -1.57079633 | 2016.09.01.13. |
| ## | 3583 | 150 | 11 | 2016.09.01 | 210 | -2.61799388 | 2016.09.01.13. |
| ## | 3584 | 15  | 12 | 2016.09.01 | 345 | -0.26179939 | 2016.09.01.13. |
| ## | 3585 | 40  | 13 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.13. |
| ## | 3586 | 120 | 14 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.13. |
| ## | 3587 | 310 | 15 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.13. |
| ## | 3588 | 230 | 16 | 2016.09.01 | 130 | 2.26892803  | 2016.09.01.13. |
| ## | 3589 | 35  | 17 | 2016.09.01 | 325 | -0.61086524 | 2016.09.01.13. |
| ## | 3590 | 40  | 18 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.13. |
| ## | 3591 | 355 | 19 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.13. |
| ## | 3592 | 70  | 20 | 2016.09.01 | 290 | -1.22173048 | 2016.09.01.13. |
| ## | 3593 | 110 | 21 | 2016.09.01 | 250 | -1.91986218 | 2016.09.01.13. |
| ## | 3594 | 80  | 22 | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.13. |
| ## | 3595 | 290 | 23 | 2016.09.01 | 70  | 1.22173048  | 2016.09.01.13. |
| ## | 3596 | 250 | 24 | 2016.09.01 | 110 | 1.91986218  | 2016.09.01.13. |
| ## | 3597 | 240 | 25 | 2016.09.01 | 120 | 2.09439510  | 2016.09.01.13. |
| ## | 3598 | 220 | 26 | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.13. |
| ## | 3599 | 265 | 27 | 2016.09.01 | 95  | 1.65806279  | 2016.09.01.13. |
| ## | 3600 | 245 | 28 | 2016.09.01 | 115 | 2.00712864  | 2016.09.01.13. |
| ## | 3601 | 250 | 29 | 2016.09.01 | 110 | 1.91986218  | 2016.09.01.13. |
| ## | 3602 | 250 | 30 | 2016.09.01 | 110 | 1.91986218  | 2016.09.01.13. |
| ## | 3603 | 260 | 31 | 2016.09.01 | 100 | 1.74532925  | 2016.09.01.13. |
| ## | 3604 | 210 | 32 | 2016.09.01 | 150 | 2.61799388  | 2016.09.01.13. |
| ## | 3605 | 310 | 33 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.13. |
| ## | 3606 | 310 | 34 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.13. |
| ## | 3607 | 300 | 35 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.13. |
| ## | 3608 | 55  | 1  | 2016.09.01 | 305 | -0.95993109 | 2016.09.01.13. |
| ## | 3609 | 30  | 2  | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.13. |
| ## | 3610 | 100 | 3  | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.13. |
| ## | 3611 | 150 | 4  | 2016.09.01 | 210 | -2.61799388 | 2016.09.01.13. |
| ## | 3612 | 85  | 5  | 2016.09.01 | 275 | -1.48352986 | 2016.09.01.13. |
| ## | 3613 | 110 | 6  | 2016.09.01 | 250 | -1.91986218 | 2016.09.01.13. |
| ## | 3614 | 60  | 7  | 2016.09.01 | 300 | -1.04719755 | 2016.09.01.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3615 | 90  | 8  | 2016.09.01 | 270 | -1.57079633 | 2016.09.01.13. |
| ## | 3616 | 60  | 9  | 2016.09.01 | 300 | -1.04719755 | 2016.09.01.13. |
| ## | 3617 | 90  | 10 | 2016.09.01 | 270 | -1.57079633 | 2016.09.01.13. |
| ## | 3618 | 120 | 11 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.13. |
| ## | 3619 | 355 | 12 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.13. |
| ## | 3620 | 345 | 13 | 2016.09.01 | 15  | 0.26179939  | 2016.09.01.13. |
| ## | 3621 | 350 | 14 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.13. |
| ## | 3622 | 0   | 15 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3623 | 0   | 16 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3624 | 345 | 17 | 2016.09.01 | 15  | 0.26179939  | 2016.09.01.13. |
| ## | 3625 | 355 | 18 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.13. |
| ## | 3626 | 0   | 19 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3627 | 5   | 20 | 2016.09.01 | 355 | -0.08726646 | 2016.09.01.13. |
| ## | 3628 | 345 | 21 | 2016.09.01 | 15  | 0.26179939  | 2016.09.01.13. |
| ## | 3629 | 0   | 22 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3630 | 330 | 23 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.13. |
| ## | 3631 | 5   | 24 | 2016.09.01 | 355 | -0.08726646 | 2016.09.01.13. |
| ## | 3632 | 340 | 25 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.13. |
| ## | 3633 | 5   | 26 | 2016.09.01 | 355 | -0.08726646 | 2016.09.01.13. |
| ## | 3634 | 355 | 27 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.13. |
| ## | 3635 | 0   | 28 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.13. |
| ## | 3636 | 340 | 1  | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.14. |
| ## | 3637 | 310 | 2  | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3638 | 350 | 3  | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.14. |
| ## | 3639 | 335 | 4  | 2016.09.01 | 25  | 0.43633231  | 2016.09.01.14. |
| ## | 3640 | 345 | 5  | 2016.09.01 | 15  | 0.26179939  | 2016.09.01.14. |
| ## | 3641 | 350 | 6  | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.14. |
| ## | 3642 | 350 | 7  | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.14. |
| ## | 3643 | 310 | 8  | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3644 | 310 | 9  | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3645 | 330 | 10 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.14. |
| ## | 3646 | 325 | 11 | 2016.09.01 | 35  | 0.61086524  | 2016.09.01.14. |
| ## | 3647 | 340 | 12 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.14. |
| ## | 3648 | 290 | 13 | 2016.09.01 | 70  | 1.22173048  | 2016.09.01.14. |
| ## | 3649 | 300 | 14 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.14. |
| ## | 3650 | 350 | 15 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.14. |
| ## | 3651 | 335 | 16 | 2016.09.01 | 25  | 0.43633231  | 2016.09.01.14. |
| ## | 3652 | 110 | 17 | 2016.09.01 | 250 | -1.91986218 | 2016.09.01.14. |
| ## | 3653 | 225 | 18 | 2016.09.01 | 135 | 2.35619449  | 2016.09.01.14. |
| ## | 3654 | 255 | 19 | 2016.09.01 | 105 | 1.83259571  | 2016.09.01.14. |
| ## | 3655 | 355 | 20 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.14. |
| ## | 3656 | 15  | 21 | 2016.09.01 | 345 | -0.26179939 | 2016.09.01.14. |
| ## | 3657 | 255 | 22 | 2016.09.01 | 105 | 1.83259571  | 2016.09.01.14. |
| ## | 3658 | 90  | 23 | 2016.09.01 | 270 | -1.57079633 | 2016.09.01.14. |
| ## | 3659 | 30  | 24 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.14. |
| ## | 3660 | 330 | 25 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.14. |
| ## | 3661 | 300 | 26 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.14. |
| ## | 3662 | 40  | 27 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3663 | 45  | 28 | 2016.09.01 | 315 | -0.78539816 | 2016.09.01.14. |
| ## | 3664 | 80  | 29 | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.14. |
| ## | 3665 | 220 | 30 | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.14. |
| ## | 3666 | 230 | 31 | 2016.09.01 | 130 | 2.26892803  | 2016.09.01.14. |
| ## | 3667 | 220 | 32 | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.14. |
| ## | 3668 | 145 | 33 | 2016.09.01 | 215 | -2.53072742 | 2016.09.01.14. |
| ## | 3669 | 100 | 34 | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.14. |
| ## | 3670 | 170 | 35 | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.14. |
| ## | 3671 | 120 | 36 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.14. |
| ## | 3672 | 270 | 37 | 2016.09.01 | 90  | 1.57079633  | 2016.09.01.14. |
| ## | 3673 | 340 | 38 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.14. |
| ## | 3674 | 345 | 39 | 2016.09.01 | 15  | 0.26179939  | 2016.09.01.14. |
| ## | 3675 | 345 | 40 | 2016.09.01 | 15  | 0.26179939  | 2016.09.01.14. |
| ## | 3676 | 0   | 41 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.14. |
| ## | 3677 | 350 | 42 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.14. |
| ## | 3678 | 310 | 43 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3679 | 340 | 44 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.14. |
| ## | 3680 | 300 | 45 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.14. |
| ## | 3681 | 340 | 46 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.14. |
| ## | 3682 | 310 | 47 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3683 | 110 | 48 | 2016.09.01 | 250 | -1.91986218 | 2016.09.01.14. |
| ## | 3684 | 280 | 49 | 2016.09.01 | 80  | 1.39626340  | 2016.09.01.14. |
| ## | 3685 | 285 | 50 | 2016.09.01 | 75  | 1.30899694  | 2016.09.01.14. |
| ## | 3686 | 0   | 51 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.14. |
| ## | 3687 | 300 | 52 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.14. |
| ## | 3688 | 265 | 53 | 2016.09.01 | 95  | 1.65806279  | 2016.09.01.14. |
| ## | 3689 | 200 | 54 | 2016.09.01 | 160 | 2.79252680  | 2016.09.01.14. |
| ## | 3690 | 80  | 55 | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.14. |
| ## | 3691 | 310 | 56 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3692 | 255 | 57 | 2016.09.01 | 105 | 1.83259571  | 2016.09.01.14. |
| ## | 3693 | 120 | 58 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.14. |
| ## | 3694 | 0   | 59 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.14. |
| ## | 3695 | 270 | 60 | 2016.09.01 | 90  | 1.57079633  | 2016.09.01.14. |
| ## | 3696 | 35  | 61 | 2016.09.01 | 325 | -0.61086524 | 2016.09.01.14. |
| ## | 3697 | 80  | 62 | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.14. |
| ## | 3698 | 100 | 63 | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.14. |
| ## | 3699 | 200 | 64 | 2016.09.01 | 160 | 2.79252680  | 2016.09.01.14. |
| ## | 3700 | 220 | 65 | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.14. |
| ## | 3701 | 80  | 66 | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.14. |
| ## | 3702 | 330 | 67 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.14. |
| ## | 3703 | 330 | 68 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.14. |
| ## | 3704 | 340 | 69 | 2016.09.01 | 20  | 0.34906585  | 2016.09.01.14. |
| ## | 3705 | 10  | 70 | 2016.09.01 | 350 | -0.17453293 | 2016.09.01.14. |
| ## | 3706 | 350 | 71 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.14. |
| ## | 3707 | 290 | 72 | 2016.09.01 | 70  | 1.22173048  | 2016.09.01.14. |
| ## | 3708 | 30  | 73 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.14. |
| ## | 3709 | 10  | 74 | 2016.09.01 | 350 | -0.17453293 | 2016.09.01.14. |
| ## | 3710 | 310 | 75 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3711 | 25  | 76 | 2016.09.01 | 335 | -0.43633231 | 2016.09.01.14. |
| ## | 3712 | 330 | 77 | 2016.09.01 | 30  | 0.52359878  | 2016.09.01.14. |
| ## | 3713 | 20  | 78 | 2016.09.01 | 340 | -0.34906585 | 2016.09.01.14. |
| ## | 3714 | 355 | 79 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.14. |
| ## | 3715 | 310 | 80 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.14. |
| ## | 3716 | 320 | 81 | 2016.09.01 | 40  | 0.69813170  | 2016.09.01.14. |
| ## | 3717 | 105 | 1  | 2016.09.01 | 255 | -1.83259571 | 2016.09.01.15. |
| ## | 3718 | 160 | 2  | 2016.09.01 | 200 | -2.79252680 | 2016.09.01.15. |
| ## | 3719 | 170 | 3  | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.15. |
| ## | 3720 | 250 | 4  | 2016.09.01 | 110 | 1.91986218  | 2016.09.01.15. |
| ## | 3721 | 165 | 5  | 2016.09.01 | 195 | -2.87979327 | 2016.09.01.15. |
| ## | 3722 | 260 | 6  | 2016.09.01 | 100 | 1.74532925  | 2016.09.01.15. |
| ## | 3723 | 95  | 7  | 2016.09.01 | 265 | -1.65806279 | 2016.09.01.15. |
| ## | 3724 | 160 | 8  | 2016.09.01 | 200 | -2.79252680 | 2016.09.01.15. |
| ## | 3725 | 100 | 9  | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.15. |
| ## | 3726 | 120 | 10 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.15. |
| ## | 3727 | 120 | 11 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.15. |
| ## | 3728 | 100 | 12 | 2016.09.01 | 260 | -1.74532925 | 2016.09.01.15. |
| ## | 3729 | 320 | 13 | 2016.09.01 | 40  | 0.69813170  | 2016.09.01.15. |
| ## | 3730 | 120 | 14 | 2016.09.01 | 240 | -2.09439510 | 2016.09.01.15. |
| ## | 3731 | 200 | 15 | 2016.09.01 | 160 | 2.79252680  | 2016.09.01.15. |
| ## | 3732 | 10  | 16 | 2016.09.01 | 350 | -0.17453293 | 2016.09.01.15. |
| ## | 3733 | 40  | 17 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.15. |
| ## | 3734 | 5   | 18 | 2016.09.01 | 355 | -0.08726646 | 2016.09.01.15. |
| ## | 3735 | 45  | 19 | 2016.09.01 | 315 | -0.78539816 | 2016.09.01.15. |
| ## | 3736 | 350 | 20 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.15. |
| ## | 3737 | 30  | 21 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.15. |
| ## | 3738 | 10  | 22 | 2016.09.01 | 350 | -0.17453293 | 2016.09.01.15. |
| ## | 3739 | 30  | 23 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.15. |
| ## | 3740 | 30  | 24 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.15. |
| ## | 3741 | 5   | 25 | 2016.09.01 | 355 | -0.08726646 | 2016.09.01.15. |
| ## | 3742 | 40  | 26 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.15. |
| ## | 3743 | 355 | 27 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.15. |
| ## | 3744 | 130 | 1  | 2016.09.01 | 230 | -2.26892803 | 2016.09.01.15. |
| ## | 3745 | 155 | 2  | 2016.09.01 | 205 | -2.70526034 | 2016.09.01.15. |
| ## | 3746 | 105 | 3  | 2016.09.01 | 255 | -1.83259571 | 2016.09.01.15. |
| ## | 3747 | 70  | 4  | 2016.09.01 | 290 | -1.22173048 | 2016.09.01.15. |
| ## | 3748 | 160 | 5  | 2016.09.01 | 200 | -2.79252680 | 2016.09.01.15. |
| ## | 3749 | 125 | 6  | 2016.09.01 | 235 | -2.18166156 | 2016.09.01.15. |
| ## | 3750 | 265 | 7  | 2016.09.01 | 95  | 1.65806279  | 2016.09.01.15. |
| ## | 3751 | 80  | 8  | 2016.09.01 | 280 | -1.39626340 | 2016.09.01.15. |
| ## | 3752 | 170 | 9  | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.15. |
| ## | 3753 | 170 | 10 | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.15. |
| ## | 3754 | 165 | 11 | 2016.09.01 | 195 | -2.87979327 | 2016.09.01.15. |
| ## | 3755 | 170 | 12 | 2016.09.01 | 190 | -2.96705973 | 2016.09.01.15. |
| ## | 3756 | 150 | 13 | 2016.09.01 | 210 | -2.61799388 | 2016.09.01.15. |
| ## | 3757 | 210 | 14 | 2016.09.01 | 150 | 2.61799388  | 2016.09.01.15. |
| ## | 3758 | 210 | 15 | 2016.09.01 | 150 | 2.61799388  | 2016.09.01.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 3759 | 220 | 16 | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.15. |
| ## 3760 | 300 | 17 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.15. |
| ## 3761 | 310 | 18 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.15. |
| ## 3762 | 220 | 19 | 2016.09.01 | 140 | 2.44346095  | 2016.09.01.15. |
| ## 3763 | 310 | 20 | 2016.09.01 | 50  | 0.87266463  | 2016.09.01.15. |
| ## 3764 | 50  | 21 | 2016.09.01 | 310 | -0.87266463 | 2016.09.01.15. |
| ## 3765 | 260 | 22 | 2016.09.01 | 100 | 1.74532925  | 2016.09.01.15. |
| ## 3766 | 10  | 23 | 2016.09.01 | 350 | -0.17453293 | 2016.09.01.15. |
| ## 3767 | 95  | 24 | 2016.09.01 | 265 | -1.65806279 | 2016.09.01.15. |
| ## 3768 | 50  | 25 | 2016.09.01 | 310 | -0.87266463 | 2016.09.01.15. |
| ## 3769 | 215 | 26 | 2016.09.01 | 145 | 2.53072742  | 2016.09.01.15. |
| ## 3770 | 300 | 27 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.15. |
| ## 3771 | 0   | 28 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.15. |
| ## 3772 | 350 | 29 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.15. |
| ## 3773 | 155 | 30 | 2016.09.01 | 205 | -2.70526034 | 2016.09.01.15. |
| ## 3774 | 230 | 31 | 2016.09.01 | 130 | 2.26892803  | 2016.09.01.15. |
| ## 3775 | 140 | 32 | 2016.09.01 | 220 | -2.44346095 | 2016.09.01.15. |
| ## 3776 | 70  | 33 | 2016.09.01 | 290 | -1.22173048 | 2016.09.01.15. |
| ## 3777 | 300 | 34 | 2016.09.01 | 60  | 1.04719755  | 2016.09.01.15. |
| ## 3778 | 260 | 35 | 2016.09.01 | 100 | 1.74532925  | 2016.09.01.15. |
| ## 3779 | 25  | 36 | 2016.09.01 | 335 | -0.43633231 | 2016.09.01.15. |
| ## 3780 | 110 | 37 | 2016.09.01 | 250 | -1.91986218 | 2016.09.01.15. |
| ## 3781 | 25  | 38 | 2016.09.01 | 335 | -0.43633231 | 2016.09.01.15. |
| ## 3782 | 0   | 39 | 2016.09.01 | 0   | 0.00000000  | 2016.09.01.15. |
| ## 3783 | 350 | 40 | 2016.09.01 | 10  | 0.17453293  | 2016.09.01.15. |
| ## 3784 | 355 | 41 | 2016.09.01 | 5   | 0.08726646  | 2016.09.01.15. |
| ## 3785 | 20  | 42 | 2016.09.01 | 340 | -0.34906585 | 2016.09.01.15. |
| ## 3786 | 20  | 43 | 2016.09.01 | 340 | -0.34906585 | 2016.09.01.15. |
| ## 3787 | 15  | 44 | 2016.09.01 | 345 | -0.26179939 | 2016.09.01.15. |
| ## 3788 | 5   | 45 | 2016.09.01 | 355 | -0.08726646 | 2016.09.01.15. |
| ## 3789 | 30  | 46 | 2016.09.01 | 330 | -0.52359878 | 2016.09.01.15. |
| ## 3790 | 60  | 47 | 2016.09.01 | 300 | -1.04719755 | 2016.09.01.15. |
| ## 3791 | 40  | 48 | 2016.09.01 | 320 | -0.69813170 | 2016.09.01.15. |
| ## 3792 | 10  | 1  | 2016.09.02 | 350 | -0.17453293 | 2016.09.02.13. |
| ## 3793 | 50  | 2  | 2016.09.02 | 310 | -0.87266463 | 2016.09.02.13. |
| ## 3794 | 345 | 3  | 2016.09.02 | 15  | 0.26179939  | 2016.09.02.13. |
| ## 3795 | 320 | 4  | 2016.09.02 | 40  | 0.69813170  | 2016.09.02.13. |
| ## 3796 | 350 | 5  | 2016.09.02 | 10  | 0.17453293  | 2016.09.02.13. |
| ## 3797 | 115 | 6  | 2016.09.02 | 245 | -2.00712864 | 2016.09.02.13. |
| ## 3798 | 290 | 7  | 2016.09.02 | 70  | 1.22173048  | 2016.09.02.13. |
| ## 3799 | 10  | 8  | 2016.09.02 | 350 | -0.17453293 | 2016.09.02.13. |
| ## 3800 | 330 | 9  | 2016.09.02 | 30  | 0.52359878  | 2016.09.02.13. |
| ## 3801 | 335 | 10 | 2016.09.02 | 25  | 0.43633231  | 2016.09.02.13. |
| ## 3802 | 150 | 11 | 2016.09.02 | 210 | -2.61799388 | 2016.09.02.13. |
| ## 3803 | 180 | 12 | 2016.09.02 | 180 | 3.14159265  | 2016.09.02.13. |
| ## 3804 | 165 | 13 | 2016.09.02 | 195 | -2.87979327 | 2016.09.02.13. |
| ## 3805 | 210 | 14 | 2016.09.02 | 150 | 2.61799388  | 2016.09.02.13. |
| ## 3806 | 150 | 15 | 2016.09.02 | 210 | -2.61799388 | 2016.09.02.13. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 3807 | 230 | 16 | 2016.09.02 | 130 | 2.26892803  | 2016.09.02.13. |
| ## 3808 | 160 | 17 | 2016.09.02 | 200 | -2.79252680 | 2016.09.02.13. |
| ## 3809 | 165 | 18 | 2016.09.02 | 195 | -2.87979327 | 2016.09.02.13. |
| ## 3810 | 180 | 19 | 2016.09.02 | 180 | 3.14159265  | 2016.09.02.13. |
| ## 3811 | 200 | 20 | 2016.09.02 | 160 | 2.79252680  | 2016.09.02.13. |
| ## 3812 | 160 | 21 | 2016.09.02 | 200 | -2.79252680 | 2016.09.02.13. |
| ## 3813 | 215 | 22 | 2016.09.02 | 145 | 2.53072742  | 2016.09.02.13. |
| ## 3814 | 165 | 23 | 2016.09.02 | 195 | -2.87979327 | 2016.09.02.13. |
| ## 3815 | 355 | 1  | 2016.09.02 | 5   | 0.08726646  | 2016.09.02.13. |
| ## 3816 | 350 | 2  | 2016.09.02 | 10  | 0.17453293  | 2016.09.02.13. |
| ## 3817 | 340 | 3  | 2016.09.02 | 20  | 0.34906585  | 2016.09.02.13. |
| ## 3818 | 0   | 4  | 2016.09.02 | 0   | 0.00000000  | 2016.09.02.13. |
| ## 3819 | 340 | 5  | 2016.09.02 | 20  | 0.34906585  | 2016.09.02.13. |
| ## 3820 | 300 | 6  | 2016.09.02 | 60  | 1.04719755  | 2016.09.02.13. |
| ## 3821 | 290 | 7  | 2016.09.02 | 70  | 1.22173048  | 2016.09.02.13. |
| ## 3822 | 310 | 8  | 2016.09.02 | 50  | 0.87266463  | 2016.09.02.13. |
| ## 3823 | 335 | 9  | 2016.09.02 | 25  | 0.43633231  | 2016.09.02.13. |
| ## 3824 | 330 | 10 | 2016.09.02 | 30  | 0.52359878  | 2016.09.02.13. |
| ## 3825 | 330 | 11 | 2016.09.02 | 30  | 0.52359878  | 2016.09.02.13. |
| ## 3826 | 305 | 12 | 2016.09.02 | 55  | 0.95993109  | 2016.09.02.13. |
| ## 3827 | 280 | 13 | 2016.09.02 | 80  | 1.39626340  | 2016.09.02.13. |
| ## 3828 | 345 | 14 | 2016.09.02 | 15  | 0.26179939  | 2016.09.02.13. |
| ## 3829 | 60  | 15 | 2016.09.02 | 300 | -1.04719755 | 2016.09.02.13. |
| ## 3830 | 75  | 16 | 2016.09.02 | 285 | -1.30899694 | 2016.09.02.13. |
| ## 3831 | 75  | 17 | 2016.09.02 | 285 | -1.30899694 | 2016.09.02.13. |
| ## 3832 | 160 | 18 | 2016.09.02 | 200 | -2.79252680 | 2016.09.02.13. |
| ## 3833 | 160 | 19 | 2016.09.02 | 200 | -2.79252680 | 2016.09.02.13. |
| ## 3834 | 30  | 20 | 2016.09.02 | 330 | -0.52359878 | 2016.09.02.13. |
| ## 3835 | 185 | 21 | 2016.09.02 | 175 | 3.05432619  | 2016.09.02.13. |
| ## 3836 | 110 | 22 | 2016.09.02 | 250 | -1.91986218 | 2016.09.02.13. |
| ## 3837 | 155 | 23 | 2016.09.02 | 205 | -2.70526034 | 2016.09.02.13. |
| ## 3838 | 210 | 24 | 2016.09.02 | 150 | 2.61799388  | 2016.09.02.13. |
| ## 3839 | 180 | 25 | 2016.09.02 | 180 | 3.14159265  | 2016.09.02.13. |
| ## 3840 | 140 | 26 | 2016.09.02 | 220 | -2.44346095 | 2016.09.02.13. |
| ## 3841 | 150 | 27 | 2016.09.02 | 210 | -2.61799388 | 2016.09.02.13. |
| ## 3842 | 180 | 28 | 2016.09.02 | 180 | 3.14159265  | 2016.09.02.13. |
| ## 3843 | 180 | 29 | 2016.09.02 | 180 | 3.14159265  | 2016.09.02.13. |
| ## 3844 | 215 | 30 | 2016.09.02 | 145 | 2.53072742  | 2016.09.02.13. |
| ## 3845 | 170 | 1  | 2016.09.06 | 190 | -2.96705973 | 2016.09.06.14. |
| ## 3846 | 170 | 2  | 2016.09.06 | 190 | -2.96705973 | 2016.09.06.14. |
| ## 3847 | 40  | 3  | 2016.09.06 | 320 | -0.69813170 | 2016.09.06.14. |
| ## 3848 | 130 | 4  | 2016.09.06 | 230 | -2.26892803 | 2016.09.06.14. |
| ## 3849 | 85  | 5  | 2016.09.06 | 275 | -1.48352986 | 2016.09.06.14. |
| ## 3850 | 110 | 6  | 2016.09.06 | 250 | -1.91986218 | 2016.09.06.14. |
| ## 3851 | 120 | 7  | 2016.09.06 | 240 | -2.09439510 | 2016.09.06.14. |
| ## 3852 | 110 | 8  | 2016.09.06 | 250 | -1.91986218 | 2016.09.06.14. |
| ## 3853 | 140 | 9  | 2016.09.06 | 220 | -2.44346095 | 2016.09.06.14. |
| ## 3854 | 195 | 10 | 2016.09.06 | 165 | 2.87979327  | 2016.09.06.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3855 | 120 | 11 | 2016.09.06 | 240 | -2.09439510 | 2016.09.06.14. |
| ## | 3856 | 150 | 12 | 2016.09.06 | 210 | -2.61799388 | 2016.09.06.14. |
| ## | 3857 | 95  | 13 | 2016.09.06 | 265 | -1.65806279 | 2016.09.06.14. |
| ## | 3858 | 100 | 14 | 2016.09.06 | 260 | -1.74532925 | 2016.09.06.14. |
| ## | 3859 | 200 | 15 | 2016.09.06 | 160 | 2.79252680  | 2016.09.06.14. |
| ## | 3860 | 115 | 16 | 2016.09.06 | 245 | -2.00712864 | 2016.09.06.14. |
| ## | 3861 | 95  | 17 | 2016.09.06 | 265 | -1.65806279 | 2016.09.06.14. |
| ## | 3862 | 165 | 18 | 2016.09.06 | 195 | -2.87979327 | 2016.09.06.14. |
| ## | 3863 | 145 | 19 | 2016.09.06 | 215 | -2.53072742 | 2016.09.06.14. |
| ## | 3864 | 120 | 20 | 2016.09.06 | 240 | -2.09439510 | 2016.09.06.14. |
| ## | 3865 | 90  | 21 | 2016.09.06 | 270 | -1.57079633 | 2016.09.06.14. |
| ## | 3866 | 210 | 22 | 2016.09.06 | 150 | 2.61799388  | 2016.09.06.14. |
| ## | 3867 | 170 | 23 | 2016.09.06 | 190 | -2.96705973 | 2016.09.06.14. |
| ## | 3868 | 100 | 24 | 2016.09.06 | 260 | -1.74532925 | 2016.09.06.14. |
| ## | 3869 | 5   | 25 | 2016.09.06 | 355 | -0.08726646 | 2016.09.06.14. |
| ## | 3870 | 30  | 26 | 2016.09.06 | 330 | -0.52359878 | 2016.09.06.14. |
| ## | 3871 | 110 | 27 | 2016.09.06 | 250 | -1.91986218 | 2016.09.06.14. |
| ## | 3872 | 175 | 28 | 2016.09.06 | 185 | -3.05432619 | 2016.09.06.14. |
| ## | 3873 | 170 | 29 | 2016.09.06 | 190 | -2.96705973 | 2016.09.06.14. |
| ## | 3874 | 160 | 30 | 2016.09.06 | 200 | -2.79252680 | 2016.09.06.14. |
| ## | 3875 | 135 | 31 | 2016.09.06 | 225 | -2.35619449 | 2016.09.06.14. |
| ## | 3876 | 170 | 32 | 2016.09.06 | 190 | -2.96705973 | 2016.09.06.14. |
| ## | 3877 | 90  | 33 | 2016.09.06 | 270 | -1.57079633 | 2016.09.06.14. |
| ## | 3878 | 175 | 34 | 2016.09.06 | 185 | -3.05432619 | 2016.09.06.14. |
| ## | 3879 | 120 | 35 | 2016.09.06 | 240 | -2.09439510 | 2016.09.06.14. |
| ## | 3880 | 130 | 36 | 2016.09.06 | 230 | -2.26892803 | 2016.09.06.14. |
| ## | 3881 | 140 | 37 | 2016.09.06 | 220 | -2.44346095 | 2016.09.06.14. |
| ## | 3882 | 175 | 38 | 2016.09.06 | 185 | -3.05432619 | 2016.09.06.14. |
| ## | 3883 | 115 | 39 | 2016.09.06 | 245 | -2.00712864 | 2016.09.06.14. |
| ## | 3884 | 125 | 40 | 2016.09.06 | 235 | -2.18166156 | 2016.09.06.14. |
| ## | 3885 | 115 | 41 | 2016.09.06 | 245 | -2.00712864 | 2016.09.06.14. |
| ## | 3886 | 145 | 42 | 2016.09.06 | 215 | -2.53072742 | 2016.09.06.14. |
| ## | 3887 | 155 | 43 | 2016.09.06 | 205 | -2.70526034 | 2016.09.06.14. |
| ## | 3888 | 165 | 44 | 2016.09.06 | 195 | -2.87979327 | 2016.09.06.14. |
| ## | 3889 | 220 | 45 | 2016.09.06 | 140 | 2.44346095  | 2016.09.06.14. |
| ## | 3890 | 145 | 46 | 2016.09.06 | 215 | -2.53072742 | 2016.09.06.14. |
| ## | 3891 | 155 | 47 | 2016.09.06 | 205 | -2.70526034 | 2016.09.06.14. |
| ## | 3892 | 155 | 48 | 2016.09.06 | 205 | -2.70526034 | 2016.09.06.14. |
| ## | 3893 | 95  | 49 | 2016.09.06 | 265 | -1.65806279 | 2016.09.06.14. |
| ## | 3894 | 70  | 50 | 2016.09.06 | 290 | -1.22173048 | 2016.09.06.14. |
| ## | 3895 | 90  | 51 | 2016.09.06 | 270 | -1.57079633 | 2016.09.06.14. |
| ## | 3896 | 120 | 52 | 2016.09.06 | 240 | -2.09439510 | 2016.09.06.14. |
| ## | 3897 | 210 | 53 | 2016.09.06 | 150 | 2.61799388  | 2016.09.06.14. |
| ## | 3898 | 105 | 54 | 2016.09.06 | 255 | -1.83259571 | 2016.09.06.14. |
| ## | 3899 | 95  | 55 | 2016.09.06 | 265 | -1.65806279 | 2016.09.06.14. |
| ## | 3900 | 140 | 56 | 2016.09.06 | 220 | -2.44346095 | 2016.09.06.14. |
| ## | 3901 | 140 | 57 | 2016.09.06 | 220 | -2.44346095 | 2016.09.06.14. |
| ## | 3902 | 160 | 58 | 2016.09.06 | 200 | -2.79252680 | 2016.09.06.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3903 | 150 | 59 | 2016.09.06 | 210 | -2.61799388 | 2016.09.06.14. |
| ## | 3904 | 80  | 60 | 2016.09.06 | 280 | -1.39626340 | 2016.09.06.14. |
| ## | 3905 | 15  | 61 | 2016.09.06 | 345 | -0.26179939 | 2016.09.06.14. |
| ## | 3906 | 60  | 62 | 2016.09.06 | 300 | -1.04719755 | 2016.09.06.14. |
| ## | 3907 | 50  | 63 | 2016.09.06 | 310 | -0.87266463 | 2016.09.06.14. |
| ## | 3908 | 140 | 1  | 2016.09.07 | 220 | -2.44346095 | 2016.09.07.14. |
| ## | 3909 | 50  | 2  | 2016.09.07 | 310 | -0.87266463 | 2016.09.07.14. |
| ## | 3910 | 45  | 3  | 2016.09.07 | 315 | -0.78539816 | 2016.09.07.14. |
| ## | 3911 | 70  | 4  | 2016.09.07 | 290 | -1.22173048 | 2016.09.07.14. |
| ## | 3912 | 125 | 5  | 2016.09.07 | 235 | -2.18166156 | 2016.09.07.14. |
| ## | 3913 | 65  | 6  | 2016.09.07 | 295 | -1.13446401 | 2016.09.07.14. |
| ## | 3914 | 60  | 7  | 2016.09.07 | 300 | -1.04719755 | 2016.09.07.14. |
| ## | 3915 | 55  | 8  | 2016.09.07 | 305 | -0.95993109 | 2016.09.07.14. |
| ## | 3916 | 70  | 9  | 2016.09.07 | 290 | -1.22173048 | 2016.09.07.14. |
| ## | 3917 | 65  | 10 | 2016.09.07 | 295 | -1.13446401 | 2016.09.07.14. |
| ## | 3918 | 340 | 11 | 2016.09.07 | 20  | 0.34906585  | 2016.09.07.14. |
| ## | 3919 | 70  | 12 | 2016.09.07 | 290 | -1.22173048 | 2016.09.07.14. |
| ## | 3920 | 120 | 13 | 2016.09.07 | 240 | -2.09439510 | 2016.09.07.14. |
| ## | 3921 | 30  | 14 | 2016.09.07 | 330 | -0.52359878 | 2016.09.07.14. |
| ## | 3922 | 85  | 15 | 2016.09.07 | 275 | -1.48352986 | 2016.09.07.14. |
| ## | 3923 | 110 | 16 | 2016.09.07 | 250 | -1.91986218 | 2016.09.07.14. |
| ## | 3924 | 155 | 17 | 2016.09.07 | 205 | -2.70526034 | 2016.09.07.14. |
| ## | 3925 | 150 | 18 | 2016.09.07 | 210 | -2.61799388 | 2016.09.07.14. |
| ## | 3926 | 155 | 19 | 2016.09.07 | 205 | -2.70526034 | 2016.09.07.14. |
| ## | 3927 | 185 | 20 | 2016.09.07 | 175 | 3.05432619  | 2016.09.07.14. |
| ## | 3928 | 160 | 21 | 2016.09.07 | 200 | -2.79252680 | 2016.09.07.14. |
| ## | 3929 | 155 | 22 | 2016.09.07 | 205 | -2.70526034 | 2016.09.07.14. |
| ## | 3930 | 110 | 23 | 2016.09.07 | 250 | -1.91986218 | 2016.09.07.14. |
| ## | 3931 | 320 | 24 | 2016.09.07 | 40  | 0.69813170  | 2016.09.07.14. |
| ## | 3932 | 340 | 25 | 2016.09.07 | 20  | 0.34906585  | 2016.09.07.14. |
| ## | 3933 | 345 | 26 | 2016.09.07 | 15  | 0.26179939  | 2016.09.07.14. |
| ## | 3934 | 280 | 27 | 2016.09.07 | 80  | 1.39626340  | 2016.09.07.14. |
| ## | 3935 | 0   | 28 | 2016.09.07 | 0   | 0.00000000  | 2016.09.07.14. |
| ## | 3936 | 290 | 29 | 2016.09.07 | 70  | 1.22173048  | 2016.09.07.14. |
| ## | 3937 | 5   | 30 | 2016.09.07 | 355 | -0.08726646 | 2016.09.07.14. |
| ## | 3938 | 295 | 31 | 2016.09.07 | 65  | 1.13446401  | 2016.09.07.14. |
| ## | 3939 | 285 | 32 | 2016.09.07 | 75  | 1.30899694  | 2016.09.07.14. |
| ## | 3940 | 335 | 33 | 2016.09.07 | 25  | 0.43633231  | 2016.09.07.14. |
| ## | 3941 | 330 | 34 | 2016.09.07 | 30  | 0.52359878  | 2016.09.07.14. |
| ## | 3942 | 350 | 35 | 2016.09.07 | 10  | 0.17453293  | 2016.09.07.14. |
| ## | 3943 | 350 | 36 | 2016.09.07 | 10  | 0.17453293  | 2016.09.07.14. |
| ## | 3944 | 320 | 37 | 2016.09.07 | 40  | 0.69813170  | 2016.09.07.14. |
| ## | 3945 | 280 | 38 | 2016.09.07 | 80  | 1.39626340  | 2016.09.07.14. |
| ## | 3946 | 345 | 39 | 2016.09.07 | 15  | 0.26179939  | 2016.09.07.14. |
| ## | 3947 | 320 | 40 | 2016.09.07 | 40  | 0.69813170  | 2016.09.07.14. |
| ## | 3948 | 320 | 41 | 2016.09.07 | 40  | 0.69813170  | 2016.09.07.14. |
| ## | 3949 | 350 | 42 | 2016.09.07 | 10  | 0.17453293  | 2016.09.07.14. |
| ## | 3950 | 10  | 43 | 2016.09.07 | 350 | -0.17453293 | 2016.09.07.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3951 | 300 | 44 | 2016.09.07 | 60  | 1.04719755  | 2016.09.07.14. |
| ## | 3952 | 345 | 45 | 2016.09.07 | 15  | 0.26179939  | 2016.09.07.14. |
| ## | 3953 | 330 | 46 | 2016.09.07 | 30  | 0.52359878  | 2016.09.07.14. |
| ## | 3954 | 310 | 47 | 2016.09.07 | 50  | 0.87266463  | 2016.09.07.14. |
| ## | 3955 | 5   | 1  | 2016.09.07 | 355 | -0.08726646 | 2016.09.07.16. |
| ## | 3956 | 15  | 2  | 2016.09.07 | 345 | -0.26179939 | 2016.09.07.16. |
| ## | 3957 | 65  | 3  | 2016.09.07 | 295 | -1.13446401 | 2016.09.07.16. |
| ## | 3958 | 70  | 4  | 2016.09.07 | 290 | -1.22173048 | 2016.09.07.16. |
| ## | 3959 | 95  | 5  | 2016.09.07 | 265 | -1.65806279 | 2016.09.07.16. |
| ## | 3960 | 105 | 6  | 2016.09.07 | 255 | -1.83259571 | 2016.09.07.16. |
| ## | 3961 | 40  | 7  | 2016.09.07 | 320 | -0.69813170 | 2016.09.07.16. |
| ## | 3962 | 90  | 8  | 2016.09.07 | 270 | -1.57079633 | 2016.09.07.16. |
| ## | 3963 | 100 | 9  | 2016.09.07 | 260 | -1.74532925 | 2016.09.07.16. |
| ## | 3964 | 50  | 10 | 2016.09.07 | 310 | -0.87266463 | 2016.09.07.16. |
| ## | 3965 | 40  | 11 | 2016.09.07 | 320 | -0.69813170 | 2016.09.07.16. |
| ## | 3966 | 15  | 12 | 2016.09.07 | 345 | -0.26179939 | 2016.09.07.16. |
| ## | 3967 | 90  | 13 | 2016.09.07 | 270 | -1.57079633 | 2016.09.07.16. |
| ## | 3968 | 300 | 14 | 2016.09.07 | 60  | 1.04719755  | 2016.09.07.16. |
| ## | 3969 | 120 | 15 | 2016.09.07 | 240 | -2.09439510 | 2016.09.07.16. |
| ## | 3970 | 65  | 16 | 2016.09.07 | 295 | -1.13446401 | 2016.09.07.16. |
| ## | 3971 | 280 | 17 | 2016.09.07 | 80  | 1.39626340  | 2016.09.07.16. |
| ## | 3972 | 330 | 18 | 2016.09.07 | 30  | 0.52359878  | 2016.09.07.16. |
| ## | 3973 | 350 | 19 | 2016.09.07 | 10  | 0.17453293  | 2016.09.07.16. |
| ## | 3974 | 350 | 20 | 2016.09.07 | 10  | 0.17453293  | 2016.09.07.16. |
| ## | 3975 | 320 | 21 | 2016.09.07 | 40  | 0.69813170  | 2016.09.07.16. |
| ## | 3976 | 320 | 22 | 2016.09.07 | 40  | 0.69813170  | 2016.09.07.16. |
| ## | 3977 | 345 | 23 | 2016.09.07 | 15  | 0.26179939  | 2016.09.07.16. |
| ## | 3978 | 335 | 24 | 2016.09.07 | 25  | 0.43633231  | 2016.09.07.16. |
| ## | 3979 | 330 | 25 | 2016.09.07 | 30  | 0.52359878  | 2016.09.07.16. |
| ## | 3980 | 295 | 1  | 2016.09.08 | 65  | 1.13446401  | 2016.09.08.14. |
| ## | 3981 | 295 | 2  | 2016.09.08 | 65  | 1.13446401  | 2016.09.08.14. |
| ## | 3982 | 240 | 3  | 2016.09.08 | 120 | 2.09439510  | 2016.09.08.14. |
| ## | 3983 | 265 | 4  | 2016.09.08 | 95  | 1.65806279  | 2016.09.08.14. |
| ## | 3984 | 250 | 5  | 2016.09.08 | 110 | 1.91986218  | 2016.09.08.14. |
| ## | 3985 | 260 | 6  | 2016.09.08 | 100 | 1.74532925  | 2016.09.08.14. |
| ## | 3986 | 320 | 7  | 2016.09.08 | 40  | 0.69813170  | 2016.09.08.14. |
| ## | 3987 | 240 | 8  | 2016.09.08 | 120 | 2.09439510  | 2016.09.08.14. |
| ## | 3988 | 280 | 9  | 2016.09.08 | 80  | 1.39626340  | 2016.09.08.14. |
| ## | 3989 | 280 | 10 | 2016.09.08 | 80  | 1.39626340  | 2016.09.08.14. |
| ## | 3990 | 220 | 11 | 2016.09.08 | 140 | 2.44346095  | 2016.09.08.14. |
| ## | 3991 | 220 | 12 | 2016.09.08 | 140 | 2.44346095  | 2016.09.08.14. |
| ## | 3992 | 300 | 13 | 2016.09.08 | 60  | 1.04719755  | 2016.09.08.14. |
| ## | 3993 | 260 | 14 | 2016.09.08 | 100 | 1.74532925  | 2016.09.08.14. |
| ## | 3994 | 290 | 15 | 2016.09.08 | 70  | 1.22173048  | 2016.09.08.14. |
| ## | 3995 | 230 | 16 | 2016.09.08 | 130 | 2.26892803  | 2016.09.08.14. |
| ## | 3996 | 240 | 17 | 2016.09.08 | 120 | 2.09439510  | 2016.09.08.14. |
| ## | 3997 | 300 | 18 | 2016.09.08 | 60  | 1.04719755  | 2016.09.08.14. |
| ## | 3998 | 230 | 19 | 2016.09.08 | 130 | 2.26892803  | 2016.09.08.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 3999 | 190 | 20 | 2016.09.08 | 170 | 2.96705973  | 2016.09.08.14. |
| ## | 4000 | 110 | 1  | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.15. |
| ## | 4001 | 190 | 2  | 2016.09.08 | 170 | 2.96705973  | 2016.09.08.15. |
| ## | 4002 | 100 | 3  | 2016.09.08 | 260 | -1.74532925 | 2016.09.08.15. |
| ## | 4003 | 175 | 4  | 2016.09.08 | 185 | -3.05432619 | 2016.09.08.15. |
| ## | 4004 | 220 | 5  | 2016.09.08 | 140 | 2.44346095  | 2016.09.08.15. |
| ## | 4005 | 205 | 6  | 2016.09.08 | 155 | 2.70526034  | 2016.09.08.15. |
| ## | 4006 | 160 | 7  | 2016.09.08 | 200 | -2.79252680 | 2016.09.08.15. |
| ## | 4007 | 160 | 8  | 2016.09.08 | 200 | -2.79252680 | 2016.09.08.15. |
| ## | 4008 | 50  | 9  | 2016.09.08 | 310 | -0.87266463 | 2016.09.08.15. |
| ## | 4009 | 110 | 10 | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.15. |
| ## | 4010 | 150 | 11 | 2016.09.08 | 210 | -2.61799388 | 2016.09.08.15. |
| ## | 4011 | 110 | 12 | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.15. |
| ## | 4012 | 165 | 13 | 2016.09.08 | 195 | -2.87979327 | 2016.09.08.15. |
| ## | 4013 | 85  | 14 | 2016.09.08 | 275 | -1.48352986 | 2016.09.08.15. |
| ## | 4014 | 50  | 15 | 2016.09.08 | 310 | -0.87266463 | 2016.09.08.15. |
| ## | 4015 | 145 | 16 | 2016.09.08 | 215 | -2.53072742 | 2016.09.08.15. |
| ## | 4016 | 330 | 17 | 2016.09.08 | 30  | 0.52359878  | 2016.09.08.15. |
| ## | 4017 | 355 | 18 | 2016.09.08 | 5   | 0.08726646  | 2016.09.08.15. |
| ## | 4018 | 325 | 19 | 2016.09.08 | 35  | 0.61086524  | 2016.09.08.15. |
| ## | 4019 | 0   | 20 | 2016.09.08 | 0   | 0.00000000  | 2016.09.08.15. |
| ## | 4020 | 20  | 21 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.15. |
| ## | 4021 | 350 | 22 | 2016.09.08 | 10  | 0.17453293  | 2016.09.08.15. |
| ## | 4022 | 20  | 23 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.15. |
| ## | 4023 | 20  | 24 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.15. |
| ## | 4024 | 20  | 25 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.15. |
| ## | 4025 | 330 | 26 | 2016.09.08 | 30  | 0.52359878  | 2016.09.08.15. |
| ## | 4026 | 355 | 27 | 2016.09.08 | 5   | 0.08726646  | 2016.09.08.15. |
| ## | 4027 | 0   | 28 | 2016.09.08 | 0   | 0.00000000  | 2016.09.08.15. |
| ## | 4028 | 5   | 29 | 2016.09.08 | 355 | -0.08726646 | 2016.09.08.15. |
| ## | 4029 | 0   | 30 | 2016.09.08 | 0   | 0.00000000  | 2016.09.08.15. |
| ## | 4030 | 30  | 31 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.15. |
| ## | 4031 | 350 | 32 | 2016.09.08 | 10  | 0.17453293  | 2016.09.08.15. |
| ## | 4032 | 110 | 33 | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.15. |
| ## | 4033 | 40  | 34 | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.15. |
| ## | 4034 | 340 | 35 | 2016.09.08 | 20  | 0.34906585  | 2016.09.08.15. |
| ## | 4035 | 40  | 36 | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.15. |
| ## | 4036 | 355 | 37 | 2016.09.08 | 5   | 0.08726646  | 2016.09.08.15. |
| ## | 4037 | 35  | 38 | 2016.09.08 | 325 | -0.61086524 | 2016.09.08.15. |
| ## | 4038 | 40  | 39 | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.15. |
| ## | 4039 | 20  | 40 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.15. |
| ## | 4040 | 60  | 41 | 2016.09.08 | 300 | -1.04719755 | 2016.09.08.15. |
| ## | 4041 | 110 | 42 | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.15. |
| ## | 4042 | 70  | 43 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## | 4043 | 160 | 44 | 2016.09.08 | 200 | -2.79252680 | 2016.09.08.15. |
| ## | 4044 | 110 | 45 | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.15. |
| ## | 4045 | 140 | 46 | 2016.09.08 | 220 | -2.44346095 | 2016.09.08.15. |
| ## | 4046 | 50  | 1  | 2016.09.08 | 310 | -0.87266463 | 2016.09.08.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 4047 | 45  | 2  | 2016.09.08 | 315 | -0.78539816 | 2016.09.08.15. |
| ## 4048 | 130 | 3  | 2016.09.08 | 230 | -2.26892803 | 2016.09.08.15. |
| ## 4049 | 100 | 4  | 2016.09.08 | 260 | -1.74532925 | 2016.09.08.15. |
| ## 4050 | 30  | 5  | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.15. |
| ## 4051 | 70  | 6  | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## 4052 | 30  | 7  | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.15. |
| ## 4053 | 80  | 8  | 2016.09.08 | 280 | -1.39626340 | 2016.09.08.15. |
| ## 4054 | 70  | 9  | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## 4055 | 85  | 10 | 2016.09.08 | 275 | -1.48352986 | 2016.09.08.15. |
| ## 4056 | 90  | 11 | 2016.09.08 | 270 | -1.57079633 | 2016.09.08.15. |
| ## 4057 | 355 | 12 | 2016.09.08 | 5   | 0.08726646  | 2016.09.08.15. |
| ## 4058 | 350 | 13 | 2016.09.08 | 10  | 0.17453293  | 2016.09.08.15. |
| ## 4059 | 280 | 14 | 2016.09.08 | 80  | 1.39626340  | 2016.09.08.15. |
| ## 4060 | 320 | 15 | 2016.09.08 | 40  | 0.69813170  | 2016.09.08.15. |
| ## 4061 | 325 | 16 | 2016.09.08 | 35  | 0.61086524  | 2016.09.08.15. |
| ## 4062 | 330 | 17 | 2016.09.08 | 30  | 0.52359878  | 2016.09.08.15. |
| ## 4063 | 320 | 18 | 2016.09.08 | 40  | 0.69813170  | 2016.09.08.15. |
| ## 4064 | 310 | 19 | 2016.09.08 | 50  | 0.87266463  | 2016.09.08.15. |
| ## 4065 | 325 | 20 | 2016.09.08 | 35  | 0.61086524  | 2016.09.08.15. |
| ## 4066 | 330 | 21 | 2016.09.08 | 30  | 0.52359878  | 2016.09.08.15. |
| ## 4067 | 320 | 22 | 2016.09.08 | 40  | 0.69813170  | 2016.09.08.15. |
| ## 4068 | 335 | 23 | 2016.09.08 | 25  | 0.43633231  | 2016.09.08.15. |
| ## 4069 | 325 | 24 | 2016.09.08 | 35  | 0.61086524  | 2016.09.08.15. |
| ## 4070 | 345 | 25 | 2016.09.08 | 15  | 0.26179939  | 2016.09.08.15. |
| ## 4071 | 350 | 26 | 2016.09.08 | 10  | 0.17453293  | 2016.09.08.15. |
| ## 4072 | 310 | 27 | 2016.09.08 | 50  | 0.87266463  | 2016.09.08.15. |
| ## 4073 | 40  | 1  | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.15. |
| ## 4074 | 350 | 2  | 2016.09.08 | 10  | 0.17453293  | 2016.09.08.15. |
| ## 4075 | 10  | 3  | 2016.09.08 | 350 | -0.17453293 | 2016.09.08.15. |
| ## 4076 | 10  | 4  | 2016.09.08 | 350 | -0.17453293 | 2016.09.08.15. |
| ## 4077 | 40  | 5  | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.15. |
| ## 4078 | 25  | 6  | 2016.09.08 | 335 | -0.43633231 | 2016.09.08.15. |
| ## 4079 | 60  | 7  | 2016.09.08 | 300 | -1.04719755 | 2016.09.08.15. |
| ## 4080 | 10  | 8  | 2016.09.08 | 350 | -0.17453293 | 2016.09.08.15. |
| ## 4081 | 80  | 9  | 2016.09.08 | 280 | -1.39626340 | 2016.09.08.15. |
| ## 4082 | 75  | 10 | 2016.09.08 | 285 | -1.30899694 | 2016.09.08.15. |
| ## 4083 | 70  | 11 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## 4084 | 70  | 12 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## 4085 | 90  | 13 | 2016.09.08 | 270 | -1.57079633 | 2016.09.08.15. |
| ## 4086 | 100 | 14 | 2016.09.08 | 260 | -1.74532925 | 2016.09.08.15. |
| ## 4087 | 70  | 15 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## 4088 | 30  | 16 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.15. |
| ## 4089 | 85  | 17 | 2016.09.08 | 275 | -1.48352986 | 2016.09.08.15. |
| ## 4090 | 45  | 18 | 2016.09.08 | 315 | -0.78539816 | 2016.09.08.15. |
| ## 4091 | 90  | 19 | 2016.09.08 | 270 | -1.57079633 | 2016.09.08.15. |
| ## 4092 | 60  | 20 | 2016.09.08 | 300 | -1.04719755 | 2016.09.08.15. |
| ## 4093 | 10  | 21 | 2016.09.08 | 350 | -0.17453293 | 2016.09.08.15. |
| ## 4094 | 120 | 22 | 2016.09.08 | 240 | -2.09439510 | 2016.09.08.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4095 | 55  | 23 | 2016.09.08 | 305 | -0.95993109 | 2016.09.08.15. |
| ## | 4096 | 10  | 24 | 2016.09.08 | 350 | -0.17453293 | 2016.09.08.15. |
| ## | 4097 | 120 | 25 | 2016.09.08 | 240 | -2.09439510 | 2016.09.08.15. |
| ## | 4098 | 330 | 26 | 2016.09.08 | 30  | 0.52359878  | 2016.09.08.15. |
| ## | 4099 | 355 | 27 | 2016.09.08 | 5   | 0.08726646  | 2016.09.08.15. |
| ## | 4100 | 70  | 28 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.15. |
| ## | 4101 | 65  | 29 | 2016.09.08 | 295 | -1.13446401 | 2016.09.08.15. |
| ## | 4102 | 160 | 30 | 2016.09.08 | 200 | -2.79252680 | 2016.09.08.15. |
| ## | 4103 | 180 | 1  | 2016.09.08 | 180 | 3.14159265  | 2016.09.08.16. |
| ## | 4104 | 95  | 2  | 2016.09.08 | 265 | -1.65806279 | 2016.09.08.16. |
| ## | 4105 | 110 | 3  | 2016.09.08 | 250 | -1.91986218 | 2016.09.08.16. |
| ## | 4106 | 230 | 4  | 2016.09.08 | 130 | 2.26892803  | 2016.09.08.16. |
| ## | 4107 | 230 | 5  | 2016.09.08 | 130 | 2.26892803  | 2016.09.08.16. |
| ## | 4108 | 240 | 6  | 2016.09.08 | 120 | 2.09439510  | 2016.09.08.16. |
| ## | 4109 | 210 | 7  | 2016.09.08 | 150 | 2.61799388  | 2016.09.08.16. |
| ## | 4110 | 280 | 8  | 2016.09.08 | 80  | 1.39626340  | 2016.09.08.16. |
| ## | 4111 | 220 | 9  | 2016.09.08 | 140 | 2.44346095  | 2016.09.08.16. |
| ## | 4112 | 150 | 10 | 2016.09.08 | 210 | -2.61799388 | 2016.09.08.16. |
| ## | 4113 | 150 | 11 | 2016.09.08 | 210 | -2.61799388 | 2016.09.08.16. |
| ## | 4114 | 130 | 12 | 2016.09.08 | 230 | -2.26892803 | 2016.09.08.16. |
| ## | 4115 | 260 | 13 | 2016.09.08 | 100 | 1.74532925  | 2016.09.08.16. |
| ## | 4116 | 70  | 14 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.16. |
| ## | 4117 | 40  | 15 | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.16. |
| ## | 4118 | 75  | 16 | 2016.09.08 | 285 | -1.30899694 | 2016.09.08.16. |
| ## | 4119 | 70  | 17 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.16. |
| ## | 4120 | 90  | 18 | 2016.09.08 | 270 | -1.57079633 | 2016.09.08.16. |
| ## | 4121 | 75  | 19 | 2016.09.08 | 285 | -1.30899694 | 2016.09.08.16. |
| ## | 4122 | 60  | 20 | 2016.09.08 | 300 | -1.04719755 | 2016.09.08.16. |
| ## | 4123 | 90  | 21 | 2016.09.08 | 270 | -1.57079633 | 2016.09.08.16. |
| ## | 4124 | 60  | 22 | 2016.09.08 | 300 | -1.04719755 | 2016.09.08.16. |
| ## | 4125 | 70  | 23 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.16. |
| ## | 4126 | 40  | 24 | 2016.09.08 | 320 | -0.69813170 | 2016.09.08.16. |
| ## | 4127 | 170 | 1  | 2016.09.08 | 190 | -2.96705973 | 2016.09.08.18. |
| ## | 4128 | 205 | 2  | 2016.09.08 | 155 | 2.70526034  | 2016.09.08.18. |
| ## | 4129 | 215 | 3  | 2016.09.08 | 145 | 2.53072742  | 2016.09.08.18. |
| ## | 4130 | 170 | 4  | 2016.09.08 | 190 | -2.96705973 | 2016.09.08.18. |
| ## | 4131 | 190 | 5  | 2016.09.08 | 170 | 2.96705973  | 2016.09.08.18. |
| ## | 4132 | 190 | 6  | 2016.09.08 | 170 | 2.96705973  | 2016.09.08.18. |
| ## | 4133 | 210 | 7  | 2016.09.08 | 150 | 2.61799388  | 2016.09.08.18. |
| ## | 4134 | 170 | 8  | 2016.09.08 | 190 | -2.96705973 | 2016.09.08.18. |
| ## | 4135 | 130 | 9  | 2016.09.08 | 230 | -2.26892803 | 2016.09.08.18. |
| ## | 4136 | 140 | 10 | 2016.09.08 | 220 | -2.44346095 | 2016.09.08.18. |
| ## | 4137 | 200 | 11 | 2016.09.08 | 160 | 2.79252680  | 2016.09.08.18. |
| ## | 4138 | 70  | 12 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.18. |
| ## | 4139 | 70  | 13 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.18. |
| ## | 4140 | 80  | 14 | 2016.09.08 | 280 | -1.39626340 | 2016.09.08.18. |
| ## | 4141 | 30  | 15 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.18. |
| ## | 4142 | 70  | 16 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.18. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4143 | 30  | 17 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.18. |
| ## | 4144 | 35  | 18 | 2016.09.08 | 325 | -0.61086524 | 2016.09.08.18. |
| ## | 4145 | 80  | 19 | 2016.09.08 | 280 | -1.39626340 | 2016.09.08.18. |
| ## | 4146 | 90  | 20 | 2016.09.08 | 270 | -1.57079633 | 2016.09.08.18. |
| ## | 4147 | 70  | 21 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.18. |
| ## | 4148 | 20  | 22 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.18. |
| ## | 4149 | 120 | 23 | 2016.09.08 | 240 | -2.09439510 | 2016.09.08.18. |
| ## | 4150 | 145 | 24 | 2016.09.08 | 215 | -2.53072742 | 2016.09.08.18. |
| ## | 4151 | 155 | 25 | 2016.09.08 | 205 | -2.70526034 | 2016.09.08.18. |
| ## | 4152 | 150 | 26 | 2016.09.08 | 210 | -2.61799388 | 2016.09.08.18. |
| ## | 4153 | 300 | 1  | 2016.09.08 | 60  | 1.04719755  | 2016.09.08.18. |
| ## | 4154 | 285 | 2  | 2016.09.08 | 75  | 1.30899694  | 2016.09.08.18. |
| ## | 4155 | 280 | 3  | 2016.09.08 | 80  | 1.39626340  | 2016.09.08.18. |
| ## | 4156 | 270 | 4  | 2016.09.08 | 90  | 1.57079633  | 2016.09.08.18. |
| ## | 4157 | 290 | 5  | 2016.09.08 | 70  | 1.22173048  | 2016.09.08.18. |
| ## | 4158 | 245 | 6  | 2016.09.08 | 115 | 2.00712864  | 2016.09.08.18. |
| ## | 4159 | 300 | 7  | 2016.09.08 | 60  | 1.04719755  | 2016.09.08.18. |
| ## | 4160 | 290 | 8  | 2016.09.08 | 70  | 1.22173048  | 2016.09.08.18. |
| ## | 4161 | 270 | 9  | 2016.09.08 | 90  | 1.57079633  | 2016.09.08.18. |
| ## | 4162 | 220 | 10 | 2016.09.08 | 140 | 2.44346095  | 2016.09.08.18. |
| ## | 4163 | 235 | 11 | 2016.09.08 | 125 | 2.18166156  | 2016.09.08.18. |
| ## | 4164 | 355 | 12 | 2016.09.08 | 5   | 0.08726646  | 2016.09.08.18. |
| ## | 4165 | 70  | 13 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.18. |
| ## | 4166 | 20  | 14 | 2016.09.08 | 340 | -0.34906585 | 2016.09.08.18. |
| ## | 4167 | 60  | 15 | 2016.09.08 | 300 | -1.04719755 | 2016.09.08.18. |
| ## | 4168 | 350 | 16 | 2016.09.08 | 10  | 0.17453293  | 2016.09.08.18. |
| ## | 4169 | 30  | 17 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.18. |
| ## | 4170 | 10  | 18 | 2016.09.08 | 350 | -0.17453293 | 2016.09.08.18. |
| ## | 4171 | 70  | 19 | 2016.09.08 | 290 | -1.22173048 | 2016.09.08.18. |
| ## | 4172 | 30  | 20 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.18. |
| ## | 4173 | 30  | 21 | 2016.09.08 | 330 | -0.52359878 | 2016.09.08.18. |
| ## | 4174 | 50  | 22 | 2016.09.08 | 310 | -0.87266463 | 2016.09.08.18. |
| ## | 4175 | 55  | 23 | 2016.09.08 | 305 | -0.95993109 | 2016.09.08.18. |
| ## | 4176 | 50  | 24 | 2016.09.08 | 310 | -0.87266463 | 2016.09.08.18. |
| ## | 4177 | 65  | 25 | 2016.09.08 | 295 | -1.13446401 | 2016.09.08.18. |
| ## | 4178 | 80  | 1  | 2016.09.12 | 280 | -1.39626340 | 2016.09.12.13. |
| ## | 4179 | 120 | 2  | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.13. |
| ## | 4180 | 355 | 3  | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.13. |
| ## | 4181 | 130 | 4  | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.13. |
| ## | 4182 | 45  | 5  | 2016.09.12 | 315 | -0.78539816 | 2016.09.12.13. |
| ## | 4183 | 355 | 6  | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.13. |
| ## | 4184 | 35  | 7  | 2016.09.12 | 325 | -0.61086524 | 2016.09.12.13. |
| ## | 4185 | 5   | 8  | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.13. |
| ## | 4186 | 10  | 9  | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.13. |
| ## | 4187 | 10  | 10 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.13. |
| ## | 4188 | 355 | 11 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.13. |
| ## | 4189 | 340 | 12 | 2016.09.12 | 20  | 0.34906585  | 2016.09.12.13. |
| ## | 4190 | 350 | 13 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4191 | 345 | 14 | 2016.09.12 | 15  | 0.26179939  | 2016.09.12.13. |
| ## | 4192 | 330 | 15 | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.13. |
| ## | 4193 | 350 | 16 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.13. |
| ## | 4194 | 330 | 17 | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.13. |
| ## | 4195 | 5   | 18 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.13. |
| ## | 4196 | 230 | 1  | 2016.09.12 | 130 | 2.26892803  | 2016.09.12.14. |
| ## | 4197 | 180 | 2  | 2016.09.12 | 180 | 3.14159265  | 2016.09.12.14. |
| ## | 4198 | 50  | 3  | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.14. |
| ## | 4199 | 30  | 4  | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.14. |
| ## | 4200 | 100 | 5  | 2016.09.12 | 260 | -1.74532925 | 2016.09.12.14. |
| ## | 4201 | 50  | 6  | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.14. |
| ## | 4202 | 70  | 7  | 2016.09.12 | 290 | -1.22173048 | 2016.09.12.14. |
| ## | 4203 | 60  | 8  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.14. |
| ## | 4204 | 75  | 9  | 2016.09.12 | 285 | -1.30899694 | 2016.09.12.14. |
| ## | 4205 | 80  | 10 | 2016.09.12 | 280 | -1.39626340 | 2016.09.12.14. |
| ## | 4206 | 110 | 11 | 2016.09.12 | 250 | -1.91986218 | 2016.09.12.14. |
| ## | 4207 | 330 | 12 | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.14. |
| ## | 4208 | 280 | 13 | 2016.09.12 | 80  | 1.39626340  | 2016.09.12.14. |
| ## | 4209 | 355 | 14 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4210 | 280 | 15 | 2016.09.12 | 80  | 1.39626340  | 2016.09.12.14. |
| ## | 4211 | 345 | 16 | 2016.09.12 | 15  | 0.26179939  | 2016.09.12.14. |
| ## | 4212 | 350 | 17 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.14. |
| ## | 4213 | 355 | 18 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4214 | 340 | 19 | 2016.09.12 | 20  | 0.34906585  | 2016.09.12.14. |
| ## | 4215 | 340 | 20 | 2016.09.12 | 20  | 0.34906585  | 2016.09.12.14. |
| ## | 4216 | 330 | 21 | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.14. |
| ## | 4217 | 340 | 22 | 2016.09.12 | 20  | 0.34906585  | 2016.09.12.14. |
| ## | 4218 | 5   | 23 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4219 | 0   | 24 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4220 | 355 | 25 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4221 | 150 | 26 | 2016.09.12 | 210 | -2.61799388 | 2016.09.12.14. |
| ## | 4222 | 115 | 27 | 2016.09.12 | 245 | -2.00712864 | 2016.09.12.14. |
| ## | 4223 | 250 | 28 | 2016.09.12 | 110 | 1.91986218  | 2016.09.12.14. |
| ## | 4224 | 300 | 1  | 2016.09.12 | 60  | 1.04719755  | 2016.09.12.14. |
| ## | 4225 | 350 | 2  | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.14. |
| ## | 4226 | 305 | 3  | 2016.09.12 | 55  | 0.95993109  | 2016.09.12.14. |
| ## | 4227 | 85  | 4  | 2016.09.12 | 275 | -1.48352986 | 2016.09.12.14. |
| ## | 4228 | 195 | 5  | 2016.09.12 | 165 | 2.87979327  | 2016.09.12.14. |
| ## | 4229 | 330 | 6  | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.14. |
| ## | 4230 | 250 | 7  | 2016.09.12 | 110 | 1.91986218  | 2016.09.12.14. |
| ## | 4231 | 60  | 8  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.14. |
| ## | 4232 | 140 | 9  | 2016.09.12 | 220 | -2.44346095 | 2016.09.12.14. |
| ## | 4233 | 155 | 10 | 2016.09.12 | 205 | -2.70526034 | 2016.09.12.14. |
| ## | 4234 | 240 | 11 | 2016.09.12 | 120 | 2.09439510  | 2016.09.12.14. |
| ## | 4235 | 115 | 12 | 2016.09.12 | 245 | -2.00712864 | 2016.09.12.14. |
| ## | 4236 | 125 | 13 | 2016.09.12 | 235 | -2.18166156 | 2016.09.12.14. |
| ## | 4237 | 200 | 14 | 2016.09.12 | 160 | 2.79252680  | 2016.09.12.14. |
| ## | 4238 | 220 | 15 | 2016.09.12 | 140 | 2.44346095  | 2016.09.12.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4239 | 60  | 16 | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.14. |
| ## | 4240 | 250 | 17 | 2016.09.12 | 110 | 1.91986218  | 2016.09.12.14. |
| ## | 4241 | 300 | 18 | 2016.09.12 | 60  | 1.04719755  | 2016.09.12.14. |
| ## | 4242 | 210 | 19 | 2016.09.12 | 150 | 2.61799388  | 2016.09.12.14. |
| ## | 4243 | 285 | 20 | 2016.09.12 | 75  | 1.30899694  | 2016.09.12.14. |
| ## | 4244 | 5   | 21 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4245 | 355 | 22 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4246 | 5   | 23 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4247 | 10  | 24 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4248 | 0   | 25 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4249 | 5   | 26 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4250 | 0   | 27 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4251 | 350 | 28 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.14. |
| ## | 4252 | 0   | 29 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4253 | 10  | 30 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4254 | 355 | 31 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4255 | 10  | 32 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4256 | 350 | 33 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.14. |
| ## | 4257 | 10  | 34 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4258 | 355 | 35 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4259 | 15  | 36 | 2016.09.12 | 345 | -0.26179939 | 2016.09.12.14. |
| ## | 4260 | 55  | 37 | 2016.09.12 | 305 | -0.95993109 | 2016.09.12.14. |
| ## | 4261 | 10  | 38 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4262 | 85  | 39 | 2016.09.12 | 275 | -1.48352986 | 2016.09.12.14. |
| ## | 4263 | 70  | 40 | 2016.09.12 | 290 | -1.22173048 | 2016.09.12.14. |
| ## | 4264 | 355 | 41 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4265 | 40  | 42 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.14. |
| ## | 4266 | 90  | 43 | 2016.09.12 | 270 | -1.57079633 | 2016.09.12.14. |
| ## | 4267 | 10  | 44 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4268 | 355 | 45 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4269 | 0   | 46 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4270 | 10  | 47 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4271 | 40  | 48 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.14. |
| ## | 4272 | 0   | 49 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4273 | 70  | 1  | 2016.09.12 | 290 | -1.22173048 | 2016.09.12.14. |
| ## | 4274 | 110 | 2  | 2016.09.12 | 250 | -1.91986218 | 2016.09.12.14. |
| ## | 4275 | 30  | 3  | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.14. |
| ## | 4276 | 90  | 4  | 2016.09.12 | 270 | -1.57079633 | 2016.09.12.14. |
| ## | 4277 | 110 | 5  | 2016.09.12 | 250 | -1.91986218 | 2016.09.12.14. |
| ## | 4278 | 130 | 6  | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.14. |
| ## | 4279 | 60  | 7  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.14. |
| ## | 4280 | 80  | 8  | 2016.09.12 | 280 | -1.39626340 | 2016.09.12.14. |
| ## | 4281 | 290 | 9  | 2016.09.12 | 70  | 1.22173048  | 2016.09.12.14. |
| ## | 4282 | 105 | 10 | 2016.09.12 | 255 | -1.83259571 | 2016.09.12.14. |
| ## | 4283 | 100 | 11 | 2016.09.12 | 260 | -1.74532925 | 2016.09.12.14. |
| ## | 4284 | 240 | 12 | 2016.09.12 | 120 | 2.09439510  | 2016.09.12.14. |
| ## | 4285 | 85  | 13 | 2016.09.12 | 275 | -1.48352986 | 2016.09.12.14. |
| ## | 4286 | 90  | 14 | 2016.09.12 | 270 | -1.57079633 | 2016.09.12.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4287 | 135 | 15 | 2016.09.12 | 225 | -2.35619449 | 2016.09.12.14. |
| ## | 4288 | 130 | 16 | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.14. |
| ## | 4289 | 5   | 17 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4290 | 355 | 18 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4291 | 130 | 19 | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.14. |
| ## | 4292 | 320 | 20 | 2016.09.12 | 40  | 0.69813170  | 2016.09.12.14. |
| ## | 4293 | 355 | 21 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4294 | 5   | 22 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4295 | 130 | 23 | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.14. |
| ## | 4296 | 10  | 24 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.14. |
| ## | 4297 | 355 | 25 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4298 | 330 | 26 | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.14. |
| ## | 4299 | 35  | 27 | 2016.09.12 | 325 | -0.61086524 | 2016.09.12.14. |
| ## | 4300 | 40  | 28 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.14. |
| ## | 4301 | 355 | 29 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4302 | 5   | 30 | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.14. |
| ## | 4303 | 0   | 31 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.14. |
| ## | 4304 | 355 | 32 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.14. |
| ## | 4305 | 220 | 33 | 2016.09.12 | 140 | 2.44346095  | 2016.09.12.14. |
| ## | 4306 | 220 | 34 | 2016.09.12 | 140 | 2.44346095  | 2016.09.12.14. |
| ## | 4307 | 250 | 35 | 2016.09.12 | 110 | 1.91986218  | 2016.09.12.14. |
| ## | 4308 | 200 | 36 | 2016.09.12 | 160 | 2.79252680  | 2016.09.12.14. |
| ## | 4309 | 225 | 37 | 2016.09.12 | 135 | 2.35619449  | 2016.09.12.14. |
| ## | 4310 | 280 | 38 | 2016.09.12 | 80  | 1.39626340  | 2016.09.12.14. |
| ## | 4311 | 240 | 39 | 2016.09.12 | 120 | 2.09439510  | 2016.09.12.14. |
| ## | 4312 | 230 | 40 | 2016.09.12 | 130 | 2.26892803  | 2016.09.12.14. |
| ## | 4313 | 240 | 41 | 2016.09.12 | 120 | 2.09439510  | 2016.09.12.14. |
| ## | 4314 | 190 | 1  | 2016.09.12 | 170 | 2.96705973  | 2016.09.12.15. |
| ## | 4315 | 160 | 2  | 2016.09.12 | 200 | -2.79252680 | 2016.09.12.15. |
| ## | 4316 | 140 | 3  | 2016.09.12 | 220 | -2.44346095 | 2016.09.12.15. |
| ## | 4317 | 140 | 4  | 2016.09.12 | 220 | -2.44346095 | 2016.09.12.15. |
| ## | 4318 | 135 | 5  | 2016.09.12 | 225 | -2.35619449 | 2016.09.12.15. |
| ## | 4319 | 210 | 6  | 2016.09.12 | 150 | 2.61799388  | 2016.09.12.15. |
| ## | 4320 | 120 | 7  | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.15. |
| ## | 4321 | 160 | 8  | 2016.09.12 | 200 | -2.79252680 | 2016.09.12.15. |
| ## | 4322 | 190 | 9  | 2016.09.12 | 170 | 2.96705973  | 2016.09.12.15. |
| ## | 4323 | 30  | 10 | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.15. |
| ## | 4324 | 215 | 11 | 2016.09.12 | 145 | 2.53072742  | 2016.09.12.15. |
| ## | 4325 | 100 | 12 | 2016.09.12 | 260 | -1.74532925 | 2016.09.12.15. |
| ## | 4326 | 130 | 13 | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.15. |
| ## | 4327 | 70  | 14 | 2016.09.12 | 290 | -1.22173048 | 2016.09.12.15. |
| ## | 4328 | 60  | 15 | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.15. |
| ## | 4329 | 260 | 16 | 2016.09.12 | 100 | 1.74532925  | 2016.09.12.15. |
| ## | 4330 | 250 | 17 | 2016.09.12 | 110 | 1.91986218  | 2016.09.12.15. |
| ## | 4331 | 250 | 18 | 2016.09.12 | 110 | 1.91986218  | 2016.09.12.15. |
| ## | 4332 | 160 | 19 | 2016.09.12 | 200 | -2.79252680 | 2016.09.12.15. |
| ## | 4333 | 95  | 20 | 2016.09.12 | 265 | -1.65806279 | 2016.09.12.15. |
| ## | 4334 | 120 | 21 | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.15. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4335 | 85  | 22 | 2016.09.12 | 275 | -1.48352986 | 2016.09.12.15. |
| ## | 4336 | 90  | 23 | 2016.09.12 | 270 | -1.57079633 | 2016.09.12.15. |
| ## | 4337 | 120 | 24 | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.15. |
| ## | 4338 | 135 | 25 | 2016.09.12 | 225 | -2.35619449 | 2016.09.12.15. |
| ## | 4339 | 355 | 26 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.15. |
| ## | 4340 | 40  | 27 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.15. |
| ## | 4341 | 50  | 28 | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.15. |
| ## | 4342 | 30  | 29 | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.15. |
| ## | 4343 | 60  | 30 | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.15. |
| ## | 4344 | 30  | 31 | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.15. |
| ## | 4345 | 60  | 32 | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.15. |
| ## | 4346 | 20  | 33 | 2016.09.12 | 340 | -0.34906585 | 2016.09.12.15. |
| ## | 4347 | 30  | 34 | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.15. |
| ## | 4348 | 30  | 35 | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.15. |
| ## | 4349 | 40  | 36 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.15. |
| ## | 4350 | 40  | 37 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.15. |
| ## | 4351 | 20  | 38 | 2016.09.12 | 340 | -0.34906585 | 2016.09.12.15. |
| ## | 4352 | 30  | 39 | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.15. |
| ## | 4353 | 10  | 40 | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.15. |
| ## | 4354 | 85  | 1  | 2016.09.12 | 275 | -1.48352986 | 2016.09.12.16. |
| ## | 4355 | 60  | 2  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## | 4356 | 60  | 3  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## | 4357 | 90  | 4  | 2016.09.12 | 270 | -1.57079633 | 2016.09.12.16. |
| ## | 4358 | 5   | 5  | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.16. |
| ## | 4359 | 60  | 6  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## | 4360 | 95  | 7  | 2016.09.12 | 265 | -1.65806279 | 2016.09.12.16. |
| ## | 4361 | 60  | 8  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## | 4362 | 30  | 9  | 2016.09.12 | 330 | -0.52359878 | 2016.09.12.16. |
| ## | 4363 | 90  | 10 | 2016.09.12 | 270 | -1.57079633 | 2016.09.12.16. |
| ## | 4364 | 50  | 11 | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.16. |
| ## | 4365 | 330 | 12 | 2016.09.12 | 30  | 0.52359878  | 2016.09.12.16. |
| ## | 4366 | 350 | 13 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |
| ## | 4367 | 355 | 14 | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.16. |
| ## | 4368 | 350 | 15 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |
| ## | 4369 | 355 | 1  | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.16. |
| ## | 4370 | 350 | 2  | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |
| ## | 4371 | 355 | 3  | 2016.09.12 | 5   | 0.08726646  | 2016.09.12.16. |
| ## | 4372 | 0   | 4  | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.16. |
| ## | 4373 | 10  | 5  | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.16. |
| ## | 4374 | 5   | 6  | 2016.09.12 | 355 | -0.08726646 | 2016.09.12.16. |
| ## | 4375 | 350 | 7  | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |
| ## | 4376 | 60  | 8  | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## | 4377 | 10  | 9  | 2016.09.12 | 350 | -0.17453293 | 2016.09.12.16. |
| ## | 4378 | 350 | 10 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |
| ## | 4379 | 60  | 11 | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## | 4380 | 280 | 12 | 2016.09.12 | 80  | 1.39626340  | 2016.09.12.16. |
| ## | 4381 | 25  | 13 | 2016.09.12 | 335 | -0.43633231 | 2016.09.12.16. |
| ## | 4382 | 350 | 14 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 4383 | 340 | 15 | 2016.09.12 | 20  | 0.34906585  | 2016.09.12.16. |
| ## 4384 | 0   | 16 | 2016.09.12 | 0   | 0.00000000  | 2016.09.12.16. |
| ## 4385 | 340 | 17 | 2016.09.12 | 20  | 0.34906585  | 2016.09.12.16. |
| ## 4386 | 350 | 18 | 2016.09.12 | 10  | 0.17453293  | 2016.09.12.16. |
| ## 4387 | 130 | 1  | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.16. |
| ## 4388 | 120 | 2  | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.16. |
| ## 4389 | 110 | 3  | 2016.09.12 | 250 | -1.91986218 | 2016.09.12.16. |
| ## 4390 | 120 | 4  | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.16. |
| ## 4391 | 105 | 5  | 2016.09.12 | 255 | -1.83259571 | 2016.09.12.16. |
| ## 4392 | 120 | 6  | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.16. |
| ## 4393 | 140 | 7  | 2016.09.12 | 220 | -2.44346095 | 2016.09.12.16. |
| ## 4394 | 130 | 8  | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.16. |
| ## 4395 | 120 | 9  | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.16. |
| ## 4396 | 120 | 10 | 2016.09.12 | 240 | -2.09439510 | 2016.09.12.16. |
| ## 4397 | 110 | 11 | 2016.09.12 | 250 | -1.91986218 | 2016.09.12.16. |
| ## 4398 | 115 | 12 | 2016.09.12 | 245 | -2.00712864 | 2016.09.12.16. |
| ## 4399 | 130 | 13 | 2016.09.12 | 230 | -2.26892803 | 2016.09.12.16. |
| ## 4400 | 150 | 14 | 2016.09.12 | 210 | -2.61799388 | 2016.09.12.16. |
| ## 4401 | 40  | 15 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.16. |
| ## 4402 | 50  | 16 | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.16. |
| ## 4403 | 40  | 17 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.16. |
| ## 4404 | 50  | 18 | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.16. |
| ## 4405 | 50  | 19 | 2016.09.12 | 310 | -0.87266463 | 2016.09.12.16. |
| ## 4406 | 40  | 20 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.16. |
| ## 4407 | 60  | 21 | 2016.09.12 | 300 | -1.04719755 | 2016.09.12.16. |
| ## 4408 | 40  | 22 | 2016.09.12 | 320 | -0.69813170 | 2016.09.12.16. |
| ## 4409 | 70  | 23 | 2016.09.12 | 290 | -1.22173048 | 2016.09.12.16. |
| ## 4410 | 350 | 1  | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.13. |
| ## 4411 | 100 | 2  | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.13. |
| ## 4412 | 130 | 3  | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.13. |
| ## 4413 | 340 | 4  | 2016.09.13 | 20  | 0.34906585  | 2016.09.13.13. |
| ## 4414 | 95  | 5  | 2016.09.13 | 265 | -1.65806279 | 2016.09.13.13. |
| ## 4415 | 85  | 6  | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.13. |
| ## 4416 | 10  | 7  | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.13. |
| ## 4417 | 310 | 8  | 2016.09.13 | 50  | 0.87266463  | 2016.09.13.13. |
| ## 4418 | 195 | 9  | 2016.09.13 | 165 | 2.87979327  | 2016.09.13.13. |
| ## 4419 | 230 | 10 | 2016.09.13 | 130 | 2.26892803  | 2016.09.13.13. |
| ## 4420 | 240 | 11 | 2016.09.13 | 120 | 2.09439510  | 2016.09.13.13. |
| ## 4421 | 185 | 12 | 2016.09.13 | 175 | 3.05432619  | 2016.09.13.13. |
| ## 4422 | 70  | 13 | 2016.09.13 | 290 | -1.22173048 | 2016.09.13.13. |
| ## 4423 | 110 | 14 | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.13. |
| ## 4424 | 270 | 15 | 2016.09.13 | 90  | 1.57079633  | 2016.09.13.13. |
| ## 4425 | 240 | 16 | 2016.09.13 | 120 | 2.09439510  | 2016.09.13.13. |
| ## 4426 | 240 | 17 | 2016.09.13 | 120 | 2.09439510  | 2016.09.13.13. |
| ## 4427 | 265 | 18 | 2016.09.13 | 95  | 1.65806279  | 2016.09.13.13. |
| ## 4428 | 340 | 19 | 2016.09.13 | 20  | 0.34906585  | 2016.09.13.13. |
| ## 4429 | 330 | 20 | 2016.09.13 | 30  | 0.52359878  | 2016.09.13.13. |
| ## 4430 | 350 | 21 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4431 | 350 | 22 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.13. |
| ## | 4432 | 330 | 23 | 2016.09.13 | 30  | 0.52359878  | 2016.09.13.13. |
| ## | 4433 | 350 | 24 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.13. |
| ## | 4434 | 345 | 25 | 2016.09.13 | 15  | 0.26179939  | 2016.09.13.13. |
| ## | 4435 | 355 | 26 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.13. |
| ## | 4436 | 350 | 27 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.13. |
| ## | 4437 | 340 | 28 | 2016.09.13 | 20  | 0.34906585  | 2016.09.13.13. |
| ## | 4438 | 355 | 29 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.13. |
| ## | 4439 | 335 | 30 | 2016.09.13 | 25  | 0.43633231  | 2016.09.13.13. |
| ## | 4440 | 320 | 31 | 2016.09.13 | 40  | 0.69813170  | 2016.09.13.13. |
| ## | 4441 | 190 | 1  | 2016.09.13 | 170 | 2.96705973  | 2016.09.13.14. |
| ## | 4442 | 235 | 2  | 2016.09.13 | 125 | 2.18166156  | 2016.09.13.14. |
| ## | 4443 | 105 | 3  | 2016.09.13 | 255 | -1.83259571 | 2016.09.13.14. |
| ## | 4444 | 220 | 4  | 2016.09.13 | 140 | 2.44346095  | 2016.09.13.14. |
| ## | 4445 | 85  | 5  | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.14. |
| ## | 4446 | 80  | 6  | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## | 4447 | 60  | 7  | 2016.09.13 | 300 | -1.04719755 | 2016.09.13.14. |
| ## | 4448 | 130 | 8  | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## | 4449 | 100 | 9  | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## | 4450 | 130 | 10 | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## | 4451 | 75  | 11 | 2016.09.13 | 285 | -1.30899694 | 2016.09.13.14. |
| ## | 4452 | 120 | 12 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## | 4453 | 100 | 13 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## | 4454 | 100 | 14 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## | 4455 | 80  | 15 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## | 4456 | 130 | 16 | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## | 4457 | 90  | 17 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4458 | 90  | 18 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4459 | 90  | 19 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4460 | 70  | 20 | 2016.09.13 | 290 | -1.22173048 | 2016.09.13.14. |
| ## | 4461 | 130 | 21 | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## | 4462 | 110 | 22 | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.14. |
| ## | 4463 | 80  | 23 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## | 4464 | 80  | 24 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## | 4465 | 150 | 25 | 2016.09.13 | 210 | -2.61799388 | 2016.09.13.14. |
| ## | 4466 | 350 | 26 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4467 | 20  | 27 | 2016.09.13 | 340 | -0.34906585 | 2016.09.13.14. |
| ## | 4468 | 10  | 28 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## | 4469 | 355 | 29 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## | 4470 | 355 | 30 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## | 4471 | 355 | 31 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## | 4472 | 350 | 32 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4473 | 10  | 33 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## | 4474 | 350 | 34 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4475 | 5   | 35 | 2016.09.13 | 355 | -0.08726646 | 2016.09.13.14. |
| ## | 4476 | 20  | 36 | 2016.09.13 | 340 | -0.34906585 | 2016.09.13.14. |
| ## | 4477 | 355 | 37 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## | 4478 | 60  | 38 | 2016.09.13 | 300 | -1.04719755 | 2016.09.13.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4479 | 5   | 39 | 2016.09.13 | 355 | -0.08726646 | 2016.09.13.14. |
| ## | 4480 | 15  | 40 | 2016.09.13 | 345 | -0.26179939 | 2016.09.13.14. |
| ## | 4481 | 10  | 41 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## | 4482 | 40  | 42 | 2016.09.13 | 320 | -0.69813170 | 2016.09.13.14. |
| ## | 4483 | 350 | 43 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4484 | 10  | 44 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## | 4485 | 10  | 45 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## | 4486 | 35  | 46 | 2016.09.13 | 325 | -0.61086524 | 2016.09.13.14. |
| ## | 4487 | 30  | 47 | 2016.09.13 | 330 | -0.52359878 | 2016.09.13.14. |
| ## | 4488 | 5   | 48 | 2016.09.13 | 355 | -0.08726646 | 2016.09.13.14. |
| ## | 4489 | 80  | 49 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## | 4490 | 50  | 50 | 2016.09.13 | 310 | -0.87266463 | 2016.09.13.14. |
| ## | 4491 | 100 | 51 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## | 4492 | 105 | 52 | 2016.09.13 | 255 | -1.83259571 | 2016.09.13.14. |
| ## | 4493 | 85  | 53 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.14. |
| ## | 4494 | 120 | 54 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## | 4495 | 95  | 55 | 2016.09.13 | 265 | -1.65806279 | 2016.09.13.14. |
| ## | 4496 | 120 | 56 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## | 4497 | 60  | 57 | 2016.09.13 | 300 | -1.04719755 | 2016.09.13.14. |
| ## | 4498 | 100 | 58 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## | 4499 | 30  | 59 | 2016.09.13 | 330 | -0.52359878 | 2016.09.13.14. |
| ## | 4500 | 350 | 60 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4501 | 50  | 61 | 2016.09.13 | 310 | -0.87266463 | 2016.09.13.14. |
| ## | 4502 | 90  | 62 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4503 | 90  | 1  | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4504 | 185 | 2  | 2016.09.13 | 175 | 3.05432619  | 2016.09.13.14. |
| ## | 4505 | 95  | 3  | 2016.09.13 | 265 | -1.65806279 | 2016.09.13.14. |
| ## | 4506 | 140 | 4  | 2016.09.13 | 220 | -2.44346095 | 2016.09.13.14. |
| ## | 4507 | 175 | 5  | 2016.09.13 | 185 | -3.05432619 | 2016.09.13.14. |
| ## | 4508 | 120 | 6  | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## | 4509 | 230 | 7  | 2016.09.13 | 130 | 2.26892803  | 2016.09.13.14. |
| ## | 4510 | 110 | 8  | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.14. |
| ## | 4511 | 160 | 9  | 2016.09.13 | 200 | -2.79252680 | 2016.09.13.14. |
| ## | 4512 | 90  | 10 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4513 | 200 | 11 | 2016.09.13 | 160 | 2.79252680  | 2016.09.13.14. |
| ## | 4514 | 190 | 12 | 2016.09.13 | 170 | 2.96705973  | 2016.09.13.14. |
| ## | 4515 | 120 | 13 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## | 4516 | 90  | 14 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## | 4517 | 70  | 15 | 2016.09.13 | 290 | -1.22173048 | 2016.09.13.14. |
| ## | 4518 | 305 | 16 | 2016.09.13 | 55  | 0.95993109  | 2016.09.13.14. |
| ## | 4519 | 220 | 17 | 2016.09.13 | 140 | 2.44346095  | 2016.09.13.14. |
| ## | 4520 | 355 | 18 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## | 4521 | 0   | 19 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## | 4522 | 350 | 20 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4523 | 350 | 21 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## | 4524 | 330 | 22 | 2016.09.13 | 30  | 0.52359878  | 2016.09.13.14. |
| ## | 4525 | 345 | 23 | 2016.09.13 | 15  | 0.26179939  | 2016.09.13.14. |
| ## | 4526 | 350 | 24 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 4527 | 5   | 25 | 2016.09.13 | 355 | -0.08726646 | 2016.09.13.14. |
| ## 4528 | 345 | 26 | 2016.09.13 | 15  | 0.26179939  | 2016.09.13.14. |
| ## 4529 | 0   | 27 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## 4530 | 0   | 28 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## 4531 | 355 | 29 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## 4532 | 355 | 30 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## 4533 | 340 | 31 | 2016.09.13 | 20  | 0.34906585  | 2016.09.13.14. |
| ## 4534 | 350 | 32 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## 4535 | 350 | 33 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## 4536 | 100 | 1  | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## 4537 | 90  | 2  | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## 4538 | 165 | 3  | 2016.09.13 | 195 | -2.87979327 | 2016.09.13.14. |
| ## 4539 | 170 | 4  | 2016.09.13 | 190 | -2.96705973 | 2016.09.13.14. |
| ## 4540 | 240 | 5  | 2016.09.13 | 120 | 2.09439510  | 2016.09.13.14. |
| ## 4541 | 70  | 6  | 2016.09.13 | 290 | -1.22173048 | 2016.09.13.14. |
| ## 4542 | 95  | 7  | 2016.09.13 | 265 | -1.65806279 | 2016.09.13.14. |
| ## 4543 | 100 | 8  | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## 4544 | 130 | 9  | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## 4545 | 135 | 10 | 2016.09.13 | 225 | -2.35619449 | 2016.09.13.14. |
| ## 4546 | 150 | 11 | 2016.09.13 | 210 | -2.61799388 | 2016.09.13.14. |
| ## 4547 | 200 | 12 | 2016.09.13 | 160 | 2.79252680  | 2016.09.13.14. |
| ## 4548 | 110 | 13 | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.14. |
| ## 4549 | 160 | 14 | 2016.09.13 | 200 | -2.79252680 | 2016.09.13.14. |
| ## 4550 | 175 | 15 | 2016.09.13 | 185 | -3.05432619 | 2016.09.13.14. |
| ## 4551 | 150 | 16 | 2016.09.13 | 210 | -2.61799388 | 2016.09.13.14. |
| ## 4552 | 160 | 17 | 2016.09.13 | 200 | -2.79252680 | 2016.09.13.14. |
| ## 4553 | 100 | 18 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## 4554 | 140 | 19 | 2016.09.13 | 220 | -2.44346095 | 2016.09.13.14. |
| ## 4555 | 0   | 20 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## 4556 | 0   | 21 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## 4557 | 350 | 22 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## 4558 | 350 | 23 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## 4559 | 15  | 24 | 2016.09.13 | 345 | -0.26179939 | 2016.09.13.14. |
| ## 4560 | 0   | 25 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## 4561 | 10  | 26 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## 4562 | 20  | 27 | 2016.09.13 | 340 | -0.34906585 | 2016.09.13.14. |
| ## 4563 | 30  | 28 | 2016.09.13 | 330 | -0.52359878 | 2016.09.13.14. |
| ## 4564 | 340 | 29 | 2016.09.13 | 20  | 0.34906585  | 2016.09.13.14. |
| ## 4565 | 20  | 30 | 2016.09.13 | 340 | -0.34906585 | 2016.09.13.14. |
| ## 4566 | 355 | 31 | 2016.09.13 | 5   | 0.08726646  | 2016.09.13.14. |
| ## 4567 | 185 | 32 | 2016.09.13 | 175 | 3.05432619  | 2016.09.13.14. |
| ## 4568 | 80  | 33 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## 4569 | 95  | 34 | 2016.09.13 | 265 | -1.65806279 | 2016.09.13.14. |
| ## 4570 | 0   | 35 | 2016.09.13 | 0   | 0.00000000  | 2016.09.13.14. |
| ## 4571 | 130 | 36 | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## 4572 | 95  | 37 | 2016.09.13 | 265 | -1.65806279 | 2016.09.13.14. |
| ## 4573 | 80  | 38 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## 4574 | 105 | 39 | 2016.09.13 | 255 | -1.83259571 | 2016.09.13.14. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 4575 | 140 | 40 | 2016.09.13 | 220 | -2.44346095 | 2016.09.13.14. |
| ## 4576 | 110 | 41 | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.14. |
| ## 4577 | 100 | 42 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## 4578 | 90  | 43 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## 4579 | 100 | 44 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## 4580 | 120 | 45 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## 4581 | 85  | 46 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.14. |
| ## 4582 | 90  | 47 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## 4583 | 110 | 48 | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.14. |
| ## 4584 | 110 | 49 | 2016.09.13 | 250 | -1.91986218 | 2016.09.13.14. |
| ## 4585 | 10  | 50 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## 4586 | 85  | 51 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.14. |
| ## 4587 | 75  | 52 | 2016.09.13 | 285 | -1.30899694 | 2016.09.13.14. |
| ## 4588 | 80  | 53 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## 4589 | 85  | 54 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.14. |
| ## 4590 | 130 | 55 | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.14. |
| ## 4591 | 100 | 56 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.14. |
| ## 4592 | 120 | 57 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## 4593 | 90  | 58 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## 4594 | 90  | 59 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.14. |
| ## 4595 | 120 | 60 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.14. |
| ## 4596 | 105 | 61 | 2016.09.13 | 255 | -1.83259571 | 2016.09.13.14. |
| ## 4597 | 10  | 62 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## 4598 | 10  | 63 | 2016.09.13 | 350 | -0.17453293 | 2016.09.13.14. |
| ## 4599 | 50  | 64 | 2016.09.13 | 310 | -0.87266463 | 2016.09.13.14. |
| ## 4600 | 80  | 65 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.14. |
| ## 4601 | 15  | 66 | 2016.09.13 | 345 | -0.26179939 | 2016.09.13.14. |
| ## 4602 | 350 | 67 | 2016.09.13 | 10  | 0.17453293  | 2016.09.13.14. |
| ## 4603 | 225 | 1  | 2016.09.13 | 135 | 2.35619449  | 2016.09.13.16. |
| ## 4604 | 215 | 2  | 2016.09.13 | 145 | 2.53072742  | 2016.09.13.16. |
| ## 4605 | 205 | 3  | 2016.09.13 | 155 | 2.70526034  | 2016.09.13.16. |
| ## 4606 | 240 | 4  | 2016.09.13 | 120 | 2.09439510  | 2016.09.13.16. |
| ## 4607 | 180 | 5  | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## 4608 | 140 | 6  | 2016.09.13 | 220 | -2.44346095 | 2016.09.13.16. |
| ## 4609 | 190 | 7  | 2016.09.13 | 170 | 2.96705973  | 2016.09.13.16. |
| ## 4610 | 155 | 8  | 2016.09.13 | 205 | -2.70526034 | 2016.09.13.16. |
| ## 4611 | 180 | 9  | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## 4612 | 180 | 10 | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## 4613 | 190 | 11 | 2016.09.13 | 170 | 2.96705973  | 2016.09.13.16. |
| ## 4614 | 150 | 12 | 2016.09.13 | 210 | -2.61799388 | 2016.09.13.16. |
| ## 4615 | 195 | 13 | 2016.09.13 | 165 | 2.87979327  | 2016.09.13.16. |
| ## 4616 | 160 | 14 | 2016.09.13 | 200 | -2.79252680 | 2016.09.13.16. |
| ## 4617 | 175 | 15 | 2016.09.13 | 185 | -3.05432619 | 2016.09.13.16. |
| ## 4618 | 85  | 16 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.16. |
| ## 4619 | 80  | 17 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.16. |
| ## 4620 | 80  | 18 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.16. |
| ## 4621 | 90  | 19 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.16. |
| ## 4622 | 75  | 20 | 2016.09.13 | 285 | -1.30899694 | 2016.09.13.16. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4623 | 100 | 21 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.16. |
| ## | 4624 | 90  | 22 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.16. |
| ## | 4625 | 120 | 23 | 2016.09.13 | 240 | -2.09439510 | 2016.09.13.16. |
| ## | 4626 | 85  | 24 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.16. |
| ## | 4627 | 70  | 25 | 2016.09.13 | 290 | -1.22173048 | 2016.09.13.16. |
| ## | 4628 | 100 | 26 | 2016.09.13 | 260 | -1.74532925 | 2016.09.13.16. |
| ## | 4629 | 85  | 27 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.16. |
| ## | 4630 | 85  | 28 | 2016.09.13 | 275 | -1.48352986 | 2016.09.13.16. |
| ## | 4631 | 90  | 29 | 2016.09.13 | 270 | -1.57079633 | 2016.09.13.16. |
| ## | 4632 | 150 | 30 | 2016.09.13 | 210 | -2.61799388 | 2016.09.13.16. |
| ## | 4633 | 220 | 31 | 2016.09.13 | 140 | 2.44346095  | 2016.09.13.16. |
| ## | 4634 | 185 | 32 | 2016.09.13 | 175 | 3.05432619  | 2016.09.13.16. |
| ## | 4635 | 190 | 33 | 2016.09.13 | 170 | 2.96705973  | 2016.09.13.16. |
| ## | 4636 | 20  | 34 | 2016.09.13 | 340 | -0.34906585 | 2016.09.13.16. |
| ## | 4637 | 15  | 35 | 2016.09.13 | 345 | -0.26179939 | 2016.09.13.16. |
| ## | 4638 | 80  | 36 | 2016.09.13 | 280 | -1.39626340 | 2016.09.13.16. |
| ## | 4639 | 70  | 37 | 2016.09.13 | 290 | -1.22173048 | 2016.09.13.16. |
| ## | 4640 | 240 | 38 | 2016.09.13 | 120 | 2.09439510  | 2016.09.13.16. |
| ## | 4641 | 160 | 39 | 2016.09.13 | 200 | -2.79252680 | 2016.09.13.16. |
| ## | 4642 | 210 | 40 | 2016.09.13 | 150 | 2.61799388  | 2016.09.13.16. |
| ## | 4643 | 220 | 41 | 2016.09.13 | 140 | 2.44346095  | 2016.09.13.16. |
| ## | 4644 | 165 | 42 | 2016.09.13 | 195 | -2.87979327 | 2016.09.13.16. |
| ## | 4645 | 180 | 1  | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## | 4646 | 250 | 2  | 2016.09.13 | 110 | 1.91986218  | 2016.09.13.16. |
| ## | 4647 | 145 | 3  | 2016.09.13 | 215 | -2.53072742 | 2016.09.13.16. |
| ## | 4648 | 255 | 4  | 2016.09.13 | 105 | 1.83259571  | 2016.09.13.16. |
| ## | 4649 | 130 | 5  | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.16. |
| ## | 4650 | 130 | 6  | 2016.09.13 | 230 | -2.26892803 | 2016.09.13.16. |
| ## | 4651 | 150 | 7  | 2016.09.13 | 210 | -2.61799388 | 2016.09.13.16. |
| ## | 4652 | 135 | 8  | 2016.09.13 | 225 | -2.35619449 | 2016.09.13.16. |
| ## | 4653 | 300 | 9  | 2016.09.13 | 60  | 1.04719755  | 2016.09.13.16. |
| ## | 4654 | 235 | 10 | 2016.09.13 | 125 | 2.18166156  | 2016.09.13.16. |
| ## | 4655 | 290 | 11 | 2016.09.13 | 70  | 1.22173048  | 2016.09.13.16. |
| ## | 4656 | 330 | 12 | 2016.09.13 | 30  | 0.52359878  | 2016.09.13.16. |
| ## | 4657 | 180 | 13 | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## | 4658 | 320 | 14 | 2016.09.13 | 40  | 0.69813170  | 2016.09.13.16. |
| ## | 4659 | 280 | 15 | 2016.09.13 | 80  | 1.39626340  | 2016.09.13.16. |
| ## | 4660 | 315 | 16 | 2016.09.13 | 45  | 0.78539816  | 2016.09.13.16. |
| ## | 4661 | 220 | 17 | 2016.09.13 | 140 | 2.44346095  | 2016.09.13.16. |
| ## | 4662 | 290 | 18 | 2016.09.13 | 70  | 1.22173048  | 2016.09.13.16. |
| ## | 4663 | 250 | 19 | 2016.09.13 | 110 | 1.91986218  | 2016.09.13.16. |
| ## | 4664 | 260 | 20 | 2016.09.13 | 100 | 1.74532925  | 2016.09.13.16. |
| ## | 4665 | 175 | 21 | 2016.09.13 | 185 | -3.05432619 | 2016.09.13.16. |
| ## | 4666 | 170 | 22 | 2016.09.13 | 190 | -2.96705973 | 2016.09.13.16. |
| ## | 4667 | 190 | 23 | 2016.09.13 | 170 | 2.96705973  | 2016.09.13.16. |
| ## | 4668 | 180 | 24 | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## | 4669 | 165 | 25 | 2016.09.13 | 195 | -2.87979327 | 2016.09.13.16. |
| ## | 4670 | 170 | 26 | 2016.09.13 | 190 | -2.96705973 | 2016.09.13.16. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4671 | 160 | 27 | 2016.09.13 | 200 | -2.79252680 | 2016.09.13.16. |
| ## | 4672 | 180 | 28 | 2016.09.13 | 180 | 3.14159265  | 2016.09.13.16. |
| ## | 4673 | 185 | 29 | 2016.09.13 | 175 | 3.05432619  | 2016.09.13.16. |
| ## | 4674 | 155 | 30 | 2016.09.13 | 205 | -2.70526034 | 2016.09.13.16. |
| ## | 4675 | 20  | 1  | 2016.09.14 | 340 | -0.34906585 | 2016.09.14.13. |
| ## | 4676 | 100 | 2  | 2016.09.14 | 260 | -1.74532925 | 2016.09.14.13. |
| ## | 4677 | 50  | 3  | 2016.09.14 | 310 | -0.87266463 | 2016.09.14.13. |
| ## | 4678 | 100 | 4  | 2016.09.14 | 260 | -1.74532925 | 2016.09.14.13. |
| ## | 4679 | 85  | 5  | 2016.09.14 | 275 | -1.48352986 | 2016.09.14.13. |
| ## | 4680 | 190 | 6  | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.13. |
| ## | 4681 | 70  | 7  | 2016.09.14 | 290 | -1.22173048 | 2016.09.14.13. |
| ## | 4682 | 120 | 8  | 2016.09.14 | 240 | -2.09439510 | 2016.09.14.13. |
| ## | 4683 | 170 | 9  | 2016.09.14 | 190 | -2.96705973 | 2016.09.14.13. |
| ## | 4684 | 90  | 10 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.13. |
| ## | 4685 | 95  | 11 | 2016.09.14 | 265 | -1.65806279 | 2016.09.14.13. |
| ## | 4686 | 50  | 12 | 2016.09.14 | 310 | -0.87266463 | 2016.09.14.13. |
| ## | 4687 | 80  | 13 | 2016.09.14 | 280 | -1.39626340 | 2016.09.14.13. |
| ## | 4688 | 85  | 14 | 2016.09.14 | 275 | -1.48352986 | 2016.09.14.13. |
| ## | 4689 | 65  | 15 | 2016.09.14 | 295 | -1.13446401 | 2016.09.14.13. |
| ## | 4690 | 80  | 16 | 2016.09.14 | 280 | -1.39626340 | 2016.09.14.13. |
| ## | 4691 | 90  | 17 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.13. |
| ## | 4692 | 355 | 18 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.13. |
| ## | 4693 | 350 | 19 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.13. |
| ## | 4694 | 0   | 20 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.13. |
| ## | 4695 | 10  | 21 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.13. |
| ## | 4696 | 45  | 22 | 2016.09.14 | 315 | -0.78539816 | 2016.09.14.13. |
| ## | 4697 | 355 | 23 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.13. |
| ## | 4698 | 0   | 24 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.13. |
| ## | 4699 | 355 | 25 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.13. |
| ## | 4700 | 340 | 26 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.13. |
| ## | 4701 | 335 | 27 | 2016.09.14 | 25  | 0.43633231  | 2016.09.14.13. |
| ## | 4702 | 345 | 28 | 2016.09.14 | 15  | 0.26179939  | 2016.09.14.13. |
| ## | 4703 | 340 | 29 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.13. |
| ## | 4704 | 340 | 30 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.13. |
| ## | 4705 | 350 | 31 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.13. |
| ## | 4706 | 235 | 1  | 2016.09.14 | 125 | 2.18166156  | 2016.09.14.14. |
| ## | 4707 | 290 | 2  | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.14. |
| ## | 4708 | 315 | 3  | 2016.09.14 | 45  | 0.78539816  | 2016.09.14.14. |
| ## | 4709 | 5   | 4  | 2016.09.14 | 355 | -0.08726646 | 2016.09.14.14. |
| ## | 4710 | 340 | 5  | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.14. |
| ## | 4711 | 315 | 6  | 2016.09.14 | 45  | 0.78539816  | 2016.09.14.14. |
| ## | 4712 | 60  | 7  | 2016.09.14 | 300 | -1.04719755 | 2016.09.14.14. |
| ## | 4713 | 275 | 8  | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.14. |
| ## | 4714 | 350 | 9  | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.14. |
| ## | 4715 | 320 | 10 | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.14. |
| ## | 4716 | 0   | 11 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.14. |
| ## | 4717 | 320 | 12 | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.14. |
| ## | 4718 | 330 | 13 | 2016.09.14 | 30  | 0.52359878  | 2016.09.14.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4719 | 280 | 14 | 2016.09.14 | 80  | 1.39626340  | 2016.09.14.14. |
| ## | 4720 | 50  | 15 | 2016.09.14 | 310 | -0.87266463 | 2016.09.14.14. |
| ## | 4721 | 0   | 16 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.14. |
| ## | 4722 | 330 | 17 | 2016.09.14 | 30  | 0.52359878  | 2016.09.14.14. |
| ## | 4723 | 345 | 18 | 2016.09.14 | 15  | 0.26179939  | 2016.09.14.14. |
| ## | 4724 | 260 | 19 | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.14. |
| ## | 4725 | 350 | 20 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.14. |
| ## | 4726 | 260 | 21 | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.14. |
| ## | 4727 | 355 | 22 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.14. |
| ## | 4728 | 275 | 23 | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.14. |
| ## | 4729 | 345 | 24 | 2016.09.14 | 15  | 0.26179939  | 2016.09.14.14. |
| ## | 4730 | 260 | 25 | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.14. |
| ## | 4731 | 340 | 26 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.14. |
| ## | 4732 | 280 | 27 | 2016.09.14 | 80  | 1.39626340  | 2016.09.14.14. |
| ## | 4733 | 240 | 28 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.14. |
| ## | 4734 | 175 | 29 | 2016.09.14 | 185 | -3.05432619 | 2016.09.14.14. |
| ## | 4735 | 175 | 30 | 2016.09.14 | 185 | -3.05432619 | 2016.09.14.14. |
| ## | 4736 | 160 | 31 | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.14. |
| ## | 4737 | 160 | 32 | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.14. |
| ## | 4738 | 190 | 33 | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.14. |
| ## | 4739 | 150 | 34 | 2016.09.14 | 210 | -2.61799388 | 2016.09.14.14. |
| ## | 4740 | 180 | 35 | 2016.09.14 | 180 | 3.14159265  | 2016.09.14.14. |
| ## | 4741 | 170 | 36 | 2016.09.14 | 190 | -2.96705973 | 2016.09.14.14. |
| ## | 4742 | 180 | 37 | 2016.09.14 | 180 | 3.14159265  | 2016.09.14.14. |
| ## | 4743 | 250 | 1  | 2016.09.14 | 110 | 1.91986218  | 2016.09.14.14. |
| ## | 4744 | 265 | 2  | 2016.09.14 | 95  | 1.65806279  | 2016.09.14.14. |
| ## | 4745 | 220 | 3  | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.14. |
| ## | 4746 | 220 | 4  | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.14. |
| ## | 4747 | 270 | 5  | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.14. |
| ## | 4748 | 275 | 6  | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.14. |
| ## | 4749 | 265 | 7  | 2016.09.14 | 95  | 1.65806279  | 2016.09.14.14. |
| ## | 4750 | 210 | 8  | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.14. |
| ## | 4751 | 265 | 9  | 2016.09.14 | 95  | 1.65806279  | 2016.09.14.14. |
| ## | 4752 | 275 | 10 | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.14. |
| ## | 4753 | 170 | 11 | 2016.09.14 | 190 | -2.96705973 | 2016.09.14.14. |
| ## | 4754 | 165 | 12 | 2016.09.14 | 195 | -2.87979327 | 2016.09.14.14. |
| ## | 4755 | 160 | 13 | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.14. |
| ## | 4756 | 170 | 14 | 2016.09.14 | 190 | -2.96705973 | 2016.09.14.14. |
| ## | 4757 | 160 | 15 | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.14. |
| ## | 4758 | 150 | 16 | 2016.09.14 | 210 | -2.61799388 | 2016.09.14.14. |
| ## | 4759 | 170 | 17 | 2016.09.14 | 190 | -2.96705973 | 2016.09.14.14. |
| ## | 4760 | 165 | 18 | 2016.09.14 | 195 | -2.87979327 | 2016.09.14.14. |
| ## | 4761 | 190 | 19 | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.14. |
| ## | 4762 | 175 | 20 | 2016.09.14 | 185 | -3.05432619 | 2016.09.14.14. |
| ## | 4763 | 165 | 21 | 2016.09.14 | 195 | -2.87979327 | 2016.09.14.14. |
| ## | 4764 | 160 | 22 | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.14. |
| ## | 4765 | 230 | 23 | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.14. |
| ## | 4766 | 70  | 24 | 2016.09.14 | 290 | -1.22173048 | 2016.09.14.14. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 4767 | 90  | 25 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.14. |
| ## 4768 | 200 | 26 | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.14. |
| ## 4769 | 340 | 27 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.14. |
| ## 4770 | 220 | 28 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.14. |
| ## 4771 | 145 | 29 | 2016.09.14 | 215 | -2.53072742 | 2016.09.14.14. |
| ## 4772 | 180 | 30 | 2016.09.14 | 180 | 3.14159265  | 2016.09.14.14. |
| ## 4773 | 320 | 31 | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.14. |
| ## 4774 | 190 | 32 | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.14. |
| ## 4775 | 145 | 33 | 2016.09.14 | 215 | -2.53072742 | 2016.09.14.14. |
| ## 4776 | 270 | 34 | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.14. |
| ## 4777 | 210 | 35 | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.14. |
| ## 4778 | 150 | 36 | 2016.09.14 | 210 | -2.61799388 | 2016.09.14.14. |
| ## 4779 | 235 | 37 | 2016.09.14 | 125 | 2.18166156  | 2016.09.14.14. |
| ## 4780 | 250 | 38 | 2016.09.14 | 110 | 1.91986218  | 2016.09.14.14. |
| ## 4781 | 195 | 39 | 2016.09.14 | 165 | 2.87979327  | 2016.09.14.14. |
| ## 4782 | 330 | 40 | 2016.09.14 | 30  | 0.52359878  | 2016.09.14.14. |
| ## 4783 | 120 | 41 | 2016.09.14 | 240 | -2.09439510 | 2016.09.14.14. |
| ## 4784 | 175 | 42 | 2016.09.14 | 185 | -3.05432619 | 2016.09.14.14. |
| ## 4785 | 60  | 43 | 2016.09.14 | 300 | -1.04719755 | 2016.09.14.14. |
| ## 4786 | 290 | 44 | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.14. |
| ## 4787 | 30  | 45 | 2016.09.14 | 330 | -0.52359878 | 2016.09.14.14. |
| ## 4788 | 60  | 1  | 2016.09.14 | 300 | -1.04719755 | 2016.09.14.14. |
| ## 4789 | 200 | 2  | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.14. |
| ## 4790 | 190 | 3  | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.14. |
| ## 4791 | 140 | 4  | 2016.09.14 | 220 | -2.44346095 | 2016.09.14.14. |
| ## 4792 | 200 | 5  | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.14. |
| ## 4793 | 180 | 6  | 2016.09.14 | 180 | 3.14159265  | 2016.09.14.14. |
| ## 4794 | 120 | 7  | 2016.09.14 | 240 | -2.09439510 | 2016.09.14.14. |
| ## 4795 | 185 | 8  | 2016.09.14 | 175 | 3.05432619  | 2016.09.14.14. |
| ## 4796 | 210 | 9  | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.14. |
| ## 4797 | 260 | 10 | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.14. |
| ## 4798 | 140 | 11 | 2016.09.14 | 220 | -2.44346095 | 2016.09.14.14. |
| ## 4799 | 200 | 12 | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.14. |
| ## 4800 | 215 | 13 | 2016.09.14 | 145 | 2.53072742  | 2016.09.14.14. |
| ## 4801 | 150 | 14 | 2016.09.14 | 210 | -2.61799388 | 2016.09.14.14. |
| ## 4802 | 5   | 15 | 2016.09.14 | 355 | -0.08726646 | 2016.09.14.14. |
| ## 4803 | 40  | 16 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.14. |
| ## 4804 | 40  | 17 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.14. |
| ## 4805 | 0   | 18 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.14. |
| ## 4806 | 30  | 19 | 2016.09.14 | 330 | -0.52359878 | 2016.09.14.14. |
| ## 4807 | 30  | 20 | 2016.09.14 | 330 | -0.52359878 | 2016.09.14.14. |
| ## 4808 | 10  | 21 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.14. |
| ## 4809 | 0   | 22 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.14. |
| ## 4810 | 340 | 23 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.14. |
| ## 4811 | 10  | 24 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.14. |
| ## 4812 | 5   | 25 | 2016.09.14 | 355 | -0.08726646 | 2016.09.14.14. |
| ## 4813 | 10  | 26 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.14. |
| ## 4814 | 350 | 27 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4815 | 350 | 28 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.14. |
| ## | 4816 | 340 | 29 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.14. |
| ## | 4817 | 10  | 30 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.14. |
| ## | 4818 | 355 | 31 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.14. |
| ## | 4819 | 0   | 32 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.14. |
| ## | 4820 | 355 | 33 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.14. |
| ## | 4821 | 270 | 34 | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.14. |
| ## | 4822 | 330 | 35 | 2016.09.14 | 30  | 0.52359878  | 2016.09.14.14. |
| ## | 4823 | 270 | 36 | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.14. |
| ## | 4824 | 50  | 37 | 2016.09.14 | 310 | -0.87266463 | 2016.09.14.14. |
| ## | 4825 | 320 | 38 | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.14. |
| ## | 4826 | 290 | 39 | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.14. |
| ## | 4827 | 110 | 40 | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.14. |
| ## | 4828 | 5   | 41 | 2016.09.14 | 355 | -0.08726646 | 2016.09.14.14. |
| ## | 4829 | 80  | 1  | 2016.09.14 | 280 | -1.39626340 | 2016.09.14.15. |
| ## | 4830 | 130 | 2  | 2016.09.14 | 230 | -2.26892803 | 2016.09.14.15. |
| ## | 4831 | 110 | 3  | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.15. |
| ## | 4832 | 130 | 4  | 2016.09.14 | 230 | -2.26892803 | 2016.09.14.15. |
| ## | 4833 | 115 | 5  | 2016.09.14 | 245 | -2.00712864 | 2016.09.14.15. |
| ## | 4834 | 145 | 6  | 2016.09.14 | 215 | -2.53072742 | 2016.09.14.15. |
| ## | 4835 | 110 | 7  | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.15. |
| ## | 4836 | 120 | 8  | 2016.09.14 | 240 | -2.09439510 | 2016.09.14.15. |
| ## | 4837 | 120 | 9  | 2016.09.14 | 240 | -2.09439510 | 2016.09.14.15. |
| ## | 4838 | 115 | 10 | 2016.09.14 | 245 | -2.00712864 | 2016.09.14.15. |
| ## | 4839 | 110 | 11 | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.15. |
| ## | 4840 | 110 | 12 | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.15. |
| ## | 4841 | 90  | 13 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.15. |
| ## | 4842 | 105 | 14 | 2016.09.14 | 255 | -1.83259571 | 2016.09.14.15. |
| ## | 4843 | 220 | 15 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.15. |
| ## | 4844 | 340 | 16 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.15. |
| ## | 4845 | 240 | 17 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.15. |
| ## | 4846 | 240 | 18 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.15. |
| ## | 4847 | 140 | 19 | 2016.09.14 | 220 | -2.44346095 | 2016.09.14.15. |
| ## | 4848 | 125 | 20 | 2016.09.14 | 235 | -2.18166156 | 2016.09.14.15. |
| ## | 4849 | 30  | 21 | 2016.09.14 | 330 | -0.52359878 | 2016.09.14.15. |
| ## | 4850 | 0   | 22 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.15. |
| ## | 4851 | 15  | 23 | 2016.09.14 | 345 | -0.26179939 | 2016.09.14.15. |
| ## | 4852 | 40  | 24 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.15. |
| ## | 4853 | 20  | 25 | 2016.09.14 | 340 | -0.34906585 | 2016.09.14.15. |
| ## | 4854 | 355 | 26 | 2016.09.14 | 5   | 0.08726646  | 2016.09.14.15. |
| ## | 4855 | 20  | 27 | 2016.09.14 | 340 | -0.34906585 | 2016.09.14.15. |
| ## | 4856 | 20  | 28 | 2016.09.14 | 340 | -0.34906585 | 2016.09.14.15. |
| ## | 4857 | 40  | 29 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.15. |
| ## | 4858 | 0   | 30 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.15. |
| ## | 4859 | 350 | 31 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.15. |
| ## | 4860 | 170 | 1  | 2016.09.14 | 190 | -2.96705973 | 2016.09.14.16. |
| ## | 4861 | 130 | 2  | 2016.09.14 | 230 | -2.26892803 | 2016.09.14.16. |
| ## | 4862 | 120 | 3  | 2016.09.14 | 240 | -2.09439510 | 2016.09.14.16. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4863 | 100 | 4  | 2016.09.14 | 260 | -1.74532925 | 2016.09.14.16. |
| ## | 4864 | 130 | 5  | 2016.09.14 | 230 | -2.26892803 | 2016.09.14.16. |
| ## | 4865 | 110 | 6  | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.16. |
| ## | 4866 | 140 | 7  | 2016.09.14 | 220 | -2.44346095 | 2016.09.14.16. |
| ## | 4867 | 110 | 8  | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.16. |
| ## | 4868 | 110 | 9  | 2016.09.14 | 250 | -1.91986218 | 2016.09.14.16. |
| ## | 4869 | 115 | 10 | 2016.09.14 | 245 | -2.00712864 | 2016.09.14.16. |
| ## | 4870 | 115 | 11 | 2016.09.14 | 245 | -2.00712864 | 2016.09.14.16. |
| ## | 4871 | 100 | 12 | 2016.09.14 | 260 | -1.74532925 | 2016.09.14.16. |
| ## | 4872 | 100 | 13 | 2016.09.14 | 260 | -1.74532925 | 2016.09.14.16. |
| ## | 4873 | 90  | 14 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.16. |
| ## | 4874 | 190 | 15 | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.16. |
| ## | 4875 | 90  | 16 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.16. |
| ## | 4876 | 130 | 17 | 2016.09.14 | 230 | -2.26892803 | 2016.09.14.16. |
| ## | 4877 | 85  | 18 | 2016.09.14 | 275 | -1.48352986 | 2016.09.14.16. |
| ## | 4878 | 150 | 19 | 2016.09.14 | 210 | -2.61799388 | 2016.09.14.16. |
| ## | 4879 | 60  | 20 | 2016.09.14 | 300 | -1.04719755 | 2016.09.14.16. |
| ## | 4880 | 80  | 21 | 2016.09.14 | 280 | -1.39626340 | 2016.09.14.16. |
| ## | 4881 | 10  | 22 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.16. |
| ## | 4882 | 65  | 23 | 2016.09.14 | 295 | -1.13446401 | 2016.09.14.16. |
| ## | 4883 | 40  | 24 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.16. |
| ## | 4884 | 290 | 1  | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.16. |
| ## | 4885 | 260 | 2  | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.16. |
| ## | 4886 | 300 | 3  | 2016.09.14 | 60  | 1.04719755  | 2016.09.14.16. |
| ## | 4887 | 275 | 4  | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.16. |
| ## | 4888 | 320 | 5  | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.16. |
| ## | 4889 | 310 | 6  | 2016.09.14 | 50  | 0.87266463  | 2016.09.14.16. |
| ## | 4890 | 310 | 7  | 2016.09.14 | 50  | 0.87266463  | 2016.09.14.16. |
| ## | 4891 | 260 | 8  | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.16. |
| ## | 4892 | 320 | 9  | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.16. |
| ## | 4893 | 90  | 10 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.16. |
| ## | 4894 | 60  | 11 | 2016.09.14 | 300 | -1.04719755 | 2016.09.14.16. |
| ## | 4895 | 40  | 12 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.16. |
| ## | 4896 | 85  | 13 | 2016.09.14 | 275 | -1.48352986 | 2016.09.14.16. |
| ## | 4897 | 35  | 14 | 2016.09.14 | 325 | -0.61086524 | 2016.09.14.16. |
| ## | 4898 | 75  | 15 | 2016.09.14 | 285 | -1.30899694 | 2016.09.14.16. |
| ## | 4899 | 70  | 16 | 2016.09.14 | 290 | -1.22173048 | 2016.09.14.16. |
| ## | 4900 | 135 | 17 | 2016.09.14 | 225 | -2.35619449 | 2016.09.14.16. |
| ## | 4901 | 25  | 18 | 2016.09.14 | 335 | -0.43633231 | 2016.09.14.16. |
| ## | 4902 | 90  | 19 | 2016.09.14 | 270 | -1.57079633 | 2016.09.14.16. |
| ## | 4903 | 40  | 20 | 2016.09.14 | 320 | -0.69813170 | 2016.09.14.16. |
| ## | 4904 | 70  | 21 | 2016.09.14 | 290 | -1.22173048 | 2016.09.14.16. |
| ## | 4905 | 35  | 22 | 2016.09.14 | 325 | -0.61086524 | 2016.09.14.16. |
| ## | 4906 | 0   | 23 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.16. |
| ## | 4907 | 0   | 24 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.16. |
| ## | 4908 | 0   | 25 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.16. |
| ## | 4909 | 5   | 26 | 2016.09.14 | 355 | -0.08726646 | 2016.09.14.16. |
| ## | 4910 | 10  | 27 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.16. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4911 | 50  | 28 | 2016.09.14 | 310 | -0.87266463 | 2016.09.14.16. |
| ## | 4912 | 10  | 29 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.16. |
| ## | 4913 | 10  | 30 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.16. |
| ## | 4914 | 10  | 31 | 2016.09.14 | 350 | -0.17453293 | 2016.09.14.16. |
| ## | 4915 | 20  | 32 | 2016.09.14 | 340 | -0.34906585 | 2016.09.14.16. |
| ## | 4916 | 0   | 33 | 2016.09.14 | 0   | 0.00000000  | 2016.09.14.16. |
| ## | 4917 | 290 | 1  | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.16. |
| ## | 4918 | 300 | 2  | 2016.09.14 | 60  | 1.04719755  | 2016.09.14.16. |
| ## | 4919 | 310 | 3  | 2016.09.14 | 50  | 0.87266463  | 2016.09.14.16. |
| ## | 4920 | 175 | 4  | 2016.09.14 | 185 | -3.05432619 | 2016.09.14.16. |
| ## | 4921 | 310 | 5  | 2016.09.14 | 50  | 0.87266463  | 2016.09.14.16. |
| ## | 4922 | 270 | 6  | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.16. |
| ## | 4923 | 330 | 7  | 2016.09.14 | 30  | 0.52359878  | 2016.09.14.16. |
| ## | 4924 | 260 | 8  | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.16. |
| ## | 4925 | 320 | 9  | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.16. |
| ## | 4926 | 275 | 10 | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.16. |
| ## | 4927 | 320 | 11 | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.16. |
| ## | 4928 | 220 | 12 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.16. |
| ## | 4929 | 325 | 13 | 2016.09.14 | 35  | 0.61086524  | 2016.09.14.16. |
| ## | 4930 | 250 | 14 | 2016.09.14 | 110 | 1.91986218  | 2016.09.14.16. |
| ## | 4931 | 310 | 15 | 2016.09.14 | 50  | 0.87266463  | 2016.09.14.16. |
| ## | 4932 | 280 | 16 | 2016.09.14 | 80  | 1.39626340  | 2016.09.14.16. |
| ## | 4933 | 320 | 17 | 2016.09.14 | 40  | 0.69813170  | 2016.09.14.16. |
| ## | 4934 | 290 | 18 | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.16. |
| ## | 4935 | 350 | 19 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.16. |
| ## | 4936 | 350 | 20 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.16. |
| ## | 4937 | 200 | 21 | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.16. |
| ## | 4938 | 240 | 22 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.16. |
| ## | 4939 | 200 | 23 | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.16. |
| ## | 4940 | 210 | 24 | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.16. |
| ## | 4941 | 220 | 25 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.16. |
| ## | 4942 | 240 | 26 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.16. |
| ## | 4943 | 230 | 27 | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.16. |
| ## | 4944 | 250 | 28 | 2016.09.14 | 110 | 1.91986218  | 2016.09.14.16. |
| ## | 4945 | 190 | 29 | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.16. |
| ## | 4946 | 260 | 30 | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.16. |
| ## | 4947 | 225 | 31 | 2016.09.14 | 135 | 2.35619449  | 2016.09.14.16. |
| ## | 4948 | 230 | 32 | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.16. |
| ## | 4949 | 215 | 33 | 2016.09.14 | 145 | 2.53072742  | 2016.09.14.16. |
| ## | 4950 | 230 | 34 | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.16. |
| ## | 4951 | 210 | 35 | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.16. |
| ## | 4952 | 275 | 36 | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.16. |
| ## | 4953 | 155 | 37 | 2016.09.14 | 205 | -2.70526034 | 2016.09.14.16. |
| ## | 4954 | 220 | 38 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.16. |
| ## | 4955 | 230 | 39 | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.16. |
| ## | 4956 | 220 | 40 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.16. |
| ## | 4957 | 260 | 41 | 2016.09.14 | 100 | 1.74532925  | 2016.09.14.16. |
| ## | 4958 | 240 | 42 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.16. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 4959 | 305 | 43 | 2016.09.14 | 55  | 0.95993109  | 2016.09.14.16. |
| ## | 4960 | 340 | 44 | 2016.09.14 | 20  | 0.34906585  | 2016.09.14.16. |
| ## | 4961 | 330 | 45 | 2016.09.14 | 30  | 0.52359878  | 2016.09.14.16. |
| ## | 4962 | 210 | 46 | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.16. |
| ## | 4963 | 350 | 47 | 2016.09.14 | 10  | 0.17453293  | 2016.09.14.16. |
| ## | 4964 | 280 | 48 | 2016.09.14 | 80  | 1.39626340  | 2016.09.14.16. |
| ## | 4965 | 250 | 49 | 2016.09.14 | 110 | 1.91986218  | 2016.09.14.16. |
| ## | 4966 | 200 | 50 | 2016.09.14 | 160 | 2.79252680  | 2016.09.14.16. |
| ## | 4967 | 290 | 51 | 2016.09.14 | 70  | 1.22173048  | 2016.09.14.16. |
| ## | 4968 | 205 | 52 | 2016.09.14 | 155 | 2.70526034  | 2016.09.14.16. |
| ## | 4969 | 165 | 53 | 2016.09.14 | 195 | -2.87979327 | 2016.09.14.16. |
| ## | 4970 | 185 | 54 | 2016.09.14 | 175 | 3.05432619  | 2016.09.14.16. |
| ## | 4971 | 165 | 1  | 2016.09.14 | 195 | -2.87979327 | 2016.09.14.17. |
| ## | 4972 | 275 | 2  | 2016.09.14 | 85  | 1.48352986  | 2016.09.14.17. |
| ## | 4973 | 255 | 3  | 2016.09.14 | 105 | 1.83259571  | 2016.09.14.17. |
| ## | 4974 | 230 | 4  | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.17. |
| ## | 4975 | 300 | 5  | 2016.09.14 | 60  | 1.04719755  | 2016.09.14.17. |
| ## | 4976 | 235 | 6  | 2016.09.14 | 125 | 2.18166156  | 2016.09.14.17. |
| ## | 4977 | 270 | 7  | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.17. |
| ## | 4978 | 230 | 8  | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.17. |
| ## | 4979 | 160 | 9  | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.17. |
| ## | 4980 | 145 | 10 | 2016.09.14 | 215 | -2.53072742 | 2016.09.14.17. |
| ## | 4981 | 190 | 11 | 2016.09.14 | 170 | 2.96705973  | 2016.09.14.17. |
| ## | 4982 | 100 | 12 | 2016.09.14 | 260 | -1.74532925 | 2016.09.14.17. |
| ## | 4983 | 280 | 13 | 2016.09.14 | 80  | 1.39626340  | 2016.09.14.17. |
| ## | 4984 | 160 | 14 | 2016.09.14 | 200 | -2.79252680 | 2016.09.14.17. |
| ## | 4985 | 210 | 15 | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.17. |
| ## | 4986 | 270 | 16 | 2016.09.14 | 90  | 1.57079633  | 2016.09.14.17. |
| ## | 4987 | 210 | 17 | 2016.09.14 | 150 | 2.61799388  | 2016.09.14.17. |
| ## | 4988 | 180 | 18 | 2016.09.14 | 180 | 3.14159265  | 2016.09.14.17. |
| ## | 4989 | 220 | 19 | 2016.09.14 | 140 | 2.44346095  | 2016.09.14.17. |
| ## | 4990 | 230 | 20 | 2016.09.14 | 130 | 2.26892803  | 2016.09.14.17. |
| ## | 4991 | 225 | 21 | 2016.09.14 | 135 | 2.35619449  | 2016.09.14.17. |
| ## | 4992 | 240 | 22 | 2016.09.14 | 120 | 2.09439510  | 2016.09.14.17. |
| ## | 4993 | 210 | 1  | 2016.09.27 | 150 | 2.61799388  | 2016.09.27.15. |
| ## | 4994 | 250 | 2  | 2016.09.27 | 110 | 1.91986218  | 2016.09.27.15. |
| ## | 4995 | 270 | 3  | 2016.09.27 | 90  | 1.57079633  | 2016.09.27.15. |
| ## | 4996 | 330 | 4  | 2016.09.27 | 30  | 0.52359878  | 2016.09.27.15. |
| ## | 4997 | 250 | 5  | 2016.09.27 | 110 | 1.91986218  | 2016.09.27.15. |
| ## | 4998 | 340 | 6  | 2016.09.27 | 20  | 0.34906585  | 2016.09.27.15. |
| ## | 4999 | 180 | 7  | 2016.09.27 | 180 | 3.14159265  | 2016.09.27.15. |
| ## | 5000 | 290 | 8  | 2016.09.27 | 70  | 1.22173048  | 2016.09.27.15. |
| ## | 5001 | 285 | 9  | 2016.09.27 | 75  | 1.30899694  | 2016.09.27.15. |
| ## | 5002 | 160 | 10 | 2016.09.27 | 200 | -2.79252680 | 2016.09.27.15. |
| ## | 5003 | 240 | 11 | 2016.09.27 | 120 | 2.09439510  | 2016.09.27.15. |
| ## | 5004 | 180 | 12 | 2016.09.27 | 180 | 3.14159265  | 2016.09.27.15. |
| ## | 5005 | 190 | 13 | 2016.09.27 | 170 | 2.96705973  | 2016.09.27.15. |
| ## | 5006 | 170 | 14 | 2016.09.27 | 190 | -2.96705973 | 2016.09.27.15. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5007 | 170 | 15 | 2016.09.27 | 190 | -2.96705973 | 2016.09.27.15. |
| ## | 5008 | 165 | 16 | 2016.09.27 | 195 | -2.87979327 | 2016.09.27.15. |
| ## | 5009 | 180 | 17 | 2016.09.27 | 180 | 3.14159265  | 2016.09.27.15. |
| ## | 5010 | 170 | 18 | 2016.09.27 | 190 | -2.96705973 | 2016.09.27.15. |
| ## | 5011 | 120 | 19 | 2016.09.27 | 240 | -2.09439510 | 2016.09.27.15. |
| ## | 5012 | 165 | 20 | 2016.09.27 | 195 | -2.87979327 | 2016.09.27.15. |
| ## | 5013 | 170 | 21 | 2016.09.27 | 190 | -2.96705973 | 2016.09.27.15. |
| ## | 5014 | 35  | 1  | 2016.09.29 | 325 | -0.61086524 | 2016.09.29.14. |
| ## | 5015 | 190 | 2  | 2016.09.29 | 170 | 2.96705973  | 2016.09.29.14. |
| ## | 5016 | 180 | 3  | 2016.09.29 | 180 | 3.14159265  | 2016.09.29.14. |
| ## | 5017 | 45  | 4  | 2016.09.29 | 315 | -0.78539816 | 2016.09.29.14. |
| ## | 5018 | 340 | 5  | 2016.09.29 | 20  | 0.34906585  | 2016.09.29.14. |
| ## | 5019 | 30  | 6  | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.14. |
| ## | 5020 | 35  | 7  | 2016.09.29 | 325 | -0.61086524 | 2016.09.29.14. |
| ## | 5021 | 70  | 8  | 2016.09.29 | 290 | -1.22173048 | 2016.09.29.14. |
| ## | 5022 | 325 | 9  | 2016.09.29 | 35  | 0.61086524  | 2016.09.29.14. |
| ## | 5023 | 110 | 10 | 2016.09.29 | 250 | -1.91986218 | 2016.09.29.14. |
| ## | 5024 | 120 | 11 | 2016.09.29 | 240 | -2.09439510 | 2016.09.29.14. |
| ## | 5025 | 140 | 12 | 2016.09.29 | 220 | -2.44346095 | 2016.09.29.14. |
| ## | 5026 | 40  | 13 | 2016.09.29 | 320 | -0.69813170 | 2016.09.29.14. |
| ## | 5027 | 20  | 14 | 2016.09.29 | 340 | -0.34906585 | 2016.09.29.14. |
| ## | 5028 | 355 | 15 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.14. |
| ## | 5029 | 40  | 16 | 2016.09.29 | 320 | -0.69813170 | 2016.09.29.14. |
| ## | 5030 | 350 | 17 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.14. |
| ## | 5031 | 340 | 18 | 2016.09.29 | 20  | 0.34906585  | 2016.09.29.14. |
| ## | 5032 | 0   | 19 | 2016.09.29 | 0   | 0.00000000  | 2016.09.29.14. |
| ## | 5033 | 100 | 1  | 2016.09.29 | 260 | -1.74532925 | 2016.09.29.14. |
| ## | 5034 | 190 | 2  | 2016.09.29 | 170 | 2.96705973  | 2016.09.29.14. |
| ## | 5035 | 190 | 3  | 2016.09.29 | 170 | 2.96705973  | 2016.09.29.14. |
| ## | 5036 | 115 | 4  | 2016.09.29 | 245 | -2.00712864 | 2016.09.29.14. |
| ## | 5037 | 70  | 5  | 2016.09.29 | 290 | -1.22173048 | 2016.09.29.14. |
| ## | 5038 | 230 | 6  | 2016.09.29 | 130 | 2.26892803  | 2016.09.29.14. |
| ## | 5039 | 130 | 7  | 2016.09.29 | 230 | -2.26892803 | 2016.09.29.14. |
| ## | 5040 | 150 | 8  | 2016.09.29 | 210 | -2.61799388 | 2016.09.29.14. |
| ## | 5041 | 90  | 9  | 2016.09.29 | 270 | -1.57079633 | 2016.09.29.14. |
| ## | 5042 | 160 | 10 | 2016.09.29 | 200 | -2.79252680 | 2016.09.29.14. |
| ## | 5043 | 210 | 11 | 2016.09.29 | 150 | 2.61799388  | 2016.09.29.14. |
| ## | 5044 | 100 | 12 | 2016.09.29 | 260 | -1.74532925 | 2016.09.29.14. |
| ## | 5045 | 70  | 13 | 2016.09.29 | 290 | -1.22173048 | 2016.09.29.14. |
| ## | 5046 | 70  | 14 | 2016.09.29 | 290 | -1.22173048 | 2016.09.29.14. |
| ## | 5047 | 110 | 15 | 2016.09.29 | 250 | -1.91986218 | 2016.09.29.14. |
| ## | 5048 | 120 | 16 | 2016.09.29 | 240 | -2.09439510 | 2016.09.29.14. |
| ## | 5049 | 105 | 17 | 2016.09.29 | 255 | -1.83259571 | 2016.09.29.14. |
| ## | 5050 | 60  | 18 | 2016.09.29 | 300 | -1.04719755 | 2016.09.29.14. |
| ## | 5051 | 30  | 19 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.14. |
| ## | 5052 | 90  | 20 | 2016.09.29 | 270 | -1.57079633 | 2016.09.29.14. |
| ## | 5053 | 350 | 21 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.14. |
| ## | 5054 | 10  | 22 | 2016.09.29 | 350 | -0.17453293 | 2016.09.29.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5055 | 345 | 23 | 2016.09.29 | 15  | 0.26179939  | 2016.09.29.14. |
| ## | 5056 | 30  | 24 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.14. |
| ## | 5057 | 20  | 25 | 2016.09.29 | 340 | -0.34906585 | 2016.09.29.14. |
| ## | 5058 | 355 | 26 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.14. |
| ## | 5059 | 5   | 27 | 2016.09.29 | 355 | -0.08726646 | 2016.09.29.14. |
| ## | 5060 | 15  | 28 | 2016.09.29 | 345 | -0.26179939 | 2016.09.29.14. |
| ## | 5061 | 5   | 29 | 2016.09.29 | 355 | -0.08726646 | 2016.09.29.14. |
| ## | 5062 | 350 | 30 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.14. |
| ## | 5063 | 350 | 31 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.14. |
| ## | 5064 | 330 | 32 | 2016.09.29 | 30  | 0.52359878  | 2016.09.29.14. |
| ## | 5065 | 60  | 33 | 2016.09.29 | 300 | -1.04719755 | 2016.09.29.14. |
| ## | 5066 | 90  | 1  | 2016.09.29 | 270 | -1.57079633 | 2016.09.29.15. |
| ## | 5067 | 70  | 2  | 2016.09.29 | 290 | -1.22173048 | 2016.09.29.15. |
| ## | 5068 | 115 | 3  | 2016.09.29 | 245 | -2.00712864 | 2016.09.29.15. |
| ## | 5069 | 65  | 4  | 2016.09.29 | 295 | -1.13446401 | 2016.09.29.15. |
| ## | 5070 | 0   | 5  | 2016.09.29 | 0   | 0.00000000  | 2016.09.29.15. |
| ## | 5071 | 100 | 6  | 2016.09.29 | 260 | -1.74532925 | 2016.09.29.15. |
| ## | 5072 | 5   | 7  | 2016.09.29 | 355 | -0.08726646 | 2016.09.29.15. |
| ## | 5073 | 105 | 8  | 2016.09.29 | 255 | -1.83259571 | 2016.09.29.15. |
| ## | 5074 | 75  | 9  | 2016.09.29 | 285 | -1.30899694 | 2016.09.29.15. |
| ## | 5075 | 60  | 10 | 2016.09.29 | 300 | -1.04719755 | 2016.09.29.15. |
| ## | 5076 | 115 | 11 | 2016.09.29 | 245 | -2.00712864 | 2016.09.29.15. |
| ## | 5077 | 0   | 12 | 2016.09.29 | 0   | 0.00000000  | 2016.09.29.15. |
| ## | 5078 | 100 | 13 | 2016.09.29 | 260 | -1.74532925 | 2016.09.29.15. |
| ## | 5079 | 10  | 14 | 2016.09.29 | 350 | -0.17453293 | 2016.09.29.15. |
| ## | 5080 | 350 | 15 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.15. |
| ## | 5081 | 355 | 16 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.15. |
| ## | 5082 | 0   | 17 | 2016.09.29 | 0   | 0.00000000  | 2016.09.29.15. |
| ## | 5083 | 0   | 18 | 2016.09.29 | 0   | 0.00000000  | 2016.09.29.15. |
| ## | 5084 | 40  | 19 | 2016.09.29 | 320 | -0.69813170 | 2016.09.29.15. |
| ## | 5085 | 0   | 20 | 2016.09.29 | 0   | 0.00000000  | 2016.09.29.15. |
| ## | 5086 | 30  | 21 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.15. |
| ## | 5087 | 20  | 22 | 2016.09.29 | 340 | -0.34906585 | 2016.09.29.15. |
| ## | 5088 | 355 | 23 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.15. |
| ## | 5089 | 30  | 24 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.15. |
| ## | 5090 | 335 | 25 | 2016.09.29 | 25  | 0.43633231  | 2016.09.29.15. |
| ## | 5091 | 20  | 26 | 2016.09.29 | 340 | -0.34906585 | 2016.09.29.15. |
| ## | 5092 | 330 | 27 | 2016.09.29 | 30  | 0.52359878  | 2016.09.29.15. |
| ## | 5093 | 355 | 28 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.15. |
| ## | 5094 | 35  | 29 | 2016.09.29 | 325 | -0.61086524 | 2016.09.29.15. |
| ## | 5095 | 70  | 1  | 2016.09.29 | 290 | -1.22173048 | 2016.09.29.15. |
| ## | 5096 | 90  | 2  | 2016.09.29 | 270 | -1.57079633 | 2016.09.29.15. |
| ## | 5097 | 90  | 3  | 2016.09.29 | 270 | -1.57079633 | 2016.09.29.15. |
| ## | 5098 | 110 | 4  | 2016.09.29 | 250 | -1.91986218 | 2016.09.29.15. |
| ## | 5099 | 60  | 5  | 2016.09.29 | 300 | -1.04719755 | 2016.09.29.15. |
| ## | 5100 | 135 | 6  | 2016.09.29 | 225 | -2.35619449 | 2016.09.29.15. |
| ## | 5101 | 160 | 7  | 2016.09.29 | 200 | -2.79252680 | 2016.09.29.15. |
| ## | 5102 | 135 | 8  | 2016.09.29 | 225 | -2.35619449 | 2016.09.29.15. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 5103 | 110 | 9  | 2016.09.29 | 250 | -1.91986218 | 2016.09.29.15. |
| ## 5104 | 75  | 10 | 2016.09.29 | 285 | -1.30899694 | 2016.09.29.15. |
| ## 5105 | 100 | 11 | 2016.09.29 | 260 | -1.74532925 | 2016.09.29.15. |
| ## 5106 | 105 | 12 | 2016.09.29 | 255 | -1.83259571 | 2016.09.29.15. |
| ## 5107 | 90  | 13 | 2016.09.29 | 270 | -1.57079633 | 2016.09.29.15. |
| ## 5108 | 85  | 14 | 2016.09.29 | 275 | -1.48352986 | 2016.09.29.15. |
| ## 5109 | 120 | 15 | 2016.09.29 | 240 | -2.09439510 | 2016.09.29.15. |
| ## 5110 | 45  | 16 | 2016.09.29 | 315 | -0.78539816 | 2016.09.29.15. |
| ## 5111 | 110 | 17 | 2016.09.29 | 250 | -1.91986218 | 2016.09.29.15. |
| ## 5112 | 320 | 18 | 2016.09.29 | 40  | 0.69813170  | 2016.09.29.15. |
| ## 5113 | 355 | 19 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.15. |
| ## 5114 | 30  | 20 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.15. |
| ## 5115 | 340 | 21 | 2016.09.29 | 20  | 0.34906585  | 2016.09.29.15. |
| ## 5116 | 20  | 22 | 2016.09.29 | 340 | -0.34906585 | 2016.09.29.15. |
| ## 5117 | 350 | 23 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.15. |
| ## 5118 | 30  | 24 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.15. |
| ## 5119 | 350 | 25 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.15. |
| ## 5120 | 5   | 26 | 2016.09.29 | 355 | -0.08726646 | 2016.09.29.15. |
| ## 5121 | 40  | 27 | 2016.09.29 | 320 | -0.69813170 | 2016.09.29.15. |
| ## 5122 | 350 | 28 | 2016.09.29 | 10  | 0.17453293  | 2016.09.29.15. |
| ## 5123 | 10  | 29 | 2016.09.29 | 350 | -0.17453293 | 2016.09.29.15. |
| ## 5124 | 355 | 30 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.15. |
| ## 5125 | 5   | 31 | 2016.09.29 | 355 | -0.08726646 | 2016.09.29.15. |
| ## 5126 | 15  | 32 | 2016.09.29 | 345 | -0.26179939 | 2016.09.29.15. |
| ## 5127 | 335 | 33 | 2016.09.29 | 25  | 0.43633231  | 2016.09.29.15. |
| ## 5128 | 30  | 34 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.15. |
| ## 5129 | 25  | 35 | 2016.09.29 | 335 | -0.43633231 | 2016.09.29.15. |
| ## 5130 | 30  | 36 | 2016.09.29 | 330 | -0.52359878 | 2016.09.29.15. |
| ## 5131 | 50  | 37 | 2016.09.29 | 310 | -0.87266463 | 2016.09.29.15. |
| ## 5132 | 355 | 38 | 2016.09.29 | 5   | 0.08726646  | 2016.09.29.15. |
| ## 5133 | 50  | 39 | 2016.09.29 | 310 | -0.87266463 | 2016.09.29.15. |
| ## 5134 | 110 | 1  | 2016.06.02 | 250 | -1.91986218 | 2016.06.02.12. |
| ## 5135 | 90  | 2  | 2016.06.02 | 270 | -1.57079633 | 2016.06.02.12. |
| ## 5136 | 90  | 3  | 2016.06.02 | 270 | -1.57079633 | 2016.06.02.12. |
| ## 5137 | 85  | 4  | 2016.06.02 | 275 | -1.48352986 | 2016.06.02.12. |
| ## 5138 | 40  | 5  | 2016.06.02 | 320 | -0.69813170 | 2016.06.02.12. |
| ## 5139 | 70  | 6  | 2016.06.02 | 290 | -1.22173048 | 2016.06.02.12. |
| ## 5140 | 275 | 7  | 2016.06.02 | 85  | 1.48352986  | 2016.06.02.12. |
| ## 5141 | 275 | 8  | 2016.06.02 | 85  | 1.48352986  | 2016.06.02.12. |
| ## 5142 | 60  | 9  | 2016.06.02 | 300 | -1.04719755 | 2016.06.02.12. |
| ## 5143 | 295 | 10 | 2016.06.02 | 65  | 1.13446401  | 2016.06.02.12. |
| ## 5144 | 320 | 11 | 2016.06.02 | 40  | 0.69813170  | 2016.06.02.12. |
| ## 5145 | 345 | 12 | 2016.06.02 | 15  | 0.26179939  | 2016.06.02.12. |
| ## 5146 | 40  | 13 | 2016.06.02 | 320 | -0.69813170 | 2016.06.02.12. |
| ## 5147 | 0   | 14 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.12. |
| ## 5148 | 10  | 15 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.12. |
| ## 5149 | 5   | 16 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.12. |
| ## 5150 | 0   | 17 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.12. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5151 | 10  | 18 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.12. |
| ## | 5152 | 0   | 19 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.12. |
| ## | 5153 | 0   | 20 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.12. |
| ## | 5154 | 10  | 21 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.12. |
| ## | 5155 | 5   | 22 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.12. |
| ## | 5156 | 0   | 23 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.12. |
| ## | 5157 | 100 | 24 | 2016.06.02 | 260 | -1.74532925 | 2016.06.02.12. |
| ## | 5158 | 90  | 25 | 2016.06.02 | 270 | -1.57079633 | 2016.06.02.12. |
| ## | 5159 | 15  | 26 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.12. |
| ## | 5160 | 65  | 1  | 2016.06.02 | 295 | -1.13446401 | 2016.06.02.13. |
| ## | 5161 | 80  | 2  | 2016.06.02 | 280 | -1.39626340 | 2016.06.02.13. |
| ## | 5162 | 205 | 3  | 2016.06.02 | 155 | 2.70526034  | 2016.06.02.13. |
| ## | 5163 | 20  | 4  | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## | 5164 | 70  | 5  | 2016.06.02 | 290 | -1.22173048 | 2016.06.02.13. |
| ## | 5165 | 300 | 6  | 2016.06.02 | 60  | 1.04719755  | 2016.06.02.13. |
| ## | 5166 | 95  | 7  | 2016.06.02 | 265 | -1.65806279 | 2016.06.02.13. |
| ## | 5167 | 115 | 8  | 2016.06.02 | 245 | -2.00712864 | 2016.06.02.13. |
| ## | 5168 | 320 | 9  | 2016.06.02 | 40  | 0.69813170  | 2016.06.02.13. |
| ## | 5169 | 100 | 10 | 2016.06.02 | 260 | -1.74532925 | 2016.06.02.13. |
| ## | 5170 | 95  | 11 | 2016.06.02 | 265 | -1.65806279 | 2016.06.02.13. |
| ## | 5171 | 355 | 12 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5172 | 5   | 13 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5173 | 0   | 14 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5174 | 0   | 15 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5175 | 5   | 16 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5176 | 20  | 17 | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## | 5177 | 350 | 18 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5178 | 5   | 19 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5179 | 0   | 20 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5180 | 20  | 21 | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## | 5181 | 15  | 22 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5182 | 5   | 23 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5183 | 15  | 24 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5184 | 0   | 25 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5185 | 355 | 26 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5186 | 5   | 27 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5187 | 355 | 28 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5188 | 0   | 29 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5189 | 350 | 30 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5190 | 350 | 31 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5191 | 10  | 32 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.13. |
| ## | 5192 | 355 | 33 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5193 | 350 | 34 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5194 | 25  | 35 | 2016.06.02 | 335 | -0.43633231 | 2016.06.02.13. |
| ## | 5195 | 30  | 36 | 2016.06.02 | 330 | -0.52359878 | 2016.06.02.13. |
| ## | 5196 | 20  | 37 | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## | 5197 | 0   | 38 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5198 | 345 | 39 | 2016.06.02 | 15  | 0.26179939  | 2016.06.02.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5199 | 10  | 40 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.13. |
| ## | 5200 | 0   | 41 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5201 | 20  | 42 | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## | 5202 | 0   | 1  | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5203 | 355 | 2  | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5204 | 0   | 3  | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5205 | 5   | 4  | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5206 | 60  | 5  | 2016.06.02 | 300 | -1.04719755 | 2016.06.02.13. |
| ## | 5207 | 5   | 6  | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5208 | 25  | 7  | 2016.06.02 | 335 | -0.43633231 | 2016.06.02.13. |
| ## | 5209 | 350 | 8  | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5210 | 15  | 9  | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5211 | 15  | 10 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5212 | 320 | 11 | 2016.06.02 | 40  | 0.69813170  | 2016.06.02.13. |
| ## | 5213 | 50  | 12 | 2016.06.02 | 310 | -0.87266463 | 2016.06.02.13. |
| ## | 5214 | 65  | 13 | 2016.06.02 | 295 | -1.13446401 | 2016.06.02.13. |
| ## | 5215 | 340 | 14 | 2016.06.02 | 20  | 0.34906585  | 2016.06.02.13. |
| ## | 5216 | 10  | 15 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.13. |
| ## | 5217 | 355 | 16 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5218 | 25  | 17 | 2016.06.02 | 335 | -0.43633231 | 2016.06.02.13. |
| ## | 5219 | 15  | 18 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5220 | 15  | 19 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5221 | 5   | 20 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5222 | 350 | 21 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5223 | 30  | 22 | 2016.06.02 | 330 | -0.52359878 | 2016.06.02.13. |
| ## | 5224 | 350 | 23 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5225 | 0   | 24 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5226 | 10  | 25 | 2016.06.02 | 350 | -0.17453293 | 2016.06.02.13. |
| ## | 5227 | 0   | 1  | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5228 | 0   | 2  | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |
| ## | 5229 | 355 | 3  | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5230 | 60  | 4  | 2016.06.02 | 300 | -1.04719755 | 2016.06.02.13. |
| ## | 5231 | 5   | 5  | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5232 | 65  | 6  | 2016.06.02 | 295 | -1.13446401 | 2016.06.02.13. |
| ## | 5233 | 345 | 7  | 2016.06.02 | 15  | 0.26179939  | 2016.06.02.13. |
| ## | 5234 | 70  | 8  | 2016.06.02 | 290 | -1.22173048 | 2016.06.02.13. |
| ## | 5235 | 20  | 9  | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## | 5236 | 30  | 10 | 2016.06.02 | 330 | -0.52359878 | 2016.06.02.13. |
| ## | 5237 | 320 | 11 | 2016.06.02 | 40  | 0.69813170  | 2016.06.02.13. |
| ## | 5238 | 50  | 12 | 2016.06.02 | 310 | -0.87266463 | 2016.06.02.13. |
| ## | 5239 | 60  | 13 | 2016.06.02 | 300 | -1.04719755 | 2016.06.02.13. |
| ## | 5240 | 335 | 14 | 2016.06.02 | 25  | 0.43633231  | 2016.06.02.13. |
| ## | 5241 | 355 | 15 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5242 | 350 | 16 | 2016.06.02 | 10  | 0.17453293  | 2016.06.02.13. |
| ## | 5243 | 15  | 17 | 2016.06.02 | 345 | -0.26179939 | 2016.06.02.13. |
| ## | 5244 | 355 | 18 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## | 5245 | 5   | 19 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## | 5246 | 0   | 20 | 2016.06.02 | 0   | 0.00000000  | 2016.06.02.13. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 5247 | 355 | 21 | 2016.06.02 | 5   | 0.08726646  | 2016.06.02.13. |
| ## 5248 | 25  | 22 | 2016.06.02 | 335 | -0.43633231 | 2016.06.02.13. |
| ## 5249 | 345 | 23 | 2016.06.02 | 15  | 0.26179939  | 2016.06.02.13. |
| ## 5250 | 5   | 24 | 2016.06.02 | 355 | -0.08726646 | 2016.06.02.13. |
| ## 5251 | 20  | 25 | 2016.06.02 | 340 | -0.34906585 | 2016.06.02.13. |
| ## 5252 | 345 | 26 | 2016.06.02 | 15  | 0.26179939  | 2016.06.02.13. |
| ## 5253 | 295 | 1  | 2016.06.14 | 65  | 1.13446401  | 2016.06.14.15. |
| ## 5254 | 350 | 2  | 2016.06.14 | 10  | 0.17453293  | 2016.06.14.15. |
| ## 5255 | 350 | 3  | 2016.06.14 | 10  | 0.17453293  | 2016.06.14.15. |
| ## 5256 | 5   | 4  | 2016.06.14 | 355 | -0.08726646 | 2016.06.14.15. |
| ## 5257 | 180 | 5  | 2016.06.14 | 180 | 3.14159265  | 2016.06.14.15. |
| ## 5258 | 30  | 6  | 2016.06.14 | 330 | -0.52359878 | 2016.06.14.15. |
| ## 5259 | 170 | 7  | 2016.06.14 | 190 | -2.96705973 | 2016.06.14.15. |
| ## 5260 | 60  | 8  | 2016.06.14 | 300 | -1.04719755 | 2016.06.14.15. |
| ## 5261 | 45  | 9  | 2016.06.14 | 315 | -0.78539816 | 2016.06.14.15. |
| ## 5262 | 30  | 10 | 2016.06.14 | 330 | -0.52359878 | 2016.06.14.15. |
| ## 5263 | 85  | 11 | 2016.06.14 | 275 | -1.48352986 | 2016.06.14.15. |
| ## 5264 | 45  | 12 | 2016.06.14 | 315 | -0.78539816 | 2016.06.14.15. |
| ## 5265 | 300 | 13 | 2016.06.14 | 60  | 1.04719755  | 2016.06.14.15. |
| ## 5266 | 300 | 14 | 2016.06.14 | 60  | 1.04719755  | 2016.06.14.15. |
| ## 5267 | 310 | 15 | 2016.06.14 | 50  | 0.87266463  | 2016.06.14.15. |
| ## 5268 | 290 | 16 | 2016.06.14 | 70  | 1.22173048  | 2016.06.14.15. |
| ## 5269 | 280 | 17 | 2016.06.14 | 80  | 1.39626340  | 2016.06.14.15. |
| ## 5270 | 275 | 18 | 2016.06.14 | 85  | 1.48352986  | 2016.06.14.15. |
| ## 5271 | 320 | 19 | 2016.06.14 | 40  | 0.69813170  | 2016.06.14.15. |
| ## 5272 | 325 | 20 | 2016.06.14 | 35  | 0.61086524  | 2016.06.14.15. |
| ## 5273 | 300 | 21 | 2016.06.14 | 60  | 1.04719755  | 2016.06.14.15. |
| ## 5274 | 325 | 22 | 2016.06.14 | 35  | 0.61086524  | 2016.06.14.15. |
| ## 5275 | 295 | 23 | 2016.06.14 | 65  | 1.13446401  | 2016.06.14.15. |
| ## 5276 | 290 | 24 | 2016.06.14 | 70  | 1.22173048  | 2016.06.14.15. |
| ## 5277 | 295 | 25 | 2016.06.14 | 65  | 1.13446401  | 2016.06.14.15. |
| ## 5278 | 280 | 26 | 2016.06.14 | 80  | 1.39626340  | 2016.06.14.15. |
| ## 5279 | 355 | 1  | 2016.06.20 | 5   | 0.08726646  | 2016.06.20.13. |
| ## 5280 | 140 | 2  | 2016.06.20 | 220 | -2.44346095 | 2016.06.20.13. |
| ## 5281 | 345 | 3  | 2016.06.20 | 15  | 0.26179939  | 2016.06.20.13. |
| ## 5282 | 340 | 4  | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.13. |
| ## 5283 | 0   | 5  | 2016.06.20 | 0   | 0.00000000  | 2016.06.20.13. |
| ## 5284 | 335 | 6  | 2016.06.20 | 25  | 0.43633231  | 2016.06.20.13. |
| ## 5285 | 340 | 7  | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.13. |
| ## 5286 | 280 | 8  | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5287 | 330 | 9  | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.13. |
| ## 5288 | 190 | 10 | 2016.06.20 | 170 | 2.96705973  | 2016.06.20.13. |
| ## 5289 | 340 | 11 | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.13. |
| ## 5290 | 320 | 12 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.13. |
| ## 5291 | 330 | 13 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.13. |
| ## 5292 | 280 | 14 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5293 | 225 | 15 | 2016.06.20 | 135 | 2.35619449  | 2016.06.20.13. |
| ## 5294 | 280 | 16 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5295 | 270 | 17 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5296 | 285 | 18 | 2016.06.20 | 75  | 1.30899694  | 2016.06.20.13. |
| ## | 5297 | 280 | 19 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## | 5298 | 290 | 20 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## | 5299 | 280 | 21 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## | 5300 | 280 | 22 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## | 5301 | 260 | 23 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## | 5302 | 270 | 24 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5303 | 250 | 25 | 2016.06.20 | 110 | 1.91986218  | 2016.06.20.13. |
| ## | 5304 | 270 | 26 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5305 | 275 | 27 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5306 | 275 | 28 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5307 | 270 | 29 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5308 | 270 | 30 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5309 | 270 | 31 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5310 | 275 | 32 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5311 | 270 | 33 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5312 | 260 | 34 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## | 5313 | 280 | 35 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## | 5314 | 285 | 36 | 2016.06.20 | 75  | 1.30899694  | 2016.06.20.13. |
| ## | 5315 | 275 | 37 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5316 | 300 | 38 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## | 5317 | 345 | 39 | 2016.06.20 | 15  | 0.26179939  | 2016.06.20.13. |
| ## | 5318 | 290 | 40 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## | 5319 | 325 | 41 | 2016.06.20 | 35  | 0.61086524  | 2016.06.20.13. |
| ## | 5320 | 270 | 42 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5321 | 275 | 43 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5322 | 225 | 44 | 2016.06.20 | 135 | 2.35619449  | 2016.06.20.13. |
| ## | 5323 | 240 | 45 | 2016.06.20 | 120 | 2.09439510  | 2016.06.20.13. |
| ## | 5324 | 260 | 46 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## | 5325 | 290 | 47 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## | 5326 | 280 | 48 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## | 5327 | 270 | 49 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5328 | 300 | 50 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## | 5329 | 250 | 51 | 2016.06.20 | 110 | 1.91986218  | 2016.06.20.13. |
| ## | 5330 | 270 | 52 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5331 | 290 | 53 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## | 5332 | 0   | 1  | 2016.06.20 | 0   | 0.00000000  | 2016.06.20.13. |
| ## | 5333 | 320 | 2  | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.13. |
| ## | 5334 | 15  | 3  | 2016.06.20 | 345 | -0.26179939 | 2016.06.20.13. |
| ## | 5335 | 315 | 4  | 2016.06.20 | 45  | 0.78539816  | 2016.06.20.13. |
| ## | 5336 | 15  | 5  | 2016.06.20 | 345 | -0.26179939 | 2016.06.20.13. |
| ## | 5337 | 345 | 6  | 2016.06.20 | 15  | 0.26179939  | 2016.06.20.13. |
| ## | 5338 | 355 | 7  | 2016.06.20 | 5   | 0.08726646  | 2016.06.20.13. |
| ## | 5339 | 200 | 8  | 2016.06.20 | 160 | 2.79252680  | 2016.06.20.13. |
| ## | 5340 | 350 | 9  | 2016.06.20 | 10  | 0.17453293  | 2016.06.20.13. |
| ## | 5341 | 0   | 10 | 2016.06.20 | 0   | 0.00000000  | 2016.06.20.13. |
| ## | 5342 | 350 | 11 | 2016.06.20 | 10  | 0.17453293  | 2016.06.20.13. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 5343 | 0   | 12 | 2016.06.20 | 0   | 0.00000000  | 2016.06.20.13. |
| ## 5344 | 15  | 13 | 2016.06.20 | 345 | -0.26179939 | 2016.06.20.13. |
| ## 5345 | 10  | 14 | 2016.06.20 | 350 | -0.17453293 | 2016.06.20.13. |
| ## 5346 | 270 | 15 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5347 | 275 | 16 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## 5348 | 300 | 17 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## 5349 | 280 | 18 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5350 | 270 | 19 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5351 | 290 | 20 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## 5352 | 300 | 21 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## 5353 | 300 | 22 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## 5354 | 275 | 23 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## 5355 | 270 | 24 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5356 | 270 | 25 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5357 | 270 | 26 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5358 | 270 | 27 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5359 | 265 | 28 | 2016.06.20 | 95  | 1.65806279  | 2016.06.20.13. |
| ## 5360 | 300 | 1  | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## 5361 | 330 | 2  | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.13. |
| ## 5362 | 340 | 3  | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.13. |
| ## 5363 | 25  | 4  | 2016.06.20 | 335 | -0.43633231 | 2016.06.20.13. |
| ## 5364 | 20  | 5  | 2016.06.20 | 340 | -0.34906585 | 2016.06.20.13. |
| ## 5365 | 70  | 6  | 2016.06.20 | 290 | -1.22173048 | 2016.06.20.13. |
| ## 5366 | 10  | 7  | 2016.06.20 | 350 | -0.17453293 | 2016.06.20.13. |
| ## 5367 | 330 | 8  | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.13. |
| ## 5368 | 10  | 9  | 2016.06.20 | 350 | -0.17453293 | 2016.06.20.13. |
| ## 5369 | 260 | 10 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## 5370 | 280 | 11 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5371 | 300 | 12 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## 5372 | 290 | 13 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## 5373 | 290 | 14 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## 5374 | 290 | 15 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## 5375 | 295 | 16 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.13. |
| ## 5376 | 270 | 17 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5377 | 280 | 18 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5378 | 270 | 19 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5379 | 285 | 20 | 2016.06.20 | 75  | 1.30899694  | 2016.06.20.13. |
| ## 5380 | 260 | 21 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## 5381 | 250 | 22 | 2016.06.20 | 110 | 1.91986218  | 2016.06.20.13. |
| ## 5382 | 320 | 23 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.13. |
| ## 5383 | 280 | 24 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5384 | 270 | 25 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## 5385 | 275 | 26 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## 5386 | 295 | 27 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.13. |
| ## 5387 | 265 | 28 | 2016.06.20 | 95  | 1.65806279  | 2016.06.20.13. |
| ## 5388 | 285 | 29 | 2016.06.20 | 75  | 1.30899694  | 2016.06.20.13. |
| ## 5389 | 280 | 30 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## 5390 | 295 | 31 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.13. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5391 | 260 | 32 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## | 5392 | 370 | 33 | 2016.06.20 | 350 | -0.17453293 | 2016.06.20.13. |
| ## | 5393 | 330 | 34 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.13. |
| ## | 5394 | 265 | 35 | 2016.06.20 | 95  | 1.65806279  | 2016.06.20.13. |
| ## | 5395 | 50  | 1  | 2016.06.20 | 310 | -0.87266463 | 2016.06.20.13. |
| ## | 5396 | 20  | 2  | 2016.06.20 | 340 | -0.34906585 | 2016.06.20.13. |
| ## | 5397 | 10  | 3  | 2016.06.20 | 350 | -0.17453293 | 2016.06.20.13. |
| ## | 5398 | 240 | 4  | 2016.06.20 | 120 | 2.09439510  | 2016.06.20.13. |
| ## | 5399 | 110 | 5  | 2016.06.20 | 250 | -1.91986218 | 2016.06.20.13. |
| ## | 5400 | 160 | 6  | 2016.06.20 | 200 | -2.79252680 | 2016.06.20.13. |
| ## | 5401 | 35  | 7  | 2016.06.20 | 325 | -0.61086524 | 2016.06.20.13. |
| ## | 5402 | 40  | 8  | 2016.06.20 | 320 | -0.69813170 | 2016.06.20.13. |
| ## | 5403 | 50  | 9  | 2016.06.20 | 310 | -0.87266463 | 2016.06.20.13. |
| ## | 5404 | 35  | 10 | 2016.06.20 | 325 | -0.61086524 | 2016.06.20.13. |
| ## | 5405 | 40  | 11 | 2016.06.20 | 320 | -0.69813170 | 2016.06.20.13. |
| ## | 5406 | 330 | 12 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.13. |
| ## | 5407 | 270 | 13 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5408 | 310 | 14 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.13. |
| ## | 5409 | 320 | 15 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.13. |
| ## | 5410 | 295 | 16 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.13. |
| ## | 5411 | 290 | 17 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## | 5412 | 275 | 18 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5413 | 310 | 19 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.13. |
| ## | 5414 | 310 | 20 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.13. |
| ## | 5415 | 245 | 21 | 2016.06.20 | 115 | 2.00712864  | 2016.06.20.13. |
| ## | 5416 | 270 | 22 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5417 | 250 | 23 | 2016.06.20 | 110 | 1.91986218  | 2016.06.20.13. |
| ## | 5418 | 265 | 24 | 2016.06.20 | 95  | 1.65806279  | 2016.06.20.13. |
| ## | 5419 | 305 | 25 | 2016.06.20 | 55  | 0.95993109  | 2016.06.20.13. |
| ## | 5420 | 290 | 26 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.13. |
| ## | 5421 | 275 | 27 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.13. |
| ## | 5422 | 300 | 28 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.13. |
| ## | 5423 | 5   | 1  | 2016.06.20 | 355 | -0.08726646 | 2016.06.20.13. |
| ## | 5424 | 340 | 2  | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.13. |
| ## | 5425 | 175 | 3  | 2016.06.20 | 185 | -3.05432619 | 2016.06.20.13. |
| ## | 5426 | 255 | 4  | 2016.06.20 | 105 | 1.83259571  | 2016.06.20.13. |
| ## | 5427 | 270 | 5  | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5428 | 150 | 6  | 2016.06.20 | 210 | -2.61799388 | 2016.06.20.13. |
| ## | 5429 | 230 | 7  | 2016.06.20 | 130 | 2.26892803  | 2016.06.20.13. |
| ## | 5430 | 90  | 8  | 2016.06.20 | 270 | -1.57079633 | 2016.06.20.13. |
| ## | 5431 | 350 | 9  | 2016.06.20 | 10  | 0.17453293  | 2016.06.20.13. |
| ## | 5432 | 110 | 10 | 2016.06.20 | 250 | -1.91986218 | 2016.06.20.13. |
| ## | 5433 | 260 | 11 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.13. |
| ## | 5434 | 280 | 12 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.13. |
| ## | 5435 | 240 | 13 | 2016.06.20 | 120 | 2.09439510  | 2016.06.20.13. |
| ## | 5436 | 270 | 14 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5437 | 245 | 15 | 2016.06.20 | 115 | 2.00712864  | 2016.06.20.13. |
| ## | 5438 | 270 | 16 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5439 | 265 | 17 | 2016.06.20 | 95  | 1.65806279  | 2016.06.20.13. |
| ## | 5440 | 270 | 18 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5441 | 270 | 19 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.13. |
| ## | 5442 | 20  | 1  | 2016.06.20 | 340 | -0.34906585 | 2016.06.20.14. |
| ## | 5443 | 355 | 2  | 2016.06.20 | 5   | 0.08726646  | 2016.06.20.14. |
| ## | 5444 | 15  | 3  | 2016.06.20 | 345 | -0.26179939 | 2016.06.20.14. |
| ## | 5445 | 50  | 4  | 2016.06.20 | 310 | -0.87266463 | 2016.06.20.14. |
| ## | 5446 | 55  | 5  | 2016.06.20 | 305 | -0.95993109 | 2016.06.20.14. |
| ## | 5447 | 120 | 6  | 2016.06.20 | 240 | -2.09439510 | 2016.06.20.14. |
| ## | 5448 | 20  | 7  | 2016.06.20 | 340 | -0.34906585 | 2016.06.20.14. |
| ## | 5449 | 180 | 8  | 2016.06.20 | 180 | 3.14159265  | 2016.06.20.14. |
| ## | 5450 | 40  | 9  | 2016.06.20 | 320 | -0.69813170 | 2016.06.20.14. |
| ## | 5451 | 60  | 10 | 2016.06.20 | 300 | -1.04719755 | 2016.06.20.14. |
| ## | 5452 | 220 | 11 | 2016.06.20 | 140 | 2.44346095  | 2016.06.20.14. |
| ## | 5453 | 40  | 12 | 2016.06.20 | 320 | -0.69813170 | 2016.06.20.14. |
| ## | 5454 | 280 | 13 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5455 | 90  | 14 | 2016.06.20 | 270 | -1.57079633 | 2016.06.20.14. |
| ## | 5456 | 280 | 15 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5457 | 310 | 16 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5458 | 290 | 17 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5459 | 330 | 18 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5460 | 290 | 19 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5461 | 270 | 20 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.14. |
| ## | 5462 | 310 | 21 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5463 | 290 | 22 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5464 | 310 | 23 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5465 | 325 | 24 | 2016.06.20 | 35  | 0.61086524  | 2016.06.20.14. |
| ## | 5466 | 300 | 25 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5467 | 320 | 26 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.14. |
| ## | 5468 | 280 | 27 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5469 | 275 | 28 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.14. |
| ## | 5470 | 310 | 29 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5471 | 290 | 30 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5472 | 330 | 31 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5473 | 320 | 32 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.14. |
| ## | 5474 | 310 | 33 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5475 | 330 | 34 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5476 | 295 | 35 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.14. |
| ## | 5477 | 295 | 36 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.14. |
| ## | 5478 | 300 | 37 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5479 | 220 | 1  | 2016.06.20 | 140 | 2.44346095  | 2016.06.20.14. |
| ## | 5480 | 125 | 2  | 2016.06.20 | 235 | -2.18166156 | 2016.06.20.14. |
| ## | 5481 | 30  | 3  | 2016.06.20 | 330 | -0.52359878 | 2016.06.20.14. |
| ## | 5482 | 5   | 4  | 2016.06.20 | 355 | -0.08726646 | 2016.06.20.14. |
| ## | 5483 | 330 | 5  | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5484 | 335 | 6  | 2016.06.20 | 25  | 0.43633231  | 2016.06.20.14. |
| ## | 5485 | 340 | 7  | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.14. |
| ## | 5486 | 45  | 8  | 2016.06.20 | 315 | -0.78539816 | 2016.06.20.14. |



|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5487 | 50  | 9  | 2016.06.20 | 310 | -0.87266463 | 2016.06.20.14. |
| ## | 5488 | 40  | 10 | 2016.06.20 | 320 | -0.69813170 | 2016.06.20.14. |
| ## | 5489 | 230 | 11 | 2016.06.20 | 130 | 2.26892803  | 2016.06.20.14. |
| ## | 5490 | 10  | 12 | 2016.06.20 | 350 | -0.17453293 | 2016.06.20.14. |
| ## | 5491 | 210 | 13 | 2016.06.20 | 150 | 2.61799388  | 2016.06.20.14. |
| ## | 5492 | 30  | 14 | 2016.06.20 | 330 | -0.52359878 | 2016.06.20.14. |
| ## | 5493 | 340 | 15 | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.14. |
| ## | 5494 | 40  | 16 | 2016.06.20 | 320 | -0.69813170 | 2016.06.20.14. |
| ## | 5495 | 350 | 17 | 2016.06.20 | 10  | 0.17453293  | 2016.06.20.14. |
| ## | 5496 | 250 | 18 | 2016.06.20 | 110 | 1.91986218  | 2016.06.20.14. |
| ## | 5497 | 290 | 19 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5498 | 300 | 20 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5499 | 300 | 21 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5500 | 295 | 22 | 2016.06.20 | 65  | 1.13446401  | 2016.06.20.14. |
| ## | 5501 | 285 | 23 | 2016.06.20 | 75  | 1.30899694  | 2016.06.20.14. |
| ## | 5502 | 270 | 24 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.14. |
| ## | 5503 | 280 | 25 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5504 | 320 | 26 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.14. |
| ## | 5505 | 280 | 27 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5506 | 270 | 28 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.14. |
| ## | 5507 | 300 | 29 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5508 | 290 | 30 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5509 | 305 | 31 | 2016.06.20 | 55  | 0.95993109  | 2016.06.20.14. |
| ## | 5510 | 310 | 32 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5511 | 280 | 33 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5512 | 305 | 34 | 2016.06.20 | 55  | 0.95993109  | 2016.06.20.14. |
| ## | 5513 | 300 | 35 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5514 | 290 | 36 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5515 | 300 | 37 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5516 | 290 | 38 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5517 | 280 | 39 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5518 | 280 | 40 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5519 | 290 | 41 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5520 | 300 | 42 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5521 | 290 | 43 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5522 | 310 | 44 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5523 | 270 | 45 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.14. |
| ## | 5524 | 300 | 46 | 2016.06.20 | 60  | 1.04719755  | 2016.06.20.14. |
| ## | 5525 | 315 | 47 | 2016.06.20 | 45  | 0.78539816  | 2016.06.20.14. |
| ## | 5526 | 15  | 48 | 2016.06.20 | 345 | -0.26179939 | 2016.06.20.14. |
| ## | 5527 | 340 | 49 | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.14. |
| ## | 5528 | 195 | 1  | 2016.06.20 | 165 | 2.87979327  | 2016.06.20.14. |
| ## | 5529 | 220 | 2  | 2016.06.20 | 140 | 2.44346095  | 2016.06.20.14. |
| ## | 5530 | 345 | 3  | 2016.06.20 | 15  | 0.26179939  | 2016.06.20.14. |
| ## | 5531 | 220 | 4  | 2016.06.20 | 140 | 2.44346095  | 2016.06.20.14. |
| ## | 5532 | 210 | 5  | 2016.06.20 | 150 | 2.61799388  | 2016.06.20.14. |
| ## | 5533 | 5   | 6  | 2016.06.20 | 355 | -0.08726646 | 2016.06.20.14. |
| ## | 5534 | 160 | 7  | 2016.06.20 | 200 | -2.79252680 | 2016.06.20.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5535 | 220 | 8  | 2016.06.20 | 140 | 2.44346095  | 2016.06.20.14. |
| ## | 5536 | 170 | 9  | 2016.06.20 | 190 | -2.96705973 | 2016.06.20.14. |
| ## | 5537 | 120 | 10 | 2016.06.20 | 240 | -2.09439510 | 2016.06.20.14. |
| ## | 5538 | 190 | 11 | 2016.06.20 | 170 | 2.96705973  | 2016.06.20.14. |
| ## | 5539 | 330 | 12 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5540 | 230 | 13 | 2016.06.20 | 130 | 2.26892803  | 2016.06.20.14. |
| ## | 5541 | 290 | 14 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5542 | 330 | 15 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5543 | 200 | 16 | 2016.06.20 | 160 | 2.79252680  | 2016.06.20.14. |
| ## | 5544 | 195 | 17 | 2016.06.20 | 165 | 2.87979327  | 2016.06.20.14. |
| ## | 5545 | 215 | 18 | 2016.06.20 | 145 | 2.53072742  | 2016.06.20.14. |
| ## | 5546 | 260 | 19 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.14. |
| ## | 5547 | 340 | 20 | 2016.06.20 | 20  | 0.34906585  | 2016.06.20.14. |
| ## | 5548 | 265 | 21 | 2016.06.20 | 95  | 1.65806279  | 2016.06.20.14. |
| ## | 5549 | 330 | 22 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5550 | 260 | 23 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.14. |
| ## | 5551 | 255 | 24 | 2016.06.20 | 105 | 1.83259571  | 2016.06.20.14. |
| ## | 5552 | 275 | 25 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.14. |
| ## | 5553 | 280 | 26 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5554 | 270 | 27 | 2016.06.20 | 90  | 1.57079633  | 2016.06.20.14. |
| ## | 5555 | 260 | 28 | 2016.06.20 | 100 | 1.74532925  | 2016.06.20.14. |
| ## | 5556 | 320 | 29 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.14. |
| ## | 5557 | 315 | 30 | 2016.06.20 | 45  | 0.78539816  | 2016.06.20.14. |
| ## | 5558 | 310 | 31 | 2016.06.20 | 50  | 0.87266463  | 2016.06.20.14. |
| ## | 5559 | 330 | 32 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5560 | 330 | 33 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5561 | 325 | 34 | 2016.06.20 | 35  | 0.61086524  | 2016.06.20.14. |
| ## | 5562 | 330 | 35 | 2016.06.20 | 30  | 0.52359878  | 2016.06.20.14. |
| ## | 5563 | 280 | 36 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5564 | 320 | 37 | 2016.06.20 | 40  | 0.69813170  | 2016.06.20.14. |
| ## | 5565 | 275 | 38 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.14. |
| ## | 5566 | 280 | 39 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5567 | 290 | 40 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5568 | 280 | 41 | 2016.06.20 | 80  | 1.39626340  | 2016.06.20.14. |
| ## | 5569 | 275 | 42 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.14. |
| ## | 5570 | 285 | 43 | 2016.06.20 | 75  | 1.30899694  | 2016.06.20.14. |
| ## | 5571 | 275 | 44 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.14. |
| ## | 5572 | 305 | 45 | 2016.06.20 | 55  | 0.95993109  | 2016.06.20.14. |
| ## | 5573 | 275 | 46 | 2016.06.20 | 85  | 1.48352986  | 2016.06.20.14. |
| ## | 5574 | 290 | 47 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5575 | 290 | 48 | 2016.06.20 | 70  | 1.22173048  | 2016.06.20.14. |
| ## | 5576 | 190 | 1  | 2016.06.21 | 170 | 2.96705973  | 2016.06.21.14. |
| ## | 5577 | 250 | 2  | 2016.06.21 | 110 | 1.91986218  | 2016.06.21.14. |
| ## | 5578 | 180 | 3  | 2016.06.21 | 180 | 3.14159265  | 2016.06.21.14. |
| ## | 5579 | 320 | 4  | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.14. |
| ## | 5580 | 175 | 5  | 2016.06.21 | 185 | -3.05432619 | 2016.06.21.14. |
| ## | 5581 | 20  | 6  | 2016.06.21 | 340 | -0.34906585 | 2016.06.21.14. |
| ## | 5582 | 235 | 7  | 2016.06.21 | 125 | 2.18166156  | 2016.06.21.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5583 | 350 | 8  | 2016.06.21 | 10  | 0.17453293  | 2016.06.21.14. |
| ## | 5584 | 180 | 9  | 2016.06.21 | 180 | 3.14159265  | 2016.06.21.14. |
| ## | 5585 | 160 | 10 | 2016.06.21 | 200 | -2.79252680 | 2016.06.21.14. |
| ## | 5586 | 190 | 11 | 2016.06.21 | 170 | 2.96705973  | 2016.06.21.14. |
| ## | 5587 | 320 | 12 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.14. |
| ## | 5588 | 240 | 13 | 2016.06.21 | 120 | 2.09439510  | 2016.06.21.14. |
| ## | 5589 | 160 | 14 | 2016.06.21 | 200 | -2.79252680 | 2016.06.21.14. |
| ## | 5590 | 265 | 15 | 2016.06.21 | 95  | 1.65806279  | 2016.06.21.14. |
| ## | 5591 | 285 | 16 | 2016.06.21 | 75  | 1.30899694  | 2016.06.21.14. |
| ## | 5592 | 275 | 17 | 2016.06.21 | 85  | 1.48352986  | 2016.06.21.14. |
| ## | 5593 | 300 | 18 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.14. |
| ## | 5594 | 260 | 19 | 2016.06.21 | 100 | 1.74532925  | 2016.06.21.14. |
| ## | 5595 | 305 | 20 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.14. |
| ## | 5596 | 310 | 21 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.14. |
| ## | 5597 | 320 | 22 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.14. |
| ## | 5598 | 290 | 23 | 2016.06.21 | 70  | 1.22173048  | 2016.06.21.14. |
| ## | 5599 | 310 | 24 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.14. |
| ## | 5600 | 270 | 25 | 2016.06.21 | 90  | 1.57079633  | 2016.06.21.14. |
| ## | 5601 | 270 | 26 | 2016.06.21 | 90  | 1.57079633  | 2016.06.21.14. |
| ## | 5602 | 350 | 1  | 2016.06.21 | 10  | 0.17453293  | 2016.06.21.14. |
| ## | 5603 | 200 | 2  | 2016.06.21 | 160 | 2.79252680  | 2016.06.21.14. |
| ## | 5604 | 210 | 3  | 2016.06.21 | 150 | 2.61799388  | 2016.06.21.14. |
| ## | 5605 | 190 | 4  | 2016.06.21 | 170 | 2.96705973  | 2016.06.21.14. |
| ## | 5606 | 225 | 5  | 2016.06.21 | 135 | 2.35619449  | 2016.06.21.14. |
| ## | 5607 | 310 | 6  | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.14. |
| ## | 5608 | 210 | 7  | 2016.06.21 | 150 | 2.61799388  | 2016.06.21.14. |
| ## | 5609 | 0   | 8  | 2016.06.21 | 0   | 0.00000000  | 2016.06.21.14. |
| ## | 5610 | 200 | 9  | 2016.06.21 | 160 | 2.79252680  | 2016.06.21.14. |
| ## | 5611 | 175 | 10 | 2016.06.21 | 185 | -3.05432619 | 2016.06.21.14. |
| ## | 5612 | 300 | 11 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.14. |
| ## | 5613 | 300 | 12 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.14. |
| ## | 5614 | 290 | 13 | 2016.06.21 | 70  | 1.22173048  | 2016.06.21.14. |
| ## | 5615 | 305 | 14 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.14. |
| ## | 5616 | 280 | 15 | 2016.06.21 | 80  | 1.39626340  | 2016.06.21.14. |
| ## | 5617 | 275 | 16 | 2016.06.21 | 85  | 1.48352986  | 2016.06.21.14. |
| ## | 5618 | 275 | 17 | 2016.06.21 | 85  | 1.48352986  | 2016.06.21.14. |
| ## | 5619 | 305 | 18 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.14. |
| ## | 5620 | 300 | 19 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.14. |
| ## | 5621 | 320 | 20 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.14. |
| ## | 5622 | 280 | 21 | 2016.06.21 | 80  | 1.39626340  | 2016.06.21.14. |
| ## | 5623 | 275 | 22 | 2016.06.21 | 85  | 1.48352986  | 2016.06.21.14. |
| ## | 5624 | 305 | 23 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.14. |
| ## | 5625 | 295 | 24 | 2016.06.21 | 65  | 1.13446401  | 2016.06.21.14. |
| ## | 5626 | 270 | 25 | 2016.06.21 | 90  | 1.57079633  | 2016.06.21.14. |
| ## | 5627 | 245 | 26 | 2016.06.21 | 115 | 2.00712864  | 2016.06.21.14. |
| ## | 5628 | 260 | 27 | 2016.06.21 | 100 | 1.74532925  | 2016.06.21.14. |
| ## | 5629 | 255 | 28 | 2016.06.21 | 105 | 1.83259571  | 2016.06.21.14. |
| ## | 5630 | 250 | 29 | 2016.06.21 | 110 | 1.91986218  | 2016.06.21.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5631 | 265 | 30 | 2016.06.21 | 95  | 1.65806279  | 2016.06.21.14. |
| ## | 5632 | 240 | 31 | 2016.06.21 | 120 | 2.09439510  | 2016.06.21.14. |
| ## | 5633 | 265 | 32 | 2016.06.21 | 95  | 1.65806279  | 2016.06.21.14. |
| ## | 5634 | 270 | 33 | 2016.06.21 | 90  | 1.57079633  | 2016.06.21.14. |
| ## | 5635 | 290 | 34 | 2016.06.21 | 70  | 1.22173048  | 2016.06.21.14. |
| ## | 5636 | 300 | 35 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.14. |
| ## | 5637 | 270 | 36 | 2016.06.21 | 90  | 1.57079633  | 2016.06.21.14. |
| ## | 5638 | 295 | 37 | 2016.06.21 | 65  | 1.13446401  | 2016.06.21.14. |
| ## | 5639 | 330 | 1  | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.14. |
| ## | 5640 | 20  | 2  | 2016.06.21 | 340 | -0.34906585 | 2016.06.21.14. |
| ## | 5641 | 350 | 3  | 2016.06.21 | 10  | 0.17453293  | 2016.06.21.14. |
| ## | 5642 | 190 | 4  | 2016.06.21 | 170 | 2.96705973  | 2016.06.21.14. |
| ## | 5643 | 265 | 5  | 2016.06.21 | 95  | 1.65806279  | 2016.06.21.14. |
| ## | 5644 | 260 | 6  | 2016.06.21 | 100 | 1.74532925  | 2016.06.21.14. |
| ## | 5645 | 265 | 7  | 2016.06.21 | 95  | 1.65806279  | 2016.06.21.14. |
| ## | 5646 | 260 | 8  | 2016.06.21 | 100 | 1.74532925  | 2016.06.21.14. |
| ## | 5647 | 285 | 9  | 2016.06.21 | 75  | 1.30899694  | 2016.06.21.14. |
| ## | 5648 | 30  | 10 | 2016.06.21 | 330 | -0.52359878 | 2016.06.21.14. |
| ## | 5649 | 15  | 11 | 2016.06.21 | 345 | -0.26179939 | 2016.06.21.14. |
| ## | 5650 | 345 | 12 | 2016.06.21 | 15  | 0.26179939  | 2016.06.21.14. |
| ## | 5651 | 330 | 13 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.14. |
| ## | 5652 | 280 | 14 | 2016.06.21 | 80  | 1.39626340  | 2016.06.21.14. |
| ## | 5653 | 300 | 15 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.14. |
| ## | 5654 | 290 | 16 | 2016.06.21 | 70  | 1.22173048  | 2016.06.21.14. |
| ## | 5655 | 275 | 17 | 2016.06.21 | 85  | 1.48352986  | 2016.06.21.14. |
| ## | 5656 | 280 | 18 | 2016.06.21 | 80  | 1.39626340  | 2016.06.21.14. |
| ## | 5657 | 330 | 19 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.14. |
| ## | 5658 | 280 | 20 | 2016.06.21 | 80  | 1.39626340  | 2016.06.21.14. |
| ## | 5659 | 325 | 21 | 2016.06.21 | 35  | 0.61086524  | 2016.06.21.14. |
| ## | 5660 | 310 | 22 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.14. |
| ## | 5661 | 310 | 23 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.14. |
| ## | 5662 | 325 | 24 | 2016.06.21 | 35  | 0.61086524  | 2016.06.21.14. |
| ## | 5663 | 310 | 25 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.14. |
| ## | 5664 | 330 | 26 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.14. |
| ## | 5665 | 325 | 27 | 2016.06.21 | 35  | 0.61086524  | 2016.06.21.14. |
| ## | 5666 | 305 | 28 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.14. |
| ## | 5667 | 245 | 1  | 2016.06.21 | 115 | 2.00712864  | 2016.06.21.15. |
| ## | 5668 | 0   | 2  | 2016.06.21 | 0   | 0.00000000  | 2016.06.21.15. |
| ## | 5669 | 205 | 3  | 2016.06.21 | 155 | 2.70526034  | 2016.06.21.15. |
| ## | 5670 | 5   | 4  | 2016.06.21 | 355 | -0.08726646 | 2016.06.21.15. |
| ## | 5671 | 25  | 5  | 2016.06.21 | 335 | -0.43633231 | 2016.06.21.15. |
| ## | 5672 | 220 | 6  | 2016.06.21 | 140 | 2.44346095  | 2016.06.21.15. |
| ## | 5673 | 40  | 7  | 2016.06.21 | 320 | -0.69813170 | 2016.06.21.15. |
| ## | 5674 | 320 | 8  | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## | 5675 | 310 | 9  | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## | 5676 | 330 | 10 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |
| ## | 5677 | 340 | 11 | 2016.06.21 | 20  | 0.34906585  | 2016.06.21.15. |
| ## | 5678 | 330 | 12 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 5679 | 330 | 13 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |
| ## 5680 | 325 | 14 | 2016.06.21 | 35  | 0.61086524  | 2016.06.21.15. |
| ## 5681 | 300 | 15 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.15. |
| ## 5682 | 310 | 16 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## 5683 | 320 | 17 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## 5684 | 320 | 18 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## 5685 | 305 | 19 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.15. |
| ## 5686 | 345 | 20 | 2016.06.21 | 15  | 0.26179939  | 2016.06.21.15. |
| ## 5687 | 0   | 21 | 2016.06.21 | 0   | 0.00000000  | 2016.06.21.15. |
| ## 5688 | 330 | 22 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |
| ## 5689 | 30  | 1  | 2016.06.21 | 330 | -0.52359878 | 2016.06.21.15. |
| ## 5690 | 230 | 2  | 2016.06.21 | 130 | 2.26892803  | 2016.06.21.15. |
| ## 5691 | 70  | 3  | 2016.06.21 | 290 | -1.22173048 | 2016.06.21.15. |
| ## 5692 | 0   | 4  | 2016.06.21 | 0   | 0.00000000  | 2016.06.21.15. |
| ## 5693 | 190 | 5  | 2016.06.21 | 170 | 2.96705973  | 2016.06.21.15. |
| ## 5694 | 120 | 6  | 2016.06.21 | 240 | -2.09439510 | 2016.06.21.15. |
| ## 5695 | 190 | 7  | 2016.06.21 | 170 | 2.96705973  | 2016.06.21.15. |
| ## 5696 | 60  | 8  | 2016.06.21 | 300 | -1.04719755 | 2016.06.21.15. |
| ## 5697 | 55  | 9  | 2016.06.21 | 305 | -0.95993109 | 2016.06.21.15. |
| ## 5698 | 80  | 10 | 2016.06.21 | 280 | -1.39626340 | 2016.06.21.15. |
| ## 5699 | 330 | 11 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |
| ## 5700 | 5   | 12 | 2016.06.21 | 355 | -0.08726646 | 2016.06.21.15. |
| ## 5701 | 105 | 13 | 2016.06.21 | 255 | -1.83259571 | 2016.06.21.15. |
| ## 5702 | 240 | 14 | 2016.06.21 | 120 | 2.09439510  | 2016.06.21.15. |
| ## 5703 | 260 | 15 | 2016.06.21 | 100 | 1.74532925  | 2016.06.21.15. |
| ## 5704 | 325 | 16 | 2016.06.21 | 35  | 0.61086524  | 2016.06.21.15. |
| ## 5705 | 300 | 17 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.15. |
| ## 5706 | 320 | 18 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## 5707 | 310 | 19 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## 5708 | 305 | 20 | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.15. |
| ## 5709 | 310 | 21 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## 5710 | 310 | 22 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## 5711 | 320 | 23 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## 5712 | 320 | 24 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## 5713 | 315 | 25 | 2016.06.21 | 45  | 0.78539816  | 2016.06.21.15. |
| ## 5714 | 335 | 26 | 2016.06.21 | 25  | 0.43633231  | 2016.06.21.15. |
| ## 5715 | 330 | 27 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |
| ## 5716 | 325 | 28 | 2016.06.21 | 35  | 0.61086524  | 2016.06.21.15. |
| ## 5717 | 340 | 29 | 2016.06.21 | 20  | 0.34906585  | 2016.06.21.15. |
| ## 5718 | 335 | 30 | 2016.06.21 | 25  | 0.43633231  | 2016.06.21.15. |
| ## 5719 | 315 | 31 | 2016.06.21 | 45  | 0.78539816  | 2016.06.21.15. |
| ## 5720 | 295 | 32 | 2016.06.21 | 65  | 1.13446401  | 2016.06.21.15. |
| ## 5721 | 340 | 33 | 2016.06.21 | 20  | 0.34906585  | 2016.06.21.15. |
| ## 5722 | 330 | 34 | 2016.06.21 | 30  | 0.52359878  | 2016.06.21.15. |
| ## 5723 | 310 | 35 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## 5724 | 340 | 36 | 2016.06.21 | 20  | 0.34906585  | 2016.06.21.15. |
| ## 5725 | 350 | 37 | 2016.06.21 | 10  | 0.17453293  | 2016.06.21.15. |
| ## 5726 | 350 | 38 | 2016.06.21 | 10  | 0.17453293  | 2016.06.21.15. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5727 | 335 | 39 | 2016.06.21 | 25  | 0.43633231  | 2016.06.21.15. |
| ## | 5728 | 300 | 40 | 2016.06.21 | 60  | 1.04719755  | 2016.06.21.15. |
| ## | 5729 | 290 | 41 | 2016.06.21 | 70  | 1.22173048  | 2016.06.21.15. |
| ## | 5730 | 310 | 42 | 2016.06.21 | 50  | 0.87266463  | 2016.06.21.15. |
| ## | 5731 | 60  | 1  | 2016.06.21 | 300 | -1.04719755 | 2016.06.21.15. |
| ## | 5732 | 60  | 2  | 2016.06.21 | 300 | -1.04719755 | 2016.06.21.15. |
| ## | 5733 | 105 | 3  | 2016.06.21 | 255 | -1.83259571 | 2016.06.21.15. |
| ## | 5734 | 45  | 4  | 2016.06.21 | 315 | -0.78539816 | 2016.06.21.15. |
| ## | 5735 | 260 | 5  | 2016.06.21 | 100 | 1.74532925  | 2016.06.21.15. |
| ## | 5736 | 70  | 6  | 2016.06.21 | 290 | -1.22173048 | 2016.06.21.15. |
| ## | 5737 | 30  | 7  | 2016.06.21 | 330 | -0.52359878 | 2016.06.21.15. |
| ## | 5738 | 60  | 8  | 2016.06.21 | 300 | -1.04719755 | 2016.06.21.15. |
| ## | 5739 | 305 | 9  | 2016.06.21 | 55  | 0.95993109  | 2016.06.21.15. |
| ## | 5740 | 320 | 10 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## | 5741 | 320 | 11 | 2016.06.21 | 40  | 0.69813170  | 2016.06.21.15. |
| ## | 5742 | 270 | 12 | 2016.06.21 | 90  | 1.57079633  | 2016.06.21.15. |
| ## | 5743 | 40  | 13 | 2016.06.21 | 320 | -0.69813170 | 2016.06.21.15. |
| ## | 5744 | 5   | 14 | 2016.06.21 | 355 | -0.08726646 | 2016.06.21.15. |
| ## | 5745 | 5   | 15 | 2016.06.21 | 355 | -0.08726646 | 2016.06.21.15. |
| ## | 5746 | 0   | 16 | 2016.06.21 | 0   | 0.00000000  | 2016.06.21.15. |
| ## | 5747 | 10  | 17 | 2016.06.21 | 350 | -0.17453293 | 2016.06.21.15. |
| ## | 5748 | 5   | 18 | 2016.06.21 | 355 | -0.08726646 | 2016.06.21.15. |
| ## | 5749 | 5   | 19 | 2016.06.21 | 355 | -0.08726646 | 2016.06.21.15. |
| ## | 5750 | 175 | 1  | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.13. |
| ## | 5751 | 155 | 2  | 2016.06.07 | 205 | -2.70526034 | 2016.06.07.13. |
| ## | 5752 | 225 | 3  | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.13. |
| ## | 5753 | 210 | 4  | 2016.06.07 | 150 | 2.61799388  | 2016.06.07.13. |
| ## | 5754 | 215 | 5  | 2016.06.07 | 145 | 2.53072742  | 2016.06.07.13. |
| ## | 5755 | 185 | 6  | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.13. |
| ## | 5756 | 160 | 7  | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.13. |
| ## | 5757 | 175 | 8  | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.13. |
| ## | 5758 | 130 | 9  | 2016.06.07 | 230 | -2.26892803 | 2016.06.07.13. |
| ## | 5759 | 350 | 10 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.13. |
| ## | 5760 | 350 | 11 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.13. |
| ## | 5761 | 0   | 12 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.13. |
| ## | 5762 | 5   | 13 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.13. |
| ## | 5763 | 350 | 14 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.13. |
| ## | 5764 | 345 | 15 | 2016.06.07 | 15  | 0.26179939  | 2016.06.07.13. |
| ## | 5765 | 350 | 16 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.13. |
| ## | 5766 | 335 | 17 | 2016.06.07 | 25  | 0.43633231  | 2016.06.07.13. |
| ## | 5767 | 5   | 18 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.13. |
| ## | 5768 | 340 | 19 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.13. |
| ## | 5769 | 310 | 20 | 2016.06.07 | 50  | 0.87266463  | 2016.06.07.13. |
| ## | 5770 | 30  | 21 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.13. |
| ## | 5771 | 0   | 22 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.13. |
| ## | 5772 | 15  | 23 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.13. |
| ## | 5773 | 10  | 24 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.13. |
| ## | 5774 | 30  | 25 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.13. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 5775 | 340 | 26 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.13. |
| ## 5776 | 355 | 27 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.13. |
| ## 5777 | 270 | 1  | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## 5778 | 220 | 2  | 2016.06.07 | 140 | 2.44346095  | 2016.06.07.14. |
| ## 5779 | 260 | 3  | 2016.06.07 | 100 | 1.74532925  | 2016.06.07.14. |
| ## 5780 | 250 | 4  | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.14. |
| ## 5781 | 190 | 5  | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.14. |
| ## 5782 | 185 | 6  | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.14. |
| ## 5783 | 175 | 7  | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.14. |
| ## 5784 | 200 | 8  | 2016.06.07 | 160 | 2.79252680  | 2016.06.07.14. |
| ## 5785 | 140 | 9  | 2016.06.07 | 220 | -2.44346095 | 2016.06.07.14. |
| ## 5786 | 160 | 10 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.14. |
| ## 5787 | 10  | 11 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 5788 | 10  | 12 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 5789 | 5   | 13 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5790 | 0   | 14 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5791 | 15  | 15 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.14. |
| ## 5792 | 0   | 16 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5793 | 5   | 17 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5794 | 15  | 18 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.14. |
| ## 5795 | 40  | 19 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.14. |
| ## 5796 | 5   | 20 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5797 | 10  | 21 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 5798 | 355 | 22 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## 5799 | 0   | 23 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5800 | 5   | 24 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5801 | 350 | 25 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## 5802 | 10  | 26 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 5803 | 0   | 27 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5804 | 15  | 28 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.14. |
| ## 5805 | 5   | 29 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5806 | 160 | 1  | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.14. |
| ## 5807 | 190 | 2  | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.14. |
| ## 5808 | 150 | 3  | 2016.06.07 | 210 | -2.61799388 | 2016.06.07.14. |
| ## 5809 | 180 | 4  | 2016.06.07 | 180 | 3.14159265  | 2016.06.07.14. |
| ## 5810 | 75  | 5  | 2016.06.07 | 285 | -1.30899694 | 2016.06.07.14. |
| ## 5811 | 240 | 6  | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.14. |
| ## 5812 | 90  | 7  | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.14. |
| ## 5813 | 200 | 8  | 2016.06.07 | 160 | 2.79252680  | 2016.06.07.14. |
| ## 5814 | 145 | 9  | 2016.06.07 | 215 | -2.53072742 | 2016.06.07.14. |
| ## 5815 | 170 | 10 | 2016.06.07 | 190 | -2.96705973 | 2016.06.07.14. |
| ## 5816 | 160 | 11 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.14. |
| ## 5817 | 145 | 12 | 2016.06.07 | 215 | -2.53072742 | 2016.06.07.14. |
| ## 5818 | 165 | 13 | 2016.06.07 | 195 | -2.87979327 | 2016.06.07.14. |
| ## 5819 | 125 | 14 | 2016.06.07 | 235 | -2.18166156 | 2016.06.07.14. |
| ## 5820 | 120 | 15 | 2016.06.07 | 240 | -2.09439510 | 2016.06.07.14. |
| ## 5821 | 30  | 16 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 5822 | 355 | 17 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |

|    |      |     |    |            |     |             |                |
|----|------|-----|----|------------|-----|-------------|----------------|
| ## | 5823 | 5   | 18 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## | 5824 | 10  | 19 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## | 5825 | 5   | 20 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## | 5826 | 240 | 1  | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.14. |
| ## | 5827 | 250 | 2  | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.14. |
| ## | 5828 | 265 | 3  | 2016.06.07 | 95  | 1.65806279  | 2016.06.07.14. |
| ## | 5829 | 270 | 4  | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## | 5830 | 265 | 5  | 2016.06.07 | 95  | 1.65806279  | 2016.06.07.14. |
| ## | 5831 | 270 | 6  | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## | 5832 | 270 | 7  | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## | 5833 | 255 | 8  | 2016.06.07 | 105 | 1.83259571  | 2016.06.07.14. |
| ## | 5834 | 260 | 9  | 2016.06.07 | 100 | 1.74532925  | 2016.06.07.14. |
| ## | 5835 | 240 | 10 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.14. |
| ## | 5836 | 250 | 11 | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.14. |
| ## | 5837 | 280 | 12 | 2016.06.07 | 80  | 1.39626340  | 2016.06.07.14. |
| ## | 5838 | 300 | 13 | 2016.06.07 | 60  | 1.04719755  | 2016.06.07.14. |
| ## | 5839 | 165 | 14 | 2016.06.07 | 195 | -2.87979327 | 2016.06.07.14. |
| ## | 5840 | 200 | 15 | 2016.06.07 | 160 | 2.79252680  | 2016.06.07.14. |
| ## | 5841 | 245 | 16 | 2016.06.07 | 115 | 2.00712864  | 2016.06.07.14. |
| ## | 5842 | 225 | 17 | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.14. |
| ## | 5843 | 275 | 18 | 2016.06.07 | 85  | 1.48352986  | 2016.06.07.14. |
| ## | 5844 | 250 | 19 | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.14. |
| ## | 5845 | 280 | 20 | 2016.06.07 | 80  | 1.39626340  | 2016.06.07.14. |
| ## | 5846 | 180 | 21 | 2016.06.07 | 180 | 3.14159265  | 2016.06.07.14. |
| ## | 5847 | 355 | 22 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5848 | 340 | 23 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5849 | 300 | 24 | 2016.06.07 | 60  | 1.04719755  | 2016.06.07.14. |
| ## | 5850 | 340 | 25 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5851 | 355 | 26 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5852 | 330 | 27 | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.14. |
| ## | 5853 | 355 | 28 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5854 | 350 | 29 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5855 | 355 | 30 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5856 | 350 | 31 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5857 | 0   | 32 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5858 | 330 | 33 | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.14. |
| ## | 5859 | 340 | 34 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5860 | 325 | 35 | 2016.06.07 | 35  | 0.61086524  | 2016.06.07.14. |
| ## | 5861 | 350 | 36 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5862 | 320 | 37 | 2016.06.07 | 40  | 0.69813170  | 2016.06.07.14. |
| ## | 5863 | 5   | 38 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## | 5864 | 340 | 39 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5865 | 350 | 40 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5866 | 0   | 41 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5867 | 0   | 42 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5868 | 330 | 43 | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.14. |
| ## | 5869 | 335 | 44 | 2016.06.07 | 25  | 0.43633231  | 2016.06.07.14. |
| ## | 5870 | 340 | 45 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 5871 | 340 | 46 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## 5872 | 340 | 47 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## 5873 | 45  | 48 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.14. |
| ## 5874 | 270 | 49 | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## 5875 | 15  | 50 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.14. |
| ## 5876 | 310 | 51 | 2016.06.07 | 50  | 0.87266463  | 2016.06.07.14. |
| ## 5877 | 5   | 52 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5878 | 0   | 53 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5879 | 10  | 54 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 5880 | 0   | 55 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5881 | 315 | 56 | 2016.06.07 | 45  | 0.78539816  | 2016.06.07.14. |
| ## 5882 | 355 | 57 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## 5883 | 250 | 58 | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.14. |
| ## 5884 | 230 | 59 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.14. |
| ## 5885 | 230 | 60 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.14. |
| ## 5886 | 250 | 61 | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.14. |
| ## 5887 | 330 | 62 | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.14. |
| ## 5888 | 90  | 63 | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.14. |
| ## 5889 | 225 | 64 | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.14. |
| ## 5890 | 215 | 65 | 2016.06.07 | 145 | 2.53072742  | 2016.06.07.14. |
| ## 5891 | 290 | 66 | 2016.06.07 | 70  | 1.22173048  | 2016.06.07.14. |
| ## 5892 | 90  | 67 | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.14. |
| ## 5893 | 240 | 68 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.14. |
| ## 5894 | 270 | 69 | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## 5895 | 300 | 70 | 2016.06.07 | 60  | 1.04719755  | 2016.06.07.14. |
| ## 5896 | 85  | 71 | 2016.06.07 | 275 | -1.48352986 | 2016.06.07.14. |
| ## 5897 | 5   | 72 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5898 | 120 | 1  | 2016.06.07 | 240 | -2.09439510 | 2016.06.07.14. |
| ## 5899 | 295 | 2  | 2016.06.07 | 65  | 1.13446401  | 2016.06.07.14. |
| ## 5900 | 120 | 3  | 2016.06.07 | 240 | -2.09439510 | 2016.06.07.14. |
| ## 5901 | 180 | 4  | 2016.06.07 | 180 | 3.14159265  | 2016.06.07.14. |
| ## 5902 | 90  | 5  | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.14. |
| ## 5903 | 185 | 6  | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.14. |
| ## 5904 | 270 | 7  | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## 5905 | 140 | 8  | 2016.06.07 | 220 | -2.44346095 | 2016.06.07.14. |
| ## 5906 | 165 | 9  | 2016.06.07 | 195 | -2.87979327 | 2016.06.07.14. |
| ## 5907 | 210 | 10 | 2016.06.07 | 150 | 2.61799388  | 2016.06.07.14. |
| ## 5908 | 150 | 11 | 2016.06.07 | 210 | -2.61799388 | 2016.06.07.14. |
| ## 5909 | 145 | 12 | 2016.06.07 | 215 | -2.53072742 | 2016.06.07.14. |
| ## 5910 | 140 | 13 | 2016.06.07 | 220 | -2.44346095 | 2016.06.07.14. |
| ## 5911 | 190 | 14 | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.14. |
| ## 5912 | 165 | 15 | 2016.06.07 | 195 | -2.87979327 | 2016.06.07.14. |
| ## 5913 | 110 | 16 | 2016.06.07 | 250 | -1.91986218 | 2016.06.07.14. |
| ## 5914 | 105 | 17 | 2016.06.07 | 255 | -1.83259571 | 2016.06.07.14. |
| ## 5915 | 180 | 18 | 2016.06.07 | 180 | 3.14159265  | 2016.06.07.14. |
| ## 5916 | 230 | 19 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.14. |
| ## 5917 | 145 | 20 | 2016.06.07 | 215 | -2.53072742 | 2016.06.07.14. |
| ## 5918 | 155 | 21 | 2016.06.07 | 205 | -2.70526034 | 2016.06.07.14. |

|    |      |     |    |            |     |             |                |
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| ## | 5919 | 95  | 22 | 2016.06.07 | 265 | -1.65806279 | 2016.06.07.14. |
| ## | 5920 | 85  | 23 | 2016.06.07 | 275 | -1.48352986 | 2016.06.07.14. |
| ## | 5921 | 155 | 24 | 2016.06.07 | 205 | -2.70526034 | 2016.06.07.14. |
| ## | 5922 | 90  | 25 | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.14. |
| ## | 5923 | 15  | 26 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.14. |
| ## | 5924 | 10  | 27 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## | 5925 | 20  | 28 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.14. |
| ## | 5926 | 5   | 29 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## | 5927 | 340 | 30 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5928 | 30  | 31 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## | 5929 | 0   | 32 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5930 | 20  | 33 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.14. |
| ## | 5931 | 5   | 34 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## | 5932 | 355 | 35 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5933 | 5   | 36 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## | 5934 | 0   | 37 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5935 | 0   | 38 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5936 | 0   | 39 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5937 | 340 | 40 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5938 | 0   | 41 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## | 5939 | 355 | 42 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5940 | 10  | 43 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## | 5941 | 350 | 44 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5942 | 25  | 45 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.14. |
| ## | 5943 | 355 | 46 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5944 | 350 | 47 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5945 | 220 | 48 | 2016.06.07 | 140 | 2.44346095  | 2016.06.07.14. |
| ## | 5946 | 50  | 49 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.14. |
| ## | 5947 | 320 | 50 | 2016.06.07 | 40  | 0.69813170  | 2016.06.07.14. |
| ## | 5948 | 340 | 51 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5949 | 320 | 52 | 2016.06.07 | 40  | 0.69813170  | 2016.06.07.14. |
| ## | 5950 | 10  | 53 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## | 5951 | 340 | 54 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## | 5952 | 30  | 55 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## | 5953 | 45  | 56 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.14. |
| ## | 5954 | 355 | 57 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5955 | 55  | 58 | 2016.06.07 | 305 | -0.95993109 | 2016.06.07.14. |
| ## | 5956 | 260 | 59 | 2016.06.07 | 100 | 1.74532925  | 2016.06.07.14. |
| ## | 5957 | 355 | 60 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5958 | 270 | 61 | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.14. |
| ## | 5959 | 355 | 62 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## | 5960 | 350 | 63 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.14. |
| ## | 5961 | 50  | 64 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.14. |
| ## | 5962 | 160 | 1  | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.14. |
| ## | 5963 | 130 | 2  | 2016.06.07 | 230 | -2.26892803 | 2016.06.07.14. |
| ## | 5964 | 155 | 3  | 2016.06.07 | 205 | -2.70526034 | 2016.06.07.14. |
| ## | 5965 | 105 | 4  | 2016.06.07 | 255 | -1.83259571 | 2016.06.07.14. |
| ## | 5966 | 130 | 5  | 2016.06.07 | 230 | -2.26892803 | 2016.06.07.14. |

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|---------|-----|----|------------|-----|-------------|----------------|
| ## 5967 | 225 | 6  | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.14. |
| ## 5968 | 200 | 7  | 2016.06.07 | 160 | 2.79252680  | 2016.06.07.14. |
| ## 5969 | 175 | 8  | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.14. |
| ## 5970 | 160 | 9  | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.14. |
| ## 5971 | 135 | 10 | 2016.06.07 | 225 | -2.35619449 | 2016.06.07.14. |
| ## 5972 | 185 | 11 | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.14. |
| ## 5973 | 160 | 12 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.14. |
| ## 5974 | 295 | 13 | 2016.06.07 | 65  | 1.13446401  | 2016.06.07.14. |
| ## 5975 | 240 | 14 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.14. |
| ## 5976 | 0   | 15 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5977 | 5   | 16 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5978 | 5   | 17 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5979 | 30  | 18 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 5980 | 0   | 19 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 5981 | 355 | 20 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## 5982 | 5   | 21 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5983 | 20  | 22 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.14. |
| ## 5984 | 20  | 23 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.14. |
| ## 5985 | 355 | 24 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.14. |
| ## 5986 | 10  | 25 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 5987 | 20  | 26 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.14. |
| ## 5988 | 25  | 27 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.14. |
| ## 5989 | 5   | 28 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.14. |
| ## 5990 | 50  | 29 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.14. |
| ## 5991 | 35  | 30 | 2016.06.07 | 325 | -0.61086524 | 2016.06.07.14. |
| ## 5992 | 30  | 31 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 5993 | 40  | 1  | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.14. |
| ## 5994 | 70  | 2  | 2016.06.07 | 290 | -1.22173048 | 2016.06.07.14. |
| ## 5995 | 85  | 3  | 2016.06.07 | 275 | -1.48352986 | 2016.06.07.14. |
| ## 5996 | 30  | 4  | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 5997 | 45  | 5  | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.14. |
| ## 5998 | 95  | 6  | 2016.06.07 | 265 | -1.65806279 | 2016.06.07.14. |
| ## 5999 | 340 | 7  | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.14. |
| ## 6000 | 65  | 8  | 2016.06.07 | 295 | -1.13446401 | 2016.06.07.14. |
| ## 6001 | 85  | 9  | 2016.06.07 | 275 | -1.48352986 | 2016.06.07.14. |
| ## 6002 | 25  | 10 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.14. |
| ## 6003 | 75  | 11 | 2016.06.07 | 285 | -1.30899694 | 2016.06.07.14. |
| ## 6004 | 130 | 12 | 2016.06.07 | 230 | -2.26892803 | 2016.06.07.14. |
| ## 6005 | 95  | 13 | 2016.06.07 | 265 | -1.65806279 | 2016.06.07.14. |
| ## 6006 | 80  | 14 | 2016.06.07 | 280 | -1.39626340 | 2016.06.07.14. |
| ## 6007 | 60  | 15 | 2016.06.07 | 300 | -1.04719755 | 2016.06.07.14. |
| ## 6008 | 30  | 16 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 6009 | 35  | 17 | 2016.06.07 | 325 | -0.61086524 | 2016.06.07.14. |
| ## 6010 | 70  | 18 | 2016.06.07 | 290 | -1.22173048 | 2016.06.07.14. |
| ## 6011 | 65  | 19 | 2016.06.07 | 295 | -1.13446401 | 2016.06.07.14. |
| ## 6012 | 30  | 20 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 6013 | 40  | 21 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.14. |
| ## 6014 | 15  | 22 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.14. |

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| ## 6015 | 50  | 23 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.14. |
| ## 6016 | 30  | 24 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 6017 | 45  | 25 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.14. |
| ## 6018 | 45  | 26 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.14. |
| ## 6019 | 30  | 27 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 6020 | 25  | 28 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.14. |
| ## 6021 | 55  | 29 | 2016.06.07 | 305 | -0.95993109 | 2016.06.07.14. |
| ## 6022 | 30  | 30 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 6023 | 30  | 31 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.14. |
| ## 6024 | 40  | 32 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.14. |
| ## 6025 | 25  | 33 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.14. |
| ## 6026 | 10  | 34 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.14. |
| ## 6027 | 55  | 35 | 2016.06.07 | 305 | -0.95993109 | 2016.06.07.14. |
| ## 6028 | 70  | 36 | 2016.06.07 | 290 | -1.22173048 | 2016.06.07.14. |
| ## 6029 | 45  | 37 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.14. |
| ## 6030 | 0   | 38 | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.14. |
| ## 6031 | 55  | 39 | 2016.06.07 | 305 | -0.95993109 | 2016.06.07.14. |
| ## 6032 | 50  | 40 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.14. |
| ## 6033 | 180 | 1  | 2016.06.07 | 180 | 3.14159265  | 2016.06.07.15. |
| ## 6034 | 185 | 2  | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.15. |
| ## 6035 | 205 | 3  | 2016.06.07 | 155 | 2.70526034  | 2016.06.07.15. |
| ## 6036 | 120 | 4  | 2016.06.07 | 240 | -2.09439510 | 2016.06.07.15. |
| ## 6037 | 130 | 5  | 2016.06.07 | 230 | -2.26892803 | 2016.06.07.15. |
| ## 6038 | 125 | 6  | 2016.06.07 | 235 | -2.18166156 | 2016.06.07.15. |
| ## 6039 | 155 | 7  | 2016.06.07 | 205 | -2.70526034 | 2016.06.07.15. |
| ## 6040 | 150 | 8  | 2016.06.07 | 210 | -2.61799388 | 2016.06.07.15. |
| ## 6041 | 195 | 9  | 2016.06.07 | 165 | 2.87979327  | 2016.06.07.15. |
| ## 6042 | 145 | 10 | 2016.06.07 | 215 | -2.53072742 | 2016.06.07.15. |
| ## 6043 | 160 | 11 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.15. |
| ## 6044 | 175 | 12 | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.15. |
| ## 6045 | 175 | 13 | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.15. |
| ## 6046 | 140 | 14 | 2016.06.07 | 220 | -2.44346095 | 2016.06.07.15. |
| ## 6047 | 180 | 15 | 2016.06.07 | 180 | 3.14159265  | 2016.06.07.15. |
| ## 6048 | 95  | 16 | 2016.06.07 | 265 | -1.65806279 | 2016.06.07.15. |
| ## 6049 | 160 | 17 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.15. |
| ## 6050 | 90  | 18 | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.15. |
| ## 6051 | 100 | 19 | 2016.06.07 | 260 | -1.74532925 | 2016.06.07.15. |
| ## 6052 | 90  | 20 | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.15. |
| ## 6053 | 70  | 21 | 2016.06.07 | 290 | -1.22173048 | 2016.06.07.15. |
| ## 6054 | 80  | 22 | 2016.06.07 | 280 | -1.39626340 | 2016.06.07.15. |
| ## 6055 | 60  | 23 | 2016.06.07 | 300 | -1.04719755 | 2016.06.07.15. |
| ## 6056 | 40  | 24 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6057 | 230 | 1  | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.15. |
| ## 6058 | 270 | 2  | 2016.06.07 | 90  | 1.57079633  | 2016.06.07.15. |
| ## 6059 | 90  | 3  | 2016.06.07 | 270 | -1.57079633 | 2016.06.07.15. |
| ## 6060 | 190 | 4  | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.15. |
| ## 6061 | 160 | 5  | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.15. |
| ## 6062 | 110 | 6  | 2016.06.07 | 250 | -1.91986218 | 2016.06.07.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 6063 | 95  | 7  | 2016.06.07 | 265 | -1.65806279 | 2016.06.07.15. |
| ## 6064 | 190 | 8  | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.15. |
| ## 6065 | 190 | 9  | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.15. |
| ## 6066 | 355 | 10 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.15. |
| ## 6067 | 345 | 11 | 2016.06.07 | 15  | 0.26179939  | 2016.06.07.15. |
| ## 6068 | 15  | 12 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.15. |
| ## 6069 | 50  | 13 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6070 | 25  | 14 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.15. |
| ## 6071 | 170 | 15 | 2016.06.07 | 190 | -2.96705973 | 2016.06.07.15. |
| ## 6072 | 100 | 16 | 2016.06.07 | 260 | -1.74532925 | 2016.06.07.15. |
| ## 6073 | 310 | 17 | 2016.06.07 | 50  | 0.87266463  | 2016.06.07.15. |
| ## 6074 | 25  | 18 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.15. |
| ## 6075 | 355 | 19 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.15. |
| ## 6076 | 340 | 20 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.15. |
| ## 6077 | 335 | 21 | 2016.06.07 | 25  | 0.43633231  | 2016.06.07.15. |
| ## 6078 | 355 | 22 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.15. |
| ## 6079 | 35  | 23 | 2016.06.07 | 325 | -0.61086524 | 2016.06.07.15. |
| ## 6080 | 5   | 24 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.15. |
| ## 6081 | 10  | 25 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.15. |
| ## 6082 | 40  | 26 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6083 | 20  | 27 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.15. |
| ## 6084 | 20  | 28 | 2016.06.07 | 340 | -0.34906585 | 2016.06.07.15. |
| ## 6085 | 25  | 29 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.15. |
| ## 6086 | 30  | 30 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.15. |
| ## 6087 | 40  | 31 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6088 | 80  | 32 | 2016.06.07 | 280 | -1.39626340 | 2016.06.07.15. |
| ## 6089 | 45  | 33 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.15. |
| ## 6090 | 50  | 34 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6091 | 25  | 35 | 2016.06.07 | 335 | -0.43633231 | 2016.06.07.15. |
| ## 6092 | 50  | 36 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6093 | 35  | 37 | 2016.06.07 | 325 | -0.61086524 | 2016.06.07.15. |
| ## 6094 | 80  | 38 | 2016.06.07 | 280 | -1.39626340 | 2016.06.07.15. |
| ## 6095 | 35  | 39 | 2016.06.07 | 325 | -0.61086524 | 2016.06.07.15. |
| ## 6096 | 65  | 40 | 2016.06.07 | 295 | -1.13446401 | 2016.06.07.15. |
| ## 6097 | 50  | 41 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6098 | 55  | 42 | 2016.06.07 | 305 | -0.95993109 | 2016.06.07.15. |
| ## 6099 | 40  | 43 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6100 | 70  | 44 | 2016.06.07 | 290 | -1.22173048 | 2016.06.07.15. |
| ## 6101 | 355 | 45 | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.15. |
| ## 6102 | 350 | 46 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.15. |
| ## 6103 | 15  | 47 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.15. |
| ## 6104 | 5   | 48 | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.15. |
| ## 6105 | 185 | 49 | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.15. |
| ## 6106 | 335 | 1  | 2016.06.07 | 25  | 0.43633231  | 2016.06.07.15. |
| ## 6107 | 15  | 2  | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.15. |
| ## 6108 | 0   | 3  | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.15. |
| ## 6109 | 330 | 4  | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.15. |
| ## 6110 | 10  | 5  | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.15. |

|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 6111 | 350 | 6  | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.15. |
| ## 6112 | 300 | 7  | 2016.06.07 | 60  | 1.04719755  | 2016.06.07.15. |
| ## 6113 | 355 | 8  | 2016.06.07 | 5   | 0.08726646  | 2016.06.07.15. |
| ## 6114 | 0   | 9  | 2016.06.07 | 0   | 0.00000000  | 2016.06.07.15. |
| ## 6115 | 310 | 10 | 2016.06.07 | 50  | 0.87266463  | 2016.06.07.15. |
| ## 6116 | 330 | 11 | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.15. |
| ## 6117 | 345 | 12 | 2016.06.07 | 15  | 0.26179939  | 2016.06.07.15. |
| ## 6118 | 340 | 13 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.15. |
| ## 6119 | 30  | 14 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.15. |
| ## 6120 | 340 | 15 | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.15. |
| ## 6121 | 40  | 16 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6122 | 45  | 17 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.15. |
| ## 6123 | 40  | 18 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6124 | 50  | 19 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6125 | 60  | 20 | 2016.06.07 | 300 | -1.04719755 | 2016.06.07.15. |
| ## 6126 | 50  | 21 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6127 | 40  | 22 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6128 | 50  | 23 | 2016.06.07 | 310 | -0.87266463 | 2016.06.07.15. |
| ## 6129 | 30  | 24 | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.15. |
| ## 6130 | 60  | 25 | 2016.06.07 | 300 | -1.04719755 | 2016.06.07.15. |
| ## 6131 | 40  | 26 | 2016.06.07 | 320 | -0.69813170 | 2016.06.07.15. |
| ## 6132 | 45  | 27 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.15. |
| ## 6133 | 45  | 28 | 2016.06.07 | 315 | -0.78539816 | 2016.06.07.15. |
| ## 6134 | 35  | 29 | 2016.06.07 | 325 | -0.61086524 | 2016.06.07.15. |
| ## 6135 | 15  | 30 | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.15. |
| ## 6136 | 10  | 31 | 2016.06.07 | 350 | -0.17453293 | 2016.06.07.15. |
| ## 6137 | 220 | 1  | 2016.06.07 | 140 | 2.44346095  | 2016.06.07.16. |
| ## 6138 | 250 | 2  | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.16. |
| ## 6139 | 230 | 3  | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.16. |
| ## 6140 | 250 | 4  | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.16. |
| ## 6141 | 140 | 5  | 2016.06.07 | 220 | -2.44346095 | 2016.06.07.16. |
| ## 6142 | 120 | 6  | 2016.06.07 | 240 | -2.09439510 | 2016.06.07.16. |
| ## 6143 | 120 | 7  | 2016.06.07 | 240 | -2.09439510 | 2016.06.07.16. |
| ## 6144 | 150 | 8  | 2016.06.07 | 210 | -2.61799388 | 2016.06.07.16. |
| ## 6145 | 125 | 9  | 2016.06.07 | 235 | -2.18166156 | 2016.06.07.16. |
| ## 6146 | 195 | 10 | 2016.06.07 | 165 | 2.87979327  | 2016.06.07.16. |
| ## 6147 | 255 | 11 | 2016.06.07 | 105 | 1.83259571  | 2016.06.07.16. |
| ## 6148 | 185 | 12 | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.16. |
| ## 6149 | 280 | 13 | 2016.06.07 | 80  | 1.39626340  | 2016.06.07.16. |
| ## 6150 | 145 | 14 | 2016.06.07 | 215 | -2.53072742 | 2016.06.07.16. |
| ## 6151 | 185 | 15 | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.16. |
| ## 6152 | 240 | 16 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6153 | 230 | 17 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.16. |
| ## 6154 | 225 | 18 | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.16. |
| ## 6155 | 230 | 19 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.16. |
| ## 6156 | 225 | 20 | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.16. |
| ## 6157 | 255 | 21 | 2016.06.07 | 105 | 1.83259571  | 2016.06.07.16. |
| ## 6158 | 245 | 22 | 2016.06.07 | 115 | 2.00712864  | 2016.06.07.16. |



|         |     |    |            |     |             |                |
|---------|-----|----|------------|-----|-------------|----------------|
| ## 6159 | 240 | 23 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6160 | 240 | 24 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6161 | 245 | 25 | 2016.06.07 | 115 | 2.00712864  | 2016.06.07.16. |
| ## 6162 | 240 | 26 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6163 | 230 | 27 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.16. |
| ## 6164 | 220 | 28 | 2016.06.07 | 140 | 2.44346095  | 2016.06.07.16. |
| ## 6165 | 240 | 29 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6166 | 225 | 30 | 2016.06.07 | 135 | 2.35619449  | 2016.06.07.16. |
| ## 6167 | 240 | 31 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6168 | 220 | 32 | 2016.06.07 | 140 | 2.44346095  | 2016.06.07.16. |
| ## 6169 | 245 | 33 | 2016.06.07 | 115 | 2.00712864  | 2016.06.07.16. |
| ## 6170 | 280 | 34 | 2016.06.07 | 80  | 1.39626340  | 2016.06.07.16. |
| ## 6171 | 250 | 35 | 2016.06.07 | 110 | 1.91986218  | 2016.06.07.16. |
| ## 6172 | 245 | 36 | 2016.06.07 | 115 | 2.00712864  | 2016.06.07.16. |
| ## 6173 | 290 | 37 | 2016.06.07 | 70  | 1.22173048  | 2016.06.07.16. |
| ## 6174 | 240 | 38 | 2016.06.07 | 120 | 2.09439510  | 2016.06.07.16. |
| ## 6175 | 210 | 39 | 2016.06.07 | 150 | 2.61799388  | 2016.06.07.16. |
| ## 6176 | 185 | 40 | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.16. |
| ## 6177 | 235 | 41 | 2016.06.07 | 125 | 2.18166156  | 2016.06.07.16. |
| ## 6178 | 160 | 42 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.16. |
| ## 6179 | 245 | 43 | 2016.06.07 | 115 | 2.00712864  | 2016.06.07.16. |
| ## 6180 | 230 | 44 | 2016.06.07 | 130 | 2.26892803  | 2016.06.07.16. |
| ## 6181 | 160 | 45 | 2016.06.07 | 200 | -2.79252680 | 2016.06.07.16. |
| ## 6182 | 235 | 46 | 2016.06.07 | 125 | 2.18166156  | 2016.06.07.16. |
| ## 6183 | 200 | 47 | 2016.06.07 | 160 | 2.79252680  | 2016.06.07.16. |
| ## 6184 | 265 | 48 | 2016.06.07 | 95  | 1.65806279  | 2016.06.07.16. |
| ## 6185 | 210 | 49 | 2016.06.07 | 150 | 2.61799388  | 2016.06.07.16. |
| ## 6186 | 265 | 50 | 2016.06.07 | 95  | 1.65806279  | 2016.06.07.16. |
| ## 6187 | 260 | 51 | 2016.06.07 | 100 | 1.74532925  | 2016.06.07.16. |
| ## 6188 | 280 | 52 | 2016.06.07 | 80  | 1.39626340  | 2016.06.07.16. |
| ## 6189 | 235 | 53 | 2016.06.07 | 125 | 2.18166156  | 2016.06.07.16. |
| ## 6190 | 220 | 54 | 2016.06.07 | 140 | 2.44346095  | 2016.06.07.16. |
| ## 6191 | 210 | 55 | 2016.06.07 | 150 | 2.61799388  | 2016.06.07.16. |
| ## 6192 | 135 | 1  | 2016.06.07 | 225 | -2.35619449 | 2016.06.07.__. |
| ## 6193 | 30  | 2  | 2016.06.07 | 330 | -0.52359878 | 2016.06.07.__. |
| ## 6194 | 5   | 3  | 2016.06.07 | 355 | -0.08726646 | 2016.06.07.__. |
| ## 6195 | 350 | 4  | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.__. |
| ## 6196 | 15  | 5  | 2016.06.07 | 345 | -0.26179939 | 2016.06.07.__. |
| ## 6197 | 325 | 6  | 2016.06.07 | 35  | 0.61086524  | 2016.06.07.__. |
| ## 6198 | 330 | 7  | 2016.06.07 | 30  | 0.52359878  | 2016.06.07.__. |
| ## 6199 | 340 | 8  | 2016.06.07 | 20  | 0.34906585  | 2016.06.07.__. |
| ## 6200 | 295 | 9  | 2016.06.07 | 65  | 1.13446401  | 2016.06.07.__. |
| ## 6201 | 305 | 10 | 2016.06.07 | 55  | 0.95993109  | 2016.06.07.__. |
| ## 6202 | 350 | 11 | 2016.06.07 | 10  | 0.17453293  | 2016.06.07.__. |
| ## 6203 | 175 | 12 | 2016.06.07 | 185 | -3.05432619 | 2016.06.07.__. |
| ## 6204 | 185 | 13 | 2016.06.07 | 175 | 3.05432619  | 2016.06.07.__. |
| ## 6205 | 190 | 14 | 2016.06.07 | 170 | 2.96705973  | 2016.06.07.__. |
| ## 6206 | 170 | 15 | 2016.06.07 | 190 | -2.96705973 | 2016.06.07.__. |

|    |         |        |    |                     |          |             |                  |    |
|----|---------|--------|----|---------------------|----------|-------------|------------------|----|
| ## | 6207    | 180    | 16 | 2016.06.07          | 180      | 3.14159265  | 2016.06.07.____. |    |
| ## | 6208    | 175    | 17 | 2016.06.07          | 185      | -3.05432619 | 2016.06.07.____. |    |
| ## | 6209    | 180    | 18 | 2016.06.07          | 180      | 3.14159265  | 2016.06.07.____. |    |
| ## | 6210    | 185    | 19 | 2016.06.07          | 175      | 3.05432619  | 2016.06.07.____. |    |
| ## | 6211    | 160    | 20 | 2016.06.07          | 200      | -2.79252680 | 2016.06.07.____. |    |
| ## | 6212    | 185    | 21 | 2016.06.07          | 175      | 3.05432619  | 2016.06.07.____. |    |
| ## | 6213    | 180    | 22 | 2016.06.07          | 180      | 3.14159265  | 2016.06.07.____. |    |
| ## | 6214    | 190    | 23 | 2016.06.07          | 170      | 2.96705973  | 2016.06.07.____. |    |
| ## | 6215    | 170    | 24 | 2016.06.07          | 190      | -2.96705973 | 2016.06.07.____. |    |
| ## | 6216    | 180    | 25 | 2016.06.07          | 180      | 3.14159265  | 2016.06.07.____. |    |
| ## | 6217    | 265    | 26 | 2016.06.07          | 95       | 1.65806279  | 2016.06.07.____. |    |
| ## | 6218    | 180    | 27 | 2016.06.07          | 180      | 3.14159265  | 2016.06.07.____. |    |
| ## | 6219    | 190    | 28 | 2016.06.07          | 170      | 2.96705973  | 2016.06.07.____. |    |
| ## | 6220    | 60     | 29 | 2016.06.07          | 300      | -1.04719755 | 2016.06.07.____. |    |
| ## | 6221    | 250    | 30 | 2016.06.07          | 110      | 1.91986218  | 2016.06.07.____. |    |
| ## | 6222    | 245    | 31 | 2016.06.07          | 115      | 2.00712864  | 2016.06.07.____. |    |
| ## | 6223    | 150    | 32 | 2016.06.07          | 210      | -2.61799388 | 2016.06.07.____. |    |
| ## | 6224    | 220    | 33 | 2016.06.07          | 140      | 2.44346095  | 2016.06.07.____. |    |
| ## | 6225    | 170    | 34 | 2016.06.07          | 190      | -2.96705973 | 2016.06.07.____. |    |
| ## | 6226    | 235    | 35 | 2016.06.07          | 125      | 2.18166156  | 2016.06.07.____. |    |
| ## | 6227    | 200    | 36 | 2016.06.07          | 160      | 2.79252680  | 2016.06.07.____. |    |
| ## | 6228    | 210    | 37 | 2016.06.07          | 150      | 2.61799388  | 2016.06.07.____. |    |
| ## | 6229    | 185    | 38 | 2016.06.07          | 175      | 3.05432619  | 2016.06.07.____. |    |
| ## | 6230    | 165    | 39 | 2016.06.07          | 195      | -2.87979327 | 2016.06.07.____. |    |
| ## | 6231    | 170    | 40 | 2016.06.07          | 190      | -2.96705973 | 2016.06.07.____. |    |
| ## | brightn | colour |    | cet_time            | sun_az   | sun_el      | sun_el_rad       | su |
| ## | 1       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 2       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 3       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 4       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 5       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 6       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 7       | l      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 8       | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 9       | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 10      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 11      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 12      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 13      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 14      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 15      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 16      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 17      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 18      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 19      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 20      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 21      | h      | u  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |
| ## | 22      | h      | g  | 2016-05-31 14:24:00 | 208.3263 | 58.81733    | 1.0265560        | -  |





|        |   |   |            |          |          |          |           |   |
|--------|---|---|------------|----------|----------|----------|-----------|---|
| ## 71  | l | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 72  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 73  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 74  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 75  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 76  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 77  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 78  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 79  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 80  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 81  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 82  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 83  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 84  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 85  | h | u | 2016-05-31 | 14:57:00 | 221.3982 | 55.83673 | 0.9745348 | - |
| ## 86  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 87  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 88  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 89  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 90  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 91  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 92  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 93  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 94  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 95  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 96  | l | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 97  | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 98  | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 99  | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 100 | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 101 | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 102 | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 103 | h | u | 2016-05-31 | 15:14:00 | 227.4222 | 53.95921 | 0.9417660 | - |
| ## 104 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 105 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 106 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 107 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 108 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 109 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 110 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 111 | l | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 112 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 113 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 114 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 115 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 116 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 117 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |
| ## 118 | h | u | 2016-05-31 | 14:18:00 | 205.7573 | 59.24757 | 1.0340651 | - |

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|    |     |   |   |            |          |          |          |           |   |
|----|-----|---|---|------------|----------|----------|----------|-----------|---|
| ## | 887 | h | u | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 888 | h | u | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 889 | h | u | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 890 | h | u | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 891 | h | u | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 892 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 893 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
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| ## | 895 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 896 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 897 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 898 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 899 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 900 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 901 | h | g | 2016-07-01 | 14:49:00 | 216.7796 | 58.18252 | 1.0154765 | - |
| ## | 902 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 903 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 904 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 905 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 906 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 907 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 908 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 909 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 910 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 911 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 912 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 913 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 914 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 915 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 916 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 917 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 918 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 919 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 920 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 921 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 922 | h | u |            | <NA>     | NA       | NA       | NA        |   |
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| ## | 924 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 925 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 926 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 927 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 928 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 929 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 930 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 931 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 932 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 933 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 934 | h | u |            | <NA>     | NA       | NA       | NA        |   |

|    |     |   |   |                     |          |          |           |    |  |
|----|-----|---|---|---------------------|----------|----------|-----------|----|--|
| ## | 935 | h | u |                     | <NA>     | NA       | NA        | NA |  |
| ## | 936 | h | u |                     | <NA>     | NA       | NA        | NA |  |
| ## | 937 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 938 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 939 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 940 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 941 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 942 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 943 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 944 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 945 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 946 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 947 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 948 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 949 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 950 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 951 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 952 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 953 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 954 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 955 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 956 | h | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 957 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 958 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 959 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 960 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 961 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 962 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 963 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 964 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 965 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 966 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 967 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 968 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 969 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 970 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 971 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 972 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 973 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 974 | l | g |                     | <NA>     | NA       | NA        | NA |  |
| ## | 975 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 976 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 977 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 978 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 979 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 980 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 981 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |
| ## | 982 | l | u | 2016-07-05 13:38:00 | 184.1899 | 61.86641 | 1.0797725 | -  |  |

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|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 2279 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2280 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2281 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2282 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2283 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2284 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2285 | l | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2286 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2287 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2288 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2289 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2290 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2291 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2292 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2293 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2294 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2295 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2296 | h | u | 2016-08-18 | 16:26:00 | 239.6198 | 37.78775 | 0.6595207 | - |
| ## | 2297 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2298 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2299 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2300 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2301 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2302 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2303 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2304 | l | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2305 | h | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2306 | h | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2307 | h | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2308 | h | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2309 | h | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2310 | h | u | 2016-08-18 | 16:27:00 | 239.8749 | 37.65118 | 0.6571370 | - |
| ## | 2311 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2312 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2313 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2314 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2315 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2316 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2317 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2318 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2319 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2320 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2321 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2322 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2323 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2324 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2325 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |
| ## | 2326 | l | u | 2016-08-19 | 14:26:00 | 202.2213 | 49.96820 | 0.8721095 | - |



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|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 2663 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2664 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2665 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2666 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2667 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2668 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2669 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2670 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2671 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2672 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2673 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2674 | h | u | 2016-08-19 | 15:33:00 | 224.7273 | 44.09856 | 0.7696651 | - |
| ## | 2675 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2676 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2677 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2678 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2679 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2680 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2681 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2682 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2683 | l | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2684 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2685 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2686 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2687 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2688 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2689 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2690 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2691 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2692 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2693 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2694 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2695 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2696 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2697 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2698 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2699 | h | u | 2016-08-19 | 15:34:00 | 225.0292 | 43.98693 | 0.7677167 | - |
| ## | 2700 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2701 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2702 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2703 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2704 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2705 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2706 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2707 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2708 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2709 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |
| ## | 2710 | l | u | 2016-08-23 | 14:01:00 | 192.8769 | 49.80199 | 0.8692087 | - |

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|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 3143 | 1 | u | 2016-08-24 | 15:41:00 | 226.4174 | 41.55728 | 0.7253114 | - |
| ## | 3144 | 1 | u | 2016-08-24 | 15:41:00 | 226.4174 | 41.55728 | 0.7253114 | - |
| ## | 3145 | 1 | u | 2016-08-24 | 15:41:00 | 226.4174 | 41.55728 | 0.7253114 | - |
| ## | 3146 | 1 | u | 2016-08-24 | 15:41:00 | 226.4174 | 41.55728 | 0.7253114 | - |
| ## | 3147 | 1 | u | 2016-08-24 | 15:41:00 | 226.4174 | 41.55728 | 0.7253114 | - |
| ## | 3148 | 1 | u | 2016-08-24 | 15:41:00 | 226.4174 | 41.55728 | 0.7253114 | - |
| ## | 3149 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3150 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3151 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3152 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3153 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3154 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3155 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3156 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3157 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3158 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3159 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3160 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3161 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3162 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3163 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3164 | 1 | u | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3165 | 1 | g | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3166 | 1 | g | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3167 | 1 | g | 2016-08-25 | 13:01:00 | 170.2619 | 49.38196 | 0.8618778 |   |
| ## | 3168 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3169 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3170 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3171 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3172 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3173 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3174 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3175 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3176 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3177 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3178 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3179 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3180 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3181 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3182 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3183 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3184 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3185 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3186 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3187 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3188 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3189 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |
| ## | 3190 | 1 | u | 2016-08-25 | 13:14:00 | 175.1676 | 49.63927 | 0.8663688 |   |

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|    |      |   |   |            |          |          |          |           |   |
|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 4151 | l | g | 2016-09-08 | 18:00:00 | 257.1945 | 17.10279 | 0.2985000 | - |
| ## | 4152 | l | g | 2016-09-08 | 18:00:00 | 257.1945 | 17.10279 | 0.2985000 | - |
| ## | 4153 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4154 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4155 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4156 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4157 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4158 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4159 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4160 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4161 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4162 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4163 | l | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4164 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4165 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4166 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4167 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4168 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4169 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4170 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4171 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4172 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4173 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4174 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4175 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4176 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4177 | h | u | 2016-09-08 | 18:21:00 | 261.4316 | 13.85026 | 0.2417326 | - |
| ## | 4178 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4179 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4180 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4181 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4182 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4183 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4184 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4185 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4186 | l | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4187 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4188 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4189 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4190 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4191 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4192 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4193 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4194 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4195 | h | g | 2016-09-12 | 13:50:00 | 189.9102 | 42.71603 | 0.7455353 | - |
| ## | 4196 | l | u | 2016-09-12 | 14:26:00 | 201.8059 | 41.15140 | 0.7182274 | - |
| ## | 4197 | l | u | 2016-09-12 | 14:26:00 | 201.8059 | 41.15140 | 0.7182274 | - |
| ## | 4198 | l | u | 2016-09-12 | 14:26:00 | 201.8059 | 41.15140 | 0.7182274 | - |

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|    |      |   |   |            |          |          |          |           |   |
|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 4823 | l | g | 2016-09-14 | 14:59:00 | 211.9817 | 37.99835 | 0.6631963 | - |
| ## | 4824 | l | g | 2016-09-14 | 14:59:00 | 211.9817 | 37.99835 | 0.6631963 | - |
| ## | 4825 | l | g | 2016-09-14 | 14:59:00 | 211.9817 | 37.99835 | 0.6631963 | - |
| ## | 4826 | l | g | 2016-09-14 | 14:59:00 | 211.9817 | 37.99835 | 0.6631963 | - |
| ## | 4827 | l | g | 2016-09-14 | 14:59:00 | 211.9817 | 37.99835 | 0.6631963 | - |
| ## | 4828 | l | g | 2016-09-14 | 14:59:00 | 211.9817 | 37.99835 | 0.6631963 | - |
| ## | 4829 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4830 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4831 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4832 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4833 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4834 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4835 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4836 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4837 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4838 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4839 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4840 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4841 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4842 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4843 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4844 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4845 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4846 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4847 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4848 | l | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4849 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4850 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4851 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4852 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4853 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4854 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4855 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4856 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4857 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4858 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4859 | h | u | 2016-09-14 | 15:47:00 | 225.5284 | 33.23996 | 0.5801468 | - |
| ## | 4860 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4861 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4862 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4863 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4864 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4865 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4866 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4867 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4868 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4869 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |
| ## | 4870 | l | u | 2016-09-14 | 16:07:00 | 230.7227 | 30.88427 | 0.5390322 | - |

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|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 4967 | h | g | 2016-09-14 | 16:54:00 | 242.0032 | 24.70077 | 0.4311098 | - |
| ## | 4968 | h | g | 2016-09-14 | 16:54:00 | 242.0032 | 24.70077 | 0.4311098 | - |
| ## | 4969 | h | g | 2016-09-14 | 16:54:00 | 242.0032 | 24.70077 | 0.4311098 | - |
| ## | 4970 | h | g | 2016-09-14 | 16:54:00 | 242.0032 | 24.70077 | 0.4311098 | - |
| ## | 4971 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4972 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4973 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4974 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4975 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4976 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4977 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4978 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4979 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4980 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4981 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4982 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4983 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4984 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4985 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4986 | l | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4987 | h | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4988 | h | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4989 | h | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4990 | h | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4991 | h | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4992 | h | u | 2016-09-14 | 17:25:00 | 248.8508 | 20.24892 | 0.3534104 | - |
| ## | 4993 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 4994 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 4995 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 4996 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 4997 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 4998 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 4999 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5000 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5001 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5002 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5003 | l | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5004 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5005 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5006 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5007 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5008 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5009 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5010 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5011 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5012 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5013 | h | u | 2016-09-27 | 15:52:00 | 225.3685 | 27.66289 | 0.4828086 | - |
| ## | 5014 | l | u | 2016-09-29 | 14:21:00 | 200.2055 | 34.68574 | 0.6053803 | - |



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|    |      |   |   |            |          |          |          |           |   |
|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 5063 | h | u | 2016-09-29 | 14:52:00 | 209.1760 | 32.63247 | 0.5695441 | - |
| ## | 5064 | h | u | 2016-09-29 | 14:52:00 | 209.1760 | 32.63247 | 0.5695441 | - |
| ## | 5065 | h | u | 2016-09-29 | 14:52:00 | 209.1760 | 32.63247 | 0.5695441 | - |
| ## | 5066 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5067 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5068 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5069 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5070 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5071 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5072 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5073 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5074 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5075 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5076 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5077 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5078 | l | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5079 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5080 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5081 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5082 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5083 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5084 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5085 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5086 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5087 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5088 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5089 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5090 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5091 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5092 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5093 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5094 | h | u | 2016-09-29 | 15:15:00 | 215.5240 | 30.68357 | 0.5355294 | - |
| ## | 5095 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5096 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5097 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5098 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5099 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5100 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5101 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5102 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5103 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5104 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5105 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5106 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5107 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5108 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5109 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |
| ## | 5110 | l | u | 2016-09-29 | 15:43:00 | 222.8706 | 27.88340 | 0.4866572 | - |

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|    |      |   |   |            |          |          |          |           |   |
|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 6023 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6024 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6025 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6026 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6027 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6028 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6029 | h | u | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6030 | l | g | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6031 | l | g | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6032 | l | g | 2016-06-07 | 14:59:00 | 222.3496 | 56.46567 | 0.9855118 | - |
| ## | 6033 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6034 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6035 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6036 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6037 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6038 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6039 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6040 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6041 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6042 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6043 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6044 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6045 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6046 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6047 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6048 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6049 | l | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6050 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6051 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6052 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6053 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6054 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6055 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6056 | h | u | 2016-06-07 | 15:24:00 | 231.0672 | 53.59245 | 0.9353646 | - |
| ## | 6057 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6058 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6059 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6060 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6061 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6062 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6063 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6064 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6065 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6066 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6067 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6068 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6069 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |
| ## | 6070 | l | u | 2016-06-07 | 15:28:00 | 232.3682 | 53.09688 | 0.9267154 | - |

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|    |      |   |   |            |          |          |          |           |   |
|----|------|---|---|------------|----------|----------|----------|-----------|---|
| ## | 6167 | h | u | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6168 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6169 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6170 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6171 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6172 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6173 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6174 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6175 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6176 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6177 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6178 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6179 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6180 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6181 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6182 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6183 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6184 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6185 | l | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6186 | h | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6187 | h | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6188 | h | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6189 | h | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6190 | h | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6191 | h | g | 2016-06-07 | 16:04:00 | 243.0603 | 48.29276 | 0.8428677 | - |
| ## | 6192 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6193 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6194 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6195 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6196 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6197 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6198 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6199 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6200 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6201 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6202 | l | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6203 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6204 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6205 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6206 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6207 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6208 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6209 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6210 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6211 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6212 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6213 | h | u |            | <NA>     | NA       | NA       | NA        |   |
| ## | 6214 | h | u |            | <NA>     | NA       | NA       | NA        |   |



|    |              |      |   |      |    |    |    |
|----|--------------|------|---|------|----|----|----|
| ## | 6215         | h    | u | <NA> | NA | NA | NA |
| ## | 6216         | h    | u | <NA> | NA | NA | NA |
| ## | 6217         | l    | g | <NA> | NA | NA | NA |
| ## | 6218         | l    | g | <NA> | NA | NA | NA |
| ## | 6219         | l    | g | <NA> | NA | NA | NA |
| ## | 6220         | l    | g | <NA> | NA | NA | NA |
| ## | 6221         | l    | g | <NA> | NA | NA | NA |
| ## | 6222         | l    | g | <NA> | NA | NA | NA |
| ## | 6223         | l    | g | <NA> | NA | NA | NA |
| ## | 6224         | l    | g | <NA> | NA | NA | NA |
| ## | 6225         | l    | g | <NA> | NA | NA | NA |
| ## | 6226         | l    | g | <NA> | NA | NA | NA |
| ## | 6227         | l    | g | <NA> | NA | NA | NA |
| ## | 6228         | l    | g | <NA> | NA | NA | NA |
| ## | 6229         | h    | g | <NA> | NA | NA | NA |
| ## | 6230         | h    | g | <NA> | NA | NA | NA |
| ## | 6231         | h    | g | <NA> | NA | NA | NA |
| ## | signed_angle |      |   |      |    |    |    |
| ## | 1            | 90   |   |      |    |    |    |
| ## | 2            | -5   |   |      |    |    |    |
| ## | 3            | -20  |   |      |    |    |    |
| ## | 4            | 40   |   |      |    |    |    |
| ## | 5            | -20  |   |      |    |    |    |
| ## | 6            | -35  |   |      |    |    |    |
| ## | 7            | -10  |   |      |    |    |    |
| ## | 8            | 15   |   |      |    |    |    |
| ## | 9            | 20   |   |      |    |    |    |
| ## | 10           | 20   |   |      |    |    |    |
| ## | 11           | 20   |   |      |    |    |    |
| ## | 12           | 30   |   |      |    |    |    |
| ## | 13           | 10   |   |      |    |    |    |
| ## | 14           | 35   |   |      |    |    |    |
| ## | 15           | 25   |   |      |    |    |    |
| ## | 16           | 30   |   |      |    |    |    |
| ## | 17           | 40   |   |      |    |    |    |
| ## | 18           | 35   |   |      |    |    |    |
| ## | 19           | 35   |   |      |    |    |    |
| ## | 20           | 5    |   |      |    |    |    |
| ## | 21           | 20   |   |      |    |    |    |
| ## | 22           | 15   |   |      |    |    |    |
| ## | 23           | 15   |   |      |    |    |    |
| ## | 24           | 35   |   |      |    |    |    |
| ## | 25           | 10   |   |      |    |    |    |
| ## | 26           | 15   |   |      |    |    |    |
| ## | 27           | 5    |   |      |    |    |    |
| ## | 28           | 125  |   |      |    |    |    |
| ## | 29           | 175  |   |      |    |    |    |
| ## | 30           | -100 |   |      |    |    |    |

|       |      |
|-------|------|
| ## 31 | -140 |
| ## 32 | -90  |
| ## 33 | 170  |
| ## 34 | -175 |
| ## 35 | 110  |
| ## 36 | -130 |
| ## 37 | -160 |
| ## 38 | -15  |
| ## 39 | -45  |
| ## 40 | -15  |
| ## 41 | -20  |
| ## 42 | -40  |
| ## 43 | -20  |
| ## 44 | -10  |
| ## 45 | -50  |
| ## 46 | -10  |
| ## 47 | -40  |
| ## 48 | -40  |
| ## 49 | -50  |
| ## 50 | -50  |
| ## 51 | -10  |
| ## 52 | -40  |
| ## 53 | -35  |
| ## 54 | -20  |
| ## 55 | -30  |
| ## 56 | -40  |
| ## 57 | -20  |
| ## 58 | -25  |
| ## 59 | -20  |
| ## 60 | -40  |
| ## 61 | -20  |
| ## 62 | -20  |
| ## 63 | 160  |
| ## 64 | -140 |
| ## 65 | 140  |
| ## 66 | -150 |
| ## 67 | -170 |
| ## 68 | -100 |
| ## 69 | -155 |
| ## 70 | -90  |
| ## 71 | -155 |
| ## 72 | 5    |
| ## 73 | -40  |
| ## 74 | -70  |
| ## 75 | -30  |
| ## 76 | -70  |
| ## 77 | -25  |
| ## 78 | -20  |

|        |      |
|--------|------|
| ## 79  | -30  |
| ## 80  | -30  |
| ## 81  | -70  |
| ## 82  | -60  |
| ## 83  | -40  |
| ## 84  | -60  |
| ## 85  | -40  |
| ## 86  | -100 |
| ## 87  | -110 |
| ## 88  | -140 |
| ## 89  | -110 |
| ## 90  | -140 |
| ## 91  | -130 |
| ## 92  | -110 |
| ## 93  | -95  |
| ## 94  | -140 |
| ## 95  | -80  |
| ## 96  | -120 |
| ## 97  | -60  |
| ## 98  | -60  |
| ## 99  | -50  |
| ## 100 | -85  |
| ## 101 | -40  |
| ## 102 | -40  |
| ## 103 | -20  |
| ## 104 | 90   |
| ## 105 | 100  |
| ## 106 | 50   |
| ## 107 | 55   |
| ## 108 | -10  |
| ## 109 | 10   |
| ## 110 | 90   |
| ## 111 | 50   |
| ## 112 | -60  |
| ## 113 | -10  |
| ## 114 | -5   |
| ## 115 | -40  |
| ## 116 | -10  |
| ## 117 | -5   |
| ## 118 | -10  |
| ## 119 | -10  |
| ## 120 | -30  |
| ## 121 | 0    |
| ## 122 | -15  |
| ## 123 | -15  |
| ## 124 | -15  |
| ## 125 | -15  |
| ## 126 | -5   |

|        |      |
|--------|------|
| ## 127 | -5   |
| ## 128 | -40  |
| ## 129 | -20  |
| ## 130 | -30  |
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| ## | 1667 | 120 |
| ## | 1668 | 40  |
| ## | 1669 | 160 |
| ## | 1670 | 175 |
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| ## | 1672 | 155 |
| ## | 1673 | 160 |
| ## | 1674 | 150 |
| ## | 1675 | 150 |
| ## | 1676 | 140 |
| ## | 1677 | 140 |
| ## | 1678 | 150 |
| ## | 1679 | 140 |
| ## | 1680 | 170 |
| ## | 1681 | 150 |
| ## | 1682 | -75 |
| ## | 1683 | -15 |
| ## | 1684 | 75  |
| ## | 1685 | 110 |
| ## | 1686 | 60  |
| ## | 1687 | 50  |
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| ## | 1689 | -35 |
| ## | 1690 | 80  |
| ## | 1691 | 80  |
| ## | 1692 | 85  |
| ## | 1693 | -80 |
| ## | 1694 | -90 |
| ## | 1695 | -20 |
| ## | 1696 | -90 |
| ## | 1697 | 85  |
| ## | 1698 | 30  |
| ## | 1699 | 95  |
| ## | 1700 | 80  |
| ## | 1701 | 45  |
| ## | 1702 | 30  |
| ## | 1703 | 10  |
| ## | 1704 | 15  |
| ## | 1705 | -30 |
| ## | 1706 | 5   |
| ## | 1707 | -5  |
| ## | 1708 | -5  |
| ## | 1709 | 0   |
| ## | 1710 | 5   |

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| ## 1718 | -10 |
| ## 1719 | -5  |
| ## 1720 | 10  |
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| ## 1724 | 10  |
| ## 1725 | -5  |
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| ## 1727 | -5  |
| ## 1728 | -10 |
| ## 1729 | -15 |
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| ## 1734 | -5  |
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| ## 1737 | 0   |
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| ## 1740 | -10 |
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| ## 1742 | 0   |
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| ## 1745 | 100 |
| ## 1746 | 50  |
| ## 1747 | 20  |
| ## 1748 | 120 |
| ## 1749 | 30  |
| ## 1750 | 100 |
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| ## 1752 | 130 |
| ## 1753 | 50  |
| ## 1754 | 55  |
| ## 1755 | 40  |
| ## 1756 | 120 |
| ## 1757 | 90  |
| ## 1758 | 130 |

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| ## 1761 | 15   |
| ## 1762 | 5    |
| ## 1763 | 5    |
| ## 1764 | 20   |
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| ## 1768 | 15   |
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| ## 1770 | 10   |
| ## 1771 | 0    |
| ## 1772 | 30   |
| ## 1773 | 15   |
| ## 1774 | 35   |
| ## 1775 | 100  |
| ## 1776 | -150 |
| ## 1777 | 90   |
| ## 1778 | 90   |
| ## 1779 | 140  |
| ## 1780 | 110  |
| ## 1781 | 145  |
| ## 1782 | 110  |
| ## 1783 | -165 |
| ## 1784 | 50   |
| ## 1785 | 105  |
| ## 1786 | 155  |
| ## 1787 | 140  |
| ## 1788 | 110  |
| ## 1789 | -150 |
| ## 1790 | -160 |
| ## 1791 | 155  |
| ## 1792 | 120  |
| ## 1793 | 175  |
| ## 1794 | 140  |
| ## 1795 | 130  |
| ## 1796 | -160 |
| ## 1797 | 150  |
| ## 1798 | 115  |
| ## 1799 | -175 |
| ## 1800 | 150  |
| ## 1801 | 145  |
| ## 1802 | 140  |
| ## 1803 | 170  |
| ## 1804 | 130  |
| ## 1805 | -50  |
| ## 1806 | 60   |

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| ## 1808 | 110  |
| ## 1809 | 130  |
| ## 1810 | 100  |
| ## 1811 | -90  |
| ## 1812 | -30  |
| ## 1813 | 140  |
| ## 1814 | 120  |
| ## 1815 | 100  |
| ## 1816 | 100  |
| ## 1817 | -80  |
| ## 1818 | -145 |
| ## 1819 | -90  |
| ## 1820 | 95   |
| ## 1821 | 5    |
| ## 1822 | 0    |
| ## 1823 | -10  |
| ## 1824 | -20  |
| ## 1825 | -10  |
| ## 1826 | 15   |
| ## 1827 | 10   |
| ## 1828 | 25   |
| ## 1829 | 5    |
| ## 1830 | 0    |
| ## 1831 | -5   |
| ## 1832 | -10  |
| ## 1833 | -10  |
| ## 1834 | 20   |
| ## 1835 | -25  |
| ## 1836 | -5   |
| ## 1837 | 0    |
| ## 1838 | -5   |
| ## 1839 | -10  |
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| ## 1841 | 20   |
| ## 1842 | -25  |
| ## 1843 | -5   |
| ## 1844 | -20  |
| ## 1845 | 170  |
| ## 1846 | -30  |
| ## 1847 | 25   |
| ## 1848 | 25   |
| ## 1849 | -110 |
| ## 1850 | -25  |
| ## 1851 | 40   |
| ## 1852 | -115 |
| ## 1853 | 75   |
| ## 1854 | 40   |



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| ## 1856 | 70   |
| ## 1857 | -130 |
| ## 1858 | -70  |
| ## 1859 | 35   |
| ## 1860 | -10  |
| ## 1861 | 45   |
| ## 1862 | -30  |
| ## 1863 | 5    |
| ## 1864 | -40  |
| ## 1865 | 5    |
| ## 1866 | 5    |
| ## 1867 | -10  |
| ## 1868 | -20  |
| ## 1869 | -30  |
| ## 1870 | -35  |
| ## 1871 | -10  |
| ## 1872 | -20  |
| ## 1873 | -50  |
| ## 1874 | -30  |
| ## 1875 | 0    |
| ## 1876 | -15  |
| ## 1877 | -5   |
| ## 1878 | -25  |
| ## 1879 | 0    |
| ## 1880 | -30  |
| ## 1881 | 5    |
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| ## 1883 | -15  |
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| ## 1885 | -15  |
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| ## 1888 | -110 |
| ## 1889 | -80  |
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| ## 1891 | -35  |
| ## 1892 | 50   |
| ## 1893 | 45   |
| ## 1894 | 15   |
| ## 1895 | -135 |
| ## 1896 | -120 |
| ## 1897 | -155 |
| ## 1898 | -105 |
| ## 1899 | -165 |
| ## 1900 | -145 |
| ## 1901 | -90  |
| ## 1902 | -110 |

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| ## | 1904 | 140  |
| ## | 1905 | -90  |
| ## | 1906 | -160 |
| ## | 1907 | -60  |
| ## | 1908 | -45  |
| ## | 1909 | -45  |
| ## | 1910 | 5    |
| ## | 1911 | -55  |
| ## | 1912 | -15  |
| ## | 1913 | -45  |
| ## | 1914 | -55  |
| ## | 1915 | -60  |
| ## | 1916 | -20  |
| ## | 1917 | -30  |
| ## | 1918 | -20  |
| ## | 1919 | 145  |
| ## | 1920 | 5    |
| ## | 1921 | 150  |
| ## | 1922 | 140  |
| ## | 1923 | 125  |
| ## | 1924 | 90   |
| ## | 1925 | 40   |
| ## | 1926 | 150  |
| ## | 1927 | 110  |
| ## | 1928 | 40   |
| ## | 1929 | -20  |
| ## | 1930 | -30  |
| ## | 1931 | 80   |
| ## | 1932 | 115  |
| ## | 1933 | 75   |
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| ## | 1935 | 85   |
| ## | 1936 | 50   |
| ## | 1937 | 55   |
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| ## | 1939 | 90   |
| ## | 1940 | 50   |
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| ## | 1945 | 20   |
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| ## | 1947 | 10   |
| ## | 1948 | 20   |
| ## | 1949 | 15   |
| ## | 1950 | 10   |

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| ## | 1952 | 15   |
| ## | 1953 | 10   |
| ## | 1954 | 45   |
| ## | 1955 | 20   |
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| ## | 1957 | 20   |
| ## | 1958 | 50   |
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| ## | 1960 | -80  |
| ## | 1961 | -100 |
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| ## | 1963 | 160  |
| ## | 1964 | 30   |
| ## | 1965 | -80  |
| ## | 1966 | -90  |
| ## | 1967 | -110 |
| ## | 1968 | -30  |
| ## | 1969 | -90  |
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| ## | 1992 | -50  |
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| ## | 1998 | -25  |

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|----|------|------|
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| ## | 2000 | 160  |
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| ## | 2003 | 60   |
| ## | 2004 | 30   |
| ## | 2005 | 15   |
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| ## | 2009 | 60   |
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| ## | 2011 | 50   |
| ## | 2012 | 50   |
| ## | 2013 | 15   |
| ## | 2014 | 5    |
| ## | 2015 | 55   |
| ## | 2016 | 35   |
| ## | 2017 | 35   |
| ## | 2018 | 10   |
| ## | 2019 | 10   |
| ## | 2020 | 0    |
| ## | 2021 | 5    |
| ## | 2022 | 10   |
| ## | 2023 | 20   |
| ## | 2024 | 50   |
| ## | 2025 | 15   |
| ## | 2026 | 20   |
| ## | 2027 | 5    |
| ## | 2028 | 20   |
| ## | 2029 | 20   |
| ## | 2030 | 30   |
| ## | 2031 | 30   |
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| ## | 2033 | -80  |
| ## | 2034 | -40  |
| ## | 2035 | -70  |
| ## | 2036 | -40  |
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|----|------|------|
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| ## | 2050 | 70   |
| ## | 2051 | 50   |
| ## | 2052 | 70   |
| ## | 2053 | 80   |
| ## | 2054 | 15   |
| ## | 2055 | 30   |
| ## | 2056 | 65   |
| ## | 2057 | 175  |
| ## | 2058 | -65  |
| ## | 2059 | -120 |
| ## | 2060 | 175  |
| ## | 2061 | 160  |
| ## | 2062 | -90  |
| ## | 2063 | -175 |
| ## | 2064 | -155 |
| ## | 2065 | -105 |
| ## | 2066 | -115 |
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| ## | 2071 | -110 |
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| ## | 2076 | 30   |
| ## | 2077 | 40   |
| ## | 2078 | 60   |
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| ## | 2085 | 35   |
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| ## | 2091 | -135 |
| ## | 2092 | -170 |
| ## | 2093 | -150 |
| ## | 2094 | 160  |

|    |      |      |
|----|------|------|
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| ## | 2100 | -50  |
| ## | 2101 | -120 |
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| ## | 2112 | 20   |
| ## | 2113 | 30   |
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| ## | 2119 | 40   |
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| ## | 2122 | -110 |
| ## | 2123 | -160 |
| ## | 2124 | 10   |
| ## | 2125 | 115  |
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| ## | 2127 | -40  |
| ## | 2128 | -50  |
| ## | 2129 | 80   |
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| ## | 2133 | 150  |
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| ## | 2135 | 130  |
| ## | 2136 | -70  |
| ## | 2137 | 150  |
| ## | 2138 | 40   |
| ## | 2139 | 10   |
| ## | 2140 | 15   |
| ## | 2141 | 70   |
| ## | 2142 | 70   |

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|----|------|------|
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| ## | 2145 | 60   |
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| ## | 2149 | 55   |
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| ## | 2152 | 80   |
| ## | 2153 | -30  |
| ## | 2154 | 15   |
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| ## | 2156 | 180  |
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| ## | 2159 | 0    |
| ## | 2160 | 115  |
| ## | 2161 | -10  |
| ## | 2162 | 80   |
| ## | 2163 | -15  |
| ## | 2164 | 90   |
| ## | 2165 | 100  |
| ## | 2166 | 30   |
| ## | 2167 | 90   |
| ## | 2168 | 40   |
| ## | 2169 | 20   |
| ## | 2170 | 40   |
| ## | 2171 | 20   |
| ## | 2172 | 70   |
| ## | 2173 | 85   |
| ## | 2174 | 90   |
| ## | 2175 | 100  |
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| ## | 2177 | 90   |
| ## | 2178 | 120  |
| ## | 2179 | 40   |
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| ## | 2188 | -75  |
| ## | 2189 | -35  |
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|    |      |      |
|----|------|------|
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| ## | 2192 | -80  |
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| ## | 2195 | -85  |
| ## | 2196 | -80  |
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| ## | 2207 | 15   |
| ## | 2208 | 20   |
| ## | 2209 | 5    |
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| ## | 2213 | -75  |
| ## | 2214 | -110 |
| ## | 2215 | 120  |
| ## | 2216 | 150  |
| ## | 2217 | 160  |
| ## | 2218 | -120 |
| ## | 2219 | -95  |
| ## | 2220 | 160  |
| ## | 2221 | -85  |
| ## | 2222 | -20  |
| ## | 2223 | 10   |
| ## | 2224 | 15   |
| ## | 2225 | -5   |
| ## | 2226 | 20   |
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| ## | 2228 | 20   |
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| ## | 2230 | 5    |
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| ## | 2232 | 20   |
| ## | 2233 | 0    |
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| ## | 2235 | -20  |
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| ## | 2237 | 180  |
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|----|------|------|
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| ## | 2240 | -170 |
| ## | 2241 | 120  |
| ## | 2242 | -170 |
| ## | 2243 | 120  |
| ## | 2244 | -130 |
| ## | 2245 | 160  |
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| ## | 2248 | -130 |
| ## | 2249 | -20  |
| ## | 2250 | 0    |
| ## | 2251 | 5    |
| ## | 2252 | 10   |
| ## | 2253 | 10   |
| ## | 2254 | 10   |
| ## | 2255 | 40   |
| ## | 2256 | 10   |
| ## | 2257 | 5    |
| ## | 2258 | 15   |
| ## | 2259 | 10   |
| ## | 2260 | -10  |
| ## | 2261 | 0    |
| ## | 2262 | -5   |
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| ## | 2264 | -5   |
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| ## | 2268 | 5    |
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| ## | 2272 | -110 |
| ## | 2273 | -70  |
| ## | 2274 | -40  |
| ## | 2275 | -110 |
| ## | 2276 | -30  |
| ## | 2277 | -40  |
| ## | 2278 | -80  |
| ## | 2279 | 180  |
| ## | 2280 | 180  |
| ## | 2281 | -110 |
| ## | 2282 | -95  |
| ## | 2283 | 150  |
| ## | 2284 | -30  |
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| ## | 2286 | -20  |

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|----|------|------|
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| ## | 2292 | 5    |
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| ## | 2294 | 5    |
| ## | 2295 | 80   |
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| ## | 2297 | -90  |
| ## | 2298 | 10   |
| ## | 2299 | -90  |
| ## | 2300 | 30   |
| ## | 2301 | -65  |
| ## | 2302 | -95  |
| ## | 2303 | 5    |
| ## | 2304 | -30  |
| ## | 2305 | 5    |
| ## | 2306 | 20   |
| ## | 2307 | 10   |
| ## | 2308 | 20   |
| ## | 2309 | 5    |
| ## | 2310 | 15   |
| ## | 2311 | -145 |
| ## | 2312 | -30  |
| ## | 2313 | -150 |
| ## | 2314 | -70  |
| ## | 2315 | -60  |
| ## | 2316 | -170 |
| ## | 2317 | 10   |
| ## | 2318 | 0    |
| ## | 2319 | -70  |
| ## | 2320 | -170 |
| ## | 2321 | -175 |
| ## | 2322 | -160 |
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| ## | 2325 | -110 |
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| ## | 2331 | -30  |
| ## | 2332 | 10   |
| ## | 2333 | 5    |
| ## | 2334 | 15   |

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| ## | 2344 | -140 |
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| ## | 2347 | -110 |
| ## | 2348 | -160 |
| ## | 2349 | 130  |
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| ## | 2351 | 25   |
| ## | 2352 | 80   |
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| ## | 2355 | 70   |
| ## | 2356 | 90   |
| ## | 2357 | 90   |
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| ## | 2359 | 85   |
| ## | 2360 | -10  |
| ## | 2361 | 0    |
| ## | 2362 | -10  |
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| ## | 2364 | -20  |
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| ## | 2367 | -5   |
| ## | 2368 | 0    |
| ## | 2369 | -50  |
| ## | 2370 | 0    |
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| ## | 2372 | 90   |
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| ## | 2376 | -110 |
| ## | 2377 | 120  |
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| ## | 2379 | -110 |
| ## | 2380 | 170  |
| ## | 2381 | 10   |
| ## | 2382 | 130  |

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|----|------|------|
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| ## | 2386 | -20  |
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| ## | 2388 | 20   |
| ## | 2389 | 5    |
| ## | 2390 | -10  |
| ## | 2391 | -40  |
| ## | 2392 | 0    |
| ## | 2393 | 0    |
| ## | 2394 | 5    |
| ## | 2395 | 0    |
| ## | 2396 | -5   |
| ## | 2397 | 0    |
| ## | 2398 | -20  |
| ## | 2399 | 10   |
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| ## 6090 | -50 |
| ## 6091 | -25 |
| ## 6092 | -50 |
| ## 6093 | -35 |
| ## 6094 | -80 |
| ## 6095 | -35 |
| ## 6096 | -65 |
| ## 6097 | -50 |
| ## 6098 | -55 |
| ## 6099 | -40 |
| ## 6100 | -70 |
| ## 6101 | 5   |
| ## 6102 | 10  |
| ## 6103 | -15 |
| ## 6104 | -5  |
| ## 6105 | 175 |
| ## 6106 | 25  |
| ## 6107 | -15 |
| ## 6108 | 0   |
| ## 6109 | 30  |
| ## 6110 | -10 |
| ## 6111 | 10  |
| ## 6112 | 60  |
| ## 6113 | 5   |
| ## 6114 | 0   |
| ## 6115 | 50  |
| ## 6116 | 30  |
| ## 6117 | 15  |
| ## 6118 | 20  |
| ## 6119 | -30 |
| ## 6120 | 20  |
| ## 6121 | -40 |
| ## 6122 | -45 |
| ## 6123 | -40 |
| ## 6124 | -50 |
| ## 6125 | -60 |
| ## 6126 | -50 |

|         |      |
|---------|------|
| ## 6127 | -40  |
| ## 6128 | -50  |
| ## 6129 | -30  |
| ## 6130 | -60  |
| ## 6131 | -40  |
| ## 6132 | -45  |
| ## 6133 | -45  |
| ## 6134 | -35  |
| ## 6135 | -15  |
| ## 6136 | -10  |
| ## 6137 | 140  |
| ## 6138 | 110  |
| ## 6139 | 130  |
| ## 6140 | 110  |
| ## 6141 | -140 |
| ## 6142 | -120 |
| ## 6143 | -120 |
| ## 6144 | -150 |
| ## 6145 | -125 |
| ## 6146 | 165  |
| ## 6147 | 105  |
| ## 6148 | 175  |
| ## 6149 | 80   |
| ## 6150 | -145 |
| ## 6151 | 175  |
| ## 6152 | 120  |
| ## 6153 | 130  |
| ## 6154 | 135  |
| ## 6155 | 130  |
| ## 6156 | 135  |
| ## 6157 | 105  |
| ## 6158 | 115  |
| ## 6159 | 120  |
| ## 6160 | 120  |
| ## 6161 | 115  |
| ## 6162 | 120  |
| ## 6163 | 130  |
| ## 6164 | 140  |
| ## 6165 | 120  |
| ## 6166 | 135  |
| ## 6167 | 120  |
| ## 6168 | 140  |
| ## 6169 | 115  |
| ## 6170 | 80   |
| ## 6171 | 110  |
| ## 6172 | 115  |
| ## 6173 | 70   |
| ## 6174 | 120  |

|         |      |
|---------|------|
| ## 6175 | 150  |
| ## 6176 | 175  |
| ## 6177 | 125  |
| ## 6178 | -160 |
| ## 6179 | 115  |
| ## 6180 | 130  |
| ## 6181 | -160 |
| ## 6182 | 125  |
| ## 6183 | 160  |
| ## 6184 | 95   |
| ## 6185 | 150  |
| ## 6186 | 95   |
| ## 6187 | 100  |
| ## 6188 | 80   |
| ## 6189 | 125  |
| ## 6190 | 140  |
| ## 6191 | 150  |
| ## 6192 | -135 |
| ## 6193 | -30  |
| ## 6194 | -5   |
| ## 6195 | 10   |
| ## 6196 | -15  |
| ## 6197 | 35   |
| ## 6198 | 30   |
| ## 6199 | 20   |
| ## 6200 | 65   |
| ## 6201 | 55   |
| ## 6202 | 10   |
| ## 6203 | -175 |
| ## 6204 | 175  |
| ## 6205 | 170  |
| ## 6206 | -170 |
| ## 6207 | 180  |
| ## 6208 | -175 |
| ## 6209 | 180  |
| ## 6210 | 175  |
| ## 6211 | -160 |
| ## 6212 | 175  |
| ## 6213 | 180  |
| ## 6214 | 170  |
| ## 6215 | -170 |
| ## 6216 | 180  |
| ## 6217 | 95   |
| ## 6218 | 180  |
| ## 6219 | 170  |
| ## 6220 | -60  |
| ## 6221 | 110  |
| ## 6222 | 115  |



```
## 6223      -150
## 6224       140
## 6225     -170
## 6226       125
## 6227       160
## 6228       150
## 6229       175
## 6230     -165
## 6231     -170
```

```
#Edrich et al. 1979
print(ed)
```

```
##          xx intensity      nn accuracy wavelength colour log10_intensity
## 1  -31.208  2.15e+09 13.010      0.14         354      UV          9.332438
## 2  -20.263  3.69e+09 14.249      0.15         535 0green          9.567026
## 3   -8.217  6.67e+09 19.135      0.20         354      UV          9.824126
## 4   -1.342  9.36e+09 15.350      0.16         535 0green          9.971276
## 5   15.118  2.10e+10 30.769      0.33         354      UV         10.322219
## 6   24.548  3.35e+10 14.731      0.16         535 0green         10.525045
## 7   32.671  4.99e+10 70.349      0.74         354      UV         10.698101
## 8   45.199  9.25e+10 37.928      0.40         535 0green         10.966142
## 9   42.996  8.30e+10 10.463      0.11         429   blue         10.919078
## 10  61.994  2.12e+11 75.030      0.79         354      UV         11.326336
## 11  69.360  3.04e+11 70.349      0.74         535 0green         11.482874
## 12  71.254  3.34e+11 48.460      0.51         429   blue         11.523746
## 13  96.412  1.15e+12 85.355      0.90         535 0green         12.060698
## 14  95.448  1.10e+12 72.208      0.76         429   blue         12.041393
## 15 104.741  1.74e+12 78.059      0.83         354      UV         12.240549
## 16 118.026  3.34e+12 80.468      0.85         429   blue         12.523746
## 17 120.986  3.86e+12 89.657      0.95         535 0green         12.586587
## 18 141.233  1.05e+13 90.189      0.95         535 0green         13.021189
## 19 141.352  1.05e+13 83.978      0.89         429   blue         13.021189
## 20 165.252  3.41e+13 91.722      0.97         535 0green         13.532754
##      logit_accuracy
## 1      -1.81528997
## 2      -1.73460106
## 3      -1.38629436
## 4      -1.65822808
## 5      -0.70818506
## 6      -1.65822808
## 7       1.04596856
## 8      -0.40546511
## 9      -2.09074110
## 10     1.32492541
## 11     1.04596856
```

|       |            |
|-------|------------|
| ## 12 | 0.04000533 |
| ## 13 | 2.19722458 |
| ## 14 | 1.15267951 |
| ## 15 | 1.58562726 |
| ## 16 | 1.73460106 |
| ## 17 | 2.94443898 |
| ## 18 | 2.94443898 |
| ## 19 | 2.09074110 |
| ## 20 | 3.47609869 |

## Session Information

► Click to expand session metadata

